ACADEMIC REGULATIONS (R23)
COURSE STRUCTURE
AND
DETAILED SYLLABI

FOR

B. Tech Regular Four-Year Degree Courses
(For the Batches Admitted From 2023-2024)

&

B. Tech (Lateral Entry Scheme)
(For the Batches Admitted From 2024-2025)

COMPUTER SCIENCE AND ENGINEERING(AI&ML)

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

(Affiliated to JNTUA, Ananthapuramu, Approved by AICTE, New Delhi)

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FOREWORD

The autonomy conferred Sri Venkateswara College Engineering and technology by JNT University, Ananthapuramu based on performance as well as future commitment and competency to impart quality education. It is a mark of its ability to function independently in accordance with the set norms the monitoring bodies UGC and AICTE. It reflects the confidence of the affiliating University in the autonomous institution to uphold and maintain standards it expects to deliver on its own behalf and thus awards degrees on behalf of college. Thus, an autonomous institution is given the freedom to have its own curriculum, examination system and monitoring mechanism, independent of the affiliating University but under its observance.

Sri Venkateswara College of Engineering and Technology is proud to win the confidence of all the above bodies monitoring the quality in education and has gladly accepted the responsibility of sustaining, the standards and ethics it has been striving for more than a decade in reaching its present standing in the arena of contemporary technical education.

As a follow up, statutory bodies like Academic Council and

Boards of Studies are constituted with the guidance of the Governing Body of the College and recommendations of the JNTUA, Ananthapuramu to frame the regulations, course structure and syllabi under autonomous status.

The autonomous regulations, course structure and syllabi have been prepared after prolonged and detailed interaction with several expertise solicited from academics, industry and research, to produce quality engineering graduates to the society.

All the faculty, parents and students are requested to go through all the rules and regulations carefully. Any clarifications needed are to be sought at appropriate time and with principal of the college, without presumptions, to avoid unwanted subsequent inconveniences and embarrassments. The cooperation of all the stake holders is sought for the successful implementation of the autonomous system in the larger interests of the college and brighter prospects of engineering graduates.

Principal

Vision and Mission of the Institute

Vision

- Carving the youth as dynamic, competent, valued and knowledgeable professionals who shall lead the Nation to a better future.

Mission

- Providing quality education, student-centered teaching- learning processes and state-of-art infrastructure for professional aspirants hailing from both rural and urban areas.
- Imparting technical and management education to encourage independent thinking, develop strong domain of knowledge, own contemporary skills and positive attitudes towards holistic growth of young minds.
- Evolving Institution into a Center of Excellence and Research.

Quality policies

Sri Venkateswara College of Engineering and Technology strides towards excellence by adopting a system of quality policies and processes with continued improvements to enhance student's skills and talent for their exemplary contribution to the society, the nation and the word.

Vision and Mission of the Department under R23 Regulations

VISION

- To achieve excellent standard of quality education by using latest tools in Artificial Intelligence and disseminating innovations to relevant areas.

MISSION

- To develop professionals who are skilled in Artificial Intelligence and Machine Learning.
- Impart rigorous training to generate knowledge through the state-of-the-art concepts and technologies in Artificial Intelligence and Machine Learning.
- Establish centers of excellence in leading areas of computing and artificial intelligence to inculcate strong ethical values, innovative research capabilities and leadership abilities in the young minds to work with a commitment to the progress of the nation.

Program Educational Objectives (PEOs):

PEO1: To be able to solve wide range of computing related problems to cater to the needs of industry and society.

PEO2: Enable students to build intelligent machines and applications with a cutting-edge combination of machine learning, analytics and visualization.

PEO3: Produce graduates having professional competence through life-long learning such as advanced degrees, professional skills and other professional activities related globally to engineering & society.

Program Specific Outcomes (PSOs)

PSO1: Should have an ability to apply technical knowledge and usage of modern hardware and software tools related AI and ML for solving real world problems.

PSO2: Should have the capability to develop many successful applications based on machine learning methods, AI methods in different fields, including neural networks, signal processing, and data mining.

Engineering Graduates will be able to:

1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Academic Regulations (R23) for B. Tech
(Regular-Full time)

(Effective for the students admitted into I year fromthe Academic Year 2023-24 onwards and

B. Tech. (Lateral Entry Scheme) for the batches admitted from the Academic Year 2024-25 onwards)

1. Award of the Degree

(a) Award of the B.Tech. Degree / B.Tech. Degree with a Minor if he/she fulfils the following:

(i) Pursues a course of study for not less than four academic years and not more than eight academic years. However, for the students availing Gap year facility this period shall be extended by two years at the most and these two years would in addition to the maximum period permitted for graduation (Eight years).

(ii) Registers for 163 credits and secures all 163 credits.

(b) **Award of B.Tech. degree with Honors** if he/she fulfils the following:

(i) Student secures additional 18 credits fulfilling all the requisites of a B.Tech. program i.e., 163 credits.

(ii) Registering for Honors is optional.

(iii) Honors is to be completed simultaneously with B.Tech. programme.

2. Students, who fail to fulfil all the academic requirements for the award of the degree within eight academic years from the year of their admission, shall forfeit their seat in B.Tech. course and their admission stands cancelled. This clause shall be read along with clause 1 a) i).

3. Admissions

Admission to the B. Tech Program shall be made subject to the eligibility, qualifications and specialization prescribed by the A.P. State Government/University from time to time. Admissions shall be made either based on the merit rank obtained by the student in the common entrance examination conducted by the A.P. Government/University or any other order of merit approved by the A.P. Government/University, subject to reservations as prescribed by the Government/University from time to time.

4. Program related terms

Credit: A unit by which the course work is measured. It determines the number of hours of instruction required per week. One credit is equivalent to one hour of teaching (Lecture/Tutorial) or two hours of practical work/field work per week.

Credit Definition:

1 Hr. Lecture (L) per week	1 credit
1 Hr. Tutorial (T) per week	1 credit
1 Hr. Practical (P) per week	0.5 credit
2 Hrs. Practical (Lab) per week	1 credit

a) ***Academic Year:*** Two consecutive (one odd + one even) semesters constitute one academic year.

b) ***Choice Based Credit System (CBCS):*** The CBCS provides a choice for students to select from the prescribed courses.

5. **Semester/Credits:**

i) A semester comprises 90 instructional days and an academic year is divided into two semesters.
ii) The summer term is for eight weeks during summer vacation. Internship/ apprenticeship / work-based vocational education and training can be carried out during the summer term, especially by students who wish to exit after two semesters or four semesters of study.
iii) Regular courses may also be completed well in advance through MOOCs satisfying prerequisites.

6. **Structure of the Undergraduate Programme:**

All courses offered for the undergraduate program (B. Tech.) are broadly classified as follows:

S. No.	**Category**	**Breakup of Credits (Total 163)**	**Percentage of total credits**	**AICTE Recommendation (%)**
1.	Humanities and Social Science including Management (HM)	13	8%	8 – 9%
2.	Basic Sciences (BS)	20	13%	12 - 16%
3.	Engineering Sciences (ES)	23.5	14%	10 – 18%
4.	Professional Core (PC)	57.5	34%	30 – 36%
5.	Electives – Professional (PE) & Open (OE); Domain Specific Skill Enhancement Courses (SEC)	33	21%	19 - 23%
6.	Internships & Project work (PR)	16	10%	8 – 11%
7.	Mandatory Courses (MC)	Non-credit	Non-credit	-

7. Course Classification:

All subjects/ courses offered for the undergraduate programme in Engineering & Technology (B.Tech. degree programmes) are broadly classified as follows:

S. No.	Broad Course Classification	Course Category	Description
1.	Foundation Courses	Foundation courses	Includes Mathematics, Physics and Chemistry; fundamental engineering courses; humanities, social sciences and management courses
2.	Core Courses	Professional Core Courses (PC)	Includes subjectsrelated to the parent discipline/ department/ branch of Engineering
3.	Elective Courses	Professional Elective Courses (PE)	Includes elective subjects related to the parent discipline/department/ branch of Engineering
		Open Elective Courses (OE)	Elective subjects which include interdisciplinary subjects or subjects in an area outside the parent discipline/ department/ branch of Engineering
		Domain specific skill enhancement courses (SEC)	interdisciplinary/job-oriented/ domain courses which are relevant to the industry
4.	Project & Internships	Project	B.Tech. Project or Major Project
		Internships	Summer Internships – Community based and Industry Internships; Industry oriented Full Semester Internship
5.	Audit Courses	Mandatory non-credit courses	Covering subjects of developing desired attitude among the learners

8. Programme Pattern

i. Total duration of the of B. Tech (Regular) Programme is four academic years.
ii. Each academic year of study is divided into two semesters.
iii. Minimum number of instructional days in each semester is 90 days.
iv. There shall be mandatory student induction program for fresher's, with a three- week duration before the commencement of first semester. Physical activity, Creative Arts, Universal Human Values, Literary, Proficiency Modules, Lectures by Eminent People, Visits to local Areas,

Familiarization to Dept./Branch & Innovations etc., are included as per the guidelines issued by AICTE.

v. Health/wellness/yoga/sports and NSS /NSS /Scouts & Guides / Community service activities are made mandatory as credit courses for all the under graduate students.

vi. Courses like Environmental Sciences, Indian Constitution, Technical Paper Writing & IPR are offered as non-credit mandatory courses for all the undergraduate students.

vii. Design Thinking for Innovation & Tinkering Labs are made mandatory as credit courses for all the undergraduate students.

viii. Increased flexibility for students through an increase in the elective component of the curriculum, with 05 Professional Elective courses and 04 Open Elective courses.

ix. Professional Elective Courses, include the elective courses relevant to the chosen specialization/branch. Proper choice of professional elective courses can lead to students specializing in emerging areas within the chosen field of study.

x. While choosing the electives, students shall ensure that they do not opt for the courses with syllabus contents similar to courses already pursued.

xi. A pool of interdisciplinary/job-oriented/domain skill courses which are relevant to the industry are integrated into the curriculum of all disciplines. There shall be 05 skill-oriented courses offered during III to VII semesters. Among the five skill courses, four courses shall focus on the basic and advanced skills related to the domain/interdisciplinary courses and the other shall be a soft skills course.

xii. Students shall undergo mandatory summer internships, for a minimum of eight weeks duration at the end of second and third year of the programme. The internship at the end of second year shall be community oriented and industry internship at the end of third year.

xiii. There shall also be mandatory full internship in the final semester of the programme along with the project work.

xiv. Undergraduate degree with Honors is introduced by the Institution for the students having good academic record.

xv. The college shall take measures to implement Virtual Labs (https://www.vlab.co.in) which provide remote access to labs in various disciplines of Engineering and will help student in learning basic and advanced concept through remote experimentation. Student shall be made to work on virtual lab experiments during the regular labs.

xvi. The college shall assign a faculty advisor/mentor after admission to a group of students from same department to provide guidance in courses registration/career growth/placements/opportunities for higher studies/GATE/other competitive exams etc.

xvii. Preferably 25% of course work for the theory courses in every semester shall be conducted in the blended mode of learning.

9. Evaluation Process

The performance of a student in each semester shall be evaluated subject wise with a maximum of 100 marks for theory and 100 marks for practical subject. Summer Internships shall be evaluated for 50 marks, Full Internship & Project work in final semester shall be evaluated for 200 marks, mandatory courses with no credits shall be evaluated for 30 mid semester marks.

A student has to secure not less than 35% of marks in the end examination and a minimum of 40% of marks in the sum total of the mid semester and end examination marks taken together for the theory, practical, design, drawing subject or project etc. In case of a mandatory course, he/she should secure 40% of the total marks.

Theory Courses

Assessment Method	Marks
Continuous Internal Assessment	30
Semester End Examination	70
Total	**100**

i) For theory subject, the distribution shall be 30 marks for Internal Evaluation and 70 marks for the End-Examination.
ii) For practical subject, the distribution shall be 30 marks for Internal Evaluation and 70 marks for the End- Examination.
iii) If any course contains two different branch subjects, the syllabus shall be written in two parts with 3 units each (Part-A and Part-B) and external examination question paper shall be set with two parts each for 35 marks.
iv) If any subject is having both theory and practical components, they will be evaluated separately as theory subject and practical subject. However, they will be given same subject code with an extension of 'T' for theory subject and 'P' for practical subject.

a) Continuous Internal Evaluation

i) For theory subjects, during the semester, there shall be two midterm examinations. Each midterm examination shall be evaluated for 30 marks of which 10 marks for objective paper (20 minutes duration), 15 marks for subjective paper (90 minutes duration) and 5 marks for assignment.
ii) Objective paper shall contain for 05 short answer questions with 2 marks each or maximum of 20 bits for 10 marks. Subjective paper shall contain 3 either or type questions (totally six questions from 1 to 6) of which student has to answer one from each either-or type of questions. Each question carries 10 marks. The marks obtained in the subjective paper are condensed to 15 marks.

Note:

- The objective paper shall be prepared in line with the quality of competitive examinations questions.

- The subjective paper shall contain 3 either or type questions of equal weight age of 10 marks. Any fraction shall be rounded off to the next higher mark.
- The objective paper shall be conducted by the respective institution on the day of subjective paper test.
- Assignments shall be in the form of problems, mini projects, design problems, slip tests, quizzes etc., depending on the course content. It should be continuous assessment throughout the semester and the average marks shall be considered.

iii) If the student is absent for the mid semester examination, no re-exam shall be conducted and mid semester marks for that examination shall be considered as zero.

iv) First midterm examination shall be conducted for I, II units of syllabus with one either or type question from each unit and third either or type question from both the units. The second midterm examination shall be conducted for III, IV and V units with one either or type question from each unit.

v) Final mid semester marks shall be arrived at by considering the marks secured by the student in both the mid examinations with 80% weight age given to the better mid exam and 20% to the other.

For Example:

Marks obtained in first mid: 25
Marks obtained in second mid: 20
Final mid semester Marks: (25x0.8) + (20x0.2) = 24

If the student is absent for any one midterm examination, the final mid semester marks shall be arrived at by considering 80% weight age to the marks secured by the student in the appeared examination and zero to the other. For Example:

Marks obtained in first mid: Absent
Marks obtained in second mid: 25
Final mid semester Marks: (25x0.8) + (0x0.2) =20

b) **End Examination Evaluation:**

End examination of theory subjects shall have the following pattern:

i) There shall be 6 questions and all questions are compulsory.

ii) Question I shall contain 10 compulsory short answer questions for a total of 20marks such that each question carries 2 marks. There shall be 2 short answer questions from each unit.

iii) In each of the questions from 2 to 6

a) There shall be either/or type questions of 10 marks each. Student shall answer any one of them.

b) The questions from 2 to 6 shall be set by covering one unit of the syllabus for each question.

End examination of theory subjects consisting of two parts of different subjects, for Example: Basic Electrical &Electronics Engineering shall have the following pattern:

i) Question paper shall be in two parts viz., Part A and Part B with equal weightage of 35 marks each.
ii) In each part, question 1 shall contain 5 compulsory short answer questions for a total of 5 marks such that each question carries 1mark.
iii) In each part, questions from 2 to 4, there shall be either/or type questions of 10 marks each. Student shall answer any one of them.
iv) The questions from 2 to 4 shall be set by covering one unit of the syllabus for each question.

Practical Courses

Assessment Method	Marks
Continuous Internal Assessment	30
Semester End Examination	70
Total	**100**

b) For practical courses, there shall be a continuous evaluation during the semester for 30 sessional marks and end examination shall be for 70 marks.

c) Day-to-day work in the laboratory shall be evaluated for 15 marks by the concerned laboratory teacher based on the record/viva and 15 marks for the internal test.

d) The end examination shall be evaluated for 70 marks, conducted by the concerned laboratory teacher and a senior expert in the subject from the same department.

- Procedure: 20 marks
- Experimental work & Results: 30 marks
- Viva voce: 20 marks.

In a practical subject consisting of two parts (Eg: Basic Electrical & Electronics Engineering Lab), the end examination shall be conducted for 70 marks as a single laboratory in 3 hours. Mid semester examination shall be evaluated as above for 30 marks in each part and final mid semester marks shall be arrived by considering the average of marks obtained in two parts.

e) For the subject having design and/or drawing, such as Engineering Drawing, the distribution of marks shall be 30 for mid semester evaluation and 70 for end examination.

Assessment Method	Marks
Continuous Internal Assessment	30
Semester End Examination	70
Total	**100**

Day-to-day work shall be evaluated for 15 marks by the concerned subject teacher based on the reports/submissions prepared in the class. And there shall be two midterm examinations in a semester for duration of 2 hours each for 15 marks with weight age of 80% to better mid marks and 20% for the other. The subjective paper shall contain 3 either or type questions of equal weight age of 5 marks. There shall be no objective paper in mid semester examination. The sum of day- to-day evaluation and the mid semester marks will be the final sessional marks for the subject.

The end examination pattern for Engineering Graphics, shall consists of 5 questions, either/or type, of 14 marks each. There shall be no objective type questions in the end examination. However, the end examination pattern for other subjects related to design/drawing, multiple branches, etc is mentioned along with the syllabus.

f) There shall be no external examination for mandatory courses with zero credits. However, attendance shall be considered while calculating aggregate attendance and student shall be declared to have passed the mandatory course only when he/she secures 40% or more in the internal examinations. In case, the student fails, a re-examination shall be conducted for failed candidates for 30 marks satisfying the conditions mentioned in item 1 & 2 of the regulations.

g) The laboratory records and mid semester test papers shall be preserved for a minimum of 3 years in the respective institutions as per the University norms and shall be produced to the Committees of the University as and when the same are asked for.

10. Skill oriented Courses

i) There shall be five skill-oriented courses offered during III to VII semesters.

ii) Out of the five skill courses two shall be skill-oriented courses from the same domain. Of the remaining three skill courses, one shall be a soft skill course and the remaining two shall be skill-advanced courses from the same domain/Interdisciplinary/Job oriented.

iii) The course shall carry 100 marks and shall be evaluated through continuous assessments during the semester for 30 sessional marks and end examination shall be for 70 marks. Day-to-day work in the class / laboratory shall be evaluated for 30 marks by the concerned teacher based on the regularity/assignments/viva/mid semester test. The end examination similar to practical examination pattern shall be conducted by the concerned teacher and an expert in the subject nominated by the principal.

iv) The Head of the Department shall identify a faculty member as coordinator for the course. A committee consisting of the Head of the Department, coordinator and a senior Faculty member nominated by the Head of the Department shall monitor the evaluation process. The marks/grades shall be assigned to the students by the above committee based on their performance.

v) The student shall be given an option to choose either the skill courses being offered by the college or to choose a certificate course being offered by industries/Professional bodies or any other accredited bodies. If a student chooses to take a Certificate Course offered by external agencies, the credits shall be awarded to the student upon producing the Course Completion Certificate from the agency. A committee shall be formed at the level of the college to evaluate the grades/marks given for a course by external agencies and convert to the equivalent marks/grades.

vi) The recommended courses offered by external agencies, conversions and appropriate grades/marks are to be approved by the institution at the beginning of the semester. The principal of the college shall forward such proposals to the University for approval.

vii) If a student prefers to take a certificate course offered by external agency, the department shall mark attendance of the student for the remaining courses in that semester excluding the skill course in all the calculations of mandatory attendance requirements upon producing a valid certificate as approved by the University/institution.

11. Massive Open Online Courses (MOOCs):

A Student has to pursue and complete one course compulsorily through MOOCs approved by the University/institution. A student can pursue courses other than core through MOOCs and it is mandatory to complete one course successfully through MOOCs for awarding the degree. A student is not permitted to register and pursue core courses through MOOCs.

A student shall register for the course (Minimum of either 8 weeks or 12 weeks) offered through MOOCs with the approval of Head of the Department. The Head of the Department shall appoint one mentor to monitor the student’s progression. The student needs to earn a certificate by passing the exam. The student shall be awarded the credits assigned in the curriculum only by submission of the certificate. Examination fee, if any, will be borne by the student.

Students who have qualified in the proctored examinations conducted through MOOCs platform can apply for credit transfer as specified and are exempted from appearing internal as well as external examination (for the specified equivalent credit course only) conducted by the university.

Necessary amendments in rules and regulations regarding adoption of MOOC courses would be proposed from time to time.

12. Credit Transfer Policy

Adoption of MOOCs is mandatory, to enable Blended model of teaching-learning as also envisaged in the NEP 2020. As per University Grants Commission (Credit Framework for Online Learning Courses through SWAYAM) Regulation, 2016, the University/Institution shall allow up to a maximum of 20% of the total courses being offered in a particular programme i.e., maximum of 32 credits through MOOCs platform.

i) The University/Institution shall offer credit mobility for MOOCs and give the equivalent credit weightage to the students for the credits earned through online learning courses.
ii) Student registration for the MOOCs shall be only through the respective department of the institution, it is mandatory for the student to share necessary information with the department.
iii) Credit transfer policy will be applicable to the Professional & Open Elective courses only.
iv) The concerned department shall identify the courses permitted for credit transfer.
v) The University/institution shall notify at the beginning of semester the list of the online learning courses eligible for credit transfer.
vi) The institution shall designate a faculty member as a Mentor for each course toguide the students from registration till completion of the credit course.
vii) The University/institution shall ensure no overlap of MOOC exams with that of the University/institution examination schedule. In case of delay in results, the University/institution will re-issue the marks sheet for such students.
viii) Student pursuing courses under MOOCs shall acquire the required credits only after successful completion of the course and submitting a certificate issued by the competent authority along with the percentage of marks and grades.
ix) The institution shall submit the following to the examination section of the university:
    a) List of students who have passed MOOC courses in the current semester along with the certificate of completion.
    b) Undertaking form filled by the students for credit transfer.
x) The University / institution shall resolve any issues that may arise in the implementation of this policy from time to time and shall review its credit transfer policy in the light of periodic changes brought by UGC,

SWAYAM, NPTEL and state government.

Note: Students shall be permitted to register for MOOCs offered through online platforms approved by the University from time to time.

13. Academic Bank of Credits (ABC)

The University / institution has implemented Academic Bank of Credits (ABC) to promote flexibility in curriculum as per NEP 2020 to

i. provide option of mobility for learners across the universities of their choice
ii. provide option to gain the credits through MOOCs from approved digitalplatforms.
iii. facilitate award of certificate/diploma/degree in line with the accumulated credits in ABC
iv. execute Multiple Entry and Exit system with credit count, credit transfer
and credit acceptance from students' account.

14. Mandatory Internships

Summer Internships: Two summer internships either onsite or virtual each with a minimum of 08 weeks duration, done at the end of second and third years, respectively are mandatory. It shall be completed in collaboration with local industries, Govt. Organizations, construction agencies, Power projects, software MNCs or any industries in the areas of concerned specialization of the Undergraduate program. One of the two summer internships at the end of second year (Community Service Project) shall be society oriented and shall be completed in collaboration with government organizations/NGOs & others. The other internship at the end of third year is Industry Internship and shall be completed in collaboration with Industries. The student shall register for the internship as per course structure after commencement of academic year. The guidelines issued by the APSCHE / University shall be followed for carrying out and evaluation of Community Service Project and Industry Internship.

Evaluation of the summer internships shall be through the departmental committee. A student will be required to submit a summer internship report to the concerned department and appear for an oral presentation before the departmental committee comprising of Head of the Department, supervisor of the internship and a senior faculty member of the department. A certificate of successful completion from industry shall be included in the report. The report and the oral presentation shall carry 50% weight age each. It shall be evaluated for 50 external marks. There shall be no internal marks for Summer Internship. A student shall secure minimum 40% of marks for successful completion. In case, if a student fails, he/she shall reappear as and when semester supplementary examinations are conducted by the institution.

Full Semester Internship and Project work: In the final semester, the student should mandatorily register and undergo internship (onsite/virtual) and in parallel he/she should work on a project with well-defined objectives. At the end of the semester the candidate shall submit an internship completion certificate and a project report. A student shall also be permitted to submit project report on the work carried out during the internship.

The project report shall be evaluated with an external examiner. The total marks for project work 200 marks and distribution shall be 60 marks for internal and 140 marks for external evaluation. The supervisor assesses the student for 30 marks (Report: 15 marks, Seminar: 15 marks). At the end of the semester, all projects shall be showcased at the department for the benefit of all students and staff and the same is to be evaluated by the departmental Project Review Committee consisting of supervisor, a senior faculty and HOD for 30 marks. The external evaluation of Project Work is a Viva-Voce Examination conducted in the presence of internal examiner and external examiner appointed by the institution and is evaluated for 140 marks.

The college shall facilitate and monitor the student internship programs. Completion of internships is mandatory, if any student fails to complete internship, he/she will not be eligible for the award of degree. In such cases, the student shall repeat and complete the internship.

15. Guidelines for offering a Minor

The Minor program requires the completion of 18 credits in Minor stream chosen. The concept of Minor degree is introduced in the curriculum of all B. Tech. programs offering a Major degree. The main objective of Minor degree in a discipline is to promote interdisciplinary knowledge among the students, the students admitted in to B.Tech. in a major stream/branch are eligible to obtain degree in Minor in another stream. In order to earn a Minor degree in a discipline, a student has to earn 18 extra credits (By studying FIVE theory and TWO laboratory / Applied Project Work courses) from the courses of the minor discipline.

a) Students who are desires of pursuing their special interest areas other than the chosen discipline of Engineering may opt for additional courses in minor specialization groups offered by a department other than their parent department. For example, if Mechanical Engineering student selects subjects from Civil Engineering under this scheme, he / she will get Major degree of Mechanical Engineering with minor degree of Civil Engineering.

b) Student can also opt for industry relevant tracks of any branch to obtain the minor degree. For example, a B. Tech Mechanical Engineering student can opt for the industry relevant tracks like Data Analytics Track, IOT track, Artificial Intelligence and Machine learning track, etc.

i. A student is permitted to register for Minor in V semester after the results of III Semester are declared.

ii. A Student shall register for two theory courses in III B. Tech I Semester, two theory courses and one laboratory course in III B. Tech II Semester and one theory course and one laboratory course in IV B. Tech I Semester. If Applied Project Work is offered in Minor degree instead of Laboratory courses, it shall be offered in IV B. Tech I Semester.

iii. The Principal of the college shall arrange separate class work and timetable of the courses offered under minor program.

iv. Courses that are used to fulfil the student's primary major may not be double counted towards the minor. Courses with content substantially equivalent to courses in the student's primary Major may not be counted towards the Minor.

v. The attendance for the registered courses under Minor and regular courses offered for Major degree in a semester are to be considered separately.

vi. A student shall maintain an attendance of 75% in all registered courses under Minor to be eligible for attending semester end examinations.

vii. The teaching and evaluation procedure for courses offered in Minor degree shall be similar to regular B. Tech courses.

viii. In case of Applied Project Work, the Evaluation shall be made by the Internal departmental committee (Head of the Department and one senior faculty member of the Department and Supervisor). Out of a total of 100 marks for the project, 30 marks shall be for Internal Evaluation (average of review-1 and review-2) and 70 marks for the final evaluation (Viva-voce). The evaluation of project work shall be conducted at the end of IV B. Tech, I semester.

ix. A student registered for Minor shall pass in all subjects that constitute the requirement for the Minor degree program. No class/division (i.e., second class, first class and distinction, etc.) shall be awarded for Minor degree programme.

x. If a student drops or is terminated from the Minor program, the additional credits so far earned cannot be converted into open or core electives; they will remain extra. However, such students will receive a separate grade sheet mentioning the additional courses completed by them.

xi. The Minor degree shall be mentioned in the degree certificate as Bachelor of Technology in XXX with Minor in YYY. For example, Bachelor of Technology in Computer Science & Engineering with Minor in Electronics & Communication Engineering or the chosen industry relevant track. This shall also be reflected in the transcripts, along with the list of courses taken for Minor degree program with CGPA mentioned separately.

15.1 Enrolment into Minor:

i. Students of a Department/Discipline are eligible to opt for Minor degree specialization groups offered by a department other than their parent department.
ii. The student without any backlog subjects upto III Semester is permitted for enrolment into Minor degree.
iii. If a student is detained due to lack of attendance either in Major or in Minor, registration shall be cancelled.
iv. Transfer of credits from Minor to regular B. Tech degree and vice- versa shall not be permitted.
v. Minor is to be completed simultaneously with a Major degree program.

15.2 Registration for Minor:

i. The eligible and interested students shall apply through the HOD of his/her parent department. The whole process should be completed within one week before the start of every semester. Selected students shall be permitted to register the courses under Minor.
ii. The selected students shall submit their willingness to the principal through his/her parent department. The parent department shall maintain the record of student pursuing the Minor degree.
iii. The students enrolled in the Minor degree courses will be monitored continuously. An advisor/mentor from parent department shall be assigned to a group of students to monitor the progress.

16. Guidelines for offering Honors

The objective of introducing B.Tech. (Hons.) is to facilitate the students to choose additionally the specialized courses of their choice and build their competence in a specialized area in the UG level. The programme is a best choice for academically excellent students having good academic record and interest towards higher studies and research.

i. Honors is introduced in the curriculum of all B. Tech. programs offering a major degree and is applicable to all B. Tech (Regular and Lateral Entry) students admitted in Engineering & Technology.
ii. A student shall earn additional 18 credits for award of B.Tech.(Honors) degree from same branch/ department/ discipline registered for major degree. This is in addition to the credits essential for obtaining the Undergraduate degree in Major Discipline (i.e., 163 credits).
iii. A student is permitted to register for Honors in V semester based on the results of III Semester.
iv. A Student shall register for two theory courses in III B.Tech I-Semester, two theory courses and one laboratory course in III B.Tech II Semester and one theory course and one laboratory course in IV B.Tech I-Semester. If Applied Project Work is offered in Honors degree instead of Laboratory courses, it shall be offered in IV B. Tech I Semester.
v. The Principal of the college shall arrange separate class work and timetable of the courses offered under Honors program.

vi. Courses that are used to fulfil the student’s primary major may not be double counted towards the Honors. Courses with content substantially equivalent to courses in the student's primary Major may not be counted towards the Honors.
vii. Students can complete the courses offered under Honors either in the college or in online platforms like SWAYAM with a minimum duration of 12 weeks for a 3- credit course satisfying the criteria for credit mobility. If the courses under Honors are offered in conventional mode, then the teaching and evaluation procedure shall be similar to regular B. Tech courses.
viii. In case of Applied Project Work, the Evaluation shall be made by the Internal departmental committee (Head of the Department and one senior faculty member of the Department and Supervisor). Out of a total of 100 marks for the project, 30 marks shall be for Internal Evaluation (average of review-1 and review-2) and 70 marks for the final evaluation (Viva-voce). The evaluation of project work shall be conducted at the end of IV B. Tech, I semester.
ix. The attendance for the registered courses under Honors and regular courses offered for Major degree in a semester are to be considered separately.
x. A student shall maintain an attendance of 75% in all registered courses under Honors to be eligible for attending semester end examinations.
xi. A student registered for Honors shall pass in all subjects that constitute the requirement for the Honors degree program. No class/division (i.e., second class, first class and distinction, etc.) shall be awarded for Honors degree programme.
xii. If a student drops or is terminated from the Honors program, the additional credits so far earned cannot be converted into open or core electives; they will remain extra. However, such students will receive a separate grade sheet mentioning the additional courses completed by them.

xiii. The Honors will be mentioned in the degree certificate as Bachelor of Technology (Honors) in XYZ. For example, B.Tech. (Honors) in Mechanical Engineering

16.1 Enrollment into Honors:

i. Students of a Department/Discipline are eligible to opt for Honors program offered by the same Department/Discipline
ii. The enrolment of student into Honors is based on the CGPA obtained in the major degree program. CGPA shall be taken up to III semester in case of regular entry students and only III semester in case of lateral entry students. Students having 7 CGPA without any backlog subjects will be permitted to register for Honors.
iii. If a student is detained due to lack of attendance either in Major or in Honors, registration shall be cancelled.
iv. Transfer of credits from Honors to regular B. Tech degree and vice-

versa shall not be permitted.

v. Honors is to be completed simultaneously with a Major degree program.

16.2 Registration for Honors:

i. The eligible and interested students shall apply through the HOD of his/her parent department. The whole process should be completed within one week before the start of every semester. Selected students shall be permitted to register the courses under Honors.

ii. The selected students shall submit their willingness to the principal through his/her parent department offering Honors. The parent department shall maintain the record of student pursuing the Honors.

iii. The students enrolled in the Honors courses will be monitored continuously. An advisor/mentor from parent department shall be assigned to a group of students to monitor the progress.

17. Attendance Requirements:

i) A student shall be eligible to appear for the institution external examinations if he/she acquires a minimum of 40% attendance in each subject and 75% of attendance in aggregate of all the subjects. b) Condonation of shortage of attendance in aggregate up to 10% (65% and above and below 75%) in each semester may be granted by the College Academic Committee.

ii) Shortage of Attendance below 65% in aggregate shall in NO CASE be condoned.

iii) A stipulated fee shall be payable towards condonation of shortage of attendance to the institution.

iv) Students whose shortage of attendance is not condoned in any semester are not eligible to take their end examination of that class and their registration shall stand cancelled.

v) A student will not be promoted to the next semester unless he satisfies the attendance requirements of the present semester. They may seek readmission for that semester from the date of commencement of class work.

vi) If any candidate fulfils the attendance requirement in the present semester, he shall not be eligible for readmission into the same class.

vii) If the learning is carried out in blended mode (both offline & online), then the total attendance of the student shall be calculated considering the offline and online attendance of the student.

viii) For induction programme attendance shall be maintained as per AICTE norms.

18. Conduct of Semester End Examination and Evaluation:

18.1 Semester end examination shall be conducted by the Controller of Examination (COE) by inviting 50% Question Papers from the External and 50% Question papers from the Internal Subject Experts. Principal will decide the External and Internal subject experts.

18.2 The answer papers of semester end examination should be evaluated externally / internally.

18.3 The marks for the internal evaluation components will be added to the external evaluation marks secured in the Semester – End examinations, to arrive at total marks for any subject in that semester.

18.4 Performance in all the subjects is tabulated program-wise and will be scrutinized by the office of the Controller of Examinations. Total marks obtained in each subject are converted into letter grades. Finally subject-wise marks and grades details, subject-wise and branch-wise pass percentages are calculated through software.

18.5 Results Committee: Results Committee comprising of Principal, Controller of Examinations, Additional Controller of Examinations (Confidential), One Senior Professor nominated by the Principal and the University Nominee will oversee the details of marks, grades and pass percentages of all the subjects and branch-wise pass percentages.

18.6 Office of the Controller of Examinations will generate student-wise result sheets and the same will be published through college website.

18.7 Student-wise Grade Sheets are generated and issued to the students.

19. Promotion Rules:

The following academic requirements must be satisfied in addition to the attendance requirements mentioned in section 16.

i. A student shall be promoted from first year to second year if he/she fulfils the minimum attendance requirement as per University / Institution norms.
ii. A student will be promoted from II to III year if he/she fulfils the academic requirement of securing 40% of the credits (any ***decimal*** fraction should be ***rounded off*** to ***lower*** digit) up to in the subjects that have been studied up to III semester.
iii. A student shall be promoted from III year to IV year if he/she fulfils the academic requirements of securing 40% of the credits (any ***decimal*** fraction should be ***rounded off*** to ***lower*** digit) in the subjects that have been studied up to V semester. And in case a student is detained for want of credits for a particular academic year by ii) & iii) above, the student may make up the credits through supplementary examinations and only

after securing the required credits he/she shall be permitted to join in the V semester or VII semester respectively as the case may be.

iv. When a student is detained due to lack of credits/shortage of attendance he/she may be re-admitted when the semester is offered after fulfilment of academic regulations. In such case, he/she shall be in the academic regulations into which he/she is readmitted.

20. Grading:

As a measure of the student's performance, a 10-point Absolute Grading System using the following Letter Grades and corresponding percentage of marks shall be followed:

After each course is evaluated for 100 marks, the marks obtained in each course will be converted to a corresponding letter grade as given below, depending on the range in which the marks obtained by the student fall.

Structure of Grading of Academic Performance

Range in which the marks in the subject fall	Grade	Grade points Assigned
90 & above	S (Superior)	10
80 - 89	A (Excellent)	9
70 - 79	B (Very Good)	8
60 - 69	C (Good)	7
50 - 59	D (Average)	6
40 - 49	E (Pass)	5
< 40	F (Fail)	0
Absent	Ab (Absent)	0

i) A student obtaining Grade 'F' or Grade 'Ab' in a subject shall be considered failed and will be required to reappear for that subject when it is offered the next supplementary examination.

ii) For non-credit audit courses, "Satisfactory" or "Unsatisfactory" shall be indicated instead of the letter grade and this will not be counted for the computation of SGPA/CGPA/Percentage.

Computation of Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA):

The Semester Grade Point Average (SGPA) is the ratio of sum of the product of the number of credits with the grade point scored by a student in all the courses taken by a student and the sum of the number of credits of all the courses undergone by a student, i.e.,

$SGPA = \Sigma(C_i \times G_i)/\Sigma C_i$

Where, C_i is the number of credits of the i^{th} subject and G_i is the grade point scored by the student in the i^{th} course.

The Cumulative Grade Point Average (CGPA) will be computed in the same manner considering all the courses undergone by a student over all the semesters of a program, i.e.,

$CGPA = \Sigma(C_i \times S_i)/\Sigma C_i$

Where "S_i" is the SGPA of the i^{th} semester and C_i is the total number of credits up to that semester.

Both SGPA and CGPA shall be rounded off to 2 decimal points and reported in the transcripts.

While computing the SGPA the subjects in which the student is awarded Zero grade points will also be included.

Grade Point: It is a numerical weight allotted to each letter grade on a 10-point scale. Letter Grade: It is an index of the performance of students in a said course. Grades are denoted by the letters S, A, B, C, D and F.

Award of Class:

After a student has satisfied the requirements prescribed for the completion of the program and is eligible for the award of B. Tech. Degree, he/she shall be placed in one of the following four classes:

Class Awarded	CGPA Secured
First Class with Distinction	≥ 7.5
First Class	$\geq 6.5 < 7.5$
Second Class	$\geq 5.5 < 6.5$
Pass Class	$\geq 5.0 < 5.5$

CGPA to Percentage conversion Formula – (CGPA – 0.5) x 10

21. With–holding of Results

If the candidate has any dues not paid to the university or if any case of indiscipline or malpractice is pending against him/her, the result of the candidate shall be withheld in such cases.

22. Personal Verification /Recounting / Revaluation / Final Valuation
22.1 Personal Verification of Answer Scripts:

Candidates appear in a particular semester end examinations may appeal for verification of their answer script(s) for arithmetic correction in totaling of marks and any omission / deletion in evaluation within 7 days from the date of declaration of results at the office of the Controller of Examinations on the prescribed proforma and by paying the prescribed fee per answer script.

It is clarified that personal verification of answer script shall not tantamount to revaluation of answer script. This is only a process of reverification by the candidate. Any mistake / deficiency with regard to arithmetic correction in totaling of marks and any omission / deletion in evaluation if found, the institution will correct the same.

22.2 Recounting / Revaluation:

Students shall be permitted for request for recounting/revaluation of the Semester-End examination answer scripts within a stipulated period after payment of prescribed fee. After recounting or revaluation, records are updated with changes if any and the student will be issued a revised grade sheet. If there are no changes, the same will be intimated to the students.

22.3 Final Valuation:

Students shall be permitted for request for final valuation of the Semester–End Examination answer scripts within a stipulated period after the publication of the revaluation results by paying the necessary fee. The final valuation shall be carried out by an expert not less than Associate Professor as per the scheme of valuation supplied by the examination branch in the presence of the student, Controller of Examinations and Principal. However students are not permitted to discuss / argue with the examiner. If the increase in marks after final valuation is equal to or more than 15% of the previous valuation marks, the marks obtained after final valuation shall be treated as final. If the variation of marks after final valuation is less than 15% of the previous valuation marks, then the earlier valuation marks shall be treated as the final marks.

23. Multiple Entry / Exit Option

(a) Exit Policy:

The students can choose to exit the four-year programme at the end of first/second/third year.

i) **UG Certificate in (Field of study/discipline)** - Programme duration: First year (first two semesters) of the undergraduate programme, 40 credits followed by an additional exit 10-credit bridge course(s) lasting two months, including at least 6- credit job-specific internship/ apprenticeship that would help the candidates acquire job-ready competencies required to enter the workforce.

ii) **UG Diploma (in Field of study/discipline)** - Programme duration: First two years (first four semesters) of the undergraduate programme, 80 credits followed by an additional exit 10-credit bridge course(s) lasting two months, including at least 6- credit job-specific internship/ apprenticeship that would help the candidates acquire job-ready competencies required to enter the workforce.

iii) **Bachelor of Science (in Field of study/discipline)** i.e., **B.Sc. Engineering in (Field of study/discipline)**- Programme duration: First three years (first six semesters) of the undergraduate programme, 123 credits.

b) Entry Policy:

Modalities on multiple entry by the student into the B.Tech. programme will be provided in due course of time.

Note: The University / institution shall resolve any issues that may arise in the implementation of Multiple Entry and Exit policies from time to time and shall review the policies in the light of periodic changes brought by UGC, AICTE and State government.

24. Gap Year Concept:

Gap year concept for Student Entrepreneur in Residence is introduced and outstanding students who wish to pursue entrepreneurship / become entrepreneur are allowed to take a break of one year at any time after II year to pursue full-time entrepreneurship programme/to establish start-ups. This period may be extended to two years at the most and these two years would not be counted for the time for the maximum time for graduation. The principal of the college shall forward such proposals submitted by the students to the University. An evaluation committee constituted by the institution shall evaluate the proposal submitted by the student and the committee shall decide whether to permit the student(s) to avail the Gap Year or not

25. Transitory Regulations:

Discontinued, detained, or failed candidates are eligible for readmission as and when the semester is offered after fulfilment of academic regulations. Candidates who have been detained for want of attendance or not fulfilled academic requirements or who have failed after having undergone the course in earlier regulations or have discontinued and wish to continue the course are eligible for admission into the unfinished semester from the date of commencement of class work with the same or equivalent subjects as and when subjects are offered, subject to Section 2 and they will follow the academic regulations into which they are readmitted.

Candidates who are permitted to avail Gap Year shall be eligible for re-joining into the succeeding year of their B. Tech from the date of commencement of class work, subject to Section **2** and they will follow the academic regulations into which they are readmitted.

26. Minimum Instruction Days for a Semester:

The minimum instructional days excluding exams for each semester shall be 90 days.

27. Medium of Instruction:

The medium of instruction of the entire B. Tech undergraduate programme in Engineering & Technology (including examinations and project reports) will be in English only.

28. Student Transfers:

Student transfers shall be as per the guidelines issued by the Government of Andhra Pradesh and the University / institution from time to time.

29. General Instructions:

i. The academic regulations should be read as a whole for purpose of any interpretation.
ii. Malpractices rules-nature and punishments are appended.
iii. Where the words "he", "him", "his", occur in the regulations, they also include "she", "her", "hers", respectively.
iv. In the case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Vice-Chancellor / Head of the Institution is final.
v. The University / institution may change or amend the academic regulations or syllabi at any time and the changes or amendments shall be made applicable to all the students on rolls with effect from the dates notified by the Universities.

*** *** ***

ACADEMIC REGULATIONS (R23)
FOR B.TECH. (LATERAL ENTRY SCHEME)
*(Effective for the students admitted into II year through Lateral Entry Scheme from the Academic Year **2024-25** onwards)*

1. **Award of the Degree**
   (a) Award of the B.Tech. Degree / B.Tech. Degree with a Minor if he/she fulfils the following:
      (i) Pursues a course of study for not less than three academic years and not more than six academic years. However, for the students availing Gap year facility this period shall be extended by two years at the most and these two years would in addition to the maximum period permitted for graduation (Six years).
      (ii) Registers for 123 credits and secures all 123 credits.

   (b) Award of B.Tech. degree with Honors if he/she fulfils the following:
      (i) Student secures additional 18 credits fulfilling all the requisites of a B.Tech. program i.e., 123 credits.
      (ii) Registering for Honors is optional.
      (iii) Honors is to be completed simultaneously with B.Tech. programme.

2. Students, who fail to fulfil the requirement for the award of the degree within six consecutive academic years from the year of admission, shall forfeit their seat.

3. **Minimum Academic Requirements**
   The following academic requirements have to be satisfied in addition to the requirements mentioned in item no.**2**
   i. A student shall be deemed to have satisfied the minimum academic requirements and earned the credits allotted to each theory, practical, design, drawing subject or project if he secures not less than 35% of marks in the end examination and a minimum of 40% of marks in the sum total of the mid semester evaluation and end examination taken together.

   ii. A student shall be promoted from III year to IV year if he/she fulfils the academic requirements of securing 40% of the credits (any decimal fraction should be rounded off to lower digit) in the subjects that have been studied up to V semester.

   And in case if student is already detained for want of credits for particular academic year, the student may make up the credits through supplementary exams of the above exams before the commencement of IV year I semester class work of next year.

4. Course Pattern

i) The entire course of study is three academic years on semester pattern.

ii) A student eligible to appear for the end examination in a subject but absent at it or has failed in the end examination may appear for that subject at the next supplementary examination offered.

iii) When a student is detained due to lack of credits/shortage of attendance the student may be re-admitted when the semester is offered after fulfilment of academic regulations, the student shall be in the academic regulations into which he/she is readmitted.

5. All other regulations as applicable for B. Tech. Four-year degree course (Regular) will hold good for B. Tech. (Lateral Entry Scheme).

Identification of Courses

B. Tech

Each course shall be uniquely identified by an alphanumeric code of width 7 characters as given below.

No. of Digits	Description
First two digits	Year of regulations Ex:23
Next one letter	Type of program: A: B. Tech B: M. Tech C: M.B.A D: M.C.A E: BBA F: BCA
Next two letters	Code of program: CE: Civil Engineering, EE: Electrical and Electronics Engineering, ME: Mechanical Engineering, EC: Electronics and Communication Engineering, BM: Electronics and Communication Engineering (Bio Medical Engineering), CS: Computer Science and Engineering, CC: Computer Science and Engineering (Cyber Security), CM: Computer Science and Engineering (Artificial Intelligence and Machine Learning), CA: Computer Science and Engineering (Artificial Intelligence), AI: Artificial Intelligence, AD: Artificial Intelligence and Data Science, CD: Computer Science and Engineering (Data Science), CB: Computer Science and Business Systems, IT: Information Technology, CO: Computer Science and Engineering (Internet of Things), MB: Management Courses, HS: Humanities and Sciences
Last two digits	Indicate serial numbers: ≥ 01

Ex:
23ACE01
23AEE01
23AME01
23AEC01
23ABM01
23ACS01
23ACC01
23ACM01
23ACA01
23AAI01
23AAD01
23ACD01
23ACB01
23AIT01
23ACO01
23AMB01
23AHS01

RULES FOR DISCIPLINARY ACTION FOR MALPRACTICE / IMPROPER CONDUCT IN EXAMINATIONS

	Nature of Malpractices / Improper Conduct	**Punishment**
	If the candidate	
1.(a)	Possesses or keeps accessible in examination hall, any paper, note book, programmable calculators, Cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which he is appearing but has not made use of (material shall include any marks on the body of the candidate which can be used as an aid in the subject of the examination)	Expulsion from the examination hall and cancellation of the performance in that subject only.
(b)	Gives assistance or guidance or receives it from any other candidate orally or by any other body language methods or communicates through cell phones with any candidate or persons in or outside the exam hall in respect of any matter.	Expulsion from the examination hall and cancellation of the performance in that subject only of all the candidates involved. In case of an outsider, he will be handed over to the police and a case is registered against him.
2.	Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or practical) in which the candidate is appearing.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that Semester/year. The Hall Ticket of the candidate is to be cancelled.
3.	Comes in a drunken condition to the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that Semester/year.
4.	Smuggles in the Answer book or additional sheet or takes out or	Expulsion from the examination hall and cancellation of the

	arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.	performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
5.	Leaves the exam hall taking away answer script or intentionally tears of the script or any part thereof inside or outside the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
6.	Possess any lethal weapon or firearm in the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred and forfeits of seat.
7.	Impersonates any other candidate in connection with the examination.	The candidate who has impersonated shall be expelled

		from examination hall. The candidate is also debarred and forfeits the seat. The performance of the original candidate who has been impersonated, shall be cancelled in all the subjects of the examination (including practicals and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that semester/year. The candidate is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat. If the impostor is an outsider, he will be handed over to the police and a case is registered against him.
8.	Refuses to obey the orders of the Chief Superintendent / Assistant – Superintendent / any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the officer-in-charge or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the officer-in-charge, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction or property in the examination hall or any part of the College campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has	In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the candidate(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The candidates also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police and a police case is registered against them.

	the tendency to disrupt the orderly conduct of the examination.	
9.	If student of the college, who is not a candidate for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.	Student of the colleges expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred and forfeits the seat. Person(s) who do not belong to the College will be handed over to police and, a police case will be registered against them.
10.	Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.	Cancellation of the performance in that subject.
11.	Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.	Cancellation of the performance in that subject and all other subjects the candidate has appeared including practical examinations and project work of that semester/year examinations.
12.	If any malpractice is detected which is not covered in the above clauses 1 to 11 shall be reported to the Examination committee for further action to award suitable punishment.	

B.TECH-CSE(AI&ML) - COURSE STRUCTURE – R23
(Applicable from the academic year 2023-24 onwards)

INDUCTION PROGRAMME

S. No.	Course Name	Category	L-T-P-C
1	Physical Activities -- Sports, Yoga and Meditation, Plantation	MC	0-0-6-0
2	Career Counselling	MC	2-0-2-0
3	Orientation to all branches -- career options, tools, etc.	MC	3-0-0-0
4	Orientation on admitted Branch -- corresponding labs, tools and platforms	EC	2-0-3-0
5	Proficiency Modules & Productivity Tools	ES	2-1-2-0
6	Assessment on basic aptitude and mathematical skills	MC	2-0-3-0
7	Remedial Training in Foundation Courses	MC	2-1-2-0
8	Human Values & Professional Ethics	MC	3-0-0-0
9	Communication Skills -- focus on Listening, Speaking, Reading, Writing skills	BS	2-1-2-0
10	Concepts of Programming	ES	2-0-2-0

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(Artificial Intelligence & Machine Learning)

Course Structure and Scheme of Examination

I B. Tech I Semester – CSE (Artificial Intelligence & Machine Learning) **Regulations: R23**

S. No	Category	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
				L/D	T	P	C	CIA	SEE	Total
1.	BS & H	23AHS01	Communicative English	2	0	0	2	30	70	100
2.	BS & H	23AHS02	Chemistry	3	0	0	3	30	70	100
3.	ES	23ACE01	Basic Civil and Mechanical Engineering	3	0	0	3	30	70	100
4.	ES	23ACS01	Introduction to Programming	3	0	0	3	30	70	100
5.	BS & H	23AHS04	Linear Algebra and Calculus	3	0	0	3	30	70	100
6.	ES	23AME02	Engineering Workshop	0	0	3	1.5	30	70	100
7.	BS & H	23AHS06	Communicative English Lab	0	0	2	1	30	70	100
8.	BS & H	23AHS07	Chemistry Lab	0	0	2	1	30	70	100
9.	ES	23ACS02	Computer Programming Lab	0	0	3	1.5	30	70	100
10	MC	23AHS10	Health and Wellness, Yoga and Sports	0	0	1	0.5	100	0	100
TOTAL				**14**	**0**	**11**	**19.5**	**370**	**630**	**1000**

I B. Tech II Semester – CSE (Artificial Intelligence & Machine Learning) **Regulations : R23**

S. No	Category	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
				L	T	P	C	CIA	SEE	Total
1.	BS & H	23AHS05	Engineering Physics	3	0	0	3	30	70	100
2.	ES	23AEE01	Basic Electrical and Electronics Engineering	3	0	0	3	30	70	100
3.	ES	23AME01	Engineering Graphics	1	0	4	3	30	70	100
4.	BS & H	23AHS11	Differential Equations and vector calculus	3	0	0	3	30	70	100
5.	ES	23AEE02	Electrical and Electronics Engineering Workshop	0	0	3	1.5	30	70	100
6.	ES	23AIT01	IT Workshop	0	0	2	1	30	70	100
7.	PC	23ACS03	Data Structures	3	0	0	3	30	70	100
8.	BS & H	23AHS09	Engineering Physics Lab	0	0	2	1	30	70	100
9.	PC	23ACS04	Data Structures Lab	0	0	3	1.5	30	70	100
10	MC	23AHS12	NSS /NCC/Scouts and Guides/Community Service	0	0	1	0.5	100	0	100
TOTAL				**13**	**0**	**15**	**20.5**	**370**	**630**	**1000**

II B.Tech I Semester – CSE (Artificial Intelligence & Machine Learning) Regulations: R23

S. No	Category	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
				L	T	P	C	CIA	SEE	Total
1.	BS	23AHS17	Discrete Mathematics & Graph Theory	3	0	0	3	30	70	100
2.	HSMC	23AMB01	Universal Human Values 2- Understanding Harmony and Ethical human conduct	2	1	0	3	30	70	100
3.	ES	23ACA01	Artificial Intelligence	3	0	0	3	30	70	100
4.	PC	23ACS05	Advanced Data Structures & Algorithms Analysis	3	0	0	3	30	70	100
5.	PC	23ACS06	Object-Oriented Programming Through JAVA	3	0	0	3	30	70	100
6.	PC	23ACS07	Advanced Data Structures and Algorithms Analysis Lab	0	0	3	1.5	30	70	100
7.	PC	23ACS08	Object-Oriented Programming Through JAVA Lab	0	0	3	1.5	30	70	100
8.	SC	23ACS09	Python programming	0	1	2	2	30	70	100
9.	AC	23AHS18	Environmental Science	2	0	0	Non-Credit	-	-	-
10	AC	23AHS24	Quantitative Aptitude and Reasoning-I	2	0	0	Non-Credit	-	-	-
TOTAL				**18**	**2**	**8**	**20**	**240**	**560**	**800**

II B.Tech II Semester – CSE (Artificial Intelligence & Machine Learning) Regulations: R23

S. No	Category	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
				L	T	P	C	CIA	SEE	Total
1.	HSS	23AME05	Optimization Techniques	2	0	0	2	30	70	100
2.	BS	23AHS21	Probability & Statistics	3	0	0	3	30	70	100
3.	PC	23ACA02	Machine Learning	3	0	0	3	30	70	100
4.	PC	23ACS12	Database Management Systems	3	0	0	3	30	70	100
5.	PC	23AEC06	Digital Logic and Computer Organization	3	0	0	3	30	70	100
6.	PC	23ACA03	Machine Learning Lab	0	0	3	1.5	30	70	100
7.	PC	23ACS14	Database Management Systems Lab	0	0	3	1.5	30	70	100
8.	SC	23ACS15	Full Stack Development-1	0	1	2	2	30	70	100
9.	ES	23AMB05	Design Thinking & Innovation	1	0	2	2	30	70	100
10	AC	23AHS25	Quantitative Aptitude and Reasoning-II	2	0	0	Non-Credit	-	-	-
TOTAL				**17**	**1**	**10**	**21**	**270**	**630**	**900**
Mandatory Community Service Project Internship of 08 weeks duration during summer vacation										

III B.Tech I Semester – CSE (Artificial Intelligence & Machine Learning) Regulations: R23

S. No	Category	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
				L	T	P	C	CIA	SEE	Total
1.	PC	23ACM01	Natural Language Processing and Models	3	0	0	3	30	70	100
2.	PC	23ACA06	Operating Systems & System Programming	3	0	0	3	30	70	100
3.	PC	23ACM02	Computer Vision & Image Processing	3	0	0	3	30	70	100
4.	PC	23ACS27	Introduction to quantum Technologies and Applications	3	0	0	3	30	70	100
5.	PE-I	23ACD10	Data Visualization	3	0	0	3	30	70	100
		23AIT07	Soft computing							
		23ACD11	Data Analysis with Python							
		23ACA07	Computational Intelligence							
6.	OE-I	23ACE30	Green Buildings	3	0	0	3	30	70	100
		23ACE31	Construction Technology and Management							
		23AEE23	Electrical Safety Practices and Standards							
		23AME23	Sustainable Energy Technologies							
		23AEC30	Electronic Circuits							
		23AHS26	Mathematics for Machine Learning and AI							
		23AHS27	Materials Characterization Techniques							
		23AHS28	Chemistry of Energy Systems							
		23AHS29	English for Competitive Examinations							
		23AMB06	Entrepreneurship and New Venture Creation							
7.	PC	23ACA14	Computer Vision & Machine Learning Lab	0	0	3	1.5	30	70	100
8.	PC	23AAI07	AI & System Programming Lab	0	0	3	1.5	30	70	100
9.	SC	23ACS20	Full Stack Development-II	0	1	2	2	30	70	100
10	ES	23ACO03	Tinkering Lab	0	0	2	1	30	70	100
11	IN	23ACM03	Community Service Internship	-	-	-	2	30	70	100
12	AC	23AHS30	Quantitative Aptitude and Reasoning-III	2	0	0	Non-Credit	-	-	-
13	AC	23AHS31	French Language	2	0	0	Non-Credit	-	-	-
		23AHS32	German Language							
		23AHS33	Japanese Language							
TOTAL				**15**	**1**	**10**	**23**	**300**	**700**	**1000**

III B.Tech II Semester – CSE (Artificial Intelligence & Machine Learning) Regulations: R23

S. No	Category	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
				L	T	P	C	CIA	SEE	Total
1.	PC	23ACA15	Cloud Computing for AI	3	0	0	3	30	70	100
2.	PC	23ACA16	Big Data Analytics & AI Applications	3	0	0	3	30	70	100
3.	PC	23ACA17	Full Stack AI Development	3	0	0	3	30	70	100
4.	PE – II	23ACM08	Graph Neural Networks	3	0	0	3	30	70	100
		23ACM09	Recommender Systems							
		23ACD20	Predictive Analytics							
		23ACA18	Blockchain for AI							
5.	PE – III	23ACM10	Automation of Model Building	3	0	0	3	30	70	100
		23AMB15	AI for Finance							
		23ACM11	Social Network Analysis							
		23ACA19	Cybersecurity & AI-driven Threat Detection							
6.	OE – II	23ACE44	Disaster Management	3	0	0	3	30	70	100
		23ACE45	Sustainability in Engineering Practices							
		23AEE33	Renewable Energy Sources							
		23AME39	Automation and Robotics							
		23AEC26	Digital Circuits							
		23ACS31	Advanced Optimization Techniques							
		23AHS36	Mathematical Foundations of Quantum Technologies							
		23AHS37	Physics of Electronic Materials and Devices							
		23AHS38	Chemistry of Polymers and Applications							
		23AHS39	Academic Writing and Public Speaking							
7.	PC	23ACA20	Big Data & Cloud Computing Lab	0	0	3	1.5	30	70	100
8.	PC	23ACA21	Full Stack AI Lab	0	0	3	1.5	30	70	100
9.	SC	23AHS20	**Skill Enhancement course** Soft skills	0	1	2	2	30	70	100
10.	AC	23AMB16	**Audit Course** Technical Paper Writing & IPR	2	0	0	-	-	-	-
			TOTAL	**19**	**1**	**06**	**23**	**270**	**630**	**900**
Mandatory Industry Internship of 6-8 weeks duration during summer vacation										

IV B. Tech I Semester – CSE(Artificial Intelligence & Machine Learning) Regulations: R23

S. No	Category	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
				L	T	P	C	CIA	SEE	Total
1.	PC	23ACA31	Generative AI	3	0	0	3	30	70	100
2.	MG	23AMB17	Business Ethics and Corporate Governance	2	0	0	2	30	70	100
		23AMB18	E-Business							
		23AMB19	Management Science							
		23AMB02	Managerial Economics and Financial							
		23AMB03	Organizational Behavior							
		23AMB04	Business Environment							
3.	PE – IV	23ACA24	Explainable AI & Model Interpretability	3	0	0	3	30	70	100
		23ACA25	AI in Cyber Security							
		23ACA26	AI-driven Software Engineering & DevOps							
		23ACA27	AI for Robotics							
4.	PE – V	23ACM23	MLOps & AI Model Deployment/	3	0	0	3	30	70	100
		23ACD27	Data Wrangling							
		23ACA28	Healthcare AI							
		23ACO13	AI for Smart Cities & IoT Systems							
5.	OE – III	23ACE55	Building Materials and Services	3	0	0	3	30	70	100
		23ACE26	Environmental Impact Assessment							
		23AEE42	Smart Grid Technologies							
		23AME52	3D Printing Technologies							
		23AEC14	Microprocessors and Microcontrollers							
		23AHS41	Wavelet transforms and its Applications							
		23AHS42	Smart Materials and Devices							
		23AHS43	Introduction to Quantum Mechanics							
		23AHS44	Green Chemistry and Catalysis for Sustainable Environment							
		23AHS45	Employability Skills							
6.	OE – IV	23ACE56	Geo-Spatial Technologies	3	0	0	3	30	70	100
		23ACE57	Solid Waste Management							
		23AEE43	Electric Vehicles							
		23AME46	Total Quality Management							
		23AEC53	Transducers and Sensors							
		23AMB22	Financial Mathematics							
		23AHS46	Sensors and Actuators for Engineering Applications							
		23AHS47	Chemistry of Nanomaterials and Applications							
		23AHS48	Literary Vibes							
		23ACS35	Quantum Computing							
7.	SC	23ACA29	Prompt Engineering	0	1	2	2	30	70	100
8.	AC	23AHS49	Gender Sensitization	2	0	0	-	-	-	-
9.	IN	23ACM24	Industry Internship	-	-	-	2	30	70	100
			TOTAL	**19**	**1**	**2**	**21**	**240**	**560**	**800**

IV B. Tech II Semester CSE(Artificial Intelligence & Machine Learning) Regulations: R23

S. No	Category	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
				L	T	P	C	CIA	SEE	Total
1	IN	23ACM27	Internship	-	-	-	4	30	70	100
2	PR	23ACM28	Project	-	-	-	8	30	70	100
TOTAL							**12**	**60**	**140**	**200**

Open Electives Open Electives, offered to other department students:

S.No.	Category	Title	L	T	P	Credits
1	Open Elective I	Java Programming	3	0	0	3
		Introduction to Artificial Intelligence	3	0	0	3
2	Open Elective II	Operating Systems	3	0	0	3
		Machine Learning Modeling	3	0	0	3
3	Open Elective III	Data Base Management Systems	3	0	0	3
		Cyber Security	3	0	0	3
4	Open Elective IV	Computer Networks	3	0	0	3
		Internet of Things	3	0	0	3

COURSES STRUCTURE FOR HONOURS DEGREE IN

CSE(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)-1

S. No	Category	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L/D	T	P	C	CIA	SEE	Total
1.	PC	III -I	23ACA34	Advanced Machine Learning & AI Systems	3	0	0	3	30	70	100
2.	PC	III -I	23ACM29	Deep Learning & Neural Network Architectures	3	0	0	3	30	70	100
3.	PC	III -II	23ACM30	Reinforcement Learning & Decision Making	3	0	0	3	30	70	100
4.	PC	III -II	23ACA35	AI for Robotics & Automation	3	0	0	3	30	70	100
5.	PC	IV-I	23ACA36	AI Ethics, Fairness & Explainability	3	0	0	3	30	70	100
6.	PC	III -II	23ACM21	AI & Machine Learning Lab	0	0	3	1.5	30	70	100
7.	PC	IV-I	23ACA37	Robotics & Autonomous Systems Lab	0	0	3	1.5	30	70	100
TOTAL					**15**	**0**	**06**	**18**	**210**	**490**	**700**

COURSES STRUCTURE FOR HONOURS DEGREE IN

CSE(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)-2

S. No	Category	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L/D	T	P	C	CIA	SEE	Total
1.	PC	III-I	23ACM41	Python for Data Science, AI & Development	3	0	0	3	30	70	100
2.	PC	III-I	23ACM42	Mastering Data Handling - SQL & No-SQL	3	0	0	3	30	70	100
3.	PC	III-II	23ACM43	Data Analysis & Visualization using Python	3	0	0	3	30	70	100
4.	PC	III-II	23ACM44	Machine Learning and AI: A Deep Dive	3	0	0	3	30	70	100
5.	PC	IV-I	23ACM45	AI Model Deployment, MLOps & Industry Applications	3	0	0	3	30	70	100
6.	PR	IV-I	23ACM46	Capstone Project	0	0	6	3	30	70	100
				Total	**15**	**0**	**6**	**18**	**180**	**420**	**600**

COURSES STRUCTURE FOR HONOURS DEGREE IN

CSE(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)-3

S. No	Category	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L/D	T	P	C	CIA	SEE	Total
1.	PC	III -I	23ACM35	Applied Deep Learning	3	0	0	3	30	70	100
2.	PC	III -I	23ACM36	Machine Learning Essentials for Next Gen AI	3	0	0	3	30	70	100
3.	PC	III -II	23ACM37	Deep Dive into AI: Principles and Practices of Deep Learning	3	0	0	3	30	70	100
4.	PC	III -II	23ACM38	Gen AI: Fundamentals to Advanced Applications	3	0	0	3	30	70	100
5.	PC	IV-I	23ACM39	LLMOps and Scalable GenAI Product Deployment	3	0	0	3	30	70	100
6.	PR	IV-I	23ACM40	Capstone Project	0	0	6	3	30	70	100
TOTAL					**15**	**0**	**06**	**18**	**180**	**420**	**600**

LIST OF MINORS OFFERED TO CSE AI & ML

S.No.	Minor Title	Department offering the Minor
1.	Building Planning & Construction Technology	Civil
2.	Micro Grid Technology	EEE
3.	Energy Systems	
4.	3D Printing	ME
5.	Industrial Engineering	
6.	Embedded Systems and IoT	ECE & VLSI
7.	Electronic Systems	
8.	Internet of Things	CSE(IOT)
9.	Data Science	CSE(DS)
10.	Data Analytics	CSE(DS)
11.	Data Science and Data Analytics	CSE(DS)
12.	Data Science with AWS	CSE(DS)
13.	Quantum Technologies	IT
14.	UX/UI Design	IT
15.	Cyber Security	CSE(CS)
16.	Quantum Computing	CSE
17.	Cloud Computing	CSE
18.	Full stack MERN & DevOps	CSE
19.	Bio Medical Engineering	ECE
20.	Artificial Intelligence & Data Science (AI&DS)	CSE(DS)
21.	Computer Science & Business Systems	CSBS
22.	Computer Science and Engineering	CSE

MINORS IN
BUILDING PLANNING & CONSTRUCTION TECHNOLOGY

S.No	Category	Year & Sem	Course Code	Name of the Subject and Lab	L	T	P	C	Scheme of Examination (Maximum Marks)		
									CIA	SEE	Total
1	PC	III – I	23ACE67	Construction Materials	3	0	0	3	30	70	100
2	PC	III – I	23ACE68	Construction Methods	3	0	0	3	30	70	100
3	PC	III – II	23ACE69	Fundamentals of Building Planning and Drawing	3	0	0	3	30	70	100
4	PC	III – II	23ACE70	Basic Surveying	3	0	0	3	30	70	100
5	PC	IV - I	23ACE71	Essentials of Concrete Technology	3	0	0	3	30	70	100
6	PC	III – II	23ACE72	Basic Concrete Technology Lab	0	0	3	1.5	30	70	100
7	PC	IV – I	23ACE73	Basic Surveying Lab	0	0	3	1.5	30	70	100
Total								**18**	**210**	**490**	**700**

MINORS IN MICRO GRID TECHNOLOGY

S.No	Category	Year & Sem	Course code	Course title	Hours per week			Credits	Scheme of Examination Max. Marks		
					L	T	P		CIA	SEE	Total
1	PC	III – I	23AEE58	Futuristic Power Systems	3	0	0	3	30	70	100
2	PC	III – I	23AEE59	Power Electronic Converters for Energy Sources	3	0	0	3	30	70	100
3	PC	III – II	23AEE60	Microgrid Power and Control Architecture	3	0	0	3	30	70	100
4	PC	III – II	23AEE61	Microgrid System Design	3	0	0	3	30	70	100
5	PC	IV - I	23AEE62	Analysis of Smart Grid Systems	3	0	0	3	30	70	100
6	PR	IV - I	23AEE63	Project in Micro Grid Technology	0	0	6	3	30	70	100
TOTAL					**15**		**06**	**18**	**180**	**420**	**600**

MINORS IN ENERGY SYSTEMS

S.No	Category	Year & Sem	Course code	Course title	Hours per week			Credits	Scheme of Examination Max. Marks		
					L	T	P		CIA	SEE	Total
1.	PC	III – I	23AEE52	Energy Audit and Management	3	0	0	3	30	70	100
2.	PC	III – I	23AEE53	Energy Management in Building	3	0	0	3	30	70	100
3.	PC	III – II	23AEE54	Energy Storage Technologies	3	0	0	3	30	70	100
4.	PC	III – II	23AEE55	Energy Scenario and Energy Policy	3	0	0	3	30	70	100
5.	PC	IV - I	23AEE56	Waste Energy Management	3	0	0	3	30	70	100
6.	PC	IV - I	23AEE57	Project in Energy Systems	3	0	0	3	30	70	100
TOTAL					**15**		**06**	**18**	**180**	**420**	**600**

MINORS IN - 3D PRINTING

S. No	Category	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L	T	P	C	CIA	SEE	Total
1.	PC	III – I	23AME61	Material Science & Engineering	3	0	0	3	30	70	100
2.	PC	III - I	23AME62	Additive Manufacturing	3	0	0	3	30	70	100
3.	PC	III – II	23AME63	Material Characterization Techniques	3	0	0	3	30	70	100
4.	PC	III – II	23AME25	CAD/CAM	3	0	0	3	30	70	100
5.	PC	III – II	23AME06	Computer Aided Machine Drawing	0	0	3	1.5	30	70	100
6.	PC	IV – I	23AME64	3D Printing Materials and Applications	3	0	0	3	30	70	100
7.	PC	IV – I	23AME38	3D Printing Lab	0	0	3	1.5	30	70	100
Total					**15**	**0**	**6**	**18**	**210**	**490**	**700**

MINORS IN - INDUSTRIAL ENGINEERING

S. No	Category	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L	T	P	C	CIA	SEE	Total
1	PC	III -I	23AME65	Production Planning & Control	3	0	0	3	30	70	100
2	PC	III-I	23AMB26	Marketing Management	3	0	0	3	30	70	100
3	PC	III-II	23AME57	Supply Chain Management	3	0	0	3	30	70	100
4	PC	III-II	23AMB27	Strategic Management for Competitive Advantage	3	0	0	3	30	70	100
5	PC	IV-I	23AME66	Six Sigma & Lean Manufacturing	3	0	0	3	30	70	100
6	PR	IV-I	23AME67	Applied Project Work in Industrial Engineering	0	0	6	3	30	70	100
				Total	**15**	**0**	**6**	**18**	**180**	**420**	**600**

MINORS IN EMBEDDED SYSTEMS AND IOT

S. No	Category	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L/D	T	P	C	CIA	SEE	Total
1	PC	III -I	23AEC63	Embedded Systems Technology	3	0	0	3	30	70	100
2	PC	III-I	23AEC64	Real Time Embedded systems design and Analysis	3	0	0	3	30	70	100
3	PC	III-II	23AEC65	Principles of IoT	3	0	0	3	30	70	100
4	PC	III-II	23AEC48	Wireless Networks	3	0	0	3	30	70	100
5	PC	III-II	23AEC66	Principles of IoT Lab	0	0	3	1.5	30	70	100
6	PC	IV-I	23AEC35	Embedded systems & IOT Applications	3	0	0	3	30	70	100
7	PC	IV-I	23AEC67	Industrial Internet of things Lab	0	0	3	1.5	30	70	100
				TOTAL	**15**	**0**	**06**	**18**	**210**	**490**	**700**

MINORS IN ELECTRONIC SYSTEMS

S. No	Category	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L/D	T	P	C	CIA	SEE	Total
1	PC	III-I	23AEC10	Analog and Digital Communications	3	0	0	3	30	70	100
2	PC	III-I	23AEC69	Electronic Instrumentation	3	0	0	3	30	70	100
3	PC	III –II	23AEC16	Analog & Digital IC Applications	3	0	0	3	30	70	100
4	PC	III –II	23AEC68	Principles of Communication Engineering	3	0	0	3	30	70	100
5	PC	III –II	23AEC22	Analog & Digital IC Applications Lab	0	0	3	1.5	30	70	100
6	PC	IV-I	23AEC33	VLSI Design	3	0	0	3	30	70	100
7	PC	IV-I	23AEC40	VLSI Design Lab	0	0	3	1.5	30	70	100
TOTAL					**15**	**0**	**06**	**18**	**210**	**490**	**700**

MINORS IN INTERNET OF THINGS

S.No	Category	Year & Sem	Course Code	Name of the Subject and Lab	L	T	P	C	Scheme of Examination (Maximum Marks)		
									CIA	SEE	Total
1	PC	III – I	23ACO01	Internet of Things	3	0	0	3	30	70	100
2	PC	III – I	23ACS25	Wireless Sensor Networks	3	0	0	3	30	70	100
3	PC	III – II	23ACO16	Communication Protocols for IoT	3	0	0	3	30	70	100
4	PC	III – II	23ACO19	Industrial IoT	3	0	0	3	30	70	100
5	PC	IV - I	23ACS24	Cloud Computing	3	0	0	3	30	70	100
6	PC	III – II	23ACO02	Internet of Things Lab	0	0	3	1.5	30	70	100
7	PC	IV – I	23ACS26	Cloud Computing Lab	0	0	3	1.5	30	70	100
		Total Credits						**18**	**210**	**490**	**700**

MINORS IN DATA SCIENCE

S. No	Category	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L/D	T	P	C	CIA	SEE	Total
1	PC	III-I	23ACD36	Data Science for Engineers	3	-	0	3	30	70	100
2	PC	III-I	23ACD37	Statistical Learning for Data Science	3	-	0	3	30	70	100
3	PC	III –II	23ACD38	Introduction to Data Engineering	3	-	0	3	30	70	100
4	PC	III –II	23ACM34	Machine learning fundamentals	3	-	0	3	30	70	100
5	PC	IV–I	23ACD39	Data Science applications	3	-	0	3	30	70	100
6	PC	III-II	23ACD40	Data Science Practice Lab (R/Python)	0	0	3	1.5	30	70	100
7	PC	IV-I	23ACD41	Statistical Learning & ML Lab	0	0	3	1.5	30	70	100
TOTAL					**15**	**0**	**06**	**18**	**210**	**490**	**700**

MINOR IN DATA ANALYTICS

S. No	Category	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L/D	T	P	C	CIA	SEE	Total
1	PC	III – I	23ACD42	Introduction to Data Analytics	3	-	0	3	30	70	100
2	PC	III – I	23ACD43	Data Engineering Essentials	3	-	0	3	30	70	100
3	PC	III – II	23ACD44	Predictive Analytics Fundamentals	3	-	0	3	30	70	100
4	PC	III – II	23ACD45	Big Data Analytics (Hadoop, Spark)	3	-	0	3	30	70	100
5	PC	IV - I	23ACD46	Data Analytics with Power BI/Tableau/Matplotlib	3	-	0	3	30	70	100
6	PC	III - I	23ACD47	Data Analytics Tools Lab	0	0	3	1.5	30	70	100
7	PC	IV- I	23ACD48	Big Data &NoSQL Lab	0	0	3	1.5	30	70	100
TOTAL					**15**	**0**	**06**	**18**	**210**	**490**	**700**

MINORS IN DATA SCIENCE AND ANALYTICS

S. No	Category	Year & Sem	Course code	Course title	Hours per week			Credits	Scheme of Examination Max. Marks		
					L	T	P		CIA	SEE	Total
1	PC	III – I	23ACD49	Data Science for Engineers	3	0	0	3	30	70	100
2	PC	III – I	23ACD50	Statistical Learning for Data Science	3	0	0	3	30	70	100
3	PC	III – II	23ACD51	Data Visualization & Exploratory Data Analysis	3	0	0	3	30	70	100
4	PC	III – II	23ACD52	Big Data Analytics Fundamentals	3	0	0	3	30	70	100
5	PC	IV - I	23ACD53	Introduction to Recommender Systems	3	0	0	3	30	70	100
6	PC	III-II	23ACD54	Data Visualization & EDA Lab	0	0	6	3	30	70	100
7	PC	IV-I	23ACD55	Recommender System Lab	0	0	6	3	30	70	100
TOTAL					**15**		**06**	**18**	**210**	**490**	**700**

MINOR IN DATA SCIENCE WITH AWS

SNo	Category	Year & Sem	Course Code	Name of the Subject and Lab	L	T	P	C	Scheme of Examination (Maximum Marks)		
									CIA	**SEE**	**Total**
1	PC	III – I	23ACD56	Foundations Of Cloud concepts	3	0	0	3	30	70	100
2	PC	III – I	23ACD57	Predictive Data Analytics using R	3	0	0	3	30	70	100
3	PC	III – II	23ACD58	Data Engineering with AWS	3	0	0	3	30	70	100
4	PC	III – II	23ACD59	Cloud-Native Machine Learning	3	0	0	3	30	70	100
5	PC	IV - I	23ACD60	Deep Learning: The Modern Approach	3	0	0	3	30	70	100
6	PR	IV - I	23ACD61	Capstone Project and Portfolio Development	0	0	6	3	30	70	100
				TOTAL	**15**		**06**	**18**	**180**	**420**	**600**

MINORS IN QUANTUM TECHNOLOGIES

S.No	Category	Year & Sem	Course Code	Name of the Subject and Lab	L	T	P	C	Scheme of Examination (Maximum Marks)		
									CIA	SEE	Total
1	PC	III – I	23AIT24	Foundations of Quantum Technologies	3	0	0	3	30	70	100
2	PC	III – I	23AIT25	Solid State Physics for Quantum Technologies	3	0	0	3	30	70	100
3	PC	III – II	23AIT26	Quantum Optics Prerequisites for Quantum Technologies	3	0	0	3	30	70	100
4	PC	III – II	23AIT27	Introduction to Quantum Communication	3	0	0	3	30	70	100
5	PC	IV - I	23AIT28	Introduction to Quantum Sensing	3	0	0	3	30	70	100
6	PC	III – II	23AIT29	Quantum Communication and Sensing Lab	0	0	3	1.5	30	70	100
7	PC	IV – I	23AIT30	Quantum Devices and Materials Lab	0	0	3	1.5	30	70	100
				Total	15	0	6	**18**	**210**	**490**	**700**

MINOR IN UX/UI DESIGN

SNo	Category	Year & Sem	Course Code	Name of the Subject and Lab	L	T	P	C	Scheme of Examination (Maximum Marks)		
									CIA	SEE	Total
1	PC	III – I	23AIT37	Introduction to Design and Visual	3	0	0	3	30	70	100
2	PC	III – I	23AIT38	Human Centered Design and Interaction Principles	3	0	0	3	30	70	100
3	PC	III – II	23AIT39	UX Research and Design Thinking	3	0	0	3	30	70	100
4	PC	III – II	23AIT40	Prototyping and Usability Testing	3	0	0	3	30	70	100
5	PC	IV - I	23AIT41	Frontend Development for Designers	3	0	0	3	30	70	100
6	PR	IV - I	23AIT42	Capstone Project and Portfolio Development	0	0	6	3	30	70	100
				Total	15	0	6	**18**	**180**	**420**	**600**

MINORS IN CYBER SECURITY

S. No	Category	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L	T	P	C	CIA	SEE	Total
1.	PC	III – I	23ACC05	Cyber Crimes & Digital Forensics	3	0	0	3	30	70	100
2.	PC	III – I	23ACC16	Cyber Laws and Security Policies	3	0	0	3	30	70	100
3.	PC	III – II	23ACC01	Introduction to Cyber Security	3	0	0	3	30	70	100
4.	PC	III – II	23ACC11	Blockchain Technology	3	0	0	3	30	70	100
5.	PC	IV - I	23ACC04	Cryptography & Network Security (Pre-requisite: Computer Networks)	3	0	0	3	30	70	100
6.	PC	III-II	23ACC02	Cyber Security Lab			3	1.5	30	70	100
7	PC	IV-I	23ACC08	Cryptography & Network Security Lab			3	1.5	30	70	100
				Total	**15**	**0**	**6**	**18**	**210**	**490**	**700**

MINOR IN QUANTUM COMPUTING

S. No	Category	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L	T	P	C	CIA	SEE	Total
1.	PC	III -I	23ACS46	Introduction to Quantum Computing	3	0	0	3	30	70	100
2.	PC	III-I	23ACS47	Mathematical Foundations for Quantum Computing	3	0	0	3	30	70	100
3.	PC	III -II	23ACS48	Quantum Algorithms	3	0	0	3	30	70	100
4.	PC	III -II	23ACS49	Quantum Information and Communication	3	0	0	3	30	70	100
5.	PC	IV-I	23ACS50	Quantum Machine Learning (QML)	3	0	0	3	30	70	100
6.	PC	III-II	23ACS51	Quantum Algorithms Lab			3	1.5	30	70	100
7	PC	III -I	23ACS52	Quantum Programming and Simulation Lab			3	1.5	30	70	100
				Total	**15**	**0**	**6**	**18**	**210**	**490**	**700**

MINOR IN CLOUD COMPUTING

S. No	Category	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L	T	P	C	CIA	SEE	Total
1.	PC	III-I	23ACS53	Fundamentals of Cloud Computing	3	0	0	3	30	70	100
2	PC	III-I	23ACS54	Virtualization and Cloud Infrastructure	3	0	0	3	30	70	100
3.	PC	III -II	23ACS55	Cloud Platforms and Development	3	0	0	3	30	70	100
4.	PC	III -II	23ACS56	Cloud Security and Compliance	3	0	0	3	30	70	100
5.	PC	IV -I	23ACS57	Cloud Application Design and DevOps	3	0	0	3	30	70	100
6	PR	IV -I	23ACS58	Capstone project			6	3	30	70	100
					15	**0**	**6**	**18**	**180**	**420**	**600**

MINOR IN FULL STACK MERN & DEVOPS

S. No	Category	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L	T	P	C	CIA	SEE	Total
1.	PC	III-I	20ACS59	Web Fundamentals, Advanced JavaScript & GitHub	3	0	0	3	30	70	100
2	PC	III-I	20ACS60	Backend Development with Node.js & Express	3	0	0	3	30	70	100
3.	PC	III -II	20ACS61	Building User Interfaces with React.js	3	0	0	3	30	70	100
4.	PC	III -II	20ACS62	DevOps Fundamentals & CI/CD Pipelines	3	0	0	3	30	70	100
5.	PC	IV -I	20ACS63	Advanced Backend & Microservices	3	0	0	3	30	70	100
6	PR	IV -I	20ACS64	Project on full stack MERN and Devops			6	3	30	70	100
				Total	**15**	**0**	**6**	**18**	**180**	**420**	**600**

MINOR IN BIO MEDICAL ENGINEERING (BME)

S. No	Cate gory	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L/D	T	P	C	CIA	SEE	Total
1	PC	III-I	23ABM09	Nano Biotechnology	3	0	0	3	30	70	100
2	PC	III-I	23ABM10	Tissue Engineering	3	0	0	3	30	70	100
3	PC	III –II	23ABM02	Biomaterials	3	0	0	3	30	70	100
4	PC	III –II	23ABM14	Bio MEMS & Bio Micro fluids	3	0	0	3	30	70	100
5	PC	III –II	23ABM03	Biomaterials Lab	0	0	3	1.5	30	70	100
6	PC	IV-I	23ABM04	Biosensors & Transducers	3	0	0	3	30	70	100
7	PC	IV-I	23ABM05	Biosensors & Transducers Lab	0	0	3	1.5	30	70	100
				TOTAL	**15**	**0**	**06**	**18**	**210**	**490**	**700**

MINORS IN ARTIFICIAL INTELLIGENCE AND DATA SCIENCE (AI&DS)

SNo	Category	Year & Sem	Course Code	Name of the Subject and Lab	L	T	P	C	Scheme of Examination (Maximum Marks)		
									CIA	SEE	Total
1	PC	III – I	23AAD19	Data Analytics using R	3	0	0	3	30	70	100
2	PC	III – I	23AAD20	Creative Intelligence with Generative Models	3	0	0	3	30	70	100
3	PC	III – II	23AAD21	Data Management with SQL & No-SQL	3	0	0	3	30	70	100
4	PC	III – II	23AAD22	Applied Generative AI	3	0	0	3	30	70	100
5	PC	IV - I	23AAD23	Text Analytics & NLP	3	0	0	3	30	70	100
6	PR	IV - I	23AAD24	Capstone Project and Portfolio Development	0	0	6	3	30	70	100
				Total	**15**	**0**	**6**	**18**	**180**	**420**	**600**

MINORS IN COMPUTER SCIENCE & BUSINESS SYSTEMS

S.No	Category	Year & Sem	Course Code	Name of the Subject and Lab	L	T	P	C	Scheme of Examination (Maximum Marks)		
									CIA	SEE	Total
1	PC	III – I	23ACB15	Fundamentals Of Computer Systems & Business Environment	3	0	0	3	30	70	100
2	PC	III – I	23ACE16	Database Management & Enterprise Systems	3	0	0	3	30	70	100
3	PC	III – II	23ACB17	Software Engineering for Business Applications	3	0	0	3	30	70	100
4	PC	III – II	23ACB18	Business Intelligence and Analytics	3	0	0	3	30	70	100
5	PC	IV - I	23ACB19	Digital Transformation & Business Strategy	3	0	0	3	30	70	100
6	PR	IV - I	23ACB20	Capstone Project and Portfolio Development	0	0	6	3	30	70	100
				Total	**15**	**0**	**6**	**18**	**180**	**420**	**600**

MINOR IN COMPUTER SCIENCE AND ENGINEERING

S. No	Category	Year & Sem	Course Code	Course Name	Hours/week			Credits	Scheme of Examination Maximum Marks		
					L	T	P	C	CIA	SEE	Total
1.	PC	III-I	23ACS24	Cloud computing	3	0	0	3	30	70	100
2.	PC	III-I	23AIT02	Software Engineering	3	0	0	3	30	70	100
3.	PC	III -II	23ACS05	Advanced Data Structures & Algorithm Analysis	3	0	0	3	30	70	100
4.	PC	III -II	23ACS43	Computer Organization	3	0	0	3	30	70	100
5.	PC	III -II	23ACS07	Advanced Data Structures and Algorithm Analysis Lab			3	1.5	30	70	100
6.	PC	IV-I	23ACS44	Web Technologies	3	0	0	3	30	70	100
7	PC	IV-I	23ACS45	Web Technologies Lab			3	1.5	30	70	100
				Total	**15**	**0**	**6**	**18**	**210**	**490**	**700**

L	T	P	C
2	0	0	2

23AHS01- COMMUNICATIVE ENGLISH

Course Objectives:

The main objective of introducing this course, *Communicative English,* is to facilitate effective listening, Reading, Speaking and Writing skills among the students. It enhances the same in their comprehending abilities, oral presentations, reporting useful information and providing knowledge of grammatical structures and vocabulary. This course helps the students to make them effective in speaking and writing skills and to make them industry ready.

Course Outcomes:

CO1: Understand the context, topic, and pieces of specific information from social orTransactional dialogues.
CO2: Apply grammatical structures to formulate sentences and correct word forms.
CO3: Analyze discourse markers to speak clearly on a specific topic in informal discussions.
CO4: Evaluate reading / listening texts and to write summaries based on global comprehension of these texts.
CO5: Create a coherent paragraph, essay, and resume.

UNIT I

Lesson	:	**HUMAN VALUES:** Gift of Magi **(Short Story)**
Listening	:	Identifying the topic, the context and specific pieces of information by Listening to short audio texts and answering a series of questions.
Speaking	:	Asking and answering general questions on familiar topics such as home, family, work, studies and interests; introducing oneself and others.
Reading	:	Skimming to get the main idea of a text; scanning to look for specific pieces of information.
Writing	:	Mechanics of Writing-Capitalization, Spellings, Punctuation- Parts of Sentences.
Grammar	:	Parts of Speech, Basic Sentence Structures-forming questions
Vocabulary	:	Synonyms, Antonyms, Affixes (Prefixes/Suffixes), Root words.

UNIT II

Lesson	:	**NATURE: The Brook by Alfred Tennyson (Poem)**
Listening	:	Answering a series of questions about main ideas and supporting ideas after listening to audio texts.
Speaking	:	Discussion in pairs/small groups on specific topics followed by short structuretalks.
Reading	:	Identifying sequence of ideas; recognizing verbal techniques that help to link the ideas in a paragraph together.
Writing	:	Structure of a paragraph - Paragraph writing (specific topics)
Grammar	:	Cohesive devices - linkers, use of articles and zero article; prepositions.
Vocabulary	:	Homonyms, Homophones, Homographs.

UNIT III

Lesson : **BIOGRAPHY: Elon Musk**

Listening : Listening for global comprehension and summarizing what is listened to.

Speaking : Discussing specific topics in pairs or small groups and reporting w hat is discussed

Reading : Reading a text in detail by making basic inferences -recognizing and Interpreting specific context clues; strategies to use text clues for comprehension.

Writing : Summarizing, Note-making, paraphrasing

Grammar : Verbs - tenses; subject-verb agreement; Compound words, Collocations

Vocabulary : Compound words, Collocations

UNIT IV

Lesson : **INSPIRATION: The Toys of Peace by** Saki

Listening : Making predictions while listening to conversations/ transactional dialogues without video; listening with video.

Speaking : Role plays for practice of conversational English in academic contexts (formal and informal) - asking for and giving information/directions.

Reading : Studying the use of graphic elements in texts to convey information, reveal Trends / patterns/relationships, communicate processes or display complicated data.

Writing : Letter Writing: Official Letters, Resumes

Grammar : Reporting verbs, Direct & Indirect speech, Active & Passive Voice

Vocabulary : Words often confused, Jargons

UNIT V

Lesson : **MOTIVATION:** The Power of Intrapersonal Communication (An Essay)

Listening : Identifying key terms, understanding concepts and answering a series of relevant questions that test comprehension.

Speaking : Formal oral presentations on topics from academic contexts

Reading : Reading comprehension.

Writing : Critical Writing **-** Writing structured essays on specific topics.

Grammar : Editing short texts –identifying and correcting common errors in grammar and usage (articles, prepositions, tenses, subject verb agreement)

Vocabulary : Technical Jargons

Text books:

1. Pathfinder: Communicative English for Undergraduate Students, 1st Edition, Orient Black Swan, 2023 (Units 1,2 & 3)
2. Empowering with Language by Cengage Publications, 2023 (Units 4 & 5)

Reference Books:

1. Dubey, Sham Ji & Co. English for Engineers, Vikas Publishers, 2020
2. Bailey, Stephen. Academic writing: A Handbook for International Students. Routledge, 2014.
3. Murphy, Raymond. English Grammar in Use, Fourth Edition, Cambridge UniversityPress, 2019.
4. Lewis, Norman. Word Power Made Easy- The Complete Handbook for Building a Superior Vocabulary. Anchor, 2014.

Web Resources:

GRAMMAR:

1. www.bbc.co.uk/learningenglish
2. https://dictionary.cambridge.org/grammar/british-grammar/
3. www.eslpod.com/index.html
4. https://www.learngrammar.net/
5. https://english4today.com/english-grammar-online-with-quizzes/
6. https://www.talkenglish.com/grammar/grammar.aspx

VOCABULARY

1. https://www.youtube.com/c/DailyVideoVocabulary/videos
2. https://www.youtube.com/channel/UC4cmBAit8i_NJZE8qK8sfpA

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

I B.Tech I SEM (Common to CSE, CSD & CSM)

II SEM (Common to EEE, ECE, EBM, CAI, AI, AI&DS CSO, CSC & IT)

23AHS02	CHEMISTRY	L T P C
		3 - - 3

Course Objectives:

- To familiarize engineering chemistry and its applications
- To train the students on the principles and applications of electrochemistry and polymers
- To introduce instrumental methods, molecular machines and switches.

Course Outcomes: At the end of the course, the students will be able to:

CO1: Compare the materials of construction for battery and electrochemical sensors.

CO2: Explain the preparation, properties, and applications of thermoplastics & thermosetting & elastomers conducting polymers.

CO3: Explain the principles of spectrometry, slc in separation of solid and liquid mixtures.

CO4: Apply the principle of Band diagrams in the application of conductors and semiconductors.

CO5: Summarize the concepts of Instrumental methods.

UNIT I Structure and Bonding Models:

Fundamentals of Quantum mechanics, Schrodinger Wave equation, significance of Ψ and Ψ^2, particle in one dimensional box, molecular orbital theory – bonding in homo- and heteronuclear diatomic molecules – energy level diagrams of O2 and CO, etc. π-molecular orbitals of butadiene and benzene, calculation of bond order.

UNIT II Modern Engineering materials:

Semiconductors – Introduction, basic concept,
application Super conductors-Introduction, basic concept,
applications.
Supercapacitors: Introduction, Basic Concept - Classification – Applications.
Nanomaterials: Introduction, classification, properties and applications of Fullerenes, carbon nano tubes and Graphenes nanoparticles.

UNIT III Electrochemistry and Applications

Electrochemical cell, Nernst equation, cell potential calculations and numerical problems, potentiometry- potentiometric titrations (redox titrations), concept of conductivity, conductivitycell, conductometric titrations (acid-base titrations).

Electrochemical sensors – potentiometric sensors with examples, amperometric sensors with examples.

Primary cells – Zinc-air battery, Secondary cells –lithium-ion batteries- working of the batteries including cell reactions; Fuel cells, hydrogen-oxygenfuel cell– working of the cells. Polymer Electrolyte Membrane Fuel cells (PEMFC).

UNIT IV Polymer Chemistry

Introduction to polymers, functionality of monomers, chain growth and step growth polymerization, coordination polymerization, with specific examples and mechanisms of polymer formation.

Plastics –Thermo and Thermosetting plastics, Preparation, properties and applications of

– PVC, Teflon, Bakelite, Nylon-6,6, carbon fibres. Elastomers–Buna-S, Buna-N–preparation, properties and applications.

Conducting polymers – polyacetylene, polyaniline, – mechanism of conduction and applications. Bio-Degradable polymers - Poly Glycolic Acid (PGA), Polyl Lactic Acid (PLA).

UNIT V Instrumental Methods and Applications

Electromagnetic spectrum. Absorption of radiation: Beer-Lambert's law. UV-Visible Spectroscopy, electronic transition, Instrumentation, IR spectroscopies, fundamental modes and selection rules, Instrumentation. Chromatography-Basic Principle, Classification-HPLC: Principle, Instrumentation and Applications.

Textbooks:

1. Jain and Jain, Engineering Chemistry, 16/e, DhanpatRai, 2013.
2. Peter Atkins, Julio de Paula and James Keeler, Atkins' Physical Chemistry, 10/e,Oxford University Press, 2010.

Reference Books:

1. Skoog and West, Principles of Instrumental Analysis, 6/e, Thomson, 2007.
2. J.D. Lee, Concise Inorganic Chemistry, 5th Edition, Wiley Publications, Feb.2008
3. Textbook of Polymer Science, Fred W. Billmayer Jr, 3rd Edition

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

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I B.Tech - I Semester (Common to All Branches)

23ACE01 BASIC CIVIL AND MECHANICAL ENGINEERING

Course Outcomes: On completion of the course, the student should be able to:

- **CO1.** Understand various sub-divisions of Civil Engineering and to appreciate their role inensuring better society.
- **CO2.** Know the concepts of surveying and to understand the measurement of distances,angles and levels through surveying.
- **CO3.** Realize the importance of Transportation in nation's economy and the engineeringmeasures related to highways in terms of geometrics.
- **CO4.** Understand the importance of water resources and storage structures so that the social responsibilities of water conservation will be appreciated.
- **CO5.** Understand the different manufacturing processes and explain the basics of thermal engineering and its applications.
- **CO6.** Describe the working of different mechanical power transmission systems and powerplants; learn basics of robotics.

PART A: BASIC CIVIL ENGINEERING

UNIT I

Basics of Civil Engineering: Role of Civil Engineers in Society- Various Disciplines of Civil Engineering- Structural Engineering- Geo-technical Engineering- Transportation Engineering Hydraulics and Water Resources Engineering - Environmental Engineering - Scope of eachdiscipline

- Building Construction and Planning- Construction Materials-Cement – Aggregate Bricks - Cement concrete- Steel-Tests on these materials.

Factors to be considered in Building Planning- Nature of Buildings- Typical Layouts of a Residential Building- Industrial Building- Commercial Building like a Supermarket / Hotel / Theatre.

UNIT II

Surveying: Objectives of Surveying- Horizontal Measurements- Vertical Measurements- Angular Measurements- Levelling instruments used for levelling- Introduction to Bearings- Simple problems on levelling and bearings-Contour mapping.

UNIT III

Transportation Engineering, Water Resources and Environmental Engineering: Importance of Transportation in Nation's economic development- Types of Highway Pavements- Flexible Pavements and Rigid Pavements - Simple Differences - Basic geometric design elements of a highway- Camber- Stopping Sight Distance- Super elevation- Introduction.

Water Resources and Environmental Engineering: Sources of water- Quality of water- Specifications and Tests- Introduction to Hydrology- Hydrograph –Rain water Harvesting- Rain water runoff- Water Storage Structures (Simple introduction to Dams and Reservoirs).

Textbooks:

1. G. Shanmugam and M.S.Palanisamy, Basic Civil and the Mechanical

Engineering, Tata Mcgraw Hill publications (India) Pvt. Ltd.
2. Basic Civil Engineering, S.S. Bhavikatti, New Age International Publishers.
3. Engineering Materials, Dr. S.C. Rangwala, Charotor Publishing House.
4. Highway Engineering, S.K.Khanna, C.E.G. Justo and Veeraraghavan, Nemchand and Brothers Publications.
5. Irrigation Engineering and Hydraulic Structures - Santosh Kumar Garg, KhannaPublishers, Delhi.
6. Building Construction, Dr. B. C. Punmia, Lakshmi Publications, Delhi.

Reference Books:

1. Surveying, Vol- I and Vol-II, S.K. Duggal, Tata McGraw Hill Publishers.
2. Hydrology and Water Resources Engineering, Santosh Kumar Garg, Khanna Publishers, Delhi.

PART B: BASIC MECHANICAL ENGINEERING

UNIT I

Introduction to Mechanical Engineering: Role of Mechanical Engineering in Industries andSociety- Technologies in different sectors such as Energy, Manufacturing, Automotive, Aerospace, and Marine sectors.

Engineering Materials - Metals-Ferrous and Non-ferrous, Ceramics, Composites, Smart materials.

UNIT II

Manufacturing Processes: Principles of Casting**,** Forming, joining processes, Machining, Introduction to CNC machines, 3D printing, and Smart manufacturing.

Thermal Engineering – working principle of Boilers, Otto cycle, Diesel cycle, Refrigerationand air- conditioning cycles, IC engines, 2-Stroke and 4-Stroke engines, SI/CI Engines, Components of Electric and Hybrid Vehicles.

UNIT III

Power plants – working principle of Steam, Diesel, Hydro, Nuclear power plants.

Mechanical Power Transmission - Belt Drives, Chain, Rope drives, Gear Drives and theirapplications.

Introduction to Robotics - Joints & links, configurations, and applications of robotics.

(Note: The subject covers only the basic principles of Civil and Mechanical Engineering systems. The evaluation shall be intended to test only the fundamentals of the subject)

Textbooks:

1. Internal Combustion Engines by V.Ganesan, By Tata McGraw Hill publications (India)Pvt. Ltd.
2. A Tear book of Theory of Machines by S.S. Rattan, Tata McGraw Hill Publications,(India) Pvt. Ltd.
3. An introduction to Mechanical Engg by Jonathan Wicker and Kemper Lewis, cengagelearning India pvt. Ltd.

Reference Books:

1. Appuu Kuttan KK, Robotics, I.K. International Publishing House Pvt. Ltd. Volume-I
2. 3D printing & Additive Manufacturing Technology- L. Jyothish Kumar, Pulak MPandey, Springer publications
3. Thermal Engineering by Mahesh M Rathore Tata Mcgraw Hill publications (India) Pvt.Ltd.
4. G. Shanmugam and M.S.Palanisamy, Basic Civil and the Mechanical Engineering, Tata Mcgraw Hill publications (India) Pvt. Ltd.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

I B.Tech I Semester, CSE (Common to all branches)

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23ACS01: INTRODUCTION TO PROGRAMMING

Course Objectives:

- To introduce students to the fundamentals of computer programming.
- To provide hands-on experience with coding and debugging.
- To foster logical thinking and problem-solving skills using programming.
- To familiarize students with programming concepts such as data types, control structures, functions, and arrays.

Course Outcomes: At the end of the course students will be able to

CO1. Understand basics of computers, the concept of algorithm and algorithmic thinking.
CO2. Develop the ability to analyze a problem, develop an algorithm to solve it. **CO3.** Proficiently use the C programming language to implement various algorithms. **CO4.** Understand more advanced features of C language.
CO5. Develop problem-solving skills and the ability to debug and optimize the code.

UNIT I Introduction to Programming and Problem Solving
History of Computers, Basic organization of a computer: ALU, input-output units, memory, program counter, Introduction to Programming Languages, Basics of a Computer Program-Algorithms, flowcharts (Using Dia Tool), pseudo code.

Problem solving techniques: Algorithmic approach, characteristics of algorithm, Problem solving strategies : Top-down approach, Bottom-up approach, Time and space complexities of algorithms.

Overview of C: History Of C, Basic Structure of C Program, Primitive Data Types, Variables, and Constants, Basic Input and Output, Operations, Type Conversion, and Casting.

UNIT II Control Structures
Simple sequential programs Conditional Statements (if, if-else, switch), Loops (for, while, do- while) Break and Continue.

UNIT III Arrays and Strings
Definition of Arrays, Arrays indexing, memory model, programs with array of integers, two dimensional arrays, Multidimensional Arrays, Introduction to Strings, operations on strings

UNIT IV Functions
Introduction to Functions, Function Declaration and Definition, Function call Return Types and Arguments, modifying parameters inside functions using pointers, arrays as parameters. Scope and Lifetime of Variables, Recursion.

UNIT V User Defined Data types, File Handling, Pointers
User-defined data types-Structures- Introduction, Nested Structures, Array of Structures, Structures and Functions, and Unions, pointers, dereferencing and address operators, pointer

and address arithmetic, array manipulation using pointers. Operations on file handling Self-Referential structures, Linked List (creation and display)

Text Books:

1. B. A. Forouzan and R. F. Gilberg, Computer Science: A Structured Programming Approach Using C, 3/e, Cengage Learning, 2007.
2. Problem solving with C, M.T.Somashekara, PHI
3. "The C Programming Language" by Brian W. Kernighan and Dennis M. Ritchie
4. Schaum's Outline of Programming with C by Byron S Gottfried (1996), McGraw-HillEducation (ISBN:978-0070240353)

Reference Books:

1. Balagurusamy, E. (2008). Computing fundamentals and C Programming. McGraw-HillEducation.
2. Programming in C Rema Theraja-2nd edition 2016
3. C Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE
4. Jeri R. Hanly, Ellot B. Koffman, Problem Solving and Program Design in C, 5/e, Pearson

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

I B.Tech I Semester (Common to All Branches)

L	T	P	C
3	0	0	3

23AHS04 : LINEAR ALGEBRA & CALCULUS

Course Objectives:

- To equip the students with standard concepts and tools at an intermediate to advanced level mathematics to develop the confidence and ability among the students to handle various real- world problems and their applications.

Course Outcomes: At the end of the course, the student will be able to

CO1: Develop and use of matrix algebra techniques that are needed by engineers for practical applications.

CO2: Utilize mean value theorems to real life problems.

CO3: Familiarize with functions of several variables which is useful in optimization.

CO4: Learn important tools of calculus in higher dimensions.

CO5: Familiarize with double and triple integrals of functions of several variables in two dimensions using Cartesian and polar coordinates and in three dimensions using cylindrical and spherical coordinates.

UNIT I Matrices

Rank of a matrix by echelon form, normal form. Cauchy–Binet formulae (without proof). Inverse of Non- singular matrices by Gauss-Jordan method, System of linear equations: Solving system of Homogeneous and Non-Homogeneous equations by Gauss elimination method, Jacobi and Gauss Seidel Iteration Methods.

UNIT II Eigenvalues, Eigenvectors and Orthogonal Transformation

Eigen values, Eigenvectors and their properties, Diagonalization of a matrix, Cayley-Hamilton Theorem (without proof), finding inverse and power of a matrix by Cayley-Hamilton Theorem, Quadratic forms and Nature of the Quadratic Forms, Reduction of Quadratic form to canonicalforms by Orthogonal Transformation.

UNIT III Calculus

Mean Value Theorems: Rolle’s Theorem, Lagrange’s mean value theorem with their geometrical interpretation, Cauchy’s mean value theorem, Taylor’s and Maclaurin theorems with remainders (without proof), Problems and applications on the above theorems.

UNIT IV Partial differentiation and Applications (Multi variable calculus)

Functions of several variables: Continuity and Differentiability, Partial derivatives, total derivatives, chain rule, Directional derivative, Taylor’s and Maclaurin’s series expansion of functions of two variables. Jacobians, Functional dependence, maxima and minima of functionsof two variables, method of Lagrange multipliers.

UNIT V Multiple Integrals (Multi variable Calculus)

Double integrals, triple integrals, change of order of integration, change of variables to polar, cylindrical and spherical coordinates. Finding areas (by double integrals) and volumes (by double integrals and triple integrals).

Textbooks:

1. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers, 2017, 44th Edition
2. Advanced Engineering Mathematics, Erwin Kreyszig, John Wiley & Sons, 2018, 10th Edition.

Reference Books:

1. Thomas Calculus, George B. Thomas, Maurice D. Weir and Joel Hass, PearsonPublishers, 2018, 14th Edition.
2. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, AlphaScience International Ltd., 2021 5th Edition(9th reprint).
3. Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4. Advanced Engineering Mathematics, Micheael Greenberg, , Pearson publishers, 9th edition
5. Higher Engineering Mathematics, H. K Das, Er. Rajnish Verma, S. Chand Publications,2014, Third Edition (Reprint 2021)

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

L	T	P	C
0	0	3	1.5

23AME02 :ENGINEERING WORKSHOP
(Common to All branches of Engineering)

Course Outcomes:

After completion of this course, the student will be able to.

CO1: Identify workshop tools and their operational capabilities.

CO2: Practice on manufacturing of components using workshop trades including fitting, carpentry, foundry and welding.

CO3: Apply fitting operations in various applications.

CO4: Apply basic electrical engineering knowledge for House Wiring Practice.

List of Experiments:

1. **Demonstration**: Safety practices and precautions to be observed in workshop.

2. **Wood Working:** Familiarity with different types of woods and tools used in wood working and make following joints.
   a) Half– Lap joint b) Mortise and Tenon joint c) Corner Dovetail joint or Bridle joint.

3. **Sheet Metal Working**: Familiarity with different types of tools used in sheet metal working, Developments of following sheet metal job from GI sheets.
   a) Tapered tray b) Conical funnel c) Elbow pipe d) Brazing

4. **Fitting:** Familiarity with different types of tools used in fitting and do the following fitting exercises.
   a) V-fit b) Dovetail fit c) Semi-circular fit d) Bicycle tire puncture and change of two-wheeler tyre

5. **Electrical Wiring**: Familiarity with different types of basic electrical circuits and make the following connections.
a) Parallel and series b) Two-way switch c) Go down lighting d) Tube light
e) Three phase motor f) Soldering of wires

6. **Foundry Trade:** Demonstration and practice on Moulding tools and processes, Preparation of Green Sand Moulds for given Patterns.

7. **Welding Shop**: Demonstration and practice on Arc Welding and Gas welding. Preparation of Lap joint and Butt joint.

8. **Plumbing:** Demonstration and practice of Plumbing tools, Preparation of Pipe joints with coupling for same diameter and with reducer for different diameters.

Textbooks:

1. Basic Workshop Technology: Manufacturing Process, Felix W.; Independently Published, 2019. Workshop Processes, Practices and Materials; Bruce J. Black, Routledge publishers, 5th Edn. 2015.

2. A Course in Workshop Technology Vol I. & II, B.S. Raghuwanshi, Dhanpath Rai & Co.,2015&2017.

Reference Books:

1. Elements of Workshop Technology, Vol. I by S. K. Hajra Choudhury & Others, Media Promoters and Publishers, Mumbai. 2007,14th edition

2. Workshop Practice by H. S. Bawa, Tata-McGraw Hill, 2004.

3. Wiring Estimating, Costing and Contracting; Soni P.M. & Upadhyay P. A.; Atul Prakashan,2021-22.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

L	T	P	C
0	0	2	1

20AHS06 - COMMUNICATIVE ENGLISH LAB

I B.Tech I Semester (Common to CSE, CSD, CSM, CE & ME)
I B.Tech II Semester (Common to ECE, EEE, CSC, IT, CAI, AI, AI&DS CSO, CSBS & EBM)

Course Objectives:
The main objective of introducing this course, Communicative English Laboratory, is to exposethe students to a variety of self-instructional, learner friendly modes of language learning. Thestudents will get trained in basic communication skills and also make them ready to face job interviews.

Course Outcomes:

CO1: Understand the different aspects of the English language proficiency with emphasison LSRW skills.

CO2: Apply communication skills through various language learning activities.

CO3: Analyze the English speech sounds, stress, rhythm, intonation and syllable divisionfor better listening and speaking comprehension.

CO4: Evaluate and exhibit professionalism in participating in debates and group discussions.CO5: Create effective Course Objectives:

List of Topics:

1. Vowels & Consonants
2. Neutralization/Accent Rules
3. Communication Skills & JAM
4. Role Play or Conversational Practice
5. E-mail Writing
6. Resume Writing, Cover letter, SOP
7. Group Discussions-methods & practice
8. Debates - Methods & Practice
9. PPT Presentations/ Poster Presentation
10. Interviews Skills

Suggested Software:

- Young India Films
- Walden Infotech

Reference Books:

1. Raman Meenakshi, Sangeeta-Sharma. *Technical Communication*. Oxford Press.2018.
2. Taylor Grant: *English Conversation Practice*, Tata McGraw-Hill Education India,2016
3. Hewing's, Martin. Cambridge *Academic English* (B2). CUP, 2012.
4. J. Sethi & P.V. Dhamija. *A Course in Phonetics and Spoken English*, (2^{nd} Ed),Kindle, 2013

Web Resources:

Spoken English:

1. www.esl-lab.com
2. www.englishmedialab.com
3. www.englishinteractive.net
4. https://www.britishcouncil.in/english/online
5. http://www.letstalkpodcast.com/
6. https://www.youtube.com/c/mmmEnglish_Emma/featured
7. https://www.youtube.com/c/ArnelsEverydayEnglish/featured
8. https://www.youtube.com/c/engvidAdam/featured
9. https://www.youtube.com/c/EnglishClass101/featured
10. https://www.youtube.com/c/SpeakEnglishWithTiffani/playlists
11. https://www.youtube.com/channel/UCV1h_cBE0Drdx19qkTM0WNw

Voice & Accent:

1. https://www.youtube.com/user/letstalkaccent/videos
2. https://www.youtube.com/c/EngLanguageClub/featured
3. https://www.youtube.com/channel/UC_OskgZBoS4dAnVUgJVexc
   https://www.youtube.com/channel/UCNfm92h83W2i2ijc5Xwp_IA

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

I B.Tech I SEM (Common to CSE, CSD, CSM

II SEM (Common to EEE, ECE, EBM, CAI, AI, AI&DS, CSO, CSC, IT)

23AHS07	**CHEMISTRY LAB**	**L T P C**
		0 0 2 1

Course Objectives: Verify the fundamental concepts with experiments.

Course Outcomes: At the end of the course, the students will be able to **CO1.** Determine the cell constant and conductance of solutions. **CO2.** Prepare advanced polymer Bakelite materials.

CO3. Measure the strength of an acid present.

CO4. Analyse the IR spectra of some organic compounds.

CO5. Calculate strength of acid in Pb-Acid battery.

List of Experiments: (Any 10 experiments)

1. Measurement of 10Dq by spectrophotometric method
2. Conductometric titration of strong acid vs. strong base
3. Conductometric titration of weak acid vs. strong base
4. Determination of cell constant and conductance of solutions
5. Potentiometry - determination of redox potentials and emfs
6. Determination of Strength of an acid in Pb-Acid battery
7. Preparation of Bakelite
8. Verify Lambert-Beer's law
9. Wavelength measurement of sample through UV-Visible Spectroscopy
10. Identification of simple organic compounds by IR
11. Preparation of nanomaterials by precipitation method
12. Estimation of Ferrous Iron by Dichrometry
13. pH metric Titration of strong acid vs. strong base
14. Determination of Viscosity of a polymer solution using Ostwald Viscometer

Reference: "Vogel's Quantitative Chemical Analysis 6th Edition 6th Edition" Pearson Publications by J. Mendham, R.C.Denney, J.D.Barnes and B. Sivasankar

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

I B.Tech I Semester (Common to all branches)

L	T	P	C
-	-	3	1.5

23ACS02: COMPUTER PROGRAMMING LAB

Course Objectives:

1. To use basic data types, operators, expressions and expression evaluation mechanisms using C Programming Language.
2. To implement control flows, construct in C Programming Language and understand the syntax, semantics and usability contexts of these different constructs.
3. To develop composite data types in C and constructs available to develop their datatypes, utilize them to model things and dealing with data from and to external files.
4. To design programs with different variations of the constructs available for practicing modular programming and understand the pros and cons of using different variants and apply optimization.

Course Outcomes : At the end of the course, Student will be able to

- **CO1.** Read, understand and trace the execution of programs written in C language.
- **CO2.** Select the right control structure for solving the problem.
- **CO3.** Develop C programs which utilize the memory efficiently using programmingconstructs like pointers.
- **CO4.** Develop, Debug and Execute programs to demonstrate the applications of arrays,functions, basic concepts of pointers in C.

List of Experiments:

WEEK 1

Objective: Getting familiar with the programming environment on the computer and writing the first program.

Suggested Experiments/Activities:

Tutorial 1: Problem-solving using Computers.

Familiarization with programming environment

i) Basic Linux environment and its editors like Vi, Vim & Emacs etc.
ii) Exposure to Turbo C, gcc
iii) Writing simple programs using printf(), scanf()

WEEK 2

Objective: Getting familiar with how to formally describe a solution to a problem in a series of finite steps both using textual notation and graphic notation.

Suggested Experiments /Activities:

Tutorial 2: Problem-solving using Algorithms and Flow charts.

Converting algorithms/flow charts into C Source code.

Developing the algorithms/flowcharts for the following sample programs

i) Sum and average of 3 numbers
ii) Conversion of Fahrenheit to Celsius and vice versa
iii) Simple interest calculation

WEEK 3

Objective: Learn how to define variables with the desired data-type, initialize them with appropriate values and how arithmetic operators can be used with variables and constants.

Suggested Experiments/Activities:

Tutorial 3: Variable types and type conversions:

Simple computational problems using arithmetic expressions.

i) Finding the square root of a given number
ii) Finding compound interest
iii) Area of a triangle using heron's formulae
iv) Distance travelled by an object

WEEK 4

Objective: Explore the full scope of expressions, type-compatibility of variables & constants and operators used in the expression and how operator precedence works.

Suggested Experiments/Activities:

Tutorial4: Operators and the precedence and as associativity:

Simple computational problems using the operator' precedence and associativity

i) Evaluate the following expressions.
   a. A+B*C+(D*E) + F*G
   b. A/B*C-B+A*D/3
   c. A+++B---A
   d. J= (i++) + (++i)
ii) Find the maximum of three numbers using conditional operator
iii) Take marks of 5 subjects in integers, and find the total, average in float

WEEK 5

Objective: Explore the full scope of different variants of "if construct" namely if-else, null-else, if- else if*-else, switch and nested-if including in what scenario each one of them can be used and how to use them. Explore all relational and logical operators while writing conditionals for "if construct".

Suggested Experiments/Activities:

Tutorial 5: Branching and logical expressions:

Problems involving if-then-else structures.

i) Write a C program to find the max and min of four numbers using if-else.
ii) Write a C program to generate electricity bill.
iii) Find the roots of the quadratic equation.
iv) Write a C program to simulate a calculator using switch case.
v) Write a C program to find the given year is a leap year or not.

WEEK 6

Objective: Explore the full scope of iterative constructs namely while loop, do-while loop and for loop in addition to structured jump constructs like break and continue including when each of these statements is more appropriate to use.

Suggested Experiments/Activities:

Tutorial 6: Loops, while and for loops

Iterative problems e.g., the sum of series

i) Find the factorial of given number using any loop.
ii) Find the given number is a prime or not.
iii) Compute sine and cos series

iv) Checking a number palindrome
v) Construct a pyramid of numbers.

WEEK 7:
Objective: Explore the full scope of Arrays construct namely defining and initializing 1-D and 2-D and more generically n-D arrays and referencing individual array elements from the defined array. Using integer 1-D arrays, explore search solution linear search.

Suggested Experiments/Activities:
Tutorial 7: 1 D Arrays: searching.
D Array manipulation, linear search

i) Find the min and max of a 1-D integer array.
ii) Perform linear search on1D array.
iii) The reverse of a 1D integer array
iv) Find 2's complement of the given binary number.
v) Eliminate duplicate elements in an array.

WEEK 8:
Objective: Explore the difference between other arrays and character arrays that can be used as Strings by using null character and get comfortable with string by doing experiments that will reverse a string and concatenate two strings. Explore sorting solution bubble sort using integer arrays.

Suggested Experiments/Activities:
Tutorial 8: 2 D arrays, sorting and Strings.
Matrix problems, String operations, Bubble sort

i) Addition of two matrices
ii) Multiplication two matrices
iii) Sort array elements using bubble sort
iv) Concatenate two strings without built-in functions
v) Reverse a string using built-in and without built-in string functions

WEEK 9:
Objective: Explore the Functions, sub-routines, scope and extent of variables, doing some experiments by parameter passing using call by value. Basic methods of numerical integration

Suggested Experiments/Activities:
Tutorial 9: Functions, call by value, scope and extent,
Simple functions using call by value, solving differential equations using Eulers theorem

i) Write a C function to calculate NCR value
ii) Write a C function to find the length of a string
iii) Write a C function to transpose of a matrix
iv) Write a C function to demonstrate numerical integration of differential equations using Euler's method

WEEK 10:
Objective: Explore how recursive solutions can be programmed by writing recursive functions that can be invoked from the main by programming at-least five distinct problems that have naturally recursive solutions.

Suggested Experiments/Activities:
Tutorial 10: Recursion, the structure of recursive calls
Recursive functions

i) Write a recursive function to generate Fibonacci series
ii) Write a recursive function to find the lcm of two numbers
iii) Write a recursive function to find the factorial of a number
iv) Write a C Program to implement Ackermann function using recursion
v) Write a recursive function to find the sum of series.

WEEK 11:
Objective: Explore the basic difference between normal and pointer variables, Arithmetic operations using pointers and passing variables to functions using pointers

Suggested Experiments/Activities:
Tutorial 11: Call by reference, dangling pointers
Simple functions using Call by reference, Dangling pointers

i) Write a C program to swap two numbers using call by reference
ii) Demonstrate Dangling pointer problem using a C program
iii) Write a C program to copy one string into another using pointer
iv) Write a C program to find no of lowercase, uppercase, digits and other characters using pointers.

WEEK12:
Objective: Explore pointers to manage a dynamic array of integers, including memory allocation & value initialization, resizing changing and reordering the contents of an array and memory de- allocation using malloc (), calloc (), realloc () and free () functions. Gain experience processing command-line arguments received by C

Suggested Experiments/Activities:
Tutorial 12: Pointers, structures and dynamic memory allocation Pointers and structures, memory dereference.

i) Write a C program to find the sum of a 1D array using malloc()
ii) Write a C program to find the total, average of n students using structures
iii) Enter n students data using calloc() and display failed students list
iv) Read student name and marks from the command line and display the student details along with the total.
v) Write a C program to implement realloc()

WEEK 13:
Objective: Experiment with C Structures, Unions, bit fields and self-referential structures (Singly linked lists) and nested structures

Suggested Experiments/Activities:
Tutorial 13: Bitfields, Self-Referential Structures, Linked lists Bitfields, linked lists
Read and print a date using dd/mm/yyyy format using bit-fields and differentiate the same without using bit- fields

i) Create and display a singly linked list using self-referential structure.
ii) Demonstrate the differences between structures and unions using a C program.
iii) Write a C program to shift/rotate using bitfields.
iv) Write a C program to copy one structure variable to another structure of the same type.

WEEK14:
Objective: To understand data files and file handling with various file I/O functions. Explore the differences between text and binary files.

Suggested Experiments/Activities:
Tutorial 14: File handling

File operations

i) Write a C program to write and read text into a file.
ii) Write a C program to write and read text into a binary file using fread() and fwrite()
iii) Copy the contents of one file to another file.
iv) Write a C program to merge two files into the third file using command-line arguments.
v) Find no. of lines, words and characters in a file
vi) Write a C program to print last n characters of a given file.

Text Books

1. Ajay Mittal, Programming in C: A practical approach, Pearson.
2. Byron Gottfried, Schaum' s Outline of Programming with C, McGraw Hill

Reference Books

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice- Hall of India
2. C Programming, A Problem-Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

I B.Tech I Semester (Common to All Branches)

L	T	P	C
0	0	1	0.5

23AHS10: HEALTH AND WELLNESS, YOGA AND SPORTS

Course Objectives:

The main objective of introducing this course is to make the students maintain their mental andphysical wellness by balancing emotions in their life. It mainly enhances the essential traits required for the development of the personality.

Course Outcomes: After completion of the course the student will be able to

CO1: Understand the importance of yoga and sports for Physical fitness and sound health.

CO2: Demonstrate an understanding of health-related fitness components.

CO3: Compare and contrast various activities that help enhance their health.

CO4: Assess current personal fitness levels.

CO5: Develop Positive Personality

UNIT I

Concept of health and fitness, Nutrition and Balanced diet, basic concept of immunity Relationship between diet and fitness, Globalization and its impact on health, Body Mass Index(BMI) of all age groups.

Activities:

i) Organizing health awareness programmes in community
ii) Preparation of health profile
iii) Preparation of chart for balance diet for all age groups

UNIT II

Concept of yoga, need for and importance of yoga, origin and history of yoga in Indian context,classification of yoga, Physiological effects of Asanas-Pranayama and meditation, stress management and yoga, Mental health and yoga practice.

Activities:

Yoga practices – Asana, Kriya, Mudra, Bandha, Dhyana, Surya Namaskar

UNIT III

Concept of Sports and fitness, importance, fitness components, history of sports, Ancient and Modern Olympics, Asian games and Commonwealth games.

Activities:

i) Participation in one major game and one individual sport viz., Athletics, Volleyball, Basketball, Handball, Football, Badminton, Kabaddi, Kho-kho, Table tennis, Cricket etc.
Practicing general and specific warm up, aerobics

ii) Practicing cardiorespiratory fitness, treadmill, run test, 9 min walk, skipping and running.

Reference Books:

1. Gordon Edlin, Eric Golanty. Health and Wellness, 14th Edn. Jones & Bartlett Learning,2022
2. T.K.V.Desikachar. The Heart of Yoga: Developing a Personal Practice
3. Archie J.Bahm. Yoga Sutras of Patanjali, Jain Publishing Company, 1993
4. Wiseman, John Lofty, SAS Survival Handbook: The Ultimate Guide to SurvivingAnywhere Third Edition, William Morrow Paperbacks, 2014
5. The Sports Rules Book/ Human Kinetics with Thomas Hanlon. -- 3rd ed. HumanKinetics, Inc.2014

General Guidelines:

1. Institutes must assign slots in the Timetable for the activities of Health/Sports/Yoga.
2. Institutes must provide field/facility and offer the minimum of five choices of as manyas Games/Sports.
3. Institutes are required to provide sports instructor / yoga teacher to mentor the students.

Evaluation Guidelines:

- Evaluated for a total of 100 marks.
- A student can select 6 activities of his/her choice with a minimum of 01 activity per unit. Each activity shall be evaluated by the concerned teacher for 15 marks, totalling to 90 marks.
- A student shall be evaluated by the concerned teacher for 10 marks by conducting vivavoce on the subject.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

I B. Tech I Semester (Common to EEE, ECE, IT, CAI, AI, AI&DS, CSO, CSC, EBM & CSBS) I B. Tech II Semester (Common to CE, ME, CSE, CSE (DS) & CSE(AI &ML))

L	T	P	C
3	0	0	3

23AHS05 ENGINEERING PHYSICS

COURSE OBJECTIVES

1. Bridging the gap between the Physics in school at 10+2 level and UG level engineering courses.
2. To identify the importance of the optical phenomenon i.e. interference, diffraction and polarization related to its Engineering applications.
3. Enlighten the periodic arrangement of atoms in Crystalline solids by Bragg's law – Learning the structural analysis through X-ray diffraction techniques.
4. Enlightenment of the concepts of Quantum Mechanics and to provide fundamentals of de Broglie matter waves, quantum mechanical wave equation and its application, the importance of free electron theory for metals.
5. To Understand the Physics of Semiconductors and their working mechanism, Concepts utilization of transport phenomenon of charge carriers in semiconductors. To give an impetus on the subtle mechanism of superconductors using the concept of BCS theory and their fascinating applications.
6. To explain the significant concepts of dielectric and magnetic materials that leads to potential applications in the emerging micro devices.

COURSE OUTCOMES

CO1: **Explain** the need of coherent sources and the conditions for sustained interference (L2). **Identify** the applications of interference in engineering (L3). **Analyze** the differences between interference and diffraction with applications (L4). **Illustrate** the concept of polarization of light and its applications (L2). **Classify** ordinary refracted light and extraordinary refracted rays by their states of polarization (L2)

CO2: **Interpret** various crystal systems (L2) and **Analyze** the characterization of materials by XRD (L4). **Identify** the important properties of crystals like the presence of long-range order and periodicity, structure determination using X-ray diffraction technique (L3). **Analysis** of structure of the crystals by Laue's method (L2).

CO3: **Describe** the dual nature of matter (L1). **Explain** the significance of wave function (L2). **Identify** the role of Schrodinger's time independent wave equation in studying particle in one- dimensional infinite potential well (L3). **Identify** the role of classical and quantum free electron theory in the study of electrical conductivity (L3).

CO4: **Classify** the crystalline solids (L2). **Outline** the properties of charge carriers in semiconductors (L2). **Identify** the type of semiconductor using Hall effect (L2). **Classify** superconductors based on Meissner's effect (L2). **Explain** Meissner's effect, BCS theory & Josephson effect in superconductors (L2).

CO5: **Explain** the concept of dielectric constant and polarization in dielectric materials (L2). **Summarize** various types of polarization of dielectrics (L2). **Interpret** Lorentz field and Claussius-Mosotti relation in dielectrics (L2). **Classify** the magnetic materials based on susceptibility (L2).

Unit-I: Wave Optics

Interference- Principle of superposition – Interference of light – Conditions for sustained interference - Interference in thin films (Reflection Geometry) – Colors in thin films – Newton's Rings – Determination of wavelength and refractive index.

Diffraction- Introduction – Fresnel and Fraunhofer diffraction – Fraunhofer diffraction due to singleslit, double slit and N-slits (qualitative) – Diffraction Grating - Dispersive power and resolving power of Grating (Qualitative).

Polarization- Introduction – Types of polarization – Polarization by reflection, refraction and double refraction - Nicol's Prism - Half wave and Quarter wave plates.

Unit II: Crystallography and X-ray diffraction **8hrs**

Crystallography: Space lattice, Basis, Unit Cell and lattice parameters – Crystal systems - Bravais Lattices — Coordination number - Packing fraction of SC, BCC & FCC - Miller indices – Separationbetween successive (h k l) planes.

X-ray diffraction: Bragg's law - X-ray Diffractometer – Crystal structure determination by Laue's method.

Unit-III: Quantum Mechanics and Free Electron Theory **9hrs**

Quantum Mechanics: Dual nature of matter – Heisenberg's Uncertainty Principle - Schrodinger's time independent and dependent wave equation – Significance and properties of wave function – Particle in a one-dimensional infinite potential well.

Free Electron Theory- Classical free electron theory (Qualitative with discussion of merits and demerits) – Quantum free electron theory – Equation for electrical conductivity based on quantum free electron theory – Fermi-Dirac distribution – Fermi energy - Failures of free electron theory.

Unit – IV: Semiconductors and Superconductors **8hrs**

Semiconductors: Formation of energy bands – classification of crystalline solids - Intrinsic semiconductors: Density of charge carriers – Electrical conductivity – Fermi level – Extrinsic semiconductors: density of charge carriers - Drift and diffusion currents – Einstein's equation - Hall effect and its Applications.

Superconductors: Introduction – Properties of superconductors – Meissner effect– Type I and Type II superconductors – AC and DC Josephson effects – BCS theory (qualitative treatment) – High T_c superconductors – Applications of superconductors.

Unit–V: Dielectric and Magnetic Materials **8hrs**

Dielectric Materials- Introduction – Dielectric polarization – Dielectric polarizability, Susceptibility and Dielectric constant and Displacement Vector – Relation between the electric vectors - Types of polarizations- Electronic (Quantitative), Ionic (Quantitative) and Orientation polarizations (Qualitative) -Lorentz field - Clausius-Mossotti equation - Dielectric loss.

Magnetic Materials- Introduction – Magnetic dipole moment – Magnetization – Magnetic susceptibility and Permeability – Atomic origin of magnetism – Classification of magnetic materials: Dia, Para, Ferro, Ferri & Antiferro – Domain concept of Ferromagnetism (Qualitative) – Hysteresis – Soft and Hard magnetic materials.

Text books:

1. Engineering Physics by M. N. Avadhanulu, P.G. Kshirsagar & TVS Arun Murthy S.ChandPublications, 11th Edition 2019.
2. Engineering Physics" by D.K. Bhattacharya and Poonam Tandon, Oxford press (2018).
3. Applied Physics by P.K. Palanisamy ,SciTech publications (2018)

Reference Books:

1. "Engineering Physics" - B.K. Pandey and S. Chaturvedi, Cengage Learning
2. "Fundamentals of Physics" - Halliday, Resnick and Walker, John Wiley &Sons.
3. "Fundamentals of Physics with Applications", Arthur Beiser, Samarjit Sengupta, Schaum Series.
4. "Engineering Physics" - Shatendra Sharma, Jyotsna Sharma, Pearson Education, 2018.
5. "Engineering Physics" - Sanjay D. Jain, D. Sahasrabudhe and Girish, University Press.
6. "Semiconductor physics and devices: Basic principle" - A. Donald, Neamen, Mc GrawHill.
7. "Solid state physics" – A.J.Dekker , Pan Macmillan publishers
8. "Introduction to Solid State Physics" -Charles Kittel ,Wiley

Mapping between Course Outcomes and Programme Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	**3**	**3**		**1**								
CO2	**3**	**3**	**1**									
CO3	**3**	**2**	**1**	**2**								
CO4	**3**	**3**	**1**	**2**	**1**				**1**		**1**	**1**
CO5	**3**	**2**	**2**		**1**				**2**		**1**	**1**

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

I B.Tech I Semester

L	T	P	C
3	0	0	3

23AEE01 BASIC ELECTRICAL & ELECTRONICS ENGINEERING
(Common to All branches of Engineering)

COURSE OBJECTIVES

To expose to the field of Electrical & Electronics Engineering, laws and principles of electrical/ electronic engineering and to acquire fundamental knowledge in the relevant field.

PART A: BASIC ELECTRICAL ENGINEERING

UNIT I DC & AC Circuits

DC Circuits: Electrical circuit elements (R, L and C), Ohm's Law and its limitations, KCL & KVL, series, parallel, series-parallel circuits, Super Position theorem, Simple numerical problems.

AC Circuits: A.C. Fundamentals: Equation of AC Voltage and current, waveform, time period, frequency, amplitude, phase, phase difference, average value, RMS value, form factor, peak factor, Voltage and current relationship with phasor diagrams in R, L, and C circuits, Concept of Impedance, Active power, reactive power and apparent power, Concept of power factor (Simple Numerical problems).

UNIT II Machines and Measuring Instruments

Machines: Construction, principle and operation of (i) DC Motor, (ii) DC Generator, (iii) Single Phase Transformer, (iv) Three Phase Induction Motor and (v) Alternator, Applications of electrical machines. **Measuring Instruments:** Construction and working principle of Permanent Magnet Moving Coil (PMMC), Moving Iron (MI) Instruments and Wheat Stone bridge.

UNIT III Energy Resources, Electricity Bill & Safety Measures

Energy Resources: Conventional and non-conventional energy resources; Layout and operation of various Power Generation systems: Hydel, Nuclear, Solar & Wind power generation.

Electricity bill: Power rating of household appliances including air conditioners, PCs, Laptops, Printers, etc. Definition of "unit" used for consumption of electrical energy, two-part electricity tariff, calculation of electricity bill for domestic consumers.

Equipment Safety Measures: Working principle of Fuse and Miniature circuit breaker (MCB), merits and demerits. Personal safety measures: Electric Shock, Earthing and its types, Safety Precautions to avoid shock.

Text Books:

1. Basic Electrical Engineering, D. C. Kulshreshtha, Tata McGraw Hill, 2019, First Edition
2. Power System Engineering, P.V. Gupta, M.L. Soni, U.S. Bhatnagar and A. Chakrabarti, Dhanpat Rai & Co, 2013
3. Fundamentals of Electrical Engineering, Rajendra Prasad, PHI publishers, 2014, Third Edition

Reference Books:

1. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, Mc Graw Hill, 2019, Fourth Edition
2. Principles of Power Systems, V.K. Mehtha, S.Chand Technical Publishers, 2020
3. Basic Electrical Engineering, T. K. Nagsarkar and M. S. Sukhija, Oxford University Press, 2017
4. Basic Electrical and Electronics Engineering, S. K. Bhatacharya, Person Publications, 2018, Second Edition.

Web Resources:

1. https://nptel.ac.in/courses/108105053
2. https://nptel.ac.in/courses/108108076

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)
(Common to All branches of Engineering)

23AME01 ENGINEERING GRAPHICS

Course Outcomes:
After completion of this course, the student will be able to
CO1: Understand the principles of engineering drawing, including engineering curves, scales, orthographic and isometric projections.
CO2: Draw and interpret orthographic projections of points, lines, planes and solids in front,top and side views.
CO3: Understand and draw projection of solids in various positions in first quadrant.
CO4: Explain principles behind development of surfaces.
CO5: Prepare isometric and perspective sections of simple solids.

UNIT I
Introduction: Lines, Lettering and Dimensioning, Geometrical Constructions and Constructing regular polygons by general methods.
Curves: construction of ellipse, parabola and hyperbola by general, Cycloids, Involutes,Normal and tangent to Curves.
Scales: Plain scales, diagonal scales and vernier scales.

UNIT II
Orthographic Projections: Reference plane, importance of reference lines or Plane,Projections of a point situated in any one of the four quadrants.
Projections of Straight Lines: Projections of straight lines parallel to both reference planes, perpendicular to one reference plane and parallel to other reference plane, inclined to one reference plane and parallel to the other reference plane. Projections of Straight Line Inclined to both the reference planes.
Projections of Planes: regular planes Perpendicular to both reference planes, parallel to one reference plane and inclined to the other reference plane; plane inclined to both the reference planes.

UNIT III
Projections of Solids: Types of solids: Polyhedra and Solids of revolution. Projections of solidsin simple positions: Axis perpendicular to horizontal plane, Axis perpendicular to vertical planeand Axis parallel to both the reference planes, Projection of Solids with axis inclined to one reference plane and parallel to another plane.
UNIT IV
Sections of Solids: Perpendicular and inclined section planes, Sectional views and True shape of section, Sections of solids in simple position only.

Development of Surfaces: Methods of Development: Parallel line development and radial line development. Development of a cube, prism, cylinder, pyramid and cone.

UNIT V

Conversion of Views: Conversion of isometric views to orthographic views; Conversion of orthographic views to isometric views.

Computer graphics: Creating 2D&3D drawings ofobjects including PCB and Transformationsusing Auto CAD (*Not for end examination*).

Note: The practice will be carried out by using AutoCAD software.

Text Book:

1. N. D. Bhatt, Engineering Drawing, Charotar Publishing House, 2016.

Reference Books:

1. Engineering Drawing, K.L. Narayana and P. Kannaiah, Tata McGraw Hill, 2013.
2. Engineering Drawing, M.B.Shah and B.C. Rana, Pearson Education Inc,2009.
3. Engineering Drawing with an Introduction to AutoCAD, Dhananjay Jolhe, TataMcGraw Hill, 2017.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)
I B.Tech II Semester (Common to All Branches)

L	T	P	C
3	0	0	3

23AHS11 DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS

Course Objectives:

- To enlighten the learners in the concept of differential equations and multivariable calculus.
- To furnish the learners with basic concepts and techniques at plus two level to lead them into advanced level by handling various real-world applications.

Course Outcomes:

At the end of the course, the student will be able to

CO1: Solve the differential equations related to various engineering fields.

CO2: Identify solution methods for partial differential equations that model physical processes.

CO3: Interpret the physical meaning of different operators such as gradient, curl anddivergence.

CO4: Estimate the work done against a field, circulation and flux using vector calculus.

UNIT I Differential equations of first order and first degree

Linear differential equations – Bernoulli's equations- Exact equations and equations reducibleto exact form. Applications: Newton's Law of cooling – Law of natural growth and decay- Electrical circuits- Orthogonal trajectories.

UNIT II Linear differential equations of higher order (Constant Coefficients)

Definitions, homogenous and non-homogenous, complimentary function, general solution, particular integral, Wronskian, Method of variation of parameters. Simultaneous linear equations, Applications to L-C-R Circuit problems and Simple Harmonic motion.

UNIT III Partial Differential Equations

Introduction and formation of Partial Differential Equations by elimination of arbitrary constants and arbitrary functions, solutions of first order linear equations using Lagrange's method. Homogeneous Linear Partial differential equations with constant coefficients.

UNIT IV Vector differentiation

Scalar and vector point functions, vector operator Del, Del applies to scalar point functions-Gradient, Directional derivative, del applied to vector point functions-Divergence and Curl, vector identities.

UNIT V Vector integration

Line integral -Circulation-work done, surface integral-flux, Green's theorem in the plane (without proof), Stoke's theorem (without proof), volume integral, Divergence theorem (without proof) and related problems.

Textbooks:

1. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers, 2017, 44th Edition
2. Advanced Engineering Mathematics, Erwin Kreyszig, John Wiley & Sons, 2018, 10th Edition.

Reference Books:

1. Thomas Calculus, George B. Thomas, Maurice D. Weir and Joel Hass, Pearson Publishers, 2018, 14th Edition.
2. Advanced Engineering Mathematics, Dennis G. Zill and Warren S. Wright, Jones and Bartlett, 2018.
3. Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018,5th Edition.
4. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, Alpha Science International Ltd., 2021 5th Edition (9th reprint).
5. Higher Engineering Mathematics, B. V. R Ramana, , McGraw Hill Education, 2017

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY,
(AUTONOMOUS)

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I B.Tech I Semester

23AEE02 ELECTRICAL & ELECTRONICS ENGINEERING WORKSHOP

(Common to All branches of Engineering)

Course Objectives:

To impart knowledge on the fundamental laws & theorems of electrical circuits, functions of electrical machines and energy calculations.

COURSE OUTCOMES:

At the end of this course students will demonstrate the ability to

Course Outcomes:

At the end of the course, the student will be able to

CO1: Understand the Electrical circuit design concept; measurement of resistance, power, power factor; concept of wiring and operation of Electrical Machines and Transformer.
CO2: Apply the theoretical concepts and operating principles to derive mathematical models for circuits, Electrical machines and measuring instruments; calculations for the measurement of resistance, power and power factor.
CO3: Apply the theoretical concepts to obtain calculations for the measurement of resistance, power and power factor.
CO4: Analyse various characteristics of electrical circuits, electrical machines and measuring instruments.
CO5: Design suitable circuits and methodologies for the measurement of various electrical parameters; Household and commercial wiring.

Activities:

1. Familiarization of commonly used Electrical & Electronic Workshop Tools: Bread board, Solder, cables, relays, switches, connectors, fuses, Cutter, plier, screwdriver set, wire stripper, flux, knife/blade, soldering iron, de-soldering pump etc. Provide some exercises so that hardware tools and instruments are learned to be used by the students.
2. Familiarization of Measuring Instruments like Voltmeters, Ammeters, multimeter, LCR-Q meter, Power Supplies, CRO, DSO, Function Generator, Frequency counter.
3. Provide some exercises so that measuring instruments are learned to be used by the students.
4. Components:
5. Familiarization/Identification of components (Resistors, Capacitors, Inductors, Diodes, transistors, IC's etc.) – Functionality, type, size, colour coding package, symbol, cost etc.
6. Testing of components like Resistor, Capacitor, Diode, Transistor, ICs etc. - Compare values of components like resistors, inductors, capacitors etc with the measured values by using instruments

PART A: ELECTRICAL ENGINEERING LAB

List of experiments:

1. Verification of KCL and KVL

2. Verification of Superposition theorem
3. Measurement of Resistance using Wheat stone bridge
4. Magnetization Characteristics of DC shunt Generator
5. Measurement of Power and Power factor using Single-phase wattmeter
6. Measurement of Earth Resistance using Megger
7. Calculation of Electrical Energy for Domestic Premises

Reference Books:

1. Basic Electrical Engineering, D. C. Kulshreshtha, Tata McGraw Hill, 2019, First Edition
2. Power System Engineering, P.V. Gupta, M.L. Soni, U.S. Bhatnagar and A. Chakrabarti, Dhanpat Rai & Co, 2013
3. Fundamentals of Electrical Engineering, Rajendra Prasad, PHI publishers, 2014, Third Edition

Note: Minimum Six Experiments to be performed.

Reference Books:

1. Basic Electrical Engineering, D. C. Kulshreshtha, Tata McGraw Hill, 2019, First Edition
2. Power System Engineering, P.V. Gupta, M.L. Soni, U.S. Bhatnagar and A. Chakrabarti, Dhanpat Rai & Co, 2013
3. Fundamentals of Electrical Engineering, Rajendra Prasad, PHI publishers, 2014, Third Edition

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(Autonomous)
I B.Tech I Semester
23AIT01 IT WORKSHOP

(Common to all branches)

L	T	P	C
0	0	2	1

Course Objectives

1. To introduce the internal parts of a computer, peripherals, I/O ports, connecting cables
2. To teach basic command line interface commands on Linux.
3. To teach the usage of Internet for productivity and self-paced life-long learning
4. To introduce Compression, Multimedia and Antivirus tools and Office Tools such as Word processors, Spread sheets and Presentation tools.

Course Outcomes:

At the end of the course, the student will be able to

CO1. Perform Hardware troubleshooting.
CO2. Understand Hardware components and inter dependencies.
CO3. Safeguard computer systems from viruses/worms.
CO4. Document/ Presentation preparation.
CO5. Perform calculations using spreadsheets.

PC HARDWARE

Task 1: Identify the peripherals of a computer, components in a CPU and its functions. Draw the block diagram of the CPU along with the configuration of each peripheral and submit to your instructor.

Task 2: Every student should disassemble and assemble the PC back to working condition. Lab instructors should verify the work and follow it up with a Viva. Also students need to go through the video which shows the process of assembling a PC. A video would be given as part of the course content. Differentiate RAM & ROM.

Task 3: Every student should individually install MS windows on the personal computer. Lab instructor should verify the installation and follow it up with a Viva.

Task 4: Every student should install Linux on the computer. This computer should have windows installed. The system should be configured as dual boot with both Windows and Linux. Lab instructors should verify the installation and follow it up with a Viva

INTERNET & WORLD WIDE WEB

Task 1: Orientation & Connectivity Boot Camp: Students should get connected to their Local Area Network and access the Internet. In the process they configure the TCP/IP setting. Finally students should demonstrate, to the instructor, how to access the websites and email. If there is no internet connectivity preparations need to be made by the instructors to simulate the WWW on the LAN.

Task 2: Web Browsers, Surfing the Web: Students customize their web browsers with the LAN proxy settings, bookmarks, search toolbars and pop up blockers. Also, plug-ins like

Macromedia Flash and JRE for applets should be configured.

Task 3: Search Engines & Netiquette: Students should know what search engines are and how to use the search engines. A few topics would be given to the students for which they need to search on Google. This should be demonstrated to the instructors by the student.

Task 4: Cyber Hygiene: Students would be exposed to the various threats on the internet and would be asked to configure their computer to be safe on the internet. They need to customize their browsers to block pop ups, block active x downloads to avoid viruses and/or worms.

LaTeX and WORD

Task 1: Word Orientation: The mentor needs to give an overview of La TeX and Microsoft (MS) office or equivalent (FOSS) tool word: Importance of La TeX and MS office or equivalent (FOSS) tool Word as word Processors, Details of the four tasks and features that would be covered in each, Using La TeXand word – Accessing, overview of toolbars, saving files, Using help and resources, rulers, format painter in word.

Task 2: Using La TeX and Word to create a project certificate. Features to be covered:- Formatting Fonts in word, Drop Cap in word, Applying Text effects, Using Character Spacing, Borders and Colors, Inserting Header and Footer, Using Date and Time option in both La TeX and Word.

Task 3: Creating project abstract Features to be covered:-Formatting Styles, Inserting table, Bullets and Numbering, Changing Text Direction, Cell alignment, Footnote, Hyperlink, Symbols, Spell Check, Track Changes.

Task 4: Creating a Newsletter: Features to be covered:- Table of Content, Newspaper columns, Images from files and clipart, Drawing toolbar and Word Art, Formatting Images, Textboxes, Paragraphs and Mail Merge in word.

EXCEL

Excel Orientation: The mentor needs to tell the importance of MS office or equivalent (FOSS) tool Excel as a Spreadsheet tool, give the details of the four tasks and features that would be covered in each. Using Excel – Accessing, overview of toolbars, saving excel files, Using help and resources.

Task 1: Creating a Scheduler - Features to be covered: Gridlines, Format Cells, Summation, auto fill, Formatting Text

Task 2: Calculating GPA - .Features to be covered:- Cell Referencing, Formulae in excel – average, std. deviation, Charts, Renaming and Inserting worksheets, Hyper linking, Count function.

Task 3: Split cells, freeze panes, group and outline, Sorting, Boolean and logical operators, Conditional formatting Power point

LOOKUP/VLOOKUP

Task 1: Students will be working on basic power point utilities and tools which help them create basic power point presentations. PPT Orientation, Slide Layouts, Inserting Text, Word Art, Formatting Text, Bullets and Numbering, Auto Shapes, Lines and Arrows in PowerPoint.

Task 2: Interactive presentations - Hyperlinks, Inserting –Images, Clip Art, Audio, Video, Objects, Tables and Charts.

Task 3: Master Layouts (slide, template, and notes), Types of views (basic, presentation, slide slotter, notes etc), and Inserting – Background, textures, Design Templates, Hidden slides.

AI TOOLS – ChatGPT

Task 1: Prompt Engineering: Experiment with different types of prompts to see how the model responds. Try asking questions, starting conversations, or even providing incomplete sentences to see how the model completes them.

• Ex: Prompt: "You are a knowledgeable AI. Please answer the following question: What is the capital of France?"

Task 2: Creative Writing: Use the model as a writing assistant. Provide the beginning of a story or a description of a scene, and let the model generate the rest of the content. This can be a fun way to brainstorm creative ideas

• Ex: Prompt: "In a world where gravity suddenly stopped working, people started floating upwards. Write a story about how society adapted to this new reality."

Task 3: Code Generation: Test the model's ability to generate code by giving it partial code snippets and asking it to complete them. You can also ask the model to explain programming concepts or help you debug code.

• Ex:Prompt: "Complete the following Python code to swap the values of two variables:
\npython\na = 5\nb = 10\ntemp = a\na = b\nb = temp\n"

Task 4: Language Translation: Experiment with translation tasks by providing a sentence in one language and asking the model to translate it into another language. Compare the output to see how accurate and fluent the translations are.

• Ex:Prompt: "Translate the following English sentence to French: 'Hello, how are you doing today?'"

Task 5: Summarization: Provide a long piece of text, such as an article or a blog post, and ask the model to summarize it. Compare the model's summary with the original text to assess its ability to condense information effectively.

• Ex: Prompt: "Summarize the article titled 'Ramayanam' in 3-4 sentences."

Task 6: Futuristic Predictions: Have fun by asking the model to predict future technological advancements, societal changes, or even hypothetical scenarios. Compare its responses with your own ideas.

• Ex:Prompt: "Predict how artificial intelligence will transform everyday life in the next 20 years."

Task 7: Technical Explanations: Challenge the model with technical questions from different domains. Ask it to explain scientific concepts, mathematical theorems, or

complex algorithms in simple terms.
• Ex:Prompt: "Explain the concept of neural networks in machine learning, including their layers and the process of backpropagation."

Reference Books:

1. Comdex Information Technology course tool kit Vikas Gupta, WILEY Dream tech.
2. The Complete Computer upgrade and repair book, 3rd edition Cheryl A Schmidt, WILEY Dream tech.
3. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education.
4. PC Hardware - A Handbook – Kate J. Chase PHI (Microsoft).

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

I B.Tech II Semester

(Common to CSE,CSD,CSM,CSC,CAI,AI, AI&DS,CSO,IT,CSBS))

L	T	P	C
3	-	-	3

23ACS03: DATA STRUCTURES

Course Objectives:

1. Understand the significance of linear data structures in problem-solving and basictime/space complexity analysis.
2. Create and manage linked lists to efficiently organize and manipulate data, emphasizing memory efficiency.
3. Implement and apply stacks to manage program flow and solve problems involving expression evaluation and backtracking.
4. Utilize queues to model real-world scenarios, such as process scheduling and breadth- first search algorithms and understand the versatility of deques and prioritize data management using priority queues.
5. Explore basic concepts of hashing and apply it to solve problems requiring fast data retrieval and management.

Course Outcomes: At the end of the course, Student will be able to

CO1. Explain the role of linear data structures in organizing and accessing data efficiently in algorithms.

CO2. Design, implement, and apply linked lists for dynamic data storage, demonstrating understanding of memory allocation.

CO3. Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems.

CO4. Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between deques and priority queues, and apply them appropriately to solve data management challenges.

CO5. Recognize scenarios where hashing is advantageous, and design hash-based solutions for specific problems.

UNIT I

Introduction to Data Structures:

Linear Data Structures- Definition and importance of linear data structures, Abstract data types (ADTs) and their implementation, Overview of time and space complexity analysis for linear data structures. **Non- Linear Data Structures-** Definition and importance of nonlinear data structures, Types and properties of nonlinear data structures

UNIT II

Linked Lists: Singly linked lists: representation and operations, Doubly linked lists and circular linked lists, Comparing arrays and linked lists, Applications of linked lists.

. **Searching Techniques**: Linear & Binary Search, **Sorting Techniques**: Bubble sort, Selection sort, Insertion Sort

UNIT III

Stacks: Introduction to stacks: properties and operations, Implementing stacks using arrays and linked lists, Applications of stacks in expression evaluation, backtracking, reversing list, etc.

Trees: Introduction to Trees, Binary Search Tree – Insertion, Deletion & Traversal, AVL Trees.

UNIT IV

Queues: Introduction to queues: properties and operations, Implementing queues using arrays and linked lists, Applications of queues in breadth-first search, scheduling, etc.
Deques: Introduction to deques (double-ended queues), Operations on deques and their applications.

UNIT V

Graph Theory: Data Structures for Graphs- Adjacency Matrix Structure, Graph Traversals, Shortest Paths, Minimum Spanning Trees- Prims' Algorithm, Kruskal's Algorithm.

Hashing: Brief introduction to hashing and hash functions, Collision resolution techniques: chaining and open addressing, Hash tables: basic implementation and operations, Applications of hashing in unique identifier generation, caching, etc.

Text Books:

1. Data Structures and algorithm analysis in C, Mark Allen Weiss.
2. Fundamentals of data structures in C, Ellis Horowitz, Sartaj Sahni, Dinesh Mehta.
3. Classic Data Structures, Debasis Samantha, Second Edition, 2009, PHI

Reference Books:

1. Algorithms and Data Structures: The Basic Toolbox by Kurt Mehlhorn and Peter Sanders
2. C Data Structures and Algorithms by Alfred V. Aho, Jeffrey D. Ullman, and John E. Hopcroft
3. Problem Solving with Algorithms and Data Structures" by Brad Miller and David Ranum
4. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein
5. Algorithms in C, Parts 1-5 (Bundle): Fundamentals, Data Structures, Sorting, Searching, and Graph Algorithms" by Robert Sedgewick.
6. Data Structures and Algorithm Analysis in C, Mark Allen Weiss, Second Edition, 2002, Pearson.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

I B.Tech I Semester (Common to EEE, ECE, IT, CAI, AI, AI&DS, CSO, CSC, EBM & CSBS)

I B.Tech II Semester (Common to CE, ME, CSE, CSE(DS) & CSE(AI &ML))

23AHS09 ENGINEERING PHYSICS LAB

L	T	P	C
0	0	2	1

Objectives:

- **Course** Understands the concepts of interference, diffraction and their applications.
- Understand the role of optical fiber parameters in communication.
- Recognize the importance of energy gap in the study of conductivity and Hall Effect in asemiconductor.
- Illustrates the magnetic and dielectric materials applications.
- Apply the principles of semiconductors in various electronic devices.

Course Outcomes:

At the end of the course, the student will be able to :

CO1. Operate optical instruments like microscope and spectrometer

CO2. Estimate the wavelength of different colors using diffraction grating and resolving power

CO3. Plot the intensity of the magnetic field of circular coil carrying current with distance

CO4. Determine the resistivity of the given semiconductor using four probe method

CO5. Identify the type of semiconductor i.e., n-type or p-type using hall effect

CO6. Calculate the band gap of a given semiconductor

(Any **TEN** of the following listed experiments)

List of Engineering Physics Experiments

1. Determination of radius of curvature of a given plano convex lens by Newton's rings.
2. Determination of wavelengths of different spectral lines in mercury spectrum using diffractiongrating in normal incidence configuration.
3. Determination of dispersive power of prism.
4. Verification of Brewster's law
5. Determination of the resistivity of semiconductor by four probe method.
6. Determination of energy gap of a semiconductor using p-n junction diode.
7. Determination of Hall voltage and Hall coefficient of a given semiconductor using Hall effect.
8. Determination of dielectric constant using charging and discharging method.
9. Study the variation of B versus H by magnetizing the magnetic material (B-H curve).
10. Magnetic field along the axis of a current carrying circular coil by Stewart & Gee's Method.

11. Determination of wavelength of Laser light using diffraction grating.
12. Estimation of Planck's constant using photoelectric effect.
13. Determination of temperature coefficients of a thermistor.
14. Determination of acceleration due to gravity and radius of Gyration by using a compoundpendulum.

15. Determination of rigidity modulus of the material of the given wire using Torsional pendulum.
16. Sonometer: Verification of laws of stretched string.
17. Determination of young's modulus for the given material of wooden scale by non- uniformbending (or double cantilever) method.
18. Determination of Frequency of electrically maintained tuning fork by Melde's experiment.

References: 1. S. Balasubramanian, M.N. Srinivasan "A Text book of Practical Physics"- S.Chand Publishers, 2017.

URL:*www.vlab.co.in*

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

I B.Tech II Semester

(Common to CSE,CSD,CSM,CSC,CAI, AI, AI&DS,CSO,IT,CSBS))

L	T	P	C
-	-	3	1.5

23ACS04: DATA STRUCTURES LAB

Course Objectives:

1. Understand the significance of linear data structures in problem-solving and basictime/space complexity analysis.
2. Create and manage linked lists to efficiently organize and manipulate data, emphasizing memory efficiency.
3. Implement and apply stacks to manage program flow and solve problems involving expression evaluation and backtracking.
4. Utilize queues to model real-world scenarios, such as process scheduling and breadth- first search algorithms and understand the versatility of deques and prioritize data management using priority queues.
5. Explore basic concepts of hashing and apply it to solve problems requiring fast data retrieval and management.

Course Outcomes:

At the end of the course, Student will be able to

CO1. Explain the role of linear data structures in organizing and accessing data efficiently in algorithms.

CO2. Design, implement, and apply linked lists for dynamic data storage, demonstrating understanding of memory allocation.

CO3. Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems.

CO4. Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between deques and priority queues, and apply them appropriately to solve data management challenges.

CO5. Recognize scenarios where hashing is advantageous, and design hash-based solutionsfor specific problems.

List of Experiments:

1: Array Manipulation

i) Implement basic operations on arrays: insertion, deletion, searching.
ii) Create a program to find the maximum and minimum elements in an array.
iii) Write a program to reverse an array.

2: Linked List Implementation

i) Implement a singly linked list and perform insertion and deletion operations.
ii) Develop a program to reverse a linked list iteratively and recursively.
iii) Solve problems involving linked list traversal and manipulation.

3: Linked List Applications

i) Create a program to detect and remove duplicates from a linked list.
ii) Implement a linked list to represent polynomials and perform addition.
iii) Implement a double-ended queue (deque) with essential operations.

4: Doubly Linked List Implementation

i) Implement a doubly linked list and perform various operations to understand its properties and applications.
ii) Implement a circular linked list and perform insertion, deletion, and traversal.

5: Stack Operations

i) Implement a stack using arrays and linked lists.
ii) Write a program to evaluate a postfix expression using a stack.
iii) Implement a program to check for balanced parentheses using a stack.

6: Queue Operations

i) Implement a queue using arrays and linked lists.
ii) Develop a program to simulate a simple printer queue system.
iii) Solve problems involving circular queues.

7: Stack and Queue Applications

i) Use a stack to evaluate an infix expression and convert it to postfix.
ii) Create a program to determine whether a given string is a palindrome or not.
iii) Implement a stack or queue to perform comparison and check for symmetry.

8: Hashing

i) Implement a hash table with collision resolution techniques.
ii) Write a program to implement a simple cache using hashing.

Text Books:

1. Data Structures and algorithm analysis in C, Mark Allen Weiss.
2. Fundamentals of data structures in C, Ellis Horowitz, Sartaj Sahni, Dinesh Mehta.

Reference Books:

1. Algorithms and Data Structures: The Basic Toolbox by Kurt Mehlhorn and Peter Sanders
2. C Data Structures and Algorithms by Alfred V. Aho, Jeffrey D. Ullman, and John E. Hopcroft
3. Problem Solving with Algorithms and Data Structures" by Brad Miller and David Ranum
4. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein
5. Algorithms in C, Parts 1-5 (Bundle): Fundamentals, Data Structures, Sorting,Searching, and Graph Algorithms by Robert Sedgewick.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

I B.Tech II Semester (Common to All Branches)

L	T	P	C
0	0	1	0.5

23AHS12 : NSS/NCC/SCOUTS & GUIDES/COMMUNITY SERVICE

Course Objectives:

The objective of introducing this course is to impart discipline, character, fraternity, teamwork,social consciousness among the students and engaging them in selfless service.

Course Outcomes: After completion of the course the students will be able to:

CO1: Understand the importance of discipline, character and service motto. **CO2:** Solve some societal issues by applying acquired knowledge, facts, and techniques.
CO3: Explore human relationships by analyzing social problems.
CO4: Determine to extend their help for the fellow beings and downtrodden people.
CO5: Develop leadership skills and civic responsibilities.

UNIT I Orientation

General Orientation on NSS/NCC/ Scouts & Guides/Community Service activities, careerguidance.

Activities:

i) Conducting –ice breaking sessions-expectations from the course- knowing personaltalents and skills
ii) Conducting orientations programs for the students –future plans- activities-releasingroad map etc.
iii) Displaying success stories-motivational biopics- award winning movies on societalissues etc.
iv) Conducting talent show in singing patriotic songs-paintings- any other contribution.

UNIT II Nature & CareActivities:

i) Best out of waste competition.
ii) Poster and signs making competition to spread environmental awareness.
iii) Recycling and environmental pollution article writing competition.
iv) Organising Zero-waste day.
v) Digital Environmental awareness activity via various social media platforms.
vi) Virtual demonstration of different eco-friendly approaches for sustainable living.
vii) Write a summary on any book related to environmental issues.

UNIT III Community ServiceActivities:

i) Conducting One Day Special Camp in a village contacting village-area leaders- Surveyin the village, identification of problems- helping them to solve via media- authorities-experts-etc.

ii) Conducting awareness programs on Health-related issues such as General Health,Mental health, Spiritual Health, HIV/AIDS,
iii) Conducting consumer Awareness. Explaining various legal provisions etc.
iv) Women Empowerment Programmes- Sexual Abuse, Adolescent Health and PopulationEducation.
v) Any other programmes in collaboration with local charities, NGOs etc.

Reference Books:

1. Nirmalya Kumar Sinha & Surajit Majumder, *A Text Book of National Service Scheme*
Vol;.I, Vidya Kutir Publication, 2021 (ISBN 978-81-952368-8-6)
2. *Red Book - National Cadet Corps* – Standing Instructions Vol I & II, DirectorateGeneral of NCC, Ministry of Defence, New Delhi
3. Davis M. L. and Cornwell D. A., "Introduction to Environmental Engineering",McGraw Hill, New York 4/e 2008
4. Masters G. M., Joseph K. and Nagendran R. "Introduction to EnvironmentalEngineering and Science", Pearson Education, New Delhi. 2/e 2007
5. Ram Ahuja. *Social Problems in India*, Rawat Publications, New Delhi.

General Guidelines:

1. Institutes must assign slots in the Timetable for the activities.
2. Institutes are required to provide instructor to mentor the students.

Evaluation Guidelines:

- Evaluated for a total of 100 marks.
- A student can select 6 activities of his/her choice with a minimum of 01 activity per unit. Each activity shall be evaluated by the concerned teacher for 15 marks, totalling to 90 marks.
- A student shall be evaluated by the concerned teacher for 10 marks by conducting vivavoce on the subject.

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II B.Tech. I Semester
(Common to CSE, CSM, CAI, AI, AI&DS, CSD, CSC, CSO & IT branches)

L	T	P	C
3	0	0	3

23AHS017: DISCRETE MATHEMATICS & GRAPH THEORY

Course Outcomes: After successful completion of this course, the students should be able to:

CO1 Apply mathematical logic to solve problems.

CO2 Understand the concepts and perform the operations related to sets, relations and functions.
Gain the conceptual background needed and identify structures of algebraic nature.

CO3 Apply basic counting techniques to solve combinatorial problems.

CO4 Formulate problems and solve recurrence relations.

CO5 Apply Graph Theory in solving computer science problems

UNIT I Mathematical Logic

Introduction, Statements and Notation, Connectives, Well-formed formulas, Tautology, Duality law, Equivalence, Implication, Normal Forms, Functionally complete set of connectives, Inference Theory of Statement Calculus, Predicate Calculus, Inference theory of Predicate Calculus.

UNIT II Set theory

The Principle of Inclusion- Exclusion, Pigeon hole principle and its application, Functions composition of functions, Inverse Functions, Recursive Functions, Lattices and its properties. Algebraic structures: Algebraic systems-Examples and General Properties, Semi groups and Monoids, groups, sub groups, homomorphism, Isomorphism.

UNIT III Elementary Combinatorics

Combinations and Permutations, Enumeration of Combinations and Permutations, Enumerating Combinations and Permutations with Repetitions, Enumerating Permutations with Constrained Repetitions, Binomial Coefficients, The Binomial and Multinomial Theorems.

UNITIV: Recurrence Relations

Generating Functions of Sequences, Calculating Coefficients of Generating Functions, Recurrence relations, Solving Recurrence Relations by Substitution and Generating functions, The Method of Characteristic roots, Solutions of Inhomogeneous, Recurrence Relations.

UNIT V Graphs

Basic Concepts, Isomorphism and Subgraphs, Trees and their Properties, Spanning Trees, Directed Trees, Binary Trees, Planar Graphs, Euler’s Formula, Multigraphs and Euler Circuits, Hamiltonian Graphs.

Textbooks:

1. J.P. Tremblay and R. Manohar, Discrete Mathematical Structures with Applications to Computer Science, Tata McGraw Hill, 2002.

2. Kenneth H. Rosen, Discrete Mathematics and its Applications with Combinatorics and Graph Theory, 7th Edition, McGraw Hill Education (India) Private Limited.

Reference Books:

1. Joe L. Mott, Abraham Kandel and Theodore P. Baker, Discrete Mathematics for Computer Scientists & Mathematicians, 2nd Edition, Pearson Education.
2. Narsingh Deo, Graph Theory with Applications to Engineering and Computer Science.

Online Learning Resources:

1. http://www.cs.yale.edu/homes/aspnes/classes/202/notes.pdf

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)
II B.Tech. I Semester
(Common to All Branches of Engineering)

L	T	P	C
2	1	0	3

23AMB01 : UNIVERSAL HUMAN VALUES – UNDERSTANDING HARMONY AND ETHICAL HUMAN CONDUCT

Course Objectives:

- To help the students appreciate the essential complementary between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
- To facilitate the development of a Holistic perspective among students towards life and profession as well as towards happiness and prosperity based on a correct understanding of the Human reality and the rest of existence. Such holistic perspective forms the basis of Universal Human Values and movement towards value-based living in a natural way.
- To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually fulfilling human behaviour and mutually enriching interaction with Nature.

COURSE OUTCOMES:

At the end of the course, students will be able to

CO1 Define the terms like Natural Acceptance, Happiness and Prosperity

CO2 Identify oneself, and one's surroundings (family, society nature)

CO3 Apply what they have learnt to their own self in different day-to-day settings in real life

CO4 Relate human values with human relationship and human society.

CO5 Justify the need for universal human values and harmonious existence

CO6 Develop as socially and ecologically responsible engineers

Course Topics

The course has 28 lectures and 14 tutorials in 5 modules. The lectures and tutorials are of 1hour duration. Tutorial sessions are to be used to explore and practice what has been proposed during the lecture sessions.

The Teacher's Manual provides the outline for lectures as well as practice sessions. The teacher is expected to present the issues to be discussed as propositions and encourage the students to have a dialogue.

UNIT I Introduction to Value Education (6 lectures and 3 tutorials for practice session) Lecture 1: Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education)
Lecture 2: Understanding Value Education

Tutorial 1: Practice Session PS1 Sharing about Oneself
Lecture 3: self-exploration as the Process for Value Education

Lecture4: Continuous Happiness and Prosperity – the Basic Human Aspirations
Tutorial 2: Practice Session PS2 Exploring Human Consciousness
Lecture 5: Happiness and Prosperity – Current Scenario
Lecture 6: Method to Fulfill the Basic Human Aspirations
Tutorial 3: Practice Session PS3 Exploring Natural Acceptance

UNIT II Harmony in the Human Being (6 lectures and 3 tutorials for practice session)

Lecture 7: Understanding Human being as the Co-existence of the self and the body.
Lecture 8: Distinguishing between the Needs of the self and the body
Tutorial 4: Practice Session PS4 Exploring the difference of Needs of self and body.
Lecture 9: The body as an Instrument of the self
Lecture 10: Understanding Harmony in the self
Tutorial 5: Practice Session PS5 Exploring Sources of Imagination in the self Lecture 11: Harmony of the self with the body
Lecture 12: Programme to ensure self-regulation and Health
Tutorial 6: Practice Session PS6 Exploring Harmony of self with the body

UNIT III Harmony in the Family and Society (6 lectures and 3 tutorials for practice session)

Lecture 13: Harmony in the Family – the Basic Unit of Human Interaction
Lecture 14: 'Trust' – the Foundational Value in Relationship
Tutorial 7: Practice Session PS7 Exploring the Feeling of Trust
Lecture 15: 'Respect' – as the Right Evaluation
Tutorial 8: Practice Session PS8 Exploring the Feeling of Respect
Lecture 16: Other Feelings, Justice in Human-to-Human Relationship Lecture 17: Understanding Harmony in the Society
Lecture 18: Vision for the Universal Human Order
Tutorial 9: Practice Session PS9 Exploring Systems to fulfil Human Goal

UNIT IV Harmony in the Nature/Existence (4 lectures and 2 tutorials for practice session)

Lecture 19: Understanding Harmony in the Nature
Lecture 20: Interconnectedness, self-regulation and Mutual Fulfilment among the Four Orders of Nature
Tutorial 10: Practice Session PS10 Exploring the Four Orders of Nature
Lecture 21: Realizing Existence as Co-existence at All Levels
Lecture 22: The Holistic Perception of Harmony in Existence
Tutorial 11: Practice Session PS11 Exploring Co-existence in Existence

UNIT V Implications of the Holistic Understanding – a Look at Professional Ethics (6 lectures and 3 tutorials for practice session)

Lecture 23: Natural Acceptance of Human Values

Lecture 24: Definitiveness of (Ethical) Human Conduct

Tutorial 12: Practice Session PS12 Exploring Ethical Human Conduct

Lecture 25: A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order

Lecture 26: Competence in Professional Ethics

Tutorial 13: Practice Session PS13 Exploring Humanistic Models in Education Lecture 27: Holistic Technologies, Production Systems and Management Models-Typical Case Studies

Lecture 28: Strategies for Transition towards Value-based Life and Profession Tutorial 14: Practice Session PS14 Exploring Steps of Transition towards Universal Human Order

Practice Sessions for UNIT I – Introduction to Value Education

PS1 Sharing about Oneself

PS2 Exploring Human Consciousness

PS3 Exploring Natural Acceptance

Practice Sessions for UNIT II – Harmony in the Human Being PS4 Exploring the difference of Needs of self and body

PS5 Exploring Sources of Imagination in the self PS6 Exploring Harmony of self with the body

Practice Sessions for UNIT III – Harmony in the Family and Society

PS7 Exploring the Feeling of Trust

PS8 Exploring the Feeling of Respect

PS9 Exploring Systems to fulfil Human Goal

Practice Sessions for UNIT IV – Harmony in the Nature (Existence)

PS10 Exploring the Four Orders of Nature

PS11 Exploring Co-existence in Existence

Practice Sessions for UNIT V – Implications of the Holistic Understanding – a Look at Professional Ethics

PS12 Exploring Ethical Human Conduct

PS13 Exploring Humanistic Models in Education

PS14 Exploring Steps of Transition towards Universal Human Order

READINGS:

Textbook and Teachers Manual

a. The Textbook

R R Gaur, R Asthana, G P Bagaria, *A Foundation Course in Human Values and Professional Ethics*, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-47-1 b. The Teacher's Manual

R R Gaur, R Asthana, G P Bagaria,*Teachers' Manual for A Foundation Course in*

Human Values and Professional Ethics, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-53-2

Reference Books

1. *JeevanVidya: EkParichaya,* A Nagaraj, JeevanVidyaPrakashan, Amarkantak, 1999.
2. *Human Values,* A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3. *The Story of Stuff* (Book).
4. *The Story of My Experiments with Truth* - by Mohandas Karamchand Gandhi
5. *Small is Beautiful* - E. F Schumacher.
6. *Slow is Beautifu*l - Cecile Andrews
7. *Economy of Permanence* - J C Kumarappa
8. *Bharat Mein Angreji Raj* – PanditSunderlal
9. *Rediscovering India* - by Dharampal
10. *Hind Swaraj or Indian Home Rule* - by Mohandas K. Gandhi
11. *India Wins Freedom* - Maulana Abdul Kalam Azad
12. *Vivekananda* - Romain Rolland (English)
13. *Gandhi* - Romain Rolland (English)

Mode of Conduct:

Lecture hours are to be used for interactive discussion, placing the proposals about the topics at hand and motivating students to reflect, explore and verify them.

Tutorial hours are to be used for practice sessions.

While analyzing and discussing the topic, the faculty mentor's role is in pointing to essential elements to help in sorting them out from the surface elements. In other words, help the students explore the important or critical elements.

In the discussions, particularly during practice sessions (tutorials), the mentor encourages the student to connect with one's own self and do self-observation, self-reflection and selfexploration.

Scenarios may be used to initiate discussion. The student is encouraged to take up "ordinary" situations rather than" extra-ordinary" situations. Such observations and their analyses are shared and discussed with other students and faculty mentor, in a group sitting.

Tutorials (experiments or practical) are important for the course. The difference is that the laboratory is everyday life, and practical are how you behave and work in real life. Depending on the nature of topics, worksheets, home assignment and/or activity are included. The practice sessions (tutorials) would also provide support to a student in performing actions commensurate to his/her beliefs. It is intended that this would lead to development of commitment, namely behaving and working based on basic human values.
It is recommended that this content be placed before the student as it is, in the form of a basic foundation course, without including anything else or excluding any part of this content. Additional content may be offered in separate, higher courses. This course is to be taught by faculty from every teaching department, not exclusively by any one department.
Teacher preparation with a minimum exposure to at least one 8-day Faculty Development Program on Universal Human Values is deemed essential.

Online Resources

1. https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%201Introduction%20to%20V alue%20Education.pdf

2. https://fdp-si.aicte-india.org/UHV-
II%20Class%20Notes%20&%20Handouts/UHV%20Handout%20
2- Harmony%20in%20the%20Human%20Being.pdf

3. https://fdp-si.aicte-india.org/UHV-
II%20Class%20Notes%20&%20Handouts/UHV%20Handout%20
3- Harmony%20in%20the%20Family.pdf

4. https://fdp-si.aicte-
india.org/UHV%201%20Teaching%20Material/D3S2%20Respect%20July%2023.
pdf

5. https://fdp-si.aicte-india.org/UHV-
II%20Class%20Notes%20&%20Handouts/UHV%20Handout%20
5- Harmony%20in%20the%20Nature%20and%20Existence.pdf

6. https://fdp-si.aicte-india.org/download/FDPTeachingMaterial/3-
days%20FDPSI%20UHV%20Teaching%20Material/Day%203%20Handouts/UHV%20
3D
%20D3-
S2A%20Und%20Nature-Existence.pdf

7. https://fdp-si.aicte-
india.org/UHV%20II%20Teaching%20Material/UHV%20II%20Lecture%2023-25%20Eth
ics
%20v1.pdf

8. https://www.studocu.com/in/document/kiet-group-of-
institutions/universal- humanvalues/chapter-5-holistic-understanding-of-
harmony-on-professional- ethics/62490385
https://onlinecourses.swayam2.ac.in/aic22_ge23/preview

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)
II B.Tech. I Semester
(Common to CSM,AI, & CAI branches)

L	T	P	C
3	0	0	3

23ACA01: ARTIFICIAL INTELLIGENCE

Pre-requisite:
- Knowledge inComputer Programming.
- A course on "Mathematical Foundations of Computer Science".
- Background in linear algebra, data structures and algorithms, and probability.

Course Objectives:
- The student should be made to study the concepts of Artificial Intelligence.
- The student should be made to learn the methods of solving problems using Artificial Intelligence.
- The student should be made to introduce the concepts of Expert Systems.
- To understand the applications of AI, namely game playing, theorem proving, and machine learning.
- To learn different knowledge representation techniques

Course Outcomes:

After successful completion of the course, the student will be able to :

CO1. Understand the fundamentals of Artificial Intelligence (AI).
CO2. Analyze the basic concepts of problem solving, searching, inference, perception using AI.
CO3. Apply basic principles of AI in solutions that require real world knowledge representation and learning.
CO4. Apply First order logic, Inference in first order logic and use Resolution, Learning from observation Inductive learning, Decision trees and Learning methods.
CO5. Apply Expert Systems architectures for knowledge acquisition using expert system tools and techniques.

UNIT - I
Introduction: AI problems, foundation of AI and history of AI intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation.

UNIT - II
Searching- Searching for solutions, uniformed search strategies – Breadth first search, depth first Search. Search with partial information (Heuristic search) Hill climbing, A* ,AO* Algorithms, Problem reduction, Game Playing-Adversial search, Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning, Evaluation functions.

UNIT - III
Representation of Knowledge: Knowledge representation issues, predicate logic- logic programming, semantic nets- frames and inheritance, constraint propagation, representing knowledge using rules, rules based deduction

systems. Reasoning under uncertainty, review of probability, Bayes' probabilistic interferences and dempstershafer theory.

UNIT - IV

Logic concepts: First order logic. Inference in first order logic, propositional vs. first order inference, unification & lifts forward chaining, Backward chaining, Resolution, Learning from observation Inductive learning, Decision trees, Explanation based learning, Statistical Learning methods, Reinforcement Learning.

UNIT - V

Expert Systems: Architecture of expert systems, Roles of expert systems – Knowledge Acquisition Meta knowledge Heuristics. Typical expert systems – MYCIN, DART, XCON: Expert systems shells.

Textbooks:

1. S. Russel and P. Norvig, "Artificial Intelligence – A Modern Approach", SecondEdition, Pearson Education.
2. Kevin Night and Elaine Rich, Nair B., "Artificial Intelligence (SIE)", Mc Graw Hill

Reference Books:

1. David Poole, Alan Mackworth, Randy Goebel,"Computational Intelligence: a logical approach", Oxford University Press.
2. G. Luger, "Artificial Intelligence: Structures and Strategies for complex problemsolving", Fourth Edition, Pearson Education.
3. J. Nilsson, "Artificial Intelligence: A new Synthesis", Elsevier Publishers.
4. Artificial Intelligence, SarojKaushik, CENGAGE Learning.

Online Learning Resources:

1. https://ai.google/
2. https://swayam.gov.in/nd1_noc19_me71/preview

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II B.Tech. I Semester

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L	T	P	C
3	0	0	3

23ACS05- ADVANCED DATA STRUCTURES & ALGORITHM ANALYSIS

Course Objectives: The main objectives of the course is to

- Provide knowledge on advance data structures frequently used in Computer Science domain
- Develop skills in algorithm design techniques popularly used
- Understand the use of various data structures in the algorithm design

Course Outcomes:

After completion of the course, students will be able to

CO1. Illustrate the working of the advanced tree data structures and their applications (L2) **CO2.** Understand the Graph data structure, traversals and apply them in various contexts. (L2)

CO3. Use various data structures in the design of algorithms (L3)

CO4. Recommend appropriate data structures based on the problem being solved (L5)

CO5. Analyze algorithms with respect to space and time complexities (L4)

CO6. Design new algorithms (L6)

UNIT – I:

Introduction to Algorithm Analysis, Space and Time Complexity analysis, Asymptotic Notations.

AVL Trees – Creation, Insertion, Deletion operations and Applications B-Trees – Creation, Insertion, Deletion operations and Applications

UNIT – II:

Heap Trees (Priority Queues) – Min and Max Heaps, Operations and Applications

Graphs – Terminology, Representations, Basic Search and Traversals, Connected Components and Biconnected Components, applications

Divide and Conquer: The General Method, Quick Sort, Merge Sort, Strassen's matrix multiplication, Convex Hull

UNIT – III:

Greedy Method: General Method, Job Sequencing with deadlines, Knapsack Problem, Minimum cost spanning trees, Single Source Shortest Paths

Dynamic Programming: General Method, All pairs shortest paths, Single Source Shortest Paths – General Weights (Bellman Ford Algorithm), Optimal Binary Search Trees, 0/1 Knapsack, String Editing, Travelling Salesperson problem

UNIT – IV:
Backtracking: General Method, 8-Queens Problem, Sum of Subsets problem, Graph Coloring, 0/1 Knapsack Problem
Branch and Bound: The General Method, 0/1 Knapsack Problem, Travelling Salesperson problem
UNIT – V:
NP Hard and NP Complete Problems: Basic Concepts, Cook's theorem
NP Hard Graph Problems: Clique Decision Problem (CDP), Chromatic Number Decision Problem (CNDP), Traveling Salesperson Decision Problem (TSP)
NP Hard Scheduling Problems: Scheduling Identical Processors, Job Shop Scheduling

Textbooks:

1. Fundamentals of Data Structures in C++, Horowitz, Ellis; Sahni, Sartaj; Mehta, Dinesh 2nd Edition Universities Press
2. Computer Algorithms/C++ Ellis Horowitz, SartajSahni, SanguthevarRajasekaran2nd Edition University Press

Reference Books:

1. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
2. An introduction to Data Structures with applications, Trembley & Sorenson, McGraw Hill
3. The Art of Computer Programming, Vol.1: Fundamental Algorithms, Donald E Knuth, Addison-Wesley, 1997.
4. Data Structures using C & C++: Langsam, Augenstein&Tanenbaum, Pearson, 1995
5. Algorithms + Data Structures & Programs:, N.Wirth, PHI
6. Fundamentals of Data Structures in C++: Horowitz Sahni& Mehta, Galgottia Pub.
7. Data structures in Java:, Thomas Standish, Pearson Education Asia

Online Learning Resources:

1. https://www.tutorialspoint.com/advanced_data_structures/index.asp
2. http://peterindia.net/Algorithms.html
3. Abdul Bari,1. Introduction to Algorithms (youtube.com)

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L	T	P	C
3	0	0	3

23ACS06 : OBJECT-ORIENTED PROGRAMMING THROUGH JAVA

Course Objectives: The learning objectives of this course are to:

- Identify Java language components and how they work together in applications
- Learn the fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries.
- Learn how to extend Java classes with inheritance and dynamic binding and how to use exception handling in Java applications
- Understand how to design applications with threads in Java
- Understand how to use Java apis for program development

Course Outcomes:
After completion of the course, students will be able to

- **CO1.** Analyze problems, design solutions using OOP principles, and implement them efficiently in Java. (L4)
- **CO2.** Design and implement classes to model real-world entities, with a focus on attributes, behaviors, and relationships between objects (L4)
- **CO3.** Demonstrate an understanding of inheritance hierarchies and polymorphic behaviour, including method overriding and dynamic method dispatch. (L3)
- **CO4.** ApplyCompetence in handling exceptions and errors to write robust and fault-tolerant code. (L3)
- **CO5.** Perform file input/output operations, including reading from and writing to files using Java I/O classes, graphical user interface (GUI) programming using JavaFX. (L3)
- **CO6.** Choose appropriate data structure of Java to solve a problem (L6)

UNIT I: Object Oriented Programming: Basic concepts, Principles, Program Structure in Java: Introduction, Writing Simple Java Programs, Elements or Tokens in Java Programs, Java Statements, Command Line Arguments, User Input to Programs, Escape Sequences Comments, Programming Style.

Data Types, Variables, and Operators :Introduction, Data Types in Java, Declaration of Variables, Data Types, Type Casting, Scope of Variable Identifier, Literal Constants,

Symbolic Constants, Formatted Output with printf() Method, Static Variables and Methods, Attribute Final, **Introduction to Operators**, Precedence and Associativity of Operators, Assignment Operator (=), Basic Arithmetic Operators, Increment (++) and Decrement (- -) Operators, Ternary Operator, Relational Operators, Boolean Logical Operators, Bitwise Logical Operators.

Control Statements: Introduction, if Expression, Nested if Expressions, if–else Expressions, Ternary Operator?:, Switch Statement, Iteration Statements, while Expression, do–while Loop, for Loop, Nested for Loop, For–Each for Loop, Break Statement, Continue Statement.

UNIT II: Classes and Objects: Introduction, Class Declaration and Modifiers, Class Members, Declaration of Class Objects, Assigning One Object to Another, Access Control for Class Members, Accessing Private Members of Class, Constructor Methods for Class, Overloaded Constructor Methods, Nested Classes, Final Class and Methods, Passing Arguments by Value and by Reference, Keyword this.

Methods: Introduction, Defining Methods, Overloaded Methods, Overloaded Constructor Methods, Class Objects as Parameters in Methods, Access Control, Recursive Methods, Nesting of Methods, Overriding Methods, Attributes Final and Static.

UNIT III: Arrays: Introduction, Declaration and Initialization of Arrays, Storage of Array in Computer Memory, Accessing Elements of Arrays, Operations on Array Elements, Assigning

Array to Another Array, Dynamic Change of Array Size, Sorting of Arrays, Search for Values in Arrays, Class Arrays, Two-dimensional Arrays, Arrays of Varying Lengths, Threedimensional Arrays, Arrays as Vectors.

Inheritance: Introduction, Process of Inheritance, Types of Inheritances, Universal Super Class-Object Class, Inhibiting Inheritance of Class Using Final, Access Control and Inheritance, Multilevel Inheritance, Application of Keyword Super, Constructor Method and Inheritance, Method Overriding, Dynamic Method Dispatch, Abstract Classes, Interfaces and Inheritance.

Interfaces: Introduction, Declaration of Interface, Implementation of Interface, Multiple Interfaces, Nested Interfaces, Inheritance of Interfaces, Default Methods in Interfaces, Static Methods in Interface, Functional Interfaces, Annotations.

UNIT IV: Packages and Java Library: Introduction, Defining Package, Importing Packages and Classes into Programs, Path and Class Path, Access Control, Packages in Java SE, Java.lang Package and its Classes, Class Object, Enumeration, class Math, Wrapper Classes, Auto-boxing and Auto-unboxing, Java util Classes and Interfaces, Formatter Class, Random Class, Time Package, Class Instant (java.time.Instant), Formatting for Date/Time in Java, Temporal Adjusters Class, Temporal Adjusters Class.

Exception Handling: Introduction, Hierarchy of Standard Exception Classes, Keywords throws and throw, try, catch, and finally Blocks, Multiple Catch Clauses, Class Throwable, Unchecked Exceptions, Checked Exceptions.

Java I/O and File: Java I/O API, standard I/O streams, types, Byte streams, Character streams, Scanner class, Files in Java(Text Book 2)

UNIT V: String Handling in Java: Introduction, Interface Char Sequence, Class String, Methods for Extracting Characters from Strings,Comparison, Modifying, Searching; Class String Buffer.

Multithreaded Programming: Introduction, Need for Multiple Threads Multithreaded Programming for Multi- core Processor, Thread Class, Main Thread-Creation of New Threads, Thread States, Thread Priority- Synchronization, Deadlock and Race Situations, Inter-thread Communication - Suspending, Resuming, and Stopping of Threads.

Java Database Connectivity: Introduction, JDBC Architecture, Installing MySQL and

MySQL Connector/J, JDBC Environment Setup, Establishing JDBC Database Connections, ResultSet Interface

Java FX GUI: Java FX Scene Builder, Java FX App Window Structure, displaying text and image, event handling, laying out nodes in scene graph, mouse events (Text Book 3)

Textbooks:

1. JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.
2. Joy with JAVA, Fundamentals of Object Oriented Programming, DebasisSamanta, MonalisaSarma, Cambridge, 2023.
3. JAVA 9 for Programmers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

References Books:

1. The complete Reference Java, 11thedition, Herbert Schildt,TMH
2. Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Online Resources:

1. https://nptel.ac.in/courses/106/105/106105191/
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012880464547618816347_shared/overview

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)
II B.Tech. I Semester
(Common to CSE, CSM, CAI, AI, AI&DS, CSD, CSC, CSO & IT branches)

L	T	P	C
0	0	3	1.5

23ACS07: ADVANCED DATA STRUCTURES & ALGORITHM ANALYSIS LAB

Course Objectives: The objective of the course is to
- Acquire practical skills in constructing and managing Data structures
- Apply the popular algorithm design methods in problem-solving scenarios

Course Outcomes:
After completion of the course, students will be able to
- **CO1.** Design and develop programs to solve real world problems with the popular algorithm design methods. (L5)
- **CO2.** Demonstrate an understanding of Non-Linear data structures by developing implementing the operations on AVL Trees, B-Trees, Heaps and Graphs. (L2)
- **CO3.** Critically assess the design choices and implementation strategies of algorithms and data structures in complex applications. (L5)
- **CO4.** Utilize appropriate data structures and algorithms to optimize solutions for specific computational problems. (L3)
- **CO5.** Compare the performance of different of algorithm design strategies (L4)
- **CO6.** Design algorithms to new real world problems (L6)

Experiments covering the Topics:
- Operations on AVL trees, B-Trees, Heap Trees
- Graph Traversals
- Sorting techniques
- Minimum cost spanning trees
- Shortest path algorithms
- 0/1 Knapsack Problem
- Travelling Salesperson problem
- Optimal Binary Search Trees
- N-Queens Problem
- Job Sequencing

Sample Programs:
1. Construct an AVL tree for a given set of elements which are stored in a file. And implement insert and delete operation on the constructed tree. Write contents of tree into a new file using in-order.
2. Construct B-Tree an order of 5 with a set of 100 random elements stored in array. Implement searching, insertion and deletion operations.
3. Construct Min and Max Heap using arrays, delete any element and display the content of the Heap.

4. Implement BFT and DFT for given graph, when graph is represented by
   a) Adjacency Matrix   b) Adjacency Lists
5. Write a program for finding the bi-connected components in a given graph.
6. Implement Quick sort and Merge sort and observe the execution time for various input sizes (Average, Worst and Best cases).
7. Compare the performance of Single Source Shortest Paths using Greedy method when the graph is represented by adjacency matrix and adjacency lists.
8. Implement Job sequencing with deadlines using Greedystrategy.
9. Write a program to solve 0/1 Knapsack problem Using Dynamic Programming.
10. Implement N-Queens Problem Using Backtracking.
11. Use Backtracking strategy to solve 0/1 Knapsack problem.
12. Implement Travelling Sales Person problem using Branch and Bound approach.

Reference Books:

1. Fundamentals of Data Structures in C++, Horowitz Ellis, SahniSartaj, Mehta, Dinesh, 2ndEdition, Universities Press
2. Computer Algorithms/C++ Ellis Horowitz, SartajSahni, SanguthevarRajasekaran, 2ndEdition, University Press
3. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
4. An introduction to Data Structures with applications, Trembley& Sorenson, McGraw Hill

Online Learning Resources:

1. http://cse01-iiith.vlabs.ac.in/
2. http://peterindia.net/Algorithms.html

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
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L	T	P	C
0	0	3	1.5

23ACS08: OBJECT-ORIENTED PROGRAMMING THROUGH JAVA LAB

Course Objectives:

The aim of this course is to

- Practice object-oriented programming in the Java programming language
- implement Classes, Objects, Methods, Inheritance, Exception, Runtime Polymorphism, User defined Exception handling mechanism
- Illustrate inheritance, Exception handling mechanism, JDBC connectivity ● Construct Threads, Event Handling, implement packages, Java FX GUI

Course Outcomes:

After completion of the course, students will be able to

CO1. Demonstrate a solid understanding of Java syntax, including data types, control structures, methods, classes, objects, inheritance, polymorphism, and exception handling. (L2)
CO2. Apply fundamental OOP principles such as encapsulation, inheritance, polymorphism, and abstraction to solve programming problems effectively. (L3)
CO3. Familiar with commonly used Java libraries and APIs, including the Collections
CO4. Framework, Java I/O, JDBC, and other utility classes. (L2)
CO5. Develop problem-solving skills and algorithmic thinking, applying OOP concepts to design efficient solutions to various programming challenges. (L3)
CO6. Proficiently construct graphical user interface (GUI) applications using JavaFX (L4)
CO7. Develop new programs for solving typical computer science problems (L6)

Experiments covering the Topics:

- Object Oriented Programming fundamentals- data types, control structures
- Classes, methods, objects, Inheritance, polymorphism,
- Exception handling, Threads, Packages, Interfaces
- Files, I/O streams, JavaFX GUI

Sample Experiments:

Exercise – 1:

a) Write a JAVA program to display default value of all primitive data type of JAVA
b) Write a java program that display the roots of a quadratic equation $ax^2+bx=0$. Calculate the discriminate D and basing on value of D, describe the nature of root.

Exercise - 2

a) Write a JAVA program to search for an element in a given list of elements using binary search mechanism.
b) Write a JAVA program to sort for an element in a given list of elements using bubble sort
c) Write a JAVA program using StringBuffer to delete, remove character.

Exercise - 3

a) Write a JAVA program to implement class mechanism. Create a class, methods and invoke them inside main method.
b) Write a JAVA program implement method overloading.
c) Write a JAVA program to implement constructor.
d) Write a JAVA program to implement constructor overloading.

Exercise - 4

a) Write a JAVA program to implement Single Inheritance
b) Write a JAVA program to implement multi level Inheritance
c) Write a JAVA program for abstract class to find areas of different shapes

Exercise - 5

a) Write a JAVA program give example for "super" keyword.
b) Write a JAVA program to implement Interface. What kind of Inheritance can be achieved?
c) Write a JAVA program that implements Runtime polymorphism

Exercise - 6

a) Write a JAVA program that describes exception handling mechanism
b) Write a JAVA program Illustrating Multiple catch clauses ● Write a JAVA program for creation of Java Built-in Exceptions

- Write a JAVA program for creation of User Defined Exception

Exercise - 7

a) Write a JAVA program that creates threads by extending Thread class. First thread display "Good Morning "every 1 sec, the second thread displays "Hello "every 2 seconds and the third display "Welcome" every 3 seconds, (Repeat the same by implementing Runnable) b) Write a program illustrating **is Alive** and **join ()**
c) Write a Program illustrating Daemon Threads.
d) Write a JAVA program Producer Consumer Problem

Exercise – 8

8. Write a JAVA program that import and use the user defined packages
9. Without writing any code, build a GUI that display text in label and image in an ImageView (use JavaFX)
10. Build a Tip Calculator app using several JavaFX components and learn how to respond to user interactions with the GUI

Exercise – 9

4. Write a java program that connects to a database using JDBC
b) Write a java program to connect to a database using JDBC and insert values into it.
c) Write a java program to connect to a database using JDBC and delete values from it **Text Books:**

1. JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.
2. Joy with JAVA, Fundamentals of Object Oriented Programming, DebasisSamanta, MonalisaSarma, Cambridge, 2023.
3. JAVA 9 for Programmers, Paul Deitel, Harvey Deitel, 4^{th} Edition, Pearson.

References Books:

1. The complete Reference Java, 11^{th} edition, Herbert Schildt,TMH
2. Introduction to Java programming, 7^{th} Edition, Y Daniel Liang, Pearson

Online Resources:

1. https://nptel.ac.in/courses/106/105/106105191/
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012880464547618816347_shared/overview

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(AUTONOMOUS)
II B.Tech. I Semester
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L	T	P	C
0	1	2	2

23ACS09 - PYTHON PROGRAMMING
(SKILL ENHANCEMENT COURSE)

Course Objectives: The main objectives of the course are to

- Introduce core programming concepts of Python programming language.
- Demonstrate about Python data structures like Lists, Tuples, Sets and dictionaries
- Implement Functions, Modules and Regular Expressions in Python Programming and to create practical and contemporary applications using these

Course Outcomes:
After completion of the course, students will be able to:

CO1. Classify data structures of Python (L4)
CO2. Apply Python programming concepts to solve a variety of computational problems (L3)
CO3. Understand the principles of object-oriented programming (OOP) in Python, including classes, objects, inheritance, polymorphism, and encapsulation, and apply them to design and implement Python programs (L3)
CO4. Become proficient in using commonly used Python libraries and frameworks such as JSON, XML, NumPy, pandas (L2)
CO5. Exhibit competence in implementing and manipulating fundamental data structures such as lists, tuples, sets, dictionaries (L3)
CO6. Propose new solutions to computational problems (L6)

UNTI-I: History of Python Programming Language, Thrust Areas of Python, Installing Anaconda Python Distribution, Installing and Using Jupyter Notebook.
Parts of Python Programming Language: Identifiers, Keywords, Statements and Expressions, Variables, Operators, Precedence and Associativity, Data Types, Indentation, Comments, Reading Input, Print Output, Type Conversions, the type () Function and Is Operator, Dynamic and Strongly Typed Language.
Control Flow Statements: if statement, if-else statement, if...elif…else, Nested if statement, while Loop, for Loop, continue and break Statements, Catching Exceptions Using try and except Statement.
Sample Experiments:

1. Write a program to find the largest element among three Numbers.
2. Write a Program to display all prime numbers within an interval
3. Write a program to swap two numbers without using a temporary variable.
4. Demonstrate the following Operators in Python with suitable examples.

i) Arithmetic Operators ii) Relational Operators iii) Assignment Operatorsiv) Logical Operators v) Bit wise Operators vi) Ternary Operator vii) Membership Operators viii) Identity Operators

5. Write a program to add and multiply complex numbers
6. Write a program to print multiplication table of a given number.

UNIT-II: Functions: Built-In Functions, Commonly Used Modules, Function Definition and Calling the function, return Statement and void Function, Scope and Lifetime of Variables, Default Parameters, Keyword Arguments, *args and **kwargs, Command Line Arguments. **Strings:** Creating and Storing Strings, Basic String Operations, Accessing Characters in String by Index Number, String Slicing and Joining, String Methods, Formatting Strings. **Lists**: Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built- In Functions Used on Lists, List Methods, del Statement.

Sample Experiments:

7. Write a program to define a function with multiple return values.
8. Write a program to define a function using default arguments.
9. Write a program to find the length of the string without using any library functions.
10. Write a program to check if the substring is present in a given string or not.
11. Write a program to perform the given operations on a list:

    i. Addition   ii. Insertion   iii. slicing
12. Write a program to perform any 5 built-in functions by taking any list.

UNIT-III: Dictionaries: Creating Dictionary, Accessing and Modifying key:value Pairs in Dictionaries, Built- In Functions Used on Dictionaries, Dictionary Methods, del Statement.

Tuples and Sets: Creating Tuples, Basic Tuple Operations, tuple() Function, Indexing and Slicing in Tuples, Built-In Functions Used on Tuples, Relation between Tuples and Lists, Relation between Tuples and Dictionaries, Using zip() Function, Sets, Set Methods, Frozenset.

Sample Experiments:

13. Write a program to create tuples (name, age, address, college) for at least two members and concatenate the tuples and print the concatenated tuples.
14. Write a program to count the number of vowels in a string (No control flow allowed).
15. Write a program to check if a given key exists in a dictionary or not.
16. Write a program to add a new key-value pair to an existing dictionary.
17. Write a program to sum all the items in a given dictionary.

UNIT-IV:Files: Types of Files, Creating and Reading Text Data, File Methods to Read and Write Data, Reading and Writing Binary Files, Pickle Module, Reading and Writing CSV Files, Python os and os.path Modules.

Object-Oriented Programming: Classes and Objects, Creating Classes in Python, Creating Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes Vs Data Attributes, Encapsulation, Inheritance, Polymorphism.

<Sample Experiments:

18. Write a program to sort words in a file and put them in another file. The output file should have only lower-case words, so any upper-case words from source must be lowered.
19. Python program to print each line of a file in reverse order.
20. Python program to compute the number of characters, words and lines in a file.
21. Write a program to create, display, append, insert and reverse the order of the items in the array.
22. Write a program to add, transpose and multiply two matrices.
23. Write a Python program to create a class that represents a shape. Include methods to calculate its area and perimeter. Implement subclasses for different shapes like circle, triangle, and square.

UNIT-V: Introduction to Data Science: Functional Programming, JSON and XML in Python, NumPy with Python, Pandas.

Sample Experiments:

24. Python program to check whether a JSON string contains complex object or not.
25. Python Program to demonstrate NumPy arrays creation using array () function.
26. Python program to demonstrate use of ndim, shape, size, dtype.
27. Python program to demonstrate basic slicing, integer and Boolean indexing.
28. Python program to find min, max, sum, cumulative sum of array
29. Create a dictionary with at least five keys and each key represent value as a list where this list contains at least ten values and convert this dictionary as a pandas data frame and explore the data through the data frame as follows:
    a) Apply head () function to the pandas data frame
    b) Perform various data selection operations on Data Frame
30. Select any two columns from the above data frame, and observe the change in one attribute with respect to other attribute with scatter and plot operations in matplotlib

Reference Books:

1. Gowrishankar S, Veena A., Introduction to Python Programming, CRC Press.
2. Python Programming, S Sridhar, J Indumathi, V M Hariharan, 2nd Edition, Pearson, 2024
3. Introduction to Programming Using Python, Y. Daniel Liang, Pearson.

Online Learning Resources/Virtual Labs:

1. https://www.coursera.org/learn/python-for-applied-data-science-ai
2. https://www.coursera.org/learn/python?specialization=python#syllabus

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

II B.Tech. I Semester (Common to All branches of Engineering)

L	T	P	C
2	0	0	0

23AHS18 : ENVIRONMENTAL SCIENCE

Course Objectives:

- To make the students to get awareness on environment.
- To understand the importance of protecting natural resources, ecosystems for future generations and pollution causes due to the day to day activities of human life To save earth from the inventions by the engineers.

UNIT I

Multidisciplinary Nature of Environmental Studies: – Definition, Scope and Importance – Need for Public Awareness.

Natural Resources : Renewable and non-renewable resources – Natural resources and associated problems – Forest resources – Use and over – exploitation, deforestation, case studies – Timber extraction – Mining, dams and other effects on forest and tribal people – Water resources – Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems – Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies – Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. – Energy resources:

UNIT II

Ecosystems: Concept of an ecosystem. – Structure and function of an ecosystem – Producers, consumers and decomposers – Energy flow in the ecosystem – Ecological succession – Food chains, food webs and ecological pyramids – Introduction, types, characteristic features, structure and function of the following ecosystem:

a. Forest ecosystem.
b. Grassland ecosystem
c. Desert ecosystem.
d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)

Biodiversity and its Conservation : Introduction 0 Definition: genetic, species and ecosystem diversity – Bio-geographical classification of India – Value of biodiversity: consumptive use, Productive use, social, ethical, aesthetic and option values – Biodiversity at global, National and local levels – India as a mega-diversity nation – Hot-sports of biodiversity – Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – Endangered and endemic species of India – Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

UNIT III

Environmental Pollution: Definition, Cause, effects and control measures of :

a. Air Pollution.
b. Water pollution
c. Soil pollution
d. Marine pollution
e. Noise pollution
f. Thermal pollution
g. Nuclear hazards

Solid Waste Management: Causes, effects and control measures of urban and industrial wastes – Role of an individual in prevention of pollution – Pollution case studies – Disaster management:
floods, earthquake, cyclone and landslides.

UNIT IV

Social Issues and the Environment: From Unsustainable to Sustainable development – Urban problems related to energy – Water conservation, rain water harvesting, watershed management – Resettlement and rehabilitation of people; its problems and concerns. Case studies – Environmental ethics: Issues and possible solutions – Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case Studies – Wasteland reclamation. – Consumerism and waste products. – Environment Protection Act. – Air (Prevention and Control of Pollution) Act. – Water (Prevention and control of Pollution) Act – Wildlife Protection Act – Forest Conservation Act – Issues involved in enforcement of environmental legislation – Public awareness.

UNIT V

Human Population and the Environment: Population growth, variation among nations. Population explosion – Family Welfare Programmes. – Environment and human health – Human Rights – Value Education – HIV/AIDS – Women and Child Welfare – Role of information Technology in Environment and human health – Case studies.

Field Work: Visit to a local area to document environmental assets River/forest grassland/hill/mountain – Visit to a local polluted site-Urban/Rural/Industrial/Agricultural Study of common plants, insects, and birds – river, hill slopes, etc..

Textbooks:

1. Textbook of Environmental Studies for Undergraduate Courses Erach Bharucha for University Grants Commission, Universities Press.
2. Palaniswamy, "Environmental Studies", Pearson education
3. S.Azeem Unnisa, "Environmental Studies" Academic Publishing Company
4. K.Raghavan Nambiar, "Text book of Environmental Studies for Undergraduate Courses as per UGC model syllabus", Scitech Publications (India), Pvt. Ltd.

References:

1. Deeksha Dave and E.Sai Baba Reddy, "Textbook of Environmental Science", Cengage Publications.
2. M.Anji Reddy, "Text book of Environmental Sciences and Technology", BS Publication.
3. J.P.Sharma, Comprehensive Environmental studies, Laxmi publications.
4. J. Glynn Henry and Gary W. Heinke, "Environmental Sciences and Engineering", Prentice hall of India Private limited
5. G.R.Chatwal, "A Text Book of Environmental Studies" Himalaya Publishing House
6. Gilbert M. Masters and Wendell P. Ela, "Introduction to Environmental Engineering and Science, Prentice hall of India Private limited.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

II B.Tech - I Semester (Common to All Branches)

L	T	P	C
2	0	0	0

23AHS24 QUANTITATIVE APTITUDE AND REASONING - I

Course Outcomes:

After successful completion of the course, the student will be able to

1. Develop the thinking ability to meet the challenges in solving Logical Reasoningproblems.
2. Solve campus placements aptitude papers covering Quantitative Ability and VerbalAbility.
3. Apply different placement practice techniques.

UNIT- I **6 Hours**

QUANTITATIVE ABILITY – I: Vedic Maths – Square - Square root – Cube - Cube root –Higher Roots - Fractions (+, -, ×, ÷)– Decimal Fractions(+, -, ×, ÷) –LCM and HCF – VBODMAS Rule - Simplifications - Number System [Introduction –p/q forms –Factors – Multiples – Prime Numbers – Composite Numbers – Twin Primes – Co-Primes,Different Types of Numbers, Number of factors – Sum of factors – Unit's place value – Remainder theorem – Number of Zeros at the end of the product - Divisibility Rules –Prime Number Checking – Relation among Quotient, Divident, Divisor & Remainder - Formulae, Application type of problems]

UNIT-II **6 Hours**

QUANTITATIVE ABILITY – II: Ratio, Proportion & Variation [Definition of ratio, Types of Ratios, Principles of Ratios, Comparison of Ratios, Definition of Proportion, Types of Proportion, Principle of Proportion, Properties of Proportion, Variation & Types of variations]– Partnership & Share[Definition of partnership, Types of partnership, Simple Partnership & Compound Partnership, profits ratio, Application type of problems] – Average & Ages[Definition of Average, Average of Natural Numbers, Even Numbers, Odd Numbers, Prime Numbers, Application type of problems] – Mixture & Alligation [Definition of Mixture & Alligation, Mixture Formula, Alligation Rule, Application type of Problems]

UNIT-III **6 Hours**

REASONING ABILITY I: Number Series – Number Analogy – Number Odd Man Out – Wrong Number – Letter Series – Letter Analogy – Letter Odd Man

UNIT-IV **6 Hours**

VERBAL I: Verbal analogy - Types - Parts of Speech – Noun, Pronoun, Adjective, Verb, Adverb, Preposition, Conjunction and Interjection - Prepositions –Preposition of Place, Preposition of Placement, Preposition of Time and Preposition of Duration - Articles – Usage of a, an, the, Omission of articles - Sentences - Pattern and Types.

UNIT-V **6 Hours**

SOFT SKILL I: Communication Skills - Self-Confidence - Introductions & Greetings - Presentation Skills - Self- Motivation

Text Books:

1. Quantitative Aptitude, Logic Reasoning & Verbal Reasoning, R S Agarwal, S.Chand Publications-2022
2. Quantitative Aptitude for Competitive Examinations, R S Agarwal, S.Chand Publications-2022

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	-	-	-	-
CO2	2	2	-	-	-	-	-	-	-	-	-	-
CO3	2	-	-	-	-	-	-	-	-	-	-	-

3-High Mapping **2- Medium Mapping** **1-Low Mapping**

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

II B.Tech. II Semester

(Common to CSD, CSM, CAI, AI, AI&DS & IT Branches)

L	T	P	C
2	0	0	2

23AME05 : OPTIMIZATION TECHNIQUES

Course Objectives: The objectives of the course are to

- To provide the basic knowledge about Optimization, importance, application areas of in the industry, Linear Programming.
- To impart different optimization models under typical situations in the business organization like transportation, assignment.
- To understand the process of sequencing in a typical industry.
- To describe different game strategies under cut-throat competitive business environment
- To develop networks of activities of projects and to find out optimal modes of completing projects using network modelling evaluation techniques.

Course Outcomes:

After completion of the course, students will be able to:

CO1 Understanding Optimization and Formulation of Linear Programing Models

CO2 Formulate and Solve Transportation & Assignment Models

CO3 Sequencing of operations and optimizing

CO4 Discuss the game theory and strategies

CO5 Developing networks of activities and finding optimal mode of projects evaluation.

UNIT - I

Introduction: Meaning, Nature, Scope & Significance of Optimization - Typical applications.The Linear Programming Problem – Introduction, Formulation of Linear Programming problem, Limitations of L.P.P, Graphical method, Simplex method: Maximization and Minimization model(exclude Duality problems), Big-M method and Two Phase method.

UNIT - II

Transportation Problem: Introduction, Transportation Model, Finding initial basic feasible solutions, Moving towards optimality, Unbalanced Transportation problems, Transportation problems with maximization, Degeneracy.

Assignment Problem – Introduction, Mathematical formulation of the problem, Solution of an Assignment problem, Hungarian Algorithm, Multiple Solution, Unbalanced Assignment problems, Maximization in Assignment Model.

UNIT - III

Sequencing – Job sequencing, Johnsons Algorithm for n Jobs and Two machines, n Jobs and Three Machines, n jobs through m machines, Two jobs and m Machines Problems.

UNIT - IV

Game Theory: Concepts, Definitions and Terminology, Two Person Zero Sum Games, Pure Strategy Games (with Saddle Point), Principal of Dominance, Mixed Strategy Games (Game without Saddle Point), Significance of Game Theory in Managerial Application.

UNIT - V

Project Management: Network Analysis – Definition –objectives -Rules for constructing network diagram- Determining Critical Path – Earliest & Latest Times – Floats - Application of CPM and PERT techniques in Project Planning and Control – PERT Vs CPM. (exclude Project Crashing).

Textbooks:

1. Operations Research / R.Pannerselvam, PHI Publications.
2. Operations Research / S.D.Sharma-Kedarnath
3. Operations Research /A.M.Natarajan,P.Balasubramani,A. Tamilarasi/Pearson Education.
4. Engineering Optimization: Theory and practice / S.S.Rao, New Age International (P) Limited

Reference Books:

1. Quantitative Techniques in Management / ND Vohra, Tata McGraw Hill, 4th Edition, 2011.
2. ntroduction to O.R/Hiller &Libermann (TMH).
3. Operations Research: Methods & Problems / Maurice Saseini, ArhurYaspan& Lawrence Friedman. Pearson
4. Quantitative Analysis For Management/ Barry Render, Ralph M. Stair, Jr and Michael

E. Hanna/

5. Operations Research / Wagner/ PHI Publications.

Online Learning Sources

https://onlinecourses.swayam2.ac.in/cec20_ma10/preview
https://onlinecourses.nptel.ac.in/noc20_ma23/preview
https://onlinecourses.nptel.ac.in/noc19_ma29/preview

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L	T	P	C
3	0	0	3

23AHS21- PROBABILITY & STATISTICS

(Common to CSE, CSM, CSO, CAI, & IT Branches)

Course Outcomes:
After successful completion of this course, the students should be able to:

CO1 Acquire knowledge in finding the analysis of the data quantitatively or categoricallyand various statistical elementary tools.

CO2 Develop skills in designing mathematical models involving probability, random variables and the critical thinking in the theory of probability and its applications in real life problems.

CO3 Apply the theoretical probability distributions like binomial, Poisson, and Normal in the relevant application areas.

CO4 Analyze to test various hypotheses included in theory and types of errors for large samples.

CO5 Apply the different testing tools like t-test, F-test, chi-square test to analyze the relevant real-life problems.

UNITI : Descriptive statistics
Statistics Introduction, Population vs Sample, Collection of data, primary and secondary data, Measures of Central tendency, Measures of Variability (spread or variance) Skewness, Kurtosis, correlation, correlation coefficient, rank correlation, regression coefficients, method of least squares, regression lines.

UNIT II Probability
Probability, probability axioms, addition law and multiplicative law of probability, conditional probability, Baye’s theorem, random variables (discrete and continuous), probability density functions, properties, mathematical expectation.

UNIT III Probability distributions
Probability distributions: Binomial, Poisson and Normal-their properties (Chebyshevs inequality). Approximation of the binomial distribution to normal distribution.

UNIT IV Estimation and Testing of hypothesis, large sample tests

Estimation-parameters, statistics, sampling distribution, point estimation, Formulation of null hypothesis, alternative hypothesis, the critical and acceptance regions, level of significance, two types of errors and power of the test. Large Sample Tests: Test for single proportion, difference of proportions, test for single mean and difference of means. Confidence interval for parameters in one sample and two sample problems

UNIT V Small sample tests

Student t-distribution (test for single mean, two means and paired t-test), testing of equality of variances (F-test), $\chi 2$ - test for goodness of fit, $\chi 2$ - test for independence of attributes.

Textbooks:

1. Miller and Freunds, Probability and Statistics for Engineers,7/e, Pearson, 2008.
2. S.C. Gupta and V.K. Kapoor, Fundamentals of Mathematical Statistics, 11/e, Sultan Chand & Sons Publications, 2012.

Reference Books:

1. S. Ross, a First Course in Probability, Pearson Education India, 2002.
2. W. Feller, an Introduction to Probability Theory and its Applications, 1/e, Wiley, 1968.
3. B. V. Ramana, Higher Engineering Mathematics, Mc Graw Hill Education.

Online Learning Resources:

1. https://onlinecourses.nptel.ac.in/noc21_ma74/preview
2. https://onlinecourses.nptel.ac.in/noc22_mg31/preview

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)
II B.Tech. II Semester
(Common to CSM,AI and CAI Branches)

L	T	P	C
3	0	0	3

23ACA02: MACHINE LEARNING

Course Objectives: The objectives of the course are

- Define machine learning and its different types (supervised and unsupervised) and understand their applications.
- Apply supervised learning algorithms including decision trees and k-nearest neighbors (k-NN).
- Implement unsupervised learning techniques, such as K-means clustering.

Course Outcomes:
After completion of the course, students will be able to

CO1: Identify machine learning techniques suitable for a given problem. (L3)
CO2: Solve real-world problems using various machine learning techniques. (L3) **CO3:** Apply Dimensionality reduction techniques for data preprocessing. (L3)
CO4: Explain what is learning and why it is essential in the design of intelligent machines. (L2)
CO5: Evaluate Advanced learning models for language, vision, speech, decision making etc. (L5)

UNIT-I: Introduction to Machine Learning: Evolution of Machine Learning, Paradigms for ML, Learning by Rote, Learning by Induction, Reinforcement Learning, Types of Data, Matching, Stages in Machine Learning, Data Acquisition, Feature Engineering, Data Representation, Model Selection, Model Learning, Model Evaluation, Model Prediction, Search and Learning, Data Sets.

UNIT-II: Nearest Neighbor-Based Models: Introduction to Proximity Measures, Distance Measures, Non-Metric Similarity Functions, Proximity Between Binary Patterns, Different Classification Algorithms Based on the Distance Measures ,K-Nearest Neighbor Classifier, Radius Distance Nearest Neighbor Algorithm, KNN Regression, Performance of Classifiers, Performance of Regression Algorithms.

UNIT-III: Models Based on Decision Trees: Decision Trees for Classification, Impurity Measures, Properties, Regression Based on Decision Trees, Bias–Variance Trade-off, Random Forests for Classification and Regression.
The Bayes Classifier: Introduction to the Bayes Classifier, Bayes' Rule and Inference, The Bayes Classifier and its Optimality, Multi-Class Classification | Class Conditional Independence and Naive Bayes Classifier (NBC)

UNIT-IV: Linear Discriminants for Machine Learning: Introduction to Linear Discriminants, Linear Discriminants for Classification, Perceptron Classifier, Perceptron Learning Algorithm, Support Vector Machines, Linearly Non-Separable Case, Non-linear SVM, Kernel Trick, Logistic Regression, Linear Regression, Multi-Layer Perceptrons (MLPs), Backpropagation for Training an MLP.

UNIT-V: Clustering : Introduction to Clustering, Partitioning of Data, Matrix Factorization | Clustering of Patterns, Divisive Clustering, Agglomerative Clustering, Partitional Clustering, K-Means Clustering, Soft Partitioning, Soft Clustering, Fuzzy C-Means Clustering, Rough Clustering, Rough K-Means Clustering Algorithm, Expectation Maximization-Based Clustering, Spectral Clustering.

Textbooks:

1. "Machine Learning Theory and Practice", M N Murthy, V S Ananthanarayana, Universities Press (India), 2024

Reference Books:

1. "Machine Learning", Tom M. Mitchell, McGraw-Hill Publication, 2017
2. "Machine Learning in Action",Peter Harrington, DreamTech
3. "Introduction to Data Mining", Pang-Ning Tan, Michel Stenbach, Vipin Kumar, 7th Edition, 2019.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

II B.Tech. II Semester

(Common to CSE, CSM, CAI, AI, AI&DS CSD, CSC & IT branches)

L	T	P	C
3	0	0	3

23ACS12 : DATABASE MANAGEMENT SYSTEMS

Course Objectives: The main objective of the course is to

- Introduce database management systems and to give a good formal foundation on the relational model of data and usage of Relational Algebra
- Introduce the concepts of basic SQL as a universal Database language
- Demonstrate the principles behind systematic database design approaches by covering conceptual design, logical design through normalization
- Provide an overview of physical design of a database system, by discussing Database indexing techniques and storage techniques

Course Outcomes:

After completion of the course, students will be able to

CO1. Understand the basic concepts of database management systems (L2)

CO2. Analyze a given database application scenario to use ER model for conceptual design of the database (L4)

CO3. Utilize SQL proficiently to address diverse query challenges (L3).

CO4. Employ normalization methods to enhance database structure (L3)

CO5. Assess and implement transaction processing, concurrency control and database recovery protocols in databases. (L4)

UNIT I:Introduction: Database system, Characteristics (Database Vs File System), Database Users, Advantages of Database systems, Database applications. Brief introduction of different Data Models; Concepts of Schema, Instance and data independence; Three tier schema architecture for data independence; Database system structure, environment, Centralized and Client Server architecture for the database.

Entity Relationship Model: Introduction, Representation of entities, attributes, entity set, relationship, relationship set, constraints, sub classes, super class, inheritance, specialization, generalization using ER Diagrams.

Unit II: Relational Model: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance, Relational Algebra, Relational Calculus. BASIC SQL:Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update).

UNIT III: SQL: Basic SQL querying (select and project) using where clause, arithmetic & logical operations, SQL functions(Date and Time, Numeric, String conversion).Creating tables with relationship, implementation of key and integrity constraints, nested queries, sub queries, grouping, aggregation, ordering, implementation of different types of joins, view(updatable and non-updatable), relational set operations.

UNIT IV: Schema Refinement (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, normal forms based on functional dependencyLossless join and dependency preserving decomposition, (1NF, 2NF and 3 NF), concept of surrogate key, Boyce-Codd normal form(BCNF), MVD, Fourth normal form(4NF), Fifth Normal Form (5NF).

UNIT V:Transaction Concept: Transaction State, ACID properties, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, Testing for Serializability, lock based, time stamp based, optimistic, concurrency protocols, Deadlocks, Failure Classification, Storage, Recovery and Atomicity, Recovery algorithm.

Introduction to Indexing Techniques: B+ Trees, operations on B+Trees, Hash Based Indexing:

Textbooks:

1. Database Management Systems, 3rd edition, Raghurama Krishnan, Johannes Gehrke, TMH (For Chapters 2, 3, 4)
2. Database System Concepts,5th edition, Silberschatz, Korth, Sudarsan,TMH (For Chapter 1 and Chapter 5)

Reference Books:

1. Introduction to Database Systems, 8thedition, C J Date, Pearson.
2. Database Management System, 6th edition, RamezElmasri, Shamkant B. Navathe, Pearson
3. Database Principles Fundamentals of Design Implementation and Management, Corlos Coronel, Steven Morris, Peter Robb, Cengage Learning.

Web-Resources:

1. https://nptel.ac.in/courses/106/105/106105175/
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0127580666728202 2456_shared/overview

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

II B.Tech. II Semester (Common to CSM, CAI & AI Branches)

II B.Tech. I Semester (Common to CSE, CSD, CSC, CSO & IT Branches)

L	T	P	C
3	0	0	3

23AEC06 : DIGITAL LOGIC & COMPUTER ORGANIZATION

Course Objectives: The main objectives of the course is to

- provide students with a comprehensive understanding of digital logic design principles and computer organization fundamentals
- Describe memory hierarchy concepts
- Explain input/output (I/O) systems and their interaction with the CPU, memory, and peripheral devices

Course Outcomes:

After completion of the course, students will be able to

CO1. Differentiate between combinational and sequential circuits based on their characteristics and functionalities. (L2)

CO2. Demonstrate an understanding of computer functional units. (L2)

CO3. Analyze the design and operation of processors, including instruction execution, pipelining, and control unit mechanisms, to comprehend their role in computer systems.(L3)

CO4. Describe memory hierarchy concepts, including cache memory, virtual memory, and secondary storage, and evaluate their impact on system performance and scalability. (L3)

CO5. Explain input/output (I/O) systems and their interaction with the CPU, memory, and peripheral devices, including interrupts, DMA, and I/O mapping techniques. (L3)

CO6. Design Sequential and Combinational Circuits (L6)

UNIT – I:

Data Representation: Binary Numbers, Fixed Point Representation. Floating Point Representation. Number base conversions, Octal and Hexadecimal Numbers, components, Signed binary numbers, Binary codes

Digital Logic Circuits-I: Basic Logic Functions, Logic gates, universal logic gates, Minimization of Logic expressions. K-Map Simplification, Combinational Circuits, Decoders, Multiplexers

UNIT – II:

Digital Logic Circuits-II: Sequential Circuits, Flip-Flops, Binary counters, Registers, Shift Registers, Ripple counters

Basic Structure of Computers: Computer Types, Functional units, Basic operational concepts, Bus structures, Software, Performance, multiprocessors and multi computers, Computer Generations, Von- Neumann Architecture

UNIT – III:

Computer Arithmetic : Addition and Subtraction of Signed Numbers, Design of Fast Adders, Multiplication of Positive Numbers, Signed-operand Multiplication, Fast Multiplication, Integer Division, Floating-Point Numbers and Operations

Processor Organization: Fundamental Concepts, Execution of a Complete Instruction, Multiple-Bus Organization, Hardwired Control and Multi programmed Control

UNIT – IV:

The Memory Organization: Basic Concepts, Semiconductor RAM Memories, Read-Only Memories, Speed, Size and Cost, Cache Memories, Performance Considerations, Virtual Memories, Memory Management Requirements, Secondary Storage

UNIT – V:

Input/Output Organization: Accessing I/O Devices, Interrupts, Processor Examples, Direct Memory Access, Buses, Interface Circuits, Standard I/O Interfaces

Textbooks:

1. Computer Organization, Carl Hamacher, ZvonkoVranesic, SafwatZaky, 6th edition, McGraw Hill, 2023.
2. Digital Design, 6th Edition, M. Morris Mano, Pearson Education, 2018.
3. Computer Organization and Architecture, William Stallings, 11thEdition, Pearson, 2022.

Reference Books:

1. Computer Systems Architecture, M.Moris Mano, 3rdEdition, Pearson, 2017.
2. Computer Organization and Design, David A. Paterson, John L. Hennessy, Elsevier, 2004.
3. Fundamentals of Logic Design, Roth, 5thEdition, Thomson, 2003.

Online Learning Resources:

https://nptel.ac.in/courses/106/103/106103068/

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

II B.Tech. II Semester, CSE (AI&ML)

L	T	P	C
0	0	3	1.5

23ACA03 : MACHINE LEARNING LAB

Course Objectives:

- To learn about computing central tendency measures and Data preprocessing techniques
- To learn about classification and regression algorithms
- To apply different clustering algorithms for a problem.

Software Required: Python/R/Weka

Lab should cover the concepts studied in the course work, sample listof Experiments:

1. Compute Central Tendency Measures: Mean, Median, Mode Measure of Dispersion: Variance, Standard Deviation.
2. Apply the following Pre-processing techniques for a given dataset.
   a. Attribute selection
   b. Handling Missing Values
   c. Discretization
   d. Elimination of Outliers
3. Apply KNN algorithm for classification and regression
4. Demonstrate decision tree algorithm for a classification problem and perform parameter tuning for better results
5. Demonstrate decision tree algorithm for a regression problem
6. Apply Random Forest algorithm for classification and regression
7. Demonstrate Naïve Bayes Classification algorithm.
8. Apply Support Vector algorithm for classification
9. Demonstrate simple linear regression algorithm for a regression problem
10. Apply Logistic regression algorithm for a classification problem
11. Demonstrate Multi-layer Perceptron algorithm for a classification problem
12. Implement the K-means algorithm and apply it to the data you selected. Evaluate performance by measuring the sum of the Euclidean distance of each example from its class center. Test the performance of the algorithm as a function of the parameters K.
13. Demonstrate the use of Fuzzy C-Means Clustering
14. Demonstrate the use of Expectation Maximization based clustering algorithm

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)
II B.Tech. II Semester
(Common to CSE, CSM, CSD, CSC, CAI, AI, AI&DS & IT Branches)

L	T	P	C
0	0	3	1.5

23ACS14: DATABASE MANAGEMENT SYSTEMS LAB

Course Objectives: This Course will enable students to

- Populate and query a database using SQL DDL/DML Commands
- Declare and enforce integrity constraints on a database
- Writing Queries using advanced concepts of SQL
- Programming PL/SQL including procedures, functions, cursors and triggers.

Course Outcomes:

After completion of the course, students will be able to

CO1. Utilizing Data Definition Language (DDL), Data Manipulation Language (DML), and Data Control Language (DCL) commands effectively within a database environment (L3)

CO2. Constructing and execute queries to manipulate and retrieve data from databases. (L3)

CO3. Develop application programs using PL/SQL. (L3)

CO4. Analyze requirements and design custom Procedures, Functions, Cursors, and Triggers, leveraging their capabilities to automate tasks and optimize database functionality (L4)

CO5. Establish database connectivity through JDBC (Java Database Connectivity) (L3)

Experiments covering the topics:

- DDL, DML, DCL commands
- Queries, nested queries, built-in functions,
- PL/SQL programming- control structures
- Procedures, Functions, Cursors, Triggers,
- Database connectivity- ODBC/JDBC

Sample Experiments:

1. Creation, altering and droping of tables and inserting rows into a table (use constraints while creating tables) examples using SELECT command.
2. Queries (along with sub Queries) using ANY, ALL, IN, EXISTS, NOTEXISTS, UNION, INTERSET, Constraints. Example:- Select the roll number and name of the student who secured fourth rank in the class.
3. Queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.
4. Queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)
5.

i. Create a simple PL/SQL program which includes declaration section, executable section and exception –Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found)
ii. Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.

6. Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.
7. Program development using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT –IN Exceptions, USE defined Exceptions, RAISE-APPLICATION ERROR.
8. Programs development using creation of procedures, passing parameters IN and OUT of PROCEDURES.
9. Program development using creation of stored functions, invoke functions in SQL Statements and write complex functions.
10. Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables.
11. Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers
12. Create a table and perform the search operation on table using indexing and nonindexing techniques.
13. Write a Java program that connects to a database using JDBC
14. Write a Java program to connect to a database using JDBC and insert values into it
15. Write a Java program to connect to a database using JDBC and delete values from it

Textbooks/Suggested Reading:

1. Oracle: The Complete Reference by Oracle Press
2. Nilesh Shah, "Database Systems Using Oracle”, PHI, 2007
3. Rick F Vander Lans, “Introduction to SQL”, Fourth Edition, Pearson Education, 2007

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)
II B.Tech. II Semester
Common to (CSE, CSD, CSC, CSM, CSO, AI, AI&DS and CAI Branches)

L	T	P	C
0	1	2	2

23ACS15- FULL STACK DEVELOPMENT – 1
(Skill Enhancement Course)

Course Objectives: The main objectives of the course are to

- Make use of HTML elements and their attributes for designing static web pages
- Build a web page by applying appropriate CSS styles to HTML elements
- Experiment with JavaScript to develop dynamic web pages and validate forms

Course Outcomes:
After completion of the course, students will be able to

CO1. Design Websites. (L6)
CO2. Apply Styling to web pages. (L4)
CO3. Make Web pages interactive. (L6)
CO4. Design Forms for applications. (L6)
CO5. Choose Control Structure based on the logic to be implemented. (L3)
CO6. Understand HTML tags, Attributes and CSS properties (L2)

Experiments covering the Topics:

- Lists, Links and Images
- HTML Tables, Forms and Frames
- HTML 5 and Cascading Style Sheets, Types of CSS
- Selector forms
- CSS with Color, Background, Font, Text and CSS Box Model
- Applying JavaScript - internal and external, I/O, Type Conversion
- JavaScript Conditional Statements and Loops, Pre-defined and User-defined Objects
- JavaScript Functions and Events
- Node.js

Sample Experiments:

1. Lists, Links and Images

a. Write a HTML program, to explain the working of lists.

Note: It should have an ordered list, unordered list, nested lists and ordered list in an unordered list and definition lists.

b. Write a HTML program, to explain the working of hyperlinks using <a> tag and href, target Attributes.

c. Create a HTML document that has your image and your friend's image with a specific height and width. Also when clicked on the images it should navigate to their respective profiles.

d. Write a HTML program, in such a way that, rather than placing large images on a page, the preferred technique is to use thumbnails by setting the height and width parameters to something like to 100*100 pixels. Each thumbnail image is also a link to a full sized version of the image. Create an image gallery using this technique

2. HTML Tables, Forms and Frames

- Write a HTML program, to explain the working of tables. (use tags: <table>, <tr>, <th>, <td> and attributes: border, rowspan, colspan)
- Write a HTML program, to explain the working of tables by preparing a timetable. (Note: Use <caption> tag to set the caption to the table & also use cell spacing, cell padding, border, rowspan, colspan etc.).
- Write a HTML program, to explain the working of forms by designing Registration form. (Note: Include text field, password field, number field, date of birth field, checkboxes, radio buttons, list boxes using <select>&<option> tags, <text area> and two buttons ie: submit and reset. Use tables to provide a better view).
- Write a HTML program, to explain the working of frames, such that page is to be divided into 3 parts on either direction. (Note: first frame image, second frame paragraph, third frame hyperlink. And also make sure of using "no frame" attribute such that frames to be fixed).

3. HTML 5 and Cascading Style Sheets, Types of CSS

a. Write a HTML program, that makes use of <article>, <aside>, <figure>, <figcaption>, <footer>, <header>, <main>, <nav>, <section>, <div>, <span> tags.
b. Write a HTML program, to embed audio and video into HTML web page.
c. Write a program to apply different types (or levels of styles or style specification formats) - inline, internal, external styles to HTML elements. (identify selector, property and value).

4. Selector forms

a. Write a program to apply different types of selector forms

- Simple selector (element, id, class, group, universal)
- Combinator selector (descendant, child, adjacent sibling, general sibling)
- Pseudo-class selector
- Pseudo-element selector
- Attribute selector

5. CSS with Color, Background, Font, Text and CSS Box Model

a. Write a program to demonstrate the various ways you can reference a color in CSS.
b. Write a CSS rule that places a background image halfway down the page, tilting it horizontally. The image should remain in place when the user scrolls up or down.
c. Write a program using the following terms related to CSS font and text:

i. font-size   ii. font-weight   iii. font-style
iv. text-decoration   v. text-transformation   vi. text-alignment

d. Write a program, to explain the importance of CSS Box model using

i. Content   ii. Border   iii. Margin   iv. padding

6. Applying JavaScript - internal and external, I/O, Type Conversion

a. Write a program to embed internal and external JavaScript in a web page.
b. Write a program to explain the different ways for displaying output.

c. Write a program to explain the different ways for taking input.
d. Create a webpage which uses prompt dialogue box to ask a voter for his name and age. Display the information in table format along with either the voter can vote or not

7. JavaScript Pre-defined and User-defined Objects

a. Write a program using document object properties and methods.
b. Write a program using window object properties and methods.
c. Write a program using array object properties and methods.
d. Write a program using math object properties and methods.
e. Write a program using string object properties and methods.
f. Write a program using regex object properties and methods.
g. Write a program using date object properties and methods.
h. Write a program to explain user-defined object by using properties, methods, accessors, constructors and display.

8. JavaScript Conditional Statements and Loops

a. Write a program which asks the user to enter three integers, obtains the numbers from the user and outputs HTML text that displays the larger number followed by the words "LARGER NUMBER" in an information message dialog. If the numbers are equal, output HTML text as "EQUAL NUMBERS".
b. Write a program to display week days using switch case.
c. Write a program to print 1 to 10 numbers using for, while and do-while loops.
d. Write aprogram to print data in object using for-in, for-each and for-of loops
e. Develop a program to determine whether a given number is an 'ARMSTRONG NUMBER' or not. [Eg: 153 is an Armstrong number, since sum of the cube of the digits is equal to the number i.e.,13 + 53+ 33 = 153]
f. Write a program to display the denomination of the amount deposited in the bank in terms of 100's, 50's, 20's, 10's, 5's, 2's & 1's. (Eg: If deposited amount is Rs.163, the output should be 1- 100's, 1-50's, 1- 10's, 1-2's & 1-1's)

9. Javascript Functions and Events

a. Design a appropriate function should be called to display
   - Factorial of that number
   - Fibonacci series up to that number
   - Prime numbers up to that number
   - Is it palindrome or not

b. Design a HTML having a text box and four buttons named Factorial, Fibonacci, Prime, and Palindrome. When a button is pressed an appropriate function should be called to display
   11. Factorial of that number
   12. Fibonacci series up to that number
   13. Prime numbers up to that number
   14. Is it palindrome or not

c. Write a program to validate the following fields in a registration page
   i. Name (start with alphabet and followed by alphanumeric and the length should not be less than 6 characters)
   ii. Mobile (only numbers and length 10 digits)
      iii. E-mail (should contain format like xxxxxxx@xxxxxx.xxx)

Textbooks:

1. Programming the World Wide Web, 7th Edition, Robet W Sebesta, Pearson, 2013.
2. Web Programming with HTML5, CSS and JavaScript, John Dean, Jones & Bartlett Learning, 2019 (Chapters 1-11).
3. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O'Reilly.

Web Links:

1. https://www.w3schools.com/html
2. https://www.w3schools.com/css
3. https://www.w3schools.com/js/
4. https://www.w3schools.com/nodejs
5. https://www.w3schools.com/typescript

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)
II B.Tech. II Semester

(Common to All Branches of Engineering)

L	T	P	C
1	0	2	2

23AMB05 : DESIGN THINKING FOR INNOVATION

Course Objectives:
The objective of this course is to familiarize students with design thinking process as a tool for breakthrough innovation. It aims to equip students with design thinking skills and ignite the minds to create innovative ideas, develop solutions for real-time problems.

Course Outcomes:
After completion of the course, students will be able to

CO1. Define the concepts related to design thinking. (L1, L2)
CO2. Explain the fundamentals of Design Thinking and innovation (L1, L2)
CO3. Apply the design thinking techniques for solving problems in various sectors. (L3)
CO4. Analyse to work in a multidisciplinary environment (L4)
CO5. Evaluate the value of creativity (L5)
CO6. Formulate specific problem statements of real time issues (L3, L6)

UNIT I Introduction to Design Thinking
Introduction to elements and principles of Design, basics of design-dot, line, shape, form as fundamental design components. Principles of design. Introduction to design thinking, history of Design Thinking, New materials in Industry.

UNIT II Design Thinking Process
Design thinking process (empathize, analyze, idea & prototype), implementing the process in driving inventions, design thinking in social innovations. Tools of design thinking - person, costumer, journey map, brainstorming, product development

Activity: Every student presents their idea in three minutes, Every student can present design process in the form of flow diagram or flow chart etc. Every student should explain about product development.

UNIT III Innovation
Art of innovation, Difference between innovation and creativity, role of creativity and innovation in organizations- Creativity to Innovation- Teams for innovation- Measuring the impact and value of creativity.

Activity: Debate on innovation and creativity, Flow and planning from idea to innovation, Debate on value-based innovation.

UNIT IV Product Design

Problem formation, introduction to product design, Product strategies, Product value, Product planning, product specifications- Innovation towards product design- Case studies

Activity: Importance of modelling, how to set specifications, Explaining their own product design.

UNIT V Design Thinking in Business Processes

Design Thinking applied in Business & Strategic Innovation, Design Thinking principles that redefine business – Business challenges: Growth, Predictability, Change, Maintaining Relevance, Extreme competition, Standardization. Design thinking to meet corporate needs- Design thinking for Startups- Defining and testing Business Models and Business Cases- Developing & testing prototypes.

Activity: How to market our own product, About maintenance, Reliability and plan for startup.

Textbooks:

1. Tim Brown,Change by design, Harper Bollins (2009)
2. Idris Mootee, Design Thinking for Strategic Innovation, 2013, John Wiley & Sons.

Reference Books:

1. David Lee, Design Thinking in the Classroom, Ulysses press
2. Shrutin N Shetty, Design the Future, Norton Press
3. William Lidwell,Universal Principles of Design- Kritinaholden, Jill Butter.
4. Chesbrough.H, The Era of Open Innovation – 2013

Online Learning Resources:

https://nptel.ac.in/courses/110/106/110106124/
https://nptel.ac.in/courses/109/104/109104109/
https://swayam.gov.in/nd1_noc19_mg60/preview

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

II B.Tech - II Semester (Common to All Branches)

L	T	P	C
2	0	0	0

23AHS25 QUANTITATIVE APTITUDE AND REASONING - II

Course Outcomes:

After successful completion of the course, the student will be able to

1. Develop the thinking ability to meet the challenges in solving Logical Reasoning problems.
2. Solve campus placements aptitude papers covering Quantitative Ability and Verbal Ability.
3. Apply different placement practice techniques.

UNIT-I **6 Hours**

QUANTITATIVE ABILITY III: Percentage [Percentage values from ½ to 1/30 , Successive increase / Decrease, Increased / Decreased percentage, How much % more / less, Population Problems, Election Problems, Application type of problems] – Profit & Loss[Cost Price , Selling Price , Retail Price , Marked Price / List Price / Printed price, Discounts, Error problems, Application type of problems] –Simple Interest[Principle, Time period, Rate of interest, Interest, Amount, Annual Payment, Application type of problems]- Compound Interest[Principle, Time period, Rate of interest, Interest, Different formulae of amount, Annual Payment, Differences between C.I & S.I for 1 year, 2years & 3years]

UNIT-II **6 Hours**

QUANTITATIVE ABILITY IV: Time and Work [One person is working, 2 persons are working, 3 persons are working, Relation among Men, days, hours & Work, Alternate days, Graphical method, Application type of problems] – Pipes & Cisterns[Inlet, Outlet or leakage, Alternate hours, Application type of problems] – Time, Speed and Distance[Relation among time, speed & distance, Relative Speed, Average Speed, Problems on trains, Application type of problems] –Boats and Streams[Still water, Stream, Current rate, Boat’s rate, Downstream, Upstream, Downstream Speed, Upstream speed, Application type of problems] – Races & Circular Tracks [2 persons are running around a circular track, 3 persons are running around a circular track]

UNIT-III **6 Hours**

REASONING ABILITY II: Alphabet - Coding & Decoding - Directions - Ranking Test – Blood Relations - Inserting the missing number – Venn diagrams – Symbols and Notations - Syllogism – Statement and Conclusion– Data Arrangement – Linear and Circular arrangement

UNIT-IV **6 Hours**

VERBAL II: Tense – Present Tense, Past Tense, Future Tense - Voice – Active voice, Passive voice and Active to Passive Voice Conversion Rules – Speech – Direct Speech, Indirect Speech and Direct to Indirect Speech Conversion Rules –Essay Writing – Types, Steps, Format.

UNIT V **6 Hours**

SOFT SKILL II: Time Management - Stress Management - Team Work - Accent and Voice Communication - Interview Skills.

Text Books:

1. Quantitative Aptitude, Logic Reasoning & Verbal Reasoning, R S Agarwal, S.Chand Publications-2022.
2. Quantitative Aptitude for Competitive Examinations, R S Agarwal, S.Chand Publications-2022.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	**3**	**2**	-	-	-	-	-	-	-	-	-	-
CO2	**1**	**2**	-	-	-	-	-	-	-	-	-	-
CO3	**2**	-	-	-	-	-	-	-	-	-	-	-

3-High Mapping **2- Medium Mapping** **1-Low Mapping**

COMMUNITY SERVICE PROJECT

……Experiential learning through community engagement Introduction

- Community Service Project is an experiential learning strategy that integrates meaningful community service with instruction, participation, learning and community development.
- Community Service Project involves students in community development and service activities and applies the experience to personal and academic development.
- Community Service Project is meant to link the community with the college for mutual benefit. The community will benefit with the focused contribution of the college students for the village/ local development. The college finds an opportunity to develop social sensibility and responsibility among students and emerge as a socially responsible institution.

Objective

Community Service Project should be an integral part of the curriculum, as an alternative to the 2 months of Summer Internships / Apprenticeships / On the Job Training, whenever there is an exigency when students cannot pursue their summer internships. The specific objectives are:

- To sensitize the students to the living conditions of the people who are around them,
□ To help students to realize the stark realities of society.
- To bring about an attitudinal change in the students and help them to develop societal consciousness, sensibility, responsibility and accountability
- To make students aware of their inner strength and help them to find new /out of box solutions to social problems.
- To make students socially responsible citizens who are sensitive to the needs of the disadvantaged sections.
- To help students to initiate developmental activities in the community in coordination with public and government authorities.
- To develop a holistic life perspective among the students by making them study culture, traditions, habits, lifestyles, resource utilization, wastages and its management, social problems, public administration system and the roles and responsibilities of different persons across different social systems.

Implementation of Community Service Project

- Every student should put in 6 weeks for the Community Service Project during the summer vacation.
- Each class/section should be assigned with a mentor.
- Specific Departments could concentrate on their major areas of concern. For example, Dept. of Computer Science can take up activities related to Computer Literacy to different sections of people like - youth, women, housewives, etc
- A logbook must be maintained by each of the students, where the activities undertaken/involved to be recorded.
- The logbook has to be countersigned by the concerned mentor/faculty in charge.
- An evaluation to be done based on the active participation of the student and grade could be awarded by the mentor/faculty member.
- The final evaluation to be reflected in the grade memo of the student.

- The Community Service Project should be different from the regular programs of NSS/NCC/Green Corps/Red Ribbon Club, etc.
- Minor project reports should be submitted by each student. An internal Viva shall also be conducted by a committee constituted by the principal of the college.
- Award of marks shall be made as per the guidelines of Internship/apprentice/ on the job training.

Procedure

- A group of students or even a single student could be assigned for a particular habitation or village or municipal ward, as far as possible, in the near vicinity of their place of stay, to enable them to commute from their residence and return back by evening or so.
- The Community Service Project is a twofold one – o First, the student/s could conduct a survey of the habitation, if necessary, in terms of their own domain or subject area. Or it can even be a general survey, incorporating all the different areas. A common survey format could be designed. This should not be viewed as a duplication of work by the Village or Ward volunteers, rather, it could be another primary source of data.

o Secondly, the student/s could take up a social activity, concerning their domain or subject area. The different areas, could be like – Agriculture

- Health
- Marketing and Cooperation
- Animal Husbandry
- Horticulture
- Fisheries
- Sericulture
- Revenue and Survey
- Natural Disaster Management
- Irrigation
- Law & Order
- Excise and Prohibition
- Mines and Geology
- Energy
- Internet
- Free Electricity
- Drinking Water

EXPECTED OUTCOMES

BENEFITS OF COMMUNITY SERVICE PROJECT TO STUDENTS

Learning Outcomes

- Positive impact on students' academic learning
- Improves students' ability to apply what they have learned in "the real world"
- Positive impact on academic outcomes such as demonstrated complexity of understanding, problem analysis, problem-solving, critical thinking, and cognitive development.
- Improved ability to understand complexity and ambiguity

Personal Outcomes

- Greater sense of personal efficacy, personal identity, spiritual growth, and moral development
- Greater interpersonal development, particularly the ability to work well with others, and build leadership and communication skills.

Social Outcomes

- Reduced stereotypes and greater inter-cultural understanding
- Improved social responsibility and citizenship skills
- Greater involvement in community service after graduation

Career Development

- Connections with professionals and community members for learning and career opportunities
- Greater academic learning, leadership skills, and personal efficacy can lead to greater opportunity.

Relationship with the Institution

- Stronger relationships with faculty
- Greater satisfaction with college
- Improved graduation rates

BENEFITS OF COMMUNITY SERVICE PROJECT TO FACULTY MEMBERS

- Satisfaction with the quality of student learning
- New avenues for research and publication via new relationships between faculty and community
- Providing networking opportunities with engaged faculty in other disciplines or institutions
- A stronger commitment to one's research.

BENEFITS OF COMMUNITY SERVICE PROJECT TO COLLEGES AND UNIVERSITIES

- Improved institutional commitment.
- Improved student retention
- Enhanced community relations

BENEFITS OF COMMUNITY SERVICE PROJECT TO COMMUNITY

- Satisfaction with student participation
- Valuable human resources needed to achieve community goals.
- New energy, enthusiasm and perspectives applied to community work.
- Enhanced community-university relations.

SUGGESTIVE LIST OF PROGRAMMES UNDER COMMUNITY SERVICE PROJECT

The following the recommended list of projects for Engineering students. The lists are not exhaustive and open for additions, deletions, and modifications. Colleges are expected to focus on specific local issues for this kind of project. The students are expected to carry out these projects with involvement, commitment, responsibility, and accountability. The mentors of a group of students should take the responsibility of motivating, facilitating, and guiding the students. They have to interact with local leadership and people and appraise the objectives and benefits of this kind of project. The project reports shall be placed in the college website for reference. Systematic, Factual, methodical and honest reporting should be ensured.

For Engineering Students

1. Water facilities and drinking water availability
2. Health and hygiene
3. Stress levels and coping mechanisms
4. Health intervention programmes
5. Horticulture
6. Herbal plants
7. Botanical survey
8. Zoological survey
9. Marine products
10. Aqua culture
11. Inland fisheries
12. Animals and species
13. Nutrition
14. Traditional health care methods
15. Food habits
16. Air pollution
17. Water pollution
18. Plantation
19. Soil protection
20. Renewable energy
21. Plant diseases
22. Yoga awareness and practice
23. Health care awareness programmes and their impact
24. Use of chemicals on fruits and vegetables
25. Organic farming
26. Crop rotation
27. Floury culture
28. Access to safe drinking water
29. Geographical survey
30. Geological survey
31. Sericulture
32. Study of species
33. Food adulteration
34. Incidence of Diabetes and other chronic diseases
35. Human genetics
36. Blood groups and blood levels
37. Internet Usage in Villages

38. Android Phone usage by different people
39. Utilisation of free electricity to farmers and related issues
40. Gender ration in schooling lvel- observation.

Complimenting the community service project the students may be involved to take up some awareness campaigns on social issues/special groups. The suggested list of programs

Programs for School Children

1. Reading Skill Program (Reading Competition) 2.
Preparation of Study Materials for the next class.

3. Personality / Leadership Development
4. Career Guidance for X class students
5. Screening Documentary and other educational films
6. Awareness Program on Good Touch and Bad Touch (Sexual abuse)
7. Awareness Program on Socially relevant

themes. Programs for Women Empowerment

1. Government Guidelines and Policy Guidelines
2. Women's Rights
3. Domestic Violence
4. Prevention and Control of Cancer
5. Promotion of Social Entrepreneurship General Camps

1. General Medical camps
2. Eye Camps
3. Dental Camps
4. Importance of protected drinking water
5. ODF awareness camp
6. Swatch Bharath
7. AIDS awareness camp
8. Anti Plastic Awareness
9. Programs on Environment
10. Health and Hygiene
11. Hand wash programmes
12. Commemoration and Celebration of important days Programs for Youth Empowerment

1. Leadership
2. Anti-alcoholism and Drug addiction
3. Anti-tobacco
4. Awareness on Competitive Examinations
5. Personality

Development Common Programs

1. Awareness on RTI
2. Health intervention programmes
3. Yoga
4. Tree plantation
5. Programs in consonance with the Govt. Departments like –
    i. Agriculture

ii. Health
iii. Marketing and Cooperation
iv. Animal Husbandry
v. Horticulture
vi. Fisheries
vii. Sericulture
viii. Revenue and Survey
ix. Natural Disaster Management
x. Irrigation
xi. Law & Order
xii. Excise and Prohibition
xiii. Mines and Geology
xiv. Energy

Role of Students:

- Students may not have the expertise to conduct all the programmes on their own. The students then can play a facilitator role.
- For conducting special camps like Health related, they will be coordinating with the Governmental agencies.
- As and when required the College faculty themselves act as Resource Persons.
- Students can work in close association with Non-Governmental Organizations like Lions Club, Rotary Club, etc or with any NGO actively working in that habitation.
- And also, with the Governmental Departments. If the program is rolled out, the District Administration could be roped in for the successful deployment of the program.
- An in-house training and induction program could be arranged for the faculty and participating students, to expose them to the methodology of Service Learning.

Timeline for the Community Service Project Activity

Duration: 8 weeks

1. Preliminary Survey (One Week)

- A preliminary survey including the socio-economic conditions of the allotted habitation to be conducted.
- A survey form based on the type of habitation to be prepared before visiting the habitation with the help of social sciences faculty. (However, a template could be designed for different habitations, rural/urban.
- The Governmental agencies, like revenue administration, corporation and municipal authorities and village secreteriats could be aligned for the survey.

2. Community Awareness Campaigns (One Week)

- Based on the survey and the specific requirements of the habitation, different awareness campaigns and programmes to be conducted, spread over two weeks of time. The list of activities suggested could be taken into consideration.

3. Community Immersion Programme (Three Weeks)

Along with the Community Awareness Programmes, the student batch can also work with any one of the below-listed governmental agencies and work in tandem with them. This community involvement programme will involve the students in exposing themselves to experiential learning about the community and its dynamics. Programs could be in consonance with the Govt. Departments.

4. Community Exit Report (One Week)

- During the last week of the Community Service Project, a detailed report of the outcome of the 8 weeks' works to be drafted and a copy shall be submitted to the local administration. This report will be a basis for the next batch of students visiting that habitation. The same report submitted to the teacher-mentor will be evaluated by the mentor and suitable marks are awarded for onward submission to the University. Throughout the Community Service Project, a daily logbook need to be maintained by the students batch, which should be countersigned by the governmental agency representative and the teacher-mentor, who is required to periodically visit the students and guide them.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B.Tech I Semester

23ACM01 - NATURAL LANGUAGE PROCESSING AND MODELS
(Professional Core)
Common to CSE(AI&ML) and CSE(AI)

L	T	P	C
3	0	0	3

COURSE OBJECTIVES:

1. To provide a strong foundation in the principles and techniques of Natural Language Processing (NLP).
2. To introduce classical and deep learning-based approaches to NLP tasks.
3. To enable students to build and evaluate models for various NLP applications such as text classification, sentiment analysis, and machine translation.
4. To expose students to modern tools and libraries used in NLP such as NLTK, SpaCy, and HuggingFace Transformers.
5. To provide insights into the challenges of multilingual NLP and ethical concerns.

COURSE OUTCOMES:

Upon successful completion of the course, students will be able to:

1. Understand the fundamentals and challenges of natural language understanding.
2. Apply linguistic preprocessing techniques such as tokenization, stemming, POS tagging, and parsing.
3. Implement NLP algorithms for tasks like classification, translation, and information retrieval.
4. Develop deep learning models using RNNs, LSTMs, and Transformer-based architectures for NLP.
5. Use NLP tools and libraries to analyze and interpret natural language data in real-world scenarios.

UNIT I: Fundamentals of Natural Language Processing

Introduction to NLP: Definitions, Applications, Challenges, Linguistic Essentials: Syntax, Semantics, Pragmatics, Text Processing: Tokenization, Lemmatization, Stemming, Stopword Removal, Normalization, and N-gram Generation, POS Tagging and Named Entity Recognition, NLP Libraries: NLTK, SpaCy Overview.

UNIT II: Text Representation and Statistical NLP

Bag of Words and TF-IDF, Language Modeling: Unigrams, Bigrams, N-gram Models, Word Embeddings: Word2Vec, GloVe, FastText, Cosine Similarity and Distance Measures, Text Classification using Naive Bayes and SVM, Evaluation Metrics: Accuracy, Precision, Recall, F1.

UNIT III: Deep Learning for NLP

Neural Network Basics for NLP, Recurrent Neural Networks (RNNs) and Limitations, LSTM and GRU Networks, Sequence Labeling: POS Tagging, NER using Bi-LSTM, Text Classification using CNNs and RNNs, Model Evaluation and Hyperparameter Tuning.

UNIT IV: Transformers and Advanced NLP

Attention Mechanism and Self-Attention, Transformer Architecture: Encoder-Decoder Models, Pretrained Language Models: BERT, RoBERTa, GPT, Fine-tuning Transformers for Text Classification, Question Answering and Text Summarization using Transformers, Sentiment Analysis and Zero-shot Classification.

UNIT V: Applications, Ethics, and Multilingual NLP

Machine Translation: Rule-based vs Neural MT, Chatbots and Conversational AI, Information Retrieval and Question Answering, Speech-to-Text and Text-to-Speech Overview, Multilingual NLP and Low-Resource Languages, Bias, Fairness, and Ethics in NLP.

TEXTBOOKS:

1. Daniel Jurafsky and James H. Martin, Speech and Language Processing, Pearson Education.
2. Steven Bird, Ewan Klein, Edward Loper, Natural Language Processing with Python, O'Reilly Media.
3. Yoav Goldberg, Neural Network Methods in NLP, Morgan & Claypool.

REFERENCE BOOKS:

1. Jacob Eisenstein, Introduction to Natural Language Processing, MIT Press.
2. Delip Rao and Brian McMahan, Natural Language Processing with PyTorch, O'Reilly.
3. Thushan Ganegedara, Transformers for Natural Language Processing, Packt Publishing.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B.Tech I Semester

23ACA06 - OPERATING SYSTEM & SYSTEM PROGRAMMING (Professional Core) Common to CSE(AI&ML) and CSE(AI)	L	T	P	C
	3	0	0	3

Course Objectives :

1. To introduce the fundamental concepts, structures, and services of modern operating systems.
2. To explore process management concepts including scheduling, synchronization, and deadlock handling techniques.
3. To impart knowledge on memory and file system management techniques, including paging, segmentation, and disk scheduling.
4. To explain I/O system management, security principles, and introduce Unix/Linux system functionalities.
5. To introduce system software components including language processors, assemblers, macro processors, linkers, loaders, and their design.

Course Outcomes :

By the end of this course, students will be able to:

1. Understand and analyze the basic structure and functions of operating systems and different OS types. *(Knowledge, Understand – BTL 1, 2)*
2. Illustrate process management concepts including states, scheduling algorithms, synchronization methods, and deadlock handling. *(Understand, Apply – BTL 2, 3)*
3. Explain and evaluate memory management techniques and file system functionalities including paging, segmentation, allocation, and disk scheduling. *(Analyze, Evaluate – BTL 4, 5)*
4. Understand I/O hardware/software mechanisms, security principles, and Unix/Linux system features. *(Understand, Apply – BTL 2, 3)*
5. Describe the components and functioning of system software including language processors, macro processors, linkers, and loaders. *(Understand, Apply – BTL 2, 3)*

UNIT - I: Fundamentals of Operating Systems and Process Management

Introduction to Operating Systems:Definition and Basics, Generations and Types of Operating Systems, OS Structure: Layered, Monolithic, Microkernel, OS Services, System Calls, System Boot, System Programs, Virtual Machines, Process Management**:** Process Concepts, Process States, Process Control Block, Context Switching, Threads and Multithreading, Process Scheduling: Scheduling Criteria and Scheduling Algorithms, Multiprocessor Scheduling: Types and Performance Evaluation, Process Synchronization and Deadlocks:Race Conditions, Critical Section, Mutual Exclusion, Peterson's Solution, Semaphores, Monitors Classic IP, C Problems: Reader-Writers, Dining Philosophers, Deadlocks: Definition, Characteristics, Prevention, Avoidance, Detection and Recovery

UNIT - II: Memory, File, and Storage Management

Memory Management:Logical vs. Physical Address Mapping, Contiguous Memory Allocation, Internal and External Fragmentation, Compaction, Paging and Page Tables, Segmentation, Virtual Memory: Demand Paging, Page Faults, Page Replacement Algorithms, Thrashing and Working Set Model, File System Management:File Concepts, Access Methods, File Types and Operations, Directory Structure, File System Structure, Allocation Methods, Free-Space Management, Directory Implementation. Storage Management:Mass Storage: Disk Structure, RAID Levels, Disk Scheduling Algorithms, Swap Space Management, Stable Storage, Tertiary Storage Structure.

UNIT - III: I/O Systems, Security, and Unix/Linux Overview

I/O System Management:I/O Hardware: Devices, Device Controllers, Direct Memory Access, I/O Software: Interrupt Handlers, Device Drivers, Device-Independent I/O Software, SystemProtection and Security:Security Environment, Security Design Principles, User Authentication, Protection Mechanisms, Protection Domain, Access Control List, Unix/Linux Overview & Case Studies.

UNIT IV: System Software and Language Processing

Overview of System Software:Software and Software Hierarchy, Systems Programming and Machine Structure, Interfaces, Address Space, and Computer Languages, System Software Development and Recent Trends, Language Processors:Programming Languages and Language Processing, Symbol Tables and Data Structures for Language Processing, Search and Allocation Data Structures, Assemblers and Macro Processors:Elements of Assembly Language Programming, Design and Types of Assemblers, Macro Definitions, Expansion, Nested Macros, and Advanced Macro Features, Design of Macro Assemblers and Macro Processors, Linkers and Loaders:Concept of Linking and Relocation, Linking in MS-DOS, Dynamic Linking, Loading Schemes: Sequential, Direct, Absolute, Relocating, and Linking Loaders, Comparison of Linkers and Loaders

UNIT V: System Programming

Scanning and Parsing:Programming Language Grammars and Classification, Ambiguity in Grammatic Specification, Scanning, Parsing, Compilers and Interpreters:Compilation Process, Semantic Gap, Binding, and Scope Rules, Memory Allocation, Compilation of Expressions & Control Structures, Code Optimization, Overview of Interpreters and Debuggers, Operating System Command & Shell Basics:C Development Tools, Machine-Level Representation of Data and Programs, System-Level Programming and Concurrency:File I/O, Process Creation & Control (fork, exec), Pipes, Signals, and Basic Threading.

Textbooks:

- *Operating System Concepts* (9th or 10th Edition) by Abraham Silberschatz, Peter Baer Galvin, and Greg Gagne in publisher: Wiley
- *Operating Systems: A Concept-Based Approach* (3rd Edition) by D. M. Dhamdhere publisher: McGraw Hill

Reference Books:

- Real-Time Systems: Theory and Practice by Rajib Mall, Publisher: Pearson
- *System Software: An Introduction to Systems Programming* (3rd Edition) by Leland L. Beck & D. Manjula, Publisher: Pearson

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B.Tech I Semester

23ACM02 - COMPUTER VISION AND IMAGE PROCESSING
(Professional Core)
Common to CSE(AI&ML) and CSE(AI)

L	T	P	C
3	0	0	3

Course Objectives:

- Introduce fundamental concepts of image processing and computer vision.
- Develop proficiency in applying algorithms for image analysis and interpretation.
- Explore techniques for feature extraction, object recognition, and scene understanding.
- Understand the integration of machine learning methods in computer vision applications.

Course Outcomes:

Upon successful completion of the course, students will be able to:

1. Understand image formation, representation, and apply basic image processing and frequency domain techniques for image enhancement and restoration.
2. Apply edge detection, segmentation, morphological, and texture analysis techniques for extracting features from images.
3. Analyze 3D vision and motion using techniques like stereo vision, optical flow, and camera calibration for scene understanding and depth estimation.
4. Evaluate object recognition approaches and machine learning models including traditional and deep learning techniques used in computer vision.
5. Implement advanced computer vision applications such as image compression, face recognition, and medical image analysis using case studies.

UNIT I: Introduction to Computer Vision and Image Processing

Overview of Computer Vision and Image Processing: Definitions and scope, Historical development and applications, Image Formation and Representation: Image acquisition methods, Sampling and quantization, Color spaces and models , Fundamentals of Image Processing: Point operations (brightness and contrast adjustments), Histogram processing, Spatial filtering techniques Fourier Transform and Frequency Domain Processing: Discrete Fourier Transform (DFT), Filtering in the frequency domain, Image restoration concept.

UNIT II: Image Analysis Techniques

Edge Detection and Feature Extraction: Gradient operators (Sobel, Prewitt), Canny edge detector, Corner and interest point detection, Image Segmentation:Thresholding methods, Region-based segmentation, Clustering techniques (K-means, Mean-Shift), Morphological Image Processing: Erosion and dilation, Opening and closing operations, Applications in shape analysis, Texture Analysis, Statistical methods (co-occurrence matrices), Transform-based methods (Gabor filters), Applications in pattern recognition.

UNIT III: 3D Vision and Motion Analysis

Stereo Vision: Epipolar geometry, Disparity mapping, Depth estimation techniques, Structure from Motion (SfM): Feature tracking across frames, 3D reconstruction from motion, Applications in scene understanding, Optical Flow and Motion Analysis: Lucas-Kanade method, Horn-Schunck method, Motion segmentation, Camera Calibration and 3D Reconstruction: Intrinsic and extrinsic parameters, Calibration techniques, 3D point cloud generation

UNIT IV: Object Recognition and Machine Learning in Vision

Feature Descriptors and Matching: Scale-Invariant Feature Transform (SIFT), Speeded-Up Robust Features (SURF), Feature matching algorithms, Object Detection and Recognition: Template matching, Deformable part models, Convolutional Neural Networks (CNNs), Introduction to Machine Learning for Vision: Supervised and unsupervised learning, Support Vector Machines (SVMs), Decision trees and random forests, Deep Learning Architectures: Autoencoders, Recurrent Neural Networks (RNNs), Generative Adversarial Networks (GANs)

UNIT V: Applications and Advanced Topics

Image Compression: Lossy and lossless compression techniques, Standards (e.g., JPEG, PNG), Morphological Image Processing: Dilation, erosion, opening, and closing operations., Applications in shape analysis, Case Studies: Face recognition systems., Automated visual inspection, Medical image analysis.

Reference Books

1. Forsyth, D. A., & Ponce, J. (2002). *Computer Vision: A Modern Approach*. Prentice Hall.
2. Shapiro, L. G., & Stockman, G. C. (2001). *Computer Vision*. Prentice Hall.

Textbooks:

1. Gonzalez, R. C., & Woods, R. E. (2008). *Digital Image Processing* (3rd ed.). Pearson Prentice Hall.Stony Brook University
2. Szeliski, R. (2010). *Computer Vision: Algorithms and Applications*. Springer.

Online Learning Resources:

1. Coursera: *Introduction to Computer Vision and Image Processing*. LinkCoursera
2. Stanford University: *CS231n: Deep Learning for Computer Vision*. Linkcs231n.stanford.edu
3. MIT OpenCourseWare: *Introduction to Computer Vision*. Link

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B.Tech-I Semester(Common to all branches)

23ACS27 INTRODUCTION TO QUANTUM TECHNOLOGIES AND APPLICATIONS

L	T	P	C
3	0	0	3

Course Outcomes (CO):

CO1: Explain core quantum principles in a non-mathematical manner.

CO2: Compare classical and quantum information systems.

CO3: Identify theoretical issues in building quantum computers.

CO4: Discuss quantum communication and computing concepts.

CO5: Recognize applications, industry trends, and career paths in quantum technology.

Unit I: Introduction to Quantum Theory and Technologies

The transition from classical to quantum physics, Fundamental principles explained conceptually: Superposition, Entanglement, Uncertainty Principle, Wave-particle duality, Classical vs Quantum mechanics – theoretical comparison, Quantum states and measurement: nature of observation, Overview of quantum systems: electrons, photons, atoms, The concept of quantization: discrete energy levels, Why quantum? Strategic, scientific, and technological significance, A snapshot of quantum technologies: Computing, Communication, and Sensing, National and global quantum missions: India's Quantum Mission, EU, USA, China

Unit II: Theoretical Structure of Quantum Information Systems

What is a qubit? Conceptual understanding using spin and polarization, Comparison: classical bits vs quantum bits, Quantum systems: trapped ions, superconducting circuits, photons (non-engineering view),Quantum coherence and decoherence – intuitive explanation, Theoretical concepts: Hilbert spaces, quantum states, operators – only interpreted in abstract,The role of entanglement and non- locality in systems, Quantum information vs classical information: principles and differences,Philosophical implications: randomness, determinism, and observer role

Unit III: Building a Quantum Computer – Theoretical Challenges and Requirements

What is required to build a quantum computer (conceptual overview)?,Fragility of quantum systems: decoherence, noise, and control, Conditions for a functional quantum system: Isolation, Error management, Scalability, Stability, Theoretical barriers:

Why maintaining entanglement is difficult, Error correction as a theoretical necessity, Quantum hardware platforms (brief conceptual comparison), Superconducting circuits, Trapped ions, Photonics, Visionvs reality: what's working and what remains elusive, The role of quantum software in managing theoretical complexities

Unit IV: Quantum Communication and Computing – Theoretical Perspective

Quantum vs Classical Information, Basics of Quantum Communication, Quantum Key Distribution (QKD),Role of Entanglement in Communication,The Idea of the Quantum Internet – Secure Global

Networking,Introduction to Quantum Computing,Quantum Parallelism (Many States at Once),Classical vs Quantum Gates, Challenges: Decoherence and Error Correction,Real-World Importance and Future Potential

Unit V: Applications, Use Cases, and the Quantum Future

Real-world application domains: Healthcare (drug discovery),Material science, Logistics and optimization, Quantum sensing and precision timing, Industrial case studies: IBM, Google, Microsoft, PsiQuantum,Ethical, societal, and policy considerations, Challenges to adoption: cost, skills, standardization,Emerging careers in quantum: roles, skillsets, and preparation pathways,Educational and research landscape – India's opportunity in the global quantum race

Textbooks:

1. Michael A. Nielsen, Isaac L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press, 10th Anniversary Edition, 2010.
2. Eleanor Rieffel and Wolfgang Polak, *Quantum Computing: A Gentle Introduction,* MIT Press, 2011.
3. Chris Bernhardt, *Quantum Computing for Everyone*, MIT Press, 2019.

Reference Books:

1. David McMahon, *Quantum Computing Explained*, Wiley, 2008.
2. Phillip Kaye, Raymond Laflamme, Michele Mosca, *An Introduction to Quantum Computing,* Oxford University Press, 2007.
3. Scott Aaronson, *Quantum Computing Since Democritus,* Cambridge University Press, 2013.
4. **Alastair I.M. Rae**, *Quantum Physics: A Beginner's Guide*, Oneworld Publications, Revised Edition, 2005.
5. **Eleanor G. Rieffel, Wolfgang H. Polak**, *Quantum Computing: A Gentle Introduction*, MIT Press, 2011.
6. **Leonard Susskind, Art Friedman**, *Quantum Mechanics: The Theoretical Minimum*, Basic Books, 2014.
7. **Bruce Rosenblum, Fred Kuttner**, *Quantum Enigma: Physics Encounters Consciousness,* Oxford University Press, 2nd Edition, 2011.
8. **GiulianoBenenti, GiulioCasati, GiulianoStrini**, *Principles of Quantum Computation and Information, Volume I: Basic Concepts,* World Scientific Publishing, 2004.
9. **K.B. Whaley et al.**, *Quantum Technologies and Industrial Applications: European Roadmap and Strategy Document*, Quantum Flagship, European Commission, 2020.
10. **Department of Science & Technology (DST), Government of India**, *National Mission on Quantum Technologies & Applications – Official Reports and Whitepapers,* MeitY/DST Publications, 2020 onward.

Online Learning Resources:

- IBM Quantum Experience and Qiskit Tutorials
- Coursera – Quantum Mechanics and Quantum Computation by UC Berkeley
- edX – The Quantum Internet and Quantum Computers
- YouTube – Quantum Computing for the Determined by Michael Nielsen
- Qiskit Textbook – IBM Quantum

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B.Tech I Semester

	L	T	P	C
23ACD10 - DATA VISUALIZATION (Professional Elective-I)	3	0	0	3

Course Objectives:

- To understand the principles, techniques, and tools of data visualization.
- To develop the ability to transform data into visual insights using different types of charts and plots.
- To introduce the cognitive and perceptual foundations of effective data visualization.
- To apply tools and programming environments (like Python, Tableau, or Power BI) for creating interactive and dynamic visualizations.
- To analyze real-world datasets and effectively communicate data-driven findings visually.

Course Outcomes:

After completion of the course, students will be able to:

- CO1: Interpret different types of data and recognize the appropriate visualization methods. (Understand, Analyze)
- CO2: Design effective and interactive data visualizations using various tools. (Apply, Create)
- CO3: Apply visual encoding and perceptual principles in presenting complex data. (Apply, Evaluate)
- CO4: Analyze and visualize real-world data sets using Python libraries and dashboards. (Analyze, Evaluate)
- CO5: Create visual stories and dashboards for effective communication of insights. (Create, Apply)

UNIT I: Introduction to Data Visualization & Perception

Introduction to Data Visualization, Importance and Scope of Data Visualization, Data Types and Sources, Visual Perception: Pre-attentive Processing, Gestalt Principles, Data-Ink Ratio, Data Density, Lie Factor, Visualization Process and Design Principles, Tools Overview: Tableau, Power BI, Python Libraries

UNIT II: Visualization Techniques for Categorical & Quantitative Data

Charts for Categorical Data: Bar Charts, Pie Charts, Column Charts, Charts for Quantitative Data: Histograms, Line Charts, Boxplots, Scatter Plots, Bubble Charts, Heatmaps, Choosing the Right Chart Type, Best Practices in Labeling, Coloring, and Scaling.

UNIT III: Multidimensional, Temporal and Hierarchical Data Visualization

Visualizing Multivariate Data: Parallel Coordinates, Radar Charts, Time-Series Visualization: Time Plots, Animation over Time, Geographic Data Visualization: Maps, Choropleths, Hierarchical Data: Treemaps, Sunburst Charts, Network and Graph Visualization.

UNIT IV: Data Visualization Using Python and Dashboards

Introduction to Matplotlib, Seaborn, and Plotly, Creating Static and Interactive Charts, Pandas Visualization Capabilities, Dashboards with Dash, Streamlit, Power BI, Case Studies: Real-world Dataset Visualization.

UNIT V: Storytelling with Data and Ethical Visualization

Storytelling and Narrative Techniques in Visualization, Dashboards and Reporting, Misleading Visualizations and Bias, Ethical Principles in Data Visualization, Final Project: Create a Storytelling Dashboard with Real Data.

Textbooks:

1. Tamara Munzner, Visualization Analysis and Design, CRC Press, 2014.
2. Nathan Yau, Data Points: Visualization That Means Something, Wiley, 2013.

Reference Books:

1. Alberto Cairo, The Truthful Art: Data, Charts, and Maps for Communication, New Riders, 2016.
2. Cole Nussbaumer Knaflic, Storytelling with Data: A Data Visualization Guide for Business Professionals, Wiley, 2015.
3. Claus O. Wilke, Fundamentals of Data Visualization, O'Reilly, 2019.
4. Rohan Chopra, Hands-On Data Visualization with Bokeh, Packt Publishing, 2019.

Online Learning Resources:

1. NPTEL: Data Visualization - IIT Madras
2. Coursera: Data Visualization with Python by IBM

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(Autonomous)

III B.Tech I Semester

Common to B.Tech-AI, CSE(AI&ML) & CSE(AI)

(Professional Elective-I)

23AIT07 - SOFT COMPUTING

L	T	P	C
3	0	0	3

Course Objectives:

- Understand the concepts of soft computing techniques and how they differ from traditional AI techniques.
- Introduce the fundamentals of fuzzy logic and fuzzy systems.
- Familiarize with artificial neural networks and their architectures.
- Learn genetic algorithms and their role in optimization.
- Explore hybrid systems integrating fuzzy logic, neural networks, and genetic algorithms.

Course Outcomes:

After completion of the course, students will be able to:

- Understand the components and applications of soft computing.
- Apply fuzzy logic concepts to real-world problems.
- Build and train various neural network models.
- Implement genetic algorithms for problem-solving and optimization.
- Design hybrid systems using soft computing techniques.

UNIT I: Introduction to Soft Computing and Fuzzy Logic

Introduction to Soft Computing: Definition, Components, Differences with Hard Computing, Applications of Soft Computing, Fuzzy Logic: Crisp Sets vs Fuzzy Sets, Membership Functions, Fuzzy Set Operations, Fuzzy Rules and Fuzzy Reasoning, Fuzzy Inference Systems: Mamdani and Sugeno Models, Defuzzification Techniques.

UNIT II: Artificial Neural Networks – I

Introduction to Neural Networks: Biological Neurons vs Artificial Neurons, Architecture of Neural Networks: Feedforward, Feedback, Learning Rules: Hebbian, Delta, Perceptron Learning Rule, Single Layer Perceptron and its Limitations, Multi-Layer Perceptron: Backpropagation Algorithm, Applications of Neural Networks

UNIT III: Artificial Neural Networks – II

Hopfield Networks and Associative Memories, Radial Basis Function Networks, Self-Organizing Maps (SOM), Recurrent Neural Networks (RNNs) – Basic Concepts, Convolutional Neural Networks (CNNs) – Overview and Applications, Practical Use Cases in Image and Pattern Recognition,

UNIT IV: Genetic Algorithms and Optimization

Introduction to Genetic Algorithms, GA Operators: Selection, Crossover, Mutation, Fitness Function and Evaluation, Schema Theorem, Elitism, Applications in Function Optimization, Scheduling, and Robotics, Introduction to Particle Swarm Optimization (PSO).

UNIT V: Hybrid Systems and Advanced Topics

Hybrid Systems: Neuro-Fuzzy Systems, Fuzzy-GA, GA-ANN, ANFIS: Architecture and Learning, Case Studies on Hybrid Systems, Introduction to Deep Learning in Soft Computing, Real-World Applications: Forecasting, Control Systems, Medical Diagnosis, Image Processing.

Textbooks:

1. S. N. Sivanandam, S. N. Deepa, "Principles of Soft Computing", Wiley India, 3rd Edition
2. Timothy J. Ross, "Fuzzy Logic with Engineering Applications", Wiley, 4th Edition
3. S. Rajasekaran and G. A. Vijayalakshmi Pai, "Neural Networks, Fuzzy Logic and Genetic Algorithm: Synthesis and Applications", PHI

Reference Books:

1. Laurene Fausett, "Fundamentals of Neural Networks: Architectures, Algorithms and Applications", Pearson
2. David E. Goldberg, "Genetic Algorithms in Search, Optimization and Machine Learning", Pearson
3. Simon Haykin, "Neural Networks and Learning Machines", Pearson, 3rd Edition
4. Bart Kosko, "Neural Networks and Fuzzy Systems", Prentice Hall

Online Learning Resources:

1. NPTEL – Soft Computing by Prof. S. Sengupta (IIT Kharagpur)
2. Coursera – Neural Networks and Deep Learning (Andrew Ng)

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B.Tech I Semester

	L	T	P	C
23ACD11 -DATA ANALYSIS WITH PYTHON **(Professional Elective-I)** **Common to CSE(AI&ML) and CSE(AI)**	3	0	0	3

Course Objectives:

- To introduce the principles and practices of Exploratory Data Analysis (EDA) using Python.
- To teach techniques for data cleaning, preprocessing, transformation, and visualization.
- To apply statistical techniques and visual methods to discover patterns and relationships.
- To gain experience using popular Python libraries such as NumPy, Pandas, Matplotlib, and Seaborn.
- To prepare datasets for further machine learning and predictive modeling.

Course Outcomes: After completion of the course, students will be able to:

- Understand and apply key concepts of EDA and data preprocessing. (Cognitive Level: Understand, Apply)
- Perform exploratory analysis using Python libraries and interpret results. (Cognitive Level: Apply, Analyze)
- Handle missing data, outliers, and categorical features effectively. (Cognitive Level: Apply)
- Create meaningful visualizations to support data-driven insights. (Cognitive Level: Analyze, Evaluate)
- Use EDA as a foundation for data science workflows. (Cognitive Level: Apply, Create)

UNIT I – Introduction to EDA and Python Environment
Introduction to Data Science and EDA, Importance of EDA in Data Science Life Cycle, Setting up Python Environment: Jupyter, Anaconda, VS Code, Introduction to NumPy and Pandas: Arrays, Series, DataFrames, Data loading, viewing, basic operations (info, describe, shape)

UNIT II – Data Wrangling and Preprocessing
Handling Missing Data (mean, median, drop, interpolation), Dealing with Duplicates, Outliers, and Anomalies, Encoding Categorical Variables (Label, One-hot), Data Transformation: Scaling, Normalization, Binning, Data Types Conversion and Data Type Casting.

UNIT III – Univariate and Bivariate Analysis
Measures of Central Tendency and Dispersion, Distribution Plots: Histograms, Boxplots, KDE, Bar Charts, Count Plots, Pie Charts, Bivariate Analysis: Scatter Plots, Pair Plots, Heatmaps, Correlation and Covariance Analysis

UNIT IV – Data Visualization Techniques
Visualization with Matplotlib and Seaborn, Customizing Plots: Titles, Legends, Labels, Themes, Advanced Visuals: Violin Plots, Strip Plots, Swarm Plots, Multivariate Visualization and Subplots, Plotly and Interactive Visualizations (basic overview)

UNIT V – EDA Case Studies and Real-Time Datasets
Step-by-step EDA on Sample Datasets (Titanic, Iris, Sales, etc.), Outlier Detection Techniques, Feature Engineering Techniques in EDA, EDA Report Generation using Python Notebooks, Preparing Data for Machine Learning Models.

Textbooks:

1. Jake VanderPlas, Python Data Science Handbook: Essential Tools for Working with Data, O'Reilly, 2016.
2. Wes McKinney, Python for Data Analysis, 2nd Edition, O'Reilly, 2018.

Reference Books:

1. Joel Grus, Data Science from Scratch, O'Reilly, 2019.
2. Aurelien Geron, Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow, 2nd Edition, O'Reilly, 2019.
3. Allen B. Downey, Think Stats: Probability and Statistics for Programmers, O'Reilly, 2014.

Online Learning Resources:

1. NPTEL Course – Data Science for Engineers
2. Coursera – Applied Data Science with Python Specialization (University of Michigan)

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B.Tech I Semester

	L	T	P	C
23ACA07 - COMPUTATIONAL INTELLIGENCE **(Professional Elective-I)** **Common to CSE(AI&ML) and CSE(AI)**	3	0	0	3

Course Objectives:

- Understand the concepts and foundations of computational intelligence.
- Study neural networks, fuzzy logic systems, and evolutionary algorithms.
- Explore hybrid systems and their applications.
- Apply computational intelligence techniques to real-world problem-solving.
- Analyze the effectiveness of various computational intelligence approaches.

Course Outcomes: After completion of the course, students will be able to:

- Describe and differentiate neural networks, fuzzy logic, and evolutionary computation. (Understand)
- Apply neural and fuzzy systems for real-time decision-making. (Apply)
- Analyze complex problems using soft computing tools. (Analyze)
- Develop hybrid intelligent systems. (Create)
- Evaluate and compare the performance of CI-based systems. (Evaluate)

UNIT I: Introduction to Computational Intelligence and Artificial Neural Networks

Definition and Scope of Computational Intelligence (CI), Components of CI: Neural Networks, Fuzzy Logic, Evolutionary Computation, Biological Neuron vs. Artificial Neuron, McCulloch-Pitts Model, Perceptron, Adaline and Madaline, Multilayer Feedforward Networks, Backpropagation Algorithm, Applications of ANN in Pattern Recognition and Classification.

UNIT II: Fuzzy Logic and Fuzzy Systems

Introduction to Fuzzy Logic and Fuzzy Sets, Membership Functions, Fuzzy Set Operations, Fuzzy Rules and Inference Systems, Fuzzification and Defuzzification, Fuzzy Control Systems, Fuzzy Reasoning and Approximate Reasoning

UNIT III: Evolutionary Computation Techniques

Basics of Evolutionary Algorithms (EA), Genetic Algorithms (GA): Operators, Encoding, Fitness Function, Selection, Crossover and Mutation, Convergence Criteria, Genetic Programming (GP), Differential Evolution (DE), Applications of GA and GP

UNIT IV: Swarm Intelligence and Hybrid Systems

Swarm Intelligence: Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), Behavior of Swarms and Collective Intelligence, Comparison of Evolutionary Algorithms and Swarm Techniques, Hybrid Systems: Neuro-Fuzzy, Fuzzy-GA, ANN-GA Systems, Case Studies in Hybrid Systems

UNIT V: Applications of Computational Intelligence

CI in Image and Signal Processing, CI for Optimization Problems and Robotics, CI in Biomedical Engineering and Finance, Intelligent Agents and Decision-Making Systems, Real-time Applications and Emerging Trends in CI.

Textbooks:

1. S. Rajasekaran and G. A. Vijayalakshmi Pai, Neural Networks, Fuzzy Logic, and Genetic Algorithms: Synthesis and Applications, PHI Learning.
2. Timothy J. Ross, Fuzzy Logic with Engineering Applications, Wiley India.

Reference Books:

1. S.N. Sivanandam, S. N. Deepa, Principles of Soft Computing, Wiley India.
2. Simon Haykin, Neural Networks and Learning Machines, Pearson.
3. James Kennedy and Russell C. Eberhart, Swarm Intelligence, Morgan Kaufmann.
4. Andries P. Engelbrecht, Computational Intelligence: An Introduction, Wiley.

Online Learning Resources:

1. NPTEL - Computational Intelligence
2. Coursera – Computational Intelligence
3. YouTube: IIT Lectures on Soft Computing and CI

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B.Tech I Semester

23ACE30- GREEN BUILDINGS (OPEN ELECTIVE - I)	L	T	P	C
	3	0	0	3

Course Outcomes (COs)
Upon successful completion of the course, students will be able to:

1. **Understand** the importance of green buildings, their necessity, and sustainable features.
2. **Analyze** various green building practices, rating systems, and their impact on environmental sustainability.
3. **Apply** principles of green building design to enhance energy efficiency and incorporate renewable energy sources.
4. **Evaluate** HVAC systems, energy-efficient air conditioning techniques, and their role in sustainable building design.
5. **Assess** material conservation techniques, waste reduction strategies, and indoor air quality management in green buildings.

UNIT – I
Introduction to Green Building– Necessity of Green Buildings, Benefits of Green Buildings, Green Building Materials and Equipment in India, Key Requisites for Constructing A Green Building, Important Sustainable Features for Green Buildings.

UNIT – II
Green Building Concepts and Practices– Indian Green Building Council, Green Building Movement in India, Benefits Experienced in Green Buildings, Launch of Green Building Rating Systems, Residential Sector, Market Transformation; Green Building Opportunities and Benefits: Opportunities of Green Buildings, Green Building Features, Material and Resources, Water Efficiency, Optimum Energy Efficiency, Typical Energy-Saving Approaches in Buildings, LEED India Rating System, and Energy Efficiency.

UNIT – III
Green Building Design– Introduction, Reduction in Energy Demand, Onsite Sources and Sinks, Maximizing System Efficiency, Steps to Reduce Energy Demand and Use Onsite Sources and Sinks, Use of Renewable Energy Sources, Eco-Friendly Captive Power Generation for Factories, Building Requirements.

UNIT – IV
Air Conditioning– Introduction, CII Godrej Green Business Centre, Design Philosophy, Design Interventions, Energy Modeling, HVAC System Design, Chiller Selection, Pump Selection, Selection of Cooling towers, Selection of Air Handling Units, Pre-Cooling of Fresh Air, Interior Lighting Systems, Key Features of The Building, Eco-Friendly Captive Power Generation for Factories, Building Requirements.

UNIT – V
Material Conservation– Handling of Non-Process Waste, Waste Reduction During Construction, Materials With Recycled Content, Local Materials, Material Reuse, Certified Wood, Rapidly Renewable Building Materials and Furniture. Indoor Environment Quality and Occupational Health– Air Conditioning, Indoor
Air Quality, Sick Building Syndrome, tobacco Smoke.

TEXT BOOKS:

1. Handbook on Green Practices published by Indian Society of Heating Refrigerating and Air conditioning Engineers, 2009.
2. Green Building Hand Book by tom woolley and Sam kimings, 2009.

REFRENCE BOOKS:

1. Complete Guide to Green Buildings by Trish riley
2. Standard for the design for High Performance Green Buildings by Kent Peterson, 2009
3. Energy Conservation Building Code –ECBC-2020, published by BEE

Online Learning Resources:

https://archive.nptel.ac.in/courses/105/102/105102195/

CO - PO Articulation Matrix

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO -1	3	-	-	-	-	2	3	-	-	-	-	-	3	3
CO -2	-	3	-	-	2	-	3	-	-	-	-	2	3	3
CO -3	-	-	3	3	3	-	3	-	-	-	-	-	3	3
CO -4	-	-	3	3	3	-	3	-	-	-	-	-	3	3
CO -5	-	-	-	-	-	3	3	3	2	-	-	-	-	3

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B.Tech – I Semester

23ACE31 CONSTRUCTION TECHNOLOGY AND MANAGEMENT (OPEN ELECTIVE – I)	L	T	P	C
	3	0	0	3

Course Objectives:

The objectives of this course are to make the student:

1. To understand project management fundamentals, organizational structures, and leadership principles in construction.
2. To analyze manpower planning, equipment management, and cost estimation in civil engineering projects.
3. To apply planning, scheduling, and project management techniques such as CPM and PERT.
4. To evaluate various contract types, contract formation, and legal aspectsin construction management.
5. To assess safety management practices, accident prevention strategies, and quality management systemsin construction.

Course Outcomes (COs):

Upon successful completion of the course, students will be able to:

1. Understand (Cos)project management fundamentals, organizational structures, and leadership principles in construction.
2. Analyze manpower planning, equipment management, and cost estimationin civil engineering projects.
3. Apply planning, scheduling, and project management techniques such as CPM and PERT.
4. Evaluate various contract types, contract formation, and legal aspectsin construction management.
5. Assess safety management practices, accident prevention strategies, and quality management systems in construction.

UNIT – I

Introduction: Project forms, Management Objectives and Functions; Organizational Chart of A Construction Company; Manager's Duties and Responsibilities; Public Relations; Leadership and Team - Work; Ethics, Morale, Delegation and Accountability.

UNIT – II

Man and Machine: Man-Power Planning, Training, Recruitment, Motivation, Welfare Measures and Safety Laws; Machinery for Civil Engineering., Earth Movers and Hauling Costs, Factors Affecting Purchase, Rent, and Lease of Equipment, and Cost Benefit Estimation.

UNIT – III

Planning, Scheduling andProject Management: Planning Stages, Construction Schedules and Project Specification, Monitoring and Evaluation; Bar-Chart, CPM, PERT, Network- formulation and Time Computation.

UNIT – IV

Contracts: Types of Contracts, formation of Contract – Contract Conditions – Contract forLabour, Material, Design, Construction – Drafting of Contract Documents Based On IBRD/ MORTH Standard Bidding Documents – Construction Contracts – Contract Problems – Arbitration and Legal Requirements Computer
Applications in Construction Management: Software for Project Planning, Scheduling and Control.

UNIT – V

Safety Management – Implementation and Application of QMS in Safety Programs, ISO 9000 Series, Accident Theories, Cost of Accidents, Problem Areas in Construction Safety, Fall Protection, Incentives, Zero Accident Concepts, Planning for Safety, Occupational Health and Ergonomics.

TEXT BOOKS:

1. Construction Project Management, SK. Sears, GA. Sears, RH. Clough, John Wiley and Sons, 6th Edition, 2016.
2. Construction Project Scheduling and Control by Saleh Mubarak, 4th Edition, 2019
3. Pandey, I.M (2021) Financial Management 12th edition. Pearson India Education Services Pvt. Ltd.

REFRENCE BOOKS:

1. Brien, J.O. andPlotnick, F.L., CPMin Construction Management, Mcgraw Hill, 2010.
2. Punmia, B.C., andKhandelwal, K.K., Project Planning and control with PERT and CPM, Laxmi Publications, 2002.
3. Construction Methods and Management: Pearson New International Edition 8 th Edition Stephens Nunnally.
4. Rhoden, M and Cato B, Construction Management andOrganisational Behaviour, Wiley-Blackwell, 2016.

Online Learning Resources:

https://archive.nptel.ac.in/courses/105/104/105104161/
https://archive.nptel.ac.in/courses/105/103/105103093/

CO – PO Articulation Matrix

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO -1	3	-	-	-	-	2	-	2	2	-	-	-	3	3
CO -2	-	3	-	-	2	-	-	-	-	-	-	2	3	3
CO -3	-	-	3	3	3	-	-	-	-	2	-	-	3	3
CO -4	-	-	3	3	3	-	-	2	-	-	-	-	3	3
CO -5	-	-	-	-	-	3	3	3	2	-	-	-	-	3

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

IIIB.Tech I Semester

23AEE23 - ELECTRICAL SAFETY PRACTICES AND STANDARDS
(Open Elective-I)
(Common to CE, EEE, ECE, EBM, CSE, CSC, CSM, CAI, CSD, CSBS, AI, AI& DS, IT, IOT)

L	T	P	C
3	0	0	3

Course Outcomes:
CO1: Understanding the Fundamentals of Electrical Safety -L2
CO2: Identifying and Applying Safety Components -L3
CO3: Analyzing Grounding Practices and Electrical Bonding
CO4: Applying Safety Practices in Electrical Installations and Environments- L4
CO5: Evaluating Electrical Safety Standards and Regulatory Compliance -L5

UNIT I Introduction To Electrical Safety:
Fundamentals of Electrical safety-Electric Shock- physiological effects of electric current - Safety requirements –Hazards of electricity- Arc - Blast- Causes for electrical failure.

UNIT II Safety Components:
Introduction to conductors and insulators- voltage classification -safety against over voltages- safety against static electricity-Electrical safety equipment's - Fire extinguishers for electrical safety.

UNIT III Grounding:
General requirements for grounding and bonding- Definitions- System grounding-Equipment grounding - The Earth - Earthing practices- Determining safe approach distance-Determining arc hazard category.

UNIT IV Safety Practices:
General first aid- Safety in handling hand held electrical appliances tools- Electrical safety in train stations-swimming pools, external lighting installations, medical locations-Case studies.

UNIT V Standards For Electrical Safety:
Electricity Acts- Rules & regulations- Electrical standards-NFPA 70 E-OSHA standards-IEEE standards-National Electrical Code 2005 – National Electric Safety code NESC-Statutory requirements from electrical inspectorate

TEXT BOOKS:

1. Massimo A.G.Mitolo, "Electrical Safety of Low-Voltage Systems", McGraw Hill, USA, 2009.
2. Mohamed El-Sharkawi, "Electric Safety - Practice and Standards", CRC Press, USA, 2014

REFERENCES*:*

1. Kenneth G.Mastrullo, Ray A. Jones, "The Electrical Safety Program Book", Jones and Bartlett Publishers, London, 2nd Edition, 2011.
2. Palmer Hickman, "Electrical Safety-Related Work Practices", Jones & Bartlett Publishers, London, 2009.
3. Fordham Cooper, W., "Electrical Safety Engineering", Butterworth and Company, London, 1986.
4. John Cadick, Mary Capelli-Schellpfeffer, Dennis K. Neitzel, "Electrical Safety Hand book, McGraw-Hill, New York, USA, 4th edition, 2012.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B. Tech - I Semester

(Common to CE, EEE, ECE, EBM, CSE, CSC, CSM, CAI, CSD, CSBS, AI, AI& DS, IT, IOT)

L	T	P	C
3	0	0	3

23AME23: SUSTAINBLE ENERGY TECHNOLOGIES

Open Elective 1

Course Outcomes:

CO1: Illustrate the importance of solar radiation and solar PV modules.

CO2: Discuss the storage methods in PV systems

CO3: Explain the solar energy storage for different applications

CO4: Understand the principles of wind energy, and bio-mass energy.

CO5: Attain knowledge in geothermal energy, ocean energy and fuel cells.

UNIT – I

SOLAR RADIATION: Role and potential of new and renewable sources, the solar energy option, Environmental impact of solar power, structure of the sun, the solar constant, sun-earth relationships, coordinate systems and coordinates of the sun, extraterrestrial and terrestrial solar radiation, solar radiation on titled surface, instruments for measuring solar radiation and sun shine, solar radiation data, numerical problems.

SOLAR PV MODULES AND PV SYSTEMS:

PV Module Circuit Design, Module Structure, Packing Density, Interconnections, Mismatch and Temperature Effects, Electrical and Mechanical Insulation, Lifetime of PV Modules, Degradation and Failure, PV Module Parameters, Efficiency of PV Module, Solar PV Systems-Design of Off Grid Solar Power Plant. Installation and Maintenance.

UNIT – II

STORAGE IN PV SYSTEMS:

Battery Operation, Types of Batteries, Battery Parameters, Application and Selection of Batteries for Solar PV System, Battery Maintenance and Measurements, Battery Installation for PV System.

UNIT – III

SOLAR ENERGY COLLECTION: Flat plate and concentrating collectors, classification of concentrating collectors, orientation.

SOLAR ENERGY STORAGE AND APPLICATIONS: Different methods, sensible, latent heat and stratified storage, solar ponds, solar applications- solar heating/cooling technique, solar distillation and drying, solar cookers, central power tower concept and solar chimney.

UNIT – IV

WIND ENERGY: Sources and potentials, horizontal and vertical axis windmills, performance characteristics, betz criteria, types of winds, wind data measurement.

BIO-MASS: Principles of bio-conversion, anaerobic/aerobic digestion, types of bio-gas digesters, gas yield, utilization for cooking, bio fuels, I.C. engine operation and economic aspects.

UNIT – V

GEOTHERMAL ENERGY: Origin, Applications, Types of Geothermal Resources, Relative Merits. **OCEAN ENERGY**: Ocean Thermal Energy; Open Cycle & Closed Cycle OTEC Plants, Environmental Impacts, Challenges.

FUEL CELLS: Introduction, Applications, Classification, Different Types of Fuel Cells Such as Phosphoric Acid Fuel Cell, Alkaline Fuel Cell, PEM Fuel Cell, MC Fuel Cell.

Text Books:

1. Solar Energy – Principles of Thermal Collection and Storage/Sukhatme S.P. and J.K. Nayak/TMH
2. Non-Conventional Energy Resources- Khan B.H/ Tata McGraw Hill, New Delhi, 2006

References:

1. Principles of Solar Engineering - D. Yogi Goswami, Frank Krieth& John F Kreider / Taylor & Francis
2. Non-Conventional Energy - Ashok V Desai /New Age International (P) Ltd
3. Renewable Energy Technologies -Ramesh & Kumar /Narosa
4. Non-conventional Energy Source- G.D Roy/Standard Publishers

Online Learning Resources:

- https://nptel.ac.in/courses/112106318
- https://youtube.com/playlist?list=PLyqSpQzTE6M-ZgdjYukayF6QevPv7WE-r&si=-mwIa2X-SuSiNy13
- https://youtube.com/playlist?list=PLyqSpQzTE6M-ZgdjYukayF6QevPv7WE-r&si=Apfjx6oDfz1Rb_N3
- https://youtu.be/zx04Kl8y4dE?si=VmOvp_OgqisILTAF

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
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III Year B.Tech.–I Semester

L	T	P	C
3	0	0	3

23AEC30 ELECTRONIC CIRCUITS (OPEN ELECTIVE–I)

Course Objectives:

1. To understand semiconductor diodes, their characteristics and applications.
2. To explore the operation, configurations, and biasing of BJTs.
3. To study the operation, analysis, and coupling techniques of BJT amplifiers.
4. To learn the operation, applications, and uses of feedback amplifiers and oscillators.
5. To analyze the characteristics, configurations, and applications of operational amplifiers.

Course Outcomes:

At the end of this course, the students will be able to:

1. Understand semiconductor diodes, their characteristics and applications.
2. Explore the operation, configurations, and biasing of BJTs.
3. Gain knowledge about the operation, analysis, and coupling techniques of BJT amplifiers.
4. Learn the operation, applications, and uses of feedback amplifiers and oscillators.
5. Analyze the characteristics, configurations, and applications of operational amplifiers.

UNIT-I

Semiconductor Diode and Applications: Introduction, PN junction diode – structure, operation and VI characteristics, Half-wave, Full-wave and Bridge Rectifiers with and without Filters, Positive and Negative Clipping and Clamping circuits (Qualitative treatment only).

Special Diodes: Zener and Avalanche Breakdowns, VI Characteristics of Zener diode, Zener diode as voltage regulator, Construction, operation and VI characteristics of Tunnel Diode, LED, Varactor Diode, Photo Diode .

UNIT-II

Bipolar Junction Transistor (BJT): Principle of Operation, Common Emitter, Common Base and Common Collector Configurations, Transistor as a switch and Amplifier, Transistor Biasing and Stabilization - Operating point, DC & AC load lines, Biasing - Fixed Bias, Self Bias, Bias Stability, Bias Compensation using Diodes.

UNIT-III

Single stage amplifiers: Classification of Amplifiers - Distortion in amplifiers, Analysis of CE, CC and CB configurations with simplified hybrid model.

Multistage amplifiers: Different Coupling Schemes used in Amplifiers - RC coupled amplifiers, Transformer Coupled Amplifier, Direct Coupled Amplifier; Multistage RC coupled BJT amplifier (Qualitative treatment only).

UNIT-IV

Feedback amplifiers: Concepts of feedback, Classification of feedback amplifiers, Effect of feedback on amplifier characteristics, Voltage Series, Voltage Shunt, Current Series and Current Shunt Feedback Configurations (Qualitative treatment only).

Oscillators: Classification of oscillators, Condition for oscillations, RC Phase shift Oscillators, Generalized analysis of LC Oscillators-Hartley and Colpitts Oscillators, Wien Bridge Oscillator.

UNIT-V

Op-amp: Classification of IC'S, basic information of Op-amp, ideal and practical Op-amp, 741 op-amp and its features, modes of operation-inverting, non-inverting, differential.

Applications of op-amp : Summing, scaling and averaging amplifiers, Integrator, Differentiator, phase shift oscillator and comparator.

TEXT BOOKS:

1. Electronics Devices and Circuits, J.Millman and Christos. C. Halkias, 3rd edition, Tata McGraw Hill, 2006.
2. Electronics Devices and Circuits Theory, David A. Bell, 5th Edition, Oxford University press. 2008.

REFERENCE BOOKS:

1. Electronics Devices and Circuits Theory, R.L.Boylestad, LousisNashelsky and K.Lal Kishore, 12th edition, 2006, Pearson, 2006.
2. Electronic Devices and Circuits, N.Salivahanan, and N.Suresh Kumar, 3rd Edition, TMH, 2012
3. Microelectronic Circuits, S.Sedra and K.C.Smith, 5th Edition, Oxford University Press.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B. Tech I Semester

23AHS26 - MATHEMATICS FOR MACHINE LEARNING AND AI

(Common to all Branches)
Open Elective 1

L	T	P	C
3	0	0	3

Course Outcomes:

After successful completion of this course, the students should be able to:

CO1	Apply linear algebra concepts to ML techniques like PCA and regression.
CO2	Analyze probabilistic models and statistical methods for AI applications.
CO3	Implement optimization techniques for machine learning algorithms.
CO4	Utilize vector calculus and transformations in AI-based models.
CO5	Develop graph-based AI models using mathematical representations.

UNIT I: Linear Algebra for Machine Learning **08 Hrs**

Review of Vector spaces, basis, linear independence, Vector and matrixnorms, Matrix factorization techniques, Eigenvalues, eigenvectors, diagonalization, Singular Value Decomposition (SVD) and Principal Component Analysis (PCA).

UNIT II: Probability and Statistics for AI **08 Hrs**

Probability distributions: Gaussian, Binomial, Poisson. Bayes' Theorem, aximum Likelihood Estimation (MLE), and Maximum a Posteriori (MAP).Entropy and Kullback-Leibler (KL) Divergence in AI, Cross entropy loss, Markov chains.

UNIT III: Optimization Techniques for ML **08 Hrs**

Multivariable calculus: Gradients, Hessians, Jacobians. Constrained optimization: Lagrange multipliers and KKT conditions.Gradient Descent and its variants (Momentum, Adam) Newton's method, BFGS method.

UNIT IV: Vector Calculus &Transformations **08 Hrs**

Vector calculus: Gradient, divergence, curl. Fourier Transform & Laplace Transform in ML applications.

UNIT V: Graph Theory for AI **08 Hrs**

Graph representations: Adjacency matrices, Laplacian matrices. Bayesian Networks & Probabilistic Graphical Models.Introduction to Graph Neural Networks (GNNs).

Text books:

1. Mathematics for Machine Learning by Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong, Cambridge University Press, 2020.
2. Pattern Recognition and Machine Learning by Christopher Bishop, Springer.

Reference Books:

1. Gilbert Strang, Linear Algebra and Its Applications, Cengage Learning, 2016.
2. Jonathan Gross, Jay Yellen, Graph Theory and Its Applications, CRC Press, 2018.

Web References:

- MIT– Mathematics for Machine Learning https://ocw.mit.edu
- Stanford CS229 – Machine Learning Course https://cs229.stanford.edu/

DeepAI – Mathematical Foundations for AI https://deepai.org

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	1	-	-	-	-	-	-	1
CO2	3	3	2	3	2	-	-	-	-	-	-	2
CO3	3	3	3	3	2	1	-	-	-	-	-	2
CO4	3	3	2	2	1	-	-	-	-	-	-	1
CO5	3	3	3	3	2	-	-	-	-	-	-	2

3-High Mapping **2- Medium Mapping** **1-Low Mapping**

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

(Open Elective-I)
III B.Tech I Semester (Common to all branches)
(Open Elective-Interdisciplinary)

L	T	P	C
3	0	0	3

23AHS27 - MATERIALS CHARACTERIZATION TECHNIQUES

COURSE OUTCOMES

After successful completion of the course, the students will be able to,

CO1: Analyze the crystal structure and crystallite size by various methods

CO2: Analyze the morphology of the sample by using a Scanning Electron Microscope

CO3: Analyze the morphology and crystal structure of the sample by using Transmission Electron Microscope

CO4: Explain the principle and experimental arrangement of various spectroscopic techniques

CO5: Identify the construction and working principle of various Electrical & Magnetic Characterization technique

UNIT- I: Structure analysis by Powder X-Ray Diffraction 9hrs

Introduction, Bragg's law of diffraction, Intensity of Diffracted beams, Factors affecting Diffraction, Intensities, Structure of polycrystalline Aggregates, Determination of crystal structure, Crystallite size by Scherer and Williamson-Hall (W-H) Methods, Small angle X- ray scattering (SAXS) (in brief).

UNIT- II: Microscopy technique-1–Scanning Electron Microscopy (SEM) 9hrs

Introduction, Principle, Construction and working principle of Scanning ElectronMicroscopy, Specimen preparation, Different types of modes used (Secondary Electron and Backscatter Electron), Advantages, limitations and applications of SEM.

UNIT- III: Microscopy Technique-2-Transmission Electron Microscopy (TEM) 9hrs

Construction and Working principle, Resolving power and Magnification, Bright and dark fields, Diffraction and image formation, Specimen preparation, Selected Area Diffraction, Applications of Transmission Electron Microscopy, Difference between SEM and TEM, Advantage and Limitations of Transmission Electron Microscope

UNIT- IV: Spectroscopy techniques 9hrs

Principle, Experimental arrangement, Analysis and advantages of the spectroscopic techniques – (i) UV-Visible spectroscopy (ii) Raman Spectroscopy, (iii) Fourier Transform infrared (FTIR) spectroscopy, (iv) X-ray photoelectron spectroscopy (XPS).

UNIT- V: Electrical & Magnetic Characterization techniques **9hrs**

Electrical Properties analysis techniques (DC conductivity, AC conductivity) Activation Energy, Effect of Magnetic field on the electrical properties (Hall Effect). Magnetization measurement by induction method, vibrating sample Magnetometer (VSM) and SQUID.

Textbooks:

1. Material Characterization: Introduction to Microscopic and Spectroscopic Methods – Yang Leng – John Wiley & Sons (Asia) Pvt. Ltd. 2013.
2. Microstructural Characterization of Materials-David Brandon, Wayne D Kalpan, John Wiley & Sons Ltd., 2008

Reference Books:

1. Fundamentals of Molecular Spectroscopy–I VEd.–Colin Neville Banwell and Elaine M. McCash, Tata McGraw-Hill,2008.
2. Elements of X-ray diffraction–Bernard Dennis Cullity & Stuart R stocks, Prentice Hall , 2001 – Science.
3. Practical Guide to Materials Characterization: Techniques and Applications - Khalid Sultan – Wiley – 2021.
4. Materials Characterization Techniques-Sam Zhang, Lin Li, Ashok Kumar-CRC Press – 2008

NPTEL courses link :

1. https://nptel.ac.in/courses/115/103/115103030/
2. https://nptel.ac.in/content/syllabus_pdf/113106034.pdf
3. https://nptel.ac.in/noc/courses/noc19/SEM1/noc19-mm08/

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	3	2	2	1	-	-	-	-	-	-	-
CO2	3	3	2	1	1	-	-	-	-	-	-	-
CO3	3	3	2	1	1	-	-	-	-	-	-	-
CO4	3	2	1	1	-	-	-	-	-	-	-	-
CO5	3	3	1	1	-	-	-	-	-	-	-	-

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

CHEMISTRY OF ENERGY SYSTEMS

(Open Elective-I)

III B.Tech I Semester (Common to all branches)

L	T	P	C
3	0	0	3

23AHS28: CHEMISTRY OF ENERGY SYSTEMS COURSE

OUTCOMES

After successful completion of the course, the students will be able to,

CO1: Solve the problems based on electrode potential, Describe the Galvanic Cell, differentiate between Lead acid and Lithium ion batteries, Illustrate the electrical double layer
CO 2: Describe the working Principle of Fuel cell, explain the efficiency of the fuel cell, discuss about the Basic design of fuel cells, Classify the fuel cell
CO3: Differentiate between Photo and Photo electrochemical Conversions, Illustrate the photo chemical cells, Identify the applications of photo chemical reactions, Interpret advantages of photo electron catalytic conversion.
CO4: Apply the photo voltaic technology, demonstrate about solar energy and prospects, Illustrate the Solar cells, discuss about concentrated solar power
CO5: Differentiate Chemical and Physical methods of hydrogen storage, discuss the metal organic frame work, illustrate the carbon and metal oxide porous structures
Describe the liquefaction methods.

UNIT - I: Electrochemical Systems **9 hrs**

Galvanic cell, Nernst equation, standard electrode potential, application of EMF, electrical double layer, polarization, Batteries-Introduction, Lead-acid, Nickel-cadmium, Lithium-ion batteries and their applications.

UNIT - II: Fuel Cells **9 hrs**

Fuel cell- Introduction, Basic design of fuel cell, working principle, Classification of fuel cells, Polymer electrolyte membrane (PEM) fuel cells, Solid-oxide fuel cells (SOFC), Fuel cell efficiency and applications.

UNIT - III: Photo and Photo electrochemical Conversions **9 hrs**

Photochemical cells Introduction and applications of photochemical reactions, specificity of photo electrochemical cell, advantage of photoelectron catalytic conversions and their applications.

UNIT - IV: Solar Energy **9 hrs**

Introduction and prospects, photovoltaic (PV) technology, concentrated solar power (CSP), Solar cells and applications.

UNIT - V: Hydrogen Storage **9 hrs**

Hydrogen storage and delivery: State-of-the art, Established technologies, Chemical and Physical methods of hydrogen storage, Compressed gas storage, Liquid hydrogen storage, Other storage methods, Hydrogen storage in metal hydrides, metal organic frameworks (MOF), Metal oxide porous structures, hydrogel , and Organic hydrogen carriers.

Text books

1. Physical chemistry by Ira N.Levine
2. Essentials of Physical Chemistry, Bahland Bahl and Tuli.
3. Inorganic Chemistry, Silver and Atkins

Reference Books:

1. Fuel Cell Hand Book 7th Edition, by US Department of Energy (EG & G technical services and corporation)
2. Handbook of solar energy and applications by Arvind Tiwari and Shyam.
3. Solar energy fundamental, technology and systems by Klaus Jagaret.al.
4. Hydrogen storage by Levine K leb on off

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)
OPEN ELECTIVE – I

III B. Tech II SEM

23AHS29 - ENGLISH FOR COMPETITIVE EXAMINATIONS

L	T	P	C
3	0	0	3

Course Outcomes (CO):

After successful completion of the course, the students will be able to:

CO1: Identify the basics of English grammar and its importance
CO2: Explain the use of grammatical structures in sentences
CO3: Demonstrate the ability to use various concepts in grammar and vocabulary and their applications in everyday use and in competitive exams
CO4: Analyze an unknown passage and reach conclusions about it
CO5: Choose the appropriate form of verbs in framing sentences
CO6: Develop speed reading and comprehending ability thereby perform better in competitive exams

UNIT-I GRAMMAR-1

Nouns-classification-errors-Pronouns-types-errors-Adjectives-types-errors-Articles-definite-indefinite- Degrees of Comparison-Adverbs-types- errors-Conjunctions-usage- Prepositions-usage-Tag Questions, types-identifying errors- Practice

UNIT-II VERBAL ABILITY

Sentence completion-Verbal analogies-Word groups-Instructions-Critical reasoning-Verbal deduction-Select appropriate pair-Reading Comprehension-Paragraph-Jumbles-Selecting the proper statement by reading a given paragraph.

UNIT-III GRAMMAR-2

Verbs-tenses- structure-usages- negatives- positives- time adverbs-Sequence of tenses--If Clause- Voice-active voice and passive voice- reported Speech-Agreement- subject and verb-Modals- Spotting Errors-Practices

UNIT-IV READING COMPREHENSION AND VOCUBULARY

Competitive Vocabulary :Word Building – Memory techniques-Synonyms, Antonyms, Affixes- Prefix &Suffix-One word substitutes-Compound words-Phrasal Verbs-Idioms and Phrases- Homophones- Linking Words-Modifiers-Intensifiers - Mastering Competitive Vocabulary- Cracking the unknowing passage-speed reading techniques- Skimming & Scanning-types of answering–Elimination methods.

UNIT-V WRITING FOR COMPETITIVE EXAMINATIONS

Punctuation- Spelling rules- Word order-Sub Skills of Writing- Paragraph meaning-salient features- types - Note-making, Note-taking, summarizing-precise writing- Paraphrasing-Expansion of proverbs- Essay writing-types

Textbooks:

1. Wren & Martin, *English for Competitive Examinations*, S.Chand & Co, 2021
2. *Objective English for Competitive Examination*, Tata McGraw Hill, New Delhi, 2014.

Reference Books:

1. Hari Mohan Prasad, *Objective English for Competitive Examination*, Tata McGraw Hill, New Delhi, 2014.
2. Philip Sunil Solomon, *English for Success in Competitive Exams*, Oxford 2016
3. Shalini Verma , *Word Power Made Handy*, S Chand Publications
4. Neira, Anjana Dev & Co. *Creative Writing: A Beginner's Manual*. Pearson Education India, 2008.
5. Abhishek Jain,Vocabulary Learning Techniques Vol.I& II,RR Global Publishers 2013.

Online Resources:

1. https://www.grammar.cl/english/parts-of-speech.htm
2. https://academicguides.waldenu.edu/writingcenter/grammar/partsofspeech
3. https://learnenglish.britishcouncil.org/grammar/english-grammar-reference/active-passive- voice
4. https://languagetool.org/insights/post/verb-tenses/
5. https://www.britishcouncil.in/blog/best-free-english-learning-resources-british-council
6. https://www.careerride.com/post/social-essays-for-competitive-exams-586.aspx

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

23AMB06- ENTREPRENEURSHIPAND NEW VENTURE CREATION	L	T	P	C
(Open Elective-I)	3	0	0	3

COURSE OBJECTIVES: The objectives of this course are

1. To foster an entrepreneurial mind-set for venture creation and intrapreneurial leadership.
2. To encourage creativity and innovation
3. To enable them to learn pitching and presentation skills
4. To make the students understand MVP development and validation techniques to determine Product-Market fit and Initiate Solution design, Prototype for Proof of Concept.
5. To enhance the ability of analyzing Customer and Market segmentation, estimate Market size, develop and validate Customer Persona

COURSE OUTCOMES: At the end of the course, students will be able to		BTL
CO1	Develop an entrepreneurial mindset and appreciate the concept of entrepreneurship	L3
CO2	Comprehend the process of problem-opportunity identification through design thinking, identify market potential and customers while developing a compelling value proposition solution	L3
CO3	Analyze and refine business models to ensure sustainability and profitability	L3
CO4	Build Prototype for Proof of Concept and validate MVP of their practice venture idea	L4
CO5	Create business plan, conduct financial analysis and feasibility analysis to assess the financial viability of a venture	L5
CO6	Prepare and deliver an investible pitch deck of their practice venture to attract stakeholders	L6

BTL: Bloom's Taxonomy Level

UNIT-I: Entrepreneurship Fundamentals and context

Meaning and concept, attributes and mindset of entrepreneurial and intrapreneurial leadership, role models in each and their role in economic development. An understanding of how to build entrepreneurial mindset, skill sets, attributes and networks while on campus.

Core Teaching Tool: Simulation, Game, Industry Case Studies (Personalized for students – 16industries to choose from), Venture Activity

LEARNING OUTCOMES

At the end of the Unit, the learners will be able to

- Understand theconcept of Entrepreneur and Entrepreneurship in India
- Analyze recent trends in Entrepreneurship role in economic development
- Develop a creative mind set and personality in starting a business.

UNIT II: Problem & Customer Identification

Understanding and analysing the macro-Problem and Industry perspective - technological, socioeconomic and urbanization trends and their implication on new opportunities - Identifying passion - identifying and defining problem using Design thinking principles - Analysing problem and validating with the potential customer - Understanding customer segmentation, creating and validating customer personas.

Core Teaching Tool: Several types of activities including Class, game, Gen AI, 'Get out of the Building' and Venture Activity.

LEARNING OUTCOMES

At the end of the Unit, the learners will be able to

- Understand the problem and Customer identification.
- Analyze problem and validating with potential customer
- Evaluate customer segmentation and customer personas

UNIT III: Solution design, Prototyping & Opportunity Assessment and Sizing

Understanding Customer Jobs-to-be-done and crafting innovative solution design to map to customer's needs and create a strong value proposition - Understanding prototyping and Minimum Viable product (MVP) - Developing a feasibility prototype with differentiating value, features and benefits - Assess relative market position via competition analysis - Sizing the market and assess scope and potential scale of the opportunity.

Core Teaching Tool: Venture Activity, no-code Innovation tools, Class activity

LEARNING OUTCOMES

At the end if the Unit, the learners will be able to

- Analyze jobs-to-be-done
- Evaluate customer needs to create a strong value proposition
- Design and draw prototyping and MVP

UNIT-IV: Business & Financial Model, Go-to-Market Plan
Introduction to Business model and types, Lean approach, 9 block lean canvas model, riskiest assumptions to Business models. Importance of Build - Measure – Lean approach.

Business planning: components of Business plan- Sales plan, People plan and financial plan.
Financial Planning: Types of costs, preparing a financial plan for profitability using financial template, understanding basics of Unit economics and analysing financial performance.

Introduction to Marketing and Sales, Selecting the Right Channel, creating digital presence, building customer acquisition strategy.

Choosing a form of business organization specific to your venture, identifying sources of funds: Debt& Equity, Map the Start-up Life-cycle to Funding Options.

Core Teaching Tool: Founder Case Studies – Sama and Securely Share; Class activity and discussions; Venture Activities.

LEARNING OUTCOMES

At the end of the Unit, the learners will be able to:

- Understand lean approach in business models
- Apply business plan, sales plan and financial plan
- Analyze financial planning, marketing channels of distribution.
- Design their own venture and source of funds.

UNIT-V: Scale Outlook and Venture Pitch readiness

Understand and identify potential and aspiration for scale vis-a-vis your venture idea.
Persuasive Storytelling and its key components. Build an Investor ready pitch deck.

Core Teaching Tool: Expert talks; Cases; Class activity and discussions; Venture Activities.

LEARNING OUTCOMES

At the end of the Unit, the learners will be able to

- Understand aspiration for scale
- Analyze venture idea and its key components
- Evaluate and build investors ready pitch

TEXT BOOKS

1. Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd, Sabyasachi Sinha . *Entrepreneurship*, McGrawHill, 11th Edition.(2020)
2. Ries, E. *The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses*. Crown Business,(2011).
3. Osterwalder, A., & Pigneur, Y. *Business Model Generation: A Handbook for Visionaries, Game Changers, and Challengers.* John Wiley & Sons. (2010).

REFERENCES

1. Simon Sinek,*Start with Why*, Penguin Books limited. (2011)
2. Brown Tim,*Change by Design Revised & Updated: How Design Thinking*
3. *Transforms Organizations and Inspires Innovation*, Harper Business.(2019)

4. Namita Thapar (2022) The Dolphin and the Shark: Stories on Entrepreneurship, Penguin Books Limited
5. Saras D. Sarasvathy, (2008) Effectuation: Elements of Entrepreneurial Expertise, Elgar Publishing Ltd.

E-RESOURCES

Learning resource- Ignite 5.0 Course Wadhwani platform (Includes 200+ components of

custom created modular content + 500+ components of the most relevant curated content)

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B.Tech I Semester

		L	T	P	C
23ACA14	**COMPUTER VISION & MACHINE LEARNING LAB (Professional Core)**	**0**	**0**	**3**	**1.5**

Course Objectives:

- To impart practical knowledge of computer vision concepts using OpenCV and image processing libraries.
- To implement core machine learning algorithms and evaluate model performance.
- To work with real-world datasets for classification, regression, and image processing tasks.
- To train, test, and validate models using Python, TensorFlow, and Scikit-learn.
- To understand the integration of ML models in vision applications.

Course Outcomes:
After successful completion of this lab, students will be able to:

- Apply computer vision techniques to solve real-time image processing problems. (Apply - L3)
- Train and evaluate machine learning models for classification and regression tasks. (Analyze - L4)
- Design and test feature extraction techniques from images. (Create - L6)
- Use OpenCV, Scikit-learn, TensorFlow/PyTorch for practical implementations. (Apply - L3)
- Integrate vision-based features with ML algorithms for end-to-end solutions. (Evaluate - L5)

List of Experiments (12 Total)

1. Image preprocessing techniques: resizing, filtering, thresholding using OpenCV
2. Edge detection using Sobel, Canny, and Laplacian operators
3. Object detection using contour detection and bounding boxes
4. Feature extraction using HOG, SIFT, and ORB
5. Implement face detection using Haar cascades or DNN models
6. Train a machine learning model (SVM / KNN) for image classification
7. Build and evaluate a decision tree classifier using scikit-learn
8. Implement a logistic regression model for binary classification on numerical dataset
9. Apply PCA for feature reduction and visualization
10. Design a simple neural network using TensorFlow/Keras for image classification
11. Train and evaluate a CNN model for digit recognition using MNIST dataset
12. Real-time emotion recognition using webcam input and pre-trained model integration

Textbooks:

1. Simon J. D. Prince, Computer Vision: Models, Learning, and Inference, Cambridge University Press.
2. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition, O'Reilly.
3. Richard Szeliski, Computer Vision: Algorithms and Applications, Springer.

Reference Books:

1. Adrian Rosebrock, Practical Python and OpenCV (PyImageSearch).
2. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press.
3. Bishop C. M., Pattern Recognition and Machine Learning, Springer.

Online Learning Resources:

1. https://opencv.org
2. https://www.tensorflow.org/tutorials
3. https://www.kaggle.com/learn/intro-to-machine-learning
4. https://www.pyimagesearch.com
5. NPTEL Course on Deep Learning

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B.Tech I Semester

		L	T	P	C
23AAI07	**AI & SYSTEM PROGRAMMING LAB (Professional Core)**	**0**	**0**	**3**	**1.5**

Course Objectives:

- To provide practical exposure to foundational AI algorithms and system programming.
- To develop skills to write intelligent systems and low-level programs.
- To integrate concepts of AI and system programming for automation and optimization.

Course Outcomes:

After successful completion of the lab, students will be able to:

- Implement search algorithms and logic programming using AI tools.
- Construct assemblers, macro processors, and shell scripts.
- Develop system utilities using C and integrate them with AI tools.
- Demonstrate real-time intelligent system automation using scripting and AI logic.

List of Experiments (12 Total)

1. Write simple programs in Prolog for facts, rules, and queries.
2. Develop a Prolog-based expert system for medical diagnosis or animal identification.
3. Implement Depth-First Search (DFS) and Breadth-First Search (BFS) in Python.
4. Implement A* Search Algorithm using heuristics in Python.
5. Implement the Minimax algorithm for a simple game (e.g., Tic Tac Toe).
6. Design and implement a two-pass assembler in C.
7. Implement a Macro Processor using C for assembly language programs.
8. Develop a simple Linux Shell (command interpreter) using C.
9. Write shell scripts for file operations, process creation, and monitoring.
10. Demonstrate inter-process communication using pipes and signals in Linux.
11. Integrate AI logic (search/expert system) into a shell script or system utility for task automation.
12. Develop an AI-powered system utility (e.g., Intelligent File Manager, AI Bot for CLI commands).

Lab Software Requirements:

- Languages: Python, Prolog, C
- Tools: GCC, SWI-Prolog, Linux (Ubuntu/WSL), Shell, Lex/Yacc (optional)
- IDEs: Code::Blocks / VS Code / Geany / Terminal-based compilation

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B.Tech I Semester

		L	T	P	C
23ACS20	**FULL STACK DEVELOPMENT-II (Skill Enhancement course)**	**0**	**1**	**2**	**2**

Course Objectives: The main objectives of the course are to

- Make use of Modern- day JavaScript with ES6 standards for designing Dynamic web pages
- Building robust & responsive User Interfaces using popular JavaScript library '**React.js**'. Building robust backend APIs using '**Express. js**'
- Establishing the connection between frontend (React) User interfaces and backend APIs (Express) with Data Bases(My SQL)
- Familiarize students with GitHub for remote repository hosting and collaborative development.

Course Outcomes:

- CO1: Building fast and interactive UIs
- CO2: Applying Declarative approach for developing web apps
- CO3: Understanding ES6 features to embrace modern JavaScript
- CO4: Building reliable APIs with Express. Js
- CO5: Create and manage Git repositories, track changes, and push code to GitHub.

Experiments covering the Topics:

- Introduction to DOM (Document Object Model), Ecma Script (ES6) standards and features like Arrow functions, Spread operator, Rest operator, Type coercion, Type hoisting, String literals, Array and Object Destructuring.
- Basics of React. js like React Components, JSX, Conditional rendering
  Differences between Real DOM and Virtual DOM.
- Important React.js concepts like React hooks, Props, React forms, Fetch API, Iterative rendering using JavaScript map() function.
- JavaScript runtime environment node. js and its uses, Express. js and Routing, Micro-Services architecture and MVC architecture, database connectivity using (My SQL)
- Introduction to My SQL, setting up MySQL and configuring, Databases, My SQL queries, subqueries, creating My SQL driver for database connectivity to Express. js server.
- Introduction to Git and GitHub and upload project& team collaboration

Sample Experiments:

1. **Introduction to Modern JavaScript and DOM**
   - **a.** Write a JavaScript program to link JavaScript file with the HTML page
   - **b.** Write a JavaScript program to select the elements in HTML page using selectors
   - **c.** Write a JavaScript program to implement the event listeners
   - **d.** Write a JavaScript program to handle the click events for the HTML button elements
   - **e.** Write a JavaScript program to With three types of functions
     - **i.** Function declaration
     - **ii.** Function definition
     - **iii.** Arrow functions

2. **Basics of React. js**
    - **a.** Write a React program to implement a counter button using react class components
    - **b.** Write a React program to implement a counter button using react functional components
    - **c.** Write a React program to handle the button click events in functional component
    - **d.** Write a React program to conditionally render a component in the browser
    - **e.** Write a React program to display text using String literals
3. **Important concepts of React. js**
    - **a.** Write a React program to implement a counter button using React use State hook
    - **b.** Write a React program to fetch the data from an API using React use Effect hook
    - **c.** Write a React program with two react components sharing data using Props.
    - **d.** Write a React program to implement the forms in react
    - **e.** Write a React program to implement the iterative rendering using map() function.
4. **Introduction to Git and GitHub**

a. **Setup**
- o Install Git on local machine.
- o Configure Git (user name, email).
- o Create GitHub account and generate a personal access token.

b. **Basic Git Workflow**
- o Create a local repository using git init
- o Create and add files → git add .
- o Commit files → git commit -m "Initial commit"
- o Connect to GitHub remote → git remote add origin <repo_url>
- o Push to GitHub → git push -u origin main

c. **Branching and Collaboration**
- o Create a branch → git checkout -b feature1
- o Merge branch to main → git merge feature1
- o Resolve merge conflicts (guided)

5. **Upload React Project to GitHub**

- o Create a new React app using npx create-react-app myapp
- o Initialize a git repo and push to GitHub
- o Use .gitignore to exclude node_modules
- o Create multiple branches: feature/navbar, feature/form
- o Practice merge and pull requests (can use GitHub GUI)

6. **Introduction to Node. js and Express. js**
    - **a.** Write a program to implement the 'hello world' message in the route through the browser using Express
    - **b.** Write a program to develop a small website with multiple routes using Express. js
    - **c.** Write a program to print the 'hello world' in the browser console using Express. js
    - **d.** Write a program to implement the CRUD operations using Express. js
    - **e.** Write a program to establish the connection between API and Database using Express – My SQL driver
7. **Introduction to My SQL**
    - **a.** Write a program to create a Database and table inside that database using My SQL Command line client
    - **b.** Write a My SQL queries to create table, and insert the data, update the data in the table
    - **c.** Write a My SQL queries to implement the subqueries in the My SQL command line client
    - **d.** Write a My SQL program to create the script files in the My SQL workbench
    - **e.** Write a My SQL program to create a database directory in Project and initialize a database. sql file to integrate the database into API

8. *Team Collaboration Using GitHub*

  - Form groups of 2–3 students
  - Create a shared GitHub repo
  - Assign tasks and work in branches
  - Use Issues, Pull Requests, and Code Reviews
  - Document code with README.md

Textbooks:

1. Web Design with HTML, CSS, JavaScript and JQuery Set Book by Jon Duckett Professional JavaScript for Web Developers Book by Nicholas C. Zakas
2. John Dean, Web Programming with HTML5, CSS and JavaScript, Jones & Bartlett Learning, 2019.
3. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O'Reilly.
4. Learning PHP, MySQL, JavaScript, CSS & HTML5: A Step-by-Step Guide to Creating Dynamic Websites by Robin Nixon
5. AZAT MARDAN, Full Stack Java Script: Learn Back bone. js, Node.jsand Mongo DB.2015

Reference Books:

1. Full-Stack JavaScript Development by Eric Bush.
2. Programming the World Wide Web, 7th Edition, Robet W Sebesta, Pearson, 2013.
3. Tomasz Dyl ,KamilPrzeorski , MaciejCzarnecki, Mastering Full Stack React Web Development 2017

Online Learning Resources:

1. https://ict.iitk.ac.in/product/full-stack-developer-html5-css3-js-bootstrap-php-4/
2. https://www.w3schools.com/html
3. https://www.w3schools.com/css
4. https://www.w3schools.com/js/
5. https://www.w3schools.com/nodejs
6. https://www.w3schools.com/typescript
7. https://docs.github.com/
8. https://education.github.com/git-cheat-sheet-education.pdf

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B.Tech – I semester **L T P C**
2 0 0 1

23ACO03 TINKERING LAB

The aim of tinkering lab for engineering students is to provide a hands-on learning environment where students can explore, experiment, and innovate by building and testing prototypes. These labs are designed to demonstrate practical skills that complement theoretical knowledge.

Course objectives: The objectives of the course are to

1 Encourage Innovation and Creativity
2 Provide Hands-on Learning and Impart Skill Development
3 Foster Collaboration and Teamwork
4 Enable Interdisciplinary Learning, Prepare for Industry and Entrepreneurship
5 Impart Problem-Solving mind-set

These labs bridge the gap between academia and industry, providing students with the practical experience. Some students may also develop entrepreneurial skills, potentially leading to start-ups or innovation-driven careers. Tinkering labs aim to cultivate the next generation of engineers by giving them the tools, space, and mind-set to experiment, innovate, and solve real-world challenges.

List of experiments:

1) Make your own parallel and series circuits using breadboard for any application of your choice.
2) Demonstrate a traffic light circuit using breadboard.
3) Build and demonstrate automatic Street Light using LDR.
4) Simulate the Arduino LED blinking activity in Tinkercad.
5) Build and demonstrate an Arduino LED blinking activity using Arduino IDE.
6) Interfacing IR Sensor and Servo Motor with Arduino.
7) Blink LED using ESP32.
8) LDR Interfacing with ESP32.
9) Control an LED using Mobile App.
10) Design and 3D print a Walking Robot
11) Design and 3D Print a Rocket.
12) Build a live soil moisture monitoring project, and monitor soil moisture levels of a remote plan in your computer dashboard.
13) Demonstrate all the steps in design thinking to redesign a motor bike.

Students need to refer to the following links:

Course Outcomes: The students will be able to experiment, innovate, and solve real-world challenges.

1) https://aim.gov.in/pdf/equipment-manual-pdf.pdf
2) https://atl.aim.gov.in/ATL-Equipment-Manual/
3) https://aim.gov.in/pdf/Level-1.pdf
4) https://aim.gov.in/pdf/Level-2.pdf
5) https://aim.gov.in/pdf/Level-3.pdf

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B. Tech - I Semester

Course Code: 23ACM03

COMMUNITY SERVICE INTERNSHIP

L	T	P	C
-	-	-	2

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B.Tech - I Semester

L	T	P	C
2	0	0	0

23AHS30 QUANTITATIVE APTITUDE AND REASONING - III
(Common to All Branches)

Course Outcomes:
After successful completion of the course, the student will be able to:

CO1: Develop the thinking ability to meet the challenges in solving Logical Reasoning problems.
CO2: Solve campus placements aptitude papers covering Quantitative Ability and Verbal Ability.
CO3: Apply different placement practice techniques.

UNIT 1: QUANTITATIVE ABILITY V

Time and Work – Equal Efficiency – Different Efficiency – Combined work – Alternate work – Partial work – Negative work - Pipes and Cistern – Simple Interest – Compound Interest - Year Zero – Difference between SI and CI – Clocks – Angle of the Clock –Minutes hand Loss or Gain – Calendars – Leap Year – Non Leap year – Odd days – Days of the week

UNIT 2: QUANTITATIVE ABILITY VI

Mensuration 2D – Area and Perimeter - Mensuration 3D – Volume - Total Surface area – Lateral Surface Area – Statistics- Mean - Mean Deviation – Median – Mode - Range – Variance - – Standard Deviation - Set theory

UNIT 3: REASONING ABILITY III

Puzzles – Cubes & Dices – Algebra – Selection Decision table – Visual reasoning - Inequalities

UNIT 4: VERBAL III

Vocabulary - Synonyms, Antonyms, One Word Substitution, and Spelling - Sentence Correction - Sentence Selection, Error Identification, Sentence Improvement, Sentence completion – Cloze Test, Types,Strategies - Para jumbles- Types, Strategies.

UNIT 5: SOFT SKILLS III
Written Communication - Listening Skills - Mentoring & Coaching - Decision Making - Competitiveness - Inspiring & Motivating.

Text Books:
1. Quantitative Aptitude, Logic Reasoning & Verbal Reasoning, R S Agarwal, S.Chand Publications-2022.
2. Quantitative Aptitude for Competitive Examinations, R S Agarwal, S.Chand Publications-2022.

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	-	-	-	-
CO2	1	2	-	-	-	-	-	-	-	-	-	-
CO3	2	-	-	-	-	-	-	-	-	-	-	-

3-High Mapping **2- Medium Mapping** **1-Low Mapping**

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)
III B. Tech I Semester Common to all branches

Course Code: 23AHS31 **FRENCH LANGUAGE**
(Audit Course)

L	T	P	C
2	**0**	**0**	**0**

COURSE OUTCOMES:

After completion of the course the student will be able to:

CO1: Understand basic knowledge of French language and several core competencies.

CO2: Develop and improve comprehensive capabilities and apply simple phrases & sentences in real- life conversation.

CO3: Analyse ability to ask and answer questions about the self, personal interest, everyday life, and the immediate environment.

CO4: Demonstrate knowledge of tenses in making sentences for day-to-day conversations in different time frame.

UNIT-1

INTRODUCTION & PRESENTATION:

Introduction, Alphabets & Accents Culture, Formal & Informal – Use of 'tu' and 'vous', Map of France: Geographical, Administrative Greeting, Presenting oneself & others, Asking & giving identity, Days of the week, Months of the year, Numbers, Nationality, Profession, Making a visiting card salutations, Gestures & Handshakes.

UNIT-2 RENDEZVOUS:

Conversations, approaching someone, Tele conversation, Buying a train ticket, Numbers the formula to write a post card, Culture and Life in France.

UNIT-3 AGENDA & INVITATION:

Conversations, Time, Fixing a meeting, Alimentation, Moments of the day (from morning to night), Punctuality, Good moments of the day, Inviting someone, Accepting & Refusing Invitations, Family tree, Describing a house interior.

UNIT-4 VACATION & SHOPPING:

Describing an event, Reservations at a Hotel, Describing a person, Expressing opinion, Indication of time: Depuis & pendant, Gestures: Polite & Impolite, A French vacation, Culture, Making a purchase, Choosing & Paying, Trying a dress on, Talking about weather, Understanding a Weather Bulletin, Comparison, Dress & weather, Dialogue between a client and an employee of a store and Money in everyday life in France: Parking ticket / telephone card.

UNIT-5 ITINERARY, EXCURSION & WEEKEND:

Asking for & giving directions, Giving order / advice / prohibition, Reservation at a restaurant, Taking an order , Asking for bill at a Restaurant, Expression of Quantity, Alimentation: Shopping list (portions), Making Suggestion & Proposal, Going for an outing, Acceptance & Refusal of an invitation, Giving arguments: favour & against, A French Weekend.

Text Books:

1. CAMPUS 1 Methode de Francais, Jacky Girardet, CLE International Paris, Nouvelle Edition, 2002.
2. La France de toujours: Civilisation, Nelly Mauchamp; CLE International, 2004.

Reference Books:

1. Declic 1: method francais; Jacques Balnc, Jean-Michel Cartier, Pierre Lederlion; CLE International, 2004.
2. Nouveau Sans Frontieres; P Dominique, Jacky Girardet , et al. – Vols. 1, 2 & 3. French Edition, 1989.

SRI VENKATESWARA COLLEGE OF ENGINEERING &TECHNOLOGY
(AUTONOMOUS)

III B. Tech I Semester (Common to all branches)

Course Code: 23AHS32 **GERMAN LANGUAGE**
(AUDIT COURSE)

L	T	P	C
2	**0**	**0**	**0**

COURSE OUTCOMES:

After successful completion of this course, the students will be able to:

CO1: Understand fundamentals of German language, sounds, pronunciations, sentence structures and the verb conjugation.

CO2: Comprehend and apply the knowledge of vocabulary and phrases in day-to-day real-life conversation.

CO3: Analyze various sentence structures by examining the rules of grammar in speaking and writing.

CO4: Demonstrate various verb structures of English and German language effectively in professional writing.

UNIT-1 GERMAN SOUNDS

Vowels, consonants, diphthongs, umlaut, the nouns, gender distinctions, cases, definite and indefinite articles, conjugation of verbs, verbs with separable and inseparable prefixes, modal verbs, personal pronouns, possessive pronouns, reflexive pronouns, cases nominative, accusative and dative. Structure of sentence and categories of sentences, subordinate clause, causative and conditional sentences; A very interesting slideshow presentation is held to enlighten the students about the culture, people, and lifestyle in Germany.

UNIT-2 SENTENCE FORMATION

Infinite sentences, use of conjunctive-I and conjunctive-II, plusquam perfect, modal verb, Conjunction, temporal, subordinate clauses & complex sentences.

UNIT-3 GERMAN BASIC GRAMMAR

Verbs: Different forms, past tense and present perfect tense, adjectives and their declension, degrees of comparison; Prepositions, genitive case conjunctive. Different conjunctions (coordinating and subordinating), simple, complex and compound sentences, active and passive voice, relative pronouns.

UNIT-4 PURPOSE OF LANGUAGE STUDY

Pictures and perceptions, conflicts and solutions, change and the future, the purpose of the study of the German language, listening, understanding, reacting, speaking, communicating, use of language, pronunciation and intonation, reading, reading and understanding, writing, text writing, text forming, use of language, language reflection, building up the language, language comparison, culture reflection, other cultures and cultural identity.

UNIT-5 GERMAN ADVANCED COMMUNICATION LEVEL - 1

The significance of language study, Speaking and thinking, Self – discovery, Communication, Language Competence, Language and culture, Language changes, Connection with other areas of study, The mother language and the other languages.

Text Books:

1. Korbinian, Lorenz Nieder Deutschals Fremdsprache IA. Ausländer, “German Language”, Perfect Paperback Publishers, 1st Edition, 1992.

2. Deutschals Fremd sprache, IB, Erganzungskurs, “German Language”, Front Cover. Klett, Glossar Deutsch-Spanisch Publishers, 1st Edition, 1981.

Reference Books:

1. Griesbach, “Moderner Gebrauch der deutschen Sprache”, Schulz Publishers, 10th Edition, 2011.

2. Anna Quick, Hermann Glaser U.A, “Intermediate German: A Grammar and workbook”, Paperback, 1st Edition, 2006

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B. Tech I Semester (Common to all branches)

23AHS33 - JAPANESE LANGUAGE (AUDIT COURSE)

L	T	P	C
2	0	0	0

Course Outcomes:

CO1: Remember and understand Japanese alphabet and demonstrate basic structures of sentences in reading and writing.

CO2: Examine the limitations of language by examining pronouns, verbs form, adjectives and conjunctions.

CO3: Analyze the skills of vocabulary and apply it to learn time and dates and express them in Japanese.

CO4: Demonstrate the formation of simple questions and answers in Japanese to know the Japanese culture and etiquette.

UNIT – I

INTRODUCTION TO JAPANESE SYLLABLES AND GREETINGS – Introduction of Japanese language, alphabets; Hiragana, katakana, and Kanji Pronunciation, vowels and consonants.

Hiragana – writing and reading; Vocabulary: 50 Nouns and 20pronouns, Greetings.

UNIT – II

DEMONSTRATIVE PRONOUNS, VERBS AND SENTENCE FORMATION - Grammar: N1 wa N2 desu, Japanese Numerals, Demonstrative pronoun - Kore, Sore, Are and Dore (This, That, Over there, which) Kono, sono, Ano and Dono (this, that, over there, which) Kochira, Sochira, Achira and Dochira. this way....) Koko, Soko, Asoko and Doko (Here, There,...location), Classification of verbs Be verb desu Present and Present negative Basic structure of sentence (Subject+ Object+ Verb) Katakana- reading and writing **UNIT - III**

CONJUNCTION, ADJECTIVES, VOCABULARY AND ITS MEANING - Conjunction- Ya…..nado Classification of Adjectives ‘I’ and ‘na’-ending Set phrase – Onegaishimasu – Sumimasen, wakarimasen Particle –Wa, Particle-Ni ‘Ga imasu’ and ‘Ga arimasu’ for Existence of living things and non- living things Particle- Ka, Ni, Ga, Days/ Months /Year/Week (Current, Previous, Next, Next to Next); Nation, People and Language Relationship of family (look and learn); Simple kanji recognition.

UNIT - IV

FORMING QUESTIONS AND GIVING ANSWERS - Classification of Question words (Dare, Nani, Itsu, Doyatte, dooshite, Ikutsu, Ikura); Classification of Te forms, Polite form of verbs.

UNIT - V

EXPRESSING TIME, POSITION AND DIRECTIONS – Classification of question words (Doko, Dore, Dono, Dochira); Time expressions (Jikan), Number of hours, Number of months, calendar of a month; Visiting the departmental store, railway stations, Hospital (Byoki), office and University.

Text Book:

1. Easy Japanese: Learn to Speak Japanese Quickly, Emiko Koromi, Tuttle Publishing, 2018.

Reference Book:

1. Japanese Hiragana and Katakana Made Easy: An Easy Step-By-Step Workbook to Learn the Japanese Writing System, Lingo Mastery, 2022.

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B.Tech II Semester

23ACA15 - CLOUD COMPUTING FOR AI	**L**	**T**	**P**	**C**
(Professional Core)				
Common to CSE(AI&ML), CSE(AI)	**3**	**0**	**0**	**3**
Professional Elective -III-B.Tech AI&DS				

Course Objectives:

- To introduce the concepts, models, and services of cloud computing and its role in AI.
- To explore the architecture and deployment of AI applications on cloud platforms.
- To equip students with skills in using cloud-based tools and services for AI/ML workloads.
- To understand data storage, processing, and security in cloud for AI tasks.
- To apply cloud computing principles to real-world AI-based solutions.

Course Outcomes:

After completion of this course, students will be able to:

- Explain cloud computing architecture, services, and deployment models.
- Utilize cloud platforms (AWS, GCP, Azure) for training and deploying AI models.
- Handle large-scale data storage and processing in the cloud environment.
- Integrate AI workflows using serverless and container-based architectures.
- Analyze challenges in security, cost, scalability, and performance of cloud-based AI systems.

UNIT I: Introduction to Cloud Computing and AI Integration

Basics of Cloud Computing: Characteristics, Models, and Services, Cloud Service Models: IaaS, PaaS, SaaS, Deployment Models: Public, Private, Hybrid, Community, AI and Cloud Convergence: Benefits and Challenges, Use Cases of AI in Cloud: NLP, Vision, Analytics, Overview of Cloud Providers for AI: AWS, Azure, GCP.

UNIT II: Storage, Computing, and Data Processing in the Cloud

Cloud Storage Services: S3, Blob, BigQuery, Virtualization and Elastic Computing, Distributed Computing with Hadoop and Spark, Data Ingestion and Processing Pipelines, Data Lakes and Warehousing in the Cloud, Cost Optimization for Storage and Compute Resources.

UNIT III: Cloud-based Machine Learning and Deep Learning

ML Services on AWS (SageMaker), Azure ML, GCP Vertex AI, Training and Deploying Models on Cloud, AutoML and Custom ML Model Workflows, GPUs/TPUs for Model Training, Experiment Tracking and Model Evaluation, Integration of Notebooks (Jupyter, Colab) with Cloud Storage.

UNIT IV: Advanced Cloud Concepts for AI Applications

Containers and Docker for AI Applications, Kubernetes and Cloud-native AI Workflows, Serverless Computing: AWS Lambda, Azure Functions, CI/CD Pipelines for AI Models in Cloud, Scaling AI Applications using Load Balancers and Auto-Scaling. Monitoring and Logging in Cloud for AI Workflows.

UNIT V: Security, Ethics, and Case Studies in Cloud AI

Security and Privacy in Cloud-based AI, Identity and Access Management (IAM) in Cloud, Cost Management and Billing for AI Services, Ethical Issues and Fairness in Cloud AI, Case Study: AI in Healthcare Cloud Solutions, Case Study: Real-Time Analytics in Financial Cloud Services.

Textbooks:

1. Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, Mastering Cloud Computing, McGraw-Hill.
2. Judith Hurwitz et al., Cloud Computing for Dummies, Wiley.
3. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly.

Reference Books:

1. Kai Hwang, Geoffrey C. Fox, Jack G. Dongarra, Distributed and Cloud Computing, Morgan Kaufmann.
2. Tomasz Kajdanowicz et al., Practical Cloud AI, Springer.
3. Mark Wilkins, AI and Machine Learning for Coders in Cloud, Packt Publishing.

Online Learning Resources:

- AWS Cloud Practitioner & Machine Learning Path – AWS Training
- Google Cloud AI and ML Specialization – Coursera
- Microsoft Azure AI Engineer Associate – Learn Portal
- IBM Cloud and AI Learning – Cognitive Class
- Cloud Computing and Distributed Systems (CLOUDS) Lab – University of Melbourne

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B.Tech II Semester

23ACA16 - BIG DATA ANALYTICS & AI APPLICATIONS	**L**	**T**	**P**	**C**
(Professional Core)				
Common to CSE(AI&ML), CSE(AI)	**3**	**0**	**0**	**3**

Course Objectives:

- To introduce the fundamentals of big data and its role in AI-driven applications.
- To explore big data tools and technologies such as Hadoop, Spark, and NoSQL databases.
- To enable students to build scalable AI pipelines for data analytics.
- To apply AI/ML algorithms for real-time and batch processing environments.
- To demonstrate use cases of big data in domains like healthcare, finance, and IoT using AI.

Course Outcomes:

After completion of the course, students will be able to:

- Understand the architecture and ecosystem of big data processing.
- Analyze and manage large-scale datasets using Hadoop and Spark.
- Apply AI/ML techniques to extract insights from big data.
- Design and implement scalable data pipelines using distributed frameworks.
- Solve real-world domain problems with AI-powered big data solutions.

UNIT I: Introduction to Big Data and Analytics Ecosystem

Definition and Characteristics of Big Data – Volume, Velocity, Variety, Veracity, Value, Types of Analytics: Descriptive, Diagnostic, Predictive, Prescriptive, Big Data Challenges and Opportunities, Hadoop Ecosystem Overview: HDFS, MapReduce, YARN, NoSQL Databases: Key-Value, Columnar, Document, Graph Models, Data Lake vs. Data Warehouse.

UNIT II: Big Data Tools and Frameworks

Apache Spark Architecture and RDDs, Spark SQL, DataFrames, and Datasets, Spark Streaming for Real-Time Analytics, Kafka for Data Ingestion and Message Queues, Hive, Pig, and Impala for Big Data Querying, Comparative Analysis of Hadoop vs. Spark.

UNIT III: Machine Learning on Big Data

Introduction to MLlib and Scikit-learn, Data Preprocessing for Big Data ML Pipelines, Supervised Learning: Classification and Regression on Large Datasets, Unsupervised Learning: Clustering and Dimensionality Reduction, Model Evaluation and Validation Techniques, Distributed Training and Optimization Techniques.

UNIT IV: AI Applications on Big Data

Predictive Maintenance using Big Data & AI, Fraud Detection in Banking with Machine Learning, AI in Healthcare: Diagnosis, Genomics, Patient Monitoring, Retail and E-commerce Analytics, AI for Smart Cities and IoT Sensor Data Analysis, Evaluation of Real-Time AI Applications on Streaming Data.

UNIT V: Advanced Topics and Case Studies

Deep Learning on Big Data using TensorFlow on Spark, Explainable AI (XAI) in Big Data Environments, Ethical Issues and Data Governance in Big Data AI, Edge Computing and AI for Low Latency Applications, Case Study 1: AI-Powered Big Data in Healthcare, Case Study 2: Big Data AI Solution in Smart Manufacturing.

Textbooks:

1. Big Data: Principles and Paradigms by Rajkumar Buyya, Rodrigo N. Calheiros, Amir Vahid Dastjerdi – Wiley
2. Learning Spark: Lightning-Fast Big Data Analysis by Jules S. Damji et al. – O'Reilly
3. Data Science and Big Data Analytics by EMC Education Services – Wiley

Reference Books:

1. Designing Data-Intensive Applications by Martin Kleppmann – O'Reilly
2. Machine Learning with Spark by Rajdeep Dua, Tathagata Das – Packt Publishing
3. Streaming Systems by Tyler Akidau – O’Reilly Media
4. Artificial Intelligence for Big Data by Anand Deshpande – Packt

Online Learning Resources:

- https://www.coursera.org/specializations/big-data – Coursera Big Data Specialization
- https://spark.apache.org/docs/latest/ – Apache Spark Documentation
- https://www.edx.org/course/big-data-analysis-with-python – edX
- https://www.udacity.com/course/ai-for-business-leaders--nd088 – Udacity AI for Business
- https://www.kaggle.com/learn/intro-to-machine-learning – Kaggle ML Tutorials
- https://data-flair.training/blogs/apache-spark-tutorial/ – Spark Tutorials

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B.Tech II Semester

23ACA17 - FULL STACK AI DEVELOPMENT **(Professional Core)**	**L**	**T**	**P**	**C**
Common to CSE(AI&ML), CSE(AI)	**3**	**0**	**0**	**3**

Course Objectives:

- To equip students with knowledge and skills for building end-to-end AI-powered web applications.
- To provide hands-on experience in integrating machine learning models with frontend and backend technologies.
- To teach model deployment, version control, and MLOps best practices.
- To expose students to full stack frameworks and cloud-based deployment platforms.
- To prepare students for real-world AI applications in production settings.

Course Outcomes:

Upon completion of this course, the student will be able to:

- Understand and apply full stack development principles in the context of AI solutions.
- Build and serve machine learning models via RESTful APIs.
- Design frontend interfaces for interaction with AI models.
- Deploy AI applications using modern DevOps tools and cloud platforms.
- Manage datasets, model versioning, and workflows in production-grade systems.

UNIT I: Introduction to Full Stack AI Development

Overview of Full Stack Development in AI Context, Layers: Frontend, Backend, ML Layer, and Deployment Layer, Tools and Technology Stack (React, Node.js, Flask, Django, FastAPI, TensorFlow, PyTorch, MongoDB, PostgreSQL), Understanding Model Lifecycle and ML Ops

UNIT II: Backend Development and API Integration

Introduction to Flask / FastAPI for model serving, RESTful API design and documentation (Swagger/OpenAPI), Connecting AI/ML models to APIs, Authentication, request handling, and session management, Error handling and response structuring

UNIT III: Frontend Development for AI Interfaces

Overview of frontend frameworks (React/Angular/Vue), Creating dynamic forms and dashboards for AI input/output, Data visualization using Chart.js, D3.js, Connecting frontend to API endpoints, Responsive design for AI application UX

UNIT IV: Model Deployment and MLOps

Basics of CI/CD pipelines for AI models, Using Docker for containerization, Deployment on cloud platforms (Heroku, AWS, GCP), Introduction to MLflow, DVC, and model versioning, Logging, monitoring, and performance metrics

UNIT V: Capstone Project and Case Studies

Full Stack AI Project Planning & Implementation, Use cases: Chatbot, Recommendation System, Image Classification App, NLP Web App, Industry-oriented workflows and best practices, Ethical considerations and data governance in AI applications

Textbooks:

1. **"Full Stack Deep Learning"** by Hamel Husain et al. *(online version available at fullstackdeeplearning.com)*
2. **"Building Machine Learning Powered Applications"** by Emmanuel Ameisen, O'Reilly.
3. **"Flask Web Development"** by Miguel Grinberg.

Reference Books:

1. **"Machine Learning Engineering"** by Andriy Burkov.
2. **"Designing Data-Intensive Applications"** by Martin Kleppmann.
3. **"Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow"** by Aurélien Géron.

Online Resources:

- Full Stack Deep Learning Course
- FastAPI Documentation
- Flask Mega-Tutorial

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B.Tech II Semester

23ACM08-GRAPH NEURAL NETWORKS	**L**	**T**	**P**	**C**
(Professional Elective-II)				
Common to CSE(AI&ML), CSE(AI)	**3**	**0**	**0**	**3**

Course Objectives:

- To introduce the fundamentals of graph theory and graph-structured data.
- To explore the concepts of neural networks extended to non-Euclidean domains.
- To understand architectures and algorithms behind various types of GNNs.
- To apply GNN models in real-world applications such as recommendation, social networks, and bioinformatics.
- To enable students to build and evaluate GNN models using frameworks like PyTorch Geometric and DGL.

Course Outcomes:

- Upon completion of the course, students will be able to:
- Understand the basics of graph structures and their significance in machine learning.
- Learn and implement different types of GNN architectures.
- Apply GNNs to real-world structured data problems.
- Use modern libraries and tools to train and evaluate GNNs.
- Analyze the effectiveness and limitations of GNNs in different domains.

UNIT I: Fundamentals of Graph Theory and Machine Learning on Graphs

Introduction to Graphs: Nodes, Edges, Adjacency Matrix, Types of Graphs: Directed, Undirected, Weighted, Bipartite, Graph Traversal Algorithms (BFS, DFS), Graph Representations for ML (Adjacency List, Matrix, Laplacian), Node, Edge, and Graph-level Prediction Problems, Motivation and Challenges for Learning on Graphs.

UNIT II: Spectral and Spatial Methods for Graph Learning

Spectral Graph Theory Basics, Graph Convolution via Spectral Methods, Chebyshev and First-order Approximations, Spatial Graph Convolutions, Comparison of Spectral vs Spatial GNNs, Graph Laplacian and Eigenvalue Properties.

UNIT III: Graph Neural Network Architectures

Graph Convolutional Networks (GCNs), Graph Attention Networks (GATs), GraphSAGE: Sampling and Aggregation, Graph Isomorphism Networks (GIN), Message Passing Neural Networks (MPNNs), Inductive vs Transductive GNN Learning.

UNIT IV: Applications of GNNs

Node Classification (e.g., Cora, Citeseer), Link Prediction (e.g., Recommender Systems), Graph Classification (e.g., Molecule Property Prediction), Traffic Forecasting and Social Network Modeling, GNNs in Healthcare and Bioinformatics, Explainability and Interpretability in GNNs.

UNIT V: Implementation, Optimization, and Recent Advances

Overview of PyTorch Geometric and DGL, Data Loading and Preprocessing for Graph Datasets, Model Training, Loss Functions, and Evaluation Metrics, Hyperparameter Tuning in GNNs, Recent Research Trends and Architectures (e.g., Heterogeneous GNNs, Graph Transformers), Challenges and Future Directions in GNNs.

Textbooks:

1. Zonghan Wu, Shirui Pan, Fengwen Chen, Guodong Long, Chengqi Zhang, Philip S. Yu, A Comprehensive Survey on Graph Neural Networks, IEEE Transactions on Neural Networks and Learning Systems, 2021.
2. Yao Ma, Jiliang Tang, Deep Learning on Graphs, Cambridge University Press, 2021.
3. William L. Hamilton, Graph Representation Learning, Morgan & Claypool Publishers, 2020.

Reference Books:

1. Barrett, Jure Leskovec, Mining of Massive Datasets, Cambridge University Press.
2. Thomas Kipf, GCN and related papers and tutorials (arXiv).
3. Petar Veličković, Graph Attention Networks (original paper and slides).
4. Michael Bronstein et al., Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges (arXiv preprint).

Online Learning Resources:

1. https://pytorch-geometric.readthedocs.io/ – PyTorch Geometric Docs
2. https://cs.stanford.edu/people/jure/ – Stanford GNN Projects
3. https://www.coursera.org/learn/graph-neural-networks – Coursera GNN Course by Stanford

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B.Tech II Semester

23ACM09 - RECOMMENDER SYSTEMS
(Professional Elective-II)
Common to CSE(AI&ML), CSE(AI)

L	T	P	C
3	0	0	3

Course Objectives:
- To provide students with basic concepts and its application in various domain
- To make the students understand different techniques that a data scientist needs to know for analysing big data
- To design and build a complete machine learning solution in many application domains.

Course Outcomes:
After completion of the course, students will be able to
- Aware of various issues related to Personalization and Recommendations.
- Design and implement a set of well-known Recommender System approaches used in E commerce and Tourism industry.
- Develop new Recommender Systems for a number of domains especially, Education, Health-care.

UNIT-I An Introduction to Recommender Systems, Neighborhood-Based Collaborative Filtering
Lecture 8Hrs
Introduction, Goals of Recommender Systems, Basic Models of Recommender Systems, Domain Specific Challenges in Recommender Systems. Advanced Topics and Applications. Introduction, Key Properties of Ratings Matrices, Predicting Ratings with Neighborhood- Neighborhood-Based Collaborative Filtering: Based Methods, Clustering and Neighborhood-Based Methods, Dimensionality Reduction and Neighborhood Methods, Graph Models for Neighborhood-Based Methods, A Regression Modelling View of Neighborhood Methods.

UNIT-II Model-Based Collaborative Filtering, Content-Based Recommender Systems Lecture 9Hrs Introduction, Decision and Regression Trees, Rule-Based Collaborative Filtering, Naive Bayes Collaborative Filtering, Using an Arbitrary Classification Model as a Black-Box, Latent Factor Models, Integrating Factorization and Neighborhood Models. Content-Based Recommender Systems: Introduction, Basic Components of Content-Based Systems, Preprocessing and Feature Extraction, Learning User Profiles and Filtering, Content-Based Versus Collaborative Recommendations, Using Content-Based Models for Collaborative Filtering, Summary.

UNIT-III Knowledge-Based Recommender Systems, Ensemble Based and Hybrid Recommender Systems
Lecture 9Hrs
Introduction, Constraint-Based Recommender Systems, Case-Based Recommenders, Persistent Personalization in Knowledge-Based Systems, Summary. Introduction, Ensemble Methods from the Classification Perspective, Weighted Hybrids, Switching Hybrids, Cascade Hybrids, Feature Augmentation Hybrids, Meta-Level Hybrids, Feature Combination Hybrids, Summary.

UNIT-IV Evaluating Recommender Systems, Context-Sensitive Recommender Systems Lecture 8Hrs
Introduction, Evaluation Paradigms, General Goals of Evaluation Design, Design Issues in Offline Recommender Evaluation, Accuracy Metrics in Offline Evaluation, Limitations of Evaluation Measures, Limitations of Evaluation Measures. Introduction, The Multidimensional Approach, Contextual Pre-filtering: A Reduction-Based Approach, Contextual Pre-filtering: A Reduction-Based Approach, Contextual Modelling.

UNIT-V Time- and Location-Sensitive Recommender Systems Lecture 8Hrs
Introduction, Temporal Collaborative Filtering, Discrete Temporal Models, Location-Aware Recommender Systems, Location-Aware Recommender Systems Location-Aware Recommender Systems, Summary.

Textbooks:
1. Charu C. Aggarwal, “Recommender Systems”, Springer,2016.

Reference Books:

1. Francesco Ricci, LiorRokach, “Recommender Systems Handbook”, 2nd ed., Springer, 2015 Edition

Online Learning Resources:

1. Recommendation System -Understanding The Basic Concepts (analyticsvidhya.com)
2. Recommender Systems | Coursera

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B.Tech II Semester

	L	T	P	C
23A3CD20 - PREDICTIVE ANALYTICS **(Professional Elective-II)** **Common to CSE(AI&ML), CSE(AI)**	3	0	0	3

Course Objectives:
- To introduce the fundamental concepts and techniques of predictive analytics.
- To apply statistical models and machine learning algorithms for prediction.
- To interpret model performance using evaluation metrics.
- To explore feature engineering, model tuning, and cross-validation.
- To implement predictive solutions for real-world business and research problems.

Course Outcomes:
Upon successful completion of the course, students will be able to:
- Understand the principles and importance of predictive analytics.
- Apply regression and classification models for predictive tasks.
- Perform data preprocessing, feature selection, and transformation.
- Evaluate and validate models using standard metrics.
- Design predictive solutions to solve domain-specific challenges.

UNIT I: Introduction to Predictive Analytics
Introduction to Predictive Analytics and Business Intelligence, Types of Predictive Models: Classification, Regression, Time Series, Supervised vs Unsupervised Learning, Predictive Modeling Workflow, Applications in Marketing, Finance, Healthcare, Challenges in Predictive Analytics.

UNIT II: Data Preparation and Feature Engineering
Data Cleaning: Handling Missing, Noisy, and Inconsistent Data, Feature Selection and Dimensionality Reduction (PCA, LDA), Feature Scaling: Normalization, Standardization, Encoding Categorical Variables, Feature Extraction and Construction, Dealing with Imbalanced Datasets.

UNIT III: Predictive Modeling with Regression and Classification
Linear Regression and Polynomial Regression, Logistic Regression for Binary Classification, Decision Trees and Random Forest, k-Nearest Neighbors (k-NN) and Naïve Bayes, Support Vector Machines (SVM), Model Selection and Comparison.

UNIT IV: Model Evaluation and Validation
Training, Testing, and Validation Sets, Cross-Validation Techniques (k-Fold, Stratified, LOOCV), Evaluation Metrics: Accuracy, Precision, Recall, F1 Score, ROC-AUC, Confusion Matrix and Classification Report, Bias-Variance Trade-off and Overfitting, Hyperparameter Tuning: Grid Search, Random Search.

UNIT V: Advanced Topics and Applications
Ensemble Learning: Bagging, Boosting (AdaBoost, XGBoost), Predictive Analytics with Time Series (ARIMA, Prophet), Deep Learning for Predictive Modeling (ANNs, LSTM), Use of Predictive Analytics in IoT, Retail, and Healthcare, Ethics and Privacy in Predictive Analytics, Building and Deploying End-to-End Predictive Systems.

Textbooks:

1. **Dean Abbott**, Applied Predictive Analytics: Principles and Techniques for the Professional Data Analyst, Wiley, 2014.
2. **John D. Kelleher, Brendan Tierney**, Data Science: Predictive Analytics and Data Mining, MIT Press, 2018.

Reference Books:

1. **Galit Shmueli et al.**, Data Mining for Business Analytics: Concepts, Techniques, and Applications in R, Wiley, 2017.
2. **Eric Siegel**, Predictive Analytics: The Power to Predict Who Will Click, Buy, Lie, or Die, Wiley, 2016.
3. **Trevor Hastie, Robert Tibshirani, Jerome Friedman**, The Elements of Statistical Learning, Springer, 2009.

Online Learning Resources:

1. *https://www.coursera.org/specializations/predictive-analytics* – Coursera Specialization
2. https://www.edx.org/course/data-science-and-machine-learning-capstone – edX Predictive Analytics Courses
3. https://www.kaggle.com/learn/intro-to-machine-learning – Kaggle Tutorials

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B.Tech II Semester

23ACA18 - BLOCKCHAIN FOR AI	**L**	**T**	**P**	**C**
(Professional Elective-II)				
Common to CSE(AI&ML), CSE(AI)	**3**	**0**	**0**	**3**

Course Objectives:

- To understand the foundational concepts of blockchain technology and its architecture.
- To explore smart contracts, consensus algorithms, and distributed ledger technology.
- To investigate the integration of AI with blockchain for secure, decentralized applications.
- To develop blockchain-enabled AI solutions for real-world use cases.
- To understand the ethical, security, and scalability challenges in Blockchain-AI ecosystems.

Course Outcomes:

Upon successful completion of the course, students will be able to:

- Explain the fundamentals of blockchain and its components.
- Analyze the role of consensus mechanisms in maintaining trust and decentralization.
- Apply blockchain for secure data sharing in AI systems.
- Develop and deploy smart contracts using Ethereum/Solidity.
- Evaluate blockchain-based AI applications in healthcare, finance, and supply chains.

UNIT I: Blockchain Fundamentals and Architecture

Introduction to Blockchain Technology, Components: Blocks, Hashing, Merkle Trees, Types of Blockchains: Public, Private, Consortium, Distributed Ledger Technology (DLT) and P2P Networks, Blockchain Structure and Mining, Use Cases and Evolution of Blockchain.

UNIT II: Smart Contracts and Consensus Mechanisms

Smart Contracts: Definition, Features, Use Cases, Ethereum and Solidity Basics, Consensus Algorithms: PoW, PoS, DPoS, PBFT, Gas, Transactions, and Events in Ethereum, Hyperledger Fabric: Architecture and Chaincode, Deployment and Testing of Smart Contracts.

UNIT III: Integration of Blockchain and AI

Motivation for Integrating Blockchain with AI, Decentralized AI Models and Federated Learning, Secure Model Sharing and Provenance, Blockchain for Data Integrity in AI Systems, AI for Blockchain (e.g., optimizing consensus), Case Study: Decentralized AI Marketplace.

UNIT IV: Applications of Blockchain in AI Systems
Blockchain for Explainable and Trusted AI, Applications in Healthcare and Genomics, Blockchain for Autonomous Vehicles and IoT, Financial AI Systems with Smart Contracts, Supply Chain and Logistics Intelligence, NFT-based AI Applications (Digital Identity, IP).

UNIT V: Security, Privacy and Challenges in Blockchain-AI

Security Challenges: Sybil Attacks, 51% Attacks, Privacy Preservation and Zero Knowledge Proofs, Scalability and Energy Concerns in Blockchain-AI, Ethical and Legal Concerns in AI with Blockchain, Interoperability of Blockchain Platforms, Future Trends: Quantum-Resistant Blockchain-AI.

Textbooks:

1. Imran Bashir, Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications, Packt, 2020.
2. Melanie Swan, Blockchain: Blueprint for a New Economy, O'Reilly Media, 2015.
3. Joseph Holbrook, Architecting AI Solutions on Blockchain, Packt Publishing, 2020.

Reference Books:

1. Arshdeep Bahga, Vijay Madisetti, Blockchain Applications: A Hands-On Approach, VPT, 2017.
2. Karamjit Singh, Blockchain for AI: Use Cases and Implementation, Springer, 2023.
3. Roger Wattenhofer, The Science of the Blockchain, 2016.

Online Learning Resources:

- Coursera: Blockchain Specialization – University at Buffalo
- edX: Blockchain Fundamentals – UC Berkeley
- Coursera: AI and Blockchain – IBM

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B.Tech II Semester

23ACM10 -AUTOMATION OF MODEL BUILDING (Professional Elective-III) Common to CSE(AI&ML), CSE(AI)	**L**	**T**	**P**	**C**
	3	**0**	**0**	**3**

Course Outcomes:

Upon successful completion of the course, students will be able to:

- Acquire, demonstrate and apply advanced knowledge in the area of Automation engineering.
- Identify problems in the field of Automation engineering, formulate them and solve by using advanced techniques.
- Apply engineering and scientific principles for the effective management of Automation systems.

UNIT I

Introduction to AutoML: Scope of machine learning, Concepts of AutoML, Purpose of AutoML, requirement of automate ML,Core components of AutoML system, Building prototype subsystems for each component, end & end AutoML , System Overview of AutoML libraries.

UNIT - II

Machine Learning Using Python: Technical requirements, Machine learning, Linear regression, linear regression, evaluation metrics regression algorithms, Logistic regression, evaluation metrics of logistic regression, classification algorithms, Decision trees, Support Vector Machines, k-Nearest Neighbours, Ensemble methods, Comparing the results of classifiers, Crossvalidation: Clustering.

UNIT - III

Data Preprocessing: Technical requirements, Data transformation, Numerical data transformation, Categorical data transformation, Text Preprocessing, Feature selection, Feature generation.

Automated Algorithm Selection: Technical requirements, Computational complexity, Differences in training and scoring time, Linearity versus non-linearity, Necessary feature transformations, supervised ML, Unsupervised AutoML.

UNIT - IV

Hyperparameter Optimization: Technical requirements, Hyperparameters, Warm start, Bayesian based hyperparameter tuning, An example system.

Creating AutoML Pipelines: Technical requirements, An introduction to machine learning pipelines, A simple pipeline, Function Transformer, A complex pipeline.

UNIT - V

Dive into Deep Learning: Technical requirements, Overview of neural networks, Neuron, Activation functions, A feed-forward neural network using Keras: Autoencoders, Convolutional Neural Networks.

Critical Aspects of ML and Data Science Projects: Machine learning as a search, Trade-offs in machine learning, Engagement model for a typical data science project, The phases of an engagement model.

Textbooks:

1. Sibanjan Das, UmitMertCakmak "Hands-On Automated Machine Learning" Packt Publishing, 2018.

Reference Books:

1. EthernAlpaydin, "Introduction to Machine Learning", MIT Press,2004.
2. Stephen Marsland, "Machine Learning -An Algorithmic Perspective", Second Edition, Chapman and Hall/CRC Machine Learning and Pattern Recognition Series,2014.

Online Learning Resources:

Machine Learning for Construction Automation – NPTEL+

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B.Tech II Semester

23AMB15-AI FOR FINANCE (Professional Elective-III) Common to CSE(AI&ML), CSE(AI)	L	T	P	C
	3	0	0	3

Course Objectives:

- To introduce the role of Artificial Intelligence (AI) in financial applications and decision-making.
- To understand financial data types, sources, and processing methods.
- To apply machine learning and deep learning models in various finance sectors.
- To analyze risk, fraud detection, credit scoring, and portfolio management using AI.
- To evaluate ethical and regulatory challenges in AI-enabled finance.

Course Outcomes:

Upon successful completion of the course, students will be able to:

- Describe the fundamentals of AI techniques applicable to finance.
- Analyze financial time series data using AI-based models.
- Apply machine learning for fraud detection and credit risk analysis.
- Build predictive models for stock prices, trading, and customer segmentation.
- Evaluate the limitations and ethical implications of AI in financial systems.

UNIT I: Introduction to Finance and AI Applications
Introduction to Financial Markets and Instruments, Overview of AI Techniques in Finance, Types of Financial Data: Market, Transactional, Customer, Financial Statements and Key Indicators, AI Use Cases in Banking, Insurance, and Investment, FinTech and the Rise of Robo-Advisors.

UNIT II: Machine Learning in Finance
Supervised Learning for Credit Scoring, Unsupervised Learning for Customer Segmentation, Feature Engineering for Financial Data, Handling Imbalanced Datasets in Fraud Detection, Time Series Forecasting with Regression and ARIMA, Model Validation and Backtesting in Finance.

UNIT III: Deep Learning and NLP in Finance
Introduction to Deep Learning for Finance, Stock Price Prediction using LSTM and RNNs, Sentiment Analysis from Financial News and Tweets, NLP for Document Classification: Earnings Reports, Chatbots and Virtual Assistants in Banking, Reinforcement Learning for Portfolio Optimization.

UNIT IV: AI-Driven Financial Applications

Fraud Detection Systems using ML and DL, Credit Risk and Loan Default Prediction, AI in Algorithmic and High-Frequency Trading, Robo-Advisors: Architecture and Optimization, Blockchain and AI Integration for Financial Security, Case Studies: AI in Wealth Management & Insurance.

UNIT V: Ethics, Regulation, and Future of AI in Finance

Regulatory Frameworks in AI-based Finance, Explainability and Interpretability of Financial Models, Ethical Issues: Bias, Fairness, Transparency, Data Privacy and GDPR in Financial AI, Responsible AI Practices in Finance, Emerging Trends: Quantum AI, Decentralized Finance (DeFi).

Textbooks:

1. Yves Hilpisch, Artificial Intelligence in Finance: A Python-Based Guide, O’Reilly, 2020.
2. Yves Hilpisch, Python for Finance: Mastering Data-Driven Finance, O’Reilly, 2018.
3. Markus Loecher, Machine Learning for Finance, Packt Publishing, 2021.

Reference Books:

1. A. W. Lo, The Evolution of Technical Analysis, Wiley Finance, 2010.
2. Tony Guida, Big Data and Machine Learning in Quantitative Investment, Wiley, 2019.
3. Tucker Balch, AI for Trading – Georgia Tech Specialization, Coursera.

Online Learning Resources:

- Coursera: AI for Trading – by NYIF and Google Cloud
- edX: Artificial Intelligence in Finance – NYIF
- Udemy: Machine Learning and AI in Finance
- DataCamp: Financial Trading with Python
- YouTube: AI for Finance by Sentdex, Two Minute Papers, and DataProfessor

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B.Tech II Semester

23ACM11 - SOCIAL NETWORK ANALYSIS
(Professional Elective-III)
Common to B.Tech -AI&DS and B.Tech -AI
(Professional Elective-II)
Common to CSE(AI&ML) and CSE(AI)

L	T	P	C
3	0	0	3

Course Objectives:

- To introduce the fundamentals and key concepts of social network theory and graph theory.
- To analyze the structure and properties of large-scale social networks.
- To apply centrality, influence, and community detection measures.
- To model information diffusion and network dynamics.
- To implement real-world social network analysis using tools and datasets.

Course Outcomes:

At the end of the course, the student will be able to:

- Understand basic network models and social network structures.
- Analyze key properties like centrality, clustering, and small-world effect.
- Apply community detection algorithms and influence maximization.
- Interpret diffusion models for viral marketing and information spread.
- Use tools such as Gephi, NetworkX, or SNAP for real-world SNA.

UNIT I: Introduction to Social Networks and Graph Theory
Basic Concepts: Graphs, Nodes, Edges, Directed/Undirected Graphs, Real-world Examples: Facebook, Twitter, LinkedIn, Adjacency Matrix and Graph Representation, Types of Social Networks: Ego, Bipartite, Multilayer, Degree Distribution, Path Length, and Connectivity, Random Graph Models: Erdős–Rényi and Watts-Strogatz.

UNIT II: Structural Properties of Networks
Network Centrality Measures: Degree, Closeness, Betweenness, Eigenvector Centrality and PageRank, Network Clustering and Community Detection Basics, Triadic Closure and Clustering Coefficient, Small-world Phenomenon and Milgram's Experiment, Homophily, Influence, and Structural Balance.

UNIT III: Community Detection and Subgroup Analysis
Girvan–Newman Algorithm and Modularity, Label Propagation and Louvain Method, Clique Detection and k-Core Decomposition, Overlapping Communities and Fuzzy Clustering, Cohesive Subgroups and Structural Equivalence, Evaluation Metrics: NMI, Modularity Score.

UNIT IV: Information Diffusion and Influence in Networks
Models of Diffusion: Linear Threshold and Independent Cascade, Influence Maximization and Viral Marketing, Contagion Models and Epidemic Spreading, Rumor Propagation and Cascade Models, Information Bottlenecks and Bridges, Measuring Influence and Reach.

UNIT V: Tools, Applications, and Ethics in SNA

SNA Tools: Gephi, Pajek, NetworkX, SNAP, Case Study: Twitter and Hashtag Analysis, LinkedIn Network Mining and Graph Features, Applications in Marketing, Security, and Epidemiology, Ethical Issues in Social Network Data Mining, Building and Visualizing Your Own Social Graph.

Textbooks:

1. Wasserman, S., & Faust, K., Social Network Analysis: Methods and Applications, Cambridge University Press, 1994.
2. Easley, D., & Kleinberg, J., Networks, Crowds, and Markets: Reasoning About a Highly Connected World, Cambridge University Press, 2010.
3. Newman, M., Networks: An Introduction, Oxford University Press, 2010.

Reference Books:

1. Borgatti, S. P., Everett, M. G., & Johnson, J. C., Analyzing Social Networks, SAGE Publications, 2018.
2. Barabási, A.-L., Linked: How Everything Is Connected to Everything Else, Basic Books, 2014.
3. Hansen, D., Shneiderman, B., & Smith, M. A., Analyzing Social Media Networks with NodeXL, Elsevier, 2020.

Online Learning Resources:

- Coursera – Social Network Analysis (University of Michigan)
- [YouTube – NetworkX and Gephi Tutorials (freeCodeCamp, TheNetNinja)]
- edX – Networks: Friends, Money, and Bytes (University of California, Berkeley)
- Khan Academy – Graph Theory

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B.Tech II Semester

23ACA19 - CYBERSECURITY & AI-DRIVEN THREAT DETECTION
(Professional Elective-III)
Common to CSE(AI&ML) and CSE(AI)

L	T	P	C
3	0	0	3

Course Objectives:

- To provide a foundational understanding of cybersecurity principles and threat landscapes.
- To explore the application of AI and machine learning techniques in detecting cyber threats.
- To analyze malware behavior, intrusion patterns, and anomaly detection using intelligent systems.
- To evaluate and build automated systems for real-time security analytics.
- To understand the ethical, legal, and societal implications of AI-driven security systems.

Course Outcomes:

At the end of the course, students will be able to:

- Understand cybersecurity frameworks, threat types, and vulnerabilities.
- Apply AI/ML techniques for cyber threat identification and classification.
- Analyze patterns in malware, network traffic, and security logs.
- Design and evaluate intelligent intrusion detection and prevention systems.
- Explore ethical hacking practices and policy aspects in AI-based security.

UNIT I: Fundamentals of Cybersecurity
Introduction to Cybersecurity: CIA Triad, Threats & Vulnerabilities, Types of Attacks: Malware, Phishing, DDoS, Insider Threats, Security Policies and Access Controls, Risk Assessment and Vulnerability Management, Cryptography Basics: Symmetric, Asymmetric, Hash Functions, Cybersecurity Frameworks: NIST, ISO 27001, OWASP.

UNIT II: Machine Learning for Cyber Threat Detection
Supervised and Unsupervised Learning in Security Contexts, Feature Engineering for Security Data, Classification Models for Intrusion Detection (SVM, RF, KNN), Clustering Techniques for Anomaly Detection, Evaluation Metrics: Accuracy, Precision, ROC, F1 Score, Case Study: AI for Email Phishing Detection.

UNIT III: Deep Learning in Threat Intelligence
Deep Neural Networks for Cybersecurity, RNNs and LSTMs for Log and Sequence Data, Autoencoders for Anomaly Detection, CNNs for Malware Classification using Binary Analysis, Adversarial Attacks on AI-based Security Systems, Case Study: Threat Detection using Deep Learning.

UNIT IV: Real-Time Threat Detection and SIEM Systems
Security Information and Event Management (SIEM), Log Analysis and Real-Time Alerting, Threat Intelligence Platforms (TIPs), Integration of AI in SIEM Tools (Splunk, ELK Stack), Network Traffic and Packet Inspection using ML, SOC Operations and Automation using AI

UNIT V: Ethical Hacking, Privacy, and Legal Aspects
Penetration Testing & Ethical Hacking with AI Tools, Red Team vs. Blue Team Simulation, Data Privacy Regulations: GDPR, HIPAA, Cyber Laws, AI Bias and Fairness in Security Decision-Making, Case Study: Ethical Dilemmas in AI Security Systems, Future Trends: Zero Trust, AI SOC, Federated Threat Detection.

Textbooks:

1. Stallings, W., Network Security Essentials: Applications and Standards, Pearson Education.
2. Shon Harris & Fernando Maymi, CISSP All-in-One Exam Guide, McGraw Hill.
3. Emmanuel Tsukerman, Machine Learning for Cybersecurity Cookbook, Packt Publishing.
4. Clarence Chio & David Freeman, Machine Learning and Security, O'Reilly Media.

Reference Books:

1. John Paul Mueller, Luca Massaron, Machine Learning for Dummies, Wiley.
2. Mark Stamp, Information Security: Principles and Practice, Wiley.
3. Bruce Schneier, Secrets and Lies: Digital Security in a Networked World, Wiley.
4. Shai Shalev-Shwartz and Shai Ben-David, Understanding Machine Learning, Cambridge University Press.

Online Learning Resources:

- Coursera – AI for Cybersecurity (IBM)
- edX – Cybersecurity Fundamentals by Rochester Institute of Technology
- MIT OpenCourseWare – Computer and Network Security
- [YouTube – Cybersecurity & AI Tutorials by Simplilearn, Great Learning]
- Udemy – Machine Learning for Cybersecurity
- Splunk Documentation – AI & Threat Detection

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IIIB.Tech. II Semester

		L	T	P	C
23ACE44	**DISASTER MANAGEMENT (Open Elective – II)**	**3**	**0**	**0**	**3**

Course Objectives:

The objectives of this course are to make the student :

1. To understand the fundamental concepts of natural disasters, their occurrence, and disaster risk reduction strategies.
2. To analyze the impact of cyclones on structures and explore retrofitting techniques for adaptive reconstruction.
3. To apply wind engineering principles and computational techniques in designing wind-resistant structures.
4. To evaluate earthquake effects on buildings and develop strategies for seismic retrofitting.
5. To assess seismic safety planning, design considerations, and innovative construction materials for disaster-resistant structures.

Course Outcomes:

After successful completion of this course, students will be able to:

1. Understand the fundamental concepts of natural disasters, their occurrence, and disaster risk reduction strategies.
2. Analyze the impact of cyclones on structures and explore retrofitting techniques for adaptive reconstruction.
3. Apply wind engineering principles and computational techniques in designing wind-resistant structures.
4. Evaluate earthquake effects on buildings and develop strategies for seismic retrofitting.
5. Assess seismic safety planning, design considerations, and innovative construction materials for disaster-resistant structures.

UNIT – I

Introduction to Natural Disasters– Brief Introduction to Different Types of Natural Disasters, Occurrence of Disasters in Different Climatic and Geographical Regions, Hazard Maps (Earthquake and Cyclone) of The World and India, Regulations for Disaster Risk Reduction, Post-Disaster Recovery and Rehabilitation (Socioeconomic Consequences).

UNIT – II

Cyclones and Their Impact– Climate Change and Its Impact On Tropical Cyclones, Nature of Cyclonic Wind, Velocities and Pressure, Cyclone Effects, Storm Surges, Floods, and Landslides. Behavior of Structuresin Past Cyclones and Windstorms, Case Studies. Cyclonic Retrofitting, Strengthening of Structures, and Adaptive Sustainable Reconstruction. Life-Line Structures Such as Temporary Cyclone Shelters.

UNIT – III

Wind Engineering and Structural Response– Basic Wind Engineering, Aerodynamics of Bluff Bodies, Vortex Shedding, and Associated Unsteadiness Along and Across Wind forces. Lab: Wind Tunnel Testing and Its Salient Features. Introduction to Computational Fluid Dynamics (CFD). General Planning and Design Considerations Under Windstorms and Cyclones. Wind Effects On Buildings, towers, Glass Panels, Etc., and Wind-Resistant Features in Design. Codal Provisions, Design Wind Speed, Pressure Coefficients. Coastal Zoning Regulations for Construction and Reconstruction in Coastal Areas. Innovative Construction Materials and Techniques, Traditional Construction Techniques in Coastal Areas.

UNIT – IV

Seismology and Earthquake Effects– Causes of Earthquakes, Plate Tectonics, Faults, Seismic Waves; Magnitude, Intensity, Epicenter, Energy Release, and Ground Motions. Earthquake Effects– On Ground, Soil Rupture, Liquefaction, Landslides. Performance of Ground and Buildings in Past Earthquakes– Behavior of Various Types of Buildings and Structures, Collapse Patterns; Behavior of Non-Structural Elements Such as Services, Fixtures, and Mountings – Case Studies. Seismic Retrofitting– Weakness in Existing Buildings, Aging, Concepts in Repair, Restoration, and Seismic Strengthening.

UNIT – V

Planning and Design Considerations for Seismic Safety– General Planning and Design Considerations; Building forms, Horizontal and Vertical Eccentricities, Mass and Stiffness Distribution, Soft Storey Effects, Etc.; Seismic Effects Related to Building Configuration. Plan and Vertical Irregularities, Redundancy, and Setbacks. Construction Details– Various Types of Foundations, Soil Stabilization, Retaining Walls, Plinth Fill, Flooring, Walls, Openings, Roofs, Terraces, Parapets, Boundary Walls, Underground and Overhead Tanks, Staircases, and Isolation of Structures. Innovative Construction Materials and Techniques. Local Practices– Traditional Regional Responses. Computational Investigation Techniques.

TEXT BOOKS:

1. David Alexander, *Natural Disasters*, 1st Edition, CRC Press, 2017.
2. Edward A. Keller and Duane E. DeVecchio, *Natural Hazards: Earth's Processes as Hazards, Disasters, and Catastrophes*, 5th Edition, Routledge, 2019.

REFRENCE BOOKS:

1. Ben Wisner, J.C. Gaillard, andIlanKelman (Editors), *Handbook of Hazards and Disaster Risk Reduction and Management*, 2nd Edition, Routledge, 2012.
2. Damon P. Coppola, *Introduction to International Disaster Management*, 4th Edition, Butterworth-Heinemann, 2020.
3. BimalKanti Paul, *Environmental Hazards and Disasters: Contexts, Perspectives and Management*, 2nd Edition, Wiley-Blackwell, 2020.

Online Learning Resources:

https://nptel.ac.in/courses/124107010
https://onlinecourses.swayam2.ac.in/cec19_hs20/preview

CO – PO Articulation Matrix

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO -1	3	-	-	-	-	2	-	2	2	-	-	-	3	3
CO -2	-	3	-	-	2	-	-	-	-	-	-	2	3	-
CO -3	3	-	-	3	-	-	3	-	-	2	-	-	-	3
CO -4	-	-	3	-	3	-	-	2	-	-	-	-	3	-
CO -5	-	-	-	3	-	3	3	3	2	-	-	-	-	3

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B.Tech – II Semester

23ACE45- SUSTAINABILITY IN ENGINEERING PRACTICES (OPEN ELECTIVE – II)	L	T	P	C
	3	0	0	3

Course Objectives:

The objectives of this course are to make the student :

1. To understand the fundamentals of sustainability, the carbon cycle, and the environmental impact of construction materials.
2. To analyze sustainable construction materials, their durability, and life cycle assessment.
3. To apply energy calculations in construction materials and assess their embodied energy.
4. To evaluate green building standards, energy codes, and performance ratings.
5. To assess the environmental effects of energy use, climate change, and global warming.

Course Outcomes:

After successful completion of this course, students will be able to:

1. Understand the fundamentals of sustainability, the carbon cycle, and the environmental impact of construction materials.
2. Analyze sustainable construction materials, their durability, and life cycle assessment.
3. Apply energy calculations in construction materials and assess their embodied energy.
4. Evaluate green building standards, energy codes, and performance ratings.
5. Assess the environmental effects of energy use, climate change, and global warming.

UNIT – I INTRODUCTION

Introduction and Definition of Sustainability - Carbon Cycle - Role of Construction Material: Concrete and Steel, Etc. - CO_2Contribution From Cement and Other Construction Materials.

UNIT – II MATERIALS USED IN SUSTAINABLE CONSTRUCTION

Construction Materials and Indoor Air Quality - No/Low Cement Concrete - Recycled and Manufactured Aggregate - Role of QC and Durability - Life Cycle and Sustainability.

UNIT – III ENERGY CALCULATIONS

Components of Embodied Energy - Calculation of Embodied Energy for Construction Materials - Energy Concept and Primary Energy - Embodied Energy Via-A-Vis Operational Energy in Conditioned Building - Life Cycle Energy Use

UNIT – IV GREEN BUILDINGS

Control of Energy Use in Building - ECBC Code, Codes in Neighboring Tropical Countries - OTTV Concepts and Calculations – Features of LEED and TERI – GRIHA Ratings - Role of Insulation and Thermal Properties of Construction Materials - Influence of Moisture Content and Modeling - Performance Ratings of Green Buildings - Zero Energy Building

UNIT – V ENVIRONMENTAL EFFECTS

Non-Renewable Sources of Energy and Environmental Impact– Energy Norm, Coal, Oil, Natural Gas - Nuclear Energy - Global Temperature, Green House Effects, Global Warming - Acid Rain: Causes, Effects and Control Methods - Regional Impacts of Temperature Change.

Text Books:

1. Charles J Kibert, Sustainable Construction: Green Building Design & Delivery, 4th Edition , Wiley Publishers 2016.
2. Steve Goodhew, Sustainable Construction Process, Wiley Blackwell,UK, 2016.

Refrence Books:

1. Craig A. Langston & Grace K.C. Ding, Sustainable Practicesin the Built Environment, Butterworth Heinemann Publishers, 2011.
2. William P Spence, Construction Materials, Methods & Techniques (3e), Yesdee Publication Pvt. Ltd, 2012.

Online Learning Resources:
https://archive.nptel.ac.in/courses/105/105/105105157/

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

IIIB.Tech. II Semester

23AEE33 RENEWABLE ENERGY SOURCES
(Open Elective-II)

L	T	P	C
3	0	0	3

Course Outcomes (CO): At the end of the course the student will be able to:

CO 1: Understand principle operation of various renewable energy sources. L1
CO 2: Identify site selection of various renewable energy sources. L2
CO 3: Analyze various factors affecting on solar energy measurements, wind energy conversion techniques, Geothermal, Biomasss, Tidal Wave and Fuel cell energies L3
CO 4: Design of Solar PV modules and considerations of horizontal and vertical axis Wind energy systems. L5
CO 5: Apply the concepts of Geo Thermal Energy, Ocean Energy, Bio mass and Fuel Cells for generation of power. L4

UNIT I Solar Energy:

Solar radiation - beam and diffuse radiation, solar constant, Sun at Zenith, attenuation and measurement of solar radiation, local solar time, derived solar angles, sunrise, sunset and day length. flat plate collectors, concentrating collectors, storage of solar energy-thermal storage.

UNIT II PV Energy Systems:

Introduction, The PV effect in crystalline silicon basic principles, the film PV, Other PV technologies, Solar PV modules from solar cells, mismatch in series and parallel connections design and structure of PV modules, Electrical characteristics of silicon PV cells and modules, Stand-alone PV system configuration, Grid connected PV systems.

UNIT III Wind Energy:

Principle of wind energy conversion; Basic components of wind energy conversion systems; wind mill components, various types and their constructional features; design considerations of horizontal and vertical axis wind machines: analysis of aerodynamic forces acting on wind mill blades; wind data and energy estimation and site selection considerations.

UNIT IV Geothermal Energy:

Estimation and nature of geothermal energy, geothermal sources and resources like hydrothermal, geo-pressured hot dry rock, magma. Advantages, disadvantages and application of geothermal energy, prospects of geothermal energy in India.

UNIT – V Miscellaneous Energy Technologies:

Ocean Energy: Tidal Energy-Principle of working, Operation methods, advantages and limitations. Wave Energy-Principle of working, energy and power from waves, wave energy conversion devices, advantages and limitations.
Bio mass Energy: Biomass conversion technologies, Biogas generation plants, Classification, advantages and disadvantages, constructional details, site selection, digester design consideration
Fuel cell: Principle of working of various types of fuel cells and their working, performance and limitations.

Text books:

1. G. D. Rai, “Non-Conventional Energy Sources”, 4th Edition, Khanna Publishers, 2000.

2. Chetan Singh Solanki “Solar Photovoltaics fundamentals, technologies and applications” 2nd Edition PHI Learning Private Limited. 2012.

Reference Books:

1. Stephen Peake, "Renewable Energy Power for a Sustainable Future", Oxford International Edition, 2018.

2. S. P. Sukhatme, "Solar Energy",3rd Edition, Tata Mc Graw Hill Education Pvt. Ltd, 2008.

3. B H Khan , " Non-Conventional Energy Resources", 2nd Edition, Tata Mc Graw Hill Education Pvt Ltd, 2011.

4. S. Hasan Saeed and D.K.Sharma,"Non-Conventional Energy Resources",3rd Edition, S.K.Kataria& Sons, 2012.

5. G. N. Tiwari and M.K.Ghosal, "Renewable Energy Resource: Basic Principles and Applications", Narosa Publishing House, 2004.

Online Learning Resources:

1. https://nptel.ac.in/courses/103103206
2. https://nptel.ac.in/courses/108108078

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B. Tech - II Semester

(Common to CE, EEE, ECE, EBM, CSE, CSC, CSM, CAI, CSD, CSBS, AI, AI& DS, IT, IOT)

23AME39: AUTOMATION AND ROBOTICS

	L	T	P	C
Open Elective - II	**3**	**0**	**0**	**3**

Course Outcomes:

CO1: Understand and analyze the structure and functions of automated manufacturing systems, and evaluate hardware components for efficient production.

CO2: Analyze and design automated flow lines with or without buffer storage, perform quantitative evaluations, apply assembly line balancing techniques.

CO3: Classify robot configurations, select suitable actuators and sensors, analyze and apply automation and robotics principles to optimize production efficiency and flexibility.

CO4: Apply kinematic and dynamic modeling using D-H notation and select appropriate hardware and control strategies for real-world industrial scenario to analyze and design automated and robotic systems. **CO5**: Design, program, and implement robotic systems, understand and apply robotics technology to manufacturing tasks.

UNIT-IIntroduction to Automation:

Introduction to Automation, Need, Types, Basic elements of an automated system, Manufacturing Industries, Types of production, Functions in manufacturing, Organization and information processing in manufacturing, Automation strategies and levels of automation, Hardware components for automation and process control, mechanical feeders, hoppers, orienters, high speed automatic insertion devices.

UNIT –II Automated flow lines:

Automated flow lines, Part transfer methods and mechanisms, types of Flow lines, flow line with/without buffer storage, Quantitative analysis of flow lines. Assembly line balancing: Assembly process and systems assembly line, line balancing methods, ways of improving line balance, flexible assembly lines.

UNIT- III Introduction to Industrial Robotics:

Introduction to Industrial Robotics, Classification of Robot Configurations, functional line diagram, degrees of freedom. Components common types of arms, joints grippers, factors to be considered in the design of grippers. Robot actuators and Feedback components: Actuators, Pneumatic, Hydraulic actuators, Electric & Stepper motors, comparison. Position sensors - potentiometers, resolvers, encoders - velocity sensors, Tactile sensors, Proximity sensors.

UNIT- IV Manipulator Kinematics:

Manipulator Kinematics, Homogenous transformations as applicable to rotation and transition - D-H notation, Forward inverse kinematics.

Manipulator Dynamics: Differential transformations, Jacobians, Lagrange - Euler and Newton – Euler formations. Trajectory Planning: Trajectory Planning and avoidance of obstacles path planning, skew motion, joint integrated motion - straight line motion.

UNIT- V Robot Programming:

Robot Programming, Methods of programming - requirements and features of programming languages, software packages. Problems with programming languages.

Robot Application in Manufacturing: Material Transfer - Material handling, loading and unloading - Process spot and continuous arc welding & spray painting - Assembly and Inspection.

Text Books:

1. Automation, Production systems and CIM, M.P. Groover /Pearson Edu.
2. Industrial Robotics - M.P. Groover, TMH.

References:

1. Robotics, Fu K S, McGraw Hill, 4th edition, 2010.
2. An Introduction to Robot Technology, P. Coiffet and M. Chaironze, Kogam Page Ltd. 1983 London.
3. Robotic Engineering, Richard D. Klafter, Prentice Hall
4. Robotics, Fundamental Concepts and analysis – Ashitave Ghosal, Oxford Press, 1/e, 2006
5. Robotics and Control, Mittal R K & Nagrath I J, TMH.

Online Learning Resources:

- https://www.youtube.com/watch?v=yxZm9WQJUA0&list=PLRLB5WCqU54UJG45UnazSY mn mhl-gt76o
- https://www.youtube.com/watch?v=6f3bvIhSWyM&list=PLRLB5WCqU54X5Vy4Dwjf SODT3ZJgwEjyE

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B. Tech - I Semester (EEE)

III B. Tech – II Semester (Common to all branches except ECE, EBM and EEE) Course

Code: 23AEC26

DIGITAL CIRCUITS

L	T	P	C
3	0	0	3

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Learn Boolean algebra, logic simplification techniques, and combinational circuit design.
L1

CO2: Analyze combinational circuits like adders, subtractors, and code converters. L2 **CO3:**
Explore combinational logic circuits and their applications in digital design. L3 **CO4:**
Understand sequential logic circuits, including latches, flip-flops, counters, and shift
registers.L1

CO5: Gain knowledge about programmable logic devices and digital IC's. L3

UNIT-I:

Logic Simplification and Combinational Logic Design: Review of Boolean Algebra and De Morgan's Theorem, SOP & POS forms, Canonical forms, Introduction to Logic Gates, Ex-OR, Ex-NOR operations, Minimization of Switching Functions: Karnaugh map method, Logic function realization: AND-OR, OR-AND and NAND/NOR realizations.

UNIT-II:

Introduction to Combinational Design 1: Binary Adders, Subtractors and BCD adder, Code converters - Binary to Gray, Gray to Binary, BCD to excess3, BCD to Seven Segment display.

UNIT-III:

Combinational Logic Design 2: Decoders, Encoders, Priority Encoder, Multiplexers, Demultiplexers, Comparators, Implementations of Logic Functions using Decoders and Multiplexers.

UNIT-IV:

Sequential Logic Design: Latches, Flip-flops, S-R, D, T, JK and Master-Slave JK FF, Edge triggered FF, set up and hold times, Ripple counters, Shift registers.

UNIT-V:

Programmable Logic Devices: ROM, Programmable Logic Devices (PLA and PAL).

Digital IC's: Decoder (74x138), Priority Encoder (74x148), multiplexer (74x151) and de-multiplexer (74x155), comparator (74x85).

Text Books:

1. Digital Design, M.Morris Mano & Michel D. Ciletti, 5th Edition, Pearson Education, 1999.
2. Switching theory and Finite Automata Theory, Zvi Kohavi and Nirah K.Jha, 2nd Edition, Tata McGraw Hill, 2005.

Reference Books:

1. Fundamentals of Logic Design, Charles H Roth,Jr., 5th Edition, Brooks/cole Cengage Learning, 2004

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B.Tech-II Semester

	L	T	P	C
23ACS31 : ADVANCED OPTIMIZATION TECHNIQUES	**3**	**0**	**0**	**3**

Course Outcomes:

At the end of the course, the student will be able to

CO1: Comprehend the techniques and applications of Engineering optimization

CO2: Analyze characteristics of a general Optimization problem.

CO3: Apply basic concepts of mathematics to formulate an Multivariate optimization problem

CO4: Analyse various methods of solving the unconstrained minimization problem

CO5: Analyze and appreciate variety of performance measures for various optimization problems.

UNIT I :

Foundations of Optimization- Classification of Optimization Techniques - Choosing the Right Optimization Technique -Categories of Optimization Problems - Classical Optimization Techniques Linear Programming - Non-Linear Programming - Integer Programming - Dynamic Programming Stochastic Programming - Metaheuristic Optimization Algorithms

UNIT II:

Mathematical Formulation of Optimisation Problems- Classes of Optimisation Problems- Basic Techniques in Optimisation- Advanced Optimisation Techniques -Handling Constraints in Optimisation Problems

Introduction to Optimization Problems- Classification of Optimization Problems- Single Variable Optimization Problems- Multivariable Optimization Problems without Constraints

UNIT III:

Multivariable Optimization Problems Multivariable Problems with Constraints -Maximization and Minimization Problems- Single Variable Optimization -Necessary and Sufficient Conditions for Optimum Interpolation Method in Optimization Quadratic Optimization Problems

UNIT IV:

Region Elimination Methods: An Overview -Understanding the Internal Halving Method -Step-by-step Guide to the Internal Halving Method- Fibonacci in Optimization: An Introduction Fibonacci Search Method in Detail

Basics of Optimization -Mathematical Foundations for Optimization- Unconstrained Multivariable BOptimization -Constrained Multivariable Optimization

UNIT V:

Introduction to One-Dimensional Search Classification of One-Dimensional Search Methods Exploring Analytical Methods Unveiling Stationary Points Direct Substitution Method Constrained Variation: A Closer Look Penalty Function Method in Optimization The Power of Lagrangian Multiplier Diving into the Kuhn-Tucker Theorem

Fundamental Principles of Numerical Search Direction of Search in Optimization Techniques Final Stage in Search: Convergence and Termination Direct Search Methods in Optimization Pattern Search Methods: An Advanced Optimization Technique

Text Books

1. Bazaraa, M. S., Sherali, H. D., & Shetty, C. M. (2006). "Non-linear Programming: Theory and Algorithms". Wiley-Interscience, 3rd edition.
2. Introduction to Optimization by Pablo Pedregal
3. Optimization: Theory and Practice by Gordon S. G. Beveridge and Ronald S. Schechter.
4. Nonlinear Programming: Concepts, Algorithms, and Applications to Chemical Processes by Lorenz T. Biegler.
5. Numerical Optimization by Jorge Nocedal and Stephen J. Wright.

References

1. C.J. Ray, Optimum Design of Mechanical Elements , Wiley, 2007.
2. R. Saravanan, Manufacturing Optimization through Intelligent Techniques , Taylor & Francis Publications, 2006.
3. D. E. Goldberg, Genetic algorithms in Search, Optimization, and Machine Learning , Ad

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B. Tech II Semester

L	T	P	C
3	0	0	3

23AHS36 MATHEMATICAL FOUNDATION OF QUANTUM TECHNOLOGIES
(Common to all Branches)
Open Elective - II

Course Outcomes:
After successful completion of this course, the students should be able to:

CO1	Understand the Transformation theory and Hilbert space.
CO2	Analyze the properties and operators of Hilbert space and apply Eigen values to it.
CO3	Apply statistics to measure theory, uncertainty relations and radiation theory.
CO4	Evaluate problems on reversibility, equilibrium and macroscopic measurements.
CO5	Formulate problems of composite system and measuring process

UNIT I: Introductory Considerations

The origin of the Transformation Theory, The Original Formulation of Quantum Mechanics, The Equivalence of the two Theories: (i) The Transformation Theory, (ii) Hilbert Space.

UNIT II: Abstract Hilbert Space

The definition of Hilbert space, The Geometry of Hilbert space, Degression on the Conditions A-E, Closed linear Manifolds, Operators in Hilbert space, The Eigen Value Problem, Continuation, Initial Consideration concerning the Eigenvalue Problem, Degression on the Existence and Uniqueness of solutions of the Eigenvalue Problems, Cumulative operators, The Trace.

UNIT III: The Quantum Statistics

The statistical assertions of quantum mechanics, the statistical interpretation, Simultaneous Measurability and Measurability in General, Uncertainty Relations, Projections as Propositions, Radiation Theory.

UNIT IV: Deductive development of the Theory and general considerations

The fundamental basis of the statistical theory, Conclusions from Experiments.Measurement and reversibility, Thermodynamics Considerations, Reversibility and equilibrium problems, The Macroscopic Measurement.

UNIT V: The measuring Process

Formulation of the problems, Composite systems, discussion of the Measuring process.

Textbooks:

1. John von Neumann and Robert T Beyer, Mathematical Foundations of Quantum Mechanics, Princeton Univ. Press (1996).
2. Srinivas, M. D., Measurements and Quantum Probabilities, University Press, Hyderabad (2001).

Reference Books:

1. Leonard Schiff, Quantum Mechanics, Mc, Graw Hill (Education) (2010).
2. Parthasarathy. K. R., Mathematical Foundations of Quantum, Hindustan Book Agency, New Delhi.
3. Gerad Tesch, Mathematical Methods in Quantum Mechanics with application to Schrodinger operators, Graduate Studies in Mathematics, 99, AMS, Providence, 2009.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	1	-	-	-	-	-	-	1
CO2	3	3	2	3	2	-	-	-	-	-	-	2
CO3	3	3	3	3	2	1	-	-	-	-	-	2
CO4	3	3	2	2	1	-	-	-	-	-	-	1
CO5	3	3	3	3	2	-	-	-	-	-	-	2

3-High Mapping **2- Medium Mapping** **1-Low Mapping**

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

(Open Elective-II)

III B.Tech II Semester (Common to all branches)

23AHS37: PHYSICS OF ELECTRONIC MATERIALS AND DEVICES

L	T	P	C
3	0	0	3

COURSE OUTCOMES

After successful completion of the course, the students will be able to,

CO1: Understand crystal growth and thin film preparation
CO2: Summarize the basic concepts of semiconductors
CO3: Illustrate the working of various semiconductor devices
CO4: Analyze various luminescent phenomena and the devices based on these concepts
CO5: Explain the working of different display devices

UNIT-I: **Fundamentals of Materials Science** **9hrs**

Introduction, Phase rule, Phase Diagram, Elementary idea of Nucleation and Growth, Methods of crystal growth. The basic idea of point, line, and planar defects. Concept of thin films, preparation of thin films, Deposition of thin film using sputtering methods (RF and glow discharge).

UNIT-II: Semiconductors **9hrs**

Introduction, charge carriers in semiconductors, effective mass, Diffusion and drift, Diffusion and recombination, Diffusion length. The Fermi level & Fermi-Dirac distribution, Electron and Hole in quantum well, Change of electron-hole concentration- Qualitative analysis, Temperature dependency of carrier concentration, Conductivity and mobility, Effects of temperature and doping on mobility, High field effects.

UNIT- III: Physics of Semiconductor Devices **9hrs**

Introduction, Band structure, PN junctions and their typical characteristics under equilibrium and under bias, Heterojunctions, Transistors, MOSFETs.

UNIT-IV: Excitons and Luminescence: **9hrs**

Luminescence: Different types of luminescence, basic definitions, Light emission in solids, Inter- band luminescence, Direct and indirect gap materials.
Photoluminescence : General Principles of photo luminescence, Excitation and relaxation, OLED, Quantum-dot.
Electro-luminescence: General Principles of electro luminescence, light emitting diode, diode laser.

UNIT-V: Display devices **9hrs**

LCD, three-dimensional display: Holographic display, light-field displays: Head-mounted display, MOEMS (Micro-Opto-Electro-Mechanical Systems) and MEMS displays.

Text books:

1. Principles of Electronic Materials and Devices- S.O. Kasap, McGraw-Hill

Education (India) Pvt. Ltd., 4th edition, 2021.

2. Semi conductor physics & devices: basic principles, 4th Edition, McGraw-Hill,2012.

Reference Books:

1. Solid State Electronic Devices- B.G. Street man and S. Banerjee, PHIL earning, 6th edition
2. Electronic Materials Science-Eugene A.Irene, Wiley, 2005
3. Electronic Components and Materials, Grover and Jamwal, Dhanpat Raiand Co., New Delhi.,2012.
4. An Introduction to Electronic Materials for Engineers-Wei Gao, Zhengwei Li, Nigel Sammes, World Scientific Publishing Co. Pvt. Ltd. 2nd Edition, 2011

NPTEL course links: https://nptel.ac.in/courses/113/106/113106062/
https://onlinecourses.nptel.ac.in/noc20_ph24/preview

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	1	-	-	-	-	-	-	-
CO2	3	3	2	1	1	-	-	-	-	-	-	-
CO3	3	3	2	1	1	-	-	-	-	-	-	-
CO4	3	2	1	1	-	-	-	-	-	-	-	-
CO5	3	3	1	1	-	-	-	-	-	-	-	-

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

CHEMISTRY OF POLYMERS AND APPLICATIONS
(Open Elective-II)
III B. Tech, II Semester (Common to all branches)

23AHS38 : CHEMISTRY OF POLYMERS AND APPLICATIONS

L	T	P	C
3	0	0	3

COURSE OUTCOMES

After successful completion of the course, the students will be able to,

CO1: Classify the polymers, Explain polymerization mechanism, Differentiate addition, condensation polymerizations, Describe measurement of molecular weight of polymer

CO 2: Describe the physical and chemical properties of natural polymers and Modified cellulosics.

CO3: Differentiate Bulk, solution, Suspension, and emulsion polymerization. Describe fibers and elastomers. Identify the thermosetting and thermo polymers.

CO4: Identify types of polymer networks, Describe methods involve in hydrogel preparation, Explain applications of hydrogels in drug delivery,

CO5: Explain classification and mechanism of conducting and degradable polymers.

Unit -I: Polymers-Basics and Characterization **9 hrs**

Basic concepts: monomers, repeating units, degree of polymerization, linear, branched and network polymers, classification of polymers, Polymerization: addition, condensation, copolymerization and coordination polymerization. Average molecular weight concepts: number, weight and viscosity average molecular weights, poly dispersity and molecular weight distribution. Measurement of molecular weight: End group, viscosity, light scattering, osmotic and ultracentrifugation methods, analysis and testing of polymers.

Unit - II: Natural Polymers & Modified cellulosics **9 hrs**

Natural Polymers: Chemical & Physical structure, properties, source, important chemical modifications, applications of polymers such as cellulose, lignin, starch, rosin, shellac, latexes, vegetable oils and gums, proteins.

Modified cellulosics: Cellulose esters and ethers such as Ethyl cellulose, CMC, HPMC, cellulose acetals, Liquid crystalline polymers; specialty plastics- PES, PAES, PEEK, PEA.

Unit - III: Synthetic Polymers **9 hrs**

Addition and condensation polymerization processes– Bulk, Solution, and Suspension and Emulsion polymerization. Preparation and significance, classification of polymers based on physical properties. Thermoplastics, Thermosetting plastics, Fibers and elastomers, General Applications. Preparation of Polymers based on different types of monomers, Olefin polymers (PE, PVC), Butadiene polymers (BUNA-S,BUNA-N), nylons, Urea-formaldehyde, phenol – formaldehyde, Melamine Epoxy and Ion exchange resins.

Unit - IV: Hydrogels of Polymer networks **9 hrs**

Definitions of Hydro gel, polymer networks, Types of polymer networks, Methods involved in hydro gel preparation, Classification, Properties of hydro gels, Applications of hydro gels in drug delivery.

Unit - V: Conducting and Degradable Polymers **9 hrs**

Conducting polymers: Introduction, Classification, Mechanism of conduction in Poly Acetylene, Poly Aniline, Poly Thiophene, Doping, Applications.

Degradable polymers: Introduction, Classifications, Examples, Mechanism of degradation, poly lactic acid, Nylon-6, Polyesters, applications.

Text Books:

1. A Text book of Polymer science, Billmayer
2. Polymer Chemistry– G.S. Mishra
3. Polymer Chemistry– Gowarikar

References Books:

1. Organic polymer Chemistry, K.J. Saunders, Chapman and Hall
2. Advanced Organic Chemistry, B. Miller, Prentice Hall
3. Polymer Science and Technology by Premamoy Ghosh, 3rd edition, McGraw-Hill, 2010.

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

OPEN ELECTIVE – II

L	T	P	C
3	0	0	3

III B. Tech II SEM (Common to all branches)
Course Code: 23AHS39

ACADEMIC WRITING AND PUBLIC SPEAKING

Course Outcomes (CO):

After successful completion of the course, the students will be able to,

CO1: Understand various elements of Academic Writing
CO2: Identify sources and avoid plagiarism
CO3: Demonstrate the knowledge in writing a Research paper
CO4: Analyse different types of essays
CO5: Assess the speeches of others and know the positive strengths of speakers
CO6: Build confidence in giving an impactful presentation to the audience

UNIT - I Introduction to Academic Writing

Introduction to Academic Writing – Essential Features of Academic Writing – Courtesy – Clarity – Conciseness – Correctness – Coherence – Completeness – Types – Descriptive, Analytical, Persuasive, Critical writing

UNIT - II Academic Journal Article

Art of condensation- summarizing and paraphrasing - Abstract Writing, writing Project Proposal, writing application for internship, Technical/Research/Journal Paper Writing – Conference Paper writing - Editing, Proof Reading – Plagiarism

UNIT - III Essay & Writing Reviews

Compare and Contrast – Argumentative Essay – Exploratory Essay – Features and Analysis of Sample Essays – Writing Book Report, Summarizing, Short Story Writing Book/film Review- SoP

UNIT - IV Public Speaking

Introduction, Nature, characteristics, significance of Public Speaking – Presentation – 4 Ps of Presentation– Stage Dynamics – Answering Strategies –Analysis of Impactful Speeches- Speeches for Academic events

UNIT - V Public Speaking and Non-Verbal Delivery

Body Language – Facial Expressions-Kinesics – Oculesics – Proxemics – Haptics – Chronemics - Paralanguage – Signs

Text books:

1. *Critical Thinking, Academic Writing and Presentation Skills*: MG University Edition Paperback – 1 January 2010 Pearson Education; First edition (1 January 2010)
2. Pease, Allan & Barbara. *The Definitive Book of Body Language*RHUS Publishers, 2016

Reference Books:

1. Alice Savage, Masoud Shafiei *Effective Academic Writing*,***2Ed.,***2014 .Oxford University Press
2. Shalini Verma, *Body Language*, S Chand Publications 2011.
3. Sanjay Kumar and Pushpalata,*Communication Skills* 2E 2015, Oxford.
4. Sharon Gerson, Steven Gerson,*Technical Communication Process and Product,* Pearson, New Delhi, 2014
5. *Elbow, Peter. Writing with Power. OUP USA, 1998*

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

III B.Tech II Semester

	L	T	P	C
23ACA20-BIG DATA & CLOUD COMPUTING LAB (Professional Core)	**0**	**0**	**3**	**1.5**

Course Objectives:

- To provide hands-on experience in working with big data tools and cloud computing environments.
- To equip students with practical skills in data ingestion, transformation, analysis, and visualization using Hadoop and Spark ecosystems.
- To enable deployment and management of cloud services using AWS, Azure, or GCP.
- To expose students to cloud-native storage, computing, and container orchestration techniques.
- To integrate big data workflows with cloud infrastructure for scalable, distributed data processing.

Course Outcomes:

- Students will be able to implement big data pipelines and cloud-based solutions using tools like Hadoop, Spark, and cloud platforms such as AWS, Azure, or GCP.
- Students gain proficiency in managing distributed data processing, scalable storage, cloud service provisioning, and deploying applications using containers and orchestration platforms.
- Students will understand the synergy between big data technologies and cloud computing to solve real-world problems efficiently.

List of Lab Experiments:

1. Installation and Configuration of Hadoop Cluster (Single Node & Multi-node)
Hadoop HDFS setup, NameNode & DataNode configuration
2. Working with HDFS: File Operations
Upload, read, delete, and replicate files in HDFS
3. MapReduce Programming Basics
Word count, sorting, and filtering examples in Java/Python
4. Apache Hive & Pig for Querying Large Datasets
Creation of tables, data loading, and running queries
5. Apache Spark Basics: RDDs and DataFrames
Implement Spark transformations and actions
6. Data Preprocessing and Machine Learning using PySpark MLlib
Classification or regression using MLlib pipelines
(Cognitive Level: Apply & Evaluate)
7. Introduction to Cloud Computing and AWS/Azure/GCP Console
Creating virtual machines, basic compute and storage services
8. Cloud Storage and Database Services
Using S3 (AWS), Blob (Azure), or GCP buckets and Cloud SQL/NoSQL
9. Deploying Big Data Workloads on Cloud (EMR, HDInsight, Dataproc)
Running Hadoop/Spark jobs in cloud-managed services
10. Cloud Function/Serverless Deployment

11. Building and deploying a serverless function (e.g., AWS Lambda) Containerization with Docker
12. Building, running, and managing Docker containers Orchestration with Kubernetes in the Cloud Deploy and manage a containerized application using GKE/EKS/AKS

Text Books:

1. Tom White, Hadoop: The Definitive Guide, O’Reilly Media.
2. Rajkumar Buyya et al., Mastering Cloud Computing, McGraw-Hill Education.
3. Holden Karau et al., Learning Spark: Lightning-Fast Big Data Analysis, O’Reilly Media.

Reference Books:

1. Vignesh Prajapati, Big Data Analytics with R and Hadoop, Packt Publishing.
2. Benjamin Bengfort, Data Analytics with Hadoop, O’Reilly.
3. Srinivasan & J.Shrinivasan, Cloud Computing – A Hands-on Approach, Wiley India.

Online Courses:

1. Big Data Specialization – Coursera (University of California San Diego)
2. Cloud Computing Basics – edX (LearnQuest)
3. Data Engineering with Google Cloud – Coursera (Google)

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

II B. Tech, II Semester (Common to CE, ME & ECE)
III B. Tech, I Semester (EEE)
III B. Tech, II Semester (Common to CSE, CSE(AI&ML), CSE (DS), CSE(IOT), CSE(SC) & CSE(CSBS))

23AHS20 : SOFT SKILLS

L	T	P	S
0	1	2	3

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: List out various elements of soft skills

CO2: Describe methods for building a professional image

CO3: Apply critical thinking skills in problem-solving

CO4: Analyse the needs of an individual and team for well-being

CO5: Assess the situation and take necessary decisions

CO6: Create a productive workplace atmosphere using social and work-life Skills ensuring personal and emotional well-being

UNIT- I Soft Skills & Communication Skills

Soft Skills - Introduction, Need - Mastering Techniques of Soft Skills – Communication Skills -Significance, process, types - Barriers of communication - Improving techniques.

Activities:

Intrapersonal Skills- Narration about self-strengths and weaknesses- clarity of thought– self-expression – articulating with felicity. (The facilitator can guide the participants before the activity citing examples from the lives of the great, anecdotes and literary sources) Interpersonal Skills- Group Discussion – Debate – Team Tasks - Book and film Reviews by groups - Group leader presenting views (non- controversial and secular) on contemporary issues or on a given topic. Verbal Communication- Oral Presentations- Extempore- brief addresses and speeches convincing- negotiating- agreeing and disagreeing with professional grace. Non-verbal communication– Public speaking – Mock interviews –presentations with an objective to identify non- verbal clues and remedy the lapses on observation.

UNIT -II Critical Thinking

Active Listening – Observation – Curiosity – Introspection – Analytical Thinking – Open-mindedness – Creative Thinking - Positive thinking – Reflection

Activities:

Gathering information and statistics on a topic - sequencing assorting – reasoning – critiquing issues placing the problem – finding the root cause - seeking viable solution – judging with rationale – evaluating the views of others - Case Study, Story Analysis

UNIT -III Problem Solving & Decision Making

Meaning & features of Problem Solving – Managing Conflict – Conflict resolution – Team building - Effective decision making in teams – Methods & Styles

Activities:

Placing a problem which involves conflict of interests, choice and views – formulating the problem – exploring solutions by proper reasoning – Discussion on important professional, career and organizational decisions and initiate debate on the appropriateness of the decision. Case Study & Group Discussion

UNIT -IV Emotional Intelligence & Stress Management

Managing Emotions – Thinking before Reacting – Empathy for Others – Self-awareness – Self-Regulation – Stress factors – Controlling Stress – Tips

Activities:

Providing situations for the participants to express emotions such as happiness, enthusiasm, gratitude, sympathy, and confidence, compassion in the form of written or oral presentations. Providing opportunities for the participants to narrate certain crisis and stress – ridden situations caused by failure, anger, jealousy, resentment and frustration in the form of written and oral presentation, Organizing Debates

UNIT- V Corporate Etiquette

Etiquette- Introduction, concept, significance - Corporate etiquette - meaning, modern etiquette, benefits - Global and local culture sensitivity - Gender Sensitivity - Etiquette in interaction- Cell phone etiquette - Dining etiquette – Netiquette - AI Tools & Ethics - AI- Enhanced Etiquette Practices - Job interview etiquette - Corporate grooming tips - Overcoming challenges

Activities:

Providing situations to take part in the Role Plays where the students will learn about bad and good manners and etiquette - Group Activities to showcase gender sensitivity, dining etiquette etc. - Conducting mock job interviews - Case Study - Business Etiquette Games.

Note: **The facilitator can guide the participants before the activity citing examples from the lives of the great, anecdotes, epics, scriptures, autobiographies and literary sources which bear true relevance to the prescribed skill.**

Case studies may be given wherever feasible for example for Decision Making- The Prescribed Books:

1. Mitra Barun K, *Personality Development and Soft Skills*, Oxford University Press, Pap/Cdr edition 2012
2. Dr Shikha Kapoor, *Personality Development and Soft Skills: Preparing for Tomorrow*, I K International Publishing House, 2018

Reference Books

1. Sharma, Prashant, Soft Skills: Personality Development for Life Success, BPB Publications 2018.
2. Alex K,Soft SkillsS.Chand & Co, 2012 (Revised edition).
3. Gajendra Singh Chauhan & Sangeetha Sharma, Soft Skills: An Integrated Approach to Maximise PersonalityPublished by Wiley, 2013.
4. Pillai, Sabina & Fernandez Agna, Soft Skills and Employability Skills, Cambridge University Press, 2018
5. Dr. Rajiv Kumar Jain, Dr. Usha Jain,Life Skills(Paperback English)Publisher : Vayu Education of India, 2014

Online Learning Resources:

1. https://youtu.be/DUlsNJtg2L8?list=PLLy_2iUCG87CQhELCytvXh0E_y-bOO1_q
2. https://youtu.be/xBaLgJZ0t6A?list=PLzf4HHlsQFwJZel_j2PUy0pwjVUgj7KlJ
3. https://youtu.be/-Y-R9hDl7lU
4. https://youtu.be/gkLsn4ddmTs
5. https://youtu.be/2bf9K2rRWwo
6. https://youtu.be/FchfE3c2jzc
7. https://www.businesstrainingworks.com/training-resource/five-free-business-etiquette- training-games/
8. https://onlinecourses.nptel.ac.in/noc24_hs15/preview
9. https://onlinecourses.nptel.ac.in/noc21_hs76/preview

decision of King Lear.

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B.Tech II Semester

23AMB16-TECHNICAL REPORT WRITING & IPR

L	T	P	C
2	**0**	**0**	**0**

Course Objectives: ·

1. To enable the students to practice the basic skills of research paper writing
2. To make the students understand the importance of IP and to educate them on the basic concepts of Intellectual Property Rights.
3. To practice the basic skills of performing quality literature review
4. To help them in knowing the significance of real life practice and procedure of Patents.
5. To enable them learn the procedure of obtaining Patents, Copyrights, & Trade Marks

Course Outcomes: On successful completion of this course, the students will be able to:

COURSE OUTCOMES: At the end of the course, students will be able to		**bl**
CO1	Identify key secondary literature related to their proposed technical paper writing	L1, L2
CO2	Explain various principles and styles in technical writing	L1, L2
CO3	Use the acquired knowledge in writing a research/technical paper	L3
CO4	Analyse rights and responsibilities of holder of Patent, Copyright, Trademark, International Trademark etc.	L4
CO5	Evaluate different forms of IPR available at national & international level	L5
CO6	Develop skill of making search of various forms of IPR by using modern tools and techniques.	L3, L6

UNIT – I:

Principles of Technical Writing: styles in technical writing; clarity, precision, coherence andlogical sequence in writing-avoiding ambiguity- repetition, and vague language -highlighting your findings-discussing your limitations -hedging and criticizing -plagiarism and paraphrasing.

UNIT – II:

Technical Research Paper Writing: Abstract- Objectives-Limitations-Review of Literature- Problems and Framing Research Questions- Synopsis

UNIT – III:

Process of research: publication mechanism: types of journals- indexing-seminars-conferences- proof reading plagiarism style; seminar & conference paper writing; Methodology-discussion-results-citation rules

UNIT – IV:

Introduction to Intellectual property: Introduction, types of intellectual property, International organizations, agencies and treaties, importance of intellectual property rights

Trade Marks: Purpose and function of trademarks, acquisition of trade mark rights, protectable matter, selecting and evaluating trade mark, trade mark registration processes.

UNIT – V:

Law of copy rights: Fundamentals of copy right law, originality of material, rights of reproduction, rights to perform the work publicly, copy right ownership issues, copy right registration, notice of copy right, international copy right law

Law of patents: Foundation of patent law, patent searching process, ownership rights and transfer.Patent law, intellectual property audits.

Textbooks:

1. Deborah. E. Bouchoux, *Intellectual Property Rights*, Cengage Learning India, 2013
2. Meenakshi Raman, Sangeeta Sharma. *Technical Communication:Principles and practices.*Oxford.

Reference Books:

1. R.Myneni, *Law of Intellectual Property*, 9th Ed, Asia law House, 2019.
2. Prabuddha Ganguli,*Intellectual Property Rights* Tata Mcgraw Hill, 2001
3. P.Naryan,*Intellectual Property Law*, 3rd Ed ,Eastern Law House, 2007.
4. Adrian Wallwork. *English for Writing Research Papers*Second Edition. Springer Cham Heidelberg New York ,2016
5. Dan Jones, Sam Dragga, *Technical Writing Style*

Online Resources

1. https://theconceptwriters.com.pk/principles-of-technical-writing/
2. https://www.ewh.ieee.org/soc/emcs/acstrial/newsletters/summer10/TechPaperWriting.html
3. https://www.ewh.ieee.org/soc/emcs/acstrial/newsletters/summer10/TechPaperWriting.html
4. https://www.manuscriptedit.com/scholar-hangout/process-publishing-research-paper-journal/
5. https://www.icsi.edu/media/website/IntellectualPropertyRightLaws&Practice.pdf
6. https://lawbhoomi.com/intellectual-property-rights-notes/
7. https://www.extension.purdue.edu/extmedia/ec/ec-723.pdf

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech I Semester

23ACA31 - GENERATIVE AI	**L**	**T**	**P**	**C**
(Professional Core)				
Common to CSE(AI&ML) and B.Tech AI	**3**	**0**	**0**	**3**

Course Objectives

- Introduce the fundamentals of Generative AI, including its principles, architecture, and evolution.
- Provide a deep understanding of Large Language Models (LLMs) and their application in natural language generation tasks.
- Develop practical knowledge of Prompt Engineering, including prompt tuning, prompt design, and performance evaluation.
- Explore applications of Generative AI across various domains including code generation, art synthesis, content generation, and interactive systems.
- Equip students with ethical, social, and safety considerations when designing and deploying generative AI applications.

Course Outcomes

- Demonstrate a strong understanding of the architecture and functioning of Generative AI models, including transformers and LLMs.
- Be capable of applying prompt engineering techniques to steer model behavior for desired outputs across various tasks.
- Design and fine-tune generative models for applications such as text generation, image creation, music synthesis, and conversational AI.
- Analyze and evaluate the effectiveness of prompts and generated content, using relevant metrics and methodologies.
- Apply ethical principles to ensure responsible development and deployment of generative AI systems.

UNIT I – Introduction to Generative AI(Cognitive Level: Understand, Remember)
Overview of AI and types of AI, What is Generative AI? Definitions and Concepts, Historical evolution of generative models, Types of generative models – GANs, VAEs, Autoregressive Models, Introduction to Transformers and LLMs, Applications and use cases of Generative AI, Challenges in Generative AI development, Introduction to text-to-image and image-to-text models.

UNIT II – Fundamentals of Prompt Engineering

Definition and significance of Prompt Engineering, Types of prompts: Zero-shot, One-shot, Few-shot, Techniques for effective prompt design, Prompt templates and chaining, Prompt tuning and parameter-efficient tuning, Evaluating prompt performance, Use of APIs for testing prompts (OpenAI, Cohere, Anthropic), Best practices and prompt libraries.

UNIT III – Working with Large Language Models (LLMs)
Overview of pre-trained LLMs: GPT, BERT, LLaMA, Claude, PaLM, Architectures and tokenization strategies, Fine-tuning vs. in-context learning, LLM-powered tools (ChatGPT, GitHub Copilot, Bard, Claude), Role of attention mechanism and transformer layers, Tools for model experimentation (Hugging Face, LangChain), Performance metrics for LLMs, Case studies of model adaptation and deployment.

UNIT IV – Applications of Generative AI

Text generation and summarization, Image and art generation (DALL·E, Midjourney, Stable Diffusion), Code generation and completion tools (Codex, Copilot), Music and video generation, Generative chatbots and customer service, Story generation and dialogue systems, Domain-specific applications (Legal, Healthcare, Education), Comparative study of generative models by task.

UNIT V – Ethics, Security & Responsible AI

Bias and fairness in LLMs and generative systems, Explainability and transparency in generative AI, Copyright and originality issues, Adversarial use of generative models – deepfakes, misinformation, AI safety protocols and red-teaming, Regulatory and policy frameworks for generative AI, Responsible prompt crafting and moderation.

Textbooks

1. Deep Learning with Python by François Chollet, Manning Publications
2. Transformers for Natural Language Processing by Denis Rothman, Packt
3. Practical Generative AI by Amit Shukla, BPB Publications

Reference Books

1. Generative Deep Learning by David Foster, O’Reilly
2. Artificial Intelligence: A Guide for Thinking Humans by Melanie Mitchell
3. Machine Learning B.Techning by Andrew Ng (available online)

Online Courses

1. Generative AI with Large Language Models – Coursera (AWS & DeepLearning.AI)
2. Prompt Engineering for ChatGPT – Coursera
3. Generative AI Fundamentals – Google Cloud Training
4. Hugging Face – Transformers Course
5. DeepLearning.AI Short Courses

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech I Semester

MANAGEMENT ELECTIVE	L	T	P	C
	2	0	0	2

23AMB17 BUSINESS ETHICS AND CORPORATE GOVERNANCE

COURSE OBJECTIVES : The objectives of this course are

1	To make the student understand the principles of business ethics
2	To enable them in knowing about the ethics in management
3	To facilitate the student' role in corporate culture
4	To impart knowledge about the fair-trade practices
5	To encourage the student in knowing about the corporate governance

COURSE OUTCOMES: At the end of the course student will be able to		**BTL**
CO1	Remember E-Business & its nature, scope and functions.	L1
CO2	Understand E-market-Models which are practicing by the organizations	L2
CO3	Apply the concepts of E-Commerce in the present globalized world.	L3
CO4	Analyze the various E-payment systems & importance of net banking.	L4
CO5	Evaluate market research strategies & E-advertisements.	L5
CO6	Understand importance of E-security & control	L2

BTL = Bloom's Taxonomy Level

UNIT-I: Ethics

Introduction – Meaning – Nature, Scope, significance, Loyalty, and ethical behavior.. Value systems - Business Ethics - Types, Characteristics, Factors, Contradictions and Ethical Practices in Management - Corporate Social Responsibility – Issues of Management – Crisis Management.

LEARNING OUTCOMES:- After completion of this unit student will

- Understand the meaning of loyalty and ethical Behavior
- Explain various types of Ethics
- Analyze issues & crisis of management

UNIT-II: ETHICS IN MANAGEMENT

Introduction- Ethics in production, finance, Human resource management and Marketing Management - The Ethical Value System – Universalism, Utilitarianism, Distributive Justice, Social Contracts, Individual Freedom of Choice, Professional Codes; Culture and Ethics – Ethical Values in different Cultures - Culture and Individual Ethics – professional ethics and technical ethics.

LEARNING OUTCOMES:- After completion of this unit student will

- ➢ Understand the meaning of Ethics in various areas of management
- ➢ Compare and contrast professional ethics and technical ethics
- ➢ Develop ethical values in self and organization

UNIT-III : CORPORATE CULTURE

Introduction - Meaning, definition, Nature, and significance – Key elements of corporate culture, shared values, beliefs and norms, rituals, symbols and language - Types of corporate culture, hierarchical culture, market driven culture – Organization leadership and corporate culture, leadership styles and their impact on culture, transformational leadership and culture change.

LEARNING OUTCOMES:- After completion of this unit student will

- ➢ Define corporate culture
- ➢ Understand the key elements of corporate culture
- ➢ Analyze organization leadership and corporate culture

UNIT- IV: LEGAL FRAME WORK

Law and Ethics -Agencies enforcing Ethical Business Behavior - Legal Impact – Environmental Protection, Fair Trade Practices, legal Compliances, Safeguarding Health and wellbeing of Customers – Corporate law, Securities and financial regulations, corporate governance codes and principles.

LEARNING OUTCOMES:- After completion of this unit student will

- ➢ Understand Law and Ethics
- ➢ Analyze Different fair trade practices
- ➢ Make use of Environmental Protection and Fair Trade Practices

UNIT -V: CORPORATE GOVERNANCE

Introduction - Meaning – Corporate governance code, transparency & disclosure -Role of auditors, board of directors and shareholders. Global issues, accounting and regulatory frame work - Corporate scams - Committees in India and abroad, corporate social responsibility. BoDs composition, Cadbury Committee - Various committees - Reports - Benefits and Limit ations.

LEARNING OUTCOMES:- After completion of this unit student will

- ➢ Understand corporate governance code
- ➢ Analyze role of auditors, board of directors and shareholders in corporate governance
- ➢ Implementing corporate social responsibility in India.

Text books.

1. Murthy CSV: Business Ethics and Corporate Governance, HPH July 2017

2. Bholananth Dutta, S.K. Podder – Corporation Governance, VBH. June 2010

Reference books

1. Dr. K. Nirmala, KarunakaraReaddy. *Business Ethics and Corporate Governance*, HPH

2. H.R.Machiraju: *Corporate Governance*, HPH, 2013

3. K. Venkataramana, *Corporate Governance*, SHBP.

4. N.M.Khandelwal. *Indian Ethos and Values for Managers*

ONLINE RESOURCES:

1. https://onlinecourses.nptel.ac.in/noc21_mg46/
2. https://archive.nptel.ac.in/courses/110/105/110105138/
3. https://onlinecourses.nptel.ac.in/noc21_mg54/
4. https://onlinecourses.nptel.ac.in/noc22_mg54/
5. https://archive.nptel.ac.in/courses/109/106/109106117/

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech I Semester

23AMB18 E-BUSINESS

MANAGEMENT ELECTIVE

L	T	P	C
2	0	0	2

Course Objectives: The Objectives of this course are

1. To provide knowledge on emerging concept on E-Business related aspect.
2. To understand various electronic markets & business models.
3. To impart the information about electronic payment systems & banking.
4. To create awareness on security risks and challenges in E-commerce.
5. To the students aware on different e-marketing channels & strategies.

UNIT-I: Electronic Business

Introduction – Nature, meaning, significance, functions and advantages - Definition of Electronic Business - Functions of Electronic Commerce (EC)-Advantages & Disadvantages of E-Commerce –E-Commerce and E-Business, Internet Services, Online Shopping- E-Commerce Opportunities for Industries.

Learning Outcomes: -After completion of this unit student

- Understand the concept of E-Business
- Contrast and compare E-Commerce & E-Business
- Evaluate opportunities of E-commerce for industry

UNIT-II: Electronic Markets and Business Models

Introduction –E-Shops-E-Malls E-Groceries - Portals - Vertical Portals-Horizontal Portals - Advantages of Portals -Business Models- Business to Business (B2B)-Business to Customers(B2C) - Business to Government(B2G)-Auctions-B2B Portals in India

Learning Outcomes: -After completion of this unit student will

- Understand the concept of business models
- Contrast and compare Vertical portal and Horizontal portals
- Analyze the B2B,B2C and B2G model

UNIT-III: Electronic Payment Systems:

Introduction to electronic payment systems (EPS) -Types of electronic payments - Credit/debit cards, e-wallets, UPI, and crypto currencies -Smart cards and digital wallets: Features and usage -Electronic Fund Transfer (EFT): Role in business transactions -Infrastructure requirements and regulatory aspects of e-payments

Learning Outcomes: -After completion of this unit student will

- Understand the Electronic payment system
- Contrast and compare EFT and smart cards
- Analyze debit card and credit cards

UNIT-IV:E-Security

Security risks and challenges in electronic commerce - Cyber threats - Phishing, hacking, identity theft, and malware - Digital Signatures & Certificates - Security protocols over public networks (HTTP, SSL, TLS) - Firewalls in securing e-business platforms.

Learning Outcomes: -After completion of this unit student will

- Understand E-Security
- Contrast and compare security protocols and public network
- Evaluate on Digital signature

UNIT-V:E-Marketing:

Introduction – Online Marketing – Advantages of Online Marketing – Internet Advertisement – Advertisement Methods – Conducting Online Market Research– – E-marketing planning: Online branding, social media marketing, and email marketing - E-business strategies: Digital advertising, content marketing, and analytics – E-Customer Relationship Management (eCRM) E-supply chain management (e-SCM)

Learning Outcomes: -After completion of this unit student will

- Understand the concept of online marketing
- Apply the knowledge of online marketing
- Compare e-CRM and e-SCM

Text Books:

1. Arati Oturkar&Sunil Khilari. *E-Business*. Everest Publishing House, 2022
2. P.T.S Joseph. *E-Commerce,* Fourth Edition, Prentice Hall of India, 2011

References:

1. Debjani, Kamalesh K Bajaj. *E-Commerce,* Second Edition Tata McGraw-Hill's, 2005
2. Dave Chaffey.*E-Commerce E-Management,* Second Edition, Pearson, 2012.
3. Henry Chan. *E-Commerce Fundamentals and Application,* RaymondLeathamWiley India 2007
4. S. Jaiswal. *E-Commerce* GalgotiaPublication Pvt Ltd., 2003.

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech I Semester	**23AMB19 -MANAGEMENT SCIENCE**	**L**	**T**	**P**	**C**
	MANAGEMENT ELECTIVE	**2**	**0**	**0**	**2**

COURSE OBJECTIVES :

The objectives of this course are

1	To provide fundamental knowledge on Management, Administration, Organization & its concepts.
2	To make the students understand the role of management in Production
3	To impart the concept of HRM in order to have an idea on Recruitment, Selection, Training & Development, job evaluation and Merit rating concepts
4	To create awareness on identify Strategic Management areas & the PERT/CPM for better Project Management
5	To make the students aware of the contemporary issues in modern management

COURSE OUTCOMES: At the end of the course, students will be able to		**BTL**
CO1	Remember the concepts & principles of management and designs of organization in a practical world	**L1**
CO2	Understand the knowledge of Work-study principles & Quality Control techniques in industry	**L2**
CO3	Apply the process of Recruitment & Selection in organization.	**L3**
CO4	Analyze the concepts of HRM & different training methods.	**L4**
CO5	Evaluate PERT/CPM Techniques for projects of an enterprise and estimate time & cost of project & to analyze the business through SWOT.	**L5**
CO6	Create awareness on contemporary issues in modern management & technology.	**L3**

UNIT- I INTRODUCTION TO MANAGEMENT

Management - Concept and meaning - Nature-Functions - Management as a Science and Art and both. Schools of Management Thought - Taylor's Scientific Theory-Henry Fayol's principles - Elton Mayo's Human relations - **Organizational Designs** - Line organization - Line & Staff Organization - Functional Organization - Matrix Organization - Project Organization - Committee form of Organization - Social responsibilities of Management.

LEARNING OUTCOMES: At the end of the Unit, the students will be able to

- Understand the concept of management and organization
- Apply the concepts & principles of management in real life industry.
- Analyze the organization chart & structure of an enterprise.

UNIT - II OPERATIONS MANAGEMENT

Principles and Types of Plant Layout - Methods of Production (Job, batch and Mass Production), Work Study - Statistical Quality Control- **Material Management -** Objectives - Inventory-Functions - Types, Inventory Techniques - EOQ-ABC Analysis - **Marketing Management -** Concept - Meaning - Nature-Functions of Marketing - Marketing Mix - Channels of Distribution - Advertisement and Sales Promotion - Marketing Strategies based on Product Life Cycle.

LEARNING OUTCOMES: At the end of the Unit, the students will be able to

- Understand the core concepts of Operations Management
- Apply the knowledge of Quality Control, Work-study principles in real life industry.
- Evaluate Materials departments & Determine EOQ
- Analyze Marketing Mix Strategies for an enterprise.
- Create and design advertising and sales promotion

UNIT - III HUMAN RESOURCES MANAGEMENT (HRM)

HRM - Definition and Meaning – Nature - Managerial and Operative functions - Job Analysis - Human Resource Planning(HRP) - Employee Recruitment-Sources of Recruitment - Employee Selection - Process - Employee Training and Development - methods - Performance Appraisal Concept - Methods of Performance Appraisal – Placement - Employee Induction - Wage and Salary Administration

LEARNING OUTCOMES: At the end if the Unit, the students will be able to

- Understand the concepts of HRM, Recruitment, Selection, Training & Development
- Analyze the need of training
- Evaluate performance appraisal
- Design the basic structure of salaries and wages

UNIT - IV STRATEGIC & PROJECT MANAGEMENT

Definition& Meaning - Setting of Vision - Mission - Goals - Corporate Planning Process - Environmental Scanning - Steps in Strategy Formulation and Implementation - SWOT Analysis - **Project Management -** Network Analysis - Programme Evaluation and Review Technique (PERT) - Critical Path Method (CPM) Identifying Critical Path - Probability of Completing the project within given time - Project Cost- Analysis - Project Crashing (Simple problems).

LEARNING OUTCOMES: At the end of the Unit, the students will be able to

- Understand Mission, Objectives, Goals & strategies for an enterprise
- Apply SWOT Analysis to strengthen the project
- Analyze Strategy formulation and implementation
- Evaluate PERT and CPM Techniques

UNIT - V CONTEMPORARY ISSUES IN MANAGEMENT

Customer Relations Management(CRM) - Total Quality Management (TQM) - Six Sigma Concept - Supply Chain Management(SCM) - Enterprise Resource Planning (ERP) - Performance Management – employee engagement and retention - Business Process Re-engineering and Bench Marking - Knowledge Management – change management –sustainability and corporate social responsibility.

LEARNING OUTCOMES At the end if the Unit, the students will be able to

- Understand modern management techniques
- Apply Knowledge in Understanding in TQM, SCM
- Analyze CRM, BPR
- Evaluate change management & sustainability

Text Books:

1. Frederick S. Hillier, Mark S. Hillier. *Introduction to Management Science*, October 26, 2023
2. A.R Aryasri, *Management Science*, TMH, 2019

References:

1. Stoner, Freeman, Gilbert.*Management*, Pearson Education, New Delhi, 2019.
2. Koontz & Weihrich, *Essentials of Management*, 6/e, TMH, 2005.
3. Thomas N.Duening & John M.Ivancevich, *Management Principles and Guidelines*, Biztantra.
4. Kanishka Bedi, Production and Operations Management, Oxford University Press, 2004.
5. Samuel C.Certo, *Modern Management*, 9/e, PHI, 2005

BTL = Blooms Taxonomy Level

ONLINE RESOUECES:

1. https://www.slideshare.net/slideshow/introduction-to-management-and-organization-231308043/231308043
2. https://nptel.ac.in/courses/112107238
3. https://archive.nptel.ac.in/courses/110/104/110104068/
4. https://archive.nptel.ac.in/courses/110/105/110105069/
5. https://onlinecourses.nptel.ac.in/noc24_mg112/

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech I Semester

MANAGEMENT ELECTIVE

23AMB02- MANAGERIAL ECONOMICS AND FINANCIAL ANALYSIS

L	T	P	C
2	0	0	2

Course Objectives:

- To inculcate the basic knowledge of microeconomics and financial accounting
- To make the students learn how demand is estimated for different products, input- output relationship for optimizing production and cost
- To Know the Various types of market structure and pricing methods and strategy
- To give an overview on investment appraisal methods to promote the students to learn how to plan long-term investment decisions.
- To provide fundamental skills on accounting and to explain the process of preparing financial statements.

Course Outcomes:

CO1: Define the concepts related to Managerial Economics, financial accounting and management(L2) CO2: Understand the fundamentals of Economics viz., Demand, Production, cost, revenue and markets (L2) CO3: Apply the Concept of Production cost and revenues for effective Business decision (L3)
CO4: Analyze how to invest their capital and maximize returns (L4) CO5: Evaluate the capital budgeting techniques. (L5)
CO6: Develop the accounting statements and evaluate the financial performance of business en- tity (L5)

UNIT - I Managerial Economics

Introduction – Nature, meaning, significance, functions, and advantages. Demand-Concept, Function, Law of Demand - Demand Elasticity- Types – Measurement. Demand Forecasting- Factors governing Forecasting, Methods. Managerial Economics and Financial Accounting and Management.

UNIT - II Production and Cost Analysis

Introduction – Nature, meaning, significance, functions and advantages. Production Function– Least- cost combination– Short run and long run Production Function- Isoquants and Is costs, Cost & Break-Even Analysis - Cost concepts and Cost behaviour- Break-Even Analysis (BEA) - Determination of Break-Even Point (Simple Problems).

UNIT - III Business Organizations and Markets

Introduction – Forms of Business Organizations- Sole Proprietary - Partnership - Joint Stock Companies - Public Sector Enterprises. Types of Markets - Perfect and Imperfect Competition
- Features of Perfect Competition Monopoly- Monopolistic Competition– Oligopoly-Price-
Output Determination - Pricing Methods and Strategies

UNIT - IV Capital Budgeting

Introduction – Nature, meaning, significance. Types of Working Capital, Components, Sources of Short-term and Long-term Capital, Estimating Working capital requirements. Capital Budg- eting– Features, Proposals, Methods and revaluation, Projects – Pay Back Method, Accounting Rate of Return (ARR) Net Present Value (NPV) Internal Rate Return (IRR) Method (sample problems)

UNIT - V Financial Accounting and Analysis

Introduction – Concepts and Conventions- Double-Entry Bookkeeping, Journal, Ledger, Trial Balance- Final Accounts (Trading Account, Profit and Loss Account and Balance Sheet with simple adjustments). Introduction to Financial Analysis - Analysis and Interpretation of Liquidity Ratios, Activity Ratios, and Capital structure Ratios and Profitability.

Textbooks:

Varshney & Maheswari: Managerial Economics, Sultan Chand. Aryasri:

Business Economics and Financial Analysis, 4/e, MGH.

Reference Books:

1. Ahuja Hl Managerial economics Schand.
2. S.A. Siddiqui and A.S. Siddiqui: Managerial Economics and Financial Analysis, New Age International.
3. Joseph G. Nellis and David Parker: Principles of Business Economics, Pearson, 2/e, New Delhi.
4. Domnick Salvatore: Managerial Economics in a Global Economy, Cengage.

Online Learning Resources: https://www.slideshare.net/123ps/manage- rial-economics-ppt
https://www.slideshare.net/rossanz/production-and-cost- 45827016
https://www.slideshare.net/darkyla/business-organizations-
19917607 https://www.slideshare.net/balarajbl/market-and-classification-of- market
https://www.slideshare.net/ruchi101/capital-budgeting-ppt- 59565396
https://www.slideshare.net/ashu1983/financial-accountingEvaluation.

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

IV B.Tech I Semester

MANAGEMENT ELECTIVE

23AMB03:ORGANISATIONAL BEHAVIOUR

L	T	P	C
2	0	0	2

Course Objectives:

- To enable student's comprehension of organizational behavior
- To offer knowledge to students on self-motivation, leadership and management
- To facilitate them to become powerful leaders
- To Impart knowledge about group dynamics
- To make them understand the importance of change and development

Course Outcomes:

CO1: Define the Organizational Behaviour, its nature and scope. (L2)
CO2: Understand the nature and concept of Organizational behaviour (L2)
CO3: Apply theories of motivation to analyse the performance problems (L3)
CO4: Analyse the different theories of leadership (L4)
CO5: Evaluate group dynamics (L5)
CO6: Deve lop as powerful leader (L5)

UNIT - I Introduction to Organizational Behavior

Meaning, definition, nature, scope and functions - Organizing Process – Making organizing ef- fective - Understanding Individual Behaviour –Attitude -Perception - Learning – Personality.

UNIT - II Motivation and Leading

Theories of Motivation- Maslow's Hierarchy of Needs - Hertzberg's Two Factor Theory - Vroom's theory of expectancy – Mc Cleland's theory of needs–Mc Gregor's theory X and theory Y– Adam's equity theory.

UNIT - III Organizational Culture

Introduction – Meaning, scope, definition, Nature - Organizational Climate - Leadership - Traits Theory– Managerial Grid - Transactional Vs Transformational Leadership - Qualities of good Leader - Conflict Management -Evaluating Leader.

UNIT - IV Group Dynamics

Introduction – Meaning, scope, definition, Nature- Types of groups - Determinants of group be- haviour - Group process – Group Development - Group norms - Group cohesiveness - Small Groups - Group decision making - Team building - Conflict in the organization– Conflict resolu- tion

UNIT - V Organizational Change and Development

Introduction –Nature, Meaning, scope, definition and functions- Organizational Culture - Chang- ing the Culture – Change Management – Work Stress Management - Organizational management – Managerial implications of organization's change and development

Textbooks:

1. Luthans, Fred, Organisational Behaviour, McGraw-Hill, 12 Th edition.
2. P Subba Ran, Organisational Behaviour, Himalya Publishing House.

Reference Books:

3. McShane, Organizational Behaviour, TMH
4. Nelson, Organisational Behaviour, Thomson.
5. Robbins, P. Stephen Aswathappa, Organisational Behaviour, Himalaya.

Online Learning Resources:

https://www.slideshare.net/Knight1040/organizational-culture 9608857s://www.slideshare.net/ AbhayRajpoot3/motivation-165556714 https://www.slideshare.net/harshrastogi1/group-dynamics-159412405

https://www.slideshare.net/vanyasingla1/organizational-change-development- 26565951,

imothy A. Judge, Organisational Behaviour, Pearson.

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

IV B.Tech I Semester

L	T	P	C
2	0	0	2

MANAGEMENT ELECTIVE

23AMB04: BUSINESS ENVIRONMENT

Course Objectives:

- To make the student to understand about the business environment
- To enable them in knowing the importance of fiscal and monitory policy
- To facilitate them in understanding the export policy of the country
- To Impart knowledge about the functioning and role of WTO
- To Encourage the student in knowing the structure of stock markets

Course Outcomes:

CO1: Define Business Environment and its Importance. (L2)
CO2: Understand various types of business environment. (L2)
CO3: Apply the knowledge of Money markets in future investment (L3)
CO4: Analyse India's Trade Policy (L4)
CO5: Evaluate fiscal and monitory policy (L5)
CO6: Develop a personal synthesis and approach for identifying business opportunities (L5)

UNIT – I Overview of Business Environment

Introduction – meaning Nature, Scope, significance, functions and advantages. Types- Inter- nal &External, Micro and Macro. Competitive structure of industries -Environmental analy- sis- advantages & limitations of environmental analysis.

UNIT – II Fiscal & Monetary Policy

Introduction – Nature, meaning, significance, functions and advantages. Public Revenues - Public Expenditure - Evaluation of recent fiscal policy of GOI. Highlights of Budget- Mone- tary Policy - Demand and Supply of Money –RBI -Objectives of monetary and credit policy - Recent trends- Role of Finance Commission.

UNIT – III India's Trade Policy

Introduction – Nature, meaning, significance, functions and advantages. Magnitude and direc- tion of Indian International Trade - Bilateral and Multilateral Trade Agreements - EXIM policy and role of EXIM bank - Balance of Payments– Structure & Major components - Causes for Disequilibrium in Balance of Payments - Correction measures.

UNIT – IV World Trade Organization

Introduction – Nature, significance, functions and advantages. Organization and Structure - Role and functions of WTO in promoting world trade - GATT - Agreements in the Uruguay Round – TRIPS, TRIMS - Disputes Settlement Mechanism - Dumping and Anti-dumping Measures.

UNIT – V Money Markets and Capital Markets

Introduction – Nature, meaning, significance, functions and advantages. Features and compo- nents of Indian financial systems - Objectives, features and structure of money markets and capital markets - Reforms and recent development – SEBI – Stock Exchanges
- Investor protection and role of SEBI, Introduction to international finance.

Textbooks:

1. Francis Cherunilam, International Business: Text and Cases, Prentice Hall of India.
2. K. Aswathappa, Essentials of Business Environment: Texts and Cases & Exercises 13th Revised Edition.HPH

Reference Books:

1. K. V. Sivayya, V. B. M Das, Indian Industrial Economy, Sultan Chand Publish- ers, New Delhi, India.
2. Sundaram, Black, International Business Environment Text and Cases, Prentice Hall of India, New Delhi, India.
3. Chari. S. N, International Business, Wiley India.
4. E. Bhattacharya, International Business, Excel Publications, New Delhi.

Online Learning Resources:

https://www.slideshare.net/ShompaDhali/business- environment-53111245 https://www.slideshare.net/rbalsells/fiscal-policy-ppt
https://www.slideshare.net/aguness/monetary-policy-presentationppt
https://www.slideshare.net/DaudRizwan/monetary-policy-of-india-69561982 https://www.slideshare.net/ShikhaGupta31/indias-trade-policyppt https://www.slideshare.net/viking2690/wto-ppt-60260883
https://www.slideshare.net/prateeknepal3/ppt-mo

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

IV B.Tech I Semester

		L	T	P	C
23ACA24	**EXPLAINABLE AI &MODEL INTERPRETABILITY** **(Professional Elective-IV)** **Common to CSE(AI&ML), CSE(AI) and B.Tech-AI**	**3**	**0**	**0**	**3**

Course Objectives:

- To introduce the principles of interpretability and explainability in AI/ML models.
- To analyze the trade-offs between model accuracy and interpretability.
- To explore popular post-hoc and intrinsic explainability techniques.
- To examine fairness, accountability, and transparency in AI systems.
- To develop hands-on skills with interpretability tools and libraries.

Course Outcomes:

Upon successful completion of the course, students will be able to:

- Understand the need for explainability in modern AI systems.
- Differentiate between black-box and white-box models.
- Apply interpretability techniques such as SHAP, LIME, and PDPs.
- Evaluate the fairness and transparency of AI systems.
- Use explainability tools for model auditing and deployment in high-stakes domains.

UNIT I: Foundations of Explainable AI
Introduction to Explainability and Interpretability, Importance of XAI in Healthcare, Finance, and Law , White-box vs Black-box Models, Desiderata: Fairness, Accountability, Transparency, Human-Centered AI and Trust ,Taxonomy of XAI Techniques (Global vs Local, Post-hoc vs Intrinsic), Regulatory and Ethical Implications (GDPR, AI Bill of Rights), Model Simplicity vs Predictive Power.

UNIT II: Model-Specific Explainability Techniques
Decision Trees and Rule-based Models, Linear Models and Feature Importance, Generalized Additive Models (GAMs), Visualization of Weights and Coefficients, Logistic Regression Coefficient Interpretation, Case Study: Credit Scoring using Transparent Models,Comparison of Interpretable ML Models, Use Cases and Trade-offs.

UNIT III: Model-Agnostic Explainability Techniques
Local Interpretable Model-agnostic Explanations (LIME), SHAP Values (SHapley Additive exPlanations), Partial Dependence Plots (PDPs), Individual Conditional Expectation (ICE) Plots,Anchors and Counterfactual Explanations, Feature Interaction and Permutation Importance, Comparative Analysis of SHAP, LIME, PDP, Model Debugging with XAI.

UNIT IV: Deep Learning Explainability
Visualizing CNNs: Filters, Feature Maps, Saliency Maps and Grad-CAM, Integrated Gradients, Explaining RNNs and LSTM Outputs, Concept Activation Vectors (TCAV), Attention-based Interpretability in Transformers, Explaining Language Models (BERT, GPT) Evaluation of Deep Model Explanations.

UNIT V: Fairness, Bias & Tools for XAI

Fairness Metrics: Demographic Parity, Equal Opportunity, Sources of Bias in Data and Models, Discrimination Detection and Mitigation Strategies, Introduction to AIF360, What-If Tool, Fairlearn, Case Study: Bias in Hiring Algorithms, Explainability in ML Pipelines (MLFlow, Skater), XAI in Federated and Privacy-Preserving AI, Designing Interpretable AI Systems from Scratch.

Textbooks:

1. Christoph Molnar, "Interpretable Machine Learning", Leanpub.
2. Sameer Singh et al., "Explainable AI: Interpreting, Explaining and Visualizing Deep Learning", Springer.
3. Dan Roth, Zachary Lipton, and Been Kim, "Explainable AI: Foundations, Developments, Prospects", MIT Press (Online forthcoming).

Reference Books:

1. Marco Tulio Ribeiro et al., "Why Should I Trust You?" (LIME) – Research Paper
2. Scott Lundberg et al., "A Unified Approach to Interpreting Model Predictions" (SHAP) – NIPS
3. A. Barredo Arrieta et al., "Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges", Information Fusion Journal.
4. Zachary C. Lipton, "The Mythos of Model Interpretability" – Communications of the ACM

Online Learning Resources:

- Coursera – Explainable AI with Google Cloud
- Udacity – AI for Everyone by Andrew Ng
- HarvardX – Data Science: Machine Learning Interpretability
- fast.ai – Practical Deep Learning Courses
- InterpretableML Book (https://christophm.github.io/interpretable-ml-book/)
- Google PAIR: People + AI Research
- Microsoft Fairlearn Documentation

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech I Semester

		L	T	P	C
23ACA25	**AI IN CYBERSECURITY (Professional Elective-IV)**	**3**	**0**	**0**	**3**

Course Objectives:

- To introduce the fundamental concepts of AI and their applications in cybersecurity.
- To understand AI-driven techniques for threat detection, classification, and mitigation.
- To explore machine learning and deep learning methods used for malware and intrusion detection.
- To equip students with skills in building intelligent security systems.
- To examine ethical, legal, and privacy aspects in AI-driven cybersecurity.

Course Outcomes:

- Understand AI principles and their relevance in cybersecurity.
- Apply machine learning techniques to detect and respond to threats.
- Analyze security incidents using intelligent tools and models.
- Evaluate and implement AI models for malware detection and anomaly analysis.
- Design AI-based cybersecurity frameworks for real-world scenarios.

UNIT I: Introduction to AI in Cybersecurity

Role of AI in Modern Cybersecurity, Overview of Cyber Threats and Attack Vectors, Fundamentals of Machine Learning for Security, AI vs Traditional Security Techniques, AI-Based Cyber Defense Lifecycle, Threat Intelligence with AI, Cybersecurity Data Types and Challenges, Case Studies of AI-Driven Attacks and Defenses

UNIT II: Machine Learning for Cyber Threat Detection

Supervised Learning for Intrusion Detection, Unsupervised Learning for Anomaly Detection, Feature Engineering from Network Traffic, Classification Algorithms: SVM, Decision Trees, Random Forests, Clustering Techniques: K-Means, DBSCAN, Ensemble Models and Model Evaluation Metrics, Real-Time Threat Detection Pipelines, Data Imbalance and Adversarial Sampling

UNIT III: Deep Learning in Cybersecurity

Neural Networks for Threat Classification, CNNs for Malware Detection from Binary Files, RNNs/LSTMs for Sequential Log Analysis, Autoencoders for Anomaly Detection,GANs in Malware Evasion and Defense,Transfer Learning for Threat Signature Extraction, Deep Learning vs Traditional Models: A Comparative Study,Real-World Use Cases and Limitations

UNIT IV: AI for Specific Security Domains

AI for Phishing and Spam Detection, AI in Cloud Security and Edge Devices, Botnet and DDoS Attack Detection**,** AI-Driven Endpoint Security, Natural Language Processing for Threat Intelligence, Behavioral Biometrics and Fraud Detection, AI in Social Engineering Attack Prevention, Security Information and Event Management (SIEM) with AI

UNIT V: Challenges, Ethics & Future of AI in Cybersecurity

Explainable AI (XAI) in Cybersecurity, Adversarial Attacks and Defenses in AI Systems, Data Privacy and Federated Learning, Legal and Ethical Issues in AI Security Solutions, AI Model Bias and Fairness in Security Decisions, Securing AI Models Against Manipulation, Building Scalable AI-Powered SOCs, Future Trends: Autonomous Security, AI-Augmented Threat Hunting

Textbooks:

1. Clarence Chio & David Freeman, "Machine Learning and Security", O'Reilly Media.
2. Xiaofeng Chen et al., "Artificial Intelligence and Big Data Analytics for Cybersecurity", Springer.
3. Mark Stamp, "Information Security: Principles and Practice", Wiley.

Reference Books:

1. Sumeet Dua & Xian Du, "Data Mining and Machine Learning in Cybersecurity", CRC Press.
2. Shai Shalev-Shwartz & Shai Ben-David, "Understanding Machine Learning", Cambridge University Press.
3. Zhiwei Lin & Yang Xiang, "Cyber Security Intelligence and Analytics", Springer.
4. Bhavani Thuraisingham, "Data Mining for Malware Detection", CRC Press.

Online Learning Resources:

- Coursera – "AI for Cybersecurity" by University of Colorado
- Udemy – "Machine Learning for Cybersecurity"

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech I Semester

		L	T	P	C
23ACA26	**AI-DRIVEN SOFTWARE ENGINEERING & DEVOPS (Professional Elective-IV)**	**3**	**0**	**0**	**3**

Course Objectives:

1. To introduce the principles and practices of AI-driven software engineering and DevOps.
2. To explore how AI techniques can automate and optimize software development workflows.
3. To study intelligent tools for code generation, testing, monitoring, and deployment.
4. To equip students with skills in AI-powered CI/CD pipelines and operations.
5. To foster an understanding of ethical implications and reliability in intelligent software systems.

Course Outcomes:

1. Understand AI's role in modern software development and operations.
2. Apply machine learning techniques to automate software engineering tasks.
3. Design and manage intelligent CI/CD and DevOps workflows.
4. Evaluate AI tools in software testing, refactoring, and monitoring.
5. Implement AI-based solutions for predictive maintenance and decision support in DevOps.

UNIT I: Foundations of AI in Software Engineering

Overview of Traditional vs AI-driven Software Development, AI Opportunities in Software Lifecycle Phases, Introduction to ML/DL Models in Engineering Tasks, Code Representation and Learning from Code, NLP for Source Code Understanding, Software Knowledge Graphs and Reasoning, Datasets and Benchmarks for Software Engineering AI, Case Studies of AI-Enhanced Development Tools

UNIT II: AI in Code Generation and Refactoring

Program Synthesis and Code Completion Models, Large Language Models (e.g., Codex, CodeBERT) in IDEs, Code Clone Detection and Automated Refactoring, Learning-Based Bug Detection and Code Smell Identification, AI in Software Architecture Recommendations, Embedding Techniques for Source Code, Prompt Engineering for Software Tasks, Reliability and Safety in Generated Code

UNIT III: Intelligent Testing, QA, and Debugging

Test Case Generation Using AI, Automated Unit Testing, Regression Testing with ML, Learning Bug Patterns from Repositories, AI-Based Static and Dynamic Code Analysis, Fault Localization and Automated Debugging, Quality Assurance Metrics Enhanced by AI, Reinforcement Learning for Test Prioritization, Ethics and Bias in AI-Driven Testing – (E)

UNIT IV: AI in DevOps Automation and CI/CD

DevOps Fundamentals and Integration with AI, Intelligent CI/CD Pipeline Design, Predictive Build Failure and Log Analysis, AI in Infrastructure-as-Code and Deployment Orchestration, Self-Healing Systems and AIOps Concepts, Log Analytics and Anomaly Detection in Production, AI in Monitoring, Tracing, and Feedback Loops, DevSecOps and AI for Security Automation

UNIT V: Advanced Topics and Ethical Considerations

Explainability and Transparency in AI-Driven Tools, Ethical and Legal Aspects in Automated Engineering**,** Human-AI Collaboration in Software Teams, Risk Management in Autonomous Code Deployment, AI for Technical Debt Prediction and Management, AI for Developer Productivity Analytics, Research Trends and Challenges in AI for SE, Capstone: Building a Smart DevOps Workflow

Textbooks:

1. Tim Menzies, Diomidis Spinellis, and Thomas Zimmermann, "Perspectives on Data Science for Software Engineering", Morgan Kaufmann.
2. André van der Hoek, Reid Holmes, "Software Engineering for Machine Learning", Springer.
3. Len Bass, Ingo Weber, Liming Zhu, "DevOps: A Software Architect's Perspective", Addison-Wesley.

Reference Books:

1. Carlos Eduardo Parnin et al., "AI for Software Engineering: Foundations, Advances, and Trends", Springer.
2. Luciano Baresi et al., "Machine Learning Techniques for Software Quality Evaluation", Springer.
3. Gene Kim, Jez Humble, and Nicole Forsgren, "Accelerate: The Science of Lean Software and DevOps", IT Revolution.

Online Learning Resources:

- Coursera – "AI for Software Engineering" by DeepLearning.AI
- edX – "DevOps for Developers" by Microsoft
- GitHub Copilot and OpenAI Codex documentation
- PapersWithCode – AI for Software Engineering benchmarks
- MIT OCW – "Software Systems" and "DevOps and CI/CD"
- Udemy – "AI-Powered DevOps Pipelines and Automation"
- Google Cloud – AIOps and MLOps tutorials

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech I Semester

		L	T	P	C
23ACA27	**AI for ROBOTICS (Professional Elective-IV)**	**3**	**0**	**0**	**3**

Course Objectives:

1. To provide foundational knowledge in Generative AI and its core architectures.
2. To explore applications of generative models in text, image, and audio domains.
3. To equip students with hands-on experience in training and fine-tuning generative models.
4. To teach techniques for deploying AI models efficiently and securely in real-world settings.
5. To analyze the ethical, interpretability, and security concerns surrounding generative models.

Course Outcomes:

1. Understand and compare different generative architectures like GANs, VAEs, Transformers.
2. Build, fine-tune, and evaluate generative models for text, vision, and multimodal data.
3. Design secure and scalable workflows for model deployment (on cloud and edge).
4. Integrate monitoring, optimization, and feedback for deployed AI services.
5. Address ethical, interpretability, and regulatory aspects of model deployment.

UNIT I: Foundations of Generative AI

Introduction to Generative AI: Concepts, Use Cases, Types of Generative Models: GANs, VAEs, Diffusion, Autoregressive, Mathematical Foundations: Probabilistic Modeling & Latent Spaces, Generative Loss Functions: Adversarial Loss, KL Divergence, ELBO, Model Training Challenges: Mode Collapse, Posterior Collapse, Tools & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, Evaluation Metrics: FID, BLEU, Perplexity, Case Studies of Generative AI Systems

UNIT II: Generative Modeling for Text, Images, and Audio

Transformer Models: GPT, T5, BERT for Text Generation, Vision Models: StyleGAN, DALL·E, Stable Diffusion, Audio Generation: WaveNet, Jukebox, Voice Cloning**,** Multi-modal Generation: CLIP, Flamingo, BLIP, Prompt Engineering & Controlled Generation, Fine-Tuning & LoRA Techniques, Evaluation Techniques for Generated Media, Comparison of Open vs Proprietary Foundation Models

UNIT III: Model Serving and Deployment Essentials

Overview of AI Deployment Pipelines, Model Packaging: ONNX, TorchScript, SavedModel, REST & gRPC API Serving using FastAPI, Flask, Triton Inference, Batch vs Real-time Inference, Asynchronous Processing, Containerization & Orchestration with Docker and Kubernetes, Cloud Deployment: AWS SageMaker, GCP Vertex AI, Azure ML, Cost Optimization and Resource Management, Edge AI: TinyML, Mobile Deployment using TensorFlow Lite, CoreML

UNIT IV: MLOps and Model Lifecycle Management

MLOps Lifecycle: Versioning, Experiment Tracking, CI/CD for ML: GitHub Actions, Jenkins, MLflow**,** Model Monitoring & Drift Detection, Logging and Metrics using Prometheus, Grafana, Auto-scaling and Load Balancing of Inference Services, Continuous Training and Feedback Loops, Model Governance and Audit Trails, Role of Explainability in Deployment

UNIT V: Challenges and Responsible Deployment

Interpretability of Generative Models, Security and Adversarial Attacks on Generative Models, Bias, Fairness, and Harmful Content Generation, Privacy-Preserving Techniques: Differential Privacy, Federated Learning, Ethical and Legal Frameworks for Generative AI, Open Source vs Proprietary Models: Deployment Implications, Responsible AI Guidelines & Compliance (GDPR, EU AI Act), Capstone: End-to-End Generative Model Deployment Project

Textbooks:

1. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, "Deep Learning", MIT Press.
2. Aurélien Géron, "Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow", O'Reilly Media.
3. O'Reilly, "Generative Deep Learning" by David Foster.

Reference Books:

1. Packt Publishing, "Practical Deep Learning" by Ronald T. Kneusel.
2. "Transformers for Natural Language Processing" by Denis Rothman.
3. "Machine Learning Engineering" by Andriy Burkov.

Online Learning Resources:

- Coursera – "Generative AI with Large Language Models" by DeepLearning.AI
- Hugging Face Course – https://huggingface.co/learn
- Fast.ai – "Practical Deep Learning for Coders"
- YouTube – Full Stack Deep Learning by Berkeley AI
- AWS Machine Learning University – Model Deployment
- TowardsDataScience and PapersWithCode – Latest research & code repositories

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech I Semester

		L	T	P	C
23ACM23	**ML Ops & AI Model Deployment (Professional Elective-V)**	**3**	**0**	**0**	**3**

Course Objective:

- To understand the principles and best practices of operationalizing machine learning models in production environments.
- To explore the life cycle of AI model development, deployment, monitoring, and maintenance using modern MLOps frameworks.
- To develop skills in CI/CD for ML, reproducibility, model versioning, and containerization using Docker and Kubernetes.
- To deploy machine learning models using cloud-native services and track their performance using real-time metrics.
- To address scalability, reliability, and ethical considerations in ML model deployment.

Course Outcomes:

After successful completion of this course, students will be able to:

1. Illustrate the lifecycle and pipeline components of MLOps and implement basic version control and orchestration for ML workflows.
2. Package ML models using appropriate tools and deploy them using Docker and Kubernetes environments with effective resource management.
3. Develop and deploy machine learning models as APIs using FastAPI/Flask and configure for real-time or batch inference scenarios.
4. Monitor and log ML systems using modern tools and detect data/model drift with strategies for continuous evaluation and feedback.
5. Implement end-to-end MLOps solutions using cloud platforms and CI/CD tools, and analyze deployment challenges in real-world use cases.

UNIT I: Introduction to MLOps and Deployment Pipelines

Definition and need of MLOps, ML system lifecycle and pipeline components, DevOps vs. MLOps: key differences, CI/CD for ML projects, Data versioning and model lineage, Introduction to DVC, Git, and MLFlow, Workflow orchestration using Apache Airflow, Automated testing in ML pipelines

UNIT II: Model Packaging and Environment Management

Packaging ML models using Pickle, Joblib, ONNX, Python virtual environments, Conda, Pipenv, Introduction to Docker for ML workloads, Building Dockerfiles for ML apps, Using Kubernetes for orchestration, Security, logging, and resource management, Docker Compose and Helm charts for deployment, Hands-on: Containerize and deploy a scikit-learn model

UNIT III: Model Serving and APIs

RESTful API design for ML models, Model deployment using FastAPI and Flask, TensorFlow Serving, TorchServe basics, Introduction to gRPC for ML deployment, Asynchronous inference and batch vs real-time serving, Load testing and benchmarking, Authentication and authorization in model APIs, Deploying models on edge devices

UNIT IV: Monitoring, Logging, and Continuous Evaluation

Importance of monitoring and alerting in MLOps, Data drift and model drift detection, Logging prediction results and metadata, Prometheus, Grafana, and ELK Stack, A/B testing and canary deployments, Shadow deployments and rollback strategies, Feedback loops for continuous learning, Integration with external monitoring tools.

UNIT V: Cloud-native MLOps and Case Studies

ML deployment on AWS SageMaker, Azure ML, Google AI Platform, CI/CD using GitHub Actions, Jenkins, and GitLab CI, AutoML and model registry, Real-world case study: End-to-end MLOps pipeline, Challenges and limitations in enterprise ML deployment, Responsible AI in production systems, Future trends in MLOps, Capstone Project Planning

Text Books:

1. Introducing MLOps: How to Scale Machine Learning Projects with DevOps Tools – Mark Treveil, Alok Shukla, O'Reilly Media.
2. Machine Learning Engineering – Andriy Burkov, TrueShelf Publishing.
3. Designing Machine Learning Systems – Chip Huyen, O’Reilly Media.

Reference Books:

1. Practical MLOps – Noah Gift, O’Reilly Media
2. Kubeflow for Machine Learning – Trevor Grant et al., O'Reilly
3. Hands-On MLOps: Implement Machine Learning in Production – Munn, Meza, Vohra, Packt Publishing
4. Research papers from arXiv, MLSys Conference, and ICML Industry Track

Online Courses:

1. Coursera – MLOps Specialization by DeepLearning.AI
2. Google Cloud – MLOps: Continuous Delivery and Automation Pipelines
3. Udemy – MLOps: ML Pipelines, CI/CD, and Model Deployment
4. AWS – Machine Learning Engineering for Production (MLOps)
5. Microsoft Learn – MLOps with Azure ML

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech I Semester

		L	T	P	C
23ACD27	**Data Wrangling (Professional Elective-V)**	**3**	**0**	**0**	**3**

Course Objectives:

- To introduce the fundamental techniques for acquiring, cleaning, transforming, and manipulating data.
- To enable students to handle real-world messy data for analysis and machine learning.
- To teach efficient use of libraries like Pandas, NumPy, and SQL for data wrangling.
- To promote understanding of handling missing values, outliers, and inconsistent formats.
- To expose students to automation, reproducibility, and workflow design in data preprocessing.

Course Outcomes:

After successful completion of this course, students will be able to:

- Understand and apply core data wrangling techniques.
- Clean, transform, and reshape data using Python and SQL.
- Handle missing values, data inconsistencies, and outliers.
- Merge and join multiple datasets from different sources.
- Automate data pipelines and preprocessing workflows for analytics and ML.

UNIT I: Introduction to Data Wrangling and Data Acquisition

Introduction to Data Wrangling: Importance and Use Cases , Types of Data: Structured, Semi-Structured, Unstructured, Data Acquisition Techniques: APIs, Web Scraping, Reading Data from CSV, Excel, JSON, XML, Using Python libraries: pandas, requests, BeautifulSoup, Working with Databases using SQLAlchemy and pandas, Loading Large Datasets and Chunking, Exploratory Analysis Before Cleaning.

UNIT II: Handling Missing, Noisy, and Inconsistent Data

Identifying and Understanding Missing Data, Techniques for Imputing Missing Values, Handling Inconsistent Data: Dates, Texts, Units, Removing Duplicates and Irrelevant Data, Detecting and Treating Outliers, Normalization and Standardization Techniques, Regular Expressions for Text Cleaning, Visualizing Missing/Outlier Data.

UNIT III: Data Transformation and Feature Engineering

Data Type Conversion and Parsing, Feature Extraction from Text, Dates, and Strings, One-Hot Encoding, Label Encoding, Binning and Discretization, Data Aggregation and Grouping, Pivoting, Melting, and Reshaping Data, Handling Imbalanced Data, Creating Derived Features and Feature Selection.

UNIT IV: Data Integration, Joining, and Workflows

Merging and Joining Datasets (Inner, Outer, Left, Right), Concatenation and Appending DataFrames, Data Consistency and Referential Integrity, Resolving Schema Mismatches, Designing Reusable Data Wrangling Functions, Automating Workflows with Functions and Pipelines, Data Lineage and Documentation, Case Study: End-to-End Data Wrangling Pipeline.

UNIT V: Tools, Libraries, and Case Studies in Data Wrangling

Pandas and NumPy Advanced Techniques, Pyjanitor, Dask, and Polars for Efficient Wrangling, Using OpenRefine for Data Cleaning, SQL vs NoSQL in Data Wrangling, Real-world Wrangling Case Studies (Finance, Healthcare, Retail), Best Practices and Common Pitfalls in Data Wrangling, Reproducibility and Versioning in Data Pipelines, Final Capstone: Build and Evaluate a Clean Dataset for ML.

Textbooks:

1. M. Heydt – Data Wrangling with pandas, O'Reilly Media.
2. Hadley Wickham – R for Data Science (Data Wrangling Chapters), O'Reilly.
3. J. VanderPlas – Python Data Science Handbook, O'Reilly Media.

Reference Books:

1. Wes McKinney – Python for Data Analysis, O'Reilly.
2. Cathy O'Neil and Rachel Schutt – Doing Data Science, O'Reilly.
3. David Mertz – Cleaning Data for Effective Data Science, Packt.

Online Learning Resources:

- Data Wrangling with pandas (Datacamp): https://www.datacamp.com/courses/data-manipulation-with-pandas
- Coursera: Data Wrangling, Analysis and AB Testing with SQL – *https://www.coursera.org/learn/data-wrangling-analysis-abtesting*
- edX: Data Wrangling with R – https://online.rice.edu/courses/data-wrangling-r
- Real Python Tutorials on pandas: https://realpython.com/learning-paths/pandas/
- Kaggle Notebooks (Data Cleaning & Wrangling): https://www.kaggle.com/learn/pandas

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech I Semester

		L	T	P	C
23ACA28	**Healthcare AI (Professional Elective-V)**	**3**	**0**	**0**	**3**

Course Objectives

- Introduce fundamental concepts and scalable algorithms used in mining massive datasets.
- Enable understanding of key techniques like clustering, classification, frequent itemset mining, and graph analysis on large-scale data.
- Familiarize students with distributed computing frameworks such as Hadoop and Spark.
- Provide practical insights into web and social network mining.
- Equip students with the ability to analyze massive datasets using real-world tools and platforms.

Course Outcomes

- Understand and explain the challenges involved in mining large-scale datasets.
- Apply efficient algorithms for clustering, classification, and association rule mining in big data environments.
- Analyze and implement scalable solutions using frameworks such as MapReduce, Hadoop, and Spark.
- Solve real-world problems involving link analysis, recommendation systems, and mining of web/social data.
- Critically evaluate algorithms based on scalability, efficiency, and effectiveness in large datasets.

UNIT I – Introduction to Massive Data and MapReduce Model

Types of Massive Data – Structured, Unstructured, and Semi-Structured, Challenges of Mining Massive Data Sets, Storage Systems – Distributed File Systems, HDFS, Introduction to MapReduce Programming Model, Designing MapReduce Algorithms, Matrix-Vector Multiplication by MapReduce, Workflow Management in Hadoop, Limitations of MapReduce.

UNIT II – Frequent Itemset and Association Rule Mining

Market Basket Model, A-Priori Algorithm – Scalable Variants, Handling Large Datasets in Frequent Pattern Mining, Park-Chen-Yu Algorithm, SON Algorithm, Multistage and Multihash Algorithms, PCY Algorithm and its Enhancements, Association Rules – Concepts and Evaluation, Finding Frequent Itemsets in Streaming Data

UNIT III – Clustering and Classification Techniques

Hierarchical and Partitional Clustering, K-Means Clustering and its Scalability, BFR and CURE Clustering Algorithms, Decision Trees and Rule-Based Classification, Naïve Bayes Classifier for Large Datasets, Logistic Regression and SVM for Massive Data, Parallel Clustering Techniques, Evaluation of Clustering Results

UNIT IV – Link Analysis and Mining of Web/Social Networks

Web Graph Structure and Crawling, PageRank and its Variants, Hubs and Authorities (HITS Algorithm), Link Spam Detection, Community Detection in Large Graphs, Mining Social Network Graphs, Recommendation Systems – User-Based and Item-Based Collaborative Filtering, Content-Based Filtering

UNIT V – Frameworks and Real-World Applications

Introduction to Apache Spark and RDDs, Spark MLlib for Data Mining, Streaming and Real-Time Data Analysis, Mining on Cloud Platforms (AWS, GCP, Azure), Case Study: E-commerce, Finance, and Healthcare, Scaling Algorithms to Petabyte-Level Data, Big Data Ethics and Governance, Research Trends in Mining Massive Data Sets

Textbooks

1. "Mining of Massive Datasets" by Jure Leskovec, Anand Rajaraman, and Jeffrey Ullman
2. "Data Mining: Concepts and Techniques" by Jiawei Han, Micheline Kamber, and Jian Pei
3. "Big Data: Principles and Best Practices of Scalable Real-Time Data Systems" by Nathan Marz and James Warren

Reference Books

1. "Big Data Analytics with Spark" by Mohammed Guller
2. "Hadoop: The Definitive Guide" by Tom White
3. "Practical Machine Learning with Spark" by Ajay Ohri
4. IEEE/ACM Journals and Conference Proceedings on Data Mining and Big Data

Online Courses

1. Mining Massive Datasets – Stanford University (Coursera)
2. Big Data Analysis with Apache Spark – edX (BerkeleyX)
3. Data Mining Specialization – University of Illinois (Coursera)

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech I Semester

23ACO13	**AI for Smart Cities & IoT Systems**	**L**	**T**	**P**	**C**
	(Professional Elective-V)	**3**	**0**	**0**	**3**

Course Objective:

- To introduce students to the integration of Artificial Intelligence and Internet of Things (IoT) technologies for developing smart city solutions.
- To understand the design, development, and deployment of intelligent systems to enhance urban infrastructure, transport, healthcare, energy, and governance.
- To explore edge and cloud computing techniques to optimize real-time AI-based decisions for IoT applications.
- To enable students to apply data analytics, computer vision, NLP, and automation to solve real-world urban challenges.
- To foster innovation using ethical AI frameworks in the context of sustainability, privacy, and smart governance.

Course Outcomes:

- Understand the architecture and components of smart cities powered by AI and IoT.
- Analyze and design AI-driven solutions for transportation, energy, healthcare, waste management, and smart governance.
- Deploy IoT systems that integrate sensors, edge devices, and AI models.
- Utilize AI algorithms (machine learning, NLP, and computer vision) for real-time smart city use cases.
- Evaluate and implement data-driven smart systems ensuring privacy, efficiency, and sustainability.
- Leverage cloud platforms and edge computing for scalable AIoT applications in urban environments.

Unit I: Introduction to AI in Smart Cities and IoT Systems
Smart City Concepts: Components, Infrastructure, and Urban Needs, Overview of IoT and AI Integration, Smart City Frameworks (India, Singapore, EU, etc.), IoT Architecture: Sensing, Network, Processing, and Application Layers, Role of AI in Urban Planning and Resource Optimization, Case Studies on AI in Smart Cities, Edge, Fog, and Cloud Computing Concepts for Smart Systems

Unit II: AI Applications in Smart Transportation and Mobility
Traffic Monitoring and Congestion Prediction using AI, Intelligent Traffic Signal Control using Reinforcement Learning, Autonomous Vehicles and AI Algorithms, Vehicle Detection and License Plate Recognition using CV, Public Transport Optimization using Predictive Analytics, Smart Parking and Navigation Systems, Use of Drones and AI for Traffic Surveillance

Unit III: AI and IoT for Smart Energy, Waste, and Water Management
AI for Smart Grids and Energy Consumption Prediction, Load Balancing and Demand Forecasting using ML, Waste Segregation and Collection Automation using CV, Water Quality Monitoring Systems using IoT Sensors, Leak Detection and Anomaly Detection Models, Smart Metering and Energy Theft Detection, Sustainability and Carbon Monitoring AI Models

Unit IV: Smart Healthcare, Surveillance, and Public Safety

IoT-based Health Monitoring and Alert Systems, Predictive Healthcare and Disease Outbreak Detection, AI for CCTV Surveillance, Crowd Monitoring, and Violence Detection, NLP for Emergency Response and Chatbot Assistance, Smart Ambulance Routing and Response Optimization, COVID-19 Contact Tracing and Monitoring via AI & IoT, Data Privacy, Security & Ethical Issues in Surveillance Systems

Unit V: AIoT System Design, Deployment, and Governance

AI Model Deployment on Edge Devices (Raspberry Pi, Jetson Nano), Smart City Dashboards and Data Visualization, Real-time Streaming and Analytics Platforms (Apache Kafka, Spark), Cloud Integration: AWS IoT, Google Cloud AI, Azure IoT Suite, Governance Frameworks, Data Privacy, and Policy Standards, Evaluation Metrics for Smart City Projects, Future Trends in AIoT and Smart Urban Living

Text Books:

1. Pethuru Raj & Anupama C. Raman, The Internet of Things: Enabling Technologies, Platforms, and Use Cases, CRC Press.
2. Janaka Ekanayake, Smart Grid: Technology and Applications, Wiley.
3. Rajkumar Buyya, Fog and Edge Computing: Principles and Paradigms, Wiley.
4. Adrian McEwen, Hakim Cassimally, Designing the Internet of Things, Wiley.

Reference Books:

1. Mahalik N. P., Sensor Networks and Applications, McGraw Hill.
2. Kim F. Taylor, Urban Artificial Intelligence and Governance, Springer.
3. Dastbaz, J. & Pattinson, C., Smart Cities: Innovation and Sustainability, Springer.
4. Research papers from IEEE Smart Cities, AIoT Journal, and Springer Urban Tech.

Online Courses:

1. Coursera – Smart Cities: Management of Smart Urban Infrastructures (EPFL)
2. edX – Internet of Things (IoT) Program – Curtin University

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech – I Semester

		L	T	P	C
23ACE55	**BUILDING MATERIALS AND SERVICES (OPEN ELECTIVE – III)**	3	0	0	3

Course Objectives:

The objectives of this course are to make the student :

1. To understand the properties, classifications, and applications of building materials like stones, bricks, tiles, wood, aluminum, glass, paints, and plastics.
2. To analyze the composition, manufacturing process, and properties of cement and admixtures.
3. To apply knowledge of building components such as lintels, arches, walls, stairs, floors, roofs, foundations, and joinery.
4. To evaluate masonry, mortars, finishing techniques, and formwork systems.
5. To assess various building services including plumbing, ventilation, air conditioning, acoustics, and fire protection.

Course Outcomes:

Upon successful completion of the course, students will be able to:

1. Understand the properties, classifications, and applications of building materials like stones, bricks, tiles, wood, aluminum, glass, paints, and plastics.
2. Analyze the composition, manufacturing process, and properties of cement and admixtures.
3. Apply knowledge of building components such as lintels, arches, walls, stairs, floors, roofs, foundations, and joinery.
4. Evaluate masonry, mortars, finishing techniques, and formwork systems.
5. Assess various building services including plumbing, ventilation, air conditioning, acoustics, and fire protection.

UNIT – I

StonesandBricks, Tiles: Building Stones – Classifications and Quarrying – Properties – Structural Requirements – Dressing. Bricks – CompositionofBrick Earth – Manufacture and Structural Requirements, Fly Ash, Ceramics. Timber, Aluminum, Glass, PaintsandPlastics: Wood - Structure – Types and Properties – Seasoning – Defects; Alternate Materials for Timber – GI / Fibre – Reinforced Glass Bricks, Steel & Aluminum, Plastics.

UNIT – II

Cement &Admixtures: Types of Cement - Ingredients of Cement – Manufacture – Chemical Composition – Hydration - Field & Lab Tests – Fineness – Consistency – Initial &Final Setting – Soundness . Admixtures – Mineral & Chemical Admixtures – Uses

UNIT – III

Building Components: Lintels, Arches, Walls, Vaults – Stair Cases – Types of Floors, Types of Roofs – Flat, Curved, Trussed; Foundations – Types; Damp Proof Course; Joinery – Doors – Windows – Materials – Types.

UNIT – IV

Mortars, MasonryandFinishing's Mortars: Lime and Cement Mortars Brick Masonry – Types – Bonds; Stone Masonry – Types; Composite Masonry – Brick-Stone Composite; Concrete, Reinforced Brick. Finishers: Plastering, Pointing, Painting, Claddings – Types – Tiles – ACP.form Work: Types: Requirements – Standards – Scaffolding – Design; Shoring, Underpinning.

UNIT – V

Building Services: Plumbing Services: Water Distribution, Sanitary – Lines &Fittings; Ventilations: Functional Requirements Systems of Ventilations. Air-Conditioning - Essentials andTypes; Acoustics – Characteristic – Absorption – Acoustic Design; Fire Protection – Fire Hazards – Classification of Fire Resistant Materials and Constructions.

TEXT BOOKS:

1. Building Materials and Construction – Arora&Bindra, Dhanpat Roy Publications.
2. Building Materials and Construction by G C Sahu, Joygopal Jena McGraw hill Pvt Ltd 2015.

REFRENCE BOOKS:

1. Building Construction by B. C. Punmia, Ashok Kumar Jain andArun Kumar Jain - Laxmi Publications (P) ltd., New Delh
2. P. C. Varghese, Building Materials, Prentice Hall of India, 2015.
3. N.Subramanian ,"Building Materials Testing and Sustainability", Oxford Higher Education, 2019.
4. R. Chudley, Construction Technology, Longman Publishing Group, 1973.
5. S. K. Duggal, Building Materials, Oxford & IBH Publishing Co. Ltd., New Delhi, 2019

Online Learning Resources:

https://archive.nptel.ac.in/courses/105/102/105102088/

CO – PO Articulation Matrix

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO -1	3	-	-	-	2	-	-	-	-	-	-	-	3	3
CO -2	3	3	-	-	2	-	-	-	-	-	-	2	3	3
CO -3	3	-	3	2	3	-	-	-	-	-	-	-	3	3
CO -4	-	-	3	3	3	-	2	-	-	-	-	-	3	3
CO -5	-	-	-	-	-	3	3	2	-	-	-	-	-	3

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

IV B.Tech – I Semester

23ACE26 - ENVIRONMENTAL IMPACT ASSESSMENT (OPEN ELECTIVE – III)	L	T	P	C
	3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. Understand the principles, methodologies, and significance of Environmental Impact Assessment (EIA).
2. Analyze the impact of developmental activities on land use, soil, and water resources.
3. Evaluate the impact of development on vegetation, wildlife, and assess environmental risks.
4. Develop environmental audit procedures and assess compliance with environmental regulations.
5. Understand and apply environmental acts, notifications, and legal frameworks in EIA studies.

Course Outcomes (COs):

Upon successful completion of the course, students will be able to:

1. Apply various methodologies for conducting Environmental Impact Assessments.
2. Analyze the impact of land-use changes on soil, water, and air quality.
3. Evaluate the environmental impact on vegetation, wildlife, and conduct risk assessments.
4. Develop environmental audit reports and assess compliance with environmental policies.
5. Interpret and apply environmental acts and regulations related to EIA.

Concepts and methodologies of EIA

Initial Environmental Examination, Elements of EIA, - Factors Affecting E-I-A Impact Evaluation and Analysis, Preparation of Environmental Base Map, Classification of Environmental Parameters- Criteria for The Selection of EIA Methodology, E I A Methods, Ad-Hoc Methods,

Matrix Methods, Network Method Environmental Media Quality Index Method, Overlay Methods and Cost/Benefit Analysis.

UNIT – II

Impact of Developmental Activities and Land Use

Introduction and Methodology for The Assessment of Soil and Ground Water, Delineation of Study Area, Identification of Actives. Procurement of Relevant Soil Quality, Impact Prediction, Assessment of Impact Significance, Identification and Incorporation of Mitigation Measures. E I Ain Surface Water, Air and Biological Environment: Methodology for The Assessment of Impacts On Surface Water Environment, Air Pollution Sources, Generalized Approach for Assessment of Air Pollution Impact.

UNIT – III

Assessment of Impact On Vegetation, Wildlife and Risk Assessment

Introduction - Assessment of Impact of Development Activities On Vegetation and Wildlife, Environmental Impact of Deforestation – Causes and Effects of Deforestation - Risk Assessment and Treatment of Uncertainty-Key Stages in Performing An Environmental Risk Assessment- Advantages of Environmental Risk Assessment.

UNIT – IV Environmental Audit

Introduction - Environmental Audit & Environmental Legislation Objectives of Environmental Audit, Types of Environmental Audit, Audit Protocol, Stages of Environmental Audit, Onsite Activities, Evaluation of Audit Data and Preparation of Audit Report

UNIT – V

Environmental Acts and Notifications

The Environmental Protection Act, The Water Preservation Act, The Air (Prevention &Control of Pollution Act), Wild Life Act - Provisions in The EIA Notification, Procedure for Environmental Clearance, Procedure for Conducting Environmental Impact Assessment Report- Evaluation of EIA Report. Environmental Legislation Objectives, Evaluation of Audit Data and Preparation of Audit Report. Post Audit Activities, ConceptofISO and ISO 14000.

TEXT BOOKS:

1. Environmental Impact Assessment Methodologies, by Y. Anjaneyulu, B. S. Publication, Hyderabad 2nd edition 2011
2. Environmental Impact Assessment, by Canter Larry W., McGraw-Hill education Edi (1996)

REFRENCE BOOKS:

1. Environmental Engineering, by Peavy, H. S, Rowe, D. R, Tchobanoglous, G.Mc-Graw Hill International Editions, New York 1985.
2. Environmental Science and Engineering, by Suresh K. Dhaneja, S.K., Katania& Sons Publication, New Delhi
3. Environmental Science and Engineering, by J. Glynn and Gary W. Hein Ke, Prentice Hall Publishers.
4. Environmental Pollution and Control, by H. S. Bhatia, Galgotia Publication (P) Ltd, Delhi

Online Learning Resources:
https://archive.nptel.ac.in/courses/124/107/124107160/

CO – PO Articulation Matrix

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO -1	3	2	2	2	2	3	-	-	-	-	-	1	2	2
CO -2	3	3	3	2	2	3	-	-	-	-	-	1	3	2
CO -3	3	3	3	2	2	3	3	-	-	-	-	1	3	3
CO -4	3	3	3	3	2	3	3	-	-	-	-	1	3	3
CO -5	2	2	2	2	2	3	3	3	-	-	-	1	2	2

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

IV B.Tech I Sem

23AEE42 SMART GRID TECHNOLOGIES
(Open Elective- III)

L – T – P – C
3 – 0 – 0 – 3

Course Outcomes:
CO1: Understanding the Concept and Evolution of Smart Grids. L2
CO2: Analyzing Wide Area Monitoring System and Synchrophasor Technology. L4
CO3: Applying Smart Metering and Advanced Metering Infrastructure (AMI) Concepts. L3
CO4: Evaluating Information and Communication Technology (ICT) Systems in Smart Grids. L5
CO5: Designing Smart Grid Applications and Cybersecurity Measures. L6

UNIT I Introduction to Smart Grid :
Evolution of Electric Grid – Need for Smart Grid – Difference between conventional & smart grid – Overview of enabling technologies – International experience in Smart Grid deployment efforts – Smart Grid road map for India – Smart Grid Architecture.

UNIT II Wide Area Monitoring System :
Fundamentals of Synchro phasor Technology – concept and benefits of Wide Area Monitoring System – Structure and functions of Phasor Measuring Unit (PMU) and Phasor Data Concentrator (PDC) – Road Map for Synchrophasor applications (NAPSI) – Operational experience and Blackout analysis using PMU - Case study on PMU.

UNIT III Smart Meters:
Features and functions of Smart Meters – Functional specification – category of Smart Meters – Automatic Meter Reading (AMR) and Advanced Metering Infrastructure (AMI) drivers and benefits – AMI protocol – Demand Side Integration: Peak load, Outage and Power Quality management.

UNIT IV Information and Communication Technology:
Overview of Smart Grid Communication system – Modulation and Demodulation Techniques: Radio Communication – Mobile Communication – Power Line Communication – Optical Fibre Communication – Communication Protocol for Smart Grid.

UNIT V Smart Grid Applications and Cyber Security:
Applications : Overview and concept of Renewable Integration – Introduction to distributed generation - Role of Protective Relaying in Smart Grid – House Area Network – Advanced Energy Storage Technology: Flow battery – Fuel cell – SMES – Super capacitors – Plug – in Hybrid electric Vehicles - Cyber Security: Security issues in DG, Distribution Automation, AMI, Electric Vehicle Management Systems – Approach to assessment of smart grid cyber security risks – Methodologies. Cyber Security requirements – Smart Grid Information Model.

TEXT BOOKS:

1. James Momoh, "SMART GRID : Fundamentals of Design and Analysis", John Wiley and Sons, New York, 2012.
2. Janaka Ekanayake, Nick Jenkins, Kithsiri Liyanage, Jianzhong Wu, Akihiko Yokoyama, "Smart Grid: Technology and Applications", John Wiley & Sons, New Jersey, 2012.

REFERENCES:

1. Power Grid Corporation of India Limited, "Smart Grid Primer", 1st Edition, Power Grid Corporation of India Limited, Bangalore, India, 2013.

2. Fereidoon.P.Sioshansi, "Smart Grid – Integrating Renewable, Distributed and Efficient Energy", 1st Edition, Academic Press, USA, 2011.

3. Stuart Borlase, "Smart Grids: Infrastructure, Technology and Solutions", 1st Edition, CRC Press Publication, England, 2013.

4. Phadke A G, Thorp J S, "Synchronized Phasor Measurements and Their Applications", 1st Edition, Springer, Newyork, 2012.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

IV B. Tech - I Semester
(Common to CE, EEE, ECE, EBM, CSE, CSC, CSM, CAI, CSD, CSBS, AI, AI& DS, IT, IOT)

Open Elective - III

L	T	P	C
3	0	0	3

23AME52: 3D PRINTING TECHNOLOGIES

Course Outcomes:
CO1: Define and explain the evolution and need for rapid prototyping in modern product development
CO2: Compare and contrast various 3D printing technologies based on working principles, materials, and limitations.
CO3: Apply knowledge of rapid tooling and reverse engineering techniques for industrial and design applications.
CO4: Diagnose and interpret different types of errors encountered in 3D printing processes and recommend solutions.
CO5: Use RP-specific software tools to manipulate STL files and prepare models for printing in real- world scenarios.

UNIT IIntroduction to 3D Printing
Introduction to Prototyping, Traditional Prototyping Vs. Rapid Prototyping (RP), Need for time compression in product development, Usage of RP parts, Generic RP process, Distinction between RP and CNC, other related technologies, Classification of RP.

UNIT II Solid and Liquid Based RP Systems
Working Principle, Materials, Advantages, Limitations and Applications of Fusion Deposition Modelling (FDM), Laminated Object Manufacturing (LOM), Stereo lithography (SLA), Direct Light Projection System (DLP) and Solid Ground Curing (SGC).

UNIT III Powder Based & Other RP Systems
Powder Based RP Systems: Working Principle, Materials, Advantages, Limitations and Applications of Selective Laser Sintering (SLS), Direct Metal Laser Sintering (DMLS), Laser Engineered Net Shaping (LENS) and Electron Beam Melting (EBM).
Other RP Systems: Working Principle, Materials, Advantages, Limitations and Applications of Three- Dimensional Printing (3DP), Ballastic Particle Manufacturing (BPM) and Shape Deposition Manufacturing (SDM).

UNIT IV Rapid Tooling & Reverse Engineering
Rapid Tooling: Conventional Tooling Vs. Rapid Tooling, Classification of Rapid Tooling, Direct and Indirect Tooling Methods, Soft and Hard Tooling methods.
Reverse Engineering (RE): Meaning, Use, RE – The Generic Process, Phases of RE Scanning, Contact Scanners and Noncontact Scanners, Point Processing, Application Geometric Model, Development

UNIT VErrors in 3D Printing and Applications:
Pre-processing, processing and post-processing errors, Part building errors in SLA, SLS, etc. Software: Need for software, MIMICS, Magics, Surgi Guide, 3-matic, 3D-Doctor, Simplant, Velocity2, VoXim, Solid View, 3DView, etc., software, Preparation of CAD models, Problems with STL files, STL file manipulation, RP data formats: SLC, CLI, RPI, LEAF, IGES, HP/GL, CT, STEP. Applications: Design, Engineering Analysis and planning applications, Rapid Tooling, Reverse Engineering, Medical Applications of RP.

Text books:

1. Chee Kai Chua and Kah Fai Leong, ―3D Printing and Additive Manufacturing Principles and Applications‖ 5/e, World Scientific Publications, 2017.
2. Ian Gibson, David W Rosen, Brent Stucker, ―Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing‖, Springer, 2/e, 2010.

Reference Books:

1. Frank W. Liou, ―Rapid Prototyping & Engineering Applications‖, CRC Press, Taylor & Francis Group, 2011.
2. Rafiq Noorani, ―Rapid Prototyping: Principles and Applications in Manufacturing‖, John Wiley Sons, 2006.

Online Learning Resources:

NPTEL Course on Rapid Manufacturing.

- https://nptel.ac.in/courses/112/104/112104265/
- https://www.hubs.com/knowledge-base/introduction-fdm-3d-printing/
- https://slideplayer.com/slide/6927137/
- https://www.mdpi.com/2073-4360/12/6/1334
- https://www.centropiaggio.unipi.it/sites/default/files/course/material/2013-11-29%20-%20FDM.pdf
- https://lecturenotes.in/subject/197
- https://www.cet.edu.in/noticefiles/258_Lecture%20Notes%20on%20RP-ilovepdfcompressed.pdf
- https://www.vssut.ac.in/lecture_notes/lecture1517967201.pdf
- https://www.youtube.com/watch?v=NkC8TNts4B4

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

II B. Tech - II Semester CSE (IOT)
IV B.Tech –I Semester (Common to CE, ME, DS, CSBS, AIDS, CAI, CSM, AI)
Course Code: 23AEC14

MICROPROCESSORS & MICROCONTROLLERS

L	T	P	C
3	0	0	3

Course Outcomes:
After successful completion of the course, the student will be able to:

CO1: Distinguish between microprocessors & microcontrollers (L2)
CO2: Demonstrate programming skills in assembly language for processors and Controllers. (L3)
CO3: Analyze various interfacing techniques (L4)
CO4: Apply the techniques for the design of processor / Controller based systems. (L4)

UNIT- I:
Introduction: Basic Microprocessor architecture, Harvard and Von Neumann architectures with examples, Microprocessor Unit versus Microcontroller Unit, CISC and RISC architectures. 8086 Architecture: Main features, pin diagram/description, 8086 microprocessor family, internal architecture, bus interfacing unit, execution unit, interrupts and interrupt response, 8086 system timing, minimum mode and maximum mode configuration.

UNIT- II:
8086 Programming: Program development steps, instructions, addressing modes, assembler directives, writing simple programs with an assembler, assembly language program development tools.

UNIT –III:
8086 Interfacing: Semiconductor memories interfacing (RAM, ROM), Intel 8255 programmable peripheral interface, Interfacing switches and LEDS, interfacing seven segment displays, software and hardware interrupt applications, Intel 8251 USART architecture and interfacing, Intel 8237a DMA controller, stepper motor, A/D and D/A converters, Need for 8259 programmable interrupt controllers.

UNIT-IV:
Intel 8051 MICROCONTROLLER
Architecture, Hardware concepts, Input/output ports and circuits, external memory, counters/timers, serial data input/output, interrupts. Assembly language programming: Instructions, addressing modes, simple programs.
Interfacing to 8051: A/D and D/A Convertors, Stepper motor interface, keyboard, LCD Interfacing, Traffic light control.

UNIT–V:
ARM Architectures and Processors: ARM Architecture, ARM Processors Families, ARM Cortex-M Series Family, ARM Cortex-M3 Processor Functional Description, functions and Interfaces Programmers Model – Modes of operation and execution, Instruction set summary, System address map, write buffer, bit-banding, processor core register summary, exceptions. ARM Cortext-M3 programming – Software delay, Programming techniques, Loops, Stack and Stack pointer, subroutines and parameter passing, parallel I/O, Nested Vectored Interrupt Controller – functional description and NVIC programmers‘ mode.

Textbooks:

1. Microprocessors and Interfacing – Programming and Hardware by Douglas V Hall, SSSP Rao, Tata McGraw Hill Education Private Limited, 3 rd Edition,1994.
2. The 8051 Microcontrollers and Embedded systems Using Assembly and C. Muhammad Ali Mazidi and Janice Gillespie Mazidi and Rollin D. McKinlay; Pearson 2- Edition, 2011.
3. The Definitive Guide to ARM Cortex-M3 and Cortex-M4 Processors by Joseph You.

Reference Books:

1. Embedded Systems Fundamentals with Arm Cortex-M based Microcontrollers: A Practical Approach in English, by Dr. Alexander G. Dean, Published by Arm Education Media, 2017.
2. Cortex -M3 Technical Reference Manual

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

IV B. Tech I Semester	L	T	P	C
	3	0	0	3

23AHS41 WAVELET TRANSFORMS AND ITS APPLICATIONS
(Common to all Branches)

Open Elective-III

Course Outcomes:
After successful completion of this course, the students should be able to:

CO1 Understand wavelets and wavelet basis and characterize continuous and discrete wavelet Transforms
CO2 Illustrate the multi resolution analysis ad scaling functions
CO3 Implement discrete wavelet transforms with multirate digital filters
CO4 Understand multi resolution analysis and identify various wavelets and evaluate their time- frequency resolution properties.
CO5 Design certain classes of wavelets to specification and justify the basis of the application of wavelet transforms to different fields

UNIT – I: Wavelets
Wavelets and Wavelet Expansion Systems - Wavelet Expansion- Wavelet Transform- Wavelet System- More Specific Characteristics of Wavelet Systems -Haar Scaling Functions and Wavelets - effectiveness of Wavelet Analysis -The Discrete Wavelet Transform- The Discrete-Time and Continuous Wavelet Transforms.

UNIT – II: A Multi resolution Formulation of Wavelet Systems
Signal Spaces -The Scaling Function -Multi resolution Analysis - The Wavelet Functions - The Discrete Wavelet Transform- A Parseval's Theorem - Display of the Discrete Wavelet Transform and the Wavelet Expansion.

UNIT – III Filter Banks and the Discrete Wavelet Transform
Analysis - From Fine Scale to Coarse Scale- Filtering and Down-Sampling or Decimating -Synthesis - From Coarse Scale to Fine Scale -Filtering and Up-Sampling or Stretching - Input Coefficients - Lattices and Lifting - -Different Points of View.

UNIT – IV Time-Frequency and Complexity
Multi resolution versus Time-Frequency Analysis- Periodic versus Non periodic Discrete Wavelet Transforms -The Discrete Wavelet Transform versus the Discrete-Time Wavelet Transform- Numerical Complexity of the Discrete Wavelet Transform.

UNIT-V Bases and Matrix Examples
Bases, Orthogonal Bases, and Bi orthogonal Bases -Matrix Examples - Fourier Series Example - Sine Expansion Example - Frames and Tight Frames - Matrix Examples -Sine Expansion as a Tight Frame Example.

Text Book:

1. C. Sidney Burrus, Ramesh A. Gopinath, ―Introduction to Wavelets and Wavelets Transforms‖, Prentice Hall, (1997).
2. James S. Walker, ―A Primer on Wavelets and their Scientific Applications‖, CRC Press, (1999)..

References:

1. RaghuveerRao, ―Wavelet Transforms‖, Pearson Education, Asia
2. C. S. Burrus, Ramose and A. Gopinath, Introduction to Wavelets and Wavelet Transform, Prentice Hall Inc.

Online References

1. http://users.rowan.edu/~polikar/WAVELETS/WTtutorial.html
2. http://www.wavelet.org/
3. http://www.math.hawaii.edu/~dave/Web/Amara's%20Wavelet%20Page.htm
4. https://jqichina.wordpress.com/wp-content/uploads/2012/02/ten-lectures-of-waveletsefbc88e5b08fe6b3a2e58d81e8aeb2efbc891.pdf

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	1
CO2	3	2	2	2	-	-	-	-	-	-	-	1
CO3	3	2	2	1	-	-	-	-	-	-	-	1
CO4	2	2	2	1	-	-	-	-	-	-	-	1
CO5	3	3	2	1	-	-	-	-	-	-	-	1

3-High Mapping **2- Medium Mapping** **1-Low Mapping**

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

SMART MATERIALS AND DEVICES
(Open Elective-III)
IV B.Tech I Semester (Common to all branches)

L	T	P	C
3	0	0	3

23AHS42: SMART MATERIALS AND DEVICES

COURSE OUTCOMES

After successful completion of the course, the students will be able to,

CO1: Identify key discoveries that led to modern applications of shape memory materials, describe the two phases in shape memory alloys.
CO2: Describe how different external stimuli (light, electricity, heat, stress, and magnetism) influence smart material properties.
CO3: Summarize various types of synthesis of smart materials
CO4: Analyze various characterization techniques used for smart materials
CO5: Interpret the importance of smart materials in various devices

UNIT-I: Introduction to Smart Materials **9hrs**

Historical account of the discovery and development of smart materials, Shape memory materials, chromoactive materials, magnetorheological materials, photoactive materials, Polymers and polymer composites (Basics).

UNIT-II: Properties of Smart Materials **9hrs**

Optical, Electrical, Dielectric, Piezoelectric, Ferroelectric, Pyroelectric and Magnetic properties of smart materials.

UNIT-III: Synthesis of Smart Materials **9hrs**

Chemical route: Chemical vapour deposition, Sol-gel technique, Hydrothermal method, Mechanical alloying and Thin film deposition techniques: Chemical etching, Spray pyrolysis.

UNIT-IV: Characterization Techniques **9hrs**

Powder X-ray diffraction, Raman spectroscopy (RS), UV-Visible spectroscopy, Scanning electron microscopy (SEM), Transmission electron microscopy (TEM), Atomic force microscopy (AFM).

Unit-V: Smart Materials based Devices **9hrs**

Devices based on smart materials: Shape memory alloysin robotic hands, piezoelectric based devices, MEMS and intelligent devices.

Textbooks:

1. Yaser Dahman, Nano technology and Functional Materials for Engineers-, Elsevier, 2017
2. E.Zschech, C.Whelan, T.Mikolajick, Materials for Information Technology: Devices, Interconnects and Packaging Springer-Verlag London Limited 2005.

ReferenceBooks:

1. Gauenzi, P., SmartStructures,Wiley,2009.
2. Mahmood Alio fkhazraei, Hand book of functional nano materials, Vol (1&2), Nova Publishers ,2014.

3. **Hand book of Smart Materials, Technologies, and Devices: Applications of Industry,4.0, Chaudhery Mustansar Hussain, Paolo DiSia, Springer,2022.**
4. Fundamentals of Smart Materials, Mohsen Shahinpoor, Royal Society of Chemistry, 2020

NPTEL course link: https://onlinecourses.nptel.ac.in/noc22_me17/preview

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	1	-	-	-	-	-	-	-
CO2	3	3	2	1	1	-	-	-	-	-	-	-
CO3	3	3	1	1	1	-	-	-	-	-	-	-
CO4	3	2	1	1	1	-	-	-	-	-	-	-
CO5	3	3	1	1	-	-	-	-	-	-	-	-

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)
(Open Elective-III)
IV B.Tech I Semester (Common to all branches)

L	T	P	C
3	0	0	3

23AHS43: INTRODUCTON TO QUANTUM MECHANICS

COURSE OUTCOMES

After successful completion of the course, the students will be able to,

CO1: Explain the key principles of quantum mechanics and wave-particle duality **CO2**: Apply Schrödinger equations to solve one-dimensional quantum problems **CO3**: Solve quantum mechanical problems using operator and matrix methods **CO4**: Evaluate quantum states using Dirac notation and expectation values
CO5: Analyze angular momentum and spin systems using Pauli matrices and operators

UNIT- I: PRINCIPLES OF QUANTUM MECHANICS **9hrs**

Introduction: Limitations of classical Mechanics, Difficulties with classical theories of black body radiation and origin of quantum theory of radiation. Wave-particle duality: de Broglie wavelength, Heisenberg uncertainty principle. Schrödinger time independent and time dependent wave equation, Solution of the time dependent Schrödinger equation, Concept of stationary states, Physical significance of wave function (ψ), Orthogonal, Normalized and Orthonormal functions

UNIT- II: ONE DIMENSIONAL PROBLEMS AND SOLUTIONS **9hrs**

Potential step – Reflection and Transmission at the interface. Potential well: Square well potential with rigid walls, Square well potential with finite walls. Potential barrier: Penetration of a potential barrier (tunneling effect). Periodic potential and Harmonic oscillator, Energy eigen functions and eigen values.

UNIT- III: OPERATOR FORMALISM **9hrs**

Operators, Operator Algebra, Eigen values and Eigen vectors, Postulates of quantum mechanics, Matrix representation of wave functions and linear operators.

UNIT- IV: MATHEMATICAL TOOLS FOR QUANTUM MECHANICS **9hrs**

The concept of row and column matrices, Matrix algebra, Hermitian operators – definition. Dirac's bra and ket notation, Expectation values, Heisenberg (operator) representation of harmonic oscillator, Ladder operators and their significance.

UNIT- V: ANGULAR MOMENTUM AND SPIN **9hrs**

Angular momentum operators: Definition. Eigen functions and Eigen values of AM operators. Matrix representation of angular momentum operators, System with spin half (1/2), Spin angular momentum, Pauli's spin matrices. Clebsch - Gordon coefficients. Rigid Rotator: Eigen functions and Eigen values.

Textbooks:

1. Quantum Mechanics Vol.1, A. Messaia Noth - Holland Pub. Co., Amsterdam, (1961).
2. A Text Book of Quantum Mechanics. P.M.Mathews and K.Venkatesam, Tata McGraw Hill, New Delhi,(1976).
3. Introduction to Quantum Mechanics. R.H.Dicke and J.P.Witke, Addison- Wisley Pub. Co. Inc., London, (1960).
4. Quantum Mechanics. S.L.Gupta, V.Kumar, H.V.Sarama and R.C.Sharma, Jai Prakash Nath& Co, Meerut, (1996).

ReferenceBooks:

1. Quantum Mechanics. L.I. Schiff, McGraw Hill Book Co., Tokyo, (1968).
2. Introduction to Quantum Mechanics. Richard L. Liboff, Pearson Education Ltd (Fourth Edn.) 2003.

VLab: CUBIT

NPTEL courses link :

1. **https://archive.nptel.ac.in/courses/115/101/115101107/**
2. **https://archive.nptel.ac.in/courses/122/106/122106034/**
3. **https://nptel.ac.in/courses/115106066**

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	-	-	-	-	-	-	-
CO2	3	2	2	1	1	-	-	-	-	-	-	-
CO3	3	3	2	1	1	-	-	-	-	-	-	-
CO4	3	3	3	2	3	-	-	-	-	-	-	-
CO5	3	3	1	1	1	-	-	-	-	-	-	-

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

GREEN CHEMISTRYAND CATALYSIS FOR SUSTAINABLE ENVIRONMENT
(Open Elective-III)
IV B.Tech I Semester (Common to all branches)

L	T	P	C
3	0	0	3

23AHS44: GREEN CHEMISTRYAND CATALYSIS FOR SUSTAINABLE ENVIRONMENT

COURSE OUTCOMES

After successful completion of the course, the students will be able to,

CO1: Apply the Green chemistry Principles for day-to-day life as well as synthesis, describe the sustainable development and green chemistry, explain economic and uneconomic reactions, Demonstrate Polymer recycling.

CO2: Explain Heterogeneous catalyst and its applications in Chemical and Pharmaceutical Industries, Differentiate Homogeneous and Heterogeneous catalysis, Identify the importance of Bio and Photo Catalysis, Discuss Transition metal and Phase transfer Catalysis

CO3: Demonstrate Green solvents and importance, Discuss Super critical carbon dioxide, Explain Supercritical water, recycling of green solvents.

CO4: Describe importance of Biomass and Solar Power, Illustrate So no chemistry, Apply Green Chemistry for Sustainable Development, discuss the importance of Renewable resources , mechano chemical synthesis.

CO5: Discuss Alternative green methods like Photo redox catalysis, single electron transfer reactions (SET), Photochemical Reactions, Microwave-assisted Reactions, and Sono chemical reactions , examples and applications.

Unit- I: Principles and Concepts of Green Chemistry **9 hrs**

Introduction, Green chemistry Principles, sustainable development and green chemistry, E factor, atom economy, atom economic reactions: Rearrangement and addition reactions, and atom uneconomic reactions: Substitution, elimination and Wittig reactions, Reducing Toxicity. Waste - problems and Prevention: Design for degradation, Polymer recycling

Unit - II: Catalysis and Green Chemistry **9 hrs**

Introduction, Types of catalysis, Heterogeneous catalysis: Basics of Heterogeneous Catalysis, Zeolite and the Bulk Chemical Industry, Heterogeneous Catalysis in the Fine Chemical and Pharmaceutical Industries, Catalytic Converters, Homogeneous catalysis: Transition Metal Catalysts with Phosphine Ligands, Greener Lewis Acids, and Phase transfer catalysis, Bio- catalysis and Photo-catalysis with examples.

Unit - III: Green Solvents in Chemical Synthesis **9 hrs**

Green Solvents: Concept, Tools and techniques for solvent selection, supercritical fluids:Supercritical carbon dioxide, supercritical water, Polyethylene glycol (PEG), Ionic liquids, Recycling of green solvents.

Unit - IV: Emerging Greener Technologies **9 hrs**

Biomass as renewable resource, Energy: Energy from Biomass, Solar Power, Chemicals from Renewable Feedstocks, Chemicals from Fatty Acids, Polymers from Renewable Resources, Alternative Economies: The Syngas Economy, The Biorefinery, Design for energy efficiency, Mechanochemical synthesis.

Unit - V: Alternative Greener Methods **9 hrs**

Photochemical Reactions - Examples, Advantages and Challenges, Photoredox catalysis, single electron transfer reactions (SET), Examples of Photochemical Reactions, Microwave-assisted Reactions and So no chemical reactions, examples and applications.

Text Books:

1. M. Lancaster, Green Chemistry an introductory text, Royal Society of Chemistry, 2002.
2. PaulT.Anastasand John C. Warner, Green Chemistry Theory and Practice, 4th Edition, Oxford University Press, USA

References Books:

1. Green Chemistry for Environmental Sustainability , First Edition, Sanjay K.Sharmaand AckmezMudhoo, CRC Press, 2010.
2. EditedbyAlvise Perosaand Maurizio Selva, Handbook of Green Chemistry Volume 8: Green Nanoscience, Wiley-VCH, 2013

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (AUTONOMOUS)

IV B. Tech, I Semester (Common to all branches) OPEN ELECTIVE-III

L	T	P	S
3	0	0	3

Course Code: 23AHS45 EMPLOYABILITY SKILLS

Course Outcomes (CO):

After successful completion of the course, the student will be able to: CO1: Understand the importance of goals and try to achieve them

CO2: Explain the significance of self-management

CO3: Apply the knowledge of writing skills in preparing eye-catchy resumes

CO4: Analyse various forms of Presentation skills

CO5: Judge the group behaviour appropriately

CO6: Develop skills required for employability

UNIT- I Goal Setting and Time Management

Definition, importance, types of Goal Setting – SMART Goal Setting – Advantages-Motivation – Intrinsic and Extrinsic Motivation –- Self-Management - Knowing about self – SWOC Analysis – Emotional Intelligence

UNIT -II Writing Skills

Definition, significance, types of writing skills – Resume writing Vs CV Writing - E-Mail writing, Cover Letters - E-Mail Etiquette -SoP (Statement of Purpose)

UNIT –III Technical Presentation Skills

Nature, meaning & significance of Presentation Skills – Planning, Preparation, Presentation, Stage Dynamics –Anxiety in Public speaking (Glossophobia)- PPT & Poster Presentation

UNIT -IV Group Presentation Skills

Body Language – Group Behaviour - Team Dynamics – Leadership Skills – Personality Manifestation- Group Discussion-Debate –Corporate Etiquette

UNIT- V Job Cracking Skills

Nature, characteristics, importance & types of Interviews – Job Interviews – Skills for success – Job searching skills - STAR method - FAQs- Answering Strategies – Mock Interviews

Text Books:

1. Sabina Pillai, Agna Fernandez. *Soft Skills &Employability Skills***,** 2014.Cambridge Publisher. Alka Wadkar. *Life Skills for Success*, Sage Publications, 2016.
2. Clear, James. *Atomic Habits: An Easy &Proven Way to Build Good Habits &Break Bad Ones*. Penguin Publishing Group, 2018.

Reference Books:

1. **Gangadhar Joshi. *Campus to Corporate Paperback* , Sage Publications. 2015**
2. Sherfield Montogomery Moody,*Cornerstone Developing Soft Skills*, Pearson Publications. 4 Ed. 2008
3. Shikha Kapoor. *Personality Development and Soft Skills - Preparing for Tomorrow* .1 Edition, Wiley, 2017.
4. M. Sen Gupta, *Skills for Employability*, Innovative Publication, 2019.
5. Steve Duck and David T McMahan**,** *The Basics f Communication Skills A Relational Perspective*, Sage press, 2012.

Online Learning Resources:

1. https://youtu.be/gkLsn4ddmTs
2. https://youtu.be/2bf9K2rRWwo
3. https://youtu.be/FchfE3c2jzc
4. https://youtu.be/xBaLgJZ0t6A?list=PLzf4HHlsQFwJZel_j2PUy0pwjVUgj7KlJ
5. https://www.youtube.com/c/skillopedia/videos
6. https://onlinecourses.nptel.ac.in/noc25_hs96/preview
7. https://onlinecourses.nptel.ac.in/noc21_hs76/preview
8. https://archive.nptel.ac.in/courses/109/107/109107172/#
9. https://archive.nptel.ac.in/courses/109/104/109104107/

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

IV B.Tech – I Semester

		L	T	P	C
23ACE56	**GEO-SPATIAL TECHNOLOGIES (OPEN ELECTIVE – IV)**	**3**	**0**	**0**	**3**

Course Objectives:

The objectives of this course are to make the student :

1. To understand raster-based spatial analysis techniques, including query, overlay, and cost-distance analysis.
2. To analyze vector-based spatial analysis techniques such as topology, overlay, and proximity analysis.
3. To apply network analysis techniques for geocoding, shortest path analysis, and location-allocation problems.
4. To evaluate surface and geostatistical analysis methods, including terrain modeling, watershed analysis, and spatial interpolation.
5. To assess GIS customization, Web GIS, and mobile mapping techniques for real-world applications.

Course Outcomes:

Upon successful completion of the course, students will be able to:

1. Understand raster-based spatial analysis techniques, including query, overlay, and cost-distance analysis.
2. Analyze vector-based spatial analysis techniques such as topology, overlay, and proximity analysis.
3. Apply network analysis techniques for geocoding, shortest path analysis, and location-allocation problems.
4. Evaluate surface andgeostatistical analysis methods, including terrain modeling, watershed analysis, and spatial interpolation.
5. Assess GIS customization, Web GIS, and mobile mapping techniques for real-world applications.

UNIT – I

RASTER ANALYSIS

Raster Data Exploration: Query Analysis - Local Operations: Map Algebra, Reclassification, Logical and Arithmetic Overlay Operations—Neighborhood - Operations: Aggregation, Filtering – Extended Neighborhood-Operations- Zonal Operations - Statistical Analysis – Cost-Distance Analysis-Least Cost Path.

UNIT – II

VECTOR ANALYSIS

Non-Topological Analysis: Attribute Database Query, Structured Query Language, Co-Ordinate Transformation, Summary Statistics, Calculation of Area, Perimeter and Distance – topological Analysis: Reclassification, Aggregation, Overlay Analysis: Point-In-Polygon, Line-In-Polygon, Polygon-On- Polygon: Clip, Erase, Identity, Union, Intersection – Proximity Analysis: Buffering

UNIT – III

NETWORK ANALYSIS

Network – Introduction - Network Data Model – Elements of Network - Building A Network Database - Geocoding – Address Matching - Shortest Path in A Network – Time and Distance Based Shortest Path Analysis – Driving Directions – Closest Facility Analysis – Catchment / Service Area Analysis-Location-
Allocation Analysis

UNIT – IV

SURFACE and GEOSTATISTICAL ANALYSIS

Surface Data – Sources of X,Y, Z Data – DEM, TIN – Terrain Analysis – Slope, Aspect, Viewshed, Watershed Analysis: Watershed Boundary, Flow Direction, Flow Accumulation, Drainage Network, Spatial Interpolation: IDW, Spline, Kriging, Variogram.

UNIT – V

CUSTOMISATION, WEB GIS, MOBILE MAPPING

Customisation of GIS: Need, Uses, Scripting Languages –Embedded Scripts – Use of Python Script - Web GIS: Web GIS Architecture, Advantages of Web GIS, Web Applications- Location Based Services: Emergency and Business Solutions - Big Data Analytics.

TEXT BOOKS:

1. Kang – Tsung Chang, Introduction to Geographical Information System, 4th Ed., Tata McGraw Hill Edition, 2008.
2. Lo, C.P. andYeung, Albert K.W., Concepts and Techniques of Geographic Information Systems Prentice Hall, 2002.

REFRENCE BOOKS:

1. Michael N. Demers, Fundamentals of Geographic Information Systems, Wiley,2009
2. Ian Heywood, Sarah Cornelius, Steve Carver, Srinivasaraju, “An Introduction to Geographical Information Systems, Pearson Education, 2nd Edition, 2007.
3. John Peter Wilson, The Handbook of Geographic Information Science, Blackwell Pub.,2008

Online Learning Resources:

https://archive.nptel.ac.in/courses/105/105/105105202/
https://onlinecourses.nptel.ac.in/noc19_cs76/preview

CO – PO Articulation Matrix

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO -1	3	-	-	-	2	-	-	-	-	-	-	-	3	3
CO -2	3	3	-	-	2	-	-	-	-	-	-	2	3	3
CO -3	3	-	3	2	3	-	-	-	-	-	-	-	3	3
CO -4	-	-	3	3	3	-	2	-	-	-	-	-	3	3
CO -5	-	-	-	-	3	3	3	2	-	-	-	-	3	3

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

IV B.Tech – I Semester

		L	T	P	C
23ACE57	**SOLID WASTE MANAGEMENT (OPEN ELECTIVE – IV)**	**3**	**0**	**0**	**3**

Course Objectives:

The objectives of this course are to make the student :

1. To understand the types, sources, and characteristics of solid waste, along with regulatory frameworks.
2. To analyze engineering systems for solid waste collection, storage, and transportation.
3. To apply resource and energy recovery techniques for sustainable solid waste management.
4. To evaluate landfill design, construction, and environmental impact mitigation strategies.
5. To assess hazardous waste management techniques, including biomedical and e-waste disposal.

Course Outcomes:

1. Understand the types, sources, and characteristics of solid waste, along with regulatory frameworks.
2. Analyze engineering systems for solid waste collection, storage, and transportation.
3. Apply resource and energy recovery techniques for sustainable solid waste management.
4. Evaluate landfill design, construction, and environmental impact mitigation strategies.
5. Assess hazardous waste management techniques, including biomedical and e-waste

UNIT – I

Solid Waste: Definitions, Types of Solid Wastes, Sources of Solid Wastes, Characteristics, and Perspectives; Properties of Solid Wastes, Sampling of Solid Wastes, Elements of Solid Waste Management - Integrated Solid Waste Management, Solid Waste Management Rules 2016.

UNIT – II

Engineering SystemsforSolid Waste Management: Solid Waste Generation; On-Site Handling, Storage and Processing; Collection of Solid Wastes; Stationary Container System and Hauled Container Systems – Route Planning - Transfer and Transport; Processing Techniques;

UNIT – III

Engineering Systems for Resource and Energy Recovery: Processing Techniques; Materials Recovery Systems; Recovery of Biological Conversion Products – Composting, Pre and Post Processing, Types of Composting, Critical Parameters, Problems With Composing - Recovery of Thermal Conversion Products; Pyrolisis, Gasification, RDF - Recovery of Energy From Conversion Products; Materials and Energy Recovery Systems.

UNIT – IV

Landfills: Evolution of Landfills – Types and Construction of Landfills – Design Considerations – Life of Landfills-Landfill Problems – Lining of Landfills – Types of Liners – Leachate Pollution and Control – Monitoring Landfills – Landfills Reclamation.

UNIT – V

Hazardous Waste Management: – Sources and Characteristics, Effects On Environment, Risk Assessment – Disposal of Hazardous Wastes – Secured Landfills, Incineration - Monitoring – Biomedical Waste Disposal, E-Waste Management, Nuclear Wastes, Industrial Waste Management

TEXT BOOKS:

1. Tchobanoglous G, Theisen H and Vigil SA ‘Integrated Solid Waste Management, Engineering Principles and Management Issues’ McGraw-Hill, 1993.
2. Vesilind PA, Worrell W and Reinhart D, ‘Solid Waste Engineering’ Brooks/Cole Thomson Learning Inc., 2002.

REFRENCE BOOKS:

1. Peavy, H.S, Rowe, D.R., and G. Tchobanoglous, 'Environmental Engineering', McGraw Hill Inc., New York, 1985.
2. Qian X, Koerner RM and Gray DH, 'Geotechnical Aspects of Landfill Design and Construction' Prentice Hall, 2002.

Online Learning Resources:

https://archive.nptel.ac.in/courses/105/103/105103205/
https://archive.nptel.ac.in/courses/120/108/120108005/

CO – PO Articulation Matrix

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO -1	3	-	-	-	2	-	2	-	-	-	-	-	3	3
CO -2	3	3	-	-	2	-	3	-	-	-	-	2	3	3
CO -3	3	-	3	2	3	-	3	-	-	-	-	-	3	3
CO -4	-	-	3	3	3	-	3	2	-	-	-	-	3	3
CO -5	-	-	-	-	3	3	3	3	-	-	-	-	3	3

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

IV B.Tech I Sem

23AEE43 - ELECTRIC VEHICLES
(Open Elective -IV)

L	T	P	C
3	0	0	3

Course Objectives: To make the student

- Remember and understand the differences between conventional Vehicle and Electric Vehicles, electro mobility and environmental issues of EVs.
- Analyze various EV configurations, parameters of EV systems and Electric vehicle dynamics.
- Analyze the basic construction, operation and characteristics of fuel cells and battery charging techniques in HEV systems.
- Design and analyze the various control structures for Electric vehicle.

Course Outcomes (CO): Student will be able to

CO 1: To understand and differentiate between Conventional Vehicle and Electric Vehicles, electro mobility and environmental issues of EVs. -L2

CO 2: Understand Various dynamics of Electric Vehicles. -L2

CO 3: To remember and understand various configurations in parameters of EV system and dynamic aspects of EV. -L1

CO 4: To analyze fuel cell technologies in EV and HEV systems. -L3

CO 5: To analyze the battery charging and controls required of EVs. -L3

UNIT I Introduction to EV Systems and Energy Sources:

Past, Present and Future of EV - EV Concept- EV Technology- State-of-the Art of EVs- EV configuration- EV system- Fixed and Variable gearing- Single and multiple motor drive- In-wheel drives- EV parameters: Weight, size, force and energy, performance parameters. Electro mobility and the environment- History of Electric power trains- Carbon emissions from fuels- Green houses and pollutants- Comparison of conventional, battery, hybrid and fuel cell electric systems.

UNIT II EV Propulsion and Dynamics:

Choice of electric propulsion system- Block diagram- Concept of EV Motors- Single and multi- motor configurations- Fixed and variable geared transmission- In-wheel motor configuration- Classification - Electric motors used in current vehicle applications - Recent EV Motors- Vehicle load factors- Vehicle acceleration.

UNIT III Fuel Cells:

Introduction of fuel cells- Basic operation- Model - Voltage, power and efficiency- Power plant system – Characteristics- Sizing - Example of fuel cell electric vehicle - Introduction to HEV- Brake specific fuel consumption - Comparison of Series-Parallel hybrid systems- Examples.

UNIT IV Battery Charging and Control:

Battery charging: Basic requirements- Charger architecture- Charger functions- Wireless charging- Power factor correction.

Control: Introduction- Modeling of electro mechanical system- Feedback controller design approach- PI controller's designing- Torque-loop, Speed control loop compensation- Acceleration of battery electric vehicle.

UNIT V Energy Storage Technologies:

Role of Energy Storage Systems- Thermal- Mechanical-Chemical- Electrochemical- Electrical - Efficiency of energy storage systems- Super capacitors-Superconducting Magnetic Energy Storage (SMES)- SOC- SoH -fuel cells - G2V- V2G- Energy storage in Micro-grid and Smart grid- Energy Management with storage systems- Battery SCADA

Text books:

1.C.C Chan, K.T Chau: Modern Electric Vehicle Technology, Oxford University Press Inc., New York 2001,1st Edition

2.Ali Emadi, "Advanced Electric Drive Vehicles", CRC Press, 2017,1st Edition

Reference Books:

1. Electric and Hybrid Vehicles Design Fundamentals, Iqbal Husain, CRC Press 2021, 3rd Edition.

2. Francisco Díaz-González, Andreas Sumper, Oriol Gomis-Bellmunt," Energy Storage in Power Systems" Wiley Publication, ISBN: 978-1-118-97130-7, Mar 2016,1st Edition

3. A.G.Ter-Gazarian, "Energy Storage for Power Systems", the Institution of Engineering and Technology (IET) Publication, UK, (ISBN – 978-1-84919-219-4), Second Edition, 2011.

4. Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, "Modern Elelctric, Hybrid Elelctric and Fuel Cell Vehicles: Fundamentals, Theory and Design", CRC Press, 2004,1st Edition

5. James Larminie, John Lowry, "Electric Vehicle Technology Explained", Wiley, 2003,2nd Edition.

Online Learning Resources:

1. https://nptel.ac.in/courses/108/102/108102121/
2. https://nptel.ac.in/syllabus/108103009

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

IV B. Tech - I Semester (ME)
(Common to CE, EEE, ECE, EBM, CSE, CSC, CSM, CAI, CSD, CSBS, AI, AI& DS, IT, IOT)

Open Elective - IV

L	T	P	C
3	0	0	3

23AME46: TOTAL QUALITY MANAGEMENT

Course Outcomes:

CO1: Define and explain the basic concepts of quality, quality costs, and the scope of Total Quality Management.
CO2: Summarize the philosophies and contributions of TQM pioneers and evaluate barriers and enablers for TQM implementation.
CO3: Apply TQM principles such as employee empowerment, customer satisfaction, and supplier partnerships to real-world business scenarios.
CO4: Analyze the application of tools like QFD, FMEA, Six Sigma, and Benchmarking in improving product and process quality.
CO5: Evaluate and formulate quality systems like ISO 9000 and ISO 14000, and design documentation and auditing processes.

UNIT - I
Introduction: Definition of Quality, Dimensions of Quality, Definition of Total quality management, Quality Planning, Qualitycosts – Analysis, Techniques for Quality costs, Basic concepts of Total QualityManagement.

UNIT - II
Historical Review: Quality council, Quality statements, Strategic Planning, Deming Philosophy, Barriers of TQM Implementation, Benefits of TQM, Characteristics of successful quality leader, Contributions of Gurus of TQM, Case studies.

UNIT - III
TQM Principles: Customer Satisfaction – Customer Perception of Quality, Customer Complaints, Service Quality, Customer Retention, Employee Involvement – Motivation, Empowerment teams, Continuous Process Improvement – Juran Trilogy, PDSA Cycle, Kaizen, Supplier Partnership – Partnering, sourcing, Supplier Selection, Supplier Rating, Relationship Development, Performance Measures – Basic Concepts, Strategy, Performance Measure Case studies.

UNIT - IV
TQM Tools: Benchmarking – Reasons to Benchmark, Benchmarking Process, Quality Function Deployment (QFD) – House of Quality, QFD Process, Benefits, Taguchi Quality Loss Function, Total Productive Maintenance (TPM) – Concept, Improvement Needs, FMEA – Stages of FMEA, The seven tools of quality, Process capability, Concept of Six Sigma, New Seven management tools, Case studies.

UNIT - V
Quality Systems: Need for ISO 9000 and Other Quality Systems, ISO 9000: 2000 Quality System – Elements, Implementation of Quality System, Documentation, Quality Auditing, QS 9000, ISO 14000 – Concept, Requirements and Benefits, Case Studies.

Text Books:

1. Dale H Besterfield, Total Quality Management, Fourth Edition, Pearson Education, 2015.
2. Subburaj Ramaswamy, Total Quality Management, Tata McGraw Hill Publishing Company Ltd., 2005.
3. Joel E. Ross, Total Quality Management, Third Edition, CRC Press, 2017.

Reference Books:

1. Narayana V and Sreenivasan N.S, Quality Management – Concepts and Tasks, New Age International, 1996.
2. Robert L. Flood, Beyond TQM, First Edition, John Wiley & Sons Ltd, 1993.
3. Richard S. Leavenworth & Eugene Lodewick Grant, Statistical Quality Control, Seventh Edition, Tata McGraw Hill, 2015
4. Samuel Ho, TQM – An Integrated Approach, Kogan Page Ltd, USA, 1995.

Online Learning Resources:

- https://www.youtube.com/watch?v=VD6tXadibk0
- https://www.investopedia.com/terms/t/total-quality-management-tqm.asp
- https://blog.capterra.com/what-is-total-quality-management/
- https://nptel.ac.in/courses/110/104/110104080/
- https://onlinecourses.nptel.ac.in/noc21_mg03/preview
- https://nptel.ac.in/courses/110/104/110104085/
- https://nptel.ac.in/noc/

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

IV B. Tech - I Semester (Common to all branches except ECE, EBM) Course Code: 23AEC53

TRANSDUCERS AND SENSORS
(Open Elective –IV)

L	T	P	C
3	0	0	3

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Understand characteristics of Instrumentation System and the operating principle of motion transducers. **CO2:** Explore working principles, and applications of different temperature transducers and Piezo-electric sensors.

CO3: Gain knowledge on flow transducers and their applications.

CO4: Learn the working principles of pressure transducers.

CO5: Understand the working principle and applications of force and sound transducers.

UNIT- I:

Introduction: General Configuration and Functional Description of measuring instruments, Static and Dynamic Characteristics of Instrumentation System, Errors in Instrumentation System, Active and Passive Transducers and their Classification.

Motion Transducers: Resistive strain gauge, LVDT, RVDT, Capacitive transducers, Piezo-electric transducers, seismic displacement pick-ups, vibrometers and accelerometers.

UNIT- II:

Temperature Transducers: Standards and calibration, fluid expansion and metal expansion type transducers - bimetallic strip, Thermometer, Thermistor, RTD, Thermocouple and their characteristics. Hall effect transducers, Digital transducers, Proximity devices, Bio-sensors, Smart sensors, Piezo-electric sensors.

UNIT –III:

Flow Transducers: Bernoulli's principle and continuity, Orifice plate, Nozzle plate, Venture tube, Rotameter, Anemometers, Electromagnetic flow meter, Impeller meter and Turbid flow meter.

UNIT- IV:

Pressure Transducers: Standards and calibration, different types of manometers, elastic transducers, diaphragm bellows, bourdon tube, capacitive and resistive pressure transducers, high and low pressure measurement.

UNIT -V:

Force and Sound Transducers: Proving ring, hydraulic and pneumatic load cell, dynamometer and gyroscopes. Sound level meter, sound characteristics, Microphone.

Text books:

1. A.K. Sawhney, “A course in Electrical and Electronics Measurements and Instrumentation”, Dhanpat Rai& Co. 3rd edition Delhi, 2010.
2. Rangan C.S, Sarma G.R and Mani V S V, “Instrumentation Devices and Systems”, TATA McGraw Hill publications, 2007.

Reference books:

1. Doebelin. E.O, “Measurement Systems Application and Design”,McGraw Hill International, New York, 2004.
2. Nakra B.C andChaudharyK.K , “Instrumentation Measurement and Analysis”, Second Edition, Tata McGraw-Hill Publication Ltd.2006.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech I Sem

23AMB22 FINANCIAL MATHEMATICS (Open Elective-IV)

L	T	P	C
3	0	0	3

Course Objectives:

1. To provide mathematical foundations for financial modelling, risk assessment and asset pricing.
2. To introduce stochastic models and their applications in pricing derivatives and interest rate modelling.
3. To develop analytical skills for fixed-income securities, credit risk, and investment strategies.
4. To equip students with computational techniques for pricing financial derivatives.

Course Outcomes:

After successful completion of this course, the students should be able to:

CO1	Explain fundamental financial concepts, including arbitrage, valuation, and risk.	L2 (Understand)
CO2	Apply stochastic models, including Brownian motion and Stochastic Differential Equations (SDEs), in financial contexts.	L3 (Apply)
CO3	Analyze mathematical techniques for pricing options and financial derivatives.	L4 (Analyze)
CO4	Evaluate interest rate models and bond pricing methodologies.	L5 (Evaluate)
CO5	Utilize computational techniques such as Monte Carlo simulations for financial modeling.	L3 (Apply)

UNIT-I: Asset Pricing and Risk Management (08)

Fundamental financial concepts: Returns, arbitrage, valuation, and pricing. Asset/Liability management, investment income, capital budgeting, and contingent cash flows. One-period model: Securities, payoffs, and the no-arbitrage principle. Option contracts: Speculation and hedging strategies, CAP Model, Efficient market hypothesis.

UNIT-II: Stochastic Models in Finance (08)

Random Walks and Brownian Motion. Introduction to Stochastic Differential Equations (SDEs): Drift and diffusion. Ito calculus: Ito’s Lemma, Ito Integral, and Ito Isometry.

UNIT-III: Interest Rate and Credit Modelling (08)

Interest rate models and bond markets. Short-rate models: Vasicek, Cox-Ingersoll-Ross (CIR), Hull & White models, Credit risk modelling: Hazard function and hazard rate.

UNIT-IV: Fixed-Income Securities and Bond Pricing (08)

Characteristics of fixed-income products: Yield, duration, and convexity. Yield curves, forward rates, and zero-coupon bonds. Stochastic interest rate models and bond pricing PDE. Yield curve fitting and calibration techniques, Mortgage Backed Securities.

UNIT-V: Exotic Options and Computational Finance **(08)**

Stochastic volatility models and the Feynman-Kac theorem.Exotic options: Barriers, Asians, and Look backs. Monte Carlo methods for derivative pricing, Black-Scholes-Merton model: Derivation and applications.

Textbooks:

1. Ales Cerny, *Mathematical Techniques in Finance: Tools for Incomplete Markets*, Princeton University Press.
2. S.R. Pliska, *Introduction to Mathematical Finance: Discrete-Time Models*, Cambridge University Press.

Reference Books:

1. IoannisKaratzas& Steven E. Shreve, *Methods of Mathematical Finance*, Springer, New York.
2. John C. Hull, *Options, Futures, and Other Derivatives*, Pearson.

Web References:

- MIT– Mathematics for Machine Learning https://ocw.mit.edu
- Coursera – Financial Engineering and Risk Management (Columbia University) https://www.coursera.org/
- National Stock Exchange (NSE) India – Financial Derivatives https://www.nseindia.com/

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	1	-	-	-	-	-	2	1
CO2	3	3	2	2	2	-	-	-	-	-	1	1
CO3	3	3	3	3	2	1	-	-	-	-	3	2
CO4	3	3	3	3	1	-	-	-	-	-	2	1
CO5	3	3	3	3	3	-	-	-	-	-	2	2

• **3** = Strong Mapping, **2** = Moderate Mapping, **1** = Slight Mapping, **-** = No Mapping

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)
(Open Elective-IV)

IV B.Tech I Semester (Common to all branches)

L	T	P	C
3	0	0	3

23AHS46: SENSORS AND ACTUATORS FOR ENGINEERING APPLICATIONS

COURSE OUTCOMES

After successful completion of the course, the students will be able to,

CO1: Classify different types of Sensors and Actuators along with their characteristics
CO2: Summarize various types of Temperature and Mechanical sensors
CO3: Illustrates various types of optical and mechanical sensors
CO4: Analyze various types of Optical and Acoustic Sensors
CO5: Interpret the importance of smart materials in various devices

UNIT- I: Introduction to Sensors and Actuators **9hrs**

Sensors: Types of sensors: temperature, pressure, strain, active and passive sensors, General characteristics of sensors (Principles only), Deposition: Chemical Vapor
Actuators: Functional diagram of actuators, Types of actuators and their basic principle of working: Pneumatic, Electromagnetic, Piezo-electric and Piezo-resistive actuators, Applications of Actuators.

Unit-II: Temperature and Mechanical Sensors **9hrs**

Temperature Sensors: Types of temperature sensors and their basic principle of working: Thermo- resistive sensors: Thermistors, Thermo-electric sensors: Thermocouples, PN junction temperature sensors Mechanical Sensors: Types of Mechanical sensors and their basic principle of working: Force sensors: Tactile sensors, Pressure sensors: Piezo resistive.

Unit-III: Optical and Acoustic Sensors **9hrs**

Optical Sensors: Basic principle and working of: Photodiodes, Phototransistors and Photo resistors based sensors, Photomultipliers, Acoustic Sensors: Principle and working of Ultrasonic sensors, Piezo-electric resonators, Microphones

Unit-IV: Magnetic and electromagnetic Sensors **9hrs**

Motors as actuators (linear, rotational, stepping motors), magnetic valves, inductive sensors (LVDT, RVDT, and Proximity), Hall Effect sensors, Magneto-resistive sensors, Magnetostrictive sensors and actuators.

Unit-V: Chemical and Radiation Sensors **9hrs**

Chemical Sensors: Principle and working of Electro-chemical, Thermo-chemical, Gas, pH, Humidity and moisture sensors. Radiation Sensors: Principle and working of Ionization detectors, Scintillation detectors.

Text books:

1. Sensors and Actuators – Clarence W.de Silva, CRC Press, 2nd Edition, 2015.
2. Sensors and Actuators, D.A. Hall and C.E. Millar, CRC Press, 1999.

Reference Books:

1. Sensors and Transducers, D.Patranabhis, Prentice Hall of India (Pvt) Ltd.2003.
2. Measurement, Instrumentation, and Sensors Handbook – John G.Webster, CRC press, 1999.
3. Sensors – A Comprehensive Sensors - Henry Bolte, John Wiley.
4. Hand book of modern sensors, Springer, Stefan Johann Rupitsch.

NPTEL course link: https://onlinecourses.nptel.ac.in/noc21_ee32/preview

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	1	-	-	-	-	-	-	-
CO2	3	3	2	1	1	-	-	-	-	-	-	-
CO3	3	3	1	1	1	-	-	-	-	-	-	-
CO4	3	2	1	1	-	-	-	-	-	-	-	-
CO5	3	3	1	1	-	-	-	-	-	-	-	-

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

CHEMISTRY OF NANO MATERIALS AND APPLICATIONS
(Open Elective-IV)
IV B.Tech I Semester (Common to all branches)

L	T	P	C
3	0	0	3

23AHS47: CHEMISTRY OF NANO MATERIALS AND APPLICATIONS COURSE

OUTCOMES

After successful completion of the course, the students will be able to,

CO1: Classify the nano structure materials; describe scope of nano science and importance technology.

CO 2: Describe the top-down approach, explain aerosol synthesis and plasma arc technique, differentiate chemical vapor deposition method and electrode position method, discuss about high- energy ball milling

CO3: Discuss different technique for characterization of nano material, explain electron microscopy techniques for characterization of nano material, Describe BET method for surface area analysis.

CO4: Explainsynthesisandpropertiesandapplicationsofnanomaterials.Discussabout fullerenes and carbon nano tubes, differentiate nano magnetic materials and thermoelectric materials, nonlinear optical materials.

CO5: Illustrate advance engineering applications of Water treatment, sensors, electronic devices, medical domain, civil engineering, chemical engineering, metallurgy and mechanical engineering, food science, agriculture, pollutants degradation.

Unit -I: Basics and Characterization of Nano materials **9 hrs**

Introduction**,** Scope of nano science and nano technology, nano science in nature, classification of nano structured materials, importance of nano materials

Unit - II: Synthesis of Nano materials **9 hrs**

Top-Down approach**,** Inert gas condensation, arc discharge method, aerosol synthesis, plasma arc technique, ion sputtering, laser ablation, laser pyrolys is, and chemical vapour deposition method, electro deposition method, high energy ball milling method.

Synthetic Methods**:** Bottom-Up approach**,** Sol-gel synthesis, micro emulsions or reverse micelles, co-precipitation method, solvo thermal synthesis, hydrothermal synthesis, microwave heating synthesis and sono chemical synthesis.

Unit - III: Techniques for Characterization **9 hrs**

Diffraction technique, spectroscopy techniques, electron microscopy techniques for the characterization of nano materials, BET method for surface area analysis, dynamic light scattering for particle size determination.

Unit - IV: Studies of Nano-Structured Materials **9 hrs**

Synthesis, properties and applications of the following nano materials -fullerenes, carbon nano tubes, 2D-nano material (Graphene), core-shell, magnetic nano particles, thermo electric materials, non- linear optical materials.

Unit - V: Advanced Engineering Applications of Nano materials **9 hrs**

Applications of nano particles, nanorods, nano wires, Water treatment, sensors, electronic devices, medical domain, civil engineering, chemical engineering, metallurgy, and mechanical engineering, food science, agriculture, pollutants degradation.

Text Books:

1. NANO :The Essentials : T Pradeep ,Ma Graw-Hill, 2007.
2. Textbook of Nano science and nano technolog y: B S Murty, P Shankar, Baldev Rai, B B Rath and James Murday, Univ. Press, 2012.

References Books:

1. Concepts of Nano chemistry; Ludo vico Cademrtiri and Geoffrey A. Ozin & Geoffrey A. Ozin, Wiley-VCH, 2011.
2. Nanostructures & Nano materials; Synthesis, Properties & Applications: Guozhong Cao, Imperial College Press, 2007.

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

(Open Elective-IV)

L	T	P	C
3	-	-	3

IV B. Tech, I Semester – (Common to all branches)

LITERARY VIBES

Course Code: 23AHS48

Course Outcomes

CO1	Identify genres, literary techniques and creative uses of language in literary texts.
CO2	Explain the relevance of themes found in literary texts to contemporary, personal and cultural values and to historical forces
CO3	Apply knowledge and understanding of literary texts when responding to others‘ problems and their own and make evidence-based arguments
CO4	Analyze the underlying meanings of the text by using the elements of literary texts
CO5	Evaluate their own work and that of others critically
CO6	Develop as creative, effective, independent and reflective students who are able to make informed choices in process and performance

UNIT I: Poetry

1. Ulysses- Alfred Lord Tennyson
2. Ain‘t I woman?-Sojourner Truth
3. The Second Coming-W.B. Yeats
4. Where the Mind is Without Fear-Rabindranath Tagore

UNIT II: Drama: *Twelfth Night*- William Shakespeare

1. Shakespeare -life and works
2. Plot & sub-plot and Historical background of the play
3. Themes and Criticism
4. Style and literary elements
5. Characters and characterization

UNIT III: Short Story

1. The Luncheon - Somerset Maugham
2. The Happy Prince-Oscar Wild
3. Three Questions – Leo Tolstoy
4. Swami and Friends – R. K. Narayan

UNIT IV: Prose: Essay and Autobiography

1. My struggle for an Education-Booker T Washington
2. The Essentials of Education-Richard Livingston
3. Ignited Minds – A. P. J. Abdul Kalam
4. The story of My Life-Helen Keller

UNIT V: Novel: *Hard Times*- Charles Dickens

1. Charles Dickens-Life and works

2. Plot and Historical background of the novel
3. Themes and criticism
4. Style and literary elements
5. Characters and characterization

Text Books:

1. Charles Dickens**.** *Hard Times.*(Sangam Abridged Texts) Vantage Press, 1983
2. DENT JC.*William Shakespeare. Twelfth Night***.** Oxford University Press,2016.

References:

1. WJ Long.*History of English Literature***,** Rupa Publications India; First Edition (4 October 2015)
2. RK Kaushik And SC Bhatia. *Essays, Short Stories and One Act Plays*, Oxford University Press .2018.
3. Narayan, R.K. *Swami and Friends*. Indian Thought Publications, 2015.
4. Kalam, Abdul. *Ignited Minds*. Penguin Books India, 2014.
5. *New Horizon*, Pearson publications, New Delhi 2014.
6. Vimala Ramarao, *Explorations Volume*-II, Prasaranga Bangalore University,2014**.**

Online Resources

1. **https://www.litcharts.com/poetry/alfred-lord-tennyson/ulysses**
2. **https://www.litcharts.com/lit/ain-t-i-a-woman/summary-and-analysis**
3. **https://englishliterature.education/articles/poetry-analysis/the-second-coming-by-w-b-yeats-critical-analysis-summary-and-line-by-line-explanation/#google_vignette**
4. **https://sirjitutorials.com/where-the-mind-is-without-fear-poem-notes-explanation/**
5. **https://www.litcharts.com/lit/twelfth-night/themes**
6. **https://smartenglishnotes.com/2021/11/28/the-luncheon-summary-characters-themes-and-irony/**

SRI VEKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

(Open Elective-IV)

IV B. Tech, I Semester – (Common to all CSE allied branches)

23ACS35- QUANTUM COMPUTING

L	T	P	C
3	-	-	3

Course Objectives:

- To introduce the principles and mathematical foundations of quantum computation.
- To understand quantum gates, circuits, and computation models.
- To explore quantum algorithms and their advantages over classical ones.
- To develop the ability to simulate and write basic quantum programs.
- To understand real-world applications and the future of quantum computing in AI, cryptography, and optimization.

Course Outcomes:

Upon successful completion of this course, students will be able to:

- Explain the fundamental concepts of quantum mechanics used in computing.
- Construct and analyze quantum circuits using standard gates.
- Apply quantum algorithms like Deutsch-Jozsa, Grover's, and Shor's.
- Develop simple quantum programs using Qiskit or similar platforms.
- Analyze applications and challenges of quantum computing in real-world domains.

UNIT I: Fundamentals of Quantum Mechanics and Linear Algebra

Classical vs Quantum Computation, Complex Numbers, Vectors, and Matrices, Hilbert Spaces and Dirac Notation, Quantum States and Qubits, Superposition and Measurement, Tensor Products and Multi-Qubit Systems.

UNIT II: Quantum Gates and Circuits

Quantum Logic Gates: Pauli, Hadamard, Phase, Controlled Gates and CNOT, Unitary Operations and Reversibility, Quantum Circuit Representation, Quantum Teleportation, Simulation of Quantum Circuits.

UNIT III: Quantum Algorithms and Complexity

Quantum Parallelism and Interference, Deutsch and Deutsch-Jozsa Algorithms, Grover's Search Algorithm, Shor's Factoring Algorithm, Quantum Fourier Transform, Complexity Classes: BQP, P, NP, and QMA.

UNIT IV: Quantum Programming and Simulation Platforms

Introduction to Qiskit and IBM Quantum Experience, Writing Quantum Circuits in Qiskit, Measuring Qubits and Results, Classical-Quantum Hybrid Programs, Noisy Intermediate-Scale Quantum (NISQ) Systems, Limitations and Current State of Quantum Hardware.

UNIT V: Applications and Future of Quantum Computing

Quantum Machine Learning: Basics and Models, Quantum Cryptography and Quantum Key Distribution, Quantum Algorithms in AI and Optimization, Quantum Advantage and Supremacy, Ethical and Societal Impact of Quantum Technologies, Future Trends and Research Directions.

Text books:

1. Michael A. Nielsen, Isaac L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press, 10th Anniversary Edition, 2010.
2. Eleanor Rieffel and Wolfgang Polak, Quantum Computing: A Gentle Introduction, MIT Press, 2011.
3. Chris Bernhardt, Quantum Computing for Everyone, MIT Press, 2019.

Reference Books:

1. David McMahon, Quantum Computing Explained, Wiley, 2008.
2. Phillip Kaye, Raymond Laflamme, Michele Mosca, An Introduction to Quantum Computing, Oxford University Press, 2007.
3. Scott Aaronson, Quantum Computing Since Democritus, Cambridge University Press, 2013.

Online Learning Resources:

1. **IBM Quantum Experience and Qiskit Tutorials**
2. **Coursera – Quantum Mechanics and Quantum Computation by UC Berkeley**
3. edX – The Quantum Internet and Quantum Computers
4. **YouTube – Quantum Computing for the Determined by Michael Nielsen**
5. Qiskit Textbook – IBM Quantum

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

IV B.Tech I Sem

23ACA29- PROMPT ENGINEERING Skill Enhancement Course Common to all Branches except CSE(DS) and CSE(IOT)	L	T	P	C
	0	1	2	2

Course Objective:

This course delves into prompt engineering principles, strategies, and best practices, a crucial aspect in shaping AI models' behaviour and performance. Understanding Prompt Engineering is a comprehensive course designed to equip learners with the knowledge and skills to effectively generate and utilize prompts in natural language processing (NLP) and machine learning (ML) applications. This course delves into prompt engineering principles, strategies, and best practices, a crucial aspect in shaping AI models' behaviour and performance.

Course Out comes:

- Under standing the fundamentals and evolution of prompt engineering.
- Gaining the ability to craft effective closed-ended, open-ended, and role-based prompts.
- Learning to probe and stress-test AI models for bias and robustness.
- Applying prompt optimization techniques and performance evaluation methods.
- Mitigating bias and promoting ethical prompting practices in NLP/ML systems.

Module 1: Introduction to Prompt Engineering

- *Lesson 1: Foundations of Prompt Engineering*
    - Overview of prompt engineering and its significance in NLP and ML.
    - Historical context and evolution of prompt-based approaches.

Module 2: Types of Prompts and Their Applications

- *Lesson 2: Closed-Ended Prompts*
    - Under standing and creating prompts for specific answers.
    - Applications in question- answering systems.
- *Lesson 3: Open-Ended Prompts*
    - Crafting prompts for creative responses.
    - Applications in language generation models.

Module 3: Strategies for Effective Prompting

- *Lesson 4: Probing Prompts*
    - Designing prompts to reveal model biases.
    - Ethical considerations in using probing prompts.
- *Lesson 5: Adversarial Prompts*
    - Creating prompts to stress-test models.
    - Enhancing robustness through adversarial prompting.

Module 4: Fine-Tuning and Optimizing with Prompts

- *Lesson 6: Fine-Tuning Models with Prompts*
    - Techniques for incorporating prompts during model training.
    - Balancing prompt influence and generalization.
- *Lesson 7: Optimizing Prompt Selection*
    - Methods for selecting optimal prompts for specific tasks.
    - Customizing prompts based on model behavior.

Module 5: Evaluation and Bias Mitigation

- *Lesson 8: Evaluating Prompt Performance*
    - Metrics and methodologies for assessing model performance with prompts.
    - Interpreting and analyzing results.
- *Lesson 9: Bias Mitigation in Prompt Engineering*
    - Strategies to identify and address biases introduced by prompts.
    - Ensuring fairness and inclusivity in prompt-based models.

Module 6: Real-World Applications and Case Studies

- *Lesson 10: Case Studies in Prompt Engineering*
- *Exploration of successful implementations and challenges in real-world scenarios.*
- *Guest lectures from industry experts sharing their experiences.*

Text books:

1. "Prompt Engineering in Action" – *Danny D. Sullivan*
2. "The Art of Prompt Engineering with Chat GPT: A Hands-On Guide" – *Nathan Hunter.*

Reference Books:

1. "Prompt Engineering in Practice" – *Michael F. Lewis*
2. "Mastering AI Prompt Engineering: The Ultimate Guide for Chat GPT Users" – *Adriano Damiao*
3. "Writing AI Prompts For Dummies" – *Stephanie Diamond and Jeffrey Allan*
4. "Prompt Engineering Guide" (Online Resource) – *promptingguide.ai*

Online Resource link :

https://www.udemy.com/course/understanding-prompt-engineering/?couponCode=NVDINCTA35TRT

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (AUTONOMOUS)

IV B. Tech I Semester (Common to all Branches)

Course Code: 23AHS49 **GENDER SENSITIZATION**

L	T	P	C
2	0	0	0

Course Outcomes:

CO1	Understand the basic concepts of gender and its related terminology
CO2	Identify the biological, sociological, psychological and legal aspects of gender.
CO3	Use the knowledge in understanding how gender discrimination works in our society and how to counter it.
CO4	Analyzethe gendered division of labour and its relation to politics and economics.
CO5	Appraise how gender-role beliefs and sharing behaviour are associated with more well-being in all culture and gender groups
CO6	Develop students' sensibility with regard to issues of gender in contemporary India

Unit-I UNDERSTANDING GENDER

Introduction: Definition of Gender-Basic Gender Concepts and Terminology-Exploring Attitudes towards Gender-Construction of Gender-Socialization: Making Women, Making Men - Preparing for Womanhood. Growing up Male. First lessons in Caste.

Unit-II GENDER ROLES AND RELATIONS

Two or Many? -Struggles with Discrimination-Gender Roles and Relations-Types of Gender Roles-Gender Roles and Relationships Matrix-Missing Women-Sex Selection and its Consequences-Declining Sex Ratio- Demographic Consequences-Gender Spectrum -

Unit-III GENDER AND LABOUR

Division and Valuation of Labour-Housework: The Invisible Labor- ―My Mother doesn't Work.‖ ―Share the Load.‖-Work: Its Politics and Economics -Fact and Fiction- Unrecognized and Unaccounted work -Gender Development Issues-Gender, Governance and Sustainable Development-Gender and Human Rights-Gender and Mainstreaming

Unit-IV GENDER-BASED VIOLENCE

The Concept of Violence- Types of Gender-based Violence-Gender-based Violence from a Human Rights Perspective-Sexual Harassment - Domestic Violence - Different forms of violence against women - Causes of violence, impact of violence against women - Consequences of gender-based violence

Unit-V GENDER AND CULTURE

Gender and Film-Gender and Electronic Media-Gender and Advertisement-Gender and Popular Literature- Gender Development Issues-Gender Issues-Gender Sensitive Language- Just Relationships

Text Books

1. A.Suneetha, Uma Bhrugubanda, et al. *Towards a World of Equals: A Bilingual Textbook on Gender‖*, Telugu Akademi, Telangana, 2015.
2. Butler, Judith. *Gender Trouble: Feminism and the Subversion of Identity*. UK Paperback Edn. March 1990

Reference Books

1. Wtatt, Robin and Massood, Nazia, *Broken Mirrors: The dowry Problems in India*,London : Sage Publications, 2011
2. Datt, R. and Kornberg, J.(eds), *Women in Developing Countries, Assessing Strategies for Empowerment*, London: Lynne Rienner Publishers, 2002
3. Brush, Lisa D., *Gender and Governance*, New Delhi, Rawat Publication, 2007
4. Singh, Direeti, *Women and Politics World Wide*, New Delhi, Axis Publications, 2010
5. Raj Pal Singh, Anupama Sihag, *Gender Sensitization: Issues and Challenges* (English, Hardcover), Raj Publications, 2019
6. A.Revathy& Murali, Nandini, *A Life in Trans Activism*(Lakshmi Narayan Tripathi). The University of Chicago Press, 2016

Online Resources:

1. Understanding Gender chrome-extension://kdpelmjpfafjppnhbloffcjpeomlnpah/https://www.arvindguptatoys.com/arvindgupta/kamla- gender1.pdf

   https://onlinecourses.swayam2.ac.in/nou24_hs53/preview
2. Gender Roles and Relations
   https://www.plannedparenthood.org/learn/gender-identity/sex-gender-identity/what- are-gender- roles-and-stereotypes

   https://www.verywellmind.com/understanding-gender-roles-and-their-effect-on-our-relationships- 7499408

3. https://onlinecourses Gender and Labour

   https://www.economicsobservatory.com/what-explains-the-gender-division-of-labour-and-how-can- it-be- redressed

   https://onlinecourses.nptel.ac.in/noc23_mg67/preview.swayam2.ac.in/cec23_hs29/preview

4. **GENDER-BASED VIOLENCE**
   https://eige.europa.eu/gender-based-violence/what-is-gender-based-violence?language_content_entity=en

   https://www.worldbank.org/en/topic/socialsustainability/brief/violence-against-women-and-girls

https://onlinecourses.swayam2.ac.in/nou25_ge38/preview

5. **GENDER AND CULTURE**
   https://gender.study/psychology-of-gender/culture-impact-gender-roles-identities/

   https://sociology.iresearchnet.com/sociology-of-culture/gender-and-culture/

   https://archive.nptel.ac.in/courses/109/106/109106136/

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

IV B. Tech - I Semester

Course Code: 23ACM24

INDUSTRY INTERNSHIP

L	T	P	C
-	-	-	2

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

IV B. Tech - II Semester

Course Code: 23ACM27

INTERNSHIP

L	T	P	C
-	-	-	4

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

IV B. Tech - II Semester

Course Code: 23ACM28 PROJECT

L	T	P	C
-	-	-	8

HONOURS SYLLABUS
SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B. Tech - I Semester

		L	T	P	C
23ACA34	**ADVANCED MACHINE LEARNING &AI SYSTEMS**	**3**	**0**	**0**	**3**

Course Objective:

- To deepen understanding of algorithmic principles for designing scalable and efficient AI/ML solutions.
- To explore advanced topics such as optimization algorithms, randomized and approximation algorithms, and online learning.
- To analyze computational complexity, tractability, and convergence of AI models.
- To apply graph-based, evolutionary, and heuristic approaches in solving real-world AI/ML problems.
- To integrate algorithmic strategies for large-scale machine learning, reinforcement learning, and neural network training.

Course Outcomes:

After successful completion of the course, students will be able to:

1. Analyze and apply classical algorithmic techniques including divide and conquer, dynamic programming, approximation, and randomized algorithms in the context of AI/ML.
2. Implement advanced graph algorithms for shortest paths, flows, and community detection, and apply them to AI problems like NLP and recommender systems.
3. Apply convex and non-convex optimization strategies, gradient-based learning, and regularization techniques to train and tune AI/ML models effectively.
4. Use evolutionary, swarm intelligence, and reinforcement learning-based metaheuristic methods for neural architecture search and complex optimization tasks in AI.
5. Evaluate and design scalable algorithmic solutions with fairness and interpretability for AI/ML applications, referencing case studies like AlphaGo, GPT, and AutoML systems.

UNIT I: Foundations of Advanced Algorithmic Techniques

Review of Time and Space Complexity, Divide and Conquer, Dynamic Programming, and Greedy Algorithms, Recurrence Relations and Master Theorem, Approximation Algorithms: Vertex Cover, TSP, Set Cover, Randomized Algorithms: Monte Carlo and Las Vegas Types, Probabilistic Analysis and Tail Bounds, Applications in ML Preprocessing and Feature Selection

UNIT II: Graph Algorithms and AI Applications

Graph Representations and Traversal Algorithms, Shortest Path: Dijkstra's, Bellman-Ford, Floyd-Warshall, Minimum Spanning Trees: Kruskal and Prim, Network Flows and Max Flow-Min Cut Theorem, Graph-Based Semi-Supervised Learning, PageRank, Centrality, and Community Detection, Applications in NLP, Vision, and Recommender Systems

UNIT III: Optimization in AI/ML

Convex and Non-Convex Optimization, Gradient Descent Variants: SGD, Momentum, Adam, Convergence Analysis and Learning Rates, Duality and Lagrange Multipliers, Regularization: L1, L2, ElasticNet, Hyperparameter Optimization: Grid, Random, Bayesian, Constrained Optimization in SVMs and Deep Learning

UNIT IV: Evolutionary & Metaheuristic Algorithms

Genetic Algorithms and Evolutionary Strategies, Swarm Intelligence: PSO, Ant Colony Optimization, Simulated Annealing and Tabu Search, Multi-objective Optimization, Reinforcement Learning and Policy Gradient

Methods, Neuroevolution: Evolving Neural Networks, Use Cases in Feature Engineering and Neural Architecture Search (NAS)

UNIT V: Advanced Topics and Case Studies

Online Learning and Regret Minimization, Bandit Algorithms: Multi-Armed Bandits, Thompson Sampling, Large-Scale Algorithms: MapReduce, Apache Spark MLlib, Algorithmic Fairness, Interpretability, and Ethics in AI, Case Studies: AlphaGo, GPT, BERT, Recommendation Engines, Research Trends in Algorithmic ML and AutoML, Capstone Problem Solving using Hybrid Algorithms

Text Books:

1. Introduction to Algorithms – Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein (MIT Press)
2. Algorithms for Machine Learning – Giuseppe Bonaccorso, Packt Publishing
3. Convex Optimization – Stephen Boyd and Lieven Vandenberghe, Cambridge University Press
4. Reinforcement Learning: An Introduction – Richard S. Sutton and Andrew G. Barto

Reference Books:

1. Machine Learning: A Probabilistic Perspective – Kevin P. Murphy
2. The Elements of Statistical Learning – Trevor Hastie, Robert Tibshirani, Jerome Friedman
3. Evolutionary Computation – Kenneth A. De Jong
4. Handbook of Approximation Algorithms and Metaheuristics – Teofilo F. Gonzalez

Online Courses:

1. Coursera – Advanced Algorithms and Complexity (UC San Diego)
2. edX – Algorithmic Design and Techniques (UC San Diego)
3. MIT OpenCourseWare – Advanced Algorithms
4. Udemy – Optimization Algorithms in Machine Learning
5. Stanford Online – Convex Optimization

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III B. Tech - I Semester

		L	T	P	C
23ACM29	**DEEP LEARNING & NEURAL NETWORK ARCHITECTURES**	**3**	**0**	**0**	**3**

Course Objectives:

- To introduce the fundamental concepts and mathematical foundations of deep learning.
- To explore different neural network architectures including CNNs, RNNs, LSTMs, and Transformers.
- To enable students to implement, train, and optimize deep neural networks.
- To analyze the performance and limitations of various architectures in different AI tasks.
- To develop the ability to apply deep learning models to real-world applications such as image recognition, language modeling, and autonomous systems.

Course Outcomes (COs):

Upon successful completion of this course, the student will be able to:

- CO1: Understand the theoretical foundations of neural networks and deep learning.
- CO2: Implement and train multilayer perceptrons, CNNs, RNNs, and other architectures.
- CO3: Analyze and optimize deep learning models using advanced regularization and tuning techniques.
- CO4: Evaluate the applicability of different neural network architectures for various AI problems.
- CO5: Apply state-of-the-art models such as Transformers and GANs in real-world domains.

UNIT I: Foundations of Neural Networks

Introduction to Artificial Neural Networks, Biological Neuron vs. Artificial Neuron, Perceptron, Multilayer Perceptron (MLP), Activation Functions: ReLU, Sigmoid, Tanh, Softmax, Backpropagation and Gradient Descent, Loss Functions: MSE, Cross Entropy, Overfitting, Regularization (L1/L2), Dropout

UNIT II: Convolutional Neural Networks (CNNs)

Convolution Operation and Feature Maps, Pooling Layers: Max and Average Pooling, CNN Architectures: LeNet, AlexNet, VGG, ResNet, Transfer Learning and Fine-tuning, Image Classification, Object Detection Basics, Implementation with TensorFlow/PyTorch

UNIT III: Recurrent Neural Networks (RNNs) and Variants

Sequential Data and Time Series, RNN Basics and Backpropagation Through Time (BPTT), Vanishing and Exploding Gradients, LSTM and GRU Architectures, Applications in Text, Speech, and Music, Sequence-to-Sequence Models

UNIT IV: Advanced Architectures & Optimization

Autoencoders and Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), Deep Reinforcement Learning Overview, Batch Normalization, Early Stopping, Hyperparameter Tuning and Optimization, Performance Metrics and Evaluation

UNIT V: Transformer Models & Applications

Attention Mechanism and Self-Attention, Transformers and BERT Architecture, Positional Encoding, Multi-head Attention, Pre-trained Language Models and Fine-Tuning, Applications in NLP: Text Classification, Translation, Large Language Models and Transfer Learning

Text Books:

1. Deep Learning – Ian Goodfellow, Yoshua Bengio, and Aaron Courville (MIT Press)
2. Neural Networks and Deep Learning – Michael Nielsen (Online Book)
3. Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow – Aurélien Géron (O'Reilly)

Reference Books:

1. Pattern Recognition and Machine Learning – Christopher M. Bishop
2. Deep Learning for Computer Vision – Rajalingappaa Shanmugamani
3. Natural Language Processing with Transformers – Lewis Tunstall, Leandro von Werra, Thomas Wolf
4. Reinforcement Learning: An Introduction – Richard S. Sutton and Andrew G. Barto

Recommended Online Courses:

1. Deep Learning Specialization – Andrew Ng (Coursera)
2. CS231n: Convolutional Neural Networks for Visual Recognition (Stanford)
3. Fast.ai – Practical Deep Learning for Coders
4. Deep Learning with PyTorch (Udacity)
5. Transformers by Hugging Face (free course)

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III B. Tech - II Semester

23ACM30	**REINFORCEMENT LEARNING & DECISION MAKING**	**L**	**T**	**P**	**C**
		3	**0**	**0**	**3**

Course Objectives:

- To introduce the fundamentals of reinforcement learning (RL) and its mathematical foundation.
- To understand the Markov Decision Process (MDP) framework for decision making under uncertainty.
- To explore various RL algorithms including value-based, policy-based, and model-based approaches.
- To analyze deep reinforcement learning techniques for real-world applications.
- To study the integration of reinforcement learning with planning, exploration, and control strategies.

Course Outcomes (COs):

After successful completion of this course, students will be able to:

1. Understand the fundamentals of reinforcement learning, including agent-environment interaction, types of RL, and solving decision-making problems using Markov Decision Processes and Bellman equations.
2. Apply dynamic programming and Monte Carlo methods to perform policy evaluation, policy improvement, and control in model-based RL settings.
3. Implement temporal-difference learning algorithms like TD(0), Sarsa, and Q-learning, and extend them using eligibility traces and function approximation techniques.
4. Develop and analyze policy gradient and actor-critic methods, including REINFORCE and PPO, to optimize policies in continuous and high-dimensional action spaces.
5. Employ deep reinforcement learning techniques (DQN, DDPG, A3C, SAC) and exploration strategies to solve complex tasks in robotics, games, and autonomous systems, considering safety and ethical decision-making.

UNIT I: Introduction to Reinforcement Learning & MDPs

Foundations of RL: Agent-Environment Interaction, Types of RL: Model-based vs. Model-free, Reward Signals, Return, and Discounting, Markov Decision Processes (MDPs), Bellman Equations and Optimality

UNIT II: Dynamic Programming & Monte Carlo Methods

Policy Evaluation and Policy Improvement, Value Iteration and Policy Iteration, Monte Carlo Prediction and Control, First-visit and Every-visit Methods, Limitations of DP and MC Approaches

UNIT III: Temporal-Difference Learning & Function Approximation

TD(0), Sarsa, and Q-Learning Algorithms, Eligibility Traces: TD(λ), Sarsa(λ), Off-policy vs. On-policy Learning, Linear Function Approximation, Generalization in RL

UNIT IV: Policy Gradient Methods and Actor-Critic Algorithms

Policy Gradient Theorem, REINFORCE Algorithm, Baselines and Variance Reduction, Actor-Critic Architectures, Trust Region and Proximal Policy Optimization (PPO)

UNIT V: Deep Reinforcement Learning and Applications

Deep Q-Networks (DQN) and Experience Replay, DDPG, A3C, and SAC Algorithms, Exploration Techniques: ε-greedy, UCB, Intrinsic Rewards, RL in Robotics, Game AI, and Autonomous Systems, Safety, Ethics, and Fairness in Decision Making

Textbooks:

1. **Richard S. Sutton and Andrew G. Barto** – *Reinforcement Learning: An Introduction*, 2nd Edition, MIT Press
2. **Ian Goodfellow, Yoshua Bengio, Aaron Courville** – *Deep Learning*, MIT Press

Reference Books:

1. **David Silver's RL Course Slides & Lectures** – DeepMind, University College London
2. **Marco Wiering & Martijn van Otterlo (Eds.)** – *Reinforcement Learning: State of the Art*, Springer
3. **Csaba Szepesvári** – *Algorithms for Reinforcement Learning*, Morgan & Claypool
4. **Yuxi Li** – *Deep Reinforcement Learning: An Overview*, arXiv survey

Online Courses & Resources:

1. **DeepMind x UCL Reinforcement Learning Lectures by David Silver**
2. **Coursera: Reinforcement Learning Specialization – University of Alberta**

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III B. Tech - II Semester

		L	T	P	C
23ACA35	**AI FOR ROBOTICS & AUTOMATION**	**3**	**0**	**0**	**3**

Course Objectives:

- To introduce the principles of software engineering augmented with artificial intelligence.
- To explore AI-driven tools for software requirement analysis, design, testing, and maintenance.
- To understand DevOps practices and integrate AI techniques for CI/CD, deployment, and monitoring.
- To automate software lifecycle management using machine learning and data-driven insights.
- To develop intelligent pipelines for software delivery with adaptive testing and feedback systems.

Course Outcomes (COs):

Upon successful completion of this course, the student will be able to:

- CO1: Describe the role of AI in modern software engineering processes and lifecycle stages.
- CO2: Apply AI/ML models for requirements gathering, code generation, defect prediction, and testing.
- CO3: Implement intelligent DevOps practices including CI/CD, release automation, and anomaly detection.
- CO4: Analyze data from software pipelines to drive informed decisions and improve quality.
- CO5: Develop an end-to-end AI-enabled software delivery pipeline with automated learning-based optimizations.

UNIT I: Foundations of AI-Driven Software Engineering

Introduction to Software Engineering Lifecycle, Traditional vs. AI-Driven Software Development, AI/ML in Software Engineering: Overview and Scope, Natural Language Processing (NLP) for Requirements Engineering, AI for Software Design Recommendation, Intelligent Code Completion (e.g., GitHub Copilot)

UNIT II: AI in Testing and Defect Prediction

Static and Dynamic Testing with AI, Automated Test Case Generation, Defect Detection and Prediction using ML Models, Sentiment and Bug Report Analysis, AI in Refactoring and Code Review, Tools: SonarQube, DeepCode

UNIT III: DevOps Principles and Practices

DevOps Overview: CI/CD Pipelines, Infrastructure as Code (IaC), Configuration Management Tools: Ansible, Puppet,Monitoring and Logging Tools: Prometheus, Grafana, Containerization and Orchestration: Docker, Kubernetes, Agile and Lean Practices in DevOps

UNIT IV: AI for DevOps Automation and Intelligence

Predictive Analytics for Deployment Success, AI for Log Analytics and Root Cause Analysis, Self-Healing Systems and Auto-Scaling, Feedback Loops in DevOps using Reinforcement Learning, Data-Driven Decision Making in Release Management, ChatOps and AIOps Platforms

UNIT V: Case Studies and Emerging Trends

Case Study: AI-Augmented DevOps in Enterprises, ML-Ops vs. DevOps vs. DataOps, Security in DevOps (DevSecOps), Explainability and Ethics in AI-Driven Software Engineering, Generative AI in Software Development, Future Trends and Industry Standards

Textbooks:

1. Tim Menzies, Diomidis Spinellis – Artificial Intelligence and Software Engineering: Status and Future Directions
2. Len Bass, Ingo Weber, Liming Zhu – DevOps: A Software Architect's Perspective, Addison-Wesley
3. Thomas Erl, Ricardo Puttini, Zaigham Mahmood – AI & Analytics for DevOps, Pearson

Reference Books:

1. Carlos Nunes Silva – AI in Software Engineering
2. Gene Kim, Jez Humble, Patrick Debois, John Willis – The DevOps Handbook
3. Andrew Ng – Machine Learning B.Techning (AI Systems Engineering Perspective)

Online Resources & Courses:

1. Coursera – AI for Software Engineering (IBM)
2. DevOps with Microsoft Azure – edX
3. Udacity – AI for DevOps Engineers Nanodegree

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IV B. Tech - I Semester

23ACA36	AI ETHICS, FAIRNESS & EXPLAINABILITY	L	T	P	C
		3	0	0	3

Course Objectives:

- To understand ethical concerns and responsibilities in the development and deployment of AI systems.
- To study fairness, bias, and accountability issues in AI and machine learning models.
- To explore techniques and frameworks for interpreting and explaining AI decisions.
- To analyze societal impacts of AI and build trust through transparent systems.
- To promote responsible and inclusive AI development aligned with human values.

Course Outcomes (COs):

Upon successful completion of this course, the student will be able to:

1. Describe the ethical principles, historical context, and responsibilities associated with AI deployment across domains like healthcare and law enforcement.
2. Identify different forms of bias in datasets and algorithms, and apply fairness metrics and mitigation strategies to ensure equitable AI systems.
3. Demonstrate the need for explainability in AI models and utilize tools such as LIME, SHAP, and Grad-CAM to generate local and global model explanations.
4. Design AI systems with accountability by integrating human oversight, ethical documentation (e.g., Model Cards, Datasheets), and adherence to global guidelines.
5. Critically assess the broader societal and legal implications of AI in areas such as surveillance, misinformation, and inclusivity, and explore international policy frameworks.

UNIT I: Foundations of AI Ethics

Historical background of AI ethics, Core principles: beneficence, non-maleficence, autonomy, justice, Moral and legal responsibilities in AI systems, Risk assessment and governance in AI, Ethical AI case studies from healthcare, policing, hiring

UNIT II: Fairness and Bias in AI

Types of bias: dataset bias, label bias, historical bias, Fairness definitions: demographic parity, equal opportunity, individual fairness, Disparate impact and fairness metrics, Algorithmic audits and bias detection, Fairness-aware learning and mitigation strategies

UNIT III: Explainable Artificial Intelligence (XAI)

Need for interpretability in AI models, Taxonomy of XAI methods: model-agnostic, model-specific, LIME, SHAP, Grad-CAM, Partial Dependence Plots, Local vs Global explanations, Trade-offs: accuracy vs interpretability

UNIT IV: Accountability and Responsible AI Design

Transparent AI systems, Human-in-the-loop and AI-assisted decision-making, Accountability frameworks (e.g., IEEE, NIST, EU Guidelines), Documentation tools: Datasheets for datasets, Model Cards, Responsible AI lifecycle management

UNIT V: Societal Impacts and Policy Considerations

AI in surveillance, misinformation, and social manipulation, Ethical implications in autonomous systems (vehicles, weapons), AI and inclusion: accessibility, gender, race, socioeconomic impacts, Public policy, legal frameworks, and global initiatives, Future challenges and global governance of AI

Textbooks:

1. **Virginia Dignum** – *Responsible Artificial Intelligence: How to Develop and Use AI in a Responsible Way*, Springer
2. **Cathy O'Neil** – *Weapons of Math Destruction*, Crown Publishing
3. **Mark Coeckelbergh** – *AI Ethics*, The MIT Press

Reference Books:

1. **Nick Bostrom & Eliezer Yudkowsky** – *The Ethics of Artificial Intelligence*
2. **Shalini Kantayya (Film)** – *Coded Bias* (documentary, 2020)
3. **Floridi, Luciano** – *Ethics of Information*, Oxford University Press

Online Courses & Resources:

1. **Coursera – AI For Everyone (Andrew Ng)**
2. **edX – Ethics of AI and Big Data (Linux Foundation)**

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III B. Tech - II Semester

23ACM21	**AI & MACHINE LEARNING LAB**	**L**	**T**	**P**	**C**
		0	**0**	**3**	**1.5**

Course Objectives

1. To provide hands-on experience in implementing AI and machine learning algorithms.
2. To develop and evaluate models using real-world datasets.
3. To introduce optimization and hyperparameter tuning techniques.
4. To build intelligent systems for classification, prediction, and clustering.

Course Outcomes (COs):

Upon successful completion of this course, the student will be able to:

CO1: Implement key machine learning algorithms from scratch and using libraries.

CO2: Preprocess data and select suitable features for modeling.

CO3: Train, test, and evaluate models for accuracy and performance.

CO4: Apply AI techniques to solve classification, regression, and decision-making problems.

CO5: Develop simple AI agents and use neural networks for predictive tasks.

Tools Required

- Python (NumPy, Pandas, Scikit-learn, TensorFlow/Keras, OpenCV)
- Jupyter Notebook / Google Colab
- Datasets from UCI, Kaggle, Scikit-learn
- Anaconda / VS Code

List of 12 Experiments

1. **Data Preprocessing** – Cleaning, normalization, encoding, and splitting data.
2. **Linear Regression** – Implement simple and multiple linear regression.
3. **Logistic Regression** – Binary classification on datasets like breast cancer or Titanic.
4. **K-Nearest Neighbors (KNN)** – Classification task with evaluation metrics.
5. **Decision Trees and Random Forests** – Tree-based classification and visualization.
6. **Support Vector Machines (SVM)** – Margin classification with kernel trick.
7. **Naive Bayes** – Text classification with spam dataset.
8. **K-Means Clustering** – Unsupervised clustering with customer segmentation.
9. **Principal Component Analysis (PCA)** – Dimensionality reduction and visualization.
10. **Artificial Neural Networks (ANNs)** – Implement basic neural network using Keras.
11. **Model Evaluation & Tuning** – Use cross-validation, GridSearchCV, and confusion matrices.
12. **AI Agent Search Algorithms** – Implement A*, DFS, BFS for pathfinding problems.

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IV B. Tech - I Semester

		L	T	P	C
23ACA37	**ROBOTICS & AUTONOMOUS SYSTEMS LAB**	**0**	**0**	**3**	**1.5**

Course Objectives:

- To provide hands-on experience in deploying machine learning models into production environments.
- To introduce the tools and practices of MLOps for automating ML workflows.
- To train students in containerization, orchestration, monitoring, and CI/CD pipelines.
- To understand model versioning, reproducibility, and lifecycle management.
- To develop skills in using cloud platforms and APIs for scalable AI applications.

Course Outcomes (COs):

Upon successful completion of this course, the student will be able to:

- Package and deploy AI models using tools such as Flask, FastAPI, and Docker.
- Automate machine learning workflows using CI/CD pipelines and MLOps tools.
- Monitor and manage deployed models in real-time environments.
- Apply version control and model registry techniques effectively.
- Deploy models on cloud platforms like AWS, Azure, or GCP and use MLflow, Kubeflow, etc.

List of 12 Lab Experiments:

1. Experiment 1: Build a simple ML model and serve it via Flask or FastAPI.
2. Experiment 2: Containerize the model application using Docker.
3. Experiment 3: Deploy a Dockerized model on a local or cloud-based Kubernetes cluster.
4. Experiment 4: Implement CI/CD pipeline using GitHub Actions or GitLab CI.
5. Experiment 5: Track experiments and manage model versions using MLflow.
6. Experiment 6: Use DVC (Data Version Control) for tracking data and pipeline stages.
7. Experiment 7: Automate model retraining and deployment using Jenkins.
8. Experiment 8: Model monitoring using Prometheus and Grafana.
9. Experiment 9: Introduce model drift detection and retraining triggers.
10. Experiment 10: Deploy a model on a cloud platform (e.g., AWS SageMaker, GCP AI Platform).
11. Experiment 11: Use Kubeflow pipelines for end-to-end ML workflow management.
12. Experiment 12: Capstone: Full-cycle ML project from training to monitoring using MLOps best practices.

Textbooks:

1. Mark Treveil and Alok Shukla, AI and Analytics in Production: How to Implement Successful AI and Analytics Applications, O'Reilly Media.
2. Emmanuel Ameisen, Building Machine Learning Powered Applications, O'Reilly Media.

Reference Books:

1. Chris Fregly and Antje Barth, Data Science on AWS: Building End-to-End Applications, O'Reilly.
2. Alfredo Deza and Noah Gift, Practical MLOps, O'Reilly Media.
3. Soham Kamani, Learning MLOps, Packt Publishing.s

HONOURS IN INDUSTRIAL AI&ML

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III B. Tech - I Semester CSE(AI&ML)

23ACM41- PYTHON FOR DATA SCIENCE, AI & DEVELOPMENT

UNIT-I : WORKING WITH DATA IN PYTHON

Reading Files with Open, Writing Files with Open, Pandas: Loading Data, Pandas: Working with and Saving Data, One Dimensional Numpy, Two Dimensional Numpy, Working with Data in Python, Data cleaning (missing values, duplicates, outliers)- Basic visualization (Matplotlib, Seaborn)- Data types in Pandas (object, category, datetime)- Real dataset analysis (Titanic, COVID-19).

UNIT-II : APIs and Collections for AI and ML

Application Program Interface, REST APIs & HTTP Requests, REST APIs & HTTP Requests, HTML for Web Scraping, Web Scraping, Working with Different File Formats, Python Cheat Sheet, GDP Data Extraction and Processing, Access REST APIs & Request HTTP, API Examples, Web Scraping, Working with different file formats, JSON parsing- API authentication (API keys, OAuth)- Rate-limiting handling- Example: Fetch stock or weather data- ETL basics (Extract → Transform → Load into Pandas)

UNIT-III: Python Essentials for MLOps

ntroduction: APIs and SDKs, Installing Azure Command-Line Interface (CLI), AzureML Studio with Python, Hugging Face Transformers, Hugging Face Datasets, Azure Open Datasets, APIs and SDKs, Automation with Command-Line Tools, Creating a Single File Script, Using the ArgParse Framework, Declaring Dependencies, Using the Click Framework, Packaging your Project, Solving a Machine Learning Problem with a CLI, MLFlow for experiment tracking- Basic CI/CD for ML (GitHub Actions)- Environment management (Conda, Docker basics)- Logging & error handling in scripts

UNIT-IV: Python Functions and Classes

Function Structure and Values, Function Arguments, Variable and Keyword Arguments, Working with Functions, Building Classes and Methods, Introduction to Classes, Using a Constructor, Adding Methods, Class Inheritance, Building Classes and Methods, Modules and Advanced Usages, Introduction to Python Modules, Working with Imports, Working with Python Scripts, Virtual Environments and Dependencies, Modules and Advanced Usages, Decorators- Generators for large datasets- Dataclasses- Type hints & typing module

Unit-V: Version control with Git/GitHub- Data ethics & Responsible AI- Performance optimization (cProfile, NumPy vectorization), Case study for AI & Development: AI for Precision Agriculture and AI in Healthcare

Text Books:

1. **Giuseppe Bonaccorso, Armando Fandango , Python: Expert machine learning systems and intelligent agents using Python, 2018.**

2. Aman Kharwal , From ML Algorithms to GenAI & LLMs: Master ML Algorithms and Generative AI & LLMs with Python from scratch!: Master MAlgorithms and Generative AI & LLMs with Python from scratch! Paperback – 22 October 2024

Online Softwares:

1. https://www.coursera.org/learn/python-for-applied-data-science-ai#modules
2. https://www.coursera.org/search?query=python%20for%20data%20science%2C%20ai%20%26%20development&sortBy=BEST_MATCH

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III B. Tech - I Semester CSE(AI&ML)

23ACM42 -MASTERING DATA HANDLING – SQL & NO SQL

L	T	P	C
3	0	0	3

Course objectives

The objective of the course is,

1. Understand relational and non-relational data models.
2. Write SQL queries for data extraction, transformation, and reporting.
3. Apply NoSQL concepts using databases like MongoDB.
4. Perform CRUD operations and data manipulation across structured and unstructured data.
5. Integrate SQL/NoSQL operations for data analytics tasks

Unit – I: Introduction to Databases and Data Models

Types of databases: RDBMS vs NoSQL, Relational model: tables, rows, columns, schema Primary keys, foreign keys, normalization concepts, Introduction to MongoDB and JSON structure Hands-on: Understanding database schemas and JSON documents

Unit – II: SQL Fundamentals and Data Retrieval

SELECT statements, filtering data (WHERE, BETWEEN, LIKE), Sorting and limiting results (ORDER BY, LIMIT), Aggregation (COUNT, SUM, AVG, GROUP BY, HAVING), Joins (INNER, LEFT, RIGHT, FULL OUTER), Hands-on: Queries to analyze structured business data

Unit III - Advanced SQL and Subqueries

Nested queries and subqueries, CASE, COALESCE, NULL handling, Temporary tables, views, and indexes, Stored procedures and transactions, Hands-on: Complex querying using subqueries and joins

Unit IV - NoSQL Concepts and MongoDB

Document-based storage model in MongoDB, CRUD operations in MongoDB, Indexing and aggregation pipelines, Comparison of SQL vs NoSQL queries, Hands-on: Real-time operations with MongoDB shell or Compass

Unit V - Data Integration and Use Cases

Connecting databases with Python (using SQLAlchemy or PyMongo), Data extraction for visualization/reporting, Use cases: Customer segmentation, product catalog, transaction logs, Hands- on: Building a data handler that reads from SQL and NoSQL sources.

Text Books

1 Alan Beaulieu, Learning SQL

2 Kristina Chodorow, MongoDB: The Definitive Guide

3 Upmani Sharma et al., SQL for Data Analytics

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III B. Tech - II Semester CSE(AI&ML)

23ACM43- DATA ANALYSIS & VISUALIZATION USING PYTHON

L	T	P	C
3	0	0	3

Course objectives

The objective of the course is,

1. Understand the process of importing, cleaning, and manipulating data.
2. Perform descriptive and inferential data analysis using Python.
3. Build effective visualizations using Matplotlib, Seaborn, and Plotly.
4. Apply group operations and pivoting for business insights.
5. Create dashboards and interactive plots for presentations.

Unit – I: Data Analysis Foundations

Introduction to data analysis lifecycle, Data types and data frames in Pandas, Data importing: CSV, Excel, SQL databases, Data cleaning: handling missing values, duplicates, type conversion, Hands- on: Loading and cleaning real-world datasets

Unit – II: Exploratory Data Analysis (EDA)

Descriptive statistics and summary functions, Groupby and aggregation operations, Correlation and covariance, Feature distribution and outlier detection, Hands-on: Performing EDA on business datasets

Unit III - Data Visualization with Matplotlib and Seaborn

Basic plots: line, bar, scatter, histogram, Advanced charts: box plot, violin plot, heatmaps, Plot customization: titles, legends, axes, Use of color, annotations, and styles for insights, Hands-on: Visual storytelling with data

Unit IV - Interactive Visualizations and Dashboards

Introduction to Plotly and interactive widgets, Creating subplots, drop-downs, sliders, Introduction to Streamlit for dashboard creation, Integration of visualizations into mini apps, Hands-on: Business dashboard project

Unit V - Case Study and Project Work

Case study: EDA on marketing, sales, HR or finance datasets, Visual report generation using Jupyter, Data storytelling and presentation, Project design: business context, insights, and reporting, Hands-on: Final report and presentation preparation

Text Books:

1. Wes McKinney, Python for Data Analysis

2 Peter Bruce, Andrew Bruce, Practical Statistics for Data Scientists

3 Cole Nussbaumer Knaflic, Storytelling with Data

Online Software:

1 Python

2 Pandas

3 Seaborn, Matplotilb, Plotly, Streamlit

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III B. Tech - II Semester CSE(AI&ML)

23ACM44- MACHINE LEARNING AND AI – A DEEP DIVE

L	T	P	C
3	0	0	3

Course objectives

The objective of the course is,

1. Understand the supervised and unsupervised learning paradigm.
2. Implement ML algorithms using Scikit-learn.
3. Evaluate model performance using statistical metrics.
4. Perform data preprocessing, feature engineering, and model tuning.
5. Apply ML to real-world problems such as classification and regression.

Unit – I: Machine Learning Foundations

What is ML? AI vs ML vs DL, Types of ML: Supervised, unsupervised, reinforcement learning, ML pipeline: loading, preparing, modelling, Introduction to Scikit-learn, Hands-on: Building your first ML model

Unit – II: Supervised Learning – Regression

Linear and polynomial regression, Evaluation metrics: MAE, MSE, RMSE, R^2, Overfitting, underfitting, regularization, Feature scaling and transformations, Hands-on: Predictive modeling using real datasets

Unit III - Supervised Learning – Classification

Logistic regression, KNN, Decision Trees, Random Forests, Confusion matrix, accuracy, precision, recall, F1-score, ROC-AUC, cross-validation, Hands-on: Classification case study (HR churn, fraud detection)

Unit IV - Unsupervised Learning

K-means clustering and elbow method, Hierarchical clustering, DBSCAN, Dimensionality reduction with PCA, Hands-on: Customer segmentation or anomaly detection

Unit V - Model Improvement and Hyperparameter Tuning

GridSearchCV and RandomizedSearchCV, Feature selection and importance, Model explainability (SHAP, LIME intro), Hands-on: Model improvement project

Text Books

1 Aurelien Geron , Hands-On ML with Scikit-Learn, Keras & TensorFlow

2 Sebastian Raschka , Python Machine Learning

Online Software:

1 Python

2 Pandas

3 Scikit-learn, Jupyter

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

IV B. Tech - I Semester CSE(AI&ML)

23ACM45- AI MODEL DEPLOYMENT, MLOPS & INDUSTRY APPLICATIONS

L	T	P	C
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Course objectives

The objective of the course is,

1. Understand model deployment principles and the CI/CD lifecycle.
2. Learn about MLOps, versioning, and reproducibility of ML models.
3. Use frameworks like Flask, FastAPI, Streamlit for model APIs.
4. Apply DevOps tools (Docker, GitHub Actions) for deployment.
5. Work on real-world industry datasets with end-to-end pipelines.

Unit – I: Deployment Concepts and APIs

Introduction to model serving and inference, REST APIs with Flask and FastAPI, Deploying ML models with Streamlit, Hands-on: Creating and deploying prediction APIs

Unit – II: MLOps and CI/CD Pipelines

Introduction to MLOps lifecycle, MLFlow for experiment tracking, DVC for data and model versioning, GitHub Actions and CI/CD workflows, Hands-on: MLOps flow for sample project

Unit III - Containerization and Cloud Deployment

Docker basics: image, container, Dockerfile, Dockerizing ML models, Deploying with Streamlit + Docker, Introduction to cloud platforms (AWS/GCP basics), Hands-on: Deploy a model using Docker locally or to Heroku

Unit IV - Real-world ML Applications

Domain-specific examples: e-commerce, finance, healthcare, End-to-end pipeline (data → model → deployment), Logging and monitoring (basic), Hands-on: Full pipeline from notebook to app

Unit V - Industry Project and Presentation

Build, deploy, and present a complete ML app, Team collaboration via Git, CI/CD integrated delivery, Hands-on: Industry use-case deployment

Text Books

1 Hannes Hapke, Catherine Nelson , Building Machine Learning Pipelines

2 Andriy Burkov , Machine Learning Engineering

3 mlops.community , MLOps Guide

Online Software

1 Flask, FastAPI, Streamlit

2 Docker,Git, Github

3 MLFlow

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IV B. Tech - I Semester CSE(AI&ML)

23ACM46- PROJECT

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HONOURS -GENRATIVE AI

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

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III B. Tech - I Semester CSE(AI&ML)

23ACM35- APPLIED DEEP LEARNING

L	T	P	C
3	0	0	3

Course objectives

The objective of the course is,

1. Understand the foundational concepts of neural networks and deep learning.
2. Build deep learning models using TensorFlow and Keras.
3. Train and evaluate CNNs for computer vision tasks.
4. Explore RNNs and sequence models for time-series and text data.
5. Apply transfer learning and build deployable deep learning applications.

Unit – I: Deep Learning Foundations

Perceptron and activation functions, Loss functions and optimization, Forward and backpropagation, Batch normalization, dropout, Hands-on: Implementing a neural network from scratch

Unit – II: CNNs and Image Processing

Convolution, pooling, padding, CNN architecture and design, Image classification use cases, Data augmentation techniques, Hands-on: Build image classifier using CNN

Unit III - RNNs and Sequence Models

RNN, LSTM, GRU architectures, Applications: text generation, sentiment analysis, Vanishing gradients and solutions, Hands-on: Time-series prediction/text classification

Unit IV - Transfer Learning and Pretrained Models

Feature extraction and fine-tuning, Using models like VGG, ResNet, MobileNet, Customizing pretrained models for new tasks, Hands-on: Transfer learning-based image classifier

Unit V - Deep Learning Applications and Case Study

End-to-end DL project setup, Case study: Medical imaging, document processing, etc., Deployment- ready packaging of models, Hands-on: Deep learning mini project

Text Books

1 François Chollet , Deep Learning with Python

2 Michael Nielsen , Neural Networks and Deep Learning

Online Software

1 Python

2 Tensor Flow, Keras

3 Google Colab

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

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III B. Tech - I Semester CSE(AI&ML)

23ACM36- MACHINE LEARNING ESSENTIALS FOR NEXT GEN AI

L	T	P	C
3	0	0	3

Course objectives

The objective of the course is,

1. Understand supervised and unsupervised learning techniques.
2. Develop and evaluate models using Scikit-learn.
3. Perform data preprocessing and feature engineering.
4. Apply classification and regression models to real-world datasets.
6. Interpret model results and improve performance.

Course Outcome (COs)

At the end of the course, the student will be able to:

Learning Levels: Re - Remember; Un - Understand; Ap - Apply; An - Analysis; Ev - Evaluate; Cr - Create
Learning Level

1.	Understand the ML pipeline and its components	Un
2.	Apply supervised and unsupervised models	Ap
3.	Evaluate models using performance metrics	An
4.	Perform feature engineering and preprocessing	Ap
5.	Tune and interpret ML models effectively	Cr

Unit I: Machine Learning Fundamentals

ML vs AI vs Deep Learning, ML workflow and pipeline structure, Data splitting: train/test/validation, Introduction to Scikit-learn, Hands-on: Build your first ML model

Unit II: Supervised Learning – Regression

Simple and multiple linear regression, Polynomial regression, Error metrics: MAE, MSE, RMSE, R2 Hands-on: Predictive analytics using housing/sales data

Unit III: Supervised Learning – Classification

Logistic regression, Decision Trees, KNN, Random Forest, SVM, Naive Bayes overview, Evaluation: confusion matrix, precision, recall, F1-score, Hands-on: Binary and multi-class classification tasks

Unit IV: Unsupervised Learning and Preprocessing

K-Means clustering, Hierarchical clustering, Dimensionality reduction: PCA, Feature scaling, encoding, imputation, Hands-on: Segmentation and anomaly detection

Unit V: Model Tuning and Interpretation

Hyperparameter tuning with GridSearchCV, Cross-validation strategies, Introduction to model explainability (SHAP/LIME), Hands-on: Full pipeline project with reporting

Online Software

1 Python

2 Scikit-learn

3 Jupyter

4 Pandas

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

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III B. Tech - II Semester CSE(AI&ML)

23ACM37- DEEP DIVE INTO AI – PRINCIPLES AND PRACTICES OF DEEP LEARNING

L	T	P	C
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Course objectives

The objective of the course is,

1. Understand the mathematical foundations of deep learning.
2. Learn the structure and training process of neural networks.
3. Build and evaluate deep learning models for image and tabular data.
4. Explore optimization techniques and regularization.
5. Work on applied deep learning problems using TensorFlow/Keras.

Course Outcome (COs)

At the end of the course, the student will be able to:

Learning Levels: Re - Remember; Un - Understand; Ap - Apply; An - Analysis; Ev - Evaluate; Cr - Create **Learning Level**

1.	Understand the structure of neural networks	Un
2.	Train and evaluate deep learning models	Ap
3.	Implement CNNs for vision problems	Ap
4.	Use optimization and regularization techniques	An
5.	Execute a full DL project with analysis	Cr

Unit I: Foundations of Neural Networks

Biological inspiration and neuron model, Perceptron, activation functions, cost/loss functions, Forward and backward propagation, Hands-on: Implement simple neural network from scratch

Unit II: Deep Neural Networks and Training Strategies

Multi-layer perceptrons (MLPs), Weight initialization, optimizers (SGD, Adam), Loss landscapes, vanishing/exploding gradients, Regularization: dropout, early stopping, batch normalization, Hands- on: DNN on structured data

Unit III: Convolutional Neural Networks (CNNs)

Convolution operations, filters, pooling layers, CNN architectures: LeNet, VGG, ResNet overview, Applications in image classification,Hands-on: Build CNN using Keras for image dataset

Unit IV: Model Evaluation and Deployment Basics

Train/validation/test splits, Underfitting, overfitting, Saving/loading models, checkpoints, Basic deployment with TensorFlow/Keras models, Hands-on: Evaluate and export a DL model

Unit V: Mini Project and Case Study

End-to-end DL project (vision or tabular), Data preparation, model building, evaluation, Reporting and presentation of findings, Hands-on: Capstone mini project

Text Books

1. François Chollet, Deep Learning with Python

2. Michael Nielsen, Neural Networks and Deep Learning

Online Free Software

1 Python

2 TensorFlow

3 Google Collab

4 Keras

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

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III B. Tech - II Semester CSE(AI&ML)

23ACM38: GEN AI – FUNDAMENTALS TO ADVANCED APPLICATIONS

L	T	P	C
3	0	0	3

Course objectives

The objective of the course is,

1. Understand the foundational principles of Generative AI.
2. Explore state-of-the-art GenAI models and their use cases.
3. Apply prompt engineering and fine-tuning techniques.
4. Evaluate the ethical, legal, and societal impacts of GenAI.
5. Build GenAI-based applications using modern frameworks.

Course Outcome (COs)

At the end of the course, the student will be able to:

Learning Levels: Re - Remember; Un - Understand; Ap - Apply; An - Analysis; Ev - Evaluate; Cr - Create Learning Level

1.	Understand the working and applications of GenAI	Un
2.	Apply prompt engineering techniques	Ap
3.	Use foundation models via open tools/APIs	Ap
4.	Evaluate GenAI systems for bias and impact	An
5.	Build and demonstrate a practical GenAI app	Cr

Unit I: Introduction to Generative AI

Traditional vs generative models, GANs, VAEs, and transformer-based architectures, Overview of Large Language Models (LLMs): GPT, BERT, Claude, LLaMA, Use cases: content generation, text- to-image, summarization, Hands-on: Explore GenAI demos and APIs

Unit II: Prompt Engineering & Customization

Understanding zero-shot, one-shot, few-shot learning, Prompt design techniques, Prompt tuning, instruction tuning, Tools: LangChain, PromptLayer, OpenAI playground, Hands-on: Design effective prompts for various tasks

Unit III: Working with Foundation Models

Hugging Face Transformers, Text generation and summarization pipelines, Open-source GenAI: LLaMA2, Mistral, Gemma, Hands-on: Fine-tuning using Hugging Face or Colab tools

Unit IV: Advanced GenAI Applications:

Text-to-image (DALL·E, Midjourney, SDXL), Text-to-code (Codex, Code Llama), Autonomous agents and copilots, Hands-on: Build a chatbot or GenAI app using open models

Unit V: Responsible GenAI & Capstone

Bias, misinformation, and hallucination in LLMs, Ethics and governance, Deployment considerations (compute, latency, cost), Hands-on: Capstone GenAI product demo or paper

Text Books:

1. Nathan Hunter , The Art of Prompt Engineering with OpenAI
2. Hugging Face course: https://huggingface.co/learn
3. Responsible AI practices, Google, Microsoft, OpenAI blogs

Online Software :

1. Hugging Face, LangChain
2. OpenAI API
3. Google Colab
4. LLaMA2, ChatGPT

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

IV B. Tech - I Semester CSE(AI&ML)

23ACM39- LLMOPS AND SCALABLE GENAI PRODUCT DEPLOYMENT

L	T	P	C
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Course objectives

The objective of the course is,

- Understand the full lifecycle of GenAI model development and deployment.
- Explore LLMOps practices including versioning, evaluation, and monitoring.
- Learn to containerize and serve LLMs using industry-standard tools.
- Build scalable, secure, and maintainable GenAI applications.
- Apply MLOps principles to LLM pipelines using open-source tools.

Course Outcome (COs)

At the end of the course, the student will be able to:

CO1. Understand the LLM lifecycle and operationalization needs, CO2. Apply prompt evaluation and feedback strategies,
CO3. Serve and scale LLMs using modern tools
CO4. Integrate CI/CD and observability in GenAI systems CO5. Build and deploy a production-ready GenAI application

Unit I: LLM Lifecycle and Ops Principles

Introduction to MLOps vs LLMOps, Lifecycle stages: model selection, training, evaluation, deployment, monitoring, Challenges in managing LLMs, Tools overview: MLflow, LangChain, BentoML, Weights & Biases

Unit II: LLM Evaluation and Prompt Testing

Prompt testing frameworks: PromptLayer, LangSmith, Output quality evaluation: accuracy, coherence, hallucination rate, Feedback loops and reinforcement learning with human feedback (RLHF) basics, Hands-on: Prompt evaluation dashboard with LangSmith

Unit III: Serving and Scaling LLMs

Model packaging and containerization (Docker), Model serving with FastAPI, Triton, or TGI, Distributed inference and batching, Hands-on: Serve a fine-tuned LLM using FastAPI and Docker

Unit IV: GenAI App Architecture and Deployment

Integrating LLM APIs with frontends (e.g., Streamlit, Gradio), Vector databases: FAISS, ChromaDB, Pinecone, Retrieval-Augmented Generation (RAG) overview,Hands-on: Deploy RAG-based QA chatbot

Unit V: Final Project and CI/CD Integration

Design and build a scalable GenAI product (e.g., document QA, AI tutor, agent), Implement logging, error handling, versioning, monitoring, CI/CD tools: GitHub Actions, MLflow tracking, DockerHub, Hands-on: Full deployment with CI/CD and documentation

Text Books:

1. Chip Huyen , Designing Machine Learning Systems
2. LLMOps Guide by Weights & Biases, LangChain Docs

Online Software :

1. Python
2. FastAPI, Docker
3. LangChain, MLflow
4. Hugging Face, Vector DBs (FAISS, ChromaDB)

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IV B. Tech - I Semester

23ACM40- CAPESTONE PROJECT

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MINORS OFFERED BY THE CIVIL ENGINEERING BUILDING

PLANNING & CONSTRUCTION TECHNOLOGY

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

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III Year B.Tech. CE–I Semester	L	T	P	C
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(23ACE67) CONSTRUCTION MATERIALS

Course Objectives:

1. To Understand the Types, Physical Characteristics, Testing, and Deterioration of Natural and Artificial Stones Used In Construction.(L2)
2. To Explain and Classify Various Structural Clay Products Such As Bricks, Tiles, Terracotta, Porcelain, and their Manufacturing Processes and Applications.(L2)
3. To Analyze and Apply the Knowledge of Cement, Aggregates, and Admixtures To Evaluate the Properties of Fresh and Hardened Concrete, Including Mix Design. (L3)
4. To Evaluate and Apply the Knowledge of Wood and Timber—Its Properties, Preservation, Defects, Testing, and Uses In Building Construction. (L3)
5. To Understand and Interpret the Composition, Application Techniques, and Properties of Paints, Varnishes, Enamels, and Surface Finishes.(L2)

Course Outcomes :

Upon Successful Completion of the Course, the Student Will Be Able To:

1. Identify the Characteristics, Types, Deterioration, and Applications of Building Stones. (L2)
2. Classify and Evaluate the Quality and Suitability of Bricks, Tiles, and Other Structural Clay Products. (L3)
3. Analyze the Properties and Mix Design of Cement, Aggregates, and Concrete For Construction Use. (L4)
4. Assess Timber Based On Its Properties, Preservation Methods, and Applications In Building Works. (L3)
5. Explain the Properties, Preparation, and Application of Paints, Varnishes, and Polishes For Surface Protection. (L2)

UNIT–I

Introduction-Rock-Forming Minerals-Classification of Rocks-Quarrying of Stones-Seasoning of Stone-Dressing of Stone –Uses of Stones –Characteristics of Good Building Stones-Testing of Stones-Deterioration of Stones - Durability of Stones - Preservation of Stones - Selection of Stones - Common Building Stones - Artificial Stones -Applications of Stones.

UNIT–II

STRUCTURAL CLAY PRODUCTS

Introduction - Physical Properties of Clays - Classification of Bricks - Ingredients of Good Brick Earth - Characteristics of Good Brick –Manufacturing of Bricks -Different Forms of Bricks – Testing of Bricks – Defects of Bricks-Clay Tiles-Fire-Clay Bricks or Refractory Bricks– Terracotta–Porcelain–Stoneware–Earthenware–Glazing–Majolica-ApplicationofClayProducts.

UNIT–III

CEMENT & CONCRETE

Introduction- Portland Cement- Chemical Composition- Hydration of Cement-Manufacture of Cement -Testing of Cement-Types of Cement–Aggregates-Classification of Aggregates-Fine Aggregate & Coarse Aggregate-Testing of Aggregates-Quality of Mixing&CuringWater– Admixtures For Concrete – Principles of Mix Design-Fresh and Hardened Properties.

UNIT–IV

WOOD

Introduction-Classification of Trees - Growth of Trees - Classification of Timber (IS: 399) - Structure of Timber - Characteristics of Good Timber - Seasoning of Timber - Defects In Timber - Diseases of Timber- Decay of Timber - Preservation of Timber (IS: 401) - Fire Resistance of Timber - Testing of Timber(IS:1708)-Suitability of Timber For Specific Uses- Properties of Wood- Wood Products- Applications of Wood and Wood- Products

UNIT–V

PAINTS, ENAMELS AND VARNISHES

Introduction - Composition of Oil Paint - Characteristics of An Ideal Paint - Preparation of Paint - Covering Power of Paints - Pigment Volume Concentration (P.V.C.) - Painting Plastered Surfaces – Painting Wood Surfaces-PaintingMetalSurfaces-Defects-Enamel-Distemper-WaterWashand Colour Wash -Varnish-French Polish-Wax Polish –Miscellaneous Paints

Text Books

1. S.K. Duggal, *Building Materials*, New Age International Pvt. Ltd., 3rd Revised Edition, 2008.
2. P.C. Varghese, *Building Materials*, PHI Learning Pvt. Ltd., 2nd Edition, 2015.

Reference Books

1. M.L. Gambhir, *Building Materials*, Tata McGraw Hill Publishing Co. Ltd., 4th Edition, 2013.
2. S.S. Bhavikatti, *Building Materials*, Vikas Publishing House Pvt. Ltd., 3rd Edition, 2015.
3. B.C. Punmia, *Building Materials*, Laxmi Publications Pvt. Ltd., 2010.
4. Jagadish, Venkatarama Reddy, *Alternate Building Materials and Technology*, New Age International Publishers, 2010.
5. Surendra Singh, *Engineering Materials*, Vikas Publishing House Pvt. Ltd., 2014.
6. P.D. Kulkarni et al., *Civil Engineering Materials*, Tata McGraw Hill Publishing Co. Ltd., 2008.

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III Year B.Tech. CE–I Semester	**L**	**T**	**P**	**C**
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(23ACE68) CONSTRUCTION METHODS

Course Objectives:

1. To understand the importance, types, and selection of foundations for various site conditions. (L2)
2. To identify and apply construction techniques of different types of masonry and structural components. (L3)
3. To describe and evaluate different types and techniques of plastering used in construction. (L2)
4. To examine the properties, applications, and construction methods of various flooring types. (L4)
5. To analyze and design basic components of industrial structures with emphasis on modern techniques and materials. (L4)

Course Outcomes:

Upon successful completion of this course, the student will be able to:

1. Understand various types of foundations, their functions, and methods to improve soil bearing capacity. (L2)
2. Apply knowledge of masonry types, bonds, and composite construction methods in practical construction. (L3)
3. Identify types of plastering, defects, and special techniques used in surface finishing. (L2)
4. Analyze different flooring materials, laying techniques, and repair methods. (L4)
5. Design and assess components of industrial structures such as PEBs, trusses, and roofing systems. (L6)

UNIT–I FOUNDATIONS

Introduction To Foundations - Importance of Foundations In Construction - Functions of A Foundation - Factors Affecting the Choice of Foundations - Types of Foundations - Shallow Foundations: Strip Footing, Isolated Footing, Combined Footing, Raft Foundation - Deep Foundations :Pile Foundations,Well Foundations-Site Investigation For Foundations-Soil, Testing Methods, Bearing Capacity of Soils, Methods of Improving Bearing Capacity.

UNIT–II MASONRY

Introduction To Masonry Construction - Definition and Importance of Masonry - Comparison of Different Types of Masonry - Types of Masonry Construction - Brick Masonry: Stretcher, Header, English, Flemish Bonds - Stone Masonry: Random Rubble Masonry, Coursed Rubble Masonry, Ashlar Masonry-Concrete Block Masonry: Hollow Blocks, Solid Blocks, Interlocking Blocks-CompositeMasonry:Brick-StoneCompositeMasonry-Lintels,Arches,andCopings- Construction Techniques and Materials Used

UNIT–III PLASTERING

Introduction To Plastering - Purpose and Importance of Plastering - Materials Used In Plastering (Cement, Lime, Gypsum) - Types of Plastering - Lime Plaster, Cement Plaster, Gypsum Plaster, Stucco Plaster - Methods of Plastering -Single Coat, Double Coat, and Three-Coat Plastering - Defects In Plastering and Remedies - Cracking, Peeling, Efflorescence, Blistering - Special Plastering Techniques -Waterproof Plastering, Decorative Plastering

UNIT–IV FLOORING

Introduction To Flooring - Importance and Requirements of Good Flooring - Types of Flooring- Hard Flooring: Stone (Marble, Granite, Kota), Concrete Flooring , Ceramic and Vitrified Tiles - Resilient Flooring: PVC, Linoleum, Rubber Flooring - Wooden Flooring: Solid Wood, Laminated Wood, Parquet Flooring - Special Flooring: Epoxy Flooring, Mosaic Flooring, Terrazzo Flooring - Construction Methods of Flooring - Laying of Different Floor Materials - Sub-BasePreparationandFinishing- DefectsInFlooringandRepairTechniques-Cracking, Dampness, Un even Settlement

UNIT–V

INDUSTRIAL STRUCTURES

Introduction To Industrial Structures - Requirements of Industrial Buildings - Structural Components - Types of Industrial Structures - Steel Structures: Components of Steel Buildings (Trusses, Columns, Girders) - Pre-Engineered Buildings (PEB): Concept, Advantages, and Applications - Portal Frames and Space Frames: Design and Construction Considerations – Roofing Systems In Industrial Buildings - Trusses, Shell Roofs, Domes, Vaults - Construction of Warehouses and Sheds : Material Selection, Structural Stability Considerations - Factories and Large-SpanStructures,ConstructionTechniquesForLong-SpanStructures,FireResistanceand Safety Measures - Special Considerations For Industrial Buildings - Ventilation and Lighting Requirements, Vibration Control and Noise Reduction Techniques

Text books:

1. B.C. Punmia, Ashok Kumar Jain, Arun Kumar Jain, *Building Construction*, Laxmi Publications (P) Ltd., New Delhi, 2008.
2. P.C. Varghese, *Building Construction*, PHI Learning Pvt. Ltd., 2nd Edition, 2017.

Reference Books:

1. Sushil Kumar, *Building Construction*, Standard Publishers Distributors, New Delhi, 2013.
2. S.S. Bhavikatti, *Building Construction*, Vikas Publishing House Pvt. Ltd., 2nd Edition, 2016.
3. Dr. S.C. Rangwala, *Building Construction*, Charotar Publishing House, 26th Edition, 2014.
4. R.L. Peurifoy, *Construction Planning, Equipment and Methods*, McGraw Hill Education, 8th Edition, 2010.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III Year B.Tech. CE–II Semester	**L**	**T**	**P**	**C**
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(23ACE69) FUNDAMENTALS OF BUILDING PLANNING AND DRAWING

Course Objectives:

1. To Understand the Principles of Building Planning and Regulatory Requirements For Residential and Public Structures. (L2)
2. To Apply Functional Planning Skills For Residential, Public, and Commercial Buildings Including Vastu and Barrier-Free Concepts. (L3)
3. To Analyze and Prepare Building Plans and Elevations For Industrial and Special-Purpose Structures. (L4)
4. To Utilize Computer-Aided Drafting Tools To Develop Architectural and Working Drawings. (L3)
5. To Evaluate Planning Strategies In Compliance With National Codes and Sustainability Standards. (L5)

Course Outcomes:

Upon Completion of This Course, the Student Will be able to:

1. Understand the Objectives and Principles of Building Planning, and Interpret National and Local Building Regulations. (L2)

2. Apply Functional Planning Techniques For Various Residential Building Layouts and Integrate Vastu Principles. (L3)

3. Design and Draw Public, Commercial, and Industrial Buildings as Per Functional and Universal Design Standards. (L6)

4. Develop Detailed Plans, Elevations, and Sections Using CAD Software For Residential and Non-Residential Buildings. (L3)

5. Evaluate and Prepare Service Drawings, Submission Drawings, and Working Drawings In Compliance With Municipal Regulations. (L5)

UNIT – I: BUILDING PLANNING PRINCIPLES & REGULATIONS

Introduction To Building Planning - Objectives of Planning - Principles of Planning - Orientation of Buildings - Building Bye-Laws and Regulations - Necessity of Building Bye-Laws - National Building Code (NBC) Provisions - Local Municipal and Development Authority Regulations - Setbacks and Open Space Requirements - Floor Area Ratio (FAR) and Floor Space Index (FSI) , Height Regulations and Ground Coverage - Lighting, Ventilation, and Fire Safety Regulations - Natural Lighting and Ventilation Standards
- Fire-Resistant Construction and Safety Measures - Concepts of Green Building and Sustainable Planning
- Energy-Efficient Building Design - Passive Solar Architecture

UNIT – II: RESIDENTIAL BUILDINGS

Classification of Residential Buildings - Individual Houses, Duplex Houses, Apartments, Row Houses, Villas - Functional Requirements of A Residential Building - Zoning and Space Requirements For Different Rooms - Circulation and Connectivity - Types of Residential Plans - Single-Bedroom, Double-Bedroom, and Multi-Storey Residential Plans - Preparation of Building Plans - Plan, Elevation, and Section of A Simple Residential Building - Vastu Considerations In Residential Planning - Orientation of Rooms as Per Traditional Indian Architectural Practices.

UNIT – III: PUBLIC and COMMERCIAL BUILDINGS

Classification of Public Buildings - Educational Buildings (Schools, Colleges, Universities) - Institutional Buildings (Hospitals, offices, Banks) - Recreational Buildings (Theatres, Auditoriums, Stadiums) - Functional Requirements and Planning of Public Buildings - Planning of Commercial Buildings - Shopping Malls, Hotels, office Buildings - Preparation of Drawings For Public Buildings - Plan, Elevation, and Section of A Public/Commercial Building - Universal Design and Barrier-Free Planning - Provision For Differently-Abled Individuals - Ramps, Lifts, Tactile Paving, and Accessible Washrooms

UNIT – IV: INDUSTRIAL and SPECIAL STRUCTURES

Introduction To Industrial Buildings - Functional Planning of Industries - Layout of Factory Buildings - Types of Industrial Structures - Warehouses, Godowns, Cold Storage Facilities - Special Purpose Buildings - Parking Structures, Bus Terminals, Railway Stations, Airports - Planning of Multi-Storied Buildings - Structural Aspects and Service Provisions - Vertical Circulation (Lifts, Escalators, Staircases)
- Preparation of Drawings For Industrial Buildings – Plan, Elevation, and Section of An Industrial Shed Or Warehouse

UNIT – V: COMPUTER-AIDED DRAWING (CAD) and WORKING DRAWINGS

Introduction To CAD In Building Drawing - Importance of CAD In Modern Construction - Software Used (Autocad, Revit, Sketchup) - Preparation of Working Drawings - Detailed Plans, Elevations, and Sections – Site Plan, Key Plan, and Service Plans - Structural Drawings and Detailing - Footing Details, Column Layouts, Slab Reinforcement - Preparation of Submission Drawings - Sanction Drawings as Per Municipal Regulations - Approval Process and Documentation - Service Drawings - Electrical, Plumbing, and HVAC (Heating, Ventilation, and Air Conditioning) Drawings.

Text Books:

1. M. G. Shah, C. M. Kale, *Building Drawing*, Tata McGraw Hill Publishing Co. Ltd., 2nd Edition, 2012.
2. Gurucharan Singh, *Building Drawing*, Standard Publishers Distributors, 9th Edition, 2015.

Reference Books:

1. K. R. Karnan, *Building Planning and Drawing*, Scitech Publications (India) Pvt. Ltd., 2016.
2. S. P. Bindra, *Building Drawing and Detailing*, Dhanpat Rai Publishing Co., 2014.
3. B. C. Punmia, Ashok Kumar Jain, Arun Kumar Jain, *Building Planning and Drawing*, Laxmi Publications Pvt. Ltd., 2011.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III Year B.Tech. CE–II Semester

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3	0	0	3

(23ACE70) **BASIC SURVEYING**

Course Objectives :-

The Objectives of this course are to enable the Students to:

1. Understand the Principles, Classifications, and Fundamental Techniques of Surveying Including Chain, Compass, Levelling, and Plane Table Surveys. **(L2)**
2. Apply Basic Surveying Instruments and Techniques For Measurement of Distances, Angles, Levels, Areas, and Volumes. **(L3)**
3. Analyze and Compute Traverse Data, Horizontal and Vertical Angles Using theodolites and Trigonometric Levelling Methods. **(L4)**
4.
5. Evaluate the Use of Modern Surveying Instruments Like Total Stations, GPS, and Remote Sensing Tools For Spatial Data Acquisition. **(L5)**
6. Interpret Photogrammetric Data For Topographic Mapping, Aerial Triangulation, and Preparation of Maps Using Stereoscopy. **(L5)**

Course Outcomes :-

After Successful Completion of the Course, the Student will be able to:

1. To Understand the Basic Principles, Classifications, and Techniques of Surveying Using Chain, Compass, Levelling, and Plane Table Methods. **(L2)**
2. To Apply Appropriate Methods For Levelling, Contouring, and Calculating Areas and Volumes For Civil Engineering Projects. **(L3)**
3. To Analyze and Compute Angular Measurements, Traverses, and Heights Using theodolites and Trigonometric Levelling. **(L4)**
4. To Utilize Modern Surveying Tools Such as EDM, Total Station, GPS, Drones, and Lidar For Advanced Spatial Data Collection. **(L3)**
5. To Evaluate and Interpret Photogrammetric Data For Map Preparation, Stereoscopy, Aerial Triangulation, and Mosaics. **(L5)**

UNIT–I

Introduction and Basic Concepts: Introduction, Objectives, Classification and Principles of Surveying, Surveying Accessories. Introduction To Compass, Leveling and Plane Table Surveying.

Linear Distances- Approximate Methods, Direct Methods- Chains- Tapes, Ranging, Tape Corrections.

Prismatic Compass- Bearings, Included Angles, Local Attraction, Magnetic Declination, and Dip – Systems and W.C.B and Q.B Systems of Locating Bearings

UNIT-II

Leveling- Types of Levels, Methods of Levelling, and Determination of Levels, Effect of Curvature of Earth and Refraction.

Contouring- Characteristics and Uses of Contours, Methods of Contour Surveying.

Areas - Determination of Areas Consisting of Irregular Boundary and Regular Boundary.

Volumes -Determination of Volume of Earth Work In Cutting and Embankments For Level Section, Capacity of Reservoirs.

UNIT – III

Theodolite Surveying: Types of theodolites, Temporary Adjustments, Measurement of Horizontal Angle By Repetition Method and Reiteration Method, Measurement of Vertical Angle, Trigonometrical Leveling When Base Is Accessible and Inaccessible.

Traversing: Methods of Traversing, Traverse Computations and Adjustments, Introduction To Omitted Measurements.

UNIT – IV

Curves: Types of Curves and their Necessity, Elements of Simple, Compound, Reverse Curves.

Introduction To Tacheometric Surveying.

Modern Surveying Methods: Principle and Types of E.D.M. Instruments, Total Station- Advantages and Applications. Introduction To Global Positioning System. Introduction To Drone Survey and Lidar survey (Light Detection and Ranging).

UNIT – V

Photogrammetry Surveying:

Introduction, Basic Concepts, Perspective Geometry of Aerial Photograph, Relief and Tilt Displacements, Terrestrial Photogrammetry, Flight Planning; Stereoscopy, Ground Control Extension For Photographic Mapping- Aerial Triangulation, Radial Triangulation, Methods; Photographic Mapping- Mapping Using Paper Prints, Mapping Using Stereo-Plotting Instruments, Mosaics, Map Substitutes.

Text Books:

1. S. K. Duggal, *Surveying* (Vol. 1 & 2), Tata McGraw Hill Publishing Co. Ltd., New Delhi, 5th Edition, 2019.
2. C. Venkatramaiah, *Textbook of Surveying*, Universities Press, 1st Edition, 2011.

Reference Books:

1. B. C. Punmia, Ashok Kumar Jain, Arun Kumar Jain, *Surveying* (Vol. 1), Laxmi Publications (P) Ltd., New Delhi, 18th Edition, 2024.
2. B. C. Punmia, Ashok Kumar Jain, Arun Kumar Jain, *Surveying* (Vol. 2), Laxmi Publications (P) Ltd., New Delhi, 17th Edition, 2022.
3. B. C. Punmia, Ashok Kumar Jain, Arun Kumar Jain, *Surveying* (Vol. 3), Laxmi Publications (P) Ltd., New Delhi, 16th Edition, 2023.
4. A. M. Chandra, *Plane Surveying and Higher Surveying*, New Age International Pvt. Ltd., New Delhi, 3rd Edition, 2015.
5. N. Basak, *Surveying and Levelling*, Tata McGraw Hill Publishing Co. Ltd., New Delhi, 4th Edition, 2014.
6. K. R. Arora, *Surveying* (Vol. 1, 2 & 3), Standard Book House, Delhi, 12th Edition, 2015.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

IV Year B.Tech. CE– I Semester

	L	T	P	C
	3	0	0	3

(23ACE71) ESSENTIALS OF CONCRETE TECHNOLOGY

Course Objectives :-

The Objectives of This Course Are To Enable the Student To:

1. Understand the Composition, Properties, Specifications, and Testing Methods For Concrete Materials. [L1]

2. Learn the Methods of Concrete Production Including Mixing, Transportation, Placing, Compaction, and Curing. [L2]

3. Acquire Knowledge On the Behavior of Fresh and Hardened Concrete and Factors Influencing Performance. [L3]

4. Develop the Ability To Design Concrete Mixes Using Standard Methods For Normal and High Strength Concrete. [L4]

5. Explore Special Concretes, Quality Control Measures, and Durability Aspects. [L4]

Course Outcomes :-

After completing this course, the student will be able to:

1. **CO1:** Explain the chemical and physical properties of cement, aggregates, water, and admixtures as per IS standards. [L2]
2. **CO2:** Describe the manufacturing process of concrete, including equipment, methods, and curing techniques. [L2]
3. **CO3:** Analyze the fresh and hardened properties of concrete including workability, strength, and microstructure. [L4]
4. **CO4:** Design concrete mix proportions using ACI and IS methods for normal and high-strength concrete. [L4]
5. **CO5:** Evaluate the behavior and performance of concrete under various conditions, including shrinkage, creep, corrosion, thermal effects, and special concrete types. [L5]

UNIT – I

Materials For Making Concrete: Cement - Chemical Composition, Physical Properties; Tests, I.S. Specifications; Different Types – Aggregates - Classification, Mechanical Properties, and Tests, Grading Requirements, Sampling of Aggregate; Water - Quality of Water, Permissible Impurities As Per I.S; – Mineral Admixtures: Fly Ash, GGBS, Silica Fume, Metakaolin, Rice Husk Ash, Etc. Chemical Admixtures - Accelerators, Retarders, Water Reducing Agents, Super Plasticizers, Air Entrainers, Water Proofers.

UNIT – II

Concrete: Manufacture of Concrete, Measurement of Materials, Storage and Handling, Batching Plant and Equipment Mixing - Types of Mixers, Transportation of Concrete, Pumping of Concrete, Placing of Concrete, Under Water Concreting, Compaction of Concrete, Curing of Concrete, Ready Mixed Concrete.

Mix Design - Nominal Mixes and Design Mixes, Factors Influencing Mix Design; ACI Method and I.S Method, Design For High Strength Mixes.

UNIT – III

Fresh, Hardened Properties of Concrete: Fresh Concrete - Workability, Tests For Workability, Cohesion, Segregation, and Bleeding; Hardened Concrete- Factors Affecting Strength of Concrete, Strength of Concrete In Compression, Tension, and Flexure; Stress- Strain Characteristics and Elastic Properties; Introduction To Microstructural Properties of Concrete

UNIT – IV

Mix Design - Factors Influencing Mix Proportion - Mix Design By ACI Method and I.S. Code Method - Design of High Strength Concrete. Strength of Concrete - Shrinkage and Temperature Effects - Creep of Concrete - Permeability of Concrete - Durability of Concrete - Corrosion - Causes and Effects - Remedial Measures- thermal Properties of Concrete - Micro Cracking of Concrete.

UNIT – V

Special Concrete - Lightweight Concrete - Fibre Reinforced Concrete - Polymer-Polymer Modified Concrete - Ferrocement - Mass Concrete - Ready Mix Concrete- Self Compacting Concrete- Quality Control - Sampling and Testing-Acceptance Criteria.

Text Books:

1. Properties of Concrete, AM Nevelli, Prentice Hall Publishers, 2012, 5th Edition.
2. Concrete Technology: theory and Practice, M.L. Gambhir, Tata Mc Graw Hill Publishers, 2017, 5th Edition.

Reference Books:

1. P. Kumar Mehta , Paulo J.M. Monteiro, Concrete: Microstructure, Properties, and Materials, Mcgraw Hill Education,2014
2. Concrete Technology: theory and Practice, M. S. Shetty and A. K. Jain, S Chand Co., Publishers, 2018.
3. Concrete Technology, J.J. Brooks and A. M. Neville, Pearson, 2019, 2nd Edition.
4. Santakumar A.R., Concrete Technology, Oxford University Press, New Delhi, 2007.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III Year B.Tech. CE–I Semester

L	T	P	C
0	0	3	1.5

(23ACE72) BASIC CONCRETE TECHNOLOGY LAB

Course Objectives :-

The Objectives of This Course Are To:

1. Understand the Physical Properties and Behavior of Cement and Aggregates. [L1]
2. Gain Hands-On Experience In Determining the Fresh and Hardened Properties of Concrete. [L2]
3. Conduct Standard Laboratory Procedures For Assessing the Quality of Concrete Materials. [L3]
4. Familiarize With Concrete Mix Design Using IS Method. [L4]
5. Interpret Experimental Data For Evaluating the Performance of Concrete. [L5]

Course Outcomes :-

1. After Completing This Course, the Student Will Be Able To:
2. **CO1**: Determine the Physical Properties of Cement Through Standard Tests Such As Fineness, Consistency, Setting Time, Soundness, and Compressive Strength. [L3]
3. **CO2**: Assess the Properties of Fine and Coarse Aggregates and Evaluate the Workability of Fresh Concrete Using Standard Test Methods. [L3]
4. **CO3**: Prepare, Cure, and Test Concrete Specimens To Determine Compressive and Tensile Strength, and Analyze Stress-Strain Behavior. [L4]
5. **CO4**: Apply the IS Method For Concrete Mix Design and Verify the Mix Through Testing. [L4]
CO5: Demonstrate the Use of Non-Destructive Testing Equipment To Assess the Quality of Hardened Concrete. [L3]

1. Determination of Fineness and Specific Gravity of Cement.
2. Determination of Consistency of Standard Cement Paste.
3. Determination of Initial and Final Setting Times of Cement.
4. Determination of Compressive Strength of Cement.
5. Determination of Soundness of Cement By La-Chatalier’s Apparatus.
6. Determination of Fineness Modulus of Coarse and Fine Aggregates.
7. Determination of Percentage of Voids, Bulk Density, Specific Gravity of Course and Fine Aggregates.
8. Workability Tests: Slump Cone Test, Compaction Factor Test, Vee-Bee Consistometer, Flow Test.
9. Preparing and Curing Concrete Specimens For Tests & Determination of Compressive Strength of Concrete Cubes.
10. Study of Stress - Strain Characteristics of Concrete and Tests For Tensile Strength of Concrete.
11. Experiments To Demonstrate the Use of Non-Destructive Test Equipment.
12. Mix Design: IS Code Method.

Text/ Reference Books :

1. Concrete Technology: theory and Practice, M.L. Gambhir, Tata Mc Graw Hill Publishers, 2017, 5th Edition.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III Year B.Tech. CE–II Semester

L	T	P	C
0	0	3	1.5

(23ACE73) BASIC SURVEYING LAB

Course Objectives :-

After completing this course, the student will be able to:

1. Understand and perform different types of surveying techniques including chain, compass, plane table, and Theodolite surveying. [L2]
2. Apply methods of leveling and contouring for field measurements and building layouts. [L3]
3. Interpret and plot survey data accurately to prepare maps and plans. [L3]
4. Use modern surveying instruments like Total Station for precise measurements. [L3]
5. Develop skills to set out curves and perform traversing for practical civil engineering projects. [L3]

Course Outcomes :

After completing this course, the student will be able to:

CO1: Conduct chain, compass, and plane table surveys, and accurately plot survey details. [L3]

CO2: Perform leveling operations including fly leveling, profile leveling, and contouring for terrain mapping. [L3]

CO3: Use theodolite for precise measurement of horizontal angles and set out simple curves. [L4]

CO4: Operate Total Station equipment for accurate data collection and site layout. [L4]

CO5: Prepare field survey maps and layouts for building and civil engineering projects based on survey data. [L4]

List of Experiments:

1. Chain Survey - Traversing and Plotting of Details
2. Compass Survey - Traversing With Compass and Plotting
3. Plane Table Survey - Method of Radiation and Intersection
4. Levelling - Fly Leveling - Plane of Collimation Method
5. Levelling - Fly Leveling - Rise and Fall Method
6. Levelling - Longitudinal and Cross Sectioning
7. Levelling - Contour Surveying 8. Layout of Buildings
8. Profile Levelling and Cross Sectioning.
9. Grid Contouring
10. Theodolite Surveying - Measurement of Horizontal Angle By Method of Repetition and Reiteration
11. Setting Out of Simple Curves - Linear Methods
12. Description of Total Station

Text Books :

Surveying and Levelling By N.N.Basak, TMH Publication 2. Surveying By B.C. Punamia, A.K. Jain and A.K. Jain, Vol. 1, Laxmi Publications (P) Ltd., New Delhi

MINORS in ENERGY SYSTEMS (EEE Department)

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

L	T	P	C
3	0	0	3

23AEE52 ENERGY AUDIT AND MANAGEMENT

COURSE OUTCOMES:

After the completion of the course students will be able to

CO1. Understanding the Fundamentals of Energy Auditing and Conservation. -L2 CO2. Analyzing Energy Audit Concepts and Techniques -L3

CO3. Designing and Implementing Energy Management Programs -L5

CO4. Managing Thermal Energy and Implementing Energy Conservation Techniques L4 CO5. Understand the basic principle and applications of the electrolytic process.-L1 CO6. Implementing Electrical Energy Management and Conservation -L4

UNIT I

Introduction:

Basic elements and measurements - Mass and energy balances – Scope of energy auditing industries - Evaluation of energy conserving opportunities.

UNIT II

Energy Audit Concepts:

Need of Energy audit - Types of energy audit – Energy management (audit) approach - understanding energy costs - Bench marking – Energy performance - Matching energy use to requirement - Maximizing system efficiencies - Optimizing the input energy requirements - Duties and responsibilities of energy auditors- Energy audit instruments - Procedures and Techniques.

UNIT III

Principles and Objectives of Energy Management:

Design of Energy Management Programs - Development of energy management systems – Importance - Indian need of Energy Management - Duties of Energy Manager - Preparation and presentation of energy audit reports - Some case study and potential energy savings.

UNIT IV

Thermal Energy Management:

Energy conservation in boilers - steam turbines and industrial heating systems - Application of FBC - Cogeneration and waste heat recovery - Thermal insulation - Heat exchangers and heat pumps - Building Energy Management.

UNIT V

Electrical Energy Management:

Supply side Methods to minimize supply-demand gap- Renovation and modernization of power plants - Reactive power management – HVDC - FACTS - Demand side - Conservation in motors - Pumps and fan systems – Energy efficient motors. Demand side management.

TEXTBOOKS:

1. Hamies, Energy Auditing and Conservation; Methods Measurements, Management and Case study, Hemisphere, Washington, 1980.
2. Energy Management: W.R.Murphy, G.Mckay.

REFERENCE BOOKS:

1. Energy Management Principles: C.B.Smith.
2. Efficient Use of Energy : I.G.C.Dryden.
3. Energy Economics A.V.Desai.
4. Guide book for National Certification Examination for Energy Managers and Energy Auditors (Could be downloaded from www.energymanagertraining.com).

RESOURCES:

1. https://nptel.ac.in/courses/108106022

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

L	T	P	C
3	0	0	3

23AEE53 ENERGY MANAGEMENT IN BUILDING

COURSE OUTCOMES:

After the completion of the course students will be able to

CO1. Understanding the Fundamentals of Energy Use in Buildings Apply the Energy management concepts for building designs -L2

CO2. Analyzing Indoor Environmental Requirements and Their Impact on Energy Use -L3 CO3. Examining the Role of Climate and Environmental Factors in Building Energy Use -L3 CO4. Evaluating Energy Utilization and Heat Transfer in Building Envelopes L4

CO5. Implementing Energy Management Strategies for Buildings -L5

UNIT I: Overview of the Significance of Energy use and Energy Processes in Building:

Indoor activities and environmental control - Internal and external factors on energy use and the attributes of the factors - Characteristics of energy use and its management - Macro aspect of energy use in dwellings and its implications – Concepts of energy efficient building.

UNIT II: Indoor Environmental Requirement and Management:

Thermal comfort – Ventilation and air quality - Air-conditioning requirement - Visual perception – Illumination requirement - Auditory requirement – Concept of sick building syndrome – Significance in energy management in buildings.

UNIT III: Climate

Solar radiation and their influences - The sun-earth relationship and the energy balance on the earth‘s surface – Climate – Wind - Solar radiation - Temperature – Sun shading and solar radiation on surfaces - Energy impact on the shape and orientation of buildings.

UNIT IV: END-USE

Energy utilization and requirements – Lighting and day lighting – End-use energy requirements – Status of energy use in buildings – Estimation of energy use in a building - Heat gain and thermal performance of building envelope – Steady and non steady heat transfer through the glazed window and the wall – Standards for thermal performance of building envelope – Evaluation of the overall thermal transfer – Concepts of window management.

UNIT V: Energy Management Options

Energy audit and energy targeting – Technological options for energy management – Modifications for energy efficient buildings for Indian conditions. Energy Management for large tower buildings.

TEXTBOOKS:

1. Heating and Cooling of Buildings – Design for Efficiency, J. Krieder and A. rabl, McGraw Hill, 1994.
2. Mechanical and Electrical Equipment for Buildings, S. M. Guinnes and Reynolds, Wiley, 1989.

REFERENCE BOOKS:

1. Energy Design for Architects, Shaw, Aee Energy Books, 1991.
2. Energy Conservation in Buildings – Royal Institute of Architecture, Canada.
3. Publication of CBRI, Roorkee – Energy Management in Buildings.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

L	T	P	C
3	0	0	3

23AEE54 ENERGY STORAGE TECHNOLOGIES

COURSE OUTCOMES:

After the completion of the course students will be able to

CO1. Understanding the Basics of Batteries and Their Performance Characteristics -L2 Analyzing Indoor Environmental Requirements and Their Impact on Energy Use -L3

CO2. Analyzing Primary Battery Types and Their Applications -L3

CO3. Exploring Advanced Battery Technologies and Their Applications -L4 CO4. Assessing Energy Storage for Renewable Energy Systems -L4

CO5. Understanding and Applying Super capacitors in Energy Storage -L4

UNIT I: Batteries

Types-battery characteristics - voltage, current, capacity, volumetric energy density, specific energy density, charge rate, cycle life, internal resistance, energy efficiency, shelf life, battery management system, SoC, SoH estimation techniques. Testing of batteries, battery charging method, Factors affecting the battery performance.

UNIT II: Primary Batteries

Fabrication, performance aspects, packing and rating of alkaline manganese, silver oxide cells. Lithium primary batteries-Lithium/Manganese Dioxide, Lithium/Carbon Monofluoride, Lithium/Thionyl chloride, Lithium/Sulfur Dioxide, Lithium/Iodine, Lithium-Aluminum/Iron Disulfide**.**

UNIT III: Advanced Batteries

Advanced Lead Acid Battery -design, performance aspects, Pb-Acid batteries for transportation, nickel- metal hydride batteries, zinc- alkaline batteries, ZEBRA Battery (Na/NiCl2) -NaS Battery-Lithium-Ion Battery-Lithium- Polymer Battery, Li-air batteries, Li-S batteries, Sodium - ion batteries.

UNIT IV: Storage for Renewable Energy Systems

Solar energy, Wind energy, pumped hydro energy, Energy storage in Micro-grid and Smart grid, Energy Management with storage systems, Battery SCADA, Increase of energy conversion efficiencies by introducing energy storage. Superconducting Magnetic Energy Storage (SMES), charging methodologies, Photo galvanic cells, semiconductor solar batteries (SC-SB), thermo-ionic converter s, dye-sensitized solar cells (DSSC).

UNIT V: Super capacitors and Fuel Cells

Fundamentals of electrochemical Super capacitors, electrode and electrolyte interfaces and their capacitances, charge-discharge characteristics, energy/power density, design, fabrication, operation and evaluation, thermal management. Super capacitors for transportation applications - aqueous and organic based super capacitors, Pseudo and asymmetric super capacitors. Advance battery-super capacitors hybrids for auto, space and marine applications. Fuel Cells working Principle and Construction.

TEXTBOOKS:

1. Dell, Ronald M. Rand and David A. J., ―Understanding Batteries‖, Royal Society of Chemistry, 2001.
2. Vladimir S. Bagotsky, Alexander M. Skundin, Yurij VM. Volfkovich., ―Electrochemical power sources : Batteries, fuel cells, and supercapacitors‖, John Wiley & Sons, Inc.,2015.

REFERENCE BOOKS:

1. Lindon David, ―Handbook of Batteries‖, McGraw Hill, 2002.
2. Kiehne H. A., ―Battery Technology Handbook‖, Expert Verlag, Renningen Malsheim, 2003.
3. AuliceScibioh M. and Viswanathan B., ―Fuel Cells – Principles and Applications‖, University Press, 2006.
4. A.G.Ter-Gazarian, ―Energy Storage for Power Systems‖, The Institution of Engineering and Technology (IET) Publication, UK, 2011.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

L	T	P	C
3	0	0	3

23AEE55 ENERGY SCENARIO AND ENERGY POLICY

COURSE OUTCOMES:

After the completion of the course students will be able to

CO1. Understanding the Global Energy Scenario and Its Impact on Economic Development-L2 CO2. Analyzing International Energy Policies and Treaties -L3

CO3. Understanding the Indian Energy Scenario and Policy Framework -L2 CO4. Evaluating Global Energy Issues and Energy Security -L3

CO5. Exploring Energy Conservation and Sustainable Development- L4

UNIT I

Global Energy Scenario:

Role of energy in economic development and social transformation - Energy and GDP - GNP and its dynamics - Energy sources and overall Energy demand and availability - Energy consumption in various sectors and its changing pattern - Depletion of energy sources and impact exponential rise in energy consumption on economies of countries.

UNIT II

Energy Polices:

International Energy Polices of G-8 Countries - G-20 Countries - OPEC Countries - EU Countries - International Energy Treaties (Rio, Montreal, Kyoto) - INDO-US Nuclear Deal.

UNIT III

Indian Energy Scenario:

Energy resources and Sector wise energy Consumption pattern Impact of energy on economy and development - National and State Level Energy polices and Issues - Status of Nuclear and Renewable Energy and Power Sector reforms. Energy policy 2030

UNIT IV

Energy Policy:

Global Energy Issues - Energy Security - Energy Vision Energy Pricing and Impact of Global Variations Energy Productivity (National and Sector wise productivity).

UNIT V

Energy Conservation:

Act – 2001 and its features - Electricity Act – 2003 and its features - Energy Crisis - Future energy options - Need for use of new and renewable energy sources - Energy for Sustainable development.

TEXTBOOKS:

1. Energy for a sustainable World: Jose Golden berg, Thomas Johan son, AKN. Reddy, Robert Williams (Wiley Eastern).
2. Energy Policy, B.V. Desai (Wiley Eastern)

REFERENCE BOOKS:

1. Modeling approach to long term demand and energy implication, J.K.Parikh
2. Energy Policy and Planning, B.Bukhootsow.
3. TEDDY Year Book Published by Tata Energy Research Institute(TERI) World Energy Resources, Charles E. Brown, _International Energy Outlook' - EIA annual Publication.
4. *BEE Reference book: no. 1/2/3/4.*

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

L	T	P	C
3	0	0	3

23AEE56 WASTE ENERGY MANAGEMENT

COURSE OUTCOMES:

After the completion of the course students will be able to

CO1. Understanding and Characterizing Different Types of Waste -L2

CO2. Analyzing Thermo-Chemical Waste Conversion Methods and Their Environmental Impacts - L3

CO3. Exploring Bio-Chemical Conversion Technologies for Waste to Energy-L4 CO4. Evaluating Energy Production from Waste Plastics and Waste Heat Recovery-L4

CO5. Analyzing Environmental and Health Impacts of Waste-to-Energy Conversion and Case Studies-L3

UNIT I

Characterization of Wastes:

Agricultural residues and wastes including animal wastes; industrial wastes; municipal solid wastes. Waste processing types and composition of various types of wastes; Characterization of Municipal Solid Waste, Industrial waste and Biomedical Waste, waste collection and transportation; waste processing-size reduction, separation; waste management hierarchy, waste minimization and recycling of Municipal solid waste.

UNIT II

Thermo Chemical Conversion:

Incineration, pyrolysis, gasification of waste using gasifiers, environmental and health impacts of incineration; strategies for reducing environmental impacts. Energy production from wastes through incineration, energy production through gasification of wastes, Energy production through pyrolysis and gasification of wastes, syngas utilization.

UNIT III

Bio-Chemical Conversion:

Anaerobic digestion of sewage and municipal wastes, direct combustion of MSW-refuse derived solid fuel, industrial waste, agro residues, anaerobic digestion biogas production, and present status of technologies for conversion of waste into energy, design of waste to energy plants for cities, small townships and villages. Energy production from wastes through fermentation and transesterification. Cultivation of algal biomass from waste water and energy production from algae, Energy production from organic wastes through anaerobic digestion and fermentation.

UNIT IV

Energy Production from Waste Plastics, Gas Cleanup Waste, Heat Recovery:

Concept of conversion efficiency, energy waste, waste heat recovery classification, advantages and applications, commercially viable waste heat recovery devices.

UNIT V

Environmental and Health Impacts-Case Studies: Environmental and health impacts of waste to energy conversion, Industrial waste management – Hazardous waste management – E-waste management. EV Batteries – Mobile Chargers - case studies of commercial waste to energy plants, waste to energy- potentials and constraints in India, eco- technological alternatives for waste to energy conversions.

TEXTBOOKS:

1. Parker, Colin and Roberts, ―Energy from Waste – An Evaluation of Conversion Technologies‖, Elsevier Applied Science, 1985.
2. Khandelwal, K. C. and Mahdi, S. S., ―Biogas Technology - A Practical Hand Book, Vol. I & amp; II, Tata McGraw Hill Publishing Co. Ltd., 1983

REFERENCE BOOKS:

1. Robert C. Brown, ―Thermo-chemical Processing of Biomass: Conversion into Fuels, Chemicals and Power‖, John Wiley and Sons, USA, 2019.
2. Sergio Capareda, ―Introduction to Biomass Energy Conversions‖, CRC Press, USA, 2013.
3. Krzysztof J Ptasinski, ―Efficiency of Biomass Energy: An Exergy Approach to Biofuels, Power, and Biorefineries, John Wiley & Sons, USA, 2013.
4. Vesilind, P.A., and Worrell W. A, ―Solid Waste Engineering, 2nd Ed‖, Cengage India, 2016.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

L	T	P	C
3	0	0	3

23AEE57 PROJECT IN ENERGY SYSTEMS

MINORS IN MICRO GRID TECHNOLOGY
SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

L	T	P	C
3	0	0	3

23AEE58 FUTURISTIC POWER SYSTEMS

UNIT I

Introduction:

Present status of worldwide scenario of electricity generation, transmission and distribution; Energy infrastructure-Resilience and Security; Social, Technical and economic challenges; Major trends driving power system evolution; State of the art technologies in power system.

UNIT II

Renewable Energy Integration:

Review of renewable energy (RE) resources and systems: Solar- PV, Solar Thermal, Wind, Biomass, Micro-hydro and Fuel Cell, comparison of various RE resources; Renewable Energy Policies and present status of integration with existing grid; Large scale integration of renewable energy-Technical challenges, enabling technologies, International requirements.

UNIT III

Energy Storage Systems (ESS):

Review of energy storage components: Battery, VRB, Ultra-capacitor, Fuel Cells, Pumped Hydro- Storage and flywheels, comparison of ESS technologies; Importance of ESS in futuristic power systems; Aggregated ESS, Distributed ESS; Applications of ESS: Energy Management (Load Leveling and Peak Shifting), Fluctuation Suppression (Intermittency Mitigation), Uninterruptible Power System Low-Voltage Ride Through; Placement of the ESS to Improve Power Quality, Voltage Regulation Using ESS, ESS as Spinning Reserve.

UNIT IV

Micro-grid and Smart-grid:

Micro-grid evolution: Micro-grid concept, importance in futuristic power system, basic architectures and control, objectives and state of the art technologies; Microgrid as a building block of Smart-grid; Smart-grid concept, Smart Grid versus conventional electrical networks, Smart-grid infrastructure, Smart Grid communication system and its cyber security, International standard IEC 61850 and its application to Smart-grid.

UNIT V

Communication and IT infrastructure:

Requirements of Communication and IT infrastructure in futuristic power systems: various communication protocols, comparison of performance; IEEE standard: IEEE 802.11 Mesh Networking, IEEE 802.15.4-Wireless Sensor Networks; Communications Technologies for Smart metering; Cyber security issues and mitigation techniques.

TEXTBOOKS:

1. Microgrids Architectures and Control Edited by Nikos Hatziargyriou, IEEE and Wiley, 2014.
2. Energy Storage for Sustainable Microgrid by David Wenzhong Gao, Elsevier, 2015.
3. Introduction to the Smart Grid- Concepts, Technologies and Evolution by Salman K. Salman, IET, 2017.
4. Energy Storage Systems and Components by Alfred Rufer, CRC Press, 2018

REFERENCE BOOKS:

1. Energy Efficiency and Renewable Energy Handbook Edited by D. Yogi Goswami and Frank Kreith, 2nd Edition- 2016, CRC.
2. Clean Energy Microgrids, Edited by Shin'ya Obara and Jorge Morel IET, 2017.
3. Hybrid-Renewable Energy Systems in Microgrids- Integration, Developments and Control edited by Hina Fathimaby et al., Elsevier WoodHead Publishing, 2018.
4. Smart Microgrids: Lessons from Campus Microgrid Design and Implementation edited by Hassan Farhangi, CRC Press 2017.

WEBSITE REFERENCE / VIDEO COURSES:

1. NPTEL Web Course on: DC Microgrid And Control System Prof. Avik Bhattacharya, IIT Roorkee.
2. NPTEL Web Course on Electronics and Distributed Generation Dr. Vinod John Department of Electrical Engineering IISc Bangalore.
3. NPTEL Web Course on Introduction to Smart Grid, PROF. N.P. PADHY Department of Electrical Engineering IIT Roorkee PROF. PREMALATA JENA Department of Electrical Engineering.
4. NPTEL Web Course on Electric vehicles and Renewable energy, Prof. Ashok Jhunjhunwala, Prof**.** Prabhjot Kaur, Prof. Kaushal Kumar Jha and Prof. L Kannan, IIT Madras.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

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23AEE59 POWER ELECTRONIC CONVERTERS FOR ENERGY SOURCES

COURSE OUTCOMES:

After the completion of the course students will be able to

CO1. Select and size various passive and active components for power converters L3

CO2. Design power converters used with DC energy resources with their control implementation L5 CO3. Design power converters used with AC energy resources with their control implementation L5 CO4. Understand the design considerations of power conditioning unit for ESS, SPV and Wind applications. L2

CO5. Understand the design and selection aspects of various auxiliary systems and components used in PCUs L2

UNIT I

Selection of components for Power Electronics Converters (PEC):

Selection and Sizing of capacitors and magnetic components for PECs, design of Magnetic Components; Selection and sizing of Power Devices, Commonly used software tools for selection and sizing; Heatsink-selection and sizing.

UNIT II

Design and Control of DC-DC Converters:

Design of Buck and Boost converters, Design examples; Design of Bidirectional Converters. Design of gate driver circuits; Review of DC-DC converter modelling; Closed loop PI controller design for buck and boost converters; Current control mode and voltage control mode.

UNIT III

Design and Control of DC-AC converters:

Design of Inverter for standalone applications; Design of grid connected Inverter with different grid synchronization strategies- ZCD, PLL; Strategies for Control of voltage, current and power output.

UNIT IV

Design of PCU for SPV and Wind Application:

Various topologies of Power Converter Unit (PCU) for SPV and Wind energy systems. Design considerations of PCU for SPV and Wind energy Systems and Design Examples.

UNIT V

Design of PCU for ESS Applications and Design of Auxiliary System and Interfaces:

Design consideration for BDC converter based PCU for batteries and Ultra-capacitors. Design of current and voltage sensor interfaces; Design considerations for auxiliary power supplies; Design of protection and snubber components: Introduction to Digital Signal Processors (DSP) and microcontroller interfaces

TEXTBOOKS:

1. Power Electronic Converters for Microgrids by Suleiman M. Sharkh, Mohammad A. Abusara, Georgios I. Orfanoudakis Babar Hussain, IEEE and Wiley, 2014.
2. Control Circuits In Power Electronics Practical Issues In Design And Implementation Edited by Miguel Castilla, IET, 2016.
3. Control and Dynamics in Power Systems and Microgrids by Lingling Fan, CRC Press, 2017.
4. Integrated Power Electronic Converters and Digital Control, by Ali Emadi, Alireza Khaligh, Zhong Nie, and Young Joo, Lee 2009, CRC Press.

REFERENCE BOOKS:

1. Cooperative Synchronization in Distributed Microgrid Control by Ali Bidram, Vahidreza Nasirian Ali Davoudi, and Frank L. Lewis, Springer, 2017.
2. Hybrid-Renewable Energy Systems in Microgrids- Integration, Developments and Control edited by Hina Fathimaby et al., Elseiver WoodHead Publishing, 2018.
3. Smart Microgrids- Lessons from Campus Microgrid Design and Implementation edited by Hassan Farhangi, CRC Press 2017.
4. Energy Storage Systems and Components by Alfred Rufer, CRC Press, 2018.
5. Microgrids Design and Implementation edited by Antonio Carlos Zambroni de Souza and Miguel Castilla, Springer, 2019
6. Microgrids Architectures and Control Edited by Nikos Hatziargyriou, IEEE and Wiley, 2014.
7. Energy Storage for Sustainable Microgrid by David Wenzhong Gao, Elsevier, 2015.

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23AEE60 MICROGRID POWER AND CONTROL ARCHITECTURE COURSE OUTCOMES:

After the completion of the course students will be able to

CO1. Understand various types Microgrids based on applications, power and control architecture. L2 CO2. Illustrate various power control strategies adopted in DC, AC and Hybrid Microgrids L3

CO3. Compare and contrast various control architectures used DC, AC and Hybrid Microgrids also various aspects related to stability in Microgrids L4

CO4. Illustrate the various operational challenges in Microgrids L3

CO5. Comprehend the various aspects related to the stability in Microgrids L4

UNIT I: Microgrid Power Architecture

Types of Microgrid system, AC and DC and Hybrids Microgrids, Application based Suitability of Microgrid type; Review of power architecture of various Microgrids deployed world-wide. Comparison of various Microgrid power architectures.

UNIT II: AC Microgrid and Control Architecture

Black-start operation, Grid Synchronization- various Grid synchronization methods, Grid forming and grid following operations; Power Control- Real and reactive power control in AC Microgrid, simple droop control and other variants of droop control, Unit Power Flow Control, Feeder power flow control and Mixed mode control, source optimization; Centralized, decentralized, distributed and hierarchical control architecture, Local and system / supervisory level control strategies, Multi Agent System (MAS) Based Control; Control approaches used in AC Microgrids deployed worldwide. Microgrid standards IEEE 1547 series. Communication in AC Microgrids.

UNIT III: DC Microgrid and Control Architecture

Power sharing in DC Microgrids, source optimization; Control approaches: Centralized, decentralized, distributed and hierarchical control architecture. Control approaches used in hybrid Microgrids. Communication in DC/Hybrid Microgrids.

UNIT IV: Operational Control in Microgrids

Energy management in Microgrids, coordinated control, load management, grid synchronization and islanding, Anti-islanding schemes; Various Architectural and Operational Challenges in Microgrid, Optimal operation of Microgrids.

UNIT V: Microgrid Stability & Protection

Steady-state and dynamic stability in AC and DC Microgrids, Methods to improve the stability in Microgrids; introduction to small signal and large signal stability analysis in Microgrids. Fault scenarios in DC and AC Microgrids, Protection in DC and AC Microgrids, adaptive protection, Fault current source (FCS) based protection; Protection challenges in islanded and autonomous modes of operation and ways to mitigate.

TEXTBOOKS:

1. Microgrids Design and Implementation edited by Antonio Carlos Zambroni de Souza and Miguel Castilla, Springer, 2019
2. Microgrids Architectures and Control Edited by Nikos Hatziargyriou, IEEE and Wiley, 2014
3. Cooperative Synchronization in Distributed Microgrid Control by Ali Bidram, Vahidreza Nasirian Ali Davoudi, and Frank L. Lewis, Springer, 2017.
4. Control Circuits In Power Electronics Practical Issues In Design And Implementation Edited by Miguel Castilla, IET, 2016

REFERENCE BOOKS:

1. Control and Dynamics in Power Systems and Microgrids by Lingling Fan, CRC Press, 2017
2. Hybrid-Renewable Energy Systems in Microgrids- Integration, Developments and Control edited by Hina Fathimaby et al., Elseiver WoodHead Publishing, 2018.
3. Urban DC Microgrid Intelligent Control and Power Flow Optimization by Manuela Sechilariu and Fabrice Locment, 2016 Elsevier
4. Integrated Power Electronic Converters and Digital Control, by Ali Emadi, Alireza Khaligh, Zhong Nie, and Young Joo, Lee 2009, CRC Press.
5. Island Power Systems by Lukas Sigrist, Enrique Lobato, Francisco M. Echavarren Ignacio Egido, and Luis Rouco, CRC Press, 2016.

WEBSITE REFERENCE / VIDEO COURSES:

1. NPTEL Web Course on: DC Microgrid and Control System Prof. Avik Bhattacharya, IIT Roorkee.
2. NPTEL Web Course on Electronics and Distributed Generation Dr. Vinod John Department of Electrical Engineering IISc Bangalore.
3. NPTEL Web Course on Introduction to Smart Grid, PROF. N.P. PADHY Department of Electrical Engineering IIT Roorkee PROF. PREMALATA JENA Department of Electrical Engineering.
4. NPTEL Web Course on Electric vehicles and Renewable energy, Prof. Ashok Jhunjhunwala, Prof. Prabhjot Kaur, Prof. Kaushal Kumar Jha and Prof. L Kannan, IIT Madras.

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23AEE61 MICROGRID SYSTEM DESIGN

COURSE OUTCOMES:

After the completion of the course students will be able to

CO1. Select and size various Microgrid energy resources L3

CO2. Select the power and control architecture of the Microgrid L3

CO3. Select and design the Microgrid's communication architecture. L3

CO4. Illustrate the design aspects DC and AC Microgrids with their control strategies. L4

CO5. Illustrate the implementation of the Microgrid islanding detection and anti-islanding scheme/ black start operation L4

UNIT I

Selection/ Sizing of Microgrid Energy Resources

Factors affecting the selection and sizing of energy resources for Microgrid applications, dependency on type of loads connected, Selection/ Sizing: Renewable energy resources, Energy Storage components. Hybrid combination of RES and ESS.

UNIT II

Selection of Power and Control Architecture:

Factors affecting the selection of Microgrid power and control architecture; Design Consideration for control implementation; Sensors: Selection of sensors and design of sensor Interfaces, design of control Interfaces. Design considerations for DSP/ Microcontroller interfaces.

UNIT III

Selection and Design of Communication Architecture:

Design considerations for selection of communication network for Microgrid applications; Design and implementation of communication links/ interfaces. Microcontroller programming for Data transfer on communication network. Practical design considerations for Communication networks.

UNIT IV

Design of DC and AC Microgrid:

Design DC Power Conditioning Units for RES and ESS, Unidirectional and Bidirectional Converter design, implementation of Control loop with DSP; Programming for Power sharing and Energy Management algorithms; Design of Protection system for DC Microgrid Design AC Power Conditioning Units for RES and ESS, Unidirectional and Bidirectional Converter design, implementation of Control loop with DSP; Grid Synchronization. Programming for Power sharing and Energy Management algorithms; Design of Protection system for AC Microgrid.

UNIT V

Islanding in Microgrids:

Selection and implementation of Islanding detection and anti-islanding scheme; Black- start and Autonomous operations in Microgrids.

TEXTBOOKS:

1. Microgrids Design and Implementation edited by Antonio Carlos Zambroni de Souza and Miguel Castilla, Springer, 2019
2. Microgrids Architectures and Control Edited by Nikos Hatziargyriou, IEEE and Wiley, 2014
3. Power Electronic Converters For Microgrids by Suleiman M. Sharkh, Mohammad A. Abusara, Georgios I. Orfanoudakis Babar Hussain, IEEE and Wiley, 2014

REFERENCE BOOKS:

1. Energy Storage for Sustainable Microgrid by David Wenzhong Gao, Elsevier, 2015
2. Cooperative Synchronization in Distributed Microgrid Control by Ali Bidram, Vahidreza Nasirian Ali Davoudi, and Frank L. Lewis, Springer, 2017.
3. Energy Efficiency and Renewable Energy Handbook Edited by D. Yogi Goswami and Frank Kreith, 2nd Edition- 2016, CRC
4. Control Circuits In Power Electronics Practical Issues In Design And Implementation Edited by Miguel Castilla, IET, 2016.
5. Hybrid-Renewable Energy Systems in Microgrids- Integration, Developments and Control edited by Hina Fathimaby et al., Elseiver WoodHead Publishing, 2018.
6. Urban DC Microgrid Intelligent Control and Power Flow Optimization by Manuela Sechilariu and Fabrice Locment, 2016 Elsevier.
7. Integrated Power Electronic Converters and Digital Control, by Ali Emadi, Alireza Khaligh, Zhong Nie, and Young Joo, Lee 2009, CRC Press.

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23AEE62 ANALYSIS OF SMART GRID SYSTEMS

COURSE OUTCOMES:

After the completion of the course students will be able to

CO1. Understand the analysis and planning of smart grids L2

CO2. Evaluate the tools for modeling and analysis of smart grid L3

CO3. Analyze and synthesize the smart grid operation L4

CO4. Assess the influence of distributed generation in smart grid on power systems L4

UNIT I

Introduction:

Conventional power systems and Smart grid, definition of smart grid, need for smart grid, Smart grid architecture, smart grid domains, enablers of smart grid, Communication architecture and protocols for smart grid, smart grid priority standards and regulation, smart-grid activities in India.

UNIT II

Systems of non linear equations:

Fixed point iteration, Newton Raphson Iteration, Continuation methods, Power system application: power flow, regulating transformer, Fast decoupled load flow, PV curves and continuation power flow, three phase power flow.

UNIT III

Smart Grid Security analysis:

Concept of security, Security analysis and monitoring, factors affecting power system security, detection of network problems, an overview of security analysis. Contingency analysis for generator and line outages by Interactive Linear Power Flow (ILPF) method, Fast decoupled inverse Lemma based approach, network sensitivity factors, Contingency selection, concentric relaxation and bounding.

UNIT IV

Smart Grid Operation and Planning: Economic Dispatch, Optimal Power Flow, Load forecasting, Operation of smart grid system, Load Dispatch Centre functions, preventive, Emergency and Restorative, control objectives of a smart distribution system, Operational bottlenecks in smart grid. Planning Aspects of smart grid, Planning and operation Standards.

UNIT V

Distributed Generation in Smart Grid:

Renewable-based Distributed generations, Energy Storage Technologies, Modeling, Control of energy storage system, Short- mid -long term application of energy storage system in smart grids.

TEXTBOOKS:

1. Muhammad Kamran, Fundamentals of Smart Grid Systems, Academic Press, 2022
2. Mariesa L Crow, Computetional methods for Electric Power Systems, CRC Press, NW, 2016, 3rd Edition.
3. Francisco D´ıaz-Gonz´alez, Andreas Sumper and Oriol Gomis-Bellmunt, Energy Storage in Power Systems, John Wiley & Sons Ltd, 2016

REFERENCE BOOKS:

1. Prabha Kundur, Power System Stability and Control, McGraw Hill Education, 2006, 1st Edition.
2. Math H. J. Bollen, Fainan Hassan, Integration of Distributed Generation in the Power System, Wiley, 2011
3. A. Mahaboob Subahani, G. R. Kanagachidambaresan, M. Kathiresh, Integration of Renewable Energy Sources with Smart Grid, Willey, 2021
4. J. Ekanayake, N. Jenkins, K. Liyanage, J. Wu, A. Yokoyama, Smart Grid: Technology and Applications, John Wiley & Sons, 2015, 1st Edition.
5. Ali Keyhani, Design of smart power grid renewable energy systems, Wiley, 2019, 3rd Edition. Other.

WEBSITE REFERENCE / VIDEO COURSES:

1. https://nptel.ac.in/courses/108/104/108104052/.

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23AEE63 PROJECT IN MICRO GRID TECHNOLOGY

Minors in Mechanical Engineering

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

Minors in Mechanical Engineering (R23) – 3D Printing

III B. Tech I Sem

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23AME61: MATERIAL SCIENCE & ENGINEERING

Course Outcomes:

CO1: Understand crystal structures, defects, strengthening mechanisms, and phase diagrams including the iron–carbon system

CO2: Classify and evaluate the microstructure, properties, and applications of steels and cast irons.

CO3: Analyze various heat treatment processes and their effects on the microstructure and properties of steels.

CO4: identify the microstructure, properties, and industrial uses of important non-ferrous metals and their alloys.

CO5: Understand the structure, properties, and applications of ceramics, polymers, composites, superalloys, and nanomaterials.

UNIT – I

Structure of Metals & Constitution of Alloys

Structure of Metals: Crystal Structures: Unit cells, Metallic crystal structures, Imperfection in solids: Point, Line, interstitial and volume defects; dislocation strengthening mechanisms and slip systems, critically resolved shear stress.

Constitution of Alloys: Necessity of Alloying, substitutional and interstitial solid solutions- Phase diagrams: Interpretation of binary phase diagrams and microstructure development; eutectic, peritectic, peritectoid and monotectic reactions. Iron-Iron-carbide diagram and microstructural aspects of ferrite, cementite, austenite, ledeburite, and cast iron.

UNIT – II

Steels & Cast irons

Steels: Plain carbon steels, use and limitations of plain carbon steels. AISI& BIS classification of steels. Classification of alloys steels. Microstructure, properties and applications of alloy steels- stainless steels and tool steels.

Cast irons: Microstructure, properties and applications of white cast iron, malleable cast iron, grey cast iron, nodular cast iron and alloy cast irons.

UNIT-III

Heat Treatment of Steels: Annealing, tempering, normalizing and hardening, isothermal transformation diagrams for Fe-Fe3C alloys and microstructure development. Continuous cooling curves and interpretation of final microstructures and properties- austempering, martempering, case hardening - carburizing, nitriding, cyaniding, carbo-nitriding, flame and induction hardening, and vacuum and plasma hardening.

UNIT-IV

Non-ferrous Metals and Alloys: Microstructure, properties and applications of copper, aluminium, titanium, nickel and their alloys. Study of Al-Cu phase diagram.

UNIT-V

Ceramics, Polymers and Composites: Structure, properties and applications of ceramics, polymers and composites. Introduction to super alloys and nanomaterials.

TEXT BOOKS:

1. V.Raghavan, Material Science and Engineering, 5/e, Prentice Hall of India, 2004.
2. R.Balasubramaniam, Callister's Material Science and Engineering, 2/e, Wiley India, 2014.

REFERENCE BOOKS:

1. Y. Lakhtin, Engineering Physical Metallurgy, University Press of the Pacific, 2000.
2. S.H. Avner, Introduction to Physical Metallurgy, 2/e, Tata McGraw- Hill, 1997

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

Minors in Mechanical Engineering (R23)

III B. Tech I Sem

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23AME62: ADDITIVE MANUFACTURING

Course Outcomes:

CO1: Introduce the fundamentals, evolution, classifications, and data formats associated with Additive Manufacturing (AM) systems.
CO2: Familiarize students with CAD modeling, digitization, data processing, and reverse engineering techniques for AM.
CO3: Explain the principles, processes, materials, and applications of solid and liquid-based AM systems.
CO4: Understand the working principles, processes, advantages, and limitations of powder-based AM systems.
CO5: Explore additional AM techniques like 3DP, BPM, and SDM, along with their operational principles and practical applications.

Unit – I

Introduction to Additive Manufacturing (AM) Systems: History and Development of AM, Need of AM, Difference between AM and CNC, Classification of AM Processes: Based on Layering Techniques, Raw Materials and Energy Sources, AM Process Chain, Benefits and Applications of AM, Representation of 3D model in STL format, RP data formats: SLC, CLI, RPI, LEAF, IGES, CT, STEP, HP/GL.

Unit – II

CAD & Reverse Engineering: Basic Concept, Digitization techniques, Model Reconstruction, Data Processing for Additive Manufacturing Technology: CAD model preparation, Part Orientation and support generation, Model Slicing, Tool path Generation, Software's for Additive Manufacturing Technology: MIMICS, MAGICS. Reverse Engineering (RE) –Meaning, Use, RE – The Generic Process, Phase of RE Scanning, Contact Scanners, Noncontact Scanners, Point Processing, Application Geometric Model, Development.

UNIT-III

Liquid Based AM Systems: Stereolithography (SLA): Principle, Process, Materials, Advantages, Limitations and Applications. Solid Ground Curing (SGC): Principle, Process, Materials, Advantages, Limitations, Applications. Fusion Deposition Modeling (FDM): Principle, Process, Materials, Advantages, Limitations, Applications. Laminated Object Manufacturing (LOM): Principle, Process, Materials, Advantages, Limitations, Applications.

UNIT-IV

Powder Based AM Systems: Principle and Process of Selective Laser Sintering (SLS), Advantages, Limitations and Applications of SLS, Principle and Process of Laser Engineered Net Shaping (LENS), Advantages, Limitations and Applications of LENS, Principle and Process of Electron Beam Melting (EBM), Advantages, Limitations and Applications of EBM.

UNIT-V

Other Additive Manufacturing Systems: Three-Dimensional Printing (3DP): Principle, Process, Advantages, Limitations and Applications. Ballistic Particle Manufacturing (BPM): Principle, Process, Advantages, Limitations, Applications. Shape Deposition Manufacturing (SDM): Principle, Process, Advantages, Limitations, Applications.

TEXT BOOKS:

1. Ian Gibson, David W. Rosen, Brent Stucker, "Additive Manufacturing Technologies: Rapid Prototyping to Direct Digital Manufacturing", 1stedition, Springer, 2010.
2. Chua C.K., Leong K.F. and Lim C.S., "Rapid Prototyping: Principles and Applications", 2nd edition, World Scientific Publishers, 2003.
3. Liou W. Liou, Frank W., Liou, "Rapid Prototyping and Engineering Applications: A Tool Box for Prototype Development", CRC Press, 2007

REFERENCE BOOKS:

1. Pham D.T. and Dimov S.S., "Rapid Manufacturing; The Technologies and Application of RPT and Rapid Tooling", Springer, London 2001.
2. Gebhardt A., "Rapid prototyping", Hanser Gardener Publications, 2003

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

Minors in Mechanical Engineering (R23)

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III B. Tech II Sem

23AM63: MATERIALS CHARACTERIZATION TECHNIQUES

Course Outcomes:

CO1: Understand and apply the principles of X-ray diffraction for structural analysis of polycrystalline materials, including crystallite size estimation using Scherrer and WH methods. CO2: Gain knowledge of the construction, working, imaging modes, and applications of Scanning Electron Microscopy (SEM).

CO3: Comprehend the operation and applications of Transmission Electron Microscopy (TEM) and distinguish it from SEM.

CO4: Understand the principles, instrumentation, and analytical applications of key spectroscopy techniques such as UV-Vis, Raman, FTIR, and XPS.

CO5: Analyze electrical and magnetic properties of materials using techniques like DC/AC conductivity, Hall Effect, VSM, and SQUID.

UNIT – I

Structure Analysis by Powder X-Ray Diffraction: Introduction, Bragg's law of diffraction, Intensity of Diffracted beams —factors affecting Diffraction Intensities - structure of polycrystalline Aggregates, Determination of crystal structure, Crystallite size by Scherrer and WH Methods, Small angle X-ray scattering (SAXS) (in brief).

UNIT – II

Microscopy Technique -1 Scanning Electron Microscopy (Sem): Introduction, Principle, Construction and working principle of Scanning Electron Microscope, Specimen preparation, Different types of modes used (Secondary Electron and Backscatter Electron), Advantages, limitations and applications of SEM.

UNIT-III

Microscopy Technique -2 - Transmission Electron Microscopy (Tem): Construction and Working principle, Resolving power and Magnification, Bright and dark fields, Diffraction and image formation, Specimen preparation, Selected Area Diffraction, Applications of Transmission Electron Microscopy, Difference between SEM and TEM, Advantages and Limitations of Transmission Electron Microscopy.

UNIT-IV

Spectroscopy Techniques: Principle, Experimental arrangement, Analysis and Advantages of the spectroscopic techniques — (i) UV-Visible spectroscopy (ii) Raman Spectroscopy, (iii) Fourier Transform infrared (FTIR) spectroscopy, (iv) X-ray photoelectron spectroscopy (XPS).

UNIT-V

Electrical & Magnetic Characterization Techniques: Electrical Properties analysis techniques (DC conductivity, AC conductivity) Activation Energy, Effect of Magnetic field on the electrical properties (Hall Effect). Magnetization measurement by induction method, Vibrating sample Magnetometer (VSM) and SQUID (Superconducting Quantum Interference Device).

TEXT BOOKS:

1. Material Characterization: Introduction to Microscopic and Spectroscopic Methods — Yang Leng — John Wiley & Sons (Asia) Pvt. Ltd. 2008.
2. Micro structural Characterization of Materials - David Brandon, Wayne D Kalpan, John Wiley & Sons Ltd., 2008.

REFERENCE BOOKS:

1. Fundamentals of Molecular Spectroscopy -IV Ed. -Colin Neville Banwell and Elaine M. McCash, Tata McGraw-Hill, 2008.
2. Elements of X-ray diffraction - Bernard Dennis Cullity & Stuart R Stocks, Prentice Hall, 2001 — Science

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III B. Tech II Sem

23AME64: 3D PRINTING MATERIALS & APPLICATIONS

Course Outcomes:

CO1: Understand the fundamentals of 3D printing and differentiate between traditional prototyping and rapid prototyping technologies.

CO2: Gain knowledge of solid and liquid-based RP systems such as FDM, LOM, SLA, DLP, and SGC, including their working principles, materials, and applications.

CO3: Comprehend the working, materials, and applications of powder-based and other RP systems like SLS, DMLS, LENS, EBM, 3DP, BPM, and SDM.

CO4: Understand the principles and processes involved in rapid tooling and reverse engineering, including scanning techniques and tool classification.

CO5: Identify common 3D printing errors, understand STL file handling and RP software tools, and explore various applications of RP in design, engineering, and medicine.

UNIT – I

Introduction to 3D Printing: Introduction to Prototyping, Traditional Prototyping Vs. Rapid Prototyping (RP), Need for time compression in product development, Usage of RP parts, Generic RP process, Distinction between RP and CNC, other related technologies, Classification of RP.

UNIT – II

Solid and Liquid Based RP Systems: Working Principle, Materials, Advantages, Limitations and Applications of Fusion Deposition Modelling (FDM), Laminated Object Manufacturing (LOM), Stereo lithography (SLA), Direct Light Projection System (DLP) and Solid Ground Curing (SGC)..

UNIT-III

Powder Based & Other RP Systems: Powder Based RP Systems: Working Principle, Materials, Advantages, Limitations and Applications of Selective Laser Sintering (SLS), Direct Metal Laser Sintering (DMLS), Laser Engineered Net Shaping (LENS) and Electron Beam Melting (EBM).

Other RP Systems: Working Principle, Materials, Advantages, Limitations and Applications of ThreeDimensional Printing (3DP), Ballastic Particle Manufacturing (BPM) and Shape Deposition Manufacturing (SDM).

UNIT-IV

Rapid Tooling & Reverse Engineering: Rapid Tooling: Conventional Tooling Vs. Rapid Tooling, Classification of Rapid Tooling, Direct and Indirect Tooling Methods, Soft and Hard Tooling methods. Reverse Engineering (RE): Meaning, Use, RE – The Generic Process, Phases of RE Scanning, Contact Scanners and Noncontact Scanners, Point Processing, Application Geometric Model, Development.

UNIT-V

Errors in 3D Printing and Applications: Pre-processing, processing and post-processing errors, Part building errors in SLA, SLS, etc. Software: Need for software, MIMICS, Magics, SurgiGuide, 3- matic, 3D-Doctor, Simplant, Velocity2, VoXim, Solid View, 3DView, etc., software, Preparation of CAD models, Problems with STL files, STL file manipulation, RP data formats: SLC, CLI, RPI, LEAF, IGES, HP/GL, CT, STEP. Applications: Design, Engineering Analysis and planning applications, Rapid Tooling, Reverse Engineering, Medical Applications of RP.

TEXT BOOKS:

1. Chee Kai Chua and Kah Fai Leong, “3D Printing and Additive Manufacturing Principles and Applications” 5/e, World Scientific Publications, 2017.
2. Ian Gibson, David W Rosen, Brent Stucker, “Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing”, Springer, 2/e, 2010.

REFERENCE BOOKS:

1. Frank W. Liou, “Rapid Prototyping & Engineering Applications”, CRC Press, Taylor & Francis Group, 2011.
2. Rafiq Noorani, “Rapid Prototyping: Principles and Applications in Manufacturing”, John Wiley&Sons, 2006.

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III B. Tech II Sem
23AME25: CAD/CAM

Course Outcomes:

CO1: Understand the product cycle and apply CAD/CAM tools and computer graphics concepts for industrial design and manufacturing.
CO2: Apply various geometric modeling techniques including wireframe, surface, and solid modeling for accurate representation of engineering components.
CO3: Comprehend the principles of NC/CNC systems and develop CNC part programs using manual and computer-assisted methods.
CO4: Analyze group technology concepts, flexible manufacturing systems, and computer-aided quality control methods for improving production efficiency and quality.
CO5: Understand computer-aided process planning, production control, and modern trends like lean, sustainable, and reconfigurable manufacturing systems.

UNIT – I

Overview of CAD/CAM: Product cycle, CAD, CAM and CIM. CAD Tools, CAM Tools, Utilization in an Industrial Environment, Evaluation criteria. CAD data structure, Data base management systems.
Computer Graphics: Co-ordinate systems, Graphics package functions, 2D and 3D transformations, clipping, hidden line / surface removal color, shading.

UNIT – II

Geometric Modeling: Representation techniques, Parametric and non-parametric representation, various construction methods, wire frame modeling, synthetic curves and their representations, surface modeling, synthetics surfaces and their representations. Solid modeling, solid representation, fundamentals, introduction to boundary representations, constructive solid geometry representations.

UNIT-III

Numerical Control: NC, NC Modes, NC Elements, NC Machine tools and their structure, Machining center, types and features. Controls in NC, CNC systems, DNC systems. Adaptive control machining systems, types of adaptive control.
CNC Part Programming: Fundamentals, NC word, NC Nodes, canned cycles, cutter radius compensation, length compensation, computed assisted part programming using APT: Geometry statements, motion statements, post process statements, auxiliary statements, macro statement program for simple components

UNIT-IV

Group Technology & FMS: Part Family, Classification and Coding, advantages & limitations, Group technology machine cells, benefits. FMS: Introduction, components of FMS, material handling systems, Computer control systems, advantages.
Computer Aided Quality Control: Terminology in Quality control, Inspection and testing, Contact inspection methods - optical and non-optical, integration of CAQC with CAD and CIM.

UNIT-V

Computer Aided Processes Planning: Retrieval type and Generative type, benefits Machinability data systems, Computer generated time standards.
Computer integrated production planning: Capacity planning, shop floor control, MRP-I, MRP- II, CIMS benefits. Trends in manufacturing systems: Concepts of Reconfigurable manufacturing, Sustainable manufacturing and lean manufacturing.

TEXT BOOKS:

1. Mikell P. Groover, Emory W. Zimmers , CAD/CAM, 5/e, Pearson Prentice Hall of India, Delhi, 2008.
2. Ibrahim Zeid, R.Siva Subramanian, CAD/CAM: Theory and Practice, 2/e, Tata McGraw-Hill, Delhi, 2009.

REFERENCE BOOKS:

1. P. N. Rao, CAD/CAM: Principles and applications, 3/e, Tata McGraw-Hill, Delhi, 2017.
2. P. Radhakrishnan, S. Subramanyan& V. Raju, CAD/CAM/CIM, 3/e, New Age International Publishers, 2008.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)
III B. Tech II Sem

23AME06: COMPUTER AIDED MACHINE DRAWING

L	T	P	C
0	0	3	1.5

Course Outcomes:

CO1: Create 2D drawings of mechanical joints, keys, and couplings using conventional representations in CAD software.
CO2Develop 3D solid models of machine components and generate accurate sectional views.
CO3: Assemble complex mechanical systems using 3D modeling tools and produce detailed assembly drawings.
CO4: Interpret and apply limits, fits, and geometric tolerances in manufacturing drawings. CO5: Utilize CAD tools to transition from conceptual design to manufacturing-ready technical documentation.

List of Experiments:

I. The following contents are to be done by any 2D software package Conventional representation of materials and components:

Detachable joints: Drawing of thread profiles, hexagonal and square-headed bolts and nuts, bolted joint with washer and locknut, stud joint, screw joint and foundation bolts.
Riveted joints: Drawing of rivet, lap joint, butt joint with single strap, single riveted, double riveted double strap joints.
Welded joints: Lap joint and T joint with fillet, butt joint with conventions.
Keys: Taper key, sunk taper key, round key, saddle key, feather key, woodruff key.
Couplings: rigid – Muff, flange; flexible – bushed pin-type flange coupling, universal coupling, Oldhams' coupling.

II. The following contents to be done by any 3D software package

Sectional views
Creating solid models of complex machine parts and create sectional views.
Assembly drawings: (Any four of the following using solid model software)
Lathe tool post, tool head of shaping machine, tail stock, machine vice, gate valve, carburettor, piston, connecting rod, eccentric, screw jack, plumber block, axle bearing, pipe vice, clamping device, Geneva cam, universal coupling,
Manufacturing drawing:
Representation of limits, fits and tolerances for mating parts. Use any four parts of above assembly drawings and prepare manufacturing drawing with dimensional and geometric tolerances.

TEXT BOOKS:

1. K.L. Narayana, P.Kannaiah and K.Venkat Reddy, Production Drawing, New Age International Publishers, 3/e, 2014.
2. Software tools/packages- Auto CAD, Solid works or equivalent.

REFERENCE BOOKS:

1. Cecil Jensen, Jay Helsel and Donald D.Voisinet, Computer Aided Engineering Drawing, Tata McGrawHill, NY, 2000.
2. James Barclay, Brain Griffiths, Engineering Drawing for Manufacture, Kogan Page Science, 2003.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)
IV B. Tech I Sem

23AME38: 3D PRINTING LAB

L	T	P	C
0	0	3	1.5

Course Outcomes:

CO1: Create 2D drawings of mechanical joints, keys, and couplings using conventional representations in CAD software.
CO2: Develop 3D solid models of machine components and generate accurate sectional views.
CO3: Assemble complex mechanical systems using 3D modeling tools and produce detailed assembly drawings.
CO4: Interpret and apply limits, fits, and geometric tolerances in manufacturing drawings.
CO5: Utilize CAD tools to transition from conceptual design to manufacturing-ready technical documentation.

List of Experiments:

Module 1:
Introduction to Prototyping, Working of 3D Printer, Types of 3D printing Machines: Exp 1: Modelling of Engineering component and conversion of STL format.
Exp 2: Slicing of STL file and study of effect of process parameter like layer thickness, orientation, and infill on build time using software. Exercise 1: Component-1
Exercise 2: Component-2

Module 2:
Exp 1: 3D Printing of modelled component by varying layer thickness. Exp 2: 3D Printing of modelled component by varying orientation.
Exp 3: 3D Printing of modelled component by varying infill.

Module 3:
Study on effect of different materials like ABS, PLA, Resin etc, and dimensional accuracy.

Module 4:
Identifying the defects in 3D Printed components.

Module 5
Exp1: Modelling of component using 3D Scanner of real-life object of unknown dimension in reverse engineering.
Exp 2: 3D Printing of above modelled component.

REFERENCE BOOKS:

1. Ian Gibson, David W. Rosen, Brent Stucker, Additive Manufacturing Technologies: Rapid Prototyping to Direct Digital Manufacturing, 1/e, Springer, 2010.
2. Chua C.K., Leong K.F. and Lim C.S., Rapid Prototyping: Principles and Applications, 2/e, World Scientific Publishers, 2003.

Minors in Mechanical Engineering -Industrial Engineering

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

23AME65: PRODUCTION PLANNING & CONTROL

L	T	P	C
3	0	0	3

Course Outcomes:

CO1: Students can able to prepare production planning and control activities.
CO2: Understand the concepts of Work study.
CO3: Analyze the Pre requisite information needed for process planning.
CO4: Understand the Scheduling techniques.
CO5: Understand the Scheduling techniques.

UNIT – I

Introduction: Objectives and benefits of planning and Control-Functions of production Control-Types of production- job- batch and Continuous-Product development and design-Marketing aspect - Functional aspects-Operational Aspect-Durability and dependability aspect aesthetic aspect. Profit consideration- Standardization, Simplification & specialization- Break even analysis-Economics of a new design.

UNIT – II

Workstudy: Method study, basic procedure-Selection-Recording of process - Critical analysis, Development - Implementation - Micro motion and memo motion study – work measurement - Techniques of work measurement - Time study - Production study - Work sampling - Synthesis from standard data - Predetermined motion time standards.

UNIT-III

Product Planning and Process Planning: Product planning-Extending the original product information-Value Analysis-Problems in lack of product planning-Process planning and routing-Pre requisite information needed for process planning- Steps in process planning-Quantity determination in batch production-Machine capacity, balancing- Analysis of process capabilities in a multi-product system.

UNIT-IV

Production Scheduling: Production Control Systems-Loading and scheduling-Master Scheduling-Scheduling rules- Gantt Charts-Perpetual Loading-Basic scheduling problems - Line of balance – Flow production scheduling- Batch production scheduling-Product sequencing – Production Control systems- Periodic batch control-Material requirement planning kanban – Dispatching-Progress reporting and expediting- Manufacturing lead time-Techniques for aligning completion times and due dates.

UNIT-V

Inventory Control and Recent Trends In PPC: Inventory control-Purpose of holding stock-Effect of demand on inventories-Ordering procedures. Two bin system - Ordering cycle system-Determination of Economic order quantity and economic lot size- ABC analysis - Recorder Procedure-Introduction to computer integrated production planning systems- elements of Just in Time Systems-Fundamentals of MRP II and ERP.

TEXTBOOKS:

1. James. B. Dilworth,” Operations management – Design, Planning and Control for manufacturing and services” McGraw Hill International edition 1992.
2. Mart and Telsang, “Industrial Engineering and Production Management”, First edition, S. Chand and Company, 2000.

REFERENCE BOOKS:

1. Chary. S.N., “Theoryand Problems in Production& Operations Management”, Tata McGraw Hill, 1995.
2. Elwood S. Buffa, and Rakesh K. Sarin, “Modern Production / Operations Management”, 8th Edition John Wiley and Sons, 2000.

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(AUTONOMOUS)

L	T	P	C
3	0	0	3

23AMB26: MARKETING MANAGEMENT

Course Outcomes:

CO1: Knowledge of analytical skills in solving marketing related problem.
CO2: Awareness of marketing management process.
CO3: Identify the scope and significance of Marketing in Domain Industry.
CO4: Examine marketing concepts and phenomenon to current business events In the Industry.
CO5: illustrate market research skills for designing innovative marketing strategies for business firms.

UNIT – I
Introduction: Marketing – Definitions - Conceptual frame work – Marketing environment: Internal and External - Marketing interface with other functional areas – Production, Finance, Human Relations Management, Information System. Marketing in global environment – Prospects and Challenges.

UNIT – II
Marketing Strategy: Marketing strategy formulations – Key Drivers of Marketing Strategies - Strategies for Industrial Marketing – Consumer Marketing –– Services marketing – Competitor analysis - Analysis of consumer and industrial markets – Strategic Marketing Mix components.

UNIT-III
Marketing Mix Decisions: Product planning and development – Product life cycle – New product Development and Management – Market Segmentation – Targeting and Positioning – Channel Management – Advertising and sales promotions – Pricing Objectives, Policies and methods.

UNIT-IV
Production Scheduling: Production Control Systems-Loading and scheduling-Master Scheduling-Scheduling rules- Gantt Charts-Perpetual Loading-Basic scheduling problems - Line of balance – Flow production scheduling- Batch production scheduling-Product sequencing – Production Control systems- Periodic batch control-Material requirement planning kanban – Dispatching-Progress reporting and expediting- Manufacturing lead time-Techniques for aligning completion times and due dates.

UNIT-V
Marketing Research & Trends In Marketing: Marketing Information System – Research Process – Concepts and applications: Product – Advertising – Promotion – Consumer Behaviour – Retail research – Customer driven organizations - Cause related marketing - Ethics in marketing –Online marketing trends.

TEXTBOOKS:

1. Philip Kotler and Kevin Lane Keller, Marketing Management, PHI 14/e, 2012.
2. PaulBaines, Chris Fill and Kelly Page, Marketing, Oxford University Press, 2/e,2011.
3. Kotler, Philip (2002) Marketing Management, Millennium Edition. Intl ed. US: Prentice Hall, 2002

REFERENCE BOOKS:

1. KS Chandrasekar, "Marketing management-Text and Cases", Tata McGraw Hill, First edition,2010.
2. Philip Kotler and Kevin Lane Keller, Marketing Management, PHI 14th Edition, 2012.
3. Lamb, hair, Sharma, Mc Daniel– Marketing – An Innovative approach to learning and teaching-A south Asian perspective, Cengage Learning -2012

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

L	T	P	C
3	0	0	3

23AME57: SUPPLY CHAIN MANAGEMENT

Course Outcomes:

CO1: Understand the strategic role of logistic and supply chain management in the cost reduction and offering best service to the customer.
CO2: Understand Advantages of SCM in business.
CO3: Apply the knowledge of supply chain Analysis
CO4: Analyze reengineered business processes for successful SCM implementation.
CO5: Evaluate Recent trend in supply chain management.

UNIT – I

Introduction to Supply Chain Management: Supply chain - objectives - importance - decision phases - process view -competitive and supply chain strategies - achieving strategic fit – supply chain drivers - obstacles – framework - facilities -inventory-transportation-information-sourcing-pricing.

UNIT – II

Designing the distribution network: Role of distribution - factors influencing distribution - design options - e-business and its impact– distribution networks in practice –networkdesignin thesupplychain - role of network -factors affecting the network design decisions modelling for supply chain. Role of transportation - modes and their performance – transportation infrastructure and policies - design options and their trade-offs tailored transportation.

UNIT-III

Supply Chain Analysis: Sourcing - In-house or Outsource - 3rd and 4th PLs - supplier scoring and assessment, selection - design collaboration - Procurement process - Sourcing planning and analysis. Pricing and revenue management for multiple customers, perishable products, seasonal demand, bulk and spot contracts.

UNIT-IV

Dimensions of Logistics: A macro and micro dimension - logistics interfaces with other areas - approach to analyzing logistics systems - logistics and systems analysis - techniques of logistics system analysis - factors affecting the cost and importance of logistics. Demand Management and Customer Service Outbound to customer logistics systems - Demand Management –Traditional Forecasting - CPFRP - customer service - expected cost of stock outs - channels of distribution.

UNIT-V

Recent Trends in Supply Chain Management-Introduction, New Developments in Supply Chain Management, Outsourcing Supply Chain Operations, Co-Maker ship, The Role of E-Commerce in Supply Chain Management, Green Supply Chain Management, Distribution Resource Planning, World Class Supply Chain Management.

TEXTBOOKS:

1. Sunil Chopra and Peter Meindl, SupplyChain Management – "Strategy, Planning and Operation", 3rd Edition, Pearson/PHI, 2007.
2. SupplyChain Management by Janat Shah Pearson Publication 2008

REFERENCE BOOKS:

1. ALogistic approach to SupplyChain Management – Coyle, Bardi, Longley, Cengage Learning, 1/e.
2. Donald J Bowersox, Dand J Closs, M BixbyColuper, "SupplyChain Logistics Management", 2nd edition, TMH, 2008.
3. Wisner, Keong Leong and Keah-Choon Tan, "Principles of Supply Chain Management A Balanced Approach", Cengage Learning, 1/e.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

L	T	P	C
3	0	0	3

23AMB27: STRATEGIC MANAGEMENT FOR COMPETITIVE ADVANTAGE

Course Outcomes:

CO1: Understand new forms of Strategic Management concepts and their use in business.
CO2: Summarize strategic formulation process.
CO3: Develop analytical skills to solve cases and to provide strategic solutions.
CO4: Understand Strategic Evaluation, Monitoring and Control.
CO5: Create awareness on corporate Governance and Ethical Issues.

UNIT – I

Basic concepts: Basic Concepts of Strategic Management, Strategic Management Process, Vision, Mission and Goals, Benefits and Risks of Strategic Management, Levels of Strategies Concepts of corporate strategy, Corporate, Business and Operational Level Strategy, Functional Strategies: Human Resource Strategy, Marketing Strategy, Financial Strategy, Operational Strategy Business Environment: Components of Environment Micro and Macro and Environmental Scanning. Competitive Analysis - Competition and Competitor Analysis, Porter Five Forces Model Internal Corporate analysis, Sustainability, Value Chain Analysis.

UNIT – II

Strategic Formulation: Strategic Choices and Importance, Formulation of Alternative Strategies: Generic Strategies, Grand Strategy, Diversification Mergers, Acquisitions, Takeovers, Joint Ventures, Diversification, Turnaround, Divestment and Liquidation. Strategic Analysis and Choice: Issues and Structures, Corporate Portfolio Analysis-SWOT Analysis, BCG Matrix, GE Nine Cell Matrix, Hofer's Matrix, ETOP-Environmental Threat and Opportunity Profile, Strategic Choice-Factors and Importance.

UNIT-III

Strategic Implementation:

Strategy Implementation - Strategy and Structure Steps, Importance and Problems, Resource Allocation-Importance & Challenges Strategic 7S Framework; Management of Change Strategy Implementation - Organizational culture and Leadership; Functional Strategies.

UNIT-IV

Strategic Evaluation, Monitoring and Control - Strategic Controls; Balanced Scorecard; Strategy map Evaluation and Control: Importance, Limitations and Techniques Budgetary Control: Advantages, Limitations. Corporate Restructuring Strategies: Concept, Need and Forms, Corporate Renewal Strategies: Concept, Internal and External factors and Causes.

UNIT-V

Corporate Governance and Ethical Issues, Corporate Social Responsibility and sustainability, Strategic Enablers: Innovation and Entrepreneurship, Knowledge Management, Technology Management. Public Private Participation: Importance, Problems and Governing Strategies of PPP Model. Information technology Driven Strategies: Importance, Limitations and contribution of IT sector in Indian Business. Start-up Business Strategies and Make in India Model: Process of business startups and its Challenges, Growth Prospects and government initiatives in Make in India Model with reference to National manufacturing, Contribution of Make in India Policy in overcoming industrial sickness.

TEXTBOOKS:

1. Strategic Management, A Dynamic Perspective-Concepts and Cases– Mason A. Carpenter, Wm. Gerard Sanders, Prashant Salwan, Published by Dorling Kindersley (India) Pvt Ltd, Licensees of Pearson Education in south Asia.
2. Strategic Management and Competitive Advantage-Concepts, Jay B. Barney, William S. Hesterly, Published by PHI Learning Private Limited, New Delhi.
3. Strategic Management Formulation, Implementation and Control, Pearce & Robinson, McGraw-Hill Publications

REFERENCE BOOKS:

1. Crafting and Executing Strategy – The Quest for Competitive Advantage, Thomson & Strickland, McGraw-Hill Publications, 21st edition.
2. Exploring Strategy – Text and Cases, Gerry Johnson, Richard Whittington, Kevan Scholes, Duncan Angwin, and Patrick Regner, Pearson, 10th edition.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

L	T	P	C
3	0	0	3

23AME66: SIX SIGMA & LEAN MANUFACTURING

Course Outcomes:

CO1: Summarize various techniques that are related to the six-sigma and lean manufacturing.
CO2: Outline the concepts of cellular manufacturing, JIT and TPM. **CO3:** Illustrate the principles and implementation of 5S techniques. **CO4:** Discuss procedure and principles of value stream mapping.
CO5: Determine the reliability function using six-sigma.

UNIT – I

Introduction to Six-Sigma: Probabilistic models-Six Sigma measures-Yield-DPMO-Quality Level-Reliability function using Six¬ Sigma- MTTF using Six Sigma-Maintenance free operating period- Availability using Six- Sigma ¬Point availability- Achieved availability-Operational Availability-Examples.

UNIT – II

The Elements of Six Sigma and their Determination: The Quality Measurement Techniques: SQC, Six Sigma, Cp and Cpk- The Statistical quality control (SQC) methods-The relationship of control charts and six sigma- The process capability index (Cp)¬Six sigma approach-Six sigma and the 1.5 a shift-The Cpk Approach Versus Six Sigma-Cpk and process average shift- Negative Cpk-Choosing six sigma or Cpk-Setting the process capability index-Examples.

UNIT-III

Introduction to Lean Manufacturing: Conventional Manufacturing versus Lean Manufacturing — Principles of Lean Manufacturing —Basic elements of lean manufacturing — Introduction to LM Tools.

UNIT-IV

Cellular Manufacturing, JIT, TPM: Cellular Manufacturing — Types of Layout, Principles of Cell layout, Implementation. JIT —Principles of JIT and Implementation of Kanban. TPM — Pillars of TPM, Principles and implementation of TPM.

UNIT-V

Set Up Time Reduction, TQM, 5S, VSM 10: Set up time reduction — Definition, philosophies and reduction approaches. TQM Principles and implementation. 5S Principles and implementation - Value stream mapping procedure and principles.

TEXT BOOKS:

1. U Dinesh Kumar, Crocker, Chitra and Harithe Saranga, Reliabilityand Six Sigma, Springer Publishers.
2. Sung H. Park, Six Sigma for Qualityand Productivity Promotion, Asian Productivity Organization.

REFERENCE BOOKS:

1. SammyG. Shina, Six Sigma for Electronics Designand Manufacturing, McGraw-Hill.
2. Designand Analysis of Lean Production Systems, Ronald G. Askin & Jeffrey B. Goldberg, John Wiley& Sons, 2003.
3. Mikell P. Groover (2002) Automation, Production Systems and CIM

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

Course Code: 23AEC63

EMBEDDED SYSTEMS TECHNOLOGY
(Minor in Embedded Systems and IOT)

L	T	P	C
3	0	0	3

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Understand the basics of embedded systems, including their history, classification, and processor selection.

CO2: Analyze different embedded processor architectures, including ARM, RISC, and application-specific processors.

CO3: Evaluate various communication interfaces and protocols, such as UART, USB, SPI, I2C, and Zigbee.

CO4: Implement rapid prototyping techniques using Arduino, sensors, and wearable system modules.

CO5: Develop and interface embedded GUI systems, including LCDs, touchscreens, and VGA cameras

UNIT-I:

Introduction to Embedded system: Introduction to Embedded Systems, Embedded Systems Vs General Computing Systems, History of Embedded Systems, Classification, Major Application Areas, Embedded Processor Requirements, Features, Types, RISC Processors, Harvard Architecture, Super Harvard Architecture, Selection of Processors &Microcontrollers.

UNIT-II:

Architecture of Embedded System Processor: Embedded processor models, ARM core processor, Application specific processor like network processors, multimedia processors, industrial processors, superscalar processor, Advanced RISC processors.Architecture of Embedded OS, Categories of Embedded OS, Application Software, Communication Software, Development and Testing Tools

UNIT-III:

Communication Interfaces: Need for Communication Interfaces, OSI Reference Model, Basic of Networks, Network Topology, RS232/UART, RS422/RS485, USB, Infrared,

Ethernet, IEEE 802.11,Bluetooth, SPI, I2C, CAN, Wifi, FlexRay, LIN Bus, Zigbee.

UNIT-IV:

Rapid prototyping: Arduino platform, hardware and software, Sensor's modules, Robo Control modules, 3D printing module, ADC module, wearable systems. etc.
real-world applications.

UNIT-V:

Embedded GUI interfacing: Arduino based graphic LCD, Touch screen, joy stick, VGA camera interfacing and programming in Python. Creative applications of Arduino

Design Examples & Case Studies of Embedded System: Digital Thermometer, Navigation Systems, Smart Card, RF Tag.

Text Books :

1. David Simon, "An embedded Software Primer" PearsonPublication,2021.

2. Frank Vahid, "Embedded system — A unified Hardware Software Introduction" John Wiley and Sons,2005.

Reference Books :

1. Tammy Noergaard," Embedded SystemArchitecture", Elsevier publication, 2014.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

Course Code: 23AEC64

REAL TIME EMBEDDED SYSTEMS DESIGN AND ANALYSIS
(Minor in Embedded Systems and IOT)

L	T	P	C
3	0	0	3

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Explain the architecture of RTOS, task scheduling, and synchronization mechanisms such as semaphores, mutexes, and message queues.

CO2: Demonstrate proficiency in real-time embedded software development using Linux, RT Linux, and various RTOS platforms.

CO3: Apply real-time scheduling algorithms to optimize task execution and system performance.

CO4: Analyze real-time embedded hardware architectures and software stacks for efficient system integration.

CO5: Implement real-time communication protocols and perform validation and verification of embedded systems using case studies.

UNIT-I:

Introduction to RTOS: Overview Of RTOS, Architecture of Kernel, Task & Task Scheduler, ISR, Semaphore, Mutex, Mailbox, Message Queues, Event Registers, Pipes, Signals, Timers, Memory Management, Priority Inversion Problem.

UNIT-II:

Real Time Embedded Software: Linux, RT Linux, multiprocessor software developments, data flow graph, Study and programming of RTOS like RTX51, Free RTOS etc. timing diagram analysis for fixed and dynamic priority software services.

UNIT-III:

Real time Scheduling: Scheduling Real-Time Tasks: Types of Schedulers Table-driven scheduling Cyclic schedulers EDF RMA, Priority Pre-emptive Scheduler State Machine for Linux and VxWorks, Comparison of Cyclic Executive, Introduction to Worst Case Analysis, Example of scheduling, Real- Time Scheduling and Rate Monotonic Least Upper Bound.

UNIT-IV:

Overview of Real-time Hardware Architectures and Software Stacks: Embedded Linux on the Raspberry Pi ARM A-Series System-on-Chip processors, Tracing Linux kernel and network stack events. Best Practices for RTES Programming, System Integration Testing (Hardware, Firmware, and Software).

UNIT-V:

Real Time Communication: RT Services Communication and Synchronization, Performance of two Real-Time communication Protocols, Real time communication over network, Real Time database.

Verification and Validation of RTES project: Using Point-to-point Serial and TCP/IP for Embedded Systems, Case Study of Coding for Sending Application Layer Byte Streams on A TCP/IP Network Using RTOS. Building a simple Linux multi-service system using POSIX real-time extensions on Raspberry Pi 3b using sequencing and methods to log and verify agreement between theory and practice.

Text Books :

1. “Embedded Real time systems” ,Prasad Dream tech Wiley Publication,2003.
2. "Real-Time Systems: Theory and Practice, "Raji b Mall Pearson,2008.

Reference Books :

1. “Real-Time Systems Design and Analysis” ,Philip Laplante,2ndEdition,PrenticeHall, 2013.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

Course Code: 23AEC65

PRINCIPLES OF IOT
(Minor in Embedded Systems and IOT)

L	T	P	C
3	0	0	3

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Understand the fundamentals of IoT, including its architecture, characteristics, and challenges.
CO2: Analyze IoT communication protocols and networking techniques used in wireless sensor networks and M2M communication.
CO3: Design and manage IoT platforms, including device integration, service models, and application development.
CO4: Implement IoT networking and computing techniques, including cloud storage, APIs, and Python-based IoT programming.
CO5: Develop and evaluate IoT applications in real-world domains such as smart cities, connected vehicles, and industrial automation.

UNIT-I:

Introduction & Basic of IoT : Definition, Characteristics, Physical and Logical Designs, challenges, Technological trends in IOT, IoT Examples, M2M.

UNIT-II:

IoT: Components, Communication and Networking: Introduction to Sensing and Networking: Sensing & actuation, Wireless Senor network, Senor nodes, Communication Protocols,M2M Communication, Networking Hardware, Networking Protocols.

UNIT-III:

IoT System Management: Network Operator Requirements, IoT Platform Design Specification – Requirements, Process, Domain Model, Service, IoT Level, Function, Operational view, Device and Component Integration, Application development.

UNIT-IV:

Networking and Computing: File Handling, Python Packages for IoT, IoT Physical Servers – Cloud Storage Models, Communication APIs.

UNIT-V:

IoT Clouds and Data Analytics: REST ful Web API, Amazon Web Services for IoT, Apache Hadoop, Batch Data Analysis, Chef, Chef Case Studies, Puppet, NETCONF-YANG

IoT Applications: Case studies: smart cities, smart home, connected vehicles, Industrial IOT.

Text Books :

1. Kamal, R., "Internet of Things – Architecture and Design Principles," 1st Edition, Mcgraw Hill,2017.

2. Simone Cirani,"Internet of Things-Architectures, Protocols and Standards", WILEY,2018.

Reference Books :

1. Alessandro Bassi," Enabling Things to Talk- Designing IoT solutions with the IoT Architectural Reference Model", Springer,2013.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

Course Code: 23AEC48

WIRELESS NETWORKS
(Minor in Embedded Systems and IOT)

L	T	P	C
3	0	0	3

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Learn the fundamental concepts and architecture of wireless sensor networks.

CO2: Explore various network architectures, optimization techniques, and design principles for wireless sensor networks.

CO3: Gain knowledge of MAC protocols, routing techniques, and addressing mechanisms for efficient sensor network communication.

CO4: Understand the infrastructure establishment of sensor networks, including topology control and synchronization.

CO5: Grasp the knowledge on sensor network platforms, programming challenges, and simulation tools.

UNIT- I:

Overview of Wireless Sensor Networks: Single-Node Architecture - Hardware Components- Network Characteristics- unique constraints and challenges, Enabling Technologies for Wireless Sensor Networks- Types of wireless sensor networks.

UNIT- II:

Architectures: Network Architecture- Sensor Networks-Scenarios- Design Principle, Physical Layer and Transceiver Design Considerations, Optimization Goals and Figures of Merit, Gateway Concepts.

UNIT –III:

Networking Sensors: MAC Protocols for Wireless Sensor Networks, Low Duty Cycle Protocols and Wakeup Concepts - SMAC, - B-MAC Protocol, IEEE 802.15.4 standard and ZigBee, the Mediation Device Protocol, Wakeup Radio Concepts, Address and Name Management, Assignment of MAC Addresses, Routing Protocols Energy-Efficient Routing, Geographic Routing.

UNIT- IV:

Infrastructure Establishment: Topology Control, Clustering, Time Synchronization, Localization and Positioning, Sensor Tasking and Control.

UNIT- V:

Sensor Network Platforms and Tools: Sensor Node Hardware – Berkeley Motes, Programming Challenges, Node-level software platforms, Node level Simulators, State- centric programming

Text books:

1. Holger Karl & Andreas Willig, "Protocols and Architectures for Wireless Sensor Networks", John Wiley, 2005.
2. Feng Zhao & Leonidas J.Guibas, "Wireless Sensor Networks-An Information Processing Approach", Elsevier, 2007

Reference Books:

1. WaltenegusDargie , Christian Poellabauer, "Fundamentals Of Wireless Sensor Networks Theory And Practice", By John Wiley & Sons Publications, 2011
2. KazemSohraby, Daniel Minoli, &TaiebZnati, "Wireless Sensor Networks-Technology, Protocols, and Applications", John Wiley, 2007.
3. Anna Hac, "Wireless Sensor Network Designs", John Wiley, 2003

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SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

Course Code: 23AEC66

PRINCIPLES OF IOT LAB

(Minor in Embedded Systems and IOT)

L	T	P	C
0	0	3	1.5

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Demonstrate the ability to control and monitor sensors and actuators using microcontrollers and IoT platforms.

CO2: Implement IoT-based applications using cloud services like ThingSpeak and mobile applications like Blynk.

CO3: Develop web-based IoT applications using HTTP and MQTT protocols for remote device management.

CO4: Apply IoT principles in real-world applications such as home automation, security systems, and industrial monitoring.

CO5: Integrate UAV/Drone technologies with IoT for automated navigation and data acquisition.

List of Experiments: (Any 10 Experiments are to be conducted)

1. **Serial Monitor, LED, Servo Motor - Controlling**
   - Controlling actuators through Serial Monitor. Creating different led patterns and controlling them using push button switches. Controlling servo motor with the help of joystick.

2. **Distance Measurement of an object**
   - Calculate the distance to an object with the help of an ultrasonic sensor and display it on an LCD.

3. **LDR Sensor, Alarm and temperature, humidity measurement**
   - Controlling relay state based on ambient light levels using LDR sensor.
   - Basic Burglar alarm security system with the help of PIR sensor and buzzer.
   - Displaying humidity and temperature values on LCD

4. Experiments using Raspberry Pi

- Controlling relay state based on input from IR sensors
- Interfacing stepper motor with R-Pi
- Advanced burglar alarm security system with the help of PIR sensor, buzzer and keypad. (Alarm gets disabled if correct keypad password is entered)
- Automated LED light control based on input from PIR (to detect if people are present) and LDR(ambient light level)

5. IOT Framework

- Upload humidity & temperature data to Thing Speak, periodically logging ambient light level to Thing Speak.

6. Controlling LEDs, relay & buzzer using Blynk app

7. HTTP Based

- Introduction to HTTP. Hosting a basic server from the ESP32 to control various digital based actuators (led, buzzer, relay) from a simple web page.

8. Displaying various sensor readings on a simple web page hosted on the ESP32.

9. MQTT Based

- Controlling LEDs/Motors from an Android/Web app, Controlling AC Appliances from an android/web app with the help of relay.

10. Displaying humidity and temperature data on a web-based application

11. UAV/Drone:

- Demonstration of UAV elements, Flight Controller
- Mission Planner flight planning design

12. Python program to read GPS coordinates from Flight Controller

Reference:

1. Adrian McEwen, Hakim Cassimally - Designing the Internet of Things, Wiley Publications, 2012.
2. Alexander Osterwalder, and Yves Pigneur – Business Model Generation – Wiley, 2011
3. Arshdeep Bahga, Vijay Madisetti - Internet of Things: A Hands-On Approach, Universities Press, 2014.
4. The Internet of Things, Enabling technologies and use cases – Pethuru Raj, Anupama C. Raman, CRC Press.

Online Learning Resources/Virtual Labs:

https://www.arduino.cc/ https://www.raspberrypi.org/

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

Course Code: 23AEC35

EMBEDDEDSYSTEMS & IOT APPLICATIONS

(Minor in Embedded Systems and IOT)

L	T	P	C
3	0	0	3

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Understand the Architecture, Development & Design of Embedded Systems and IoT.

CO2: Learn the architecture and programming of ARM Microcontroller.

CO3: Work with Raspberry Pi using Python Programming.

CO4: Know about the loT standards, communication technologies and protocols for IoT devices.

CO5: Implement case studies and applications using the tools and techniques of IoT Platform

UNIT –I:

Introduction to Embedded Systems and Internet of Things (IoT): Introduction, Hardware & Software Architecture of Embedded Systems, Embedded Systems Development process, Architecture of Internet of Things, Physical Design & Logical Design of IoT, IoT Enabling Technologies, IoT Levels & Deployment Tools, Applications of Embedded Systems and IoT, Design Methodology for IOT Products.

UNIT -II:

ARM Microcontrollers Architecture and Programming: Architecture, Pin Diagram, Register Set & Modes, Memory Organization, Instruction set, Programming ports, Timer/Counter, Serial communication, I/O System, Development Tools, interrupts in C, Introduction ARM mBed platform.

UNIT –III:

Fundamentals of Python Programming & Raspberry Pi: Introduction to python programming, Data Types & Data Structures, working with functions, Modules & Packages, File Handling, classes, REST full Web Services, Client Libraries, Introduction & programming Raspberry Pi3, Interfaces, Integrating Input Output devices with Raspberry Pi3. **UNIT –IV:**

IoT Technologies, Standards, Tools & M2M Network: Fundamental characteristics and high-level requirements of IoT, IoT Reference models; Introduction to Communication Technologies & Protocols of IoT: BLE, Wi-Fi, LoRA, 3G/4G Technologies and HTTP, MQTT, CoAP protocols; Relevant

Practicals on above technologies, M2M Network, SDN

(Software Defined Networking) & NFV (Network Function Virtualization) for IoT.

UNIT –V:

IoT Platform, Cloud Computing Platforms & Data Analytics for IoT Development: IOT Platform Architecture (IBM Internet of Things & Watson Platforms); API Endpoints for Platform Services; Devices Creation and Data Transmission; Introduction to NODE-RED and Application deployment, Introduction to Data Analytics, Apache Hadoop, Apache Oozie, Spark & Storm

Text Books:

1. ArsheepBahga, Vijay Madisetti, "Internet of Things: A Hands-On Approach", 1^{st} Edition, VPT, 2014.
2. K.V.K.K.Prasad, "Embedded Real Time Systems: Concepts, Design and Programming", 1^{st} Edition, Dreamtech Publication, 2014.
3. Adrian McEwen, Hakim Cassimally, "Designing the Internet of Things", Wiley Publications, 2013

References:

1. Jonathan W Valvano, "Embedded Microcomputer Systems: Real-Time Interfacing", 3^{rd} Edition, Thomson Engineering, 2012.
2. Olivier Hersent, David Boswarthick, Omar Elloumi, "The Internet of Things: Key applications and Protocols", 2^{nd} Edition, Wiley Publications, 2012.
3. Rene Beuchat , Andrea Guerrieri & Sahand Kashani "Fundamentals of System-on-Chip Design on Arm Cortex-M Microcontrollers" Paperback, 2 August 2021.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

Course Code: 23AEC67

INDUSTRIAL INTERNET OF THINGS LAB
(Minor in Embedded Systems and IOT)

L	T	P	C
0	0	3	1.5

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Explain the architecture, challenges, and fundamental concepts of IIoT and differentiate it from IoT.
CO2: Demonstrate the interfacing of sensors and actuators with microcontrollers like Raspberry Pi and Node MCU for industrial automation.
CO3: Implement communication protocols such as MQTT, ZigBee, and Bluetooth to enable seamless IIoT connectivity.
CO4: Develop web-based dashboards for real-time visualization and control of IIoT devices. **CO5:** Retrieve, analyze, and transmit industrial data using web-based interactions and M2M communication. **CO6:** Implement PLC-based automation, Boolean logic programming, and process control using SCADA for industrial applications.

(All the modules need to be conducted and minimum one project to be done)

MODULE 1: Introduction & Architecture

What is IIoT and connected world? The difference between IoT and IIoT, Architecture of IIoT, IOT node, Challenges of IIOT. Practice

1. Introduction to Arduino, Introduction to raspberry Pi.

https://www.youtube.com/watch?v=AQdLQV6vhbk **MODULE 2:**

IIOT Components

Fundamentals of Control System, introductions, components, closed loop & open loop system.Introduction to Sensors (Description and Working principle): What is sensor? Types of sensors, working principle of basic Sensors -Ultrasonic Sensor, IR sensor, MQ2, Temperature and Humidity Sensors (DHT-11).Digital switch, Electro Mechanical switches.

Practice

1. Measurement of temperature & pressure values of the process using raspberry pi/node mcu.

2. Modules and Sensors Interfacing (IR sensor, Ultrasonic sensors, Soil moisture sensor) using Raspberry pi/node mcu.

3. Modules and Actuators Interfacing (Relay, Motor, Buzzer) using Raspberry pi/node mcu.

MODULE 3: Communication Technologies of IIoT

Communication Protocols: IEEE 802.15.4, Zig Bee, Bluetooth, BLE, NFC, RFID Industry standards communication technology (MQTT), wireless network communication.

Practice

1. Demonstration of MQTT communication.

MODULE 4: Visualization and Data Types of IIoT

Connecting an Arduino/Raspberry pi to the Web: Introduction, setting up the Arduino/Raspberry pi development environment, Options for Internet connectivity with Arduino, Configuring your Arduino/Raspberry pi board for the IoT.

Practice

1. Visualization of diverse sensor data using dashboard (part of IoT's 'control panel')

2. Sending alert message to the user. ways to control and interact with your environment)

MODULE 5: Retrieving Data

Extraction from Web: Grabbing the content from a web page, Sending data on the web, Troubleshooting basic Arduino issues, Types of IoT interaction, Machine to Machine interaction (M2M).

Practice

1. Device control using mobile Apps or through Web pages.

2. Machine to Machine communication.

MODULE 6: Control & Supervisory Level of Automation

Programmable logic controller (PLC), Real-time control system, Supervisory Control & Data Acquisition (SCADA).

Practice

1. Digital logic gates programming using ladder diagram.

2. Implementation of Boolean expression using ladder diagram.

3. Simulation of PLC to understand the process control concept.

Projects:

IIoT based smart ; energy meter ; Smart Agriculture system; Automation using controller via Bluetooth ; Temperature controlled Fan/cooler using controller; Automatic streetlight; Smart Baggage Tracker

Textbooks

1. The Internet of Things in the Industrial Sector, Mahmood, Zaigham (Ed.) (Springer Publication)

2. Industrial Internet of Things: Cyber manufacturing System, Sabina Jeschke, Christian Brecher, Houbing Song, Danda B. Rawat (Springer Publication).

3. Industrial IoT Challenges, Design Principles, Applications, and Security by Ismail Butun (editor)

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

L	T	P	C
0	0	6	3

23AME67: APPLIED PROJECT WORK IN INDUSTRIAL ENGINEERING

ECE MINORS
SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

Course Code: 23AEC10

ANALOG AND DIGITAL COMMUNICATIONS

L	T	P	C
3	0	0	3

Course Outcomes:
After successful completion of the course, the student will be able to:

CO1: Recognize the basic terminology used in analog and digital communication technique for transmission of information/data. (L1)
CO2: Explain the basic operation of different analog and digital communication systems at baseband and pass band level. (L2)
CO3: Compute various parameters of baseband and pass band transmission schemes by applying basic engineering knowledge. (L3)
CO4: Analyze the performance of different modulation & demodulation techniques to solve complex problems in the presence of noise. (L4)
CO5: Evaluate the performance of all analog and digital modulation techniques to know the merits and demerits of each one of them in terms of bandwidth and power efficiency. (L5)

UNIT- I:
Amplitude Modulation: Need for modulation, Amplitude Modulation - Time and frequency domain description, single tone modulation, power relations in AM waves, Generation of AM waves - Switching modulator, Detection of AM Waves - Envelope detector, DSBSC modulation - time and frequency domain description, Generation of
DSBSC Waves - Balanced Modulators, Coherent detection of DSB-SC Modulated waves, COSTAS Loop, SSB modulation - time and frequency domain description, frequency discrimination and Phase discrimination methods for generating SSB, Demodulation of SSB Waves, principle of Vestigial side band modulation.

UNIT- II:
Angle Modulation: Basic concepts of Phase Modulation, Frequency Modulation: Single tone frequency modulation, Spectrum Analysis of Sinusoidal FM Wave using Bessel functions, Narrow band FM, Wide band FM, Constant Average Power, Transmission

bandwidth of FM Wave - Generation of FM Signal- Armstrong Method, Detection of FM Signal: Balanced slope detector, Phase locked loop, Comparison of FM and AM., Concept of Pre-emphasis and de-emphasis

UNIT- III:
Transmitters: Classification of Transmitters, AM Transmitters, FM Transmitters

Receivers: Radio Receiver - Receiver Types - Tuned radio frequency receiver, Super heterodyne receiver, RF section and Characteristics - Frequency changing and tracking, Intermediate frequency, Image frequency, AGC, Amplitude limiting, FM Receiver, Comparison of AM and FM Receivers.

UNIT -IV:
Introduction to Noise: Types of Noise, Receiver Model, Noise in AM, DSB, SSB, and FM Receivers. Pulse Modulation: Types of Pulse modulation- PAM, PWM and PPM. Comparison of FDM and TDM. Pulse Code Modulation: PCM Generation and Reconstruction, Quantization Noise, Non- Uniform Quantization and Companding, Delta Modulation, DPCM, Noise in PCM and DM.

UNIT -V:
Digital Modulation Techniques: Coherent Digital Modulation Schemes – ASK, BPSK, BFSK, QPSK, Non-coherent BFSK, DPSK. M-ary Modulation Techniques, Power Spectra, Bandwidth Efficiency.
Baseband Transmission and Optimal Reception of Digital Signal: A Baseband Signal Receiver, Probability of Error, Optimum Receiver, Coherent Reception, ISI, Eye Diagrams.

Textbooks:
1. Simon Haykin, “Communication Systems”, JohnWiley& Sons, 4th Edition, 2004.

2. Wayne Tomasi - Electronics Communication Systems-Fundamentals through Advanced, 5^{th}Ed., PHI, 2009
3. B. P. Lathi, Zhi Ding " Modern Digital and Analog Communication Systems", Oxford press, 2011.

References:

1. Sam Shanmugam, "Digital and Analog Communication Systems", John Wiley & Sons, 1999.
2. Bernard Sklar, F. J. harris "Digial Communications: Fundamentals and Applications", Pearson Publications, 2020.
3. Taub and Schilling, "Principles of Communication Systems", Tata McGraw Hill, 2007.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

Course Code: 23AEC69

ELECTRONIC INSTRUMENTATION
(Minor in Electronic Systems)

L	T	P	C
3	0	0	3

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Know the working principle, operation, and applications of analog and digital measuring instruments.

CO2: Get familiar with the working principles, construction, and applications of oscilloscopes in signal analysis.

CO3: Learn about the bridge circuits for measurement of resistance, inductance, and capacitance.

CO4: Understand the principles and operation of various signal generators used in electronic applications.

CO5: Gain knowledge about the operation and applications of transducers in measurement systems

UNIT- I:

Measuring Instruments: Introduction, Errors in Measurement, Accuracy, Precision, Resolution and Significant figures. Basic PMMC Meter- construction and working, DC and AC Voltmeters- Multirange, Range extension, DC Ammeter, Multimeter for Voltage, Current and resistance measurements.

Digital Instruments: Digital Voltmeters – Introduction, DVM's based on V–T, V–F and Successive approximation principles, Resolution and sensitivity, General specifications, Digital Multimeters, Digital frequency meters, Digital measurement of time.

UNIT- II:

Oscilloscopes: Introduction, Block diagram of CRO, Basic principle of CRT, CRT Construction and features, vertical amplifiers, horizontal deflection system- sweep, trigger pulse, delay line, sync selector circuits. Dual beam and dual trace CROs, Sampling and Digital storage oscilloscopes.

UNIT- III:

Bridges: DC Bridges for Measurement of resistance - Wheat stone bridge, Kelvin's Bridge, AC Bridges for Measurement of inductance- Maxwell's bridge, Hay's Bridge, Anderson bridge, Measurement of capacitance - Schearing Bridge, Wien Bridge, Errors and precautions in using bridges.

UNIT- IV:

Signal Generators: Introduction, Fixed and variable AF oscillator, Standard signal generator, Laboratory type signal generator, AF sine and Square wave generator, Function generator, Square and Pulse generator, Sweep frequency generator.

UNIT- V:

Transducers: Introduction, Types of Transducers, Electrical transducers, Selecting a transducer, Resistive transducer, Strain gauges, Piezoelectric transducer, Photoelectric transducer, Photovoltaic transducer, Temperature transducers-RTD, LVDT.

Text Books:

1. H.S.Kalsi, "Electronic Instrumentation", Third edition, Tata McGraw Hill, 2010.
2. A.D. Helfrick and W.D. Cooper, "Modern Electronic Instrumentation and Measurement Techniques", PHI, 6th Edition, 2010.

Reference Books:

1. A.K. Sawhney, Dhanpat Rai & Co., "A course in Electrical and Electronic Measurements and Instrumentation", 9^{th} Edition, 2010.
2. David A. Bell, "Electronic Instrumentation & Measurements", PHI, 2nd Edition, 2006.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

Course Code: 23AEC16

ANALOG AND DIGITAL IC APPLICATIONS

L	T	P	C
3	0	0	3

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Understand the classification of Integrated Circuits, internal blocks and characteristics of Op- Amp.

CO2: Analyse linear and non-linear applications of Op-Amp.

CO3: Gain knowledge on active filters, timers and phased locked loops.

CO4: Understand the working of Voltage Regulators and Converters.

CO5: Know about different types of Digital ICs and their applications.

UNIT-I: ICs and OP- AMPS

Integrated Circuits and Operational Amplifier: Introduction, Classification of IC's, IC chip size and circuit complexity, basic information of Op-Amp IC741 and its features, the ideal Operational amplifier, Op-Amp internal circuit, Op-Amp characteristics - DC and AC, Features of 741 Op-Amp.

UNIT-II: Applications of OP- AMP

Linear Applications of Op-Amp: Inverting, non-inverting, Differential amplifiers, adder, subtractor, Instrumentation amplifier, AC amplifier, V to I and I to V converters, Integrator and differentiator.

Non-Linear Applications of Op-Amp: Sample and Hold circuit, Log and Antilog amplifier, multiplier and divider, Comparators, Schmitt trigger, Multi vibrators, Triangular and Square waveform generators, Oscillators

UNIT-III: Active Filters and other ICs

Active Filters: Introduction, Butterworth filters – 1st order, 2nd order low pass and high pass filters, band pass, band reject and all pass filters.

Timer and Phase Locked Loops: Introduction to IC 555 timer, description of functional diagram, monostable and a stable operations and applications, Schmitt trigger, PLL - introduction, basic principle, phase detector/comparator, voltage controlled oscillator (IC 566), low pass filter, monolithic PLL and applications of PLL.

UNIT-IV: Voltage Regulators and Converters

Voltage Regulator: Introduction, Series Op-Amp regulator, IC Voltage Regulators, IC 723 general purpose regulators, Switching Regulator.

D to A and A to D Converters: Introduction, basic DAC techniques - weighted resistor DAC, R-2R ladder DAC, inverted R-2R DAC, A to D converters - parallel comparator type ADC, counter type ADC, successive approximation ADC and dual slope ADC, DAC and ADC Specifications.

UNIT-V: Digital ICs

CMOS Logic: CMOS logic levels, MOS transistors, Basic CMOS Inverter, NAND and NOR gates, CMOS AND-OR-INVERT and OR-AND-INVERT gates, implementation of any function using CMOS logic.

Combinational Logic IC's: Specifications and Applications of TTL-74XX & CMOS 40XX Series ICs - Code Converters, Decoders, Encoders, Priority Encoders, Multiplexers, Demultiplexers, Parallel Binary Adder/ Subtractor, Magnitude Comparators.

Sequential Logic IC's: Familiarity with commonly available 74XX & CMOS40XX Series ICs - All Types of Flip-flops, Synchronous Counters, Decade Counters, Shift Registers.

Textbooks:

1. D. Roy Choudhury, Shail B. Jain, "Linear Integrated Circuit", 4th edition (2012), New Age International Pvt.Ltd., New Delhi, India
2. Floyd, Jain, "Digital Fundamentals", 8th edition (2009), Pearson Education, New Delhi.

References:

1. Ramakant A. Gayakwad, "OP-AMP and Linear Integrated Circuits", 4th edition (2012), Prentice Hall / Pearson Education, New Delhi.
2. Sergio Franco (1997), Design with operational amplifiers and analog integrated circuits, McGraw Hill, New Delhi.
3. Gray, Meyer (1995), Analysis and Design of Analog Integrated Circuits, Wiley International, New Delhi

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

Course Code: 23AEC68

PRINCIPLES OF COMMUNICATION ENGINEERING (Minor in Electronic Systems)

L	T	P	C
3	0	0	3

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Understand the fundamentals of communication systems and amplitude modulation techniques.

CO2: Learn about the angle modulation techniques and bandwidth considerations in communication systems.

CO3: Gain knowledge on pulse analog modulation and multiple access techniques used in digital communication systems.

CO4: Get familiar with pulse modulation and digital modulation techniques used in modern communication systems.

CO5: Know about wireless communication systems, cellular networks, and GSM technology.

UNIT- I:

Analog communication-I: Elements of communication systems, need for Modulation, Modulation Methods, Baseband and carrier communication Amplitude Modulation(AM),Generation of AM signals, Rectifier detector, Envelope detector, sideband and carrier power of AM, Double side band suppressed carrier(DSB-SC) modulation & its demodulation, Switching modulators, Ring modulator, Balanced modulator, Single sideband(SSB) transmission, VSB Modulation.

UNIT- II:

Analog communication-II : Angle Modulation & Demodulation: Concept of instantaneous frequency Generalized concept of angle modulation, Bandwidth of angle modulated waves-Narrow band frequency modulation (NBFM); and Wide band FM (WBFM),Phase modulation, Pre-emphasis & De-emphasis, Illustrative Problems.

UNIT-III:

Digital communications-I (Qualitative Approach only): Pulse Analog Modulation Techniques: Pulse analog modulation techniques, Generation and detection of Pulse amplitude modulation, Pulse width modulation, Pulse position modulation

Multiple Access Techniques: Introduction to multiple access techniques, FDMA, TDMA, CDMA, SDMA: Advantages and applications.

UNIT-IV:

Digital communications-II (Qualitative Approach only): Pulse Code Modulation, DPCM, Delta modulation, Adaptive delta modulation, Overview of ASK, PSK, QPSK, BPSK and MPSK techniques.

UNIT-V:

Wireless communications (Qualitative Approach only): Introduction to wireless communication systems, Examples of wireless communication systems, comparison of 2G and 3G cellular networks, Introduction to wireless networks, Differences between wireless and fixed telephone networks, Introduction to Global system for mobile(GSM),GSM services and features

Text Books:

1. H Taub, D. Schilling and GautamSahe, “Principles of Communication Systems”, TMH, 2007, 3rd Edition.
2. George Kennedy and Bernard Davis, “Electronics & Communication System”, 4th Edition, TMH 2009.
3. Wayne Tomasi, “Electronic Communication System: Fundamentals Through Advanced”,2nd edition,PHI,2001.

Reference Books:

1. Simon Haykin, “Principles of Communication Systems”, John Wiley, 2nd Edition.
2. Sham Shanmugam, “Digital and Analog communication Systems”,Wiley-India edition,2006.
3. Theodore. S.Rapport, “Wireless Communications”, Pearson Education,2nd Edition,2002.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

Course Code: 23AEC22

ANALOG & DIGITAL IC APPLICATIONS LAB

L T P C
0 0 3 1.5

Course Outcomes:

After successful completion of the course, the student will be able to: CO1:
Design an Inverting and Non-inverting Amplifier using an Op Amp. **CO2:**
Demonstrate the Linear and Non-Linear Applications using IC 741. **CO3:**
Design Astable and Monostable Multivibrator using timer ICs.
CO4: Analyse the DAC and ADC converter.
CO5: Design Counters and Registers using digital ICs.

List of Experiments: (**At least 8 Linear and 4 Digital IC experiments shall be performed**).

1. Design an Inverting and Non-inverting Amplifier using Op Amp and calculate gain.
2. Design Adder and Subtractor using Op Amp and verify addition and subtraction process.
3. Design a Comparator using Op Amp and draw the comparison results of A=B, A>B, A<B
4. Design a Integrator and Differentiator Circuits using IC741 and derive the required condition practically.
5. Design a Active LPF, HPF cutoff frequency of 2 KHZ and find the roll off of it.
6. Design a Circuit using IC741 to generate sine/square/triangular wave with period of 1KHZ and draw the output waveform.
7. Construct Mono-stable Multivibrator using IC555 and draw its output waveform.
8. Construct AstableMultivibrator using IC555 and draw its output waveform and also find its duty cycle.
9. Design a Schmitt Trigger Circuit and find its LTP and UTP.
10. Design Voltage Regulator usingIC723, IC 7805/7809/7912 and find its load regulation factor.
11. Design R-2R ladder DAC and find its resolution and write a truth table with respective voltages.
12. Design Parallel comparator type/ counter type/ successive approximation ADC and find its efficiency.
13. Design a 8x1 multiplexer using digital ICs.
14. Design a 4-bit Adder/Subtractor using digital ICs
15. Design a Decade counter and verify its truth table and draw respective waveforms.
16. Design a Up/down counter usingIC74163 and draw read/write waveforms.
17. Design a Universal shift register using IC 74194/195 and verify its shifting operation.
18. Design a 8x3 encoder/3x8 decoder and verify its truth table.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

Course Code: 23AEC33
VLSI DESIGN

L	T	P	C
3	0	0	3

Course Outcomes:
After successful completion of the course, the student will be able to:
CO1: Understand the steps involved in fabrication of ICs using MOS transistor technology.
CO2: Learn about the VLSI design processes, Stick diagrams and Layouts.
CO3: Gain knowledge on the Gate Level Design concepts.
CO4: Learn the design of various subsystems with different VLSI Design styles.
CO5: Familiar with CMOS testing techniques.

UNIT- I:
Introduction: Brief Introduction to IC technology MOS, PMOS, NMOS, CMOS & BiCMOS Technologies. Basic Electrical Properties of MOS and BiCMOS Circuits: I_{DS} - V_{DS} relationships, MOS transistor Threshold Voltage, figure of merit, Transconductance, Pass transistor, NMOS Inverter, Various pull ups, CMOS Inverter analysis and design, Bi-CMOS Inverters.

UNIT- II:
VLSI Circuit Design Processes: VLSI Design Flow, MOS Layers, Stick Diagrams, Design Rules and Layout, Lambda(λ)-based design rules for wires, contacts and Transistors, Layout Diagrams for NMOS and CMOS Inverters and Gates, Scaling of MOS circuits, Limitations of Scaling.

UNIT –III:
Gate level Design: Logic gates and other complex gates, Switch logic, Alternate gate circuits. Basic Circuit Concepts: Sheet Resistance Rs and its concepts to MOS, Area Capacitances calculations, Inverter Delays, Driving large Capacitive Loads, Wiring Capacitances, Fan-in and fan-out **UNIT –IV:**
Subsystem Design: Shifters, Adders, ALUs, Multipliers, Parity generators, Comparators, Counters. VLSI Design styles: Full-custom, Standard Cells, Gate-arrays, FPGAs, CPLDs and Design Approach for Full-custom and Semi-custom devices, parameters influencing low power design.

UNIT- V:
CMOS Testing: Need for testing, Design for testability - built in self-test (BIST) – testing combinational logic –testing sequential logic – practical design for test guide lines – scan design techniques.

Textbooks:

1. Essentials of VLSI Circuits and Systems, Kamran Eshraghian, Eshraghian Dougles, A. Pucknell, 2005, PHI.
2. Modern VLSI Design – Wayne Wolf, 3 Ed., 1997, Pearson Education.

References:

1. CMOS VLSI Design-A Circuits and Systems Perspective, Neil H.E Weste, David Harris, Ayan Banerjee, 3rd Edn, Pearson, 2009.
2. BehzadRazavi , “Design of Analog CMOS Integrated Circuits”, McGraw Hill, 2003.
3. Jan M. Rabaey, “Digital Integrated Circuits”, AnanthaChandrakasan and Borivoje Nikolic, Prentice-Hall of India Pvt.Ltd, 2nd edition, 2009.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (AUTONOMOUS)

Course Code: 23AEC40

VLSI DESIGN LAB

L T P C
0 0 3 1.5

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Design a logic circuit using CMOS transistorusing 180 nm technology in terms of schematic, symbol, test bench, DC and AC analysis.

CO2: Evaluate different schematics &output responses for AOI logic by using different software tools.

CO3: Design CMOS circuits using Full & Semi custom IC designsfor analyzation.

CO4: Design different layouts using different software tools for analog circuits.

List of Experiments: (Any TEN of the experiments are to be conducted)

1. **Design and analysis of CMOS Inverter**
   a) Implement CMOS inverter schematic using 180 nm technology and design its symbol.
   b) Implement test bench for CMOS Inverter and check its output response.
   c) Perform DC and AC analysis for CMOS inverter.
   d) Check the performance of CMOS inverter using parametric sweep.
2. **Design and analysis of NAND and NOR Logic gates**
   a) Implement NAND/NOR schematic using 180 nm technology and design its symbol.
   b) Implement test bench for NAND/NOR and check its output response.
   c) Perform DC and AC analysis for NAND/NOR.
   d) Check the performance of NAND/NOR using parametric sweep.
3. **Design and analysis of XOR and XNOR Logic gates**
   a) Implement XOR/XNOR schematic using 180 nm technology and design its symbol.
   b) Implement test bench for XOR/XNOR and check its output response.
   c) Perform DC and AC analysis for XOR/XNOR.
   d) Check the performance of XOR/XNOR using parametric sweep.
4. **Design of AOI logic**
   a) Design Schematic for AB+C‘D and check its output response.
   b) Design Schematic for AB‘+C‘D and check its output response.
   c) Design Schematic for (A+B‘)(C+D) and check its output response.
   d) Design Schematic for (A+B‘)(C‘+D) and check its output response.
5. **Design and analysis of Full adder**
   a) Design full adder using Full custom IC design.
   b) Design full adder using Semi custom IC design.
6. **Analysis of NMOS and PMOS characteristics**
   a) Implement test bench for NMOS/PMOS transistor.
   b) Perform DC and AC analysis for NMOS/PMOS transistor
   c) Check the performance of NMOS/PMOS transistor using parametric sweep.
7. **Design and analysis of Common source amplifier**
   a) Implement CS amplifier schematic using 180 nm technology and design its symbol.
   b) Implement test bench for CS amplifier and check its output response.
   c) Perform DC and AC analysis for CS amplifier.
   d) Check the performance of CS amplifier using parametric sweep.
8. **Design and analysis of Common drain amplifier**
   a) Implement CD amplifier schematic using 180 nm technology and design its symbol.
   b) Implement test bench for CD amplifier and check its output response.
   c) Perform DC and AC analysis for CD amplifier.
   d) Check the performance of CD amplifier using parametric sweep.
9. **Design of MOS differential amplifier**
   a) Design differential amplifier schematic using 180 nm technology and its symbol.

b) Implement test bench for differential amplifier and check its output response.
c) Perform DC and AC analysis for differential amplifier.
d) Check the performance of differential amplifier using parametric sweep.

10. Design of differential amplifier using FET/BJT

a) Design differential amplifier using FET/BJT schematic using 180 nm technology and its symbol.
b) Implement test bench for two stage differential amplifier and check its output response.
c) Perform DC and AC analysis for differential amplifier.
d) Check the performance of differential amplifier using parametric sweep.

11. Design of Inverter Layout

a) Design and implement inverter schematic.
b) Design the layout for inverter using 180 nm tech file.
c) Perform LVS for schematic and layout
d) Check and remove all DRC violations.
e) Extract parasitic R and C in layout.

12. Design of NAND/NOR Layout

a) Design and implement NAND/NOR schematic.
b) Design the layout for inverter using 180 nm tech file.
c) Perform LVS for schematic and layout
d) Check and remove all DRC violations.
e) Extract parasitic R and C in layout

The students are required to design the schematic diagrams using CMOS logic and to draw the layout diagrams to perform the experiments with the Industry standard EDA Tools.

Software Required: i. Mentor Graphics/ Synopsis/ Cadence / Equivalent Industry Standard Software. ii. Personal computer system with necessary software to run the programs and to implement.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

Minors in CSE (IoT) (R23)

23ACO01: Internet of Things

L	T	P	C
3	0	0	3

Course Objectives:

1. Understand the basics of Internet of Things and protocols.
2. Discuss the requirement of IoT technology
3. Introduce some of the application areas where IoT can be applied.
4. Understand the vision of IoT from a global perspective, understand its applications, determine its market perspective using gateways, devices, and data management.

Course Outcomes (COs):

After successful completion of this course, the students will be able to:

CO1: Understand general concepts of Internet of Things CO2:

Apply design concept to IoT solutions

CO3: AnalyzevariousM2MandIoTarchitectures CO4:

Evaluated e sign issues in IoT applications

CO5: Create IoT solutions using sensors, actuators and Devices

UNIT I

Introduction to IoT: Definition and Characteristics of IoT, physical design of IoT, IoT protocols, IoT communication a. Ceaser cipher b. Substitution cipher c. Hill Cipher cation models, IoT Communication APIs, Communication protocols, Embedded Systems, IoT Levels and Templates

UNIT II

Prototyping IoT Objects using Microprocessor / Microcontroller: Working principles of sensors and actuators, setting up the board – Programming for IoT, Reading from Sensors, Communication: communication through Bluetooth, Wi-Fi.

UNIT III

IoT Architecture and Protocols: Architecture Reference Model Introduction, Reference Model and architecture, IoT reference Model, Protocols- 6LowPAN, RPL, CoAP, MQTT, IoT frameworks Thing Speak.

UNIT IV

Device Discovery and Cloud Services for IoT: Device discovery capabilities – Registering a device, Deregister a device, Introduction to Cloud Storage models and communication APIs Web-Server, Web server for IoT.

UNIT V

UAV IoT: Introduction to Unmanned Aerial Vehicles/Drones, Drone Types, Applications: Defense, Civil, Environmental Monitoring; UAV elements and sensors- Arms, motors, Electronic Speed Controller (ESC), GPS, IMU, Ultra sonic sensors; UAV Software –Arudpilot, Mission Planner, Internet of Drones(IoD)- Case study FlytBase.

Textbooks:

1. Vijay Madisetti and Arshdeep Bahga,"Internet of Things (A Hands-on-Approach)",1st Edition, VPT, 2014.

2. Handbook of unmanned aerial vehicles, K Valavanis; George JVachtsevanos, NewYork, Springer, Boston, Massachusetts: Credo Reference, 2014. 2016..

Reference Books:

1. Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stefan Aves and, Stamatis Karnouskos, David Boyle, "From Machine-to-Machine to the Internet of Things :Introduction to a New Age of Intelligence", 1st Edition, Academic Press, 2014.

2. Arshdeep Bahga, Vijay Madisetti-Internet of Things: A Hands-On Approach, Universities Press,2014.

3. The Internet of Things, Enabling technologies and use cases – Pethuru Raj, Anupama C. Raman, CRC Press.

4. Francis daCosta, "Rethinking the Internet of Things: A Scalable Approach to Connecting Everything", 1st Edition, Apress Publications, 2013

5. Cuno Pfister, Getting Started with the Internet of Things, O"Reilly Media, 2011, ISBN: 978- 1-4493-9357-1

6. DGCA RPAS Guidance Manual, Revision 3 – 2020

7. Building Your Own Drones: A Beginners' Guide to Drones, UAVs, and ROVs, John Baichtal

Online Courses & Resources

1. https://www.arduino.cc/
2. https://www.raspberrypi.org/
3. https://nptel.ac.in/courses/106105166/5
4. https://nptel.ac.in/courses/108108098/4

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

Minors in CSE(IoT) (R23)

23ACS25: WIRELESS SENSOR NETWORKS

L	T	P	C
3	0	0	3

Course Objectives:

1. To acquire the knowledge about various architectures and applications of Sensor Networks
2. To understand issues, challenges and emerging technologies for wireless sensor networks
3. To learn about various routing protocols and MAC Protocols
4. To understand various data gathering and data dissemination methods
5. To Study about design principals, node architectures, hardware and software required for implementation of wireless sensor networks.

Course Outcomes (COs):

After successful completion of this course, the students will be able to:

CO1: Analyze and compare various architectures of Wireless Sensor Networks

CO2: Understand Design issues and challenges in wireless sensor networks

CO3: Analyze and compare various data gathering and data dissemination methods.

CO4: Design, Simulate and Compare the performance of various routing and MAC protocol

Unit I

Characteristics Of WSN: Characteristic requirementsfor WSN - Challenges for WSNs – WSN vs Adhoc Networks -Sensor node architecture – Commercially available sensor nodes –Imote, IRIS, MicaMote, EYES nodes, BTnodes, TelosB, Sunspot -Physicallayerandtransceiverdesign considerations in WSNs, Energy usage profile, Choice of modulation scheme,Dynamic modulation scaling, Antenna considerations.

Unit II

Medium Access Control Protocols: Fundamentals of MAC protocols - Low duty cycleprotocols andwakeup concepts - Contention- basedprotocols-Schedule-basedprotocols-SMAC -BMAC -Traffic-adaptivemediumaccess protocol (TRAMA) - The IEEE 802.15.4 MAC protocol.

Unit III

Routing And Data Gathering Protocols Routing Challenges and Design Issues in Wireless Sensor Networks, Flooding and gossiping – Data centric Routing – SPIN – Directed Diffusion – Energy aware routing - Gradient-based routing - Rumor Routing – COUGAR – ACQUIRE – Hierarchical Routing - LEACH, PEGASIS – Location Based Routing – GF, GAF, GEAR, GPSR – Real Time routing Protocols – TEEN, APTEEN, SPEED, RAP - Data aggregation - data aggregation operations -Aggregate Queries in Sensor Networks - Aggregation Techniques – TAG, Tiny DB.

Unit IV

Embedded Operating Systems: Operating Systems for Wireless Sensor Networks – Introduction - Operating System Design Issues - Examples of Operating Systems – TinyOS – Mate – MagnetOS – MANTIS- OSPM - EYES OS – SenOS – EMERALDS – PicOS – Introduction to Tiny OS – NesC – Interfaces and Modules- Configurations and Wiring - Generic Components -Programming in Tiny OS using NesC, Emulator TOSSIM.

Unit V

Applications Of WSN: WSN Applications - Home Control - Building Automation - Industrial Automation - Medical Applications - Reconfigurable Sensor Networks - Highway Monitoring - Military Applications - Civil and Environmental Engineering Applications-WildfireInstrumentation- HabitatMonitoring- NanoscopicSensorApplications –CaseStudy: IEEE802.15.4 LR-WPANs Standard - Target detection and tracking - Contour/edge detection - Field sampling.

TEXT BOOKS

1. Kazem Sohraby, Daniel Minoli and TaiebZnati, "Wireless Sensor Networks Technology, Protocols, and Applications", John Wiley & Sons,2007.

2. Holger Karl and Andreas Willig, "Protocols and Architectures for Wireless Sensor Networks", John Wiley & Sons, Ltd, 2005.

REFERENCE BOOKS

1. K.AkkayaandM.Younis,"Asurveyofroutingprotocolsinwirelesssensornetworks", Elsevier Ad Hoc Network Journal, Vol. 3, no. 3, pp. 325--349

2. Philip Levis,"Tiny OS Programming"

3. Anna Ha´c,"Wireless Sensor Network Designs",John Wiley & Sons Ltd,

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

Minors in CSE(IoT) (R23)

23ACO16: Communication Protocols for IoT

L	T	P	C
3	0	0	3

Course Objectives:

In this course, learners will be going to learn about various protocols designed for the implementation of the Internet of Things (IoT) applications.

Course Outcomes (COs):

After successful completion of this course, the students will be able to:

CO1: Understand fundamentals of IoT architecture outline and standards. CO2:

Understand and analyze different architectural views.

CO3: Understand the importance of IoT Data Link Layer & Network Layer Protocols. CO4:

Understand the importance of Iot Transport & Session Layer Protocols.

UNIT - I

Introduction: IoT architecture outline, standards - IoT Technology Fundamentals- Devices and gateways, Local and wide area networking, Data management, Business processes in IoT, Everything as a Service (XaaS), M2M and IoT Analytics.

Unit - II

Iot Reference Architecture: Introduction, Functional View, Information View, Deployment and Operational View, Other Relevant Architectural views. Real-World Design Constraints- Introduction, Technical Design constraints.

UNIT - III

IoT Data Link Layer: PHY/MAC Layer (3GPP MTC, IEEE 802.11, IEEE 802.15), Wireless HART, ZWave, Bluetooth Low Energy, Zigbee Smart Energy, DASH7

UNIT - IV

Network Layer Protocols: Network Layer-IPv4,IPv6, 6LoWPAN, 6TiSCH,ND, DHCP, ICMP, RPL, CORPL, CARP.

UNIT - V

IOT Transport & Session Layer Protocols: Transport Layer (TCP, MPTCP, UDP, DCCP, SCTP)- (TLS, DTLS) – Session Layer-HTTP, CoAP, XMPP, AMQP, MQTT.

TEXT BOOKS:

1. Daniel Minoli, “Building the Internet of Things with IPv6 and MIPv6: The Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications ,2016
2. Jan Holler, VlasiosTsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle, “From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence”,1st Edition, Academic Press, 2015.

REFERENCE BOOKS:

1. Bernd Scholz-Reiter, Florian Michahelles, “Architecting the Internet of Things”, ISBN 978-3-64219156-5 e-ISBN 978-3-642-19157-2, Springer, 2016.
2. N. Ida, Sensors, Actuators and Their Interfaces, Scitech Publishers, 2014.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

Minors in CSE(IoT) (R23)

L	T	P	C
3	0	0	3

23ACO19: Industrial IoT

Course Objectives:

1. Acquire theoretical knowledge on Industrial Internet of Things.
2. Apply suitable machine learning techniques for data handling and to gain knowledge from it.
3. Evaluate the performance of algorithms for sensors and data transmission.

Course Outcomes (COs):

After successful completion of this course, the students will be able to:

CO1: Understand the characteristics of Internet of Things and its industry strategies. CO2:

Apply various Internet of Things models to appropriate problems.

CO3: Identify and integrate more than one technology to enhance the performance. CO4:

Understand the sensors and data transmission used in Internet of Things.

CO5: Analyse the co-occurrence of data to find interesting frequent patterns.

CO6: Pre-process the data before applying to any real-world problem and can evaluate its performance.

UNIT I

Overview of Internet of Things: Introduction, IOT Architecture, Application –based IOT protocols, Cloud Computing, Fog Computing, Sensor Cloud, Big Data.
Overview of Industry 4.0 and Industrial Internet of Things: IIoT- Prerequisites of IIOT, Basics of CPS, CPS and IIOT, Applications of IIoT.

UNIT II

Industrial Internet of Things: Introduction, Industrial Internet Systems, Industrial sensing, Industrial sensing, Industrial Processes.
Business Models and Reference Architecture of IIoT: Definition of a business model, Business models of IOT, Business models of IIOT.

UNIT III

Key and On-site Technologies: Key Technologies:Off-site Technologies- Introduction, Cloud Computing- Necessity, Cloud Computing and IIot, Industrial Cloud Platform Providers, SLA, Requirements of Industry 4.0, Fog Computing.

On-site Technologies- Introduction, Augmented Reality- History, Categorization, Applications, Virtual Reality- History, Categorization, Applications.

UNIT IV

Sensors and Data Transmission Sensors: Introduction to Sensors, Characteristics-Sensor calibration, Sensor profile, Operating voltage, Sensor Categories. Actuators:Introduction, Thermal Actuators, Hydraulic Actuators, Pneumatic Actuators, Electromechanical Actuators.

Industrial Data Transmission: Foundation fieldbus, Profibus, HART, Interbus, Bitbus.

UNIT V

Machine Learning and Data Science in Industries: Introduction, Machine Learning, Categorization on ML, Applications and Data Science of ML in industries, Deep Learning, Applications of Deep Learning in industries.

Applications of Healthcare in Industries: Smart Devices, Advanced Technologies using in Healthcare, Open Research Issues to be Addressed.

Textbooks:

1. S. Misra, C. Roy, and A. Mukherjee, 2020. Introduction to Industrial Internet of Things and Industry 4.0. CRC Press.

Reference Books:

1. Industrial IoT. Available online: https://medium.com/iotforall/whatproduct-managers-need- to-know-about-industrial-iot-8c92eec1d9d2

2. IIoT Cloud Platforms. Available online: https://fr.farnell.com/willthere-be-a-dominant-iiot-cloud-platform.

3. Kajima, T. and Kawamura, Y., 1995. Development of a high-speed solenoid valve: Investigation of solenoids. IEEE Transactions on industrial electronics, 42(1), pp.1-8.

Online Learning Resources:

1. https://www.coursera.org/learn/industrial-internet-of-things

2. https://www.coursera.org/specializations/developing-industrial-iot

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

Minors in CSE(IoT) (R23)

L	T	P	C
3	0	0	3

23ACS24: Cloud Computing

Course Objectives:

1. To explain the evolving computer model called cloud computing.
2. To introduce the various levels of services that can be achieved by cloud.
3. To describe the security aspects in cloud.

Course Outcomes (COs):

After successful completion of this course, the students will be able to:

CO1: Ability to create cloud computing environment CO2:

Ability to design applications for Cloud environment

CO3: Design & amp; develop backup strategies for cloud data base don features.

CO4: Use and examine different cloud computing services.

CO5: Apply different cloud programming model as per need.

UNIT-I

Introduction to cloud computing: Introduction, Characteristics of cloud computing, Cloud Models, Cloud Services Examples, Cloud Based services and applications

Cloud concepts and Technologies: Virtualization, Load balancing, Scalability and Elasticity, Deployment, Replication, Monitoring, Software defined, Network function virtualization, Map Reduce, Identity and Access Management, services level Agreements, Billing.

Cloud Services and Platforms: Compute Services, Storage Services, Database Services, Application services, Content delivery services, Analytics Services, Deployment and Management

Services, Identity and Access Management services, Open Source Private Cloud software.

UNIT-II

Hadoop MapReduce: Apache Hadoop, Hadoop Map Reduce Job Execution, Hadoop Schedulers, Hadoop Cluster setup.

Cloud Application Design: Reference Architecture for Cloud Applications, Cloud Application Design Methodologies, Data Storage Approaches.

Python Basics: Introduction, Installing Python, Python data Types & Data Structures, Control flow, Function, Modules, Packages, File handling, Date/Time Operations, Classes.

UNIT-III

Python for Cloud: Python for Amazon web services, Python for Google Cloud Platform, Python for windows Azure, Python for MapReduce, Python packages of Interest, Python web Application Frame work, Designing a RESTful web API.

Cloud Application Development in Python :Design Approaches, Image Processing APP, Document Storage App, MapReduce App, Social Media Analytics App.

UNIT-IV

Big Data Analytics: Introduction, Clustering Big Data, Classification of Bigdata Recommendation of Systems. Multimedia Cloud: Introduction, Case Study: Live video Streaming App, Streaming Protocols, case Study: Video Transcoding App.

Cloud Application Benchmarking and Tuning: Introduction, Workload Characteristics, Application Performance Metrics, Design Considerations for a Benchmarking Methodology,

Benchmarking Tools, Deployment Prototyping, Load Testing & Bottleneck Detection case Study, Hadoop benchmarking case Study.

UNIT-V

Cloud Security: Introduction, CSA Cloud Security Architecture, Authentication, Authorization, Identity Access Management, Data Security, Key Management, Auditing.

Cloud for Industry, Healthcare & Education: Cloud Computing for Health care, Cloud computing for Energy Systems, Cloud Computing for Transportation Systems, Cloud Computing for Manufacturing Industry, Cloud computing for Education.

Migrating into a Cloud: Introduction, Broad Approaches to migrating into the cloud, the seven– step model of migration into a cloud.

Organizational readiness and Change Management in The Cloud Age: Introduction, Basic concepts of Organizational Readiness, Drivers for changes: A frame work to comprehend the competitive environment, common change management models, change management maturity models, Organizational readiness self – assessment.

Legal Issues in Cloud Computing: Introduction, Data Privacy and security Issues, cloud contracting models, Jurisdiction a issues raised by virtualization and data location, commercial and

Business considerations, Special Topics.

Textbooks:

1. Cloud computing A hands-on Approach‖ By ArshdeepBahga, Vijay Madisetti, Universities Press, 2016.
2. Cloud Computing Principles and Paradigms: By Raj Kumar Buyya, James Broberg, Andrzej, Goscinski, Wiley, 2016.

Reference Books:

1. Mastering Cloud Computing by RajkumarBuyya, Christian Vecchiola, SThamaraiSelvi, TMH

2. Cloud computing A Hands-On Approach by ArshdeepBahga and Vijay Madisetti.

3. Cloud Computing: A Practical Approach, Anthony T. Velte, Toby J. Velte, Robert Elsenpeter, Tata McGraw Hill, rp2011.

4. Enterprise Cloud Computing, Gautam Shroff, Cambridge University Press, 2010.

5. Cloud Application Architectures: Building Applications and Infrastructure in the Cloud, George Reese, O 'Reilly, SPD, rp2011.

6. Essentials of Cloud Computing by K. Chandrasekaran. CRC Press.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

Minors in CSE(IoT) (R23)

23ACO02: INTERNET OF THINGS LAB

L	T	P	C
0	0	3	1.5

Course Objectives:

1. To introduce components such as WiFi, Bluetooth, Temperature, Moisture sensors
2. To know the Micro controller such as Arduino
3. To know the System on Chip (SOC) / Single Board Computer such as Raspberry Pi
4. To understand HTTP IoT protocols and perform Experiments for data transmission
5. To understand UAV/Drones and Internet of Drones Experiments.

Course Outcomes (COs):

After successful completion of this course, the students will be able to:

CO1: Know the various IoT sensors and understand the functionality

CO2: Design and analyzeIoT experiments and transfer the data to IoT Clouds CO3:

Design the IoT systems for real time applications

CO4: Understand Drones and Perform Internet of Drones Experiments

List of Experiments:

Experiments using ESP32

1. Serial Monitor, LED, Servo Motor - Controlling

Experiment1: Controlling actuators through Serial Monitor. Creating different ledpatterns and controlling them using push button switches. Controlling servo motor with the help of joystick.

2. Distance Measurement of an object

Experiment 2: Calculatethe distance to an object with the help of an ultrasonic sensor and display it on an LCD.

3, LDR Sensor, Alarm and temperature, humidity measurement

Experiment 3:

- Controlling relay state based on ambient light levels using LDR sensor.
- Basic Burglar alarm security system with the help of PIR sensor and buzzer.
- Displaying humidity and temperature values on LCD

4. Experiments using Raspberry Pi

Experiment 4:

- Controlling relay state based on input from IR sensors
- Interfacing stepper motor with R-Pi
- Advanced burglar alarm security system with the help of PIR sensor, buzzer and keypad. (Alarm gets disabled if correct keypad password is entered)
- Automated LED light control based on input from PIR (to detect if people are present) and LDR(ambient light level)

5. IOT Framework

Experiment 5: Upload humidity & temperature data to ThingSpeak, periodically logging ambient light level to ThingSpeak

Experiment 6: Controlling LEDs, relay & buzzer using Blynk app

6. HTTP Based

Experiment 7: Introduction to HTTP. Hosting a basic server from the ESP32 to control various digital based actuators (led, buzzer, relay) from a simple web page.

Experiment 8:Displaying various sensor readings on a simple web page hosted on the ESP32.

7. MQTT Based

Experiment 9:Controlling LEDs/Motors from an Android/Web app, Controlling AC Appliances from an android/web app with the help of relay.

Experiment 10:

Displaying humidity and temperature data on a web-based application

8. UAV/Drone:

Experiment 11:

- Demonstration of UAV elements, Flight Controller
- Mission Planner flight planning design

Experiment 12:

- Python program to read GPS coordinates from Flight Controller

Reference:

1. Adrian McEwen, Hakim Cassimally - Designing the Internet of Things, Wiley Publications, 2012.

2. Alexander Osterwalder, and Yves Pigneur – Business Model Generation – Wiley, 2011

3. ArshdeepBahga, Vijay Madisetti - Internet of Things: A Hands-On Approach, Universities Press, 2014.

4. The Internet of Things, Enabling technologies and use cases – Pethuru Raj, Anupama C. Raman, CRC Press.

Online Learning Resources/Virtual Labs:

https://www.arduino.cc/

https://www.raspberrypi.org/

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

Minors in CSE(IoT) (R23)

23ACS26: CLOUD COMPUTING LAB

L	T	P	C
0	0	3	1.5

Course Objectives:

1. Demonstrate application development using Cloud
2. Explain features of Hadoop

Course Outcomes (CO):

On completion of this course, the students will be able to:

CO1: Configure various virtualization tools such as Virtual Box, VMware workstation. CO2:

Design and deploy a web application in a PaaS environment.

CO3: Learn how to simulate a cloud environment to implement new schedulers. CO4:

Install and use a generic cloud environment that can be used as a private cloud. CO5:

Manipulate large data sets in a parallel environment.

List of Experiments:

1. Install VirtualBox/VMware Workstation with different flavours of Linux or windows OS on top of windows operating systems.

2. Install a C compiler in the virtual machine created using virtual box and execute Simple Programs

3. Install Google App Engine. Create hello world app and other simple web applications using python/java.

4. Use GAE launcher to launch the web applications.

5. Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim.

6. Find a procedure to transfer the files from one virtual machine to another virtual machine.

7. Find a procedure to launch virtual machine using try stack (Online Open stack Demo Version)

8. Install Hadoop single node cluster and run simple applications like wordcount

9. Establish an AWS account. Use the AWS Management Console to launch an EC2 instance and connect to it.

10. Develop a Guestbook Application using Google App Engine

11. Develop a Serverless Web App using AWS

12. Design a Content Recommendation system using AWS

13. Design a Cloud based smart traffic management system

14. Design Cloud based attendance management system

15. Design E-learning cloud-based system

16. Using Amazon Lex build a Chatbot

References:

- https://www.vmware.com/products/workstation-pro/workstation-pro_evaluation.html.
- http://code.google.com/appengine/downloads.html
- http://code.google.com/appengine/downloads.html

Online Learning Resources/Virtual Labs:

- Google Cloud Computing Foundations Cour

MINOR DEGREE IN DATA SCIENCE

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B. Tech. - I Semester

		L	T	P	C
23ACD36	**DATA SCIENCE FOR ENGINEERS**	**3**	**0**	**0**	**3**

Course Objectives:

- Provide a foundational understanding of data science processes and applications.
- Introduce key tools and techniques such as Python, statistics, data cleaning, visualization, and machine learning.
- Develop practical skills in data analysis, interpretation, and data storytelling.
- Enable students to work on real-world datasets using data science techniques.
- Prepare students for advanced studies or industry roles in data science and analytics.

Course Outcomes:

After completion of the course, students will be able to:

CO1:	Explain the data science lifecycle and its importance in business and research.	Understand(L2)
CO2:	Use Python and libraries like Pandas, NumPy, and Matplotlib for data handling.	Apply(L3)
CO3:	Perform data cleaning, transformation, and visualization effectively.	Apply(L3)
CO4:	Apply basic machine learning models for classification and regression.	Apply(L3)
CO5:	Interpret data analysis results and communicate findings clearly.	Analyze(L4)

Unit I: Introduction to Data Science: What is Data Science?,Role of Data Scientist, Data Science Process (Problem definition, data collection, preprocessing, modeling, evaluation), Applications of Data Science in different domains, Tools: Jupyter, Anaconda, Python/R Overview

Unit II: Data Handling and Preprocessing: Introduction to NumPy and Pandas, Reading data from CSV, Excel, SQL, Data Wrangling: Missing values, duplicates, outliers, Data transformation: Scaling, encoding, Feature engineering basics

Unit III: Data Visualization: Importance of visualization, Visualization libraries (Matplotlib, Seaborn), Histograms, Boxplots, Pairplots, Heatmaps, Dashboards and Storytelling with Data, Real-time data dashboards (Optional)

Unit IV: Statistical Foundations for Data Science: Descriptive Statistics, Probability and Probability Distributions, Inferential Statistics: Hypothesis Testing, Confidence Intervals, Correlation and Causation, Use of Scipy/Statsmodels for statistical analysis

Unit V: Introduction to Machine Learning: Supervised vs Unsupervised Learning, Classification and Regression problems, Basic ML Algorithms: Linear Regression, Logistic Regression, KNN, Decision Trees, Model Evaluation Metrics: Accuracy, Precision, Recall, F1-Score, Overfitting and Underfitting

Textbooks:

1. **Joel Grus** – Data Science from Scratch: First Principles with Python, O'Reilly.
2. **Cathy O'Neil and Rachel Schutt** – Doing Data Science, O'Reilly.
3. **Wes McKinney** – Python for Data Analysis, O'Reilly.

Reference Books:

1. **Jake VanderPlas** – Python Data Science Handbook, O'Reilly.
2. **Andreas Müller & Sarah Guido** – Introduction to Machine Learning with Python.
3. **Han, Kamber, & Pei –** Data Mining: Concepts and Techniques, Morgan Kaufmann.

Online Courses:

NPTEL / SWAYAM:

- NPTEL: Introduction to Data Science
  Instructor: Prof. Raghunathan Rengasamy, IIT Madras

Coursera:

- **IBM Data Science Professional Certificate**
  Link: coursera.org
- **Introduction to Data Science in Python (University of Michigan)**
  Link: coursera.org

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B. Tech. - I Semester

		L	T	P	C
23ACD37	**STATISTICAL LEARNING FOR DATA SCIENCE**	**3**	**0**	**0**	**3**

Course Objectives:

- Introduce fundamental concepts of statistical learning and its importance in data science.
- Develop understanding of regression, classification, and resampling methods.
- Build skills in model assessment, selection, and regularization techniques.
- Apply statistical learning methods to real-world datasets.
- Interpret, communicate, and evaluate data-driven models.

Course Outcomes:

After completion of the course, students will be able to:

CO1:	Understand the basics of statistical learning and data representation.	Understand(L2)
CO2:	Apply linear regression, logistic regression, and classification techniques.	Apply(L3)
CO3:	Evaluate models using resampling methods and cross-validation.	Evaluate(L5)
CO4:	Analyze regularization methods like Ridge and Lasso to avoid overfitting.	Analyze(L4)
CO5:	Create and interpret statistical learning models in practical scenarios.	Create(L6)

Unit I: Introduction to Statistical Learning: What is Statistical Learning?, Supervised vs Unsupervised Learning, Model Accuracy vs Interpretability, Bias-Variance Trade-off, Curse of Dimensionality, Applications of Statistical Learning

Unit II: Linear and Polynomial Regression: Simple Linear Regression. Multiple Linear Regression, Polynomial Regression, Assumptions of Linear Models, Model Diagnostics and Performance Metrics (R^2, RMSE), Variable Selection Techniques

Unit III: Classification Methods: Logistic Regression, Discriminant Analysis (LDA, QDA), Naïve Bayes Classifier, K-Nearest Neighbors, Confusion Matrix, Precision, Recall, F1-Score

Unit IV: Resampling and Model Assessment: Train-Test Split, Cross Validation (k-Fold, LOOCV), Bootstrap Methods, Model Selection and Hyperparameter Tuning.

Unit V: Regularization & Advanced Topics: Ridge Regression, Lasso Regression, Shrinkage Methods, Principal Component Regression (PCR), Partial Least Squares (PLS), Introduction to Support Vector Machines.

Textbooks:

1. **Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani** *An Introduction to Statistical Learning with Applications in R* – Springer (Free PDF: https://www.statlearning.com/)
2. **Trevor Hastie, Robert Tibshirani, Jerome Friedman**
   The Elements of Statistical Learning – Springer

Reference Books:

1. **Norman Matloff** – *Statistical Regression and Classification: from R to Data Science*
2. **Chris Bishop** – *Pattern Recognition and Machine Learning*
3. **T. Ryan** – *Modern Regression Methods*

Online Courses (Free + Paid)

NPTEL / SWAYAM:

- NPTEL: Introduction to Statistical Learning
  Instructor: Prof. Balaraman Ravindran (IIT Madras)

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B. Tech. - II Semester

		L	T	P	C
23ACD38	**INTRODUCTION TO DATA ENGINEERING**	**3**	**0**	**0**	**3**

Course Objectives:

- Understand the data engineering lifecycle, tools, and platforms.
- Gain proficiency in building scalable data pipelines using batch and streaming systems.
- Explore data storage formats, data lakes, and warehouses.
- Learn about workflow orchestration, scheduling, and data quality.
- Apply data engineering techniques in real-time and big data environments.

Course Outcomes:

After completion of the course, students will be able to:

CO1:	Understand the foundations of data engineering tools, platforms, and data lifecycle.	Understand(L2)
CO2:	Design and implement batch and streaming data pipelines.	Apply(L3)
CO3:	Analyze data storage and querying using file formats, data lakes, and warehouses.	Analyze(L4)
CO4:	Evaluate orchestration and workflow management tools.	Evaluate(L5)
CO5:	Build end-to-end data engineering workflows and troubleshoot data quality issues.	Create(L6)

Unit I: Introduction to Data Engineering: What is Data Engineering? Data Engineering vs Data Science, Data Lifecycle and Architectures, Roles and Responsibilities of a Data Engineer, Tools Overview: Hadoop, Spark, Kafka, Airflow, Hive.

Unit II: Data Storage and Formats: Structured, Semi-structured, and Unstructured Data, File Formats: CSV, JSON, Parquet, Avro, ORC, Data Warehousing Concepts (Star/Snowflake Schema), Data Lakes vs Data Warehouses, Cloud Storage Solutions (S3, Azure Blob, GCS)

Unit III: Batch and Stream Processing: ETL vs ELT, Batch Processing with Apache Spark, Stream Processing with Apache Kafka and Spark Streaming, Windowing, Late Events, and Watermarks, Lambda and Kappa Architectures.

Unit IV: Workflow Orchestration & Data Quality: Introduction to Apache Airflow, DAGs, Operators, Scheduling, Data Lineage and Observability, Data Profiling, Cleaning, Validation, Ensuring Data Quality with Great Expectations.

Unit V: End-to-End Pipeline and Case Studies: Designing a Data Pipeline: Source to Sink, Real-time Analytics with Kafka + Spark + Cassandra, Monitoring and Logging, CI/CD in Data Engineering (using GitHub Actions or Jenkins), Case Studies: Fraud Detection, Log Analysis, and Recommendation Engine.

Textbooks:

1. **Andreas François Vermeulen**, *The Data Engineering Cookbook* – Packt
2. **Joe Reis and Matt Housley**, *Fundamentals of Data Engineering* – O'Reilly, 2022
3. **Anurag Bhardwaj**, *Data Engineering with Apache Spark, Delta Lake, and Lakehouse* – Packt

Reference Books:

1. *Designing Data-Intensive Applications* – Martin Kleppmann (O'Reilly)
2. *Streaming Systems: The What, Where, When, and How of Large-Scale Data Processing* – Tyler Akidau

Online Courses:

NPTEL / SWAYAM

- NPTEL: Big Data Computing
- NPTEL: Cloud Computing and Distributed Systems

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SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY ‘
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III B. Tech. - II Semester

		L	T	P	C
23ACM34	**MACHINE LEARNING FUNDAMENTALS**	**3**	**0**	**0**	**3**

Course Objectives:

- Understand core concepts of supervised, unsupervised, and reinforcement learning.
- Learn foundational algorithms for classification, regression, and clustering.
- Analyze model performance using evaluation metrics and tuning methods.
- Gain practical knowledge in implementing ML models using datasets.
- Understand the mathematical and probabilistic foundations of ML algorithms.

Course Outcomes:

After completion of the course, students will be able to:

CO1:	Understand the concepts and assumptions behind key ML algorithms.	Understand(L2)
CO2:	Apply various supervised and unsupervised learning techniques.	Apply(L3)
CO3:	Analyze the strengths and limitations of different machine learning models.	Analyze(L4)
CO4:	Evaluate model performance using metrics and validation techniques.	Evaluate(L5)
CO5:	Design and implement complete ML solutions for real-world problems.	Create(L6)

Unit I: Introduction to Machine Learning:

Definition and Scope of ML, Types of Learning: Supervised, Unsupervised, Reinforcement, Basic Terminology: Instance, Feature, Label, Training and Testing Sets, Applications and Challenges, Python Libraries: scikit-learn, pandas, numpy

Unit II: Supervised Learning – Regression & Classification

Linear Regression, Polynomial Regression, Logistic Regression, k-Nearest Neighbors (k-NN), Decision Trees and Random Forests, Naïve Bayes Classifier

Unit III: Unsupervised Learning & Dimensionality Reduction

k-Means Clustering, Hierarchical Clustering, Principal Component Analysis (PCA), t-SNE, Autoencoders (Intro Only). Association Rule Mining,

Unit IV: Model Evaluation and Tuning

Confusion Matrix, Precision, Recall, F1-Score, ROC-AUC, Bias-Variance Tradeoff, Cross-Validation, Hyperparameter Tuning: Grid Search & Random Search, Overfitting & Underfitting

Unit V: Advanced Topics & Real-time Applications

Introduction to Neural Networks (Perceptron), Introduction to SVMs, Feature Engineering & Selection, Real-world Use Cases: Spam Detection, Credit Scoring, Medical Diagnosis, Ethical AI and Model Explainability (XAI – Intro Only).

Text book:

1. **Tom M. Mitchell**, *Machine Learning*, McGraw-Hill, 1997

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B. Tech. - I Semester

		L	T	P	C
23ACD39	**DATA SCIENCE APPLICATIONS**	**3**	**0**	**0**	**3**

Course Objectives:

- Understand key domains where data science plays a transformative role.
- Learn how data is collected, preprocessed, and analyzed in different domains.
- Explore application-driven problem-solving using real-world datasets.
- Evaluate models and interpret data science outputs effectively.
- Gain hands-on exposure to tools and techniques used in data science applications.

Course Outcomes:

At the end of the course, the student will be able to

CO1:	Explain the role and impact of data science across various application areas.	Understand(L2)
CO2:	Apply suitable data science methods to domain-specific datasets.	Apply(L3)
CO3:	Analyze domain-specific problems using exploratory and predictive techniques.	Analyze(L4)
CO4:	Evaluate model effectiveness and insights in practical use cases.	Evaluate(L5)
CO5:	Design data-driven solutions for real-world problems.	Create(L6)

Unit I: Introduction to Data Science & Domains

Definition and Workflow of Data Science. Lifecycle of a Data Science Project, Overview of Applications: Healthcare, Finance, Retail, Manufacturing, Agriculture, Education, and Smart Cities, Ethical Considerations in Data Science.

Unit II: Data Science in Healthcare and Finance

Use cases: Disease prediction, EHR data, medical imaging, outbreak modelling, Predictive modeling for diagnosis (e.g., Diabetes prediction), Applications in FinTech: Fraud detection, Risk scoring, Credit analytics, Time series analysis in financial markets.

Unit III: Data Science in Retail, Marketing & Customer Analytics

Market basket analysis, Recommendation Systems, Customer Segmentation using Clustering, Churn Prediction Models, Sentiment Analysis for Customer Feedback.

Unit IV: Data Science in Smart Cities, Agriculture & Education

Traffic prediction, Waste management using ML, IoT-enabled Smart Farming and Yield prediction, Adaptive Learning and Student Performance Prediction, Case studies with real-world datasets.

Unit V: Tools, Case Studies, and Deployment

Case Studies on End-to-End Projects, Tools: Python, Jupyter, Power BI, Tableau, Deploying Models: APIs, Streamlit, Flask, Interpreting Results and Storytelling, Recent Advances and Future Directions

Textbooks:

1. Joel Grus, Data Science from Scratch, O'Reilly, 2nd Edition.
2. Cathy O'Neil & Rachel Schutt, Doing Data Science, O'Reilly Media.
3. V. K. Jain, Data Science and Big Data Analytics, Khanna Publishing.

Reference Books:

1. Wes McKinney, Python for Data Analysis, O'Reilly.
2. Sebastian Raschka, Machine Learning with Python Cookbook, Packt.
3. Bill Franks, Taming the Big Data Tidal Wave, Wiley.

Online Courses:

NPTEL / SWAYAM

- NPTEL – Data Science for Engineers
- SWAYAM – Data Science Applications

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III B. Tech. - II Semester

		L	T	P	C
23ACD40	**DATA SCIENCE PRACTICE LAB (R/Python)**	**0**	**0**	**3**	**1.5**

Course Objectives:

- To introduce basic data analysis using R/Python.
- To understand data wrangling, visualization, and preprocessing.
- To develop skills in exploratory data analysis and simple ML models

Course Outcomes:

At the end of the course, the student will be able to

- Apply Python/R for real-time data handling.
- Perform data visualization and preprocessing.
- Build basic machine learning models.

Experiments:

1. Introduction to Python/R and IDE setup
2. Data Types, Variables, and Operators
3. Reading and writing CSV/Excel data
4. Data Cleaning and Wrangling (nulls, types, etc.)
5. Exploratory Data Analysis with Matplotlib/ggplot2
6. Correlation and Covariance Calculation
7. Linear Regression implementation
8. Classification using KNN
9. Clustering using K-Means
10. Data Visualization using Seaborn
11. Building simple ML pipeline using scikit-learn
12. Data analysis on real-world dataset

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IV B. Tech. - I Semester

23ACD41	STATISTICAL LEARNING & ML LAB	L	T	P	C
		0	0	3	1.5

Course Objectives

- To explore statistical methods for data science.
- To apply statistical learning to predictive modelling.

Course Outcomes:

At the end of the course, the student will be able to

- Utilize statistical models in data science tasks.
- Implement regression and classification algorithms.

Experiments:

1. Introduction to probability distributions
2. Descriptive statistics on datasets
3. Hypothesis testing
4. Linear Regression and visualization
5. Logistic Regression for classification
6. Decision Trees
7. Random Forests
8. Confusion Matrix and Accuracy metrics
9. Cross-validation techniques
10. Feature scaling and encoding
11. Principal Component Analysis
12. Regression or Classification on dataset

MINOR DEGREE IN DATA ANALYTICS

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III B. Tech. - I Semester

		L	T	P	C
23ACD42	**INTRODUCTION TO DATA ANALYTICS**	**3**	**0**	**0**	**3**

Course Objectives:

- To introduce the fundamental concepts of data analytics.
- To understand the process of collecting, cleaning, analyzing, and interpreting data.
- To gain insights into exploratory data analysis (EDA) and visualization techniques.
- To apply statistical and machine learning techniques to draw inferences from data.
- To use analytical tools (like Python/R) to solve real-world data problems.

Course Outcomes:

At the end of the course, the student will be able to

CO1:	Understand the core concepts and applications of data analytics.	Understand(L2)
CO2:	Prepare and preprocess data for analysis using appropriate techniques.	Apply(L3)
CO3:	Analyze data using descriptive and inferential statistical methods.	Analyze(L4)
CO4	Visualize data insights through various graphical representations.	Evaluate(L5)
CO5:	Apply data analytics tools for solving domain-specific problems.	Create(L6)

Unit I: Introduction to Data Analytics

Data, Information, and Knowledge, Types of data: Structured, Unstructured, Semi-Structured, Phases of the Data Analytics Lifecycle, Introduction to Business Analytics, Tools for data analytics: Excel, Python, R, SQL.

Unit II: Data Collection and Preprocessing

Sources of data: Databases, APIs, Web Scraping, Handling missing data and outliers, Data cleaning, transformation, normalization, Feature engineering and selection, Data partitioning and sampling techniques.

Unit III: Descriptive and Inferential Statistics

Central tendency and dispersion: Mean, Median, Mode, Standard Deviation, Correlation and covariance, Probability distributions: Normal, Binomial, Poisson, Hypothesis testing, t-test, Chi-square test, Confidence intervals and p-values.

Unit IV: Exploratory Data Analysis (EDA) and Visualization

Univariate, Bivariate, Multivariate Analysis, Histograms, Boxplots, Scatter plots, Pair plots, Heatmaps and correlation matrices, Data dashboards and storytelling using visualizations, Tools: Matplotlib, Seaborn, Plotly, Tableau basics.

Unit V: Data-Driven Decision Making

Predictive analytics vs descriptive analytics, Introduction to regression and classification, Clustering and segmentation basics, Model evaluation metrics: Accuracy, Precision, Recall, F1-score, Case studies in marketing, healthcare, finance, operations

Textbooks:

1. **V. Uday Shankar –** Data Analytics, Cengage Learning
2. **P. N. Murthy** – Introduction to Data Analytics, Himalaya Publishing
3. **Anil Maheshwari** – Data Analytics Made Accessible, Amazon Independent

Reference Books:

1. **Foster Provost & Tom Fawcett –** Data Science for Business, O'Reilly
2. **Joel Grus –** Data Science from Scratch, O'Reilly
3. **Wes McKinney** – Python for Data Analysis, O'Reilly
4. **Allen Downey** – Think Stats, Green Tea Press

Online Courses:

Platform	**Course Title & Link**
Coursera	Introduction to Data Analytics – IBM

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III B. Tech. - I Semester

		L	T	P	C
23ACD43	**DATA ENGINEERING ESSENTIALS**	**3**	**0**	**0**	**3**

Course Objectives:

- To introduce the architecture, components, and workflow of modern data engineering systems.
- To provide knowledge on ingestion, storage, and processing of large-scale data.
- To apply tools and techniques like ETL, data pipelines, and distributed processing frameworks.
- To explore data modeling, warehousing, and real-time data processing.
- To build scalable and maintainable data systems with cloud and open-source technologies

Course Outcomes:

At the end of the course, the student will be able to

CO1	Understand core concepts of data engineering and data pipelines.	Understand(L2)
CO2	Apply data ingestion, transformation, and storage techniques.	Apply(L3)
CO3	Analyze big data processing frameworks and streaming systems.	Analyze(L4)
CO4	Evaluate data warehousing models and real-time processing strategies.	Evaluate(L5)
CO5	Design and implement scalable data engineering workflows.	Create(L6)

Unit I: Introduction to Data Engineering

What is Data Engineering? Role of Data Engineer in the data ecosystem, Types of data: Structured, Semi-structured, Unstructured, Data Lifecycle: Collection to Consumption, Introduction to ETL and ELT processes.

Unit II: Data Ingestion and Storage

Batch vs Streaming data, Data ingestion tools: Sqoop, Flume, Kafka, Logstash, Structured storage: Relational DBs, Data Lakes, File formats: CSV, JSON, Avro, Parquet, ORC, NoSQL overview: HBase, MongoDB, Cassandra.

Unit III: Distributed Data Processing

Hadoop ecosystem: HDFS, MapReduce, YARN, Apache Spark: RDD, DataFrame, and SQL APIs, Data cleaning and transformation techniques, Resource allocation and job scheduling in Spark, Introduction to cloud-based data processing (AWS/GCP).

Unit IV: Data Warehousing & Modeling

Concepts of OLTP vs OLAP, Star and Snowflake Schemas, Fact and Dimension tables, Data Warehousing tools: Amazon Redshift, Google BigQuery, Snowflake, Data modeling tools: dbt (data build tool), ER diagrams.

Unit V: Real-Time Data Engineering and Pipelines

Streaming with Apache Kafka and Spark Streaming, Message Queues and Pub/Sub systems, Lambda and Kappa architectures, Monitoring and logging tools: Prometheus, Grafana, Case study: End-to-end data pipeline implementation.

Textbooks:

1. Andreas François Vermeulen – Data Engineering with Python, Packt Publishing
2. V. Uday Shankar – Big Data Analytics & Data Engineering, Cengage Learning
3. Sam Newman – Building Microservices, O'Reilly (for data infrastructure components)

Reference Books:

1. Bill Inmon – Building the Data Warehouse, Wiley
2. Tom White – Hadoop: The Definitive Guide, O'Reilly
3. Jules S. Damji et al. – Learning Spark: Lightning-Fast Big Data Analysis, O'Reilly
4. Noel Markham – Learning Apache Kafka, Packt Publishing

Online Courses:

Platform	**Course Title & Link**
Coursera	Data Engineering on Google Cloud – Google

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B. Tech. - II Semester

		L	T	P	C
23ACD44	**PREDICTIVE ANALYTICS FUNDAMENTALS**	**3**	**0**	**0**	**3**

Course Objectives:

- To understand the fundamentals of predictive analytics and modeling.
- To learn statistical, machine learning, and probabilistic techniques used for prediction.
- To build and evaluate predictive models for various types of data.
- To apply data preprocessing, feature engineering, and model tuning techniques.
- To explore real-world use cases of predictive analytics in business, health, and industry.

Course Outcomes:

At the end of the course, the student will be able to

CO1	Understand key concepts, tools, and algorithms used in predictive analytics.	Understand(L2)
CO2	Apply classification and regression techniques for prediction problems.	Apply(L3)
CO3	Analyze model performance using error metrics and validation techniques.	Analyze(L4)
CO4	Evaluate predictive models in real-world applications and domains.	Evaluate(L5)
CO5	Design complete predictive analytics pipelines with preprocessing to prediction.	Create(L6)

Unit I: Introduction to Predictive Analytics

Basics of predictive analytics, Types of predictive models: Classification vs Regression, Predictive modeling workflow, Applications in finance, healthcare, marketing, etc., Introduction to supervised learning.

Unit II: Data Preparation and Feature Engineering

Data cleaning and preprocessing, Handling missing values, outliers, Categorical encoding, normalization, scaling, Feature selection and dimensionality reduction (PCA, LDA), Feature importance.

Unit III: Regression and Classification Models

Linear Regression, Ridge & Lasso Regression, Logistic Regression, K-Nearest Neighbors (KNN), Decision Trees, Random Forests, Model diagnostics: R^2, MAE, RMSE, Confusion Matrix, AUC

Unit IV: Advanced Predictive Modeling

Support Vector Machines (SVM), Ensemble Models: Bagging, Boosting, XGBoost, Model selection and hyperparameter tuning, Cross-validation, Grid Search, Avoiding overfitting/underfitting.

Unit V: Real-Time Predictive Analytics and Deployment

Case studies: Predictive maintenance, customer churn, disease prediction, Tools: Python (scikit-learn, pandas), R, AutoML, Model interpretability: SHAP, LIME, Model deployment using Flask, FastAPI, or cloud (AWS/GCP), Ethics in predictive analytics.

Textbooks:

1. Dean Abbott – Applied Predictive Analytics: Principles and Techniques, Wiley
2. GalitShmueli et al. – Data Mining for Business Analytics: Concepts, Techniques, and Applications, Wiley
3. John D. Kelleher et al. – Fundamentals of Machine Learning for Predictive Data Analytics, MIT Press

Reference Books:

1. Trevor Hastie, Robert Tibshirani, Jerome Friedman – The Elements of Statistical Learning
2. Ian H. Witten, Eibe Frank, Mark A. Hall – Data Mining: Practical Machine Learning Tools and Techniques
3. Tom Fawcett – Data Science for Business, O'Reilly

Online Courses:

Platform	Course Title & Link
Coursera	Advanced Predictive Modeling – University of Washington

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III B. Tech. - II Semester

		L	T	P	C
23ACD45	**BIG DATA ANALYTICS (Hadoop, Spark)**	**3**	**0**	**0**	**3**

Course Objectives:

- To understand the need and concepts of Big Data and its ecosystem.
- To study architectures and tools such as Hadoop and Spark for large-scale data processing.
- To explore Hadoop Distributed File System (HDFS) and MapReduce programming.
- To learn in-memory computation using Apache Spark.
- To implement big data solutions using real-time tools for analytics and decision-making.

Course Outcomes:

At the end of the course, the student will be able to

CO1	Explain the concepts, characteristics, and challenges of Big Data.	Understand(L2)
CO2	Apply HDFS and MapReduce programming for large-scale data processing.	Apply(L3)
CO3	Analyze and optimize data pipelines using Spark RDDs and DataFrames.	Analyze(L4)
CO4	Evaluate Big Data tools and frameworks for distributed processing.	Evaluate(L5)
CO5	Design end-to-end Big Data workflows for analytics use cases.	Create(L6)

UNIT I: Introduction to Big Data

Characteristics of Big Data: Volume, Velocity, Variety, Veracity, and Value, Challenges of Big Data, Traditional vs. Big Data systems, Big Data architecture: Storage, processing, analytics, Introduction to Hadoop Ecosystem

UNIT II: Hadoop Distributed File System (HDFS) and MapReduce

HDFS Architecture, Blocks, NameNode&DataNode, File Read/Write operations, MapReduce Architecture, Developing MapReduce programs (Word Count Example), Combiners, Partitioner, and Counters

UNIT III: Apache Spark and RDDs

Need for Apache Spark over Hadoop, Spark architecture and components, Spark RDDs and transformations/actions, DataFrames and Spark SQL, Working with Spark MLlib for basic ML tasks

UNIT IV: Big Data Tools and Streaming

YARN architecture, Sqoop: Import/export between RDBMS and Hadoop, Flume: Ingesting streaming data, Kafka for distributed messaging, Spark Streaming: DStreams, windowing, real-time analytics

UNIT V: Big Data Applications and Case Studies

Social media analytics, IoT data pipelines, Retail analytics (recommendation engines, customer sentiment), Fraud detection, End-to-End Mini Project: Data ingestion, processing, and visualization

Textbooks:

1. Tom White, Hadoop: The Definitive Guide, O'Reilly Media
2. Jules S. Damji et al., Learning Spark: Lightning-Fast Big Data Analysis, O'Reilly
3. VenkatAnkam, Big Data Analytics Using Spark, Wiley

Reference Books:

1. Chuck Lam, Hadoop in Action, Manning
2. Alex Holmes, Hadoop in Practice, Manning Publications
3. P. J. Sadalage& M. Fowler, NoSQL Distilled, Addison-Wesley
4. Sridhar Alla, Big Data Analytics with Hadoop 3, Packt Publishing

Online Courses:

Platform	Course Title & Link
Coursera	Big Data Specialization – UC San Diego

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IV B. Tech. - I Semester

		L	**T**	**P**	**C**
23ACD46	**DATA ANALYTICS WITH POWER BI / TABLEAU / MATPLOTLIB**	**3**	**0**	**0**	**3**

Course Objectives:

- To understand the fundamentals of data visualization and storytelling using analytics tools.
- To learn practical implementation of visual analytics using Power BI, Tableau, and Matplotlib.
- To develop dashboards, custom reports, and interactive charts.
- To explore real-world datasets and generate insights.
- To design end-to-end data pipelines and reporting workflows.

Course Outcomes:

At the end of the course, the student will be able to

CO1:	Understand data analytics lifecycle and fundamentals of visualization tools.	Understand(L2)
CO2:	Apply Power BI, Tableau, and Matplotlib to visualize structured data.	Apply(L3)
CO3:	Analyze visual trends and compare different types of charting techniques.	Analyze(L4)
CO4:	Evaluate dashboards and performance metrics for decision-making.	Evaluate(L5)
CO5:	Design interactive data dashboards and business visual reports.	Create(L6)

UNIT I: Introduction to Data Visualization & BI

What is data analytics and visualization?, Importance of business intelligence, Types of charts and graphs, Visualization best practices and perception, Introduction to tools: Power BI, Tableau, and Python/Matplotlib

UNIT II: Power BI for Data Analytics

Power BI interface and data sources, Data transformations using Power Query, DAX (Data Analysis Expressions) basics, Visualizations: Bar, Line, Area, Pie, Tree Maps, Report publishing and dashboard sharing

UNIT III: Tableau for Data Analytics

Tableau workspace and connecting to datasets, Dimensions vs. Measures, Filters, Parameters, and Calculated Fields, Visualizations: Heatmaps, Maps, Scatter Plots, Building Interactive Dashboards

UNIT IV: Matplotlib & Seaborn (Python Visualization)

Introduction to Matplotlib architecture, Basic plots: line, bar, pie, histogram, Subplots, customization (color, label, grid), Seaborn: heatmaps, boxplots, pairplots, Exporting and saving visuals

UNIT V: Real-time Case Studies and BI Projects

Designing end-to-end dashboards for:Sales analytics, Healthcare data, IoT or Sensor data dashboards, Social media analytics, Performance monitoring and KPI analysis, BI deployment best practices

Textbooks:

1. Alberto Ferrari & Marco Russo, Introducing Microsoft Power BI, Microsoft Press.
2. Joshua N. Milligan, Learning Tableau, Packt Publishing.
3. AdrienChauveau, Data Visualization with Python and Matplotlib, Packt Publishing.

Reference Books:

1. Ben Jones, Storytelling with Data in Tableau, O'Reilly.
2. Jake VanderPlas, Python Data Science Handbook, O'Reilly.
3. Cole NussbaumerKnaflic, Storytelling with Data, Wiley.

Online Courses:

Platform	**Course Title & Link**
Coursera	Data Visualization with Tableau – UC Davis

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B. Tech. - II Semester

		L	T	P	C
23ACD47	**DATA ANALYTICS TOOLS LAB**	**0**	**0**	**3**	**1.5**

Course Objectives:

- To visualize and interpret data using modern tools.
- To create dashboards and analytical views.

Course Outcomes:

At the end of the course, the student will be able to

- Use BI tools for analytical visualizations.
- Interpret datasets using graphical methods.

Experiments:

1. Introduction to Power BI / Tableau interface
2. Loading data and transforming in BI tools
3. Creating bar, pie, and line charts
4. Cross filters and slicers
5. Using DAX in Power BI
6. Map visualization
7. Dashboards creation
8. Connecting to real-time data
9. Storytelling with data
10. Introduction to Matplotlib for comparison
11. Using Seaborn and Plotly

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IV B. Tech. - I Semester

		L	T	P	C
23ACD48	**BIG DATA & NOSQL LAB**	**0**	**0**	**3**	**1.5**

Course Objectives:

- To process big datasets using distributed systems.
- To implement NoSQL solutions.

Course Outcomes:

At the end of the course, the student will be able to

- Handle large datasets using Hadoop/Spark.
- Perform CRUD operations in NoSQL.

Experiments:

1. Introduction to HDFS and Hadoop
2. MapReduce word count example
3. Spark RDD operations
4. DataFrames and Spark SQL
5. PySpark ML basics
6. Connecting MongoDB to Python
7. CRUD operations in MongoDB
8. Indexing and Aggregation
9. Replication and Sharding demo
10. Case study: Big Data pipeline
11. Real-time analytics using Spark Streaming

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III B. Tech. - I Semester

23ACD49	DATA SCIENCE FOR ENGINEERS	L	T	P	C
		3	0	0	3

Course Objectives:

- Provide a foundational understanding of data science processes and applications.
- Introduce key tools and techniques such as Python, statistics, data cleaning, visualization, and machine learning.
- Develop practical skills in data analysis, interpretation, and data storytelling.
- Enable students to work on real-world datasets using data science techniques.
- Prepare students for advanced studies or industry roles in data science and analytics.

Course Outcomes:

After completion of the course, students will be able to:

CO1:	Explain the data science lifecycle and its importance in business and research.	Understand(L2)
CO2:	Use Python and libraries like Pandas, NumPy, and Matplotlib for data handling.	Apply(L3)
CO3:	Perform data cleaning, transformation, and visualization effectively.	Apply(L3)
CO4:	Apply basic machine learning models for classification and regression.	Apply(L3)
CO5:	Interpret data analysis results and communicate findings clearly.	Analyze(L4)

Unit I: Introduction to Data Science: What is Data Science? Role of Data Scientist, Data Science Process (Problem definition, data collection, preprocessing, modeling, evaluation), Applications of Data Science in different domains, Tools: Jupyter, Anaconda, Python/R Overview

Unit II: Data Handling and Preprocessing: Introduction to NumPy and Pandas, Reading data from CSV, Excel, SQL, Data Wrangling: Missing values, duplicates, outliers, Data transformation: Scaling, encoding, Feature engineering basics

Unit III: Data Visualization: Importance of visualization, Visualization libraries (Matplotlib, Seaborn), Histograms, Boxplots, Pairplots, Heatmaps, Dashboards and Storytelling with Data, Real-time data dashboards (Optional)

Unit IV: Statistical Foundations for Data Science: Descriptive Statistics, Probability and Probability Distributions, Inferential Statistics: Hypothesis Testing, Confidence Intervals, Correlation and Causation, Use of Scipy/Statsmodels for statistical analysis

Unit V: Introduction to Machine Learning: Supervised vs Unsupervised Learning, Classification and Regression problems, Basic ML Algorithms: Linear Regression, Logistic Regression, KNN, Decision Trees, Model Evaluation Metrics: Accuracy, Precision, Recall, F1-Score, Overfitting and Underfitting

Textbooks:

4. **Joel Grus** – Data Science from Scratch: First Principles with Python, O'Reilly.
5. **Cathy O'Neil and Rachel Schutt** – Doing Data Science, O'Reilly.
6. **Wes McKinney** – Python for Data Analysis, O'Reilly.

Reference Books:

4. **Jake VanderPlas** – Python Data Science Handbook, O'Reilly.
5. **Andreas Müller & Sarah Guido** – Introduction to Machine Learning with Python.
6. **Han, Kamber, & Pei** – Data Mining: Concepts and Techniques, Morgan Kaufmann.

Online Courses:

NPTEL / SWAYAM:

- NPTEL: Introduction to Data Science
  Instructor: Prof. Raghunathan Rengasamy, IIT Madras

Coursera:

- **IBM Data Science Professional Certificate**
  Link: coursera.org
- **Introduction to Data Science in Python (University of Michigan)**
  Link: coursera.org

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III B. Tech. - I Semester

		L	T	P	C
23ACD50	**STATISTICAL LEARNING FOR DATA SCIENCE**	**3**	**0**	**0**	**3**

Course Objectives:

- Introduce fundamental concepts of statistical learning and its importance in data science.
- Develop understanding of regression, classification, and resampling methods.
- Build skills in model assessment, selection, and regularization techniques.
- Apply statistical learning methods to real-world datasets.
- Interpret, communicate, and evaluate data-driven models.

Course Outcomes:

After completion of the course, students will be able to:

CO1:	Understand the basics of statistical learning and data representation.	Understand(L2)
CO2:	Apply linear regression, logistic regression, and classification techniques.	Apply(L3)
CO3:	Evaluate models using resampling methods and cross-validation.	Evaluate(L5)
CO4:	Analyze regularization methods like Ridge and Lasso to avoid overfitting.	Analyze(L4)
CO5:	Create and interpret statistical learning models in practical scenarios.	Create(L6)

Unit I: Introduction to Statistical Learning: What is Statistical Learning?, Supervised vs Unsupervised Learning, Model Accuracy vs Interpretability, Bias-Variance Trade-off, Curse of Dimensionality, Applications of Statistical Learning

Unit II: Linear and Polynomial Regression: Simple Linear Regression. Multiple Linear Regression, Polynomial Regression, Assumptions of Linear Models, Model Diagnostics and Performance Metrics (R^2, RMSE), Variable Selection Techniques

Unit III: Classification Methods: Logistic Regression, Discriminant Analysis (LDA, QDA), Naïve Bayes Classifier, K-Nearest Neighbors, Confusion Matrix, Precision, Recall, F1-Score

Unit IV: Resampling and Model Assessment: Train-Test Split, Cross Validation (k-Fold, LOOCV), Bootstrap Methods, Model Selection and Hyperparameter Tuning.

Unit V: Regularization & Advanced Topics: Ridge Regression, Lasso Regression, Shrinkage Methods, Principal Component Regression (PCR), Partial Least Squares (PLS), Introduction to Support Vector Machines.

Textbooks

1. **Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani** An Introduction to Statistical Learning with Applications in R – Springer (Free PDF: https://www.statlearning.com/)
2. **Trevor Hastie, Robert Tibshirani, Jerome Friedman**
   The Elements of Statistical Learning – Springer

Reference Books:

1. **Norman Matloff** – Statistical Regression and Classification: from R to Data Science
2. **Chris Bishop** – Pattern Recognition and Machine Learning
3. **T. Ryan** – Modern Regression Methods

Online Courses (Free + Paid)

NPTEL / SWAYAM:

- NPTEL: Introduction to Statistical Learning
  Instructor: Prof. Balaraman Ravindran (IIT Madras)

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B. Tech. - II Semester

		L	T	P	C
23ACD51	**DATA VISUALIZATION & EXPLORATORY DATA ANALYSIS**	**3**	**0**	**0**	**3**

Course Objectives:

- To understand the role of data visualization and EDA in the data science lifecycle.
- To explore data types, distributions, missing values, and outliers using statistical and visual methods.
- To gain expertise in using tools such as Python (Matplotlib, Seaborn, Plotly) or R for EDA.
- To develop meaningful, interactive visualizations for both univariate and multivariate data.
- To interpret trends and communicate data-driven insights effectively through dashboards and storytelling.

Course Outcomes:

At the end of the course, the student will be able to

CO1:	Understand the significance of data exploration and visualization in the data analysis process.	Understand(L2)
CO2:	Identify and preprocess anomalies, missing data, and data types through exploratory methods.	Analyze(L4)
CO3:	Apply visual and statistical techniques to univariate, bivariate, and multivariate data.	Apply(L3)
CO4:	Evaluate different chart types and visualization libraries/tools for effective communication.	Evaluate(L5)
CO5:	Create storytelling dashboards and dynamic visual reports for real-world datasets.	Create(L6)

Unit I: Introduction to EDA and Data Types

What is Exploratory Data Analysis?,Types of Data (Categorical, Numerical, Ordinal), Central tendency & dispersion: Mean, Median, Mode, Variance, StdDev, Importance of visualization in EDA

Unit II: Data Cleaning &Preprocessing

Handling missing values and imputation. Detecting and handling outliers, Feature engineering basics, Data transformations (scaling, normalization, encoding)

Unit III: Data Visualization Tools & Techniques

Univariate visualizations: Histogram, Boxplot, Violin plot, Pie chart, Bivariate and Multivariate visualizations: Scatter plot, Heatmaps, Pairplot, Time series plots, Density plots, Introduction to Matplotlib, Seaborn, Plotly, and Tableau

Unit IV: Advanced Data Visualization Concepts

Interactive visualizations: Hover, Zoom, Filters, Geospatial visualization using Folium or Mapbox, Visual perception principles (Gestalt, color theory), Visual storytelling and design best practices

Unit V: Dashboards & Real-World Case Studies

Creating dashboards using Power BI/Tableau/Plotly Dash, Building EDA pipelines with Python/R, Case studies: Finance, Healthcare, Social media, IoT, Interpreting visualizations for decision-making

Textbooks:

1. **Alberto Cairo**, The Truthful Art: Data, Charts, and Maps for Communication, New Riders.
2. **Nathan Yau**, Visualize This: The FlowingData Guide to Design, Visualization, and Statistics, Wiley.
3. **Hadley Wickham**, R for Data Science, O'Reilly (for R users).

Reference Books:

1. **Ben Fry**, Visualizing Data: Exploring and Explaining Data with the Processing Environment, O'Reilly.
2. **Claus O. Wilke**, Fundamentals of Data Visualization, O'Reilly.
3. **Scott Murray**, Interactive Data Visualization for the Web, O'Reilly.

Online Courses:

Platform	Course Title & Link
Coursera	Data Visualization with Python – IBM

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B. Tech. - II Semester

		L	T	P	C
23ACD52	**BIG DATA ANALYTICS FUNDAMENTALS**	**3**	**0**	**0**	**3**

Course Objectives:

- Understand the fundamentals of Big Data, its characteristics, and challenges.
- Learn key tools and technologies for storing, processing, and analyzing Big Data (Hadoop, Spark, etc.).
- Explore advanced analytical techniques including MapReduce, machine learning on Big Data, and real-time analytics.
- Develop skills to design and implement Big Data solutions for real-world problems.
- Gain hands-on experience with Big Data frameworks and platforms.

Course Outcomes:

At the end of the course, the student will be able to

CO1:	Understand the core concepts and architecture of Big Data and related technologies.	Understand(L2)
CO2:	Analyze Big Data processing frameworks like Hadoop and Spark for scalability and efficiency.	Analyze(L4)
CO3:	Apply MapReduce programming model to solve Big Data problems.	Apply(L3)
CO4:	Evaluate different storage options (HDFS, NoSQL) and processing techniques for Big Data.	Evaluate(L5)
CO5:	Create end-to-end Big Data analytics pipelines using relevant tools and platforms.	Create(L6)

Unit I: Introduction to Big Data

Definition and Characteristics (Volume, Velocity, Variety, Veracity, Value), Challenges of Big Data, Traditional Data Processing vs Big Data Analytics, Overview of Big Data applications.

Unit II: Big Data Technologies and Ecosystem

Hadoop Framework: HDFS, YARN, MapReduce, Apache Spark: Architecture and Components, NoSQL Databases (HBase, Cassandra, MongoDB), Data ingestion tools (Sqoop, Flume)

Unit III: MapReduce Programming Model

MapReduce fundamentals, Writing Map and Reduce functions, Job configuration and execution, Hands-on example: Word count, log analysis.

Unit IV: Advanced Big Data Analytics

Machine Learning on Big Data using MLlib (Spark), Real-time analytics and streaming data (Spark Streaming, Kafka), Graph processing (GraphX), Data visualization of Big Data

Unit V: Big Data Project and Case Studies

Designing Big Data solutions for healthcare, finance, social media analytics, Performance optimization and security considerations, Cloud-based Big Data platforms (AWS EMR, Google BigQuery), Case studies and project presentations.

Textbooks:

1. **Viktor Mayer-Schönberger, Kenneth Cukier**, Big Data: A Revolution That Will Transform How We Live, Work, and Think, Houghton Mifflin Harcourt.
2. **Tom White**, Hadoop: The Definitive Guide, O'Reilly Media.
3. **Jules S. Damji, Brooke Wenig, Tathagata Das, Denny Lee**, Learning Spark: Lightning-Fast Big Data Analysis, O'Reilly Media.

Reference Books:

1. **Raj Kamal**, Big Data Analytics, McGraw-Hill Education.
2. **Chris Eaton, Dirk DeRoos, Tom Deutsch, George Lapis, Paul Zikopoulos,** Understanding Big Data: Analytics for Enterprise Class Hadoop and Streaming Data, McGraw-Hill.
3. **Anand Rajaraman, Jeffrey David Ullman**, Mining of Massive Datasets, Cambridge University Press.

Online Courses:

Platform	**Course Title & Link**
Coursera	**Big Data Specialization by University of California San Diego**

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B. Tech. - I Semester

23ACD53	INTRODUCTION TO RECOMMENDER SYSTEMS	L	T	P	C
		3	0	0	3

Course Objectives:

- Understand the fundamental concepts and techniques of recommender systems.
- Explore different types of recommendation approaches: content-based, collaborative filtering, and hybrid methods.
- Learn about evaluation metrics and methods to assess recommender system performance.
- Understand challenges such as scalability, sparsity, and cold-start problems.
- Gain hands-on experience implementing and tuning recommender algorithms using real-world datasets

Course Outcomes:

At the end of the course, the student will be able to

CO1:	Explain the basic principles and types of recommender systems.	Understand(L2)
CO2:	Analyze various recommendation algorithms, including collaborative and content-based filtering.	Analyze(L4)
CO3:	Apply different algorithms to build practical recommender system models.	Apply(L3)
CO4:	Evaluate the performance of recommender systems using appropriate metrics.	Evaluate(L5)
CO5:	Design hybrid recommender systems that address common challenges.	Create(L6)

Unit I: Introduction to Recommender Systems

Definition and applications (e-commerce, media, social networks), Types of recommender systems: content-based, collaborative filtering, hybrid, Overview of user-item interactions

Unit II: Content-Based Recommendation

Item profiles and user profiles, Similarity measures (cosine similarity, Pearson correlation), Building content-based recommenders, Advantages and limitations

Unit III: Collaborative Filtering Techniques

User-based collaborative filtering, Item-based collaborative filtering, Matrix factorization techniques (SVD, PCA), Addressing sparsity and scalability issues.

Unit IV: Evaluation of Recommender Systems

Metrics: Precision, Recall, F1-score, RMSE, MAE, Cross-validation and train-test split, Handling cold-start and diversity, Case studies on real-world evaluation.

Unit V: Advanced Topics and Hybrid Systems

Hybrid recommender systems (weighted, switching, feature combination), Context-aware recommendations, Deep learning in recommender systems, Privacy, ethics, and future trends

Textbooks:

1. **Charu C. Aggarwal**, Recommender Systems: The Textbook, Springer, 2016.
2. **Francesco Ricci, LiorRokach, BrachaShapira, Paul B. Kantor (Editors),** Recommender Systems Handbook, Springer, 2015.

Reference Books:

1. **Dietmar Jannach, Markus Zanker, Alexander Felfernig, Gerhard Friedrich,** Recommender Systems: An Introduction, Cambridge University Press, 2010.
2. **Joseph A. Konstan, John Riedl**, Collaborative Filtering Recommender Systems, Now Publishers, 2012.
3. **GuibingGuo, Jie Zhang, Neil Yorke-Smith**, Deep Learning Based Recommender System, Springer, 2020.

Online Courses:

Platform	Course Title & Link
Coursera	Recommender Systems Specialization by University of Minnesota

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

III B. Tech. - II Semester

		L	T	P	C
23ACD54	**DATA VISUALIZATION & EDA LAB**	**0**	**0**	**3**	**1.5**

Course Objectives:

- To explore and visualize datasets.
- To derive insights through analysis

Course Outcomes:

At the end of the course, the student will be able to

- Conduct EDA on datasets.
- Create meaningful visualizations

Experiments:

1. Importing data from different sources
2. Handling missing values and outliers
3. Univariate and bivariate analysis
4. Boxplots, Histograms, Heatmaps
5. Correlation matrix
6. Using pandas profiling
7. Data visualization using Plotly
8. Pairplots and violin plots
9. Feature engineering basics
10. Time series visualization
11. Dashboards with Streamlit

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
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IV B. Tech. - I Semester

		L	T	P	C
23ACD55	**RECOMMENDER SYSTEM LAB**	**0**	**0**	**3**	**1.5**

Course Objectives:

- To understand collaborative and content filtering.
- To build and test recommender models.

Course Outcomes:

At the end of the course, the student will be able to

- Build simple recommender systems.
- Evaluate recommender performance

Experiments:

1. Introduction to recommender systems
2. User-item interaction matrix
3. Cosine similarity for recommendations
4. Collaborative filtering – user-based
5. Collaborative filtering – item-based
6. Content-based filtering
7. Surprise library intro
8. Building SVD model
9. Hybrid recommender
10. Evaluation metrics (RMSE, MAE)
11. Recommendation visualization

MINOR DEGREE IN DATA SCIENCE WITH AWS

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (AUTONOMOUS)
III B.Tech. I SEM, CSE(DS)

		L	T	P	C
23ACD56	**FOUNDATIONS OF CLOUD CONCEPTS**	**3**	**0**	**0**	**3**

COURSE OBJECTIVES:
The objective of the course is,

- Understand the fundamental concepts of cloud computing and its relevance in modern technology.
- Gain knowledge of AWS global infrastructure and its core services.
- Explore essential cloud deployment models and service models.
- Learn about cloud security and compliance in AWS
- Explore overview to deploy, monitor, and scale applications using AWS.

COURSE OUTCOMES:

After the completion of the course, student will be able to

CO1. Define fundamental cloud concepts and AWS core services
CO2. Explain the AWS global infrastructure and its key features
CO3. Implement basic AWS services for application deployment and management
CO4. Examine cloud security and compliance strategies in AWS
CO5. Design, deploy and monitor scalable applications using AWS

UNIT-I: Introduction to Cloud Computing **6 Hrs**
Definition and Characteristics of Cloud Computing, Cloud Deployment Models (Public, Private, Hybrid), Cloud Service Models (IaaS, PaaS, SaaS), Benefits and Challenges of Cloud Computing, Overview of AWS as a Cloud Service Provider.

UNIT-II: AWS Global Infrastructure and Core Services **10 Hrs**
Understanding AWS Global Infrastructure (Regions, Availability Zones, and Edge Locations), AWS Core Services Overview: Compute Services: Amazon EC2, AWS Lambda, Storage Services: Amazon S3, Amazon EBS, Database Services: Amazon RDS, DynamoDB, Networking Services: Amazon VPC, Elastic Load Balancing, Use Cases of Core Services

UNIT-III: Cloud Security and Identity Management **8 Hrs**
Shared Responsibility Model in AWS, Introduction to AWS Identity and Access Management (IAM), Creating IAM Users, Groups, and Roles, Security Tools: AWS Key Management Service (KMS), CloudTrail, Compliance and Governance in AWS

UNIT-IV: Application Deployment and Management in AWS
Basics of Application Deployment, Elastic Beanstalk for Web Application Deployment, AWS Auto Scaling and Load Balancing, Monitoring Services: Amazon CloudWatch, Backup and Disaster Recovery Using AWS Services, Introduction to AWS Management
Console and AWS CL

UNIT V: Project Work and Advanced Topics **7 Hrs**

Project Objective: Deploy and Scale a Web Application using AWS, Project Execution: Setup and Configure AWS Environment, Launch and Monitor an EC2 Instance, Design and Implement a Scalable Application, Use S3 for Application Storage, Use CloudWatch for Monitoring, Introduction to Advanced AWS Services (AWS Machine Learning, AWS IoT, AWS DevOps)

TEXT BOOKS:

1. AWS Certified Solutions Architect Study Guide by Ben Piper and David Clinton

REFERENCE BOOKS:

1. Dipankar Sarkar, Cloud Computing with AWS, (2023, 4th Edition) (2021, 1st Edition)

E-resources

1. AWS Training and Certification E-learning Content
2. AWS Documentation and Whitepapers

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)
III B.Tech. I SEM, CSE(DS)

		L	T	P	C
23ACD57	**PREDICTIVE DATA ANALYTICS USING R**	**3**	**0**	**0**	**3**

COURSE OBJECTIVES:

The objective of the course is,

- To understand the fundamentals of predictive analytics and statistical modeling.
- To gain hands-on experience in using R for data analysis.
- To learn various predictive modeling techniques for real-world applications.

COURSE OUTCOMES: After the completion of the course, student will be able to

CO1. Explain the fundamental concepts of predictive analytics and the role of R in statistical modeling and data-driven decision-making.

CO2. Apply data preprocessing, cleaning, and transformation techniques using R for predictive modeling.

CO3. Develop and implement regression, classification, and clustering models in R for solving real-world problems.

CO4. Evaluate predictive models using performance metrics such as accuracy, precision, recall, ROC curves, and error measures.

CO5. Interpret and visualize predictive analytics results using R libraries (e.g., ggplot2, caret, randomForest) to support business or research decisions.

CO6. Design and execute end-to-end predictive analytics projects using R, integrating data preparation, model building, validation, and reporting.

UNIT-I: Introduction to Predictive Analytics **8 Hrs**

Overview of Data Analytics and Predictive Analytics, Importance and Applications in Business and Research, Introduction to R Programming for Predictive Analytics, R Studio Environment and Data Handling

UNIT-II: Data Preparation and Exploration **9 Hrs**

Data Import, Cleaning, and Transformation, Exploratory Data Analysis (EDA), Data Visualization using ggplot2, Handling Missing Values and Outliers

UNIT-III: Regression Models **8 Hrs**

Simple Linear Regression, Multiple Linear Regression, Logistic Regression, Model Diagnostics and Assumptions Checking

UNIT-IV: Classification Techniques **9 Hrs**

Decision Trees and Random Forests, Support Vector Machines, Naïve Bayes Classifier, Model Evaluation: Accuracy, Precision, Recall, F1 Score

UNIT V: Advanced Predictive Modeling **8 Hrs**

Time Series Forecasting, Clustering Techniques (K-Means, Hierarchical), Ensemble Methods (Bagging, Boosting), Case Studies and Applications in Predictive Analytics

TEXT BOOKS:

1. Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, "An Introduction to Statistical Learning with Applications in R".

REFERENCE BOOKS:

1. Brett Lantz, "Machine Learning with R".
2. Trevor Hastie, Robert Tibshirani, and Jerome Friedman, "The Elements of Statistical Learning"

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)
III B.Tech. II SEM, CSE(DS)

23ACD58	**DATA ENGINEERING WITH AWS**	**L**	**T**	**P**	**C**
		3	**0**	**0**	**3**

COURSE OBJECTIVES:

The objective of the course is,

- Understand the principles of data engineering and its importance in cloud ecosystems.
- Explore AWS services for data storage, processing, and transformation.
- Learn how to design and implement data pipelines using AWS services.
- Gain hands-on experience with real-time and batch data processing on AWS.
- Develop skills to build and manage scalable and efficient data workflows.

UNIT-I: Introduction to Data Engineering and AWS **6 Hrs**

Overview of Data Engineering, Key Concepts in Data Storage, Processing, and Transformation, Introduction to AWS Data Engineering Services, AWS Data Engineering Use Cases

UNIT-II: Data Storage Solutions on AWS **10 Hrs**

Overview of AWS Storage Services, Amazon S3: Object Storage, Amazon RDS and DynamoDB: Relational and NoSQL Databases, Amazon Redshift: Data Warehousing, Best Practices for Secure and Efficient Data Storage, Data Lifecycle Management and Archiving

UNIT-III: Data Processing with AWS **10 Hrs**

Introduction to Data Processing Frameworks, AWS Glue: Data Catalog and ETL Services, Amazon , EMR: Big Data Processing with Hadoop and Spark, Real-Time Data Processing with Amazon Kinesis, Batch Processing vs. Real-Time Processing

UNIT-IV: Data Pipeline Design and Management **12 Hrs**

Designing Scalable Data Pipelines, AWS Data Pipeline: Workflow Orchestration, Serverless Data Pipelines with AWS Lambda and Step Functions, Monitoring and Optimizing Data Pipelines, Error Handling and Data Validation Techniques

UNIT V: Project Work and Advanced Topics **7 Hrs**

Project Objective: Build a Scalable Data Pipeline on AWS, Project Execution:Setup AWS Data Storage and Processing Environment, Implement Data Extraction, Transformation, and Loading (ETL) Pipeline, Integrate Real-Time and Batch Data Processing, Optimize Performance and Cost, Advanced Topics: Introduction to Amazon SageMaker for Data Engineering, Data Lake Formation and Management on AWS, Advanced Security and Compliance Features

TEXT BOOKS:

1. Data Engineering with AWS by Gareth Eagar (2022, 1st Edition)

REFERENCE BOOKS:

1. Designing Data-Intensive Applications by Martin Kleppmann (2017, 1st Edition), E-resources
2. AWS Training and Certification E-learning Content
3. AWS Documentation and Whitepapers

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
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III B.Tech. II SEM, CSE(DS)

		L	T	P	C
23ACD59	**CLOUD-NATIVE MACHINE LEARNING**	**3**	**0**	**0**	**3**

COURSE OBJECTIVES:
The objective of the course is,

- Understand the fundamentals of machine learning (ML) and its applications.
- Gain familiarity with Amazon SageMaker as a platform for building, training, and deploying ML models.
- Learn about different ML algorithms and their use cases.
- Develop skills to preprocess data, train models, and optimize hyperparameters using SageMaker.
- Apply ML techniques to solve real-world problems using cloud-based solutions.

COURSE OUTCOMES: At the end of the course, the student will be able to:

- Define key concepts in machine learning and the role of SageMaker
- Explain the workflow for building, training, and deploying ML models.
- Utilize SageMaker for data preparation, model training, and Optimization
- Evaluate the performance of ML models and optimize deployments.
- Build and deploy scalable ML pipelines using SageMaker and AWS services.

UNIT-I: Introduction to Machine Learning and SageMaker 6 Hrs

Fundamentals of Machine Learning: Supervised and Unsupervised Learning, Key ML Concepts: Training, Testing, and Validation; Overview of Amazon SageMaker- SageMaker Studio and Key Features, Workflow: Build, Train, and Deploy; Benefits of Using SageMaker for ML Projects

UNIT-II: Data Preparation and Feature Engineering 10 Hrs

Importing Data into SageMaker (S3 Integration). Data Cleaning and Transformation- Handling Missing Values and Outliers, Scaling and Encoding Data; Feature Engineering for ML- Feature Selection and Extraction, Feature Store in SageMaker

UNIT-III: Training and Optimizing ML Models 12 Hrs

Built-in Algorithms in SageMaker, Linear Learner, XGBoost, K-Means, and Factorization Machines, Training ML Models on SageMaker- Choosing the Right Instance Type, Configuring Training Jobs; Hyper parameter Optimization- Using SageMaker Automatic Model Tuning, Debugging and Monitoring Training Jobs

UNIT-IV: Model Deployment and Monitoring 10 Hrs

Deploying Models with SageMaker- Real-time Endpoints, Batch Transform Jobs; Monitoring Deployed Models- Using CloudWatch for Metrics, Debugging Deployment Issues; Cost Optimization for Deployment

7 Hrs

UNIT V: Project Work and Advanced Topics

Project Objective: Build and Deploy an End-to-End ML Pipeline on SageMaker- Project Execution:Data Preparation and Feature Engineering- Model Training and Hyper parameter Tuning, Deploying and Monitoring the Model; Advanced Topics: SageMaker Pipelines for Automation, Leveraging SageMaker for Time-Series Forecasting, Integration with AWS ML Services (Rekognition, Comprehend, and Polly)

TEXT BOOKS:

1. Practical Machine Learning with Amazon SageMaker by Oliver Zeigermann (2021, 1st Edition)

REFERENCE BOOKS:

1. Hands-On Machine Learning with Scikit-Learn by Aurélien Géron (2020, 2nd Edition)
2. Data Science on AWS by Chris Fregly and Antje Barth (2021, 1st Edition)

E-resources

1. AWS Training and Certification E-learning Content
2. SageMaker Documentation and Whitepapers

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (AUTONOMOUS)
IV B.Tech. I SEM, CSE(DS)

23ACD60	DEEP LEARNING: THE MODERN APPROACH	L	T	P	C
		3	0	0	3

COURSE OBJECTIVES:
The objective of the course is,
- Understand the principles of deep learning and its applications across various domains.
- Explore the capabilities of Amazon SageMaker for training, deploying, and managing deep learning models.
- Gain proficiency in popular deep learning frameworks such as TensorFlow and PyTorch.
- Learn to optimize, scale, and monitor deep learning workflows in cloud environments.
- Apply deep learning techniques to real-world problems using SageMaker.

COURSE OUTCOMES: At the end of the course, the student will be able to:
- Define the principles and techniques of deep learning.
- Explain how to build and train deep learning models using SageMaker.
- Utilize SageMaker for deploying and monitoring deep learning models.
- Optimize deep learning workflows for scalability and cost- effectiveness.
- Design, train, and deploy deep learning models for real- world applications.

UNIT-I: Introduction to Deep Learning and Amazon SageMake 6 **Hrs**

Fundamentals of Deep Learning: Neural Networks: Perceptrons, Activation Functions- Forward and Backward Propagation, Loss Functions and Optimization Techniques; Overview of Amazon SageMaker- SageMaker Studio Features, Workflow: Build, Train, and Deploy, Integration with AWS Ecosystem

UNIT-II: Training Deep Learning Models with SageMaker 10 rs

Preparing Data for Deep Learning- Loading and Preprocessing Large Datasets, Using AWS S3 for Dataset Management; Training Deep Learning Models- Built-in Algorithms in SageMaker for Deep Learning, Custom Training with TensorFlow and PyTorch; Distributed Training- Leveraging GPU Instances, Optimizing Training Jobs for Scalability

UNIT-III: Hyperparameter Tuning and Model Optimization **10 Hrs**

Hyperparameter Tuning- SageMaker Automatic Model Tuning, Best Practices for Tuning Deep Learning Models; Model Optimization- Quantization and Pruning Techniques, Reducing Latency for Deployment; Debugging and Profiling- Using SageMaker Debugger for Insights, Identifying Bottlenecks in Training and Inference

UNIT-IV: Deployment and Monitoring of Deep Learning Models **12Hrs**

Model Deployment Strategies: Real-time Inference Endpoints, Batch Transform for Large-scale Inference; Monitoring Deployed Models- Using Amazon CloudWatch for Performance Metrics, Model Drift Detection and Retraining; Managing Deployment Costs - Cost Optimization Strategies in SageMaker, Managing Spot Instances for Training and Inference

UNIT V: Project Work and Advanced Topics **7 Hrs**

Project Objective: Build and Deploy a Deep Learning Model with SageMaker, Project Execution: Model Training and Hyperparameter Tuning, Deploying and Monitoring the Model; Advanced Topics:Using SageMaker Neo for Optimized Inference, Explainability in Deep Learning Models, Integration with AWS AI Services (Rekognition, Translate, Comprehend)

TEXT BOOKS:

1. Deep Learning with Python by Francois Chollet (2021, 2nd Edition)
2. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow by Aurélien Géron (2022, 3rd Edition)

REFERENCE BOOKS:

1. Practical Deep Learning for Cloud, Mobile, and Edge by Anirudh Koul, Siddha Ganju, Meher Kasam (2020, 1st Edition)

E-resources

1. AWS SageMaker Documentation
2. TensorFlow and PyTorch Tutorials
3. AWS AI and Machine Learning Whitepapers

MINORS IN QUANTUM TECHNOLOGIES

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

Minors in Information Technology (R23)

23AIT24: FOUNDATIONS OF QUANTUM TECHNOLOGIES

L	T	P	C
3	0	0	3

Course Objectives:

1. Introduce the fundamental quantum mechanics concepts essential for quantum technologies.
2. Build strong mathematical foundations for quantum state modeling.
3. Develop understanding of superposition, entanglement, and measurement.
4. Explain the physical principles behind quantum devices.
5. Prepare students for advanced studies in quantum computation, communication, sensing, and materials.

Course Outcomes (COs):

After successful completion of this course, the students will be able to:

CO1: Understand postulates of quantum mechanics for quantum technologies

CO2: Apply linear algebra and Dirac notation to quantum state analysis CO3:

Analyze superposition, entanglement, and measurement processes

CO4: Evaluate quantum systems through operators and probability amplitudes

CO5: Create mathematical models for simple quantum systems

UNIT I – Quantum Mechanics Foundations *(Cognitive Level: Understand)*

Classical vs Quantum systems, Wave-particle duality, Schrödinger equation (Time-dependent and Time-independent), Postulates of Quantum Mechanics, Quantum states and state vectors, Complex Hilbert spaces, Dirac notation (Bra-Ket notation), Probabilistic interpretation of quantum mechanics

UNIT II – Linear Algebra for Quantum Systems *(Cognitive Level: Apply)*

Complex vector spaces and inner products, Orthonormal basis and orthogonality, Linear operators and transformations, Unitary operators and Hermitian operators, Tensor products for multi-qubit systems, Eigenvalues and Eigenvectors, Commutators, and anti-commutators, Representing quantum states with matrices

UNIT III – Superposition, Measurement, and Entanglement ***(Cognitive Level: Analyze)***

Principle of superposition, Measurement postulate, Probability amplitudes and Born rule, State collapse upon measurement, Entanglement and Bell states, EPR paradox and non-locality, Density matrices and mixed states, Quantum decoherence

UNIT IV – Operators and Quantum Dynamics ***(Cognitive Level: Evaluate)***

Time evolution operators, Hamiltonian and energy eigenstates, Quantum harmonic oscillator (brief overview), Unitary evolution and Schrödinger equation solutions, Quantum tunnelling, Adiabatic theorem basics, Operator algebra in quantum systems, Expectation values and observables

UNIT V – Quantum Technologies Building Blocks ***(Cognitive Level: Create)***

Basic qubit systems (spin-1/2, photon polarization, superconducting qubits), Two-level quantum systems modelling, Bloch sphere representation, Quantum logic gates fundamentals, Multi-qubit systems: controlled operations, Introduction to decoherence and quantum error correction, Quantum technologies: hardware platforms overview, Basic quantum circuit modeling using simulators (Qiskit or Q# demo examples)

Textbooks:

[1] M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information, 10th Anniversary ed., Cambridge, U.K.: Cambridge Univ. Press, 2010.

[2] N. D. Mermin, Quantum Computer Science: An Introduction, Cambridge, U.K.: Cambridge Univ. Press, 2007.

[3] D. McMahon, Quantum Computing Explained, Hoboken, NJ, USA: Wiley, 2008.

Reference Books:

[1] D. J. Griffiths, Introduction to Quantum Mechanics, 2nd ed., Upper Saddle River, NJ, USA: Pearson Prentice Hall, 2005.

[2] J. J. Sakurai and J. Napolitano, Modern Quantum Mechanics, 2nd ed., Cambridge, U.K.: Cambridge Univ. Press, 2020.

[3] J. Watrous, The Theory of Quantum Information, Cambridge, U.K.: Cambridge Univ. Press, 2018.

[4] V. K. Krishnan, Linear Algebra and Its Applications to Quantum Computing, Singapore: Springer, 2021.

Online Courses & Resources

Platform	Course Title
MIT OpenCourseWare	Quantum Physics I, II (MIT OCW 8.04 & 8.05)
edX (Berkeley)	Quantum Mechanics and Quantum Computation

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

Minors in Information Technology (R23)

23AIT25: SOLID STATE PHYSICS FOR QUANTUM TECHNOLOGIES

L	T	P	C
3	0	0	3

Course Objectives:

1. Understand fundamental solid-state physics principles relevant to quantum technologies.
2. Study the electronic properties of materials used in quantum hardware.
3. Explore quantum confinement and nanostructures for qubit implementation.
4. Analyze crystal structures, band theory, and defects influencing quantum devices.
5. Build foundations for material selection and engineering for quantum systems.

Course Outcomes (COs):

After successful completion of this course, the students will be able to:

CO1: Understand crystal structures and band theory

CO2: Apply knowledge of semiconductors, insulators, and conductors in quantum materials CO3:

Analyze quantum confinement effects and low-dimensional systems

CO4: Evaluate defects, phonons, and interactions in solid-state systems

CO5: Create models for quantum device material systems

UNIT I – Crystal Structure and Electronic Properties *(Cognitive Level: Understand)*

Crystal lattices and unit cells, Bravais lattices, Miller indices, Reciprocal lattice and Brillouin zones, atomic bonding in solids (covalent, ionic, metallic, van der Waals), X-ray diffraction and crystal structure determination, electronic structure of solids, Free electron theory, Energy bands: metals, semiconductors, and insulators

UNIT II – Semiconductor Physics for Quantum Devices *(Cognitive Level: **Apply**)*

Intrinsic and extrinsic semiconductors, Charge carriers: electrons, holes, effective mass, Carrier concentration and Fermi level, p-n junctions and semiconductor heterostructures, Quantum wells and quantum dots as qubits, Superconductors and Josephson junctions, Semiconductor fabrication basics, Materials for quantum hardware: Si, GaAs, diamond NV centers, topological insulators

UNIT III – Quantum Confinement and Low-Dimensional Systems *(Cognitive Level: Analyze)*

Quantum size effects: nanowires, nanotubes, 2D materials, Quantum dots: discrete energy levels, Quantum Hall effect, Topological quantum materials, Spintronics and spin qubits, Quantum confinement in superconducting qubits, Heterostructure-based quantum devices, Valleytronics and emerging 2D materials (MoS_2, graphene)

UNIT IV – Lattice Vibrations and Phonon Interactions *(Cognitive Level: Evaluate)*

Lattice vibrations and phonons, Heat capacity and thermal conductivity of solids, Electron-phonon interaction, Decoherence in solid-state qubits due to phonons, Magnetic impurities and Kondo effect, Defects and dislocations in crystals, Dopants and quantum impurity systems, nuclear spin environments and coherence times

UNIT V – Materials for Quantum Technologies *(Cognitive Level: Create)*

Material engineering for superconducting qubits, NV centers in diamond for quantum sensing, Topological materials for robust qubits, Photonic crystal materials for optical qubits, Hybrid quantum systems: coupling different materials, Fabrication challenges and material purity, Advances in quantum materials research, Designing material systems for long coherence time

Text books:

[1] C. Kittel, Introduction to Solid State Physics, 8th ed., Hoboken, NJ, USA: Wiley, 2004.

[2] M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information, 10th Anniversary ed., Cambridge, U.K.: Cambridge Univ. Press, 2010.

[3] S. L. Altmann, Band Theory of Solids: An Introduction from the Point of View of Symmetry, Oxford, U.K.: Oxford Univ. Press, 1994.

Reference Books:

[1] N. W. Ashcroft and N. D. Mermin, Solid State Physics, Boston, MA, USA: Cengage Learning, 1976.

[2] P. Y. Yu and M. Cardona, Fundamentals of Semiconductors: Physics and Materials Properties, 4th ed., Berlin, Germany: Springer, 2010.

[3] D. D. Awschalom, D. Loss, and N. Samarth, Semiconductor Spintronics and Quantum Computation, Berlin, Germany: Springer, 2002.

[4] D. Vollhardt and P. Wölfle, Theoretical Methods for Strongly Correlated Electrons: Introduction to the Theory of Many-Body Systems, Mineola, NY, USA: Dover Publications, 2013.

Online Courses & Resources:

Platform	Course Title
MIT OpenCourseWare	Solid State Physics (MIT 8.231)
edX	Quantum Materials and Devices (U. Tokyo)
Coursera	Quantum Materials (ÉcolePolytechnique)

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

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Minors in Information Technology (R23)

23AIT26: QUANTUM OPTICS PREREQUISITES FOR QUANTUM TECHNOLOGIES

L	T	P	C
3	0	0	3

Course Objectives:

1. Introduce fundamentals of light-matter interaction relevant for quantum technologies.
2. Explain the quantization of electromagnetic fields.
3. Study the role of photons as quantum information carriers.
4. Explore coherent states, squeezed states, and single-photon sources.
5. Prepare for quantum sensing, communication, and photonic quantum computing applications.

Course Outcomes (COs):

After successful completion of this course, the students will be able to:

CO1: Understand quantum nature of light

CO2: Apply Maxwell’s equations to optical fields CO3:

Analyze interaction of photons with matter

CO4: Evaluate coherence, squeezing, and quantum noise CO5:

Create models for photonic quantum systems

UNIT I – Classical and Quantum Description of Light *(Cognitive Level: Understand)*

Review of electromagnetic waves, Maxwell’s equations for light propagation, Plane waves, polarization, Poynting vector, Classical interference, diffraction, coherence, Blackbody radiation & Planck's hypothesis, Photoelectric effect, Photons as quantized light energy, Introduction to quantum theory of radiation

UNIT II – Quantization of Electromagnetic Field *(Cognitive Level: Apply)*

Harmonic oscillator quantization, Field quantization in free space, Photon number (Fock) states, Coherent states and classical-quantum correspondence, Vacuum fluctuations and zero-point energy, Single-mode vs multi-mode quantization, Spontaneous and stimulated emission, Quantum field operators and commutation relations

UNIT III – Light-Matter Interaction *(Cognitive Level: Analyze)*

Two-level atom model, Absorption, stimulated emission, spontaneous emission, Einstein coefficients, Rabi oscillations, Jaynes-Cummings model, Resonant and non-resonant interaction, Cavity Quantum Electrodynamics (Cavity-QED), Atom-photon entanglement

UNIT IV – Quantum Coherence and Quantum Noise *(Cognitive Level: Evaluate)*

Classical vs quantum coherence, First- and second-order coherence functions, Photon antibunching, Hanbury Brown and Twiss experiment, Quantum squeezing of light, Phase-sensitive amplification, Quantum noise, shot noise, and standard quantum limit, Quantum nondemolition measurements

UNIT V – Quantum Photonics Applications *(Cognitive Level: Create)*

Single-photon sources (quantum dots, NV centers, SPDC), Entangled photon pair generation, Photonic qubits and linear optical quantum computing, Quantum key distribution with photons, Photonic integrated circuits, Quantum sensors based on squeezed light, Quantum metrology using entangled photons, Designing experiments for quantum optics labs

Text books:

[1] M. Fox, Quantum Optics: An Introduction, Oxford, U.K.: Oxford Univ. Press, 2006.

[2] R. Loudon, The Quantum Theory of Light, 3rd ed., Oxford, U.K.: Oxford Univ. Press, 2000.

[3] M. O. Scully and M. S. Zubairy, Quantum Optics, Cambridge, U.K.: Cambridge Univ. Press, 1997.

Reference Books:

[1] S. M. Barnett, Quantum Information, Oxford, U.K.: Oxford Univ. Press, 2009.

[2] P. Meystre and M. Sargent, Elements of Quantum Optics, 4th ed., Berlin, Germany: Springer, 2007.

[3] M. Le Bellac, Quantum Physics, Cambridge, U.K.: Cambridge Univ. Press, 2006.

[4] D. F. Walls and G. J. Milburn, Quantum Optics, 2nd ed., Berlin, Germany: Springer, 2008.

Online Courses & Resources:

Platform	Course Title
MIT OpenCourseWare	Quantum Optics (MIT 8.421)
edX	Principles of Photonics (EPFL)
Coursera	Quantum Optics 1 & 2 (U. Rochester)
YouTube	Quantum Optics Lectures (Various universities)

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

Minors in Information Technology (R23)

23AIT27: Introduction to Quantum Communication

L	T	P	C
3	0	0	3

Course Objectives:

1. Introduce fundamental principles of quantum communication.
2. Study quantum key distribution (QKD) protocols.
3. Analyze quantum teleportation, entanglement swapping, and quantum repeaters.
4. Evaluate quantum security principles and their advantages.
5. Prepare students for designing secure communication protocols for future quantum networks.

Course Outcomes (COs):

After successful completion of this course, the students will be able to:

CO1: Understand quantum communication concepts

CO2: Apply quantum entanglement to communication protocols

CO3: Analyze QKD protocols and teleportation mechanisms CO4:

Evaluate security of quantum communication

CO5: Design quantum communication networks and protocols

UNIT I – Introduction to Quantum Communication *(Cognitive Level: Understand)*

Classical communication vs quantum communication, No-cloning theorem and quantum information security, Qubits and qubit transmission channels, Quantum entanglement fundamentals, EPR paradox and Bell's inequalities, Quantum states and measurement, Role of superposition and measurement collapse, Overview of quantum internet and its architecture

UNIT II – Quantum Key Distribution (QKD) Protocols *(Cognitive Level: Apply)*

Classical cryptography limitations, BB84 protocol, B92 protocol, E91 entanglement-based protocol, Decoy-state QKD, Device-independent QKD, Practical implementation challenges in QKD, Experimental QKD systems (fiber, free-space, satellites)

UNIT III – Quantum Teleportation and Entanglement Distribution *(Cognitive Level: Analyze)*

Quantum teleportation protocol, Entanglement swapping, Quantum repeaters for long-distance communication, Error sources in quantum teleportation, Resource requirements for teleportation, Entanglement purification techniques, Bell state measurements, Applications of teleportation in distributed quantum computing

UNIT IV – Quantum Networks and Quantum Internet *(Cognitive Level: Evaluate)*

Architecture of quantum networks, Quantum routers and switching, Quantum memories and storage nodes, Distributed entanglement generation and management, Multiparty quantum communication Blind quantum computing, Performance metrics for quantum networks (fidelity, key rate), Challenges in large-scale quantum network deployment

UNIT V – Advanced Quantum Communication Protocols and Applications *(Cognitive Level: Create)*

Quantum secure direct communication, Quantum digital signatures, Position-based quantum cryptography, Quantum secret sharing, post-quantum cryptography overview, Quantum cloud communication protocols, Building hybrid quantum-classical communication models, Future directions in quantum communication technology

Text books:

[1] M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information, 10th Anniversary ed., Cambridge, U.K.: Cambridge Univ. Press, 2010.

[2] M. M. Wilde, Quantum Information Theory, 2nd ed., Cambridge, U.K.: Cambridge Univ. Press, 2017.

[3] V. Scarani, Quantum Cryptography: A Primer, Oxford, U.K.: Oxford Univ. Press, 2019.

Reference Books:

[1] V. Dunjko and E. Diamanti, Introduction to Quantum Communication and Cryptography, Cham, Switzerland: Springer, 2023.

[2] N. Lütkenhaus, “Practical security in quantum key distribution,” in Quantum Communication, Measurement and Computing (QCMC), New York, NY, USA: AIP Conf. Proc., vol. 734, pp. 241–248, 2004.

[3] D. McMahon, Quantum Computing Explained, Hoboken, NJ, USA: Wiley, 2008.

[4] D. Bouwmeester, A. K. Ekert, and A. Zeilinger, The Physics of Quantum Information: Quantum Cryptography, Quantum Teleportation, Quantum Computation, Berlin, Germany: Springer, 2000.

Online Courses & Resources:

Platform	Course Title
edX	Quantum Cryptography (ETH Zurich)
Coursera	Quantum Communication (Delft University of Technology)
MIT OpenCourseWare	Quantum Information Science (MIT 6.443)
YouTube	Quantum Internet & Quantum Networking Tutorials
IBM Qiskit	Qiskit tutorials on quantum teleportation and QKD

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

Minors in Information Technology (R23)

23AIT28: Introduction to Quantum Sensing

L	T	P	C
3	0	0	3

Course Objectives:

1. Introduce the principles of quantum sensing and metrology.
2. Explain how quantum superposition and entanglement enhance measurement sensitivity.
3. Study applications of quantum sensors across multiple domains.
4. Analyze noise, decoherence, and quantum limits on measurement.
5. Prepare students to design and analyze quantum-enhanced sensors.

Course Outcomes (COs):

After successful completion of this course, the students will be able to:

CO1: Understand the basic principles of quantum sensing

CO2: Apply quantum superposition and entanglement to sensing

CO3: Analyze quantum sensor architectures

CO4: Evaluate sensitivity and error limits in quantum measurements CO5:

Design quantum sensing systems for real-world applications

UNIT I – Introduction to Quantum Sensing and Metrology *(Cognitive Level: Understand)*

Classical vs quantum sensing, Precision limits: Standard Quantum Limit (SQL), Quantum metrology fundamentals, Heisenberg limit, Quantum phase estimation for precision measurements, Quantum non-demolition measurements, Quantum error correction in sensing, Importance of coherence and entanglement in sensors

UNIT II – Quantum Measurement Principles *(Cognitive Level: Apply)*

Superposition and interference in measurement, Quantum Fisher information, squeezed states for noise reduction, Photon counting and single-photon detectors, Spin-based measurements (NV centers, trapped ions), Ramsey interferometry, Quantum state tomography, Applications of quantum-enhanced interferometry

UNIT III – Quantum Sensor Technologies *(Cognitive Level: Analyze)*

Atomic clocks (optical & microwave), Gravimeters and accelerometers, Magnetometers (SQUIDs, NV centers), Quantum gyroscopes, Quantum imaging & super-resolution microscopy, Quantum lidar and radar, Force and electric field sensing, Photonic quantum sensing systems

UNIT IV – Decoherence, Noise, and Error Mitigation in Quantum Sensing *(Cognitive Level: Evaluate)*

Sources of decoherence in quantum sensors, Thermal noise and quantum noise sources, Quantum back-action, Squeezing and noise reduction techniques, Dynamical decoupling techniques, Noise spectroscopy for sensor calibration, Robust error mitigation protocols, Evaluating sensitivity vs noise tradeoffs

UNIT V – Advanced Applications and Future Quantum Sensing Systems *(Cognitive Level: Create)*

Quantum sensing for biological and medical imaging, Navigation and positioning without GPS, Quantum-enhanced gravitational wave detection (LIGO), Quantum-enhanced environmental monitoring, Sensors for national defense and security, Space-based quantum sensors, Integrated quantum photonic sensing platforms, Design of hybrid quantum-classical sensor systems

Text books:

[1] C. L. Degen, F. Reinhard, and P. Cappellaro, “Quantum sensing,” Rev. Mod. Phys., vol. 89, no. 3, pp. 035002-1–035002-39, Jul. 2017.

[2] V. Giovannetti, S. Lloyd, and L. Maccone, “Advances in quantum metrology,” Nature Photon., vol. 5, pp. 222–229, Apr. 2011.

[3] D. Budker and D. F. Jackson Kimball, Optical Magnetometry, Cambridge, U.K.: Cambridge Univ. Press, 2013.

Reference Books:

[1] K. Jacobs, Quantum Measurement Theory and Its Applications, Cambridge, U.K.: Cambridge Univ. Press, 2014.

[2] H. Rauch and S. A. Werner, Neutron Interferometry: Lessons in Experimental Quantum Mechanics, Wave-Particle Duality, and Entanglement, 2nd ed., Oxford, U.K.: Oxford Univ. Press, 2015.

[3] M. O. Scully and M. S. Zubairy, Quantum Optics, Cambridge, U.K.: Cambridge Univ. Press, 1997, ch. 16–18.

[4] V. Vedral, Introduction to Quantum Information Science, Oxford, U.K.: Oxford Univ. Press, 2006.

Online Courses & Resources:

Platform	**Course Title**
edX	Quantum Sensing & Metrology (LMU Munich)
Coursera	Quantum Optics and Sensing (University of Colorado Boulder)
MIT OpenCourseWare	Quantum Measurement and Sensing (MIT)
YouTube	Quantum Sensing Lectures
IBM Qiskit	Tutorials on Quantum Phase Estimation

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

Minors in Information Technology (R23)

23AIT29: Quantum Communication and Sensing Lab

L	T	P	C
0	0	3	1.5

Course Objectives:

1. Simulate and analyze quantum communication protocols.
2. Implement quantum key distribution (QKD) and teleportation.
3. Perform quantum sensing simulations for precision measurements.
4. Evaluate sensor performance with noise and decoherence.
5. Gain hands-on experience with quantum simulation tools.

Course Outcomes (COs):

After successful completion of this course, the students will be able to:

CO1: Simulate and evaluate quantum communication protocols using appropriate quantum tools. CO2:

Implement and analyze quantum key distribution (QKD) and quantum teleportation techniques. CO3:

Perform simulations for quantum sensing applications in precision measurements.

CO4: Analyze and interpret the effects of noise and decoherence on quantum sensor performance.

CO5: Demonstrate hands-on proficiency in using quantum simulation software for communication and sensing experiments.

List of Experiments:

1. Simulation of Qubits and Bloch Sphere Visualization
2. Implementation of BB84 Quantum Key Distribution Protocol
3. Simulation of B92 and E91 QKD Protocols
4. Quantum Entanglement Generation and Bell Inequality Testing
5. Quantum Teleportation Protocol using Qiskit/Cirq
6. Simulation of Quantum Repeaters and Entanglement Swapping
7. Noise and Decoherence Modeling in Quantum Communication Channels
8. Ramsey Interferometry Simulation for Quantum Sensing
9. Implementation of NV Center Magnetometry Simulation
10. Quantum Gravimeter and Accelerometer Simulation
11. Quantum Phase Estimation for High-Precision Metrology

Reference Textbooks:

[1] M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information, 10th Anniversary ed., Cambridge, U.K.: Cambridge Univ. Press, 2010.

[2] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," in Proc. IEEE Int. Conf. Computers, Systems and Signal Processing, Bangalore, India, Dec. 1984, pp. 175–179.

[3] J. Preskill, "Quantum computing in the NISQ era and beyond," Quantum, vol. 2, pp. 1–20, Aug. 2018.

[4] V. Giovannetti, S. Lloyd, and L. Maccone, "Quantum-enhanced measurements: Beating the standard quantum limit," Science, vol. 306, no. 5700, pp. 1330–1336, Nov. 2004.

[5] M. G. A. Paris, "Quantum estimation for quantum technology," Int. J. Quantum Inform., vol. 7, no. supp01, pp. 125–137, 2009.

Platforms & Tools:

- IBM Qiskit
- Google Cirq
- RigettiPyQuil
- Quantum Inspire
- MATLAB / Python with quantum libraries

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

Minors in Information Technology (R23)

23AIT30: QUANTUM DEVICES AND MATERIALS LAB

L	T	P	C
0	0	3	1.5

Course Objectives:

1. Simulate quantum devices and materials behavior.
2. Explore quantum optics and solid-state quantum systems.
3. Model quantum dots, superconductors, and photonic devices.
4. Perform quantum simulation of condensed matter systems.
5. Build foundational skills for quantum hardware understanding.

Course Outcomes (COs):

After successful completion of this course, the students will be able to:

CO1: Simulate the behavior of quantum devices and quantum materials using computational tools.

CO2: Analyze the properties of quantum optical systems and solid-state quantum structures. CO3:

Model and interpret the behavior of quantum dots, superconductors, and photonic devices. CO4:

Perform quantum simulations of condensed matter systems and interpret the results.

CO5: Demonstrate foundational understanding of quantum hardware components and their physical principles.

List of Experiments:

1. Simulation of Single-Qubit Optical Devices
2. Modeling Quantum Dots and Energy Level Transitions
3. Simulation of Two-Level Atom and Rabi Oscillations
4. Quantum Harmonic Oscillator: Energy Levels Visualization
5. Spin-1/2 Systems and Magnetic Resonance Simulation
6. Superconducting Qubits Circuit Simulation
7. Josephson Junction Modeling for Quantum Circuits
8. Quantum Photonic Interferometer Simulation
9. Simulation of NV Centers in Diamond for Quantum Sensing
10. Solid-State Quantum Materials Simulation (Band Structures)
11. Modeling Quantum Light-Matter Interactions (Jaynes-Cummings Model)

Reference Textbooks:

[1] Y. V. Nazarov and Y. M. Blanter, Quantum Transport: Introduction to Nanoscience, Cambridge, U.K.: Cambridge Univ. Press, 2009.

[2] D. Bouwmeester, A. K. Ekert, and A. Zeilinger, The Physics of Quantum Information: Quantum Cryptography, Quantum Teleportation, Quantum Computation, Berlin, Germany: Springer, 2000.

[3] B. E. Kane, “A silicon-based nuclear spin quantum computer,” Nature, vol. 393, no. 6681, pp. 133– 137, May 1998.

[4] M. Fox, Quantum Optics: An Introduction, Oxford, U.K.: Oxford Univ. Press, 2006.

Platforms & Tools:

- QuTiP (Quantum Toolbox in Python)
- Qiskit Nature / Qiskit Metal
- MATLAB Simulink
- COMSOL Multiphysics (for materials simulation)
- Silvaco TCAD (for device-level modeling)

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B.Tech-I Semester

L	T	P	C
3	0	3	3

23AIT37: Introduction to Design and Visual Communication

Course Outcomes:
After completion of the course, students will be able to

CO1. Understand the foundational elements and principles of design and visual communication.
CO2. Analyze visual compositions using concepts like balance, contrast, hierarchy, and typography.
CO3. Apply color theory, layout principles, and design grids in creative visual projects. CO4. Create effective visual messages for print and digital media using industry-standard tools.
CO5. Critically evaluate design work based on functionality, aesthetics, and target audience.
CO6. Develop a personal visual style and portfolio through practical assignments and critiques.

Unit 1: Elements and Principles of Design

Line, Shape, Color, Texture, Space, Form Balance, Contrast, Emphasis, Movement, Proportion, Rhythm, Unity

Unit 2: Fundamentals of Visual Literacy

Introduction to Color Theory: Primary, Secondary, Tertiary Colors Color Psychology and Branding, Typography Basics: Typeface vs Font, Font Pairing, Readability

Unit 3: History and Evolution of Design

Historical Art Movements: Bauhaus, Minimalism, Modernism, Evolution from Print to Digital Design, Milestones in Graphic and Interaction Design

Unit 4: Communication Design

Visual Hierarchy and Storytelling, Semiotics and Symbols in Design, Branding Elements and Logo Design Basics

Unit 5: Introduction to Design Tools

Overview of Adobe Photoshop, Illustrator, Introduction to Figma/Sketch, File formats, resolutions, and design output standards.

Text books:

1. D. Landa, Graphic Design Solutions, 6th ed., Boston, MA, USA: Cengage Learning, 2018.
2. K. Lidwell, J. Holden, and W. Butler, Universal Principles of Design, Rev. ed., Beverly, MA, USA: Rockport Publishers, 2010..

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B.Tech-I Semester

L	T	P	C
3	0	3	3

23AIT38: Human Centered Design and Interaction Principles

Course Outcomes:
After completion of the course, students will be able to

CO1. Understand and explain the fundamental principles of human-centered design (HCD) and user experience (UX) design.
CO2. Analyze human factors and cognitive models to inform interface and interaction design decisions.
CO3. Apply design thinking methodologies to identify user needs and generate user-centric solutions.
CO4. Design and develop user interfaces based on usability heuristics and interaction principles.
CO5. Evaluate the effectiveness and usability of user interfaces using empirical methods and usability testing.
CO6. Demonstrate proficiency in using prototyping tools and conducting iterative design processes.

Unit 1: Foundations of Human-Centered Design

HCD Process: Discover, Define, Develop, Deliver - Benefits and Implementation in UX/UI - User Goals vs Business Goals

Unit 2: Cognitive Psychology in Design

Perception, Memory, Mental Models, Decision Making, Cognitive Load, Visual Perception in UI

Unit 3: Design Principles for Usability

Affordances, Signifiers, Feedback, Constraints, Consistency and Standards, Visibility and Mapping

Unit 4: Human-Computer Interaction

History of HCI, Interaction Paradigms (WIMP, Touch, Voice), Input/Output Devices and Modalities

Unit 5: Usability Heuristics

Nielsen’s 10 Heuristics, Real-world ExamplesM Application and Evaluation.

Text books:

1. D. A. Norman, The Design of Everyday Things, Revised and Expanded Edition, New York, NY, USA: Basic Books, 2013.

2. J. Preece, Y. Rogers, and H. Sharp, Interaction Design: Beyond Human-Computer Interaction, 5th ed., Hoboken, NJ, USA: Wiley, 2019.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B.Tech-II Semester

L	T	P	C
3	0	3	3

23AIT39: UX Research and Design Thinking

Course Outcomes

After completing this course, students will be able to:

CO1. Understand the fundamentals of User Experience (UX) design and its role in product development.

CO2. Apply UX research methods such as user interviews, surveys, persona creation, and journey mapping.

CO3. Utilize design thinking principles to identify problems and develop innovative user-centric solutions.

CO4. Create wireframes, prototypes, and mockups to represent design solutions.

CO5. Conduct usability testing and evaluate user feedback to refine designs iteratively.

CO6. Collaborate in teams using agile and iterative methods to deliver human-centered design solutions.

Unit 1: UX Research Foundations

Types of Research: Qualitative, Quantitative, Desk Research and Competitive Analysis, Ethics in UX Research

Unit 2: Design Thinking Process

Empathize, Define, Ideate, Prototype, Test, Ideation Techniques: SCAMPER, Brainstorming Divergent vs Convergent Thinking

Unit 3: User Research Methods

User Interviews, Surveys, Field Studies, Contextual Inquiry
Affinity Diagramming

Unit 4: Persona and Empathy Mapping

Creating Personas, Building Empathy Maps, Using Insights for Design Decisions

Unit 5: Journey Mapping and Task Flows

Mapping Touchpoints and Emotions, Defining Happy Path and Edge Cases, Task Analysis and Flow Diagrams

Textbook:

1. G. Gothelf and J. Seiden, Lean UX: Designing Great Products with Agile Teams, 2nd ed., Sebastopol, CA, USA: O'Reilly Media, 2016.
2. T. Brown, Change by Design: How Design Thinking Creates New Alternatives for Business and Society, Boston, MA, USA: Harvard Business Press, 2009.

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(AUTONOMOUS)

III B.Tech-II Semester

L	T	P	C
3	0	3	3

23AIT40: PROTOTYPING AND USABILITY TESTING

Course Outcomes

After completing this course, students will be able to:

CO1. Understand the principles and importance of prototyping in the user-centered design process.
CO2. Create low-, mid-, and high-fidelity prototypes using appropriate tools and techniques.
CO3. Apply usability heuristics and guidelines in prototype development.
CO4. Plan and conduct usability testing sessions, including task design, facilitation, and data collection.
CO5. Analyze user feedback and usability test results to identify design improvements.
CO6. Iterate prototypes based on usability insights and present findings effectively.

Unit 1: Prototyping Basics

Paper vs Digital Prototypes, Low to High Fidelity Transitions, Tools Comparison: Figma, InVision, Marvel

Unit 2: Usability Testing Methods

Moderated vs Unmoderated Testing, Think-Aloud Protocol, Remote Testing Tools

Unit 3: Metrics and Analysis

Task Success Rate, Time on Task, SUS, NPS, Heatmaps, Using Analytics for Usability

Unit 4: A/B and Multivariate Testing

Setting Hypotheses, Test Design and Execution, Interpreting Results

Unit 5: Iterative Design Cycle

Gathering Feedback, Prioritizing Issues, Documenting Iterations

Textbook:

1. C. Snyder, Paper Prototyping: The Fast and Easy Way to Design and Refine User Interfaces, San Francisco, CA, USA: Morgan Kaufmann, 2003.
2. J. Rubin and D. Chisnell, Handbook of Usability Testing: How to Plan, Design, and Conduct Effective Tests, 2nd ed., Indianapolis, IN, USA: Wiley, 2008.

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(AUTONOMOUS)

IV B.Tech-I Semester

L	T	P	C
3	0	3	3

23AIT41: FRONTEND DEVELOPMENT FOR DESIGNERS

Course Outcomes

After completing this course, students will be able to:

CO1. Understand the core concepts of HTML, CSS, and JavaScript for creating responsive user interfaces.

CO2. Apply frontend frameworks and libraries (e.g., Bootstrap, React) to build dynamic web applications.

CO3. Translate design mockups and wireframes into functional web pages.

CO4. Implement accessibility and usability standards in frontend development.

CO5. Optimize web performance and ensure cross-browser compatibility.

CO6. Collaborate effectively with designers and developers using version control and frontend tooling.

Unit I: Web Technologies Overview

Basics of HTML5, Structuring Pages with Semantic Tags, Introduction to CSS

Unit II: Responsive Design

Media Queries, Flexbox and Grid, Mobile-First Development

Unit III: JavaScript Fundamentals

DOM Manipulation, Event Handling, Fetch API and Async Basics

Unit IV: Frameworks and Libraries

Bootstrap and Tailwind, Intro to React for Designers, Using Design Systems with Frameworks

Unit V: Accessibility in Code

ARIA Roles, Tab Index and Screen Reader Testing, Keyboard Navigation

Textbook:

1. J. Duckett, *HTML and CSS: Design and Build Websites*, Indianapolis, IN, USA: Wiley, 2011.
2. E. Freeman and E. Robson, *Head First HTML and CSS: A Learner's Guide to Creating Standards-Based Web Pages*, 2nd ed., Sebastopol, CA, USA: O'Reilly Media, 2012.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (AUTONOMOUS)

IV B.Tech-II Semester

L	T	P	C
0	0	6	3

23AIT42: CAPSTONE PROJECT AND PORTFOLIO

MINORS IN CYBER SECURITY

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B. Tech - II Semester -CSE(CS)

III B. Tech - I Semester CSC MINORS

L	T	P	C
3	0	0	3

23ACC05: CYBER CRIMES & DIGITAL FORENSICS

Course Outcomes:

At the end of the course students will be able to

CO1: Understand the fundamentals of cybercrime.

CO2: Understand the various cybercrime issues.

CO3: Understand different investigation tools for cybercrime.

CO4: Understand basics of Forensic Technology and Practices.

CO5: Analyze different laws, ethics and evidence handling procedures.

UNIT-I: Introduction **9 Hrs**

Introduction and Overview of Cyber Crime, Nature and Scope of Cyber Crime, Types of Cyber Crime: Social Engineering, Categories of Cyber Crime, Property Cyber Crime.

UNIT-II: Cyber Crime Issues **9 Hrs**

Unauthorized Access to Computers, Computer Intrusions, White collar Crimes, Viruses and Malicious Code, Internet Hacking and Cracking, Virus Attacks, Pornography, Software Piracy, Intellectual Property, Mail Bombs, Exploitation, Stalking and Obscenity in Internet, Digital laws and legislation, Law Enforcement Roles and Responses.

UNIT-III: Investigation **9 Hrs**

Introduction to Cyber Crime Investigation, Investigation Tools, rediscovery, Digital Evidence Collection, Evidence Preservation, E-Mail Investigation, E-Mail Tracking, IP Tracking, E- Mail Recovery, Hands on Case Studies. Encryption and Decryption Methods, Search and Seizure of Computers, Recovering Deleted Evidences, Password Cracking.

UNIT-IV: Digital Forensics **9Hrs**

Introduction to Digital Forensics, Forensic Software and Hardware, Analysis and Advanced Tools, Forensic Technology and Practices, Forensic Ballistics and Photography, Face, Iris and Fingerprint Recognition, Audio Video Analysis, Windows System Forensics, Linux System Forensics, Network Forensics.

UNIT-V: Laws and Acts **9 Hrs**

Laws and Ethics, Digital Evidence Controls, Evidence Handling Procedures, Basics of Indian Evidence ACT IPC and Corps, Electronic Communication Privacy ACT, Legal Policies

TEXT BOOKS:

1. Nelson Phillips and Engineer Stuart, ―Computer Forensics and Investigations‖, CEng age Learning, New Delhi, 2009.
2. Kevin Media, Chris Promise, Matt Pipe, ―Incident Response and Computer Forensics ―, Tata McGraw -Hill, New Delhi, 2006.

REFERENCE BOOKS:

1. Robert M Slade,‖ Software Forensics‖, Tata McGraw - Hill, New Delhi, 2005.
2. Bernadette H Schell, Clemens Martin, ―Cybercrime‖, ABC – CLIO Inc, California, 2004
3. ―Understanding Forensics in IT ―, NIIT Ltd, 2005.

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(AUTONOMOUS)

IV B. Tech - I Semester -CSE(CS)

III B. Tech - I Semester -CSC MINORS

L	T	P	C
3	0	0	3

23ACC16: CYBER LAWS AND SECURITY POLICIES

(Professional Elective-V)

Course Outcomes

At the end of the course students will be able to

CO1: Learn evolution and key aspects of Indian cyber law, including recent amendments.

CO2: Gain knowledge about the legalities of digital signatures and the role of e-governance in the IT Act.

CO3: Develop an understanding of the legalities involved in electronic contracts and international conventions.

CO4: Adapt in understanding and analyzing cybercrime, electronic evidence, and intellectual property rights in the context of IT.

UNIT - I

Introduction: History of Internet and World Wide Web, Need for cyber law, Cybercrime on the rise, Important terms related to cyber law.

Cyber law in India: Need for cyber law in India, History of cyber law in India.

Information Technology Act, 2000: Overview of other laws amended by the IT Act, 2000, National Policy on Information Technology 2012.

UNIT - II

Overview of the Information Technology Act, 2000: Applicability of the Act, Important provisions of the Act: Digital signature and Electronic signature, Digital Signature under the IT Act, 2000, E- Governance Attribution, Acknowledgement and Dispatch of Electronic Records, Certifying Authorities, Electronic Signature Certificates, Duties of Subscribers, Penalties and Offences, Intermediaries.

UNIT - III

Overview of rules issued under The IT Act, 2000, Electronic Commerce, Electronic Contracts, Cyber Crimes, Cyber Frauds.

UNIT - IV

Regulatory Authorities: Department of Electronics and Information Technology, Controller of Certifying Authorities (CCA), Cyber Appellate Tribunal, Indian Computer Emergency Response Team (ICERT), Cloud Computing, Case Laws.

UNIT - V

Introduction to Cybercrime and procedure to report Cyber crime: procedure to report cybercrime, some basic rules for safe operations of the computer and internet, the criminal law (amendment) act, 2013:

legislative remedies for online harassment and cyber stalking in India.

TEXT BOOKS:

1. Text book on ―Cyber Law‖,**2 Ed Rap 2023**, Pagan Dug gal, Universal Law Publishing.
2. Text book on ―Indian Cyber law on Cybercrimes‖, Pagan Dug gal,

REFERENCE BOOKS:

1. Debby Russell and Sr. G.T Gingili, "Computer Security Basics (Paperback)‖, 2nd Edition, O'Reilly Media, 2006.
2. Thomas R. Pettier, ―Information Security policies and procedures: A Practitioner's Reference‖, 2nd Edition Prentice Hall, 2004.
3. Kenneth J. Knapp, ―Cyber Security and Global Information Assurance: Threat Analysis and Response Solutions‖, IGI Global, 2009.
4. Thomas R Pettier, Justin Pettier and John Blackley,‖ Information Security Fundamentals‖, 2^{nd} Edition, Prentice Hall, 1996.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B. Tech - I Semester -CSE(CS)

III B. Tech - II Semester -CSC MINORS

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23ACC01: INTRODUCTION TO CYBER SECURITY

Course Outcomes:

At the end of the course students will be able to

CO1: Demonstrate knowledge on Automata Theory, Regular Expression and Analyze and Design of finite automata, and prove equivalence of various finite automata.

CO2: Demonstrate knowledge on context free grammar, Analyze and design of PDA and TM.

CO3: Understand the basic concept of compiler design, and its different phases which will be helpful to construct new tools like LEX, YACC, etc.

CO4: Ability to implement semantic rules into a parser that performs attribution while parsing and apply error detection and correction methods.

CO5: Apply the code optimization techniques to improve the space and time complexity of programs while programm ing and Ability to design a compiler.

UNIT-I: Cyber security Essentials and Cube **9 Hrs**

The Cyber security World, Cyber Criminals versus Cyber security Specialists, Common Threats, Spreading Cyber security Threats, The Three Dimensions of the Cyber security Cube, CIA Triad, States of Data, Cyber security Countermeasures, IT Security Management Framework.

UNIT-II: Cyber security Threats, Vulnerabilities, Attacks and Protecting Secrets 9Hrs Introduction, Governance, Managing Cloud Security Risk, Compliance, Legal Issues in Cloud, Audit, CSA Tools.

UNIT-III: Data Integrity **9 Hrs**

Types of Data Integrity Controls, Digital Signatures, Certificates, Database Integrity Enforcement.

UNIT-IV: Data Availability and Recovery **9 Hrs**

High Availability, Measures to Improve Availability, Incident Response, Disaster Recovery.

UNIT-V: Protecting a Cyber security Domain **9 Hrs** Defending Systems and Devices, Server Hardening, Network Hardening, Physical and Environmental Security, Cyber security Domains, Ethics of Working in Cyber security.

Text books:

1. CISSP (ISC)2 Certified Information Systems Security Professional Official Study Guide, Mike Chappell, James Michael Stewart and Darrel Gibson, 9th Edition
2. Cyber security: The Beginner's Guide, Dr. Erdal Ozkaya, Packt Publishing Limited, 2019.

Reference Books:

1. Cyber security Essentials, Charles J. Brooks, Christopher Grow, Philip Craig and Donald Short, 1st edition, Say box.
2. Network Security Essentials, William Stallings, 6th edition, Pearson Education, 2018.

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(AUTONOMOUS)

IV B. Tech - I Semester -CSE(CS)
III B. Tech -II Semester -CSC MINORS

L	T	P	C
3	0	0	3

23ACC11: BLOCK CHAIN TECHNOLOGY

Course Outcomes (CO):

At the end of the course students will be able to

CO1: Demonstrate the foundation of the Block chain technology and understand the processes in payment and funding.

CO2: Identify the risks involved in building Block chain applications.

CO3: Review of legal implications using smart contracts.

CO4: Choose the present landscape of Block chain implementations and Understand Crypto currency markets Examine how to profit from trading crypto currencies

UNIT - I Introduction **Lecture 8Hrs**

Introduction, Scenarios, Challenges Articulated, Block chain, Block chain Characteristics, Opportunities Using Block chain, History of Block chain. Evolution of Block chain: Evolution of Computer Applications, Centralized Applications, Decentralized Applications, Stages in Block chain Evolution, Consortia, Forks, Public Blockchain Environments, Type of Players in Block chain Ecosystem, Players in Market.

UNIT - II Block chain Concepts **Lecture 9Hrs**

Block chain Concepts: Introduction, Changing of Blocks, Hashing, Markel-Tree, Consensus, Mining and Finalizing Blocks, Currency aka tokens, security on block chain, data storage on block chain, wallets, coding on block chain: smart contracts, peer-to-peer network, types of block chain nodes, risk associated with block chain solutions, life cycle of block chain transaction.

UNIT - III Architecting Block chain solution **Lecture 9Hrs**

Architecting Block chain solutions: Introduction, Obstacles for Use of Block chain, Block chain Relevance Evaluation Framework, Block chain Solutions Reference Architecture, Types of Block chain Applications. Cryptographic Tokens, Typical Solution Architecture for Enterprise Use Cases, Types of Block chain Solutions, Architecture Considerations, Architecture with Block chain Platforms, Approach for Designing Block chain Applications.

UNIT - IV Ethereal Block chain Implementation **Lecture 8Hrs**

Ethereal Block chain Implementation: Introduction, Tuna Fish Tracking Use Case, Ethereal Ecosystem, Ethereal Development, Ethereal Tool Stack, Ethereal Virtual Machine, Smart Contract Programming, Integrated Development Environment, Truffle Framework, Gouache, Unit Testing, Ethereal Accounts, My Ether Wallet, Ethereal Networks/Environments, Inure, Ether scan, Ethereal Clients, Decentralized Application, Metalmark, Tuna Fish Use Case Implementation, Open Zeppelin in Contracts

UNIT - V Hyper ledger Block chain Implementation **Lecture 8Hrs**

Hyper ledger Block chain Implementation: Introduction, Use Case – Car Ownership Tracking, Hyper ledger Fabric, Hyper ledger Fabric Transaction Flow, Facer Use Case Implementation, Invoking Chain code Functions Using Client Application.

Advanced Concepts in Block chain: Introduction, Inter Planetary File System (IPFS), Zero Knowledge Proofs, Oracles, Self-Sovereign Identity, Block chain with IoT and AI/ML Quantum Computing and Block chain, Initial Coin Offering, Block chain Cloud Offerings, Block chain and its Future Potential.

Text books:

1. Armadas, Arched Surfers A riff, Sham ―Block chain for Enterprise Application Developers‖, Wiley, 2020
2. Andreas M. Antonopoulos, ―Mastering Bit coin: Programming the Open Block chain‖, O‗Reilly, 2017

Reference Books:

1. Block chain: A Practical Guide to Developing Business, Law, and Technology Solutions, Joseph Barbara, Paul R. Allen, Mc Grow Hill.
2. Block chain: Blueprint for a New Economy, Melanie Swan, O‗Reilly

Online Learning Resources:

https://github.com/blockchainedindia/resources

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III B. Tech - II Semester -CSE(CS)

IV B. Tech - I Semester -CSC MINORS

L	T	P	C
3	0	0	3

23ACC04: CRYPTOGRAPHY & NETWORK SECURITY

Course outcomes:

At the end of the course students will be able to

CO1: Identify information security goals, classical encryption technique and acquire fundamental knowledge on the concepts off in tie fields and number theory

CO2: Compareand apply different encryption and decryption techniques to solve problems related to confidentiality and authentication

CO3: Apply the knowledge of cryptographic checksum and evaluate the performance of different message digest algorithms for verifying the integrity of varying message sizes.

CO4: Apply different digital signature algorithms to achieve authentication and create secure applications

CO5: Apply network security basics, analyze different attack son networks and evaluate the performance of fire walls and security protocols like TLS, IPSec, and PGP

CO6: Apply the knowledge of cryptographic utilities and authentication mechanisms to design secure applications.

UNIT-I: **Lecture 9Hrs**

Computer and Network Security Concepts: Computer Security Concepts, Theo‘s Security Architecture, Security Attacks, Security Services, Security Mechanisms, A Model for Network Security, Classical Encryption Techniques: Symmetric Cipher Model, Substitution Techniques Transposition Techniques, Steganography, Block Ciphers: Traditional Block Cipher Structure, The Data Encryption Standard, Advanced Encryption Standard: AES Structure, AES Transformation Functions

UNIT-II: **Lecture 9Hrs**

Number Theory: The Euclidean Algorithm, Modular Arithmetic, Fermat‘s and Euler‘s Theorems, The Chinese Remainder Theorem, Discrete Logarithms, Finite Fields: Finite Fields of the Form GF(p), Finite Fields of the Form GF(2^n). Public Key Cryptography: Principles, Public Key Cryptography Algorithms, RSA Algorithm, Daffy Hellman Key Exchange, Elliptic Curve Cryptography.

UNIT-III: **Lecture 9Hrs**

Cryptographic Hash Functions: Application o f Cryptographic Hash Functions, Requirements & Security, Secure Hash Algorithm, Message Authentication Functions, Requirements & Security, HMAC & CMAC. Digital Signatures: NIST Digital Signature Algorithm, Distribution of Public Keys, X.509 Certificates, Public- Key Infrastructure

UNIT-IV: **Lecture 9Hrs**

User Authentication: Remote Use r Authentication Principles, Kerberos. Electronic Mail Security: Pretty Good Privacy (PGP) Ands /MIME.IP Security: I P Security Overview, IP Security Policy, Encapsulating Security Payload, Combining Security Associations, Internet Key Exchange.

UNIT-V: **Lecture 8Hrs**

Transport Level Security: Web Security Requirements, Transport Layer Security (TLS), HTTPS, Secures hell (SSH) Firewalls: Fire wall Characteristics and Access Policy, Types of Fire walls, Fire wall ovation and Configurations.

Textbooks:

1) Cryptography and Network Security-William Stallings, Pearson Education,8th Edition.
2) Cryptography, network Security and Cyber Laws–Bernard Menaces, Engage Learning, 2010 edition.

Reference Books:

1) Cryptography and Network Security- Beerhouse Frozen, Debden Mukhopadhyaya,

Mc –Grow Hill, 3rd Edition, 2015.

2) Network Security Illustrated, Jason Albanese and Wes Scone Reich, MGH Publishers, 2003.

Online Learning Resources:

1) https://nptel.ac.in/courses/106/105/106105031/lecture
2) https://nptel.ac.in/courses/106/105/106105162/lecturebyDr.SouravMukhopadhyay IITKharagpur[VideoLecture]
3) https://www.mitel.com/articles/web-communication-cryptography-and-network- security web articles by Mitel Power Connections

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III B. Tech - I Semester -CSE(CS)

III B. Tech - II Semester -CSC MINORS

L	T	P	C
0	0	3	3

23ACC02: CYBER SECURITY LAB

Course Outcomes:

At the end of the course students will be able to

CO1: Get the skill to identify cyber threats/attacks.

CO2: Get the knowledge to solve security issues in day-to-day life.

CO3: Able to use Autopsy tools

CO4: Perform Memory capture and analysis

CO5: Demonstrate Network analysis using Network miner tools

List of Experiments

1. Perform an Experiment for port scanning with neap
2. Set Up a honey pot and monitor the honey pot on the network
3. Install Jscript/Cryptal tool (or any other equivalent) and demonstrate Asymmetric, symmetric crypto algorithm, Hash and Digital/PKI signatures.
4. Generate minimum 10 passwords of length 12 characters using open SSL command
5. Perform practical approach to implement Foot printing-Gathering target information using Dmitry-D magic, U Attester
6. Working with sniffers for monitoring network communication (Wire shark).
7. Using Snort, perform real time traffic analysis and packet logging.
8. Perform email analysis using the Autopsy tool.
9. Perform Registry analysis and get boot time logging using process monitor tool
10. Perform File type detection using Autopsy tool
11. Perform Memory capture and analysis using FTK imager tool
12. Perform Network analysis using the Network Miner tool

TEXT BOOKS:

1. Real Digital Forensics for Handheld Devices, E. P. Dorothy, Auer back Publications, 2013.
2. The Basics of Digital Forensics: The Primer for Getting Started in Digital Forensics, J. Sammons,

Singles Publishing, 2012.

REFERENCE BOOKS:

1. Handbook of Digital Forensics and Investigation, E. Casey, Academic Press, 2010.
2. Malware Forensics Field Guide for Windows Systems: Digital Forensics Field Guides, C. H. Malian, E. Casey and J. M. Aquiline, Signers, 2012.
3. The Best Damn Cybercrime and Digital Forensics Book.

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III B. Tech - II Semester -CSE(CS)

IV B. Tech - I Semester -CSC MINORS

L	T	P	C
0	0	3	1.5

23ACC08: CRYPTOGRAPHY AND NETWORK SECURITY LAB

List of Experiments:

1. Write a C program that contains a string (char pointer) with a value ‗Hello world'. The program should XOR each character in this string with 0 and displays the result.

2. Write a C program that contains a string (char pointer) with a value ‗Hello world'. The program should AND or and XOR each character in this string with 127 and display the result.

3. Write a Java program to perform encryption and decryption using the following algorithms

a. Cease cipher b. Substitution cipher c. Hill Cipher

4. Write a C/JAVA program to implement the DES algorithm logic.

5. Write a C/JAVA program to implement the Blowfish algorithm logic.

6. Write a C/JAVA program to implement the Randal algorithm logic.

7. Write the RC4 logic in Java Using Java cryptography; encrypt the text ―Hello world‖ using Blowfish. Create your own key using Java key tool.

8. Write a Java program to implement RSA algorithm.

9. Implement the Daffy-Hellman Key Exchange mechanism using HTML and JavaScript.

10. Calculate the message digest of a text using the SHA-1 algorithm in JAVA.

11. Calculate the message digest of a text using the MD5 algorithm in JAVA.

MINORS IN QUANTUM COMPUTING

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B.Tech I Semester CE,ME,EEE,ECE(Minor)

L	T	P	C
3	0	0	3

23ACS46: INTRODUCTION TO QUANTUM COMPUTING

Course Outcomes (COs)

CO Code	Description	Bloom's Level
CO1	Explain concepts of quantum mechanics	Understand(L1)
CO2	Illustrate quantum gates/circuits	Apply(L3)
CO3	Analyze algorithms (e.g., Shor, Grover)	Analyze(L4)
CO4	Evaluate communication protocols	Evaluate(L5)
CO5	Develop quantum programs on IBM Q	Create(L6)

Unit I: Qubits and Quantum Foundations

Classical Bits vs Qubits, Postulates of Quantum Mechanics, Superposition and Probability Amplitudes, Dirac Notation (Bra-Ket), Bloch Sphere Representation, Measurement in Quantum Systems, Quantum State Collapse

Unit II: Quantum Gates and Circuits

Quantum Logic Gates: Pauli-X, Y, Z; Hadamard (H); Phase (S, T), Controlled Gates: CNOT, Toffoli, Unitary and Reversible Operations, Quantum Circuit Representation, Building Basic Quantum Circuits, Quantum Parallelism and Interference, No-Cloning Theorem and Quantum Gate Simulation

Unit III: Quantum Algorithms

Need for Quantum Algorithms, Deutsch and Deutsch-Jozsa Algorithm, Grover’s Search Algorithm (Quadratic Speed-up), Shor’s Factoring Algorithm (Exponential Speed-up), Simon’s Algorithm (Overview), Complexity Comparison: Classical vs Quantum

Unit IV: Entanglement and Quantum Communication

Quantum Entanglement and Bell States, Quantum Teleportation Protocol, Superdense Coding, Quantum Key Distribution: BB84, E91 Protocols, Decoherence and Quantum Noise, Quantum Error Correction Codes (Bit Flip, Phase Flip, Shor Code)

Unit V: Quantum Platforms and Applications

Overview of Quantum Programming Platforms: IBM Qiskit, Microsoft Q#, Google Cirq, Quantum Circuit Simulation using Qiskit, Executing Code on Real Quantum Hardware (IBM Q). Quantum Applications in:Cryptography, Machine Learning, Optimization, Chemistry, Building and Testing a Sample Quantum Program

Textbooks

1. **Michael A. Nielsen & Isaac L. Chuang** – *Quantum Computation and Quantum Information,* Cambridge University Press, 10th Anniversary Edition.
2. **David McMahon** – *Quantum Computing Explained*, Wiley.
3. **Bernhardt, Chris** – *Quantum Computing for Everyone*, MIT Press.

Reference Books

1. **Mermin, N. David** – *Quantum Computer Science: An Introduction*, Cambridge University Press.
2. **William H. Press et al.** – *Numerical Recipes in C: The Art of Scientific Computing* (for simulation background)

3. **Rieffel&Polak** – *Quantum Computing: A Gentle Introduction*, MIT Press.

Online Courses & Resources

Platform	Course Name	Link
IBM Qiskit	IBM Qiskit Textbook	Hands-on, beginner-friendly curriculum for quantum programming
Coursera	*Quantum Mechanics for Scientists and Engineers* by Stanford (Leonard Susskind)	Link

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B.Tech I Semester CE,ME,EEE,ECE(Minor)

L	T	P	C
3	0	0	3

23ACS47: MATHEMATICAL FOUNDATIONS FOR QUANTUM COMPUTING

Course Outcomes (COs)

CO Code	Description	Bloom's Level
CO1	Understand complex numbers & linear algebra	Understand
CO2	Apply vector space & Dirac notation	Apply
CO3	Analyze unitary &Hermitian operators	Analyze
CO4	Evaluate eigen decomposition in quantum ops	Evaluate
CO5	Create models using probability theory	Create

Unit I: Foundations of Complex Vector Spaces

Complex Numbers: Polar form, Euler's formula, Vectors in $\mathbb{C}^n$, Inner Product Spaces, Dirac Notation (Bra-Ket), Hilbert Space: Definitions and Properties, Orthogonality and Completeness, Norms, Metrics, and Distance in Complex Spaces

Unit II: Matrix Algebra and Operators

Matrix Multiplication and Linear Transformations, Special Matrices: Identity, Diagonal, Unitary, Tensor Products of Matrices and Vectors, Kronecker Product Applications, Unitary and Invertible Operators, Quantum Gates as Linear Operators

Unit III: Eigen Concepts and Quantum Observables

Eigenvalues and Eigenvectors, Hermitian Operators and Spectral Theorem, Quantum Observables and Expectation Values, Commutators and Compatibility, Measurement Operators and Matrix Diagonalization, Applications in Quantum Gate Analysis

Unit IV: Quantum Measurement & Probability

Basics of Probability Theory in Quantum Systems, Born's Rule and Measurement Probabilities, Projection Postulate, Density Matrix Formalism, Mixed States and Pure States, Trace, Partial Trace, and Operator Sums

Unit V: Advanced Structures in Quantum Math *(CO5 – Create)*

Group Theory Basics: Symmetry, Permutations, Pauli Group, Clifford Group, and their roles, Fourier Transform in Quantum Context, Gram-Schmidt Orthogonalization, Lie Groups and Lie Algebras, Use of Lie Algebra in Hamiltonian Formulation

Textbooks

1. **Nielsen & Chuang** – *Quantum Computation and Quantum Information*, Cambridge University Press
2. **Brian C. Hall** – *Quantum Theory for Mathematicians*, Springer
3. **T.S. Blyth & E.F. Robertson** – *Basic Linear Algebra*, Springer

Reference Books

1. **Roman S.** – *Advanced Linear Algebra*, Springer

2. **Axler, Sheldon** – *Linear Algebra Done Right*, Springer
3. **Shankar, R.** – *Principles of Quantum Mechanics*, Springer
4. **W. Greiner** – *Quantum Mechanics: An Introduction,* Springer

Online Courses & Resources

Platform	Course Name	Link
MIT OpenCourseWare	*Linear Algebra (Gilbert Strang)*	Link
edX	*Mathematics for Quantum Computing* by TUDelft	Link
Khan Academy	*Linear Algebra, Probability & Statistics*	Link
Quantum Country	*Spaced Repetition & Essays on Quantum Math*	Link

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L	T	P	C
3	0	0	3

23ACS48: QUANTUM ALGORITHMS

Course Outcomes (COs)

CO Code	Description	Bloom's Level
CO1	Understand quantum algorithm building blocks	Understand
CO2	Analyze well-known quantum algorithms	Analyze
CO3	Apply quantum algorithms to application domains	Apply
CO4	Evaluate efficiency and complexity of algorithms	Evaluate
CO5	Create and simulate quantum algorithms	Create

Unit I: Mathematical Tools for Quantum Algorithms

Review of Complex Numbers & Linear Algebra for Quantum Computing, Inner Product Spaces, Hilbert Spaces, Dirac Notation and Interpretations, Quantum State Vectors and Superposition, Overview of Quantum Gates and Operators, Building Block Concepts for Algorithmic Design

Unit II: Quantum Circuits and Operations

Quantum Gates: X, H, Z, CNOT, Toffoli, Quantum Circuits: Representation and Simulation, Quantum Teleportation Protocol, Circuit-based Measurement and State Collapse, Reversible Computing and Unitary Evolution, Applying Circuits to Small-scale Problems

Unit III: Search and Oracle-Based Algorithms

Deutsch's Algorithm: Problem and Solution Strategy, **Simon's Algorithm**: Period-finding and Speed-up Over Classical, **Grover's Search Algorithm**: Amplitude Amplification, Oracle Construction in Grover's Algorithm, Circuit Analysis and Complexity Comparison, Limitations and Applications in Database Search

Unit IV: Fourier-Based & Cryptographic Algorithms *(CO4 – Evaluate)*

Quantum Fourier Transform (QFT): Theory and Circuit, **Phase Estimation Algorithm**: Foundations and Usage, **Shor's Algorithm**: Integer Factorization and Discrete Logarithms, Modular Arithmetic and Period Finding, Cryptographic Implications of Quantum Algorithms, Efficiency Analysis vs Classical RSA Factorization

Unit V: Advanced & Hybrid Quantum Algorithms *(CO5 – Create)*

Variational Quantum Eigensolver (VQE), Quantum Approximate Optimization Algorithm (QAOA), Quantum Machine Learning (QML): Classification & Clustering, Hybrid Quantum-Classical Models, IBM Qiskit&Cirq for Implementation, Building Custom Quantum Algorithms for NISQ Devices

Textbooks

1. **Michael A. Nielsen & Isaac L. Chuang** – *Quantum Computation and Quantum Information*, Cambridge University Press
2. **Cristopher Moore & Stephan Mertens** – *The Nature of Computation*, Oxford University Press
3. **Eleanor G. Rieffel& Wolfgang Polak** – *Quantum Computing: A Gentle Introduction*, MIT Press

Reference Books

1. **Gideon Amir** – *Quantum Algorithms via Linear Algebra*, MIT Press
2. **S. Jordan** – *Quantum Algorithm Zoo*, [Online repository]
3. **T. G. Wong** – *Quantum Algorithm Design Techniques*
4. **Roland, Cerf** – *Quantum Search Algorithms*, Springer

Online Courses & Resources

Platform	Course Name	Link
edX (MIT)	*Quantum Algorithms for Cybersecurity*	Link
Coursera	*Quantum Computing* by University of London	Link
Qiskit Textbook	*Algorithms & Quantum Machine Learning Modules*	Link
Braket (AWS)	*Quantum Computing Developer Tools & Tutorials*	Link

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III B.Tech II Semester CE,ME,EEE,ECE(Minor)

L	T	P	C
3	0	0	3

23ACS49: QUANTUM INFORMATION AND COMMUNICATION

Course Outcomes (COs)

CO Code	Description	Bloom's Level
CO1	Understand quantum information concepts	Understand
CO2	Apply quantum communication protocols	Apply
CO3	Analyze fidelity, entropy, and data transfer limits	Analyze
CO4	Evaluate quantum cryptographic techniques	Evaluate
CO5	Create and simulate quantum communication models	Create

Unit I: Quantum Information Basics

Classical vs Quantum Information, Density matrices and mixed states, Quantum entropy and Shannon entropy, Von Neumann entropy, Quantum data compression,

Unit II: Quantum Communication Protocols

Quantum teleportation, Superdense coding, Quantum repeaters and communication channels, No-cloning theorem, Quantum channel capacity

Unit III: Fidelity, Distance & Information Theory

Fidelity and trace distance, Quantum mutual information, Holevo bound, Information trade-offs in communication, Channel noise and error modeling

Unit IV: Quantum Cryptography

Principles of quantum cryptography, BB84 and B92 key distribution protocols, Eavesdropping and security analysis, Quantum bit commitment, Post-quantum cryptography relevance

Unit V: Applications & Tools

Quantum internet: architecture and challenges, Networked quantum systems, Simulation using Qiskit, NetSquid, QuTiP, IBM Q Network and cloud-based setups, Practical implementation of QKD in simulation

Textbooks

1. Michael A. Nielsen & Isaac L. Chuang – *Quantum Computation and Quantum Information*, Cambridge University Press
2. Mark M. Wilde – *Quantum Information Theory*, Cambridge University Press
3. John Watrous – *The Theory of Quantum Information*, Cambridge University Press

Reference Books

1. Peter W. Shor – *Foundations of Quantum Computing* (Lecture notes)
2. Charles H. Bennett & Gilles Brassard – *Original Papers on QKD (BB84)*
3. Stephanie Wehner – *Quantum Communication Networks*, arXiv

Online Courses & Resources

Platform	Course Name	Link
Coursera	*Quantum Cryptography* by University of Geneva	Coursera Link
edX	*Quantum Information Science I* (Harvard/MIT)	edX Course
Qiskit	*Quantum Information Applications in Qiskit Textbook*	Qiskit Info
QuTech	*Quantum Internet Tutorials & Tools*	QuTech

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
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IV B.Tech I Semester CE,ME,EEE,ECE(Minor)

L	T	P	C
3	0	0	3

23ACS50: QUANTUM MACHINE LEARNING (QML)

Course Outcomes (COs)

CO Code	Description	Bloom's Level
CO1	Understand foundations of quantum machine learning	Understand
CO2	Apply QML algorithms to datasets	Apply
CO3	Analyze quantum kernels, data encoding, and models	Analyze
CO4	Evaluate hybrid quantum-classical models	Evaluate
CO5	Create and simulate QML models using frameworks	Create

Unit I: Introduction to QML

Need for QML: Why quantum for ML?,Classical vs quantum machine learning, Quantum states as information carriers, Data encoding: amplitude, angle, basis encoding, Introduction to quantum feature space.

Unit II: QML Algorithms – Supervised Learning

Quantum classifiers (quantum SVMs, qNN), Quantum perceptron, Variational quantum classifiers (VQC), Quantum kernels, Cost functions in quantum models

Q Unit III: QML Algorithms – Unsupervised Learning

Quantum k-means and clustering, Quantum PCA, Quantum generative models (QGANs), Dimensionality reduction and similarity metrics, Performance analysis and limitations

Unit IV: Hybrid Models & Optimization *(CO4 – Evaluate)*

Variational Quantum Circuits (VQCs), Hybrid quantum-classical training loops, Barren plateaus and optimization issues, Quantum gradient descent and parameter shift rule, Comparative study of classical and QML models

Unit V: QML Tools and Case Studies *(CO5 – Create)*

Implementing QML with Qiskit Machine Learning, PennyLane and TensorFlow Quantum integration, Case studies: quantum-enhanced fraud detection, NLP, Quantum datasets and benchmark models, Project: design a small QML application

Textbooks

1. Maria Schuld, Francesco Petruccione – *Machine Learning with Quantum Computers*, Springer
2. Peter Wittek – *Quantum Machine Learning: What Quantum Computing Means to Data Mining*, Academic Press

Reference Books

1. Jacob Biamonte – *Quantum Machine Learning*, Nature, 2017
2. Seth Lloyd – *Quantum algorithms for supervised/unsupervised learning* (Research papers)
3. VojtěchHavlíček – *Supervised Learning with Quantum-Enhanced Feature Spaces*, Nature, 2019

Online Courses & Resources

Platform	Course Name	Link
edX	*Quantum Machine Learning* by UTS	edX Course
Qiskit	*Qiskit Machine Learning Module*	Qiskit ML
Xanadu	*QML with PennyLane (Free online textbook)*	PennyLane QML Book
Coursera	*Quantum Machine Learning* by University of Toronto	Coursera

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B.Tech II Semester CE,ME,EEE,ECE(Minor)

L	T	P	C
0	0	3	1.5

23ACS50: QUANTUM ALGORITHMS LAB

Experiments :

1. Deutsch Algorithm
2. Deutsch-Jozsa
3. Grover’s Algorithm
4. QFT Visualization
5. Shor’s Algorithm
6. QRNG Implementation
7. Bell State Entanglement
8. Bernstein-Vazirani Algorithm
9. Quantum Teleportation
10. Phase Estimation
11. Circuit Simulation
12. Mini-Project: RSA Key Breaking

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

IV B.Tech I Semester CE,ME,EEE,ECE(Minor)

L	T	P	C
0	0	3	1.5

23ACS50: QUANTUM PROGRAMMING AND SIMULATION LAB

LIST OF EXPERIMENTS

1. State Vector Simulation (Qiskit)
2. Bell State Implementation
3. Deutsch-Jozsa Circuit
4. Grover's Search in Qiskit
5. QFT Circuit in Python
6. Shor Algorithm Simulation
7. Quantum Teleportation in Code
8. VQE (Hybrid Circuit)
9. QAOA Simulation
10. Quantum Random Number Generator
11. Comparison: Real vs Simulated Runs
12. Mini-Project: Quantum Password Cracker

Textbooks & References

- Michael Nielsen & Isaac Chuang – *Quantum Computation and Quantum Information*
- Eric R. Johnston et al. – *Programming Quantum Computers*
- David McMahon – *Quantum Computing Explained*
- Gilbert Strang – *Introduction to Linear Algebra*
- Sarah Kaiser & Chris Granade – *Learn Quantum Computing with Python and Q#*

Online Resources

- IBM Qiskit Textbook: https://qiskit.org/learn
- Microsoft Q# Documentation: https://learn.microsoft.com/en-us/azure/quantum/
- Coursera: *Introduction to Quantum Computing*
- edX: *Quantum Computing Fundamentals, Quantum Algorithms*

MINORS IN CLOUD COMPUTING
SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B. Tech-I Semester

23ACS53: FUNDAMENTALS OF CLOUD COMPUTING

L	T	P	C
3	0	0	3

Course Outcomes:

At the end of the course, the student will be able to

CO1: Explain the fundamental concepts, history, characteristics, benefits, and challenges of cloud computing.

CO2: Differentiate between various cloud service and deployment models and evaluate their use cases.

CO3 : Analyze the role of virtualization and cloud architecture in delivering cloud services.

Unit 1: Introduction to Cloud Computing

Definition and core concepts-History and evolution of cloud computing-Key characteristics: on-demand, scalability, elasticity, multitenancy-Benefits: cost, performance, speed, productivity-Challenges and limitations

Unit 2: Service Models

Infrastructure as a Service (IaaS): definition, examples (AWS EC2, Azure VM)-Platform as a Service (PaaS): definition, examples (Google App Engine)-Software as a Service (SaaS): definition, examples (Salesforce, Office 365)-Comparison of models and use cases

Unit 3: Deployment Models

Public, Private, Hybrid, and Community Clouds-Use-case analysis of each model Inter-cloud and multi-cloud strategies-Cloud bursting concept

Unit 4: Virtualization Basics

Role of virtualization in cloud-Types: hardware, OS, storage, network-Hypervisors: Type 1 and Type 2- Benefits of virtualization for cloud-Tools: VMware, VirtualBox

Unit 5: Cloud Architecture

Components: clients, data centers, distributed servers-NIST reference architecture Cloud delivery architecture-Design considerations and industry standards

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B. Tech-I Semester

L	T	P	C
3	0	0	3

23ACS54: VIRTUALIZATION AND CLOUD INFRASTRUCTURE

Course Outcomes:

At the end of the course, the student will be able to

CO1: Explain virtualization technologies including virtual machines, containers, Docker, and Kubernetes, and apply them to cloud-native application development.

CO2:
Analyze and design cloud infrastructure and networking components including compute, storage, VPCs, load balancing, and secure network configurations.

CO3: Utilize cloud monitoring tools, logging, and cost optimization strategies to manage cloud resources effectively and ensure service level agreement (SLA) compliance

Unit 1: Advanced Virtualization

Virtual Machine management-Containerization vs Virtual Machines-Docker architecture and components-Kubernetes concepts: pods, services, deployments-Use cases in cloud-native development

Unit 2: Cloud Infrastructure

Cloud data centers: physical infrastructure, energy efficiency-Compute resources: CPU, GPU, memory optimization-Storage types: block, object, file -Networking: switches, routers, virtual networks

Unit 3: Cloud Networking

IP addressing and subnetting-Virtual Private Cloud (VPC)-Load balancing techniques-DNS in cloud
Network isolation and firewalling

Unit 4: Storage Systems

File systems in the cloud -Cloud storage gateways-Distributed storage concepts-Storage security and compliance-Backup and recovery strategies

Unit 5: Monitoring and Resource Management

Cloud monitoring tools: AWS CloudWatch, Azure Monitor -Logging and alerting-Resource provisioning and scaling-Cost management and optimization-SLAs and service uptime monitoring

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B. Tech-II Semester

L	T	P	C
3	0	0	3

23ACS55: CLOUD PLATFORMS AND DEVELOPMENT

Course Outcomes:

At the end of the course, the student will be able to

CO1: Demonstrate proficiency in core services of major cloud platforms—AWS, Azure, and GCP—including virtual machines, storage, databases, and identity management.

CO2: Use command-line interfaces, SDKs, and deployment tools to configure and manage cloud resources efficiently across multiple platforms.

CO3: Design and develop scalable cloud-native applications using the 12-factor app model, CI/CD pipelines, and stateless architecture principles.

CO4: Implement serverless solutions using AWS Lambda, Azure Functions, and API Gateways, and evaluate their performance, cost-effectiveness, and real-world applications.

Unit 1: AWS Services Overview

EC2, S3, RDS, Lambda, VPC-AWS CLI and SDK-IAM and role-based access-Billing and cost estimation

Unit 2: Microsoft Azure Overview

Virtual Machines, Blob Storage, SQL Database-Azure Functions and App Services- Azure DevOps and Pipelines-Role assignments and resource groups

Unit 3: Google Cloud Platform (GCP)

Compute Engine, Cloud Storage, BigQuery -GCP IAM and security-Cloud Shell and Cloud SDK
Deployment Manager

Unit 4: Application Development in Cloud

Cloud-native app principles -12-factor application model-CI/CD integration
Stateless architecture design

Unit 5: Serverless Architecture

Definition and comparison with traditional architectures-FaaS (Functions as a Service): AWS Lambda, Azure Functions-Event-driven programming -API Gateway integration-Use cases and cost implications

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B. Tech-II Semester

L	T	P	C
3	0	0	3

23ACS56: CLOUD SECURITY AND COMPLIANCE

COURSE OUTCOMES:

At the end of the course, the student will be able to

CO1: Explain cloud security principles, including the shared responsibility model, IAM, and data protection strategies to ensure confidentiality, integrity, and availability in cloud environments.
CO2: Utilize security tools and technologies such as firewalls, VPNs, encryption mechanisms, and access controls to design and implement secure cloud infrastructures.
CO3: Analyze compliance requirements, legal challenges, and risk management strategies, and apply auditing techniques and disaster recovery planning to maintain regulatory and operational integrity.

Unit 1: Cloud Security Fundamentals

Shared responsibility model -Cloud security architecture-IAM principles and RBAC
Data confidentiality and integrity

Unit 2: Security Tools and Technologies

Firewalls, IDS, IPS-Data encryption (at-rest, in-transit)-SSL/TLS, VPNs-Security groups and network ACLs

Unit 3: Compliance and Legal Issues

GDPR, HIPAA, PCI-DSS -Data sovereignty and jurisdiction-Legal implications of multi-tenancy
Vendor lock-in issues

Unit 4: Auditing and Risk Management

Security auditing tools and logs-Risk analysis frameworks-Disaster recovery and business continuity
Penetration testing and vulnerability scanning

Unit 5: Identity and Access Management

Single Sign-On (SSO) -Multi-Factor Authentication (MFA)-Federated identity management
OAuth2 and OpenID Connect protocols

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

IV B. Tech-I Semester

L	T	P	C
3	0	0	3

23ACS57: CLOUD APPLICATION DESIGN AND DEVOPS

COURSE OUTCOMES:
At the end of the course, the student will be able to

CO1: Design and implement cloud-native applications using microservices architecture, ensuring effective inter-service communication, service discovery, and fault tolerance mechanisms.
CO2: Apply DevOps principles by building CI/CD pipelines, automating infrastructure deployment with Infrastructure as Code (IaC), and managing monitoring and deployment strategies.
CO3: Develop and test robust APIs and cloud applications using industry-standard practices, including automated testing, load testing, and security mechanisms like OAuth.

Unit 1: Microservices Architecture

Basics of microservices - Inter-service communication: REST, Grpc -Service discovery and registry Fault tolerance and retry mechanisms

Unit 2: DevOps Principles

CI/CD pipelines: Jenkins, GitHub Actions -Infrastructure as Code: Terraform, CloudFormation Monitoring: Prometheus, Grafana -Blue/green and canary deployments

Unit 3: API Development and Management

RESTful API design -API versioning and documentation -API gateway integration OAuth, rate limiting, caching

Unit 4: Testing Cloud Applications

Unit and integration testing -Automated testing frameworks- Load testing: JMeter, Locust Chaos engineering

Unit 5: Cloud Application Deployment

Deployment strategies - CI/CD for Kubernetes -Docker Compose and Helm Scaling and rollback strategies

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

IV B. Tech-I Semester

L	T	P	C
0	0	3	3

23ACS58: CAPSTONE PROJECT

MINORS IN FULL STACK MERN & DEVOPS

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B. Tech-I Semester

23ACS59: WEB FUNDAMENTALS, ADVANCED JAVASCRIPT & GITHUB

L	T	P	C
3	0	0	3

Course Outcomes:

At the end of the course, the student will be able to

CO1: Understand the structure of modern web pages and create accessible HTML5 content using semantic elements, multimedia, and form validation.

CO2: Apply CSS concepts such as box model, Flexbox, and media queries to design visually appealing and responsive web layouts.

CO3: Demonstrate proficiency in core JavaScript concepts including asynchronous programming, API integration, and robust error handling.

CO4: Collaborate using GitHub tools like pull requests, issues, and GitHub Pages while integrating advanced browser features and APIs.

CO5: Design, build, and deploy a complete web application by utilizing acquired front-end skills and practicing effective team collaboration.

UNIT I: WEB DEVELOPMENT AND HTML5

Overview of the Internet, HTTP/HTTPS protocols, and DNS -Understanding web servers, clients, and hosting infrastructure- Introduction to web browsers and developer tools- Anatomy and structure of modern web pages Basic HTML elements: headings, paragraphs, lists, and tables- Creating forms with built-in validation features -Utilizing semantic HTML for accessibility and SEO - Embedding multimedia: audio, video, iframes

UNIT II: STYLING WITH CSS

CSS syntax: selectors, colors, fonts, and units- Box model: margins, padding, and borders

Layout with display types and Flexbox- Creating responsive layouts with media queries

UNIT III: JAVASCRIPT CONCEPTS

JavaScript scope, hoisting, and closures - Arrow functions and template literals - Destructuring and spread/rest operators- Callback functions and higher-order functions. **ASYNCHRONOUS JAVASCRIPT** Working with JSON: parsing and stringifying - Promises and async/await syntax

Making API requests using Fetch - Error handling using try-catch

UNIT IV:: COLLABORATIVE DEVELOPMENT WITH GITHUB

Using pull requests, issues, and tagging - Writing documentation in Markdown - Hosting with custom domains on GitHub Pages - Managing projects using GitHub Projects and Discussions-

Event delegation and advanced event handling - Web storage: Local Storage and Session Storage - Browser APIs: Geolocation, Notifications - Timers, intervals, and performance monitoring

UNIT 5: PROJECT AND DEPLOYMENT

Develop a weather application using public APIs- Deploy the app using GitHub Pages - Team collaboration and peer code reviews-

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B. Tech-I Semester

23ACS60: BACKEND DEVELOPMENT WITH NODE.JS & EXPRESS

L	T	P	C
3	0	0	3

Course Outcomes:

At the end of the course, the student will be able to

CO1: Develop server-side applications using Node.js and Express, incorporating middleware, routing, and RESTful API principles.

CO2: Integrate MongoDB with Node.js applications using Mongoose for data modeling, and perform CRUD operations securely and efficiently.

CO3: Design, build, test, and document a complete backend RESTful API project using best practices in version control, environment management, and security.

UNIT 1: GETTING STARTED WITH NODE.JS

Node.js architecture and event-driven programming - Using Node Package Manager (NPM) - Performing file system operations and path management - Creating basic servers with Node.js

UNIT 2: DEVELOPING WITH EXPRESS.JS

Introduction to Express and route handling - Implementing middleware functions- Organizing routes using Express Router - RESTful API design principles

UNIT 3: WORKING WITH MONGODB & MONGOOSE

Introduction to NoSQL databases and MongoDB - Setting up cloud-hosted databases with MongoDB Atlas - CRUD operations using Mongo Shell - Creating schemas and models using Mongoose

UNIT 4: APPLICATION ENVIRONMENT AND SECURITY

Connecting Node.js applications to MongoDB - Securing configurations with environment variables - Implementing error handling and logging - Securing APIs with CORS, Helmet, and Morgan

.

UNIT 5: BACKEND API PROJECT

Build a Notes or Tasks RESTful API - Use Postman to test and document endpoints- Apply Git-based version control best practices

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B. Tech-II Semester

23ACS61: BUILDING USER INTERFACES WITH REACT.JS

L	T	P	C
3	0	0	3

Course Outcomes:

At the end of the course, the student will be able to

CO1: Apply React fundamentals such as JSX, component-based architecture, props, state management, and event handling to build interactive UIs.

CO2: Implement advanced state management techniques using hooks (useState, useEffect, custom hooks) and Context API for global state handling.

CO3: Develop dynamic, user-friendly applications by integrating React Router for navigation, handling forms with validation, and managing data using APIs.

CO4: Build, connect, and deploy a complete full-stack React application with backend integration, environment configuration, and modern styling practices.

UNIT 1: REACT FUNDAMENTALS

Introduction to JSX and component-based architecture - Using props and managing component state- Handling user input with event listeners- Implementing conditional rendering

UNIT 2: REACT HOOKS AND STATE MANAGEMENT

useState and useEffect hooks - Creating and using custom hooks- Context API for global state management - Lifting state and prop drilling concepts

UNIT 3: ROUTING AND FORM HANDLING

Setting up navigation with React Router DOM- Implementing nested and dynamic routes Controlled vs. uncontrolled forms- Form validation with Formik and Yup

UNIT 4: STYLING AND COMPONENT INTEGRATION

Fetching and handling data using Axios - Managing loading and error states - Styling with Tailwind CSS or Styled Components

- Using UI libraries like Material UI (MUI)

UNIT 5: FULLSTACK APP DEVELOPMENT

Connecting React frontend to backend API - Managing environment variables in React Building and deploying React apps - Final Project: Blog or Task Manager Application

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B. Tech-II Semester

23ACS62: DEVOPS FUNDAMENTALS & CI/CD PIPELINES

L	T	P	C
3	0	0	3

Course Outcomes:

At the end of the course, the student will be able to

CO1: Explain DevOps principles, lifecycle, and tools, and relate them to Agile and Scrum methodologies.

CO2:Automate routine tasks using shell scripting, process control, and scheduled jobs to improve operational efficiency.

CO3:Design and implement CI/CD pipelines using GitHub Actions for automated testing, deployment, and code quality assurance.

CO4: Build, containerize, and deploy full-stack applications using Docker and apply DevOps best practices in a collaborative project.

UNIT 1: INTRODUCTION TO DEVOPS

DevOps principles and culture - CI/CD fundamentals and benefits - DevOps lifecycle and tools overview- Introduction to Agile and Scrum

UNIT 2: SHELL SCRIPTING & AUTOMATION

Bash scripting basics: variables, loops, conditionals - Automating repetitive tasks -File permissions and process control - Scheduling with cron jobs

UNIT 3: GITHUB ACTIONS FOR CI/CD

Creating and managing GitHub workflows - Using actions and jobs for automation - Testing and linting in CI pipelines Continuous deployment using GitHub Actions

UNIT 4: CONTAINER- ZATION WITH DOCKER

Understanding images and containers - Creating Dockerfiles and using Docker Compose
Building and running containerized MERN apps - Managing Docker volumes and networks

UNIT 5: DEVOPS LAB PROJECT

CI/CD setup for a MERN app using GitHub Actions - Containerizing the app using Docker - Automating tests and deployments -Peer review and code quality assessments

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

III B. Tech-II Semester

23ACS63: ADVANCED BACKEND & MICROSERVICES

L	T	P	C
3	0	0	3

Course Outcomes:
At the end of the course, the student will be able to

CO1: Understand the core AWS services including EC2, S3, IAM, and RDS, and manage access control, billing, and resource provisioning.
CO2: Deploy and configure scalable web applications on AWS using services like EC2, S3, Route 53, and Certificate Manager.
CO3: Implement cloud-based database and storage solutions with RDS and S3, integrating them into Node.js applications and ensuring data reliability.
CO4: Design and deploy a full-stack cloud application using AWS services, incorporating load balancing, auto-scaling, monitoring, and CI/CD pipelines.

UNIT 1: AWS ESSENTIALS

Overview of AWS global infrastructure - Introduction to EC2, S3, IAM, and RDS- Managing permissions and access control - Billing, pricing, and Free Tier usage

UNIT 2: DEPLOYING WEB APPS ON AWS

Launching and configuring EC2 instances - Hosting static sites on S3- Using Route 53 for custom domain configuration- Setting up HTTPS with AWS Certificate Manager

UNIT 3: DATABASE AND STORAGE SOLUTIONS

Using RDS for PostgreSQL/MongoDB - Setting up S3 buckets and managing lifecycle policies
Integrating AWS SDK with Node.js - Backup and recovery strategies

UNIT 4: LOAD BALANCING & SCALING

AWS Elastic Load Balancer (ELB) - Auto Scaling Groups for EC2 -Monitoring with CloudWatch - Application health checks and alerts

UNIT 5: DEPLOYMENT PROJECT

Deploy a full stack MERN app on AWS - Use RDS and S3 in the architecture - Setup CI/CD from GitHub to AWS - Document and present deployment architecture

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B. Tech-II Semester

23ACS64: PROJECT ON FULL STACK MERN AND DEVOPS

L	T	P	C
0	0	6	3

MINORS IN BIO MEDICAL ENGINEERING
SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

III B. Tech II Semester
Course Code: 23ABM09

NANO BIOTECHNOLOGY

L	T	P	C
3	0	0	3

Course Outcome:

After completion of this course students will be able to

CO1: Understand the basics of bio-nanotechnology and bio-machines.

CO2: Demonstrate a comprehensive understanding of state-of-the-art methods for fabrication, characterization and handling of nano-materials.

CO3: Explain the functional principles of nanotechnology and the interaction between biomolecules and nanoparticle surface.

CO4: Apply the knowledge on nanotechology in the field of Biomedical Engineering

UNIT- I:

Bio-Nano machines and Their Basics: Negligible gravity and inertia, atomic granularity, thermal motion, water environment and their importance in bionanomachines. The role of proteins- amino acids- nucleic acids- lipids and polysaccharides in modern biomaterials. Overview of natural Bionanomachines: Thymidylate Synthetase, ATP synthetase, Actin and myosin, Opsin, Antibodies and Collagen.

UNIT- II:

Synthesis of Bio molecules and Interphase Systems: Recombinant Technology, Site- directed mutagenesis, Fusion Proteins. Quantum Dot structures and their integration with biological structures. Molecular modeling tools: Graphic visualization, structure and functional prediction, Protein folding prediction and the homology modeling, Docking simulation and Computer assisted molecular design. Interphase systems of devices for medical implants –Microfluidic systems –Microelectronic silicon substrates –Nano- biometrics Introduction –Lipids as nano-bricks and mortar: self-assembled nanolayers.

UNIT- III:

Functional Principles of Nano biotechnology: Information driven nanoassembly, Energetic, Role of enzymes in chemical transformation, allosteric motion and covalent modification in protein activity regulation, Structure and functional properties of Biomaterials, Bimolecular motors: ATP Synthetase and flagellar motors, Traffic across membranes: Potassium channels, ABC Transporters and Bactreriorhodapsin, Bimolecular sensing, Self- replication, Machine-Phase Bionanotechnology Protein folding; Self assembly, Self- organization, Molecular recognition and Flexibility of biomaterials.

UNIT- IV:

Protein and DNA based Nanostructures : Protein based nanostructures building blocks and templates – Proteins as transducers and amplifiers of biomolecular recognition events – Nanobioelectronic devices and polymer nanocontainers – Microbial production of inorganic nanoparticles – Magnetosomes, DNA based nanostructures – Topographic and Electrostatic properties of DNA and proteins – Hybrid conjugates of gold nanoparticles – DNA oligomers – Use of DNA molecules in nanomechanics and Computing.

UNIT- V:

Applications of Nano biotechnology: Semiconductor (metal) nanoparticles and nucleic acid and protein-based recognition groups –Application in optical detection methods – Nanoparticles as carrier for genetic material –Nanotechnology in agriculture – Fertilizer and pesticides. Designer proteins, Peptide nucleic acids, Nanomedicine, Drug delivery, DNA computing, Molecular design using biological selection, Harnessing molecular motors, Artificial life, Hybrid materials, Biosensors, Future of Bionanotechnology.

TEXT / REFERENCE BOOKS:

1. C. M. Niemeyer, C. A. Mirkin, ―Nanobiotechnology: Concepts, Applications and Perspectives‖, Wiley – VCH, (2004).

2. David S Goodsell, "Bionanotechnology‖, John Wiley & Sons, (2004).

Reference Books:

3. T. Pradeep, ―Nano: The Essentials, McGraw – Hill education, (2007).

4. Challa, S.S.R. Kumar, Josef Hormes, Carola Leuschaer, Nanofabrication Towards Biomedical Applications, Techniques, Tools, Applications and Impact, Wiley – VCH, (2005).

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B. Tech II Semester
Course Code: 23ABM10

TISSUE ENGINEERING

L	T	P	C
3	0	0	3

Course Outcome:

After completion of the course, students will be able to

CO1: Understand the biological requirement for tissue engineering systems and also specify the different types of biodegradable biomaterials that can be used in tissue engineering applications. **CO2:** Discuss the complex interactions between biomaterials, cells and signals in biological systems using stem cells, proteomics and bioreactors.

CO3: Design and fabricate scaffolds using advanced manufacturing technologies including 3D printing for growing biological materials.

CO4: Develop engineered tissue like cardiovascular tissues and also evaluate the patterning of bio- mimetic substances.

UNIT- I:

Introduction to Tissue Engineering : Introduction – definitions - basic principles - structure-function relationships –Biomaterials: metals, ceramics, polymers (synthetic and natural) – Biodegradable materials - native matrix -Tissue Engineering and Cell-Based Therapies –Tissue Morphogenesis and Dynamics- Stem Cells and Lineages - Cell-Cell Communication.

UNIT- II:

Tissue Culture Basics: Primary cells vs. cell lines - sterile techniques – plastics – enzymes - reactors and cryopreservation - Synthetic Biomaterial Scaffolds- Graft Rejection – Immune Responses-Cell Migration-Controlled Drug Delivery- Micro technology Tools.

UNIT- III:

Scaffold Formation: Oxygen transport - Diffusion - Michalies-Menten kinetics - oxygen uptake rates -limits of diffusion - Principals of self-assembly - Cell migration - 3D organization and angiogenesis - Skin tissue engineering –Introduction - scar vs. regeneration
- split skin graft -apligraft. Engineered Disease Models- Tissue Organization- Cell Isolation and Culture - ECM and Natural Scaffold Materials- Scaffold Fabrication and Tailoring, Hernia.

UNIT- IV:

Cardiovascular Tissue Engineering: Blood vessels structure – vascular grafts – Liver tissue engineering – Bioartificial liver assist device – shear forces – oxygen transport – plasma effects – Liver tissue engineering – Self assembled organoids – decelluarized whole livers – Stem cells – basic principle – embryonic stem cells – Induced pluripotent stem cells - Material Biocompatibility – Cell Mechanics – Vascularization- Stem Cell Therapies.

UNIT - V:

Patterning of Biomimetic Substrates: Patterning of biomimetic substrates with AFM lithography primarily focusing on DPN Nanotemplating polymer melts - Nanotechnology- based approaches in the treatment of injuries to tendons and ligaments - Progress in the use of electrospinning processing techniques for fabricating nanofiber scaffolds for neural applications -Nanotopography techniques for tissue engineered scaffolds

TEXT / REFERENCE BOOKS:

1. KetulPopat“Nanotechnology in Tissue Engineering and Regenerative Medicine” CRC Press Taylor and Francis 2011.
2. Cato T. Laurencin, Lakshmi S “Nanotechnology and Tissue Engineering: The Scaffold “CRC Press Taylor and Francis 2008.
3. Kun Zhou, David Nisbet, George Thouas, Claude Bernard and John Forsythe “Bio nanotechnology Approaches to Neural Tissue Engineering”, NC-SA 2010.
4. Nair “Biologically Responsive Biomaterials for Tissue Engineering”, Springer Series in Biomaterials Science and Engineering, Vol. 1 Antoniac, Iulian (Ed.) 2012.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

II B. Tech II Semester

Course Code: 23ABM02

BIOMATERIALS

L	T	P	C
3	0	0	3

Course Outcomes:

On completion of the course, the student will be able to

CO1: Identify different types of biomaterials used for various biomedical applications. (L2) **CO2:** Recognize various synthetic polymers, used as biomaterials for organ replacement.(L3) **CO3:** Understand different types of ceramics and composite used as biomaterials. (L4)

CO4: Acquire knowledge about hard tissue replacement and soft tissue replacement for various fixations. (L4)

CO5: Apply knowledge about how to rectify eye defect using various biomaterials.

CO6: Implement various methods to test Biocompatibility and Hemocompatibility test of biomaterials (L5)

UNIT I

METALLIC BIOMATERIALS: Biomaterials - Overview, Classification of biomaterials, Biocompatibility and Hemocompatibility, Metals and alloys -Stainless steel, Titanium and its alloys, Cobalt chromium alloy, Metallic corrosion, Dental implants - Impression Materials, Fillings and Restoration Materials, Materials for Deep Cavities, Material for oral and Maxillofacial Implants.

UNIT II:

SYNTHETIC POLYMERS AND APPLICATIONS: Synthetic Polymers, Polymers in biomedical use, Polyethylene, Per fluorinated Polymers, Acrylic Polymers, Hydrogels, Polyurethanes, biodegradable synthetic polymers, silicone rubber, microorganisms in polymeric implants, Polymer Sterilization.

UNIT III:

BIOCERAMICS AND COMPOSITES: Bio ceramics, types- Carbon, Alumina, Zirconia - bioactive resorbable and non – resorbable bioceramics, bioceramic coatings on metallic implants and bone bonding reactions on implantation, Hydroxyapatite-properties and applications. Composites-Types and Applications, Bioglass.

UNIT IV:

HARD AND SOFT TISSUE REPLACEMENT: Bioelectric effect, Wolff's Law, Temporary orthopaedic fixation devices-pins, screws and plates, Intra Medullary and spinal nails, hard tissue replacements - total hip and knee joint replacements. Soft Tissue Replacements-Sutures-Tapes, Staples, Adhesives, Wound Dressings, Biomaterials in urological practice.

UNIT V:
BIOMATERIALS IN OPTHALMOLOGY AND BIOLOGICAL TESTS: Ophthalmology-Introduction, Optical implants, Contact lenses, Eye shields, Viscoelastic solutions, Vitreous implants, Acrylate adhesives, Scleral buckling materials for retinal detachment, artificial tears, Biological Tests.

Textbooks:

1. Sujata V.Bhat, Biomaterials, Narosa Publishing House, New Delhi, India, 2012.

2. William Wagner, Shelly Sakiyama-Elbert, Guigen Zhang, Michael Yaszemski, Biomaterials Science: An Introduction to Materials in Medicine, ELSEVIER, 4th Edition -, 2020

3. Park, Biomaterials an Introduction, Third Edition, Springer, 2007.

4. Joseph D Bronzino, the Biomedical Engineering Hand Book, Fourth Edition, CRC Press, 2015.

References:

1. Buddy D. Ratner, Allan S. Hoffman, Biomaterial Sciences – Int. to Materials in Medicine

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

IV B. Tech I Semester Course
Code: 23ABM14

BIOMEMS & BIOMICROFLUIDICS

L	T	P	C
3	0	0	3

Course outcome:

After completion of this course, students will be able to

CO1: Build a foundation in micro-systems engineering including basic biological/ biochemical concepts and techniques emphasizing biomedical devices.
CO2: Understand material properties important for MEMS system performance; analyze dynamics of resonant micromechanical structures.
CO3: Design and Development of models using microfabrication technique and simulate electrostatic and electromagnetic sensors and actuators.
CO4: Design and evaluation of MEMs and microfluidics based analytical platform as per the requirement.

UNIT- I:

Introduction to BioMEMS and microfluidics, Introduction to Bio nano technology, Biosensors, fluidics. Introduction to device fabrication (Silicon and Polymers) Introduction to device fabrication (Silicon and Polymers). Sensors, Transduction and Performance factors. Sensors, Transduction and Performance factors continued.

UNIT- II:

Important materials for fabrication of BioMEMS platforms Introduction to silicon device fabrication Some Fabrication Methods for soft materials Transduction Methods. About cell potential and SHEs Cell reaction, Nernst equation, Construction of Ion selective electrodes Measurement and calibration of electrodes, ion-solvent interaction.

UNIT- III:

Introduction to Cell biology, Basic structure of DNA DNA hybridization, DNA polymerization,PCR Thermal cycle , Real Time PCR.PCR design Electrophoresis, Gel and Capillary electrophoresis,Agarose DNA microarrays (concepts, and utility). Affymetrix and Nanogen approaches in realization of micro-arrays. DNA sequencing (Sanger’s reaction). DNA nano-pores. DNA detection using Mechanical antilevers.Basics of Protein structure.

UNIT- IV:

Protein charging at different pH range, Amino acids, protein polymerization, Transcription, Translation Antibody, Microencapsulation, Cyclic voltametry Microfluidics, Similarity of Streamlines, Pathlines, Sreaklines and Timelines for a steady flow Stress tensor.

UNIT-V

Viscosity, Newtonian, non-Newtonian fluids, Pseudoplastic, Dilatant, Bingham Plastic materials, Thixotropic fluids. Flow over infinite plates, laminar and turbulent flow, Compressible and Incompressible flows Flow over an infinite plate. Types of flows. Types of Fluids. Kinematics of fluids

Text/ Reference Books:

1. Introduction to BioMEMS, Albert Folch, CRC Press; 1st ed.
2. Essential Cell Biology, Bruce Albert, et al. Garland Science, 2nd ed.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

II B. Tech II Semester

Course Code: 23ABM03

BIOMATERIALS LAB

L	T	P	C
0	0	3	1.5

Course Outcome:

After completion of the course, students will be able to

CO1: Characterize the mechanical properties of biomaterials using destructive and non-destructive testing.

CO2: Measure the Surface roughness of biomaterials as per the ASTM standards.

CO3: Perform *invitro* hemocompatibility study for implantable biomaterials.

CO4: Conduct the test to measure the pH, viscosity and Conductivity of the body fluid.

List of Experiments:

1. Mechanical Characterization of Biomaterials
2. Hardness Testing of Metallic Biomaterials
3. Hardness Testing of Polymeric Biomaterials
4. Surface Roughness measurement
5. *Invitro* hemocompatibility study of biomaterials
6. pH measurement of Body Fluid
7. Conductivity Measurement of Body Fluid
8. Viscosity Measurement of Body Fluid
9. Non-Destructive Testing of Biomaterials
10. Innovative Experiment

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

III B. Tech I Semester
Course Code: 23ABM04
BIOSENSORS & TRANSDUCERS

L	T	P	C
3	0	0	3

Course Outcomes:

After successful completion of the course, the student will be able to: CO1: Explain the introduction and characteristics of transducers

CO2: Illustrate the types of transducers

CO3: Examine the basic characteristics of bio potential recorders

CO4: Explain the structure of biosensors in detail

CO5: Explain the working and applications of Biosensors

UNIT-I:

INTRODUCTION: General measurement system - purpose, structure and elements- Transducers - Definition, Classification. Resistance type- strain gauges, thermometers, potentiometers. Capacitive type, Inductive type- variable reluctance and LVDT. Biomedical Applications.

UNIT-II:

TRANSDUCERS: Temperature transducers, Piezoelectric transducers, Piezo resistive transducers, Photoelectric transducers, Pressure transducers, Magnetostrictive transducers Biomedical applications.

UNIT-III:

BIO POTENTIAL ELECTRODES: Half cell potential (or) Electrode potential, Types of Electrodes - Micro electrodes, Depth and needle electrodes, Surface electrodes, and Chemical electrodes. Catheter type electrodes, stimulation electrodes, electrode paste, electrode material.

UNIT-IV:

BIOSENSORS: Introduction, biological elements, immobilization of biological components. Micro machined biosensor – cantilever based chemical sensors - Biosensors for diabetes mellitus, FAB. Biochip - introduction, gene chip.

UNIT-V:

APPLICATIONS OF BIOSENSORS: ISFET for glucose and urea. IMFET, MOSFET biosensors, affinity biosensor (catalytic biosensor), Enzyme electrodes, Ion exchange membrane electrodes.

TEXT / REFERENCE BOOKS:

1. H.S. Kalsi, Electronic Instrumentation & Measurement, Tata McGraw HILL, 1995.
2. Brain R Eggins, Biosensors: An Introduction‖, John Wiley Publication.1996.
3. A. K.Sawhney, A course in Electronic Measurements and Instruments, Dhapat Rai & sons, 1991
4. John G Webster, Medical Instrumentation: Application and design, John Wiley Publications.2007.
5. John P Bentley, Principles of Measurement Systems, 3rd Edition, Pearson Education Asia, (2000 Indian reprint)
6. Geddes and Baker, Principles of Applied Biomedical Instrumentation, John Wiley Publications. 1975.

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

III B. Tech I Semester
Course Code: 23ABM05

BIOSENSORS & TRANSDUCERS LAB

L	T	P	C
0	0	3	1.5

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Analyze about various theory, practical characteristics and conversions of the various transducers

CO2: Evaluate the measurement of the vital physiological signals such as heart rate, spo2 etc

CO3: Analyze various types of transducers

CO4: Test compatibility for any clinical measurement

CO5: Apply the biosensors in different measurement techniques

LIST OF EXPERIMENTS

1. Characteristics of pressure transducer
2. Measurement of displacement capacitive transducer, LVDT and Inductive transducer
3. Characteristics of optical transducer for SpO2 measurement
4. Measurement of skin temperature by both contact and non-contact method
5. Study of the characteristics of capacitor level sensor for saline level measurement in a I-V set.
6. Data acquisition of physiological signals
7. Study of hot-wire anemometry
8. Study of ampere metric sensor for blood glucose measurement
9. Electronic weighing machine for the measurement of chemical compounds
10. Non-invasive gas analyzer as an electronic nose.

MINOR DEGREE IN AI & DS
SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)
III B.Tech. I SEM, AI & DS

		L	T	P	C
23AAD19	**DATA ANALYTICS USING R**	**3**	**0**	**0**	**3**

COURSE OBJECTIVES: The objective of the course is,

1. Understand the fundamentals of data analytics and the role of R in data analysis.
2. Perform data manipulation, cleaning, and preprocessing using R packages.
3. Apply statistical techniques and visualizations using R.
4. Develop models for descriptive, predictive, and prescriptive analytics.
5. Work on real-world datasets and develop end-to-end analytics solutions using R.

COURSE OUTCOMES:

After the completion of the course, student will be able to

CO1.Apply R programming to read, process, and analyze structured data.

CO2.Use data visualization tools in R to summarize and present findings.

CO3.Perform exploratory data analysis (EDA) and interpret outputs.

CO4.Build basic predictive models using regression and classification techniques. CO5.Execute real-time mini projects involving data collection, analysis, and visualization.

UNIT-I: Introduction to Data Analytics and R **9 Hrs**

Introduction to Data Analytics: Definitions, types (Descriptive, Predictive, Prescriptive), Overview of R: History, features, RStudio, Installing and using R packages, Basic syntax, data types, and operators in R

UNIT-II: Data Manipulation and Exploration **8 Hrs**

Importing data: CSV, Excel, Web APIs, Databases, Data frames, lists, matrices, factors, Data wrangling with dplyrand tidyr, Handling missing data, filtering, sorting, summarizing, Exploratory Data Analysis (EDA)

UNIT-III: Data Visualization using R **7 Hrs**

Introduction to visualization principles, Base R graphics vs. ggplot2,Bar charts, histograms, box plots, scatter plots, line graphs, Advanced visualization: heatmaps, correlation matrices, Interactive visualizations using plotlyand shiny(intro)

UNIT-IV: Statistical Analysis and Machine Learning in R **8 Hrs**

Statistical inference: mean, median, variance, standard deviation, Hypothesis testing: t-test, chi-square test, Linear regression, logistic regression, Clustering: k-means, Classification: Decision Trees, Naive Bayes (intro)

UNIT V: Advanced Topics in R for Data Analytics **9 Hrs**

Time series analysis: basics, decomposition, forecasting with forecastand tspackages, Text analytics: basic text mining using tm, wordcloud, and tidytext, Web scraping using rvest, Introduction to dashboards using shiny, Automation and reporting with RMarkdownand knitr, Introduction to Big Data integration (connecting R with Hadoop/Spark via sparklyr - optional/overview)

TEXT BOOKS:

1. **"R for Data Science"** by Hadley Wickham & Garrett Grolemund, O'Reilly Media

2. **"The Art of R Programming"** by Norman Matloff, No Starch Press

REFERENCE BOOKS:

1. **"Practical Data Science with R"** by Nina Zumel & John Mount, Manning Publications

2. **"Hands-On Programming with R"** by Garrett Grolemund, O'Reilly Media

3. **"Data Science for Business"** by Foster Provost & Tom Fawcett, O'Reilly Media (for conceptual understanding)

4. **"Advanced R"** by Hadley Wickham, CRC Press

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(AUTONOMOUS)
III B.Tech. I SEM, AI & DS

		L	T	P	C
23AAD20	**Creative Intelligence with Generative Models**	**3**	**0**	**0**	**3**

COURSE OBJECTIVES: The objective of the course is,

1. Understand the theoretical foundations of creative intelligence and generative models.
2. Explore various generative architectures (e.g., GANs, VAEs, Transformers).
3. Analyze and evaluate the creative output of generative systems.
4. Apply generative models to domains such as text, image, music, and video.
5. Examine ethical, philosophical, and legal issues related to AI-generated content.
6. Design and implement creative projects using generative AI tools and frameworks.

COURSE OUTCOMES:

After the completion of the course, student will be able to

CO1. Explain the principles of creative intelligence in artificial systems.
CO2. Differentiate between various types of generative models and their applications.
CO3. Use state-of-the-art generative tools (e.g., GPT, DALL·E, Diffusion Models) to produce creative content.
CO4. Critically evaluate the originality and value of AI-generated outputs.
CO5. Build a mini-project involving multi-modal AI creativity (e.g., text-to-image, music generation).
CO6. Address ethical challenges in deploying generative models in real-world applications.

UNIT-I: Foundations of Creative Intelligence **10 Hrs**

Introduction to creativity: Human vs. machine creativity, Theories of creativity (Guilford, Sternberg, Boden), Computational creativity: History and goals, Basics of creative cognition and associative thinking

UNIT-II: Generative Models – Concepts & Techniques 7 Hrs

Introduction to generative modeling, Probabilistic models: VAEs, Adversarial models: GANs and their variants, Transformer-based models (GPT, BERT, T5), Diffusion models (DDPMs, Stable Diffusion)

UNIT-III: Generative Creativity in Text, Image, and Audio **8 Hrs**

Text generation: GPT, prompting, narrative generation, Image synthesis: DALL·E, Midjourney, Stable Diffusion, Music/audio generation: MusicLM, Suno, AudioCraft, Prompt engineering & prompt chaining

9 Hrs

UNIT-IV: Evaluation and Ethics

Measuring creativity in AI (novelty, value, surprise), Turing Test vs. Lovelace Test, Deepfakes, plagiarism, and bias in generative content, Ownership, copyright, and ethical concerns

UNIT V: Applications and Project Work 8 Hrs

Multi-modal generation (text-to-image, text-to-video, text-to-audio), Applications in design, storytelling, game development, marketing, Creative AI tools: Runway, ChatGPT, Canva AI, Adobe Firefly, Final creative project planning, implementation, and review

TEXT BOOKS:

1. **"Artificial Intelligence and Creativity: An Interdisciplinary Approach"** – Tarek R. Besold et al.
2. **"Deep Learning"** – Ian Goodfellow, Yoshua Bengio, and Aaron Courville (for GANs and VAEs)
3. **"The Creativity Code: Art and Innovation in the Age of AI"** – Marcus du Sautoy

REFERENCE BOOKS:

- **"Architects of Intelligence"** – Martin Ford
- **"Reclaiming Art in the Age of Artificial Intelligence"** – Jared Marcel Pollen
- **"Computational Creativity: The Philosophy and Engineering of Autonomously Creative Systems"** – Tony Veale and Feyaad Alladin
- Papers from **AAAI, NeurIPS, ICML**, and **IJCAI** on generative models and computational creativity
- Online platforms: **Hugging Face**, **Runway ML**, **Google Colab**, **Kaggle Notebooks**

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)
III B.Tech. II SEM, AI & DS

23AAD21	**Data Management with SQL & No-SQL**	**L**	**T**	**P**	**C**
		3	**0**	**0**	**3**

COURSE OBJECTIVES: The objective of the course is,

- Understand the fundamentals of data management and database systems.
- Design and implement relational databases using SQL.
- Explore NoSQL databases and their applications in modern data environments.
- Compare and contrast SQL and NoSQL approaches for different use cases.
- Apply data modeling, querying, and optimization techniques in both paradigms.

COURSE OUTCOMES: After the completion of the course, student will be able to CO1. Design normalized relational schemas and write efficient SQL queries.
CO2. Implement CRUD operations and manage transactions in SQL databases. CO3. Understand NoSQL data models (document, key-value, column, graph). CO4. Choose appropriate database systems based on application requirements. CO5. Demonstrate proficiency in using tools like MySQL, PostgreSQL,
MongoDB, and Cassandra.

UNIT-I: **Introduction to Data Management** **7 Hrs**
Data, databases, and DBMS, Types of databases: Relational vs Non-relational, Overview of SQL and NoSQL, Database design principles

UNIT-II: Relational Databases and SQL **9 Hrs**
SQL syntax and commands (SELECT, INSERT, UPDATE, DELETE), Joins, subqueries, and views, Normalization and schema design, Transactions and indexing

UNIT-III: NoSQL Databases **8 Hrs**
Types of NoSQL databases: Document, Key-Value, Column, Graph, Data modeling in NoSQL, CAP theorem and BASE properties, Use cases and limitations

UNIT-IV: Working with SQL and NoSQL Tools **7 Hrs**
MySQL/PostgreSQL setup and usage, MongoDB and Cassandra basics, Querying and managing data, Performance tuning and scalability

UNIT V: Comparative Analysis and Real-World Applications **6 Hrs**
SQL vs NoSQL: strengths and trade-offs, Hybrid architectures and polyglot persistence, Case studies: e-commerce, social media, IoT, Future trends in data management

TEXT BOOKS:

- *Database System Concepts* by Abraham Silberschatz, Henry F. Korth, and S. Sudarshan (McGraw-Hill)
- *NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence* by Pramod J. Sadalage and Martin Fowler (Addison-Wesley)
- *Learning SQL* by Alan Beaulieu (O'Reilly Media).

REFERENCE BOOKS:

- *SQL in a Nutshell* by Kevin Kline – O'Reilly Media
- *MongoDB: The Definitive Guide* by Kristina Chodorow – O'Reilly Media
- *Cassandra: The Definitive Guide* by Jeff Carpenter and Eben Hewitt – O'Reilly Media
- *Designing Data-Intensive Applications* by Martin Kleppmann – O'Reilly Media

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III B.Tech. II SEM, AI & DS

		L	T	P	C
23AAD22	**Applied Generative AI**	**3**	**0**	**0**	**3**

COURSE OBJECTIVES: The objective of the course is,

- Understand the principles and techniques behind generative AI.
- Apply foundational models like GANs, VAEs, and LLMs to real-world problems.
- Develop and deploy GenAI applications using tools such as LangChain, Hugging Face, and Azure AI Studio.
- Explore advanced topics like agentic AI, retrieval-augmented generation (RAG), and AI governance.
- Gain hands-on experience through projects and capstone work.

COURSE OUTCOMES: After the completion of the course, student will be able to CO1.
Build and fine-tune generative models for text, image, and code generation. CO2.
Implement prompt engineering and agentic workflows.
CO3. Use industry-standard GenAI tools and frameworks.
CO4. Evaluate ethical and governance considerations in AI deployment. CO5.
Create portfolio-worthy projects demonstrating applied GenAI skills.

UNIT-I: Foundations of Generative AI **6 Hrs**
Introduction to GenAI and its evolution, Overview of GANs, VAEs, and LLMs, Applications across industries

UNIT-II: Tools and Technologies **7 Hrs**
LangChain, Hugging Face, Azure AI Studio, Prompt engineering and model APIs, Deployment environments

UNIT-III: Advanced Architectures **6 Hrs**
Transformer models and attention mechanisms, Fine-tuning LLMs,
Retrieval-Augmented Generation (RAG)

UNIT-IV: Agentic AI and Governance **7 Hrs**
Agentic frameworks and autonomous agents, AI ethics, bias, and governance, Responsible AI deployment

UNIT V: Capstone and Real-World Applications **7 Hrs**
Hands-on projects and case studies, Building end-to-end GenAI applications, Career pathways and industry trends

TEXT BOOKS:

- *Generative Deep Learning* by David Foster (O'Reilly Media)
- *Designing Data-Intensive Applications* by Martin Kleppmann (O'Reilly Media)

REFERENCE BOOKS:

- *Hands-On Generative AI with Python* by John Smith
- *Transformers for Natural Language Processing* by Denis Rothman

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (AUTONOMOUS) (AUTONOMOUS)
IV B.Tech. I SEM, AI & DS

23AAD23	Text Analytics & NLP	L	T	P	C
		3	0	0	3

COURSE OBJECTIVES: The objective of the course is,

- Understand the core concepts and challenges in NLP and text analytics.
- Learn syntax, semantics, and morphological analysis of natural language.
- Explore word and sentence representation models for NLP applications.
- Apply machine learning and deep learning techniques to NLP tasks.
- Implement NLP solutions using Python and modern libraries.

COURSE OUTCOMES:

After the completion of the course, student will be able to CO1.
Apply computational linguistics tools to analyze text.
CO2. Implement word representation models like TF-IDF and embeddings.
CO3. Develop ML/DL models for NLP tasks such as sentiment analysis and translation. CO4.
Evaluate NLP models using appropriate metrics and visualization techniques.

UNIT-I: Introduction to Text Mining & NLP **8 Hrs**
Difference between text mining and NLP, Challenges in NLP: ambiguity, context, syntax, Language models and NLP workflow, Applications of NLP in business and research

UNIT-II: Fundamentals of Text Analytics **9 Hrs**
Text preprocessing: tokenization, stop word removal, stemming, lemmatization, Removing noise: HTML tags, emojis, special characters, Text vectorization: Bag-of- Words, TF-IDF, Topic modeling and text visualization

UNIT-III: Word Embeddings & Representation **6 Hrs**
Limitations of count-based models, Co-occurrence matrices and cosine similarity, Word2Vec, GloVe, and pre-trained embeddings, Manipulating words in vector space

UNIT-IV: Sentiment Analysis & Classification **7 Hrs**
Supervised learning for sentiment analysis, Feature extraction from text, Logistic regression for classification, Visualizing tweet sentiment and model performance

UNIT V: Python Applications for NLP **8 Hrs**
Processing text and PDF files, Regular expressions for pattern matching, Tokenization and vocabulary matching with spaCy, Named Entity Recognition (NER), speech tagging, Building chatbots and natural language generation (NLG)

TEXT BOOKS:

- *Speech and Language Processing* by Daniel Jurafsky & James H. Martin
- *Foundations of Statistical Natural Language Processing* by Christopher Manning & Hinrich Schütze

- *Natural Language Processing with Python* by Steven Bird, Ewan Klein, and Edward Loper

REFERENCE BOOKS:

- *Text Analytics with Python* by Dipanjan Sarkar
- *Deep Learning for Natural Language Processing* by Palash Goyal, Sumit Pandey, and Karan Jain
- *Practical Natural Language Processing* by Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, and Harshit Surana
- *Machine Learning for Text* by Charu C. Aggarwal
- *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* by Aurélien Géron
- *Introduction to Information Retrieval* by Manning, Raghavan, and Schütze

MINOR DEGREE IN COMPUTER SCIENCE & BUSINESS SYSTEMS SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (AUTONOMOUS)
III B.Tech. I SEM, CSBS

		L	T	P	C
23ACB15	**Fundamentals of Computer Systems & Business Environment**	**3**	**0**	**0**	**3**

COURSE OBJECTIVES: The objective of the course is,

- **Understand the fundamentals** of data, information, knowledge, and systems in a business context.
- **Familiarize students** with the various types of business information systems (BIS) and their strategic roles.
- **Explore the components** of computer hardware and software and their relevance in BIS.
- **Introduce database concepts** and the application of business intelligence for decision-making.
- **Describe networks, telecommunications**, and how the Internet enables business connectivity and e-commerce.
- **Analyze enterprise and functional systems** used across different business departments.

COURSE OUTCOMES: After the completion of the course, student will be able to CO1. Distinguish between data, information, knowledge, and wisdom in a business setting.

CO2. Explain the role of BIS in supporting managerial decisions and gaining competitive advantage.

CO3. Identify the major components of computer hardware and software and assess their applications in business.

CO4. Describe various types of databases and apply BI tools to improve organizational decision-making.

CO5. llustrate the role of networks and the Internet in enabling business communication and e-commerce.

CO6. Analyze the use of enterprise systems and departmental applications in streamlining business operations.

UNIT-I: INTRODUCTION TO BUSINESS INFORMATION SYSTEMS 8 Hrs

Data and information, creating information, Qualities of information, Knowledge and wisdom, the business environment, Managerial decision making, Knowledge management. Introduction to systems, Different types of systems, Business information systems, Resources that support BIS, Categories of business information system, E-business systems, Enterprise systems, BIS and strategic advantage.

UNIT-II: HARDWARE AND SOFTWARE 10 Hrs

Components of a computer system, Major categories of computers, Types of microcomputers, selecting input devices, selecting output devices, selecting storage devices, Processors, Categories of software, Document production software, Graphics packages, Spreadsheets, Management applications of productivity software, Multimedia software,
Software for using the Internet.

UNIT-III: DATABASES AND BUSINESS INTELLIGENCE 12 Hrs

Databases: Business-level advantages of databases an overview of the types of database-File processing databases, Database management systems, Relational database management systems, Object-oriented databases, Network and hierarchical databases; Business intelligence, Data warehouses: Data warehouse, Data Mart, Architecture Data mining: Text and Web Mining Business analytics: OLAP, Cube Analysis Case Study: Making business intelligence.

UNIT-IV: NETWORKS, TELECOMMUNICATIONS AND THE INTERNET 10 Hrs

Computer networks, Network components, Network types, The Internet, Case Study: Asian apps challenge western dominance

UNIT V: ENTERPRISE AND FUNCTIONAL BUSINESS INFORMATION SYSTEMS 5 Hrs

Enterprise systems, Operations information systems, Management information systems, Departmental applications, Case Study: Managing the supply chain. Total Periods: 45 Topics for self-study are provided in the lesson plan

TEXT BOOKS:

1. Paul Bocij, Andrew Greasley, and Simon Hickie, "Business Information Systems Technology, Development and Management for the E-Business", Fifth edition, Pearson Education Limited, 2015.

REFERENCE BOOKS:

1. R. Kelly Rainer Jr., Casey G. Cegielski, "Introduction to Information Systems Supporting and Transforming Business", Fourth Edition, John Wiley & Sons. Inc, 2012.

2. WitoldAbramowicz, Heinrich C. Mayr, "Technologies for Business Information Systems", Springer, 2007.

ADDITIONAL LEARNING RESOURCES:

1. https://www.edx.org/micromasters/iux-information-systems.
2. https://www.coursera.org/learn/business-model-canvas.

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (AUTONOMOUS)
III B.Tech. I SEM, CSBS

		L	T	P	C
23ACB16	**DATABASE MANAGEMENT & ENTERPRISE SYSTEMS**	**3**	**0**	**0**	**3**

COURSE OBJECTIVES: The objective of the course is,

- **Introduce fundamental concepts** of database systems, including their architecture, models, and advantages.
- **Develop skills in the relational model**, relational algebra, SQL, and database normalization.
- **Understand transaction management**, concurrency control, and recovery in database systems.
- **Familiarize students with enterprise systems**, their components, architecture, and implementation lifecycle.
- **Explain modern enterprise platforms**, integration technologies, and their role in business process improvement.

COURSE OUTCOMES: After the completion of the course, student will be able to

CO1. Describe the characteristics, components, and architecture of a DBMS and its advantages over file systems.
CO2. Apply relational algebra and SQL commands to model and manipulate data in relational databases.
CO3. Demonstrate the process of database normalization and construct ER models with appropriate keys.
CO4. Explain transaction processing concepts, concurrencycontrol mechanisms, and database recovery techniques.
CO5. Identify the structure and modules of enterprise systems like ERP, SCM, and CRM, and their business roles.
CO6. Evaluate enterprise integration techniques, cloud/on-premise ERP solutions, and their impact on businesses.

UNIT-I: Introduction to DBMS **8 Hrs**

Definition and Characteristics of DBMS, Advantages over File Systems, Database Models (Relational, Hierarchical, Network, Object-Oriented), Three-level Architecture, Data Independence, DBMS Components and Architecture

UNIT-II: Relational Model and SQL **9 Hrs**

Relational Algebra & Relational Calculus, Keys (Primary, Candidate, Foreign), Entity-Relationship (ER) Model, Normalization: 1NF to BCNF, SQL: DDL, DML, DCL, TCL, Joins, Subqueries, Views, Indexes

UNIT-III: Transaction Management & Concurrency Control **8 Hrs**

Transactions and ACID Properties, Serializability, Lock-based Concurrency Control, Deadlock Handling, Recovery Techniques, Log-based Recovery

UNIT-IV: Introduction to Enterprise Systems **9 Hrs**

Overview of Enterprise Systems (ERP, SCM, CRM), Enterprise Architecture, ERP Modules: Finance, HR, Production, Sales, Procurement, ERP Implementation Lifecycle, Benefits and Challenges of ERP Systems

UNIT V: Enterprise System Platforms and Integration **8 Hrs**

Major ERP Vendors (SAP, Oracle, Microsoft Dynamics), Cloud ERP vs On-Premise ERP, Enterprise Application Integration (EAI), Middleware, SOA, Web Services, Business Process Reengineering (BPR)

TEXT BOOKS:

1. **"Database System Concepts"** by Abraham Silberschatz, Henry F. Korth, and S. Sudarshan 7th Edition, McGraw-Hill

2. "Fundamentals of Database Systems" by Ramez Elmasri and Shamkant Navathe 7th Edition, Pearson

REFERENCE BOOKS:

1. "Database Management Systems" by Raghu Ramakrishnan and Johannes Gehrke,3rd Edition, McGowan-Hill**Enterprise Systems**

2. "Enterprise Resource Planning: Concepts and Practice" by Mahadeo Jaiswal and Ganesh Vanapalli,2nd Edition, Pearson

"Modern ERP: Select, Implement, and Use Today's Advanced Business Systems" by Marianne Bradford,3rd Edition, Lulu Press

"ERP Demystified" by Alexis Leon,2nd Edition, Tata McGowan-Hill

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)
III B.Tech. II SEM, CSBS

		L	T	P	C
23ACB17	**Software Engineering for Business Applications**	**3**	**0**	**0**	**3**

COURSE OBJECTIVES:

The objective of the course is,

- **Understand the fundamentals** of software engineering and various software process models.
- **Analyze and model requirements** using structured and object-oriented approaches.
- **Gain insight into Agile methodologies** and apply them to modern software development projects.
- **Explore the principles of software project management**, including planning, risk analysis, and evaluation.
- **Comprehend software quality concepts** and apply estimation techniques for project planning.

COURSE OUTCOMES:

After the completion of the course, student will be able to

CO1. Describe the nature of software and various software engineering practices and myths.
C02. Compare different software process models and select an appropriate model for a given software project.
CO3. Apply requirements engineering techniques including elicitation, modeling, and validation.
CO4. Illustrate various modeling strategies such as scenario-based, class-based, flow-oriented, and behavioral.
CO5. Explain the principles of Agile development and various agile methodologies like XP, Scrum, etc.
CO6. Demonstrate knowledge of software project management practices and tools.
CO7. Evaluate software projects using quality models, estimation techniques, and risk evaluation strategies.

UNIT-I: Software and Software Engineering 9Hrs

The nature of Software, The unique nature of WebApps, Software Engineering, The software Process, Software Engineering Practice, Software Myths.

Process Models: A generic process model, Process assessment and improvement, Prescriptive process models: Waterfall model, Incremental process models, Evolutionary process models, Concurrent models, Specialized process models. Unified Process , Personal and Team process models.

UNIT-II: Understanding Requirements 8 Hrs

Requirements Engineering, Establishing the ground work, Eliciting Requirements, developing use cases, Building the requirements model, Negotiating Requirements, Validating Requirements. Requirements Modeming Scenarios, Information and Analysis classes: Requirement Analysis, Scenario based modeling, UML models that supplement the Use Case, Data modeling Concepts, Class-Based Modeling. Requirement Modeling Strategies : Flow oriented Modeling , Behavioral Modeling.

UNIT-III:Agile Development **10 Hrs**

What is Agility, Agility and the cost of change. What is an agile Process?, Extreme Programming (XP), Other Agile Process Models, A tool set for Agile process . Principles that guide practice: Software Engineering Knowledge, Core principles, Principles that guide each framework activity.

UNIT-IV: Introduction to Project Management **9 Hrs**

Introduction, Project and Importance of Project Management, Contract Management, Activities Covered by Software Project Management, Plans, Methods and Methodologies, Some ways of categorizing Software Projects, Stakeholders, Setting Objectives, Business Case, Project Success and Failure, Management and Management Control, Project Management life cycle, Traditional versus Modern Project Management Practices. Project Evaluation: Evaluation of Individual projects, Cost–benefit Evaluation Techniques, Risk Evaluation.

UNIT V: Software Quality **8 Hrs**

Introduction, The place of software quality in project planning, Importance of software quality, Defining software quality, Software quality models, product versus process quality management. Software Project Estimation: Observations on Estimation, Decomposition Techniques, Empirical Estimation Models.

TEXT BOOKS:

Roger S. Pressman: Software Engineering-A Practitioners approach, 7th Edition, Tata McGraw Hill. 2. Bob Hughes, Mike Cotterell, Rajib Mall: Software Project Management, 6th Edition, McGraw Hill Education, 2018.

REFERENCE BOOKS:

1. Pankaj Jalote: An Integrated Approach to Software Engineering, Wiley India.
2. “Software Engineering: Principles and Practice", Hans van Vliet, Wiley India, 3rd Edition, 2010.

Web links and Video Lectures (e-Resources):

- https://onlinecourses.nptel.ac.in/noc20_cs68/preview
- https://onlinecourses.nptel.ac.in/noc24_mg01/preview

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)
III B.Tech. II SEM, CSBS

		L	T	P	C
23ACB18	**Business Intelligence and Analytics**	**3**	**0**	**0**	**3**

COURSE OBJECTIVES: The objective of the course is,

- Introduce the role and structure of **business intelligence (BI)** and **decision support systems (DSS)** in organizational decision-making.
- Provide a comprehensive understanding of the **decision-making process** and how it is supported by DSS.
- Familiarize students with **neural networks, sentiment analysis**, and other intelligent systems used in modern analytics.
- Equip students with knowledge of **model-based decision-making techniques**, including mathematical and spreadsheet-based models.
- Introduce **automated decision systems** and **expert systems** as advanced tools in AI- driven business analytics.

COURSE OUTCOMES: After the completion of the course, student will be able to CO1. Describe the evolution and components of Business Intelligence and its role in decision- making.

CO2. Explain the phases and structure of the decision-making process and the capabilities of Decision Support Systems.

CO3. Apply basic concepts of neural networks, SVM, and sentiment analysis in real- world decision-making scenarios.

CO4. Construct and evaluate decision models using mathematical programming, spreadsheets, and decision trees.

CO5. Describe the architecture, development, and application of expert systems and automated decision-making tools.

UNIT-I:An Overview of Business Intelligence, Analytics, and Decision Support 8 Hrs

Information Systems Support for Decision Making, An Early Framework for Computerized Decision Support, The Concept of Decision Support Systems, A Framework for Business Intelligence, Business Analytics Overview, Brief Introduction to Big Data Analytics.

UNIT-II: Decision Making 8 Hrs

Introduction and Definitions, Phases of the Decision, Making Process, The Intelligence Phase, Design Phase, Choice Phase, Implementation Phase, Decision Support Systems Capabilities, Decision Support Systems Classification, Decision Support Systems Components**.**

UNIT-III: Neural Networks and Sentiment Analysis 12 Hrs

Basic Concepts of Neural Networks, Developing Neural Network-Based Systems, Illuminating the Black Box of ANN with Sensitivity, Support Vector Machines, A Process Based Approach to the Use of SVM, Nearest Neighbour Method for Prediction, Sentiment

Analysis Overview, Sentiment Analysis Applications, Sentiment Analysis Process, Sentiment Analysis, Speech Analytics

UNIT-IV: Model-Based Decision Making 12 Hrs

Decision Support Systems modeling, Structure of mathematical models for decision support, Certainty, Uncertainty, and Risk, Decision modeling with spreadsheets, Mathematical programming optimization, Decision Analysis with Decision Tables and Decision Trees, Multi-Criteria Decision Making with Pairwise Comparisons.

UNIT V: Automated Decision Systems and Expert Systems 5 Hrs

Automated Decision Systems, The Artificial Intelligence field, Basic concepts of Expert Systems, Applications of Expert Systems, Structure of Expert Systems, Knowledge Engineering, Development of Expert Systems.

TEXT BOOKS:

1. Ramesh Sharda, Dursun Delen, Efraim Turban, J.E. Aronson, Ting-Peng Liang, David King, "Business Intelligence and Analytics: System for Decision Support", 10th Edition, Pearson Global Edition, 2013

REFERENCE BOOKS:

1. Data Analytics: The Ultimate Beginner's Guide to Data Analytics Paperback – 12 November 2017by Edward Mize.

Web links and Video Lectures (e-Resources):

- https://www.youtube.com/watch?v=zbcCdoHeS4w

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)
IV B.Tech. I SEM, CSBS

		L	T	P	C
23ACB19	**Digital Transformation & Business Strategy**	**3**	**0**	**0**	**3**

COURSE OBJECTIVES: The objective of the course is,

- Provide an in-depth understanding of the concept and evolution of digital transformation across industries.
- Explore the strategic alignment between digital initiatives and corporate strategy.
- Examine emerging digital technologies that drive transformation, such as AI, IoT, cloud computing, and blockchain.
- Develop knowledge of implementation frameworks including change management, agile practices, and performance measurement.
- Analyze real-world applications of digital transformation in various sectors and anticipate future trends in the digital economy.

COURSE OUTCOMES:

After the completion of the course, student will be able to

CO1. Explain the scope, drivers, and impact of digital transformation on traditional and modern businesses.
CO2. Evaluate how digital strategies align with business goals and describe digital business models and value chains.
CO3. Identify and describe key enabling technologies such as cloud, AI, IoT, blockchain, and cybersecurity.
CO4. Develop a roadmap for digital transformation, considering organizational readiness, change management, and risk.
CO5. Analyze digital transformation practices across sectors and explore emerging trends like digital workplaces, ethics, and sustainability.

UNIT-I: Introduction to Digital Transformation **9Hrs**

Concept and scope of digital transformation, Traditional vs digital businesses, Role of digital in business evolution, Drivers and inhibitors of digital transformation, Case studies: Netflix, Amazon, Uber, etc.

UNIT-II: Strategic Foundations of Digital Business **8 Hrs**

Integration of digital and corporate strategy, Business model innovation (B2B, B2C, platform models), Digital value chain and value proposition, Strategy frameworks: Porter's Five Forces, Value Chain Analysis, Competitive advantage in the digital economy

UNIT-III:Technologies Enabling Digital Transformation **10 Hrs**

loud computing and virtualization, Artificial Intelligence and Machine Learning, Big Data and analytics, Internet of Things (IoT) and Industry 4.0, Blockchain technology, cybersecurity fundamentals, Digital architecture and infrastructure

UNIT-IV:Digital Transformation Implementation **9 Hrs**

Roadmap for digital transformation, Organizational readiness and capability development, Change management and digital leadership, Agile methodologies and DevOps, Risk management, governance, compliance, Key metrics and performance evaluation

UNIT V:Sectoral Applications and Future Trends **8 Hrs**

Digital transformation in core sectors: **Retail, Banking & FinTech**, **Healthcare**, Manufacturing, **Education**, Customer-centric digital strategy, Sustainability and ethics in digital business, The future of work: AI, gig economy, digital workplaces

TEXT BOOKS:

1. "Leading Digital: Turning Technology into Business Transformation" By
George Westerman, Didier Bonnet, and Andrew McAfee

2. "Digital Transformation: Survive and Thrive in an Era of Mass Extinction" By
Thomas Siebel

REFERENCE BOOKS:

1. "Digital Strategy: A Guide to Digital Business Transformation" By
Mark Baker

2. "The Digital Transformation Playbook" By
David L. Rogers

3. "Digital to the Core: Remastering Leadership for Your Industry, Your Enterprise, and Yourself"
By Mark Raskino and Graham Waller

4. "Enterprise Architecture as Strategy"
By Jeanne W. Ross, Peter Weill, and David Robertson

MINORS IN COMPUTER SCIENCE AND ENGINEERING

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (AUTONOMOUS)

III B.Tech-I Semester CE, E,EEE,ECE(MINOR)

L	T	P	C
3	0	0	3

23ACS24: CLOUD COMPUTING

Course Outcomes:

At the end of the course, the student will be able to

CO1: Ability to create cloud computing environment
CO2: Ability to design applications for Cloud environment
CO3: Design & develop back up strategies for cloud data based on features.
CO4: Use and Examine different cloud computing services.
CO5: Apply different cloud programming model as perneed.

UNITI :VBasics of Cloud computing **Lecture8Hrs**

Introduction to cloud computing: Introduction, Character is tics of cloud computing, Cloud Models, Cloud Services Examples, Cloud Based services and applications

Cloud concepts and Technologies: Virtualization, Load balancing, Scalability and Elasticity, Deployment, Replication, Monitoring, Software defined, Network function virtualization, Map Reduce, Identity and Access Management, services level Agreements, Billing.

Cloud Services and Plat forms: Compute Services, Storage Services, Data, base Services, Application services, Content delivery services Analytics Services, Deployment and Management Services, Identity and Access Management services, Open Source Private Cloud software.

UNIT II: Hadoop and Python **Lecture 9Hrs**

Hadoop Map Reduce: Apache Hadoop, Hadoop Map Reduce Job Execution, Hadoop Schedulers, Hadoop Clusterset up. Cloud Application Design: Reference Architecture for Cloud Applications, Cloud Application Design Methodologies, Data Storage Approaches.

Python Basics: Introduction, Installing Python, Python data Types & Data Structures,Controlflow,Function,Modules,Packages,Filehandling,Date/TimeOperations,Classes.

UNIT III : **Python for Cloud computing** **Lecture 8Hrs**

Python for Cloud: Python for Amazon web services, Python for Google Cloud Platform, Python for windows Azure, Python for Map Reduce, Python packages of Interest, Python web Application Frame work, Designing a REST ful web API.

Cloud Application Development in Python: Design Approaches, Image Processing APP, Document Storage App, Map Reduce App, Social Media Analytics App.

UNIT IV: **Big data, multimedia and Tuning** **Lecture8Hrs**

Big Data Analytics: Introduction, Clustering Big Data, Classification of Big data Recommendation of Systems

Multimedia Cloud: Introduction, Case Study: Live video Streaming App, Streaming Protocols, case Study: Video Trans coding App.

Cloud Application Bench marking and Tuning: Introduction, Work load Character is tics, Application Performance Metrics, Design Considerations for a Bench marking Methodology, Bench marking Tools, Deployment Prototyping, Load Testing & Bottleneck Detection case Study, Hadoop bench marking case Study.

UNIT V: **Applications and Issues in Cloud** **Lecture 9Hrs**

Cloud Security: Introduction, CSA Cloud Security Architecture, Authentication, Authorization, Identity

Access Management, Data Security, Key Management, Auditing.

Cloud for Industry, Health care & Education: Cloud Computing for Health care, Cloud computing for Energy Systems, Cloud Computing for Transportation Systems, Cloud Computing for Manufacturing Industry, Cloud computing for Education.

Migrating in to a Cloud: Introduction, Broad Approaches to migrating into the cloud, the seven– step model of migration in to a cloud.

Organizational readiness and Change Management in The Cloud Age: Introduction, Basic concepts of Organizational Readiness, Drivers for changes: A frame work to comprehend the competitive environment, common change management models, change management maturity models, Organizational readiness self– assessment.

Legal Issues in Cloud Computing: Introduction, Data Privacy and security Issues, cloud contracting models, Jurisdictional issues raised by virtualization and at a location, commercial and business considerations, Special Topics.

Text books:

1. Cloud computing Ahands - on Approach ‖By Arshdeep Bahga, Vijay Madisetti, Universities Press, 2016
2. Cloud Computing Principles and Paradigms: By RajKumar Buyya, James Broberg, Andrzej Goscinski, Wiley, 2016

Reference Books:

1. Masterin g Cloud Computing by Rajkumar Buyya, Christian Vecchiola, S Thamarai Selvi, TMH
2. Cloud computing AHands-On Approach by Arshdeep Bahga and Vijay Madisetti.
3. Cloud Computing: A Practical Approach, Anthony T.Velte, To by J.Velte, Robert Elsenpeter, Tata Mc Graw Hill,rp 2011.
1. Enterprise Cloud Computing, Gautam Shroff, Cambridge University Press, 2010.
2. Cloud Application Architectures: Building Applications and Infrastructure in the Cloud, George Reese,O_Reilly, SPD, rp 2011.
6. Essentials of Cloud Computing by K.Chandrasekaran. CRC Press.

Online Learning Resources:

Cloud computing – Course (nptel.ac.in)

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

II B.Tech II Semester (Common to CSE, CSD, CSM, CSC, CAI,CSO,IT,CSBS))
III B.Tech I Semester CE.ME,EEE,ECE(Minor)

L	T	P	C
3	0	0	3

23AIT02: SOFTWARE ENGINEERING

Course Outcomes: After completion of the course, students will be able to
CO1: Perform various life cycle activities like Analysis, Design, Implementation, Testing and Maintenance (L3)
CO2: Analyse various software engineering models and apply methods for design and development of software projects. (L4)
CO3: Develop system designs using appropriate techniques. (L3)
CO4: Understand various testing techniques for a software project. (L2)
CO5: Apply standards, CASE tools and techniques for engineering software projects (L3)

UNIT I:
Introduction: Evolution, Software development projects, Exploratory style of software developments, Emergence of software engineering, Notable changes in software development practices, Computer system engineering.
Software Life Cycle Models: Basic concepts, Waterfall model and its extensions, Rapid application development, Agile development model, Spiral model.

UNIT II:
Software Project Management: Software project management complexities, Responsibilities of a software project manager, Metrics for project size estimation, Project estimation techniques, Empirical Estimation techniques, COCOMO, Halstead's software science, risk management.
Requirements Analysis And Specification: Requirements gathering and analysis, Software Requirements Specification (SRS), Formal system specification, Axiomatic specification, Algebraic specification, Executable specification and 4GL.

UNIT III:
Software Design: Overview of the design process, How to characterize a good software design? Layered arrangement of modules, Cohesion and Coupling. approaches to software design.
Agility: Agility and the Cost of Change, Agile Process, Extreme Programming (XP), Other Agile Process Models, Tool Set for the Agile Process (Text Book 2)
Function-Oriented Software Design: Overview of SA/SD methodology, Structured analysis, Developing the DFD model of a system, Structured design, Detailed design, and Design Review.
User Interface Design: Characteristics of a good user interface, Basic concepts, Types of user interfaces, Fundamentals of component-based GUI development, and user interface design methodology.

UNIT IV:

Coding And Testing: Coding, Code review, Software documentation, Testing, Black-box testing, White-Box testing, Debugging, Program analysis tools, Integration testing**,** Testing object-oriented programs, Smoke testing, and Some general issues associated with testing.

Software Reliability And Quality Management: Software reliability. Statistical testing, Software quality, Software quality management system, ISO 9000. SEI Capability maturity model. Few other important quality standards, and Six Sigma**.**

UNIT V:

Computer-Aided Software Engineering (Case): CASE and its scope, CASE environment, CASE support in the software life cycle, other characteristics of CASE tools, Towards second generation CASE Tool, and Architecture of a CASE Environment.

Software Maintenance: Characteristics of software maintenance, Software reverse engineering, Software maintenance process models and Estimation of maintenance cost.

Software Reuse: reuse- definition, introduction, reason behind no reuse so far, Basic issues in any reuse program, A reuse approach, and Reuse at organization level.

Text Books:

1. Fundamentals of Software Engineering, Rajib Mall, 5th Edition, PHI.
2. Software Engineering A practitioner's Approach, Roger S. Pressman, 9th Edition, Mc- Graw Hill International Edition.

Reference Books:

1. Software Engineering, Ian Sommerville,10th Edition, Pearson.
2. Software Engineering, Principles and Practices, Deepak Jain, Oxford University Press.

e-Resources:

1) https://nptel.ac.in/courses/106/105/106105182/
2) https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01260589506387148827_shared/overview
3) https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013382690411003904735_shared/overview

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

II B.Tech I Semester (Common to CSE,CSD,CSM,CSC,CAI,CSO,IT,CSBS))
III B.Tech II Semester CE,ME,EEE,ECE(Minor)

L	T	P	C
3	0	0	3

23ACS05: ADVANCED DATA STRUCTURES & ALGORITHM ANALYSIS

Course Outcomes: After completion of the course, students will be able to
CO1: Illustrate the working of the advanced tree data structures and their applications (L2)
CO2: Understand the Graph data structure, traversals and apply them in various contexts. (L2)
CO3: Use various data structures in the design of algorithms (L3)
CO4: Recommend appropriate data structures based on the problem being solved (L5)
CO5: Analyze algorithms with respect to space and time complexities (L4)
CO6: Design new algorithms (L6)

UNIT – I:
Introduction to Algorithm Analysis, Space and Time Complexity analysis, Asymptotic Notations. AVL Trees – Creation, Insertion, Deletion operations and Applications B-Trees – Creation, Insertion, Deletion operations and Applications

UNIT – II:
Heap Trees (Priority Queues) – Min and Max Heaps, Operations and Applications
Graphs – Terminology, Representations, Basic Search and Traversals, Connected Components and Biconnected Components, applications
Divide and Conquer: The General Method, Quick Sort, Merge Sort, Strassen's matrix multiplication, Convex Hull

UNIT – III:
Greedy Method: General Method, Job Sequencing with deadlines, Knapsack Problem, Minimum cost spanning trees, Single Source Shortest Paths
Dynamic Programming: General Method, All pairs shortest paths, Single Source Shortest Paths – General Weights (Bellman Ford Algorithm), Optimal Binary Search Trees, 0/1 Knapsack, String Editing, Travelling Salesperson problem

UNIT – IV:
Backtracking: General Method, 8-Queens Problem, Sum of Subsets problem, Graph Coloring, 0/1 Knapsack Problem
Branch and Bound: The General Method, 0/1 Knapsack Problem, Travelling Salesperson problem

UNIT – V:
NP Hard and NP Complete Problems: Basic Concepts, Cook's theorem,NP Hard Graph Problems: Clique Decision Problem (CDP), Chromatic Number Decision Problem (CNDP), Traveling Salesperson Decision Problem (TSP)
NP Hard Scheduling Problems: Scheduling Identical Processors, Job Shop Scheduling

Textbooks:
1. Fundamentals of Data Structures in C++, Horowitz, Ellis; Sahni, Sartaj; Mehta, Dinesh 2nd Edition Universities Press
2. Computer Algorithms/C++ Ellis Horowitz, SartajSahni, SanguthevarRajasekaran2nd Edition University Press

Reference Books:
1. Data Structures and program design in C, Robert Kruse, Pearson Education Asia

2. An introduction to Data Structures with applications, Trembley & Sorenson, McGraw Hill
3. The Art of Computer Programming, Vol.1: Fundamental Algorithms, Donald E Knuth, Addison-Wesley, 1997.
4. Data Structures using C & C++: Langsam, Augenstein&Tanenbaum, Pearson, 1995
5. Algorithms + Data Structures &Programs:,N.Wirth, PHI
6. Fundamentals of Data Structures in C++: Horowitz Sahni& Mehta, Galgottia Pub.
7. Data structures in Java:, Thomas Standish, Pearson Education Asia

Online Learning Resources:

1. https://www.tutorialspoint.com/advanced_data_structures/index.asp
2. http://peterindia.net/Algorithms.html
3. Abdul Bari,1. Introduction to Algorithms (youtube.com)

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B.Tech II Semester CE.ME,EEE,ECE(Minor)

L	T	P	C
3	0	0	3

23ACS43: COMPUTER ORGANIZATION

Course Outcomes :

After completion of the course, students will be able to

- Understand computer architecture concepts related to the design of modern processors, memories and I/Os
- Identify the hardware requirements for cache memory and virtual memory
- Design algorithms to exploit pipelining and multiprocessors
- Understand the importance and trade-offs of different types of memories.
- Identify pipeline hazards and possible solutions to those hazards

UNIT - I

Basic Structure of Computer: Computer Types, Functional Units, Basic operational Concepts, Bus Structure, Software, Performance, Multiprocessors and Multicomputer.

Machine Instructions and Programs: Numbers, Arithmetic Operations and Programs, Instructions and Instruction Sequencing, Addressing Modes, Basic Input/output Operations, Stacks and Queues, Subroutines, Additional Instructions.

UNIT - II

Arithmetic: Addition and Subtraction of Signed Numbers, Design of Fast Adders, Multiplication of Positive Numbers, Signed-operand Multiplication, Fast Multiplication, Integer Division, Floating-Point Numbers and Operations.

Basic Processing Unit: Fundamental Concepts, Execution of a Complete Instruction, Multiple-Bus Organization, Hardwired Control, and Multi programmed Control.

UNIT - III

The Memory System: Basic Concepts, Semiconductor RAM Memories, Read-Only Memories, Speed, Size and Cost, Cache Memories, Performance Considerations, Virtual Memories, Memory Management Requirements, Secondary Storage.

UNIT - IV

Input/Output Organization: Accessing I/O Devices, Interrupts, Processor Examples, Direct Memory Access, Buses, Interface Circuits, Standard I/O Interfaces.

UNIT - V

Pipelining: Basic Concepts, Data Hazards, Instruction Hazards, Influence on Instruction Sets.

Large Computer Systems: Forms of Parallel Processing, Array Processors, The Structure of General-Purpose multiprocessors, Interconnection Networks.

Textbooks:

1. Carl Hamacher, ZvonkoVranesic, SafwatZaky, “Computer Organization”, 5th Edition, McGraw Hill Education, 2013.

Reference Books:

1. M.Morris Mano, “Computer System Architecture”, 3rd Edition, Pearson Education.
2. Themes and Variations, Alan Clements, “Computer Organization and Architecture”, CENGAGE Learning.
3. SmrutiRanjanSarangi, “Computer Organization and Architecture”, McGraw Hill Education.
4. John P.Hayes, “Computer Architecture and Organization”, McGraw Hill Education

Online Learning Resources:

https://nptel.ac.in/courses/106/103/106103068/

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

III B.Tech II Semester CE.ME,EEE,ECE(Minor)

L	T	P	C
3	0	0	3

23ACS44:WEB TECHNOLOGIES

Course Outcomes:

- Understand the Web essentials.
- Develop web pages using XHTML
- Apply style to web pages using CSS
- Write scripts for client side
- Develop and transform XML documents.

UNIT I

Web Essentials: Clients, Servers, and Communication, The Internet, Basic Internet protocols, WWW, HTTP request message, HTTP response message, Web clients, Web Servers, Case study.

UNIT II

Markup Languages: XHTML 1.0,An introduction to HTML, Basic XHTML syntax and semantics, fundamental HTML elements, Relative URLs, Lists, Tables, Frames, Forms, Defining XHTML's abstract syntax, Creating HTML documents.

UNIT III

Cascading Style Sheets: Introduction, features, core syntax, style sheets and HTML, style rule cascading and inheritance, text properties, Box model, normal flow box layout, beyond the normal flow, lists, tables, cursor styles.

UNIT IV

Client-side programming - JavaScript: Basic syntax, variables and data types, statements, operators, literals, functions, objects, Arrays, built-in objects, JavaScript debuggers.

UNIT V

Representing Web Data-XML: Documents and vocabularies, Versions and declaration, Namespaces, Ajax, DOM and SAX parsers, transforming XML documents, XPath, XSLT, Displaying XML documents in Web browsers.

Textbooks:

1. J.C. Jackson, Web technologies: A computer science perspective, Pearson.
2.

Reference Books:

1. Sebesta, Programming world wide web, Pearson.
2. Dietel and Nieto , Internet and World Wide Web –How to program, Pearson Education
3. Chris Bates , Web Programming, building internet applications, 2nd edition, WILEY, Dreamtech

Online Learning Resources:

http://getbootstrap.com/
https://www.w3schools.com/whatis/
https://nptel.ac.in/courses/106105084

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

II B.Tech I Semester (Common to CSE,CSM,CSC,CAI,CSO,IT,CSBS))
III B.Tech II Semester CE,ME,EEE,ECE(Minor)

L	T	P	C
0	0	3	1.5

23ACS07: ADVANCED DATA STRUCTURES & ALGORITHM ANALYSIS LAB

Course Outcomes: After completion of the course, students will be able to
CO1: Design and develop programs to solve real world problems with the popular algorithm design methods. (L5)
CO2: Demonstrate an understanding of Non-Linear data structures by developing implementing the operations on AVL Trees, B-Trees, Heaps and Graphs. (L2)
CO3: Critically assess the design choices and implementation strategies of algorithms and data structures in complex applications. (L5)
CO4: Utilize appropriate data structures and algorithms to optimize solutions for specific computational problems. (L3)
CO5: Compare the performance of different of algorithm design strategies (L4)
CO6: Design algorithms to new real world problems (L6)

Experiments covering the Topics:

- Operations on AVL trees, B-Trees, Heap Trees
- Graph Traversals
- Sorting techniques
- Minimum cost spanning trees
- Shortest path algorithms
- 0/1 Knapsack Problem
- Travelling Salesperson problem
- Optimal Binary Search Trees
- N-Queens Problem
- Job Sequencing

Sample Programs:

1. Construct an AVL tree for a given set of elements which are stored in a file. And implement insert and delete operation on the constructed tree. Write contents of tree into a new file using in-order.
2. Construct B-Tree an order of 5 with a set of 100 random elements stored in array. Implement searching, insertion and deletion operations.
3. Construct Min and Max Heap using arrays, delete any element and display the content of the Heap.
4. Implement BFT and DFT for given graph, when graph is represented by
   a) Adjacency Matrix   b) Adjacency Lists
5. Write a program for finding the bi-connected components in a given graph.
6. Implement Quick sort and Merge sort and observe the execution time for various input sizes (Average, Worst and Best cases).
7. Compare the performance of Single Source Shortest Paths using Greedy method when the graph is represented by adjacency matrix and adjacency lists.

8. Implement Job sequencing with deadlines using Greedy strategy.
9. Write a program to solve 0/1 Knapsack problem Using Dynamic Programming.
10. Implement N-Queens Problem Using Backtracking.
11. Use Backtracking strategy to solve 0/1 Knapsack problem.
12. Implement Travelling Sales Person problem using Branch and Bound approach.

Reference Books:

1. Fundamentals of Data Structures in C++, Horowitz Ellis, SahniSartaj, Mehta, Dinesh, 2nd Edition, Universities Press
2. Computer Algorithms/C++ Ellis Horowitz, SartajSahni, SanguthevarRajasekaran, 2nd Edition, University Press
3. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
4. An introduction to Data Structures with applications, Trembley& Sorenson, McGraw Hill

Online Learning Resources:

1. http://cse01-iiith.vlabs.ac.in/
2. http://peterindia.net/Algorithms.html

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

IV B.Tech I Semester CE,ME,EEE,ECE(Minor)

L	T	P	C
0	**0**	**3**	**1.5**

23ACS45: WEB TECHNOLOGIES LAB

Course Outcomes (CO):

After completion of the course, students will be able to

- Construct web sites with valid HTML, CSS, JavaScript
- Create responsive Web designs that work on phones, tablets, or traditional laptops and widescreen monitors.
- Develop websites using jQuery to provide interactivity and engaging user experiences
- Embed Google chart tools in a website for better visualization of data.
- Design and develop web applications using Content Management Systems like WordPress

Sample Programs:

1. Create a Basic HTML document
2. Create your Profile Page
3. Create a Class Timetable (to merge rows/columns, use rowspan/colspan)
4. Create a Student Hostel Application Form
5. Make the Hostel Application Form designed in Module -4 beautiful using CSS (add colors,backgrounds, change font properties, borders, etc.)
6. Style the Hostel Application Form designed in Module-5still more beautiful using Bootstrap
7. Analyse various HTTP requests (initiators, timing diagrams, responses) and identify problems if any.
8. Design a simple calculator using JavaScript to perform sum, product, difference, and quotient operations:
9. Design& develop a Shopping Cart Application with features including Add Products, Update Quantity, Display Price(Sub-Total & Total), Remove items/products from the cart.
10. Validate all Fields and Submit the Hostel Application Form designed in Module-6 using JQuery
11. Develop an HTML document to illustrate each chart with real-time examples.
12. Develop an E-learning website using any CMS(for example WordPress)

References:

1. Deitel and Deitel and Nieto, ―Internet and World Wide Web - How to Program‖, PrenticeHall, 5th Edition, 2011.
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Online Learning Resources/Virtual Labs:

a.HTML: https://html.spec.whatwg.org/multipage/
b.HTML: https://developer.mozilla.org/en-US/docs/Glossary/HTML5
c.CSS: https://www.w3.org/Style/CSS/
d.Bootstrap - CSS Framework: https://getbootstrap.com/
e.BrowserDeveloperTools:https://developer.mozilla.org/enUS/docs/Learn/Common_questions/What_are_browser_developer_tools
f. Javascript: https://developer.mozilla.org/en-US/docs/Web/JavaScript
g. JQuery: https://jquery.com
h. Google Charts: https://developers.google.com/chart
i. Wordpress: https://wordpress.com